

Solutions to

Fumio Hayashi

Econometrics

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Chapters 1–10

Solutions to all the review questions, analytical exercises, empirical exercises and Monte Carlo exercises, with the numerical parts computed by the Python programs in `code/`.

The exercise statements are not reproduced here: read them in Hayashi's *Econometrics*.

Each solution is headed by the number of its exercise in the book; notation, the part letters (a), (b), ... and labels such as (*) or (2) follow the statements there.

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Chapter 1

Finite-Sample Properties of OLS

Solutions to all the exercises of Chapter 1 of Fumio Hayashi, *Econometrics* (Princeton University Press, 2000): the 46 review questions of Sections 1.1–1.7, the seven analytical exercises, the empirical exercise on Nerlove’s data, and the Monte Carlo exercise. Assumptions, propositions and equations cited as ASSUMPTION 1.4, PROPOSITION 1.1 or (1.2.15) refer to the book. The numerical parts are computed by `code/ch1.py`.

Section 1.1 The Classical Linear Regression Model

Review Question 1.1.1

Write $WAGE_i^c = 100 \cdot WAGE_i$ for the wage in cents. Since $\log(100 \cdot WAGE_i) = \log(100) + \log(WAGE_i)$,

$$\log(WAGE_i^c) = [\beta_1 + \log(100)] + \beta_2 S_i + \beta_3 TENURE_i + \beta_4 EXPR_i + \varepsilon_i.$$

Only the intercept changes, from β_1 to $\beta_1 + \log(100) = \beta_1 + 4.6052$. The slope coefficients, which are semi-elasticities (a one-unit change in a regressor changes the wage by a fixed *percentage*, and percentages do not depend on the unit of account), are unchanged, and so is the error term. Because the dependent variable has only been shifted by a constant, the OLS residuals, SSR, s^2 and R^2 from a sample are also unchanged; only the estimated intercept moves by $\log(100)$.

Review Question 1.1.2

Fix $i \neq j$. By the law of iterated expectations, conditioning first on the larger information set $(\mathbf{X}, \varepsilon_j)$,

$$E(\varepsilon_i \varepsilon_j \mid \mathbf{X}) = E[E(\varepsilon_i \varepsilon_j \mid \mathbf{X}, \varepsilon_j) \mid \mathbf{X}] = E[\varepsilon_j E(\varepsilon_i \mid \mathbf{X}, \varepsilon_j) \mid \mathbf{X}],$$

because ε_j is a function of the conditioning variables in the inner expectation. Under random sampling $(\varepsilon_i, \mathbf{x}_i)$ is independent of $(\varepsilon_j, \mathbf{x}_1, \dots, \mathbf{x}_{i-1}, \mathbf{x}_{i+1}, \dots, \mathbf{x}_n)$, so conditioning ε_i on $(\mathbf{X}, \varepsilon_j)$ is the same as conditioning on $\mathbf{x}_i$ alone: $E(\varepsilon_i \mid \mathbf{X}, \varepsilon_j) = E(\varepsilon_i \mid \mathbf{x}_i)$. That quantity is a function of $\mathbf{x}_i$, hence known given $\mathbf{X}$, and can be taken out of the outer expectation:

$$E(\varepsilon_i \varepsilon_j \mid \mathbf{X}) = E(\varepsilon_i \mid \mathbf{x}_i) E(\varepsilon_j \mid \mathbf{X}) = E(\varepsilon_i \mid \mathbf{x}_i) E(\varepsilon_j \mid \mathbf{x}_j),$$

the last step using the same independence argument for ε_j . In particular, if $E(\varepsilon_i \mid \mathbf{x}_i) = 0$ (Assumption 1.2 for a random sample), then the errors are uncorrelated across observations conditional on $\mathbf{X}$, which is the “no correlation between observations” half of ASSUMPTION 1.4: for a random sample that half is implied, not assumed.

Review Question 1.1.3

($\Rightarrow$) By linearity $y_i = \mathbf{x}_i' \beta + \varepsilon_i$. Since $\mathbf{x}_i$ is a row of $\mathbf{X}$, $\mathbf{x}_i' \beta$ is a function of the conditioning variable, so

$$E(y_i | \mathbf{X}) = \mathbf{x}_i' \beta + E(\varepsilon_i | \mathbf{X}) = \mathbf{x}_i' \beta,$$

by strict exogeneity. This is (1.1.20).

($\Leftarrow$) Suppose $E(y_i | \mathbf{X}) = \mathbf{x}_i' \beta$ and define $\varepsilon_i \equiv y_i - \mathbf{x}_i' \beta$. Then $y_i = \mathbf{x}_i' \beta + \varepsilon_i$ identically, which is Assumption 1.1, and

$$E(\varepsilon_i | \mathbf{X}) = E(y_i | \mathbf{X}) - \mathbf{x}_i' \beta = 0,$$

which is Assumption 1.2. So the pair of assumptions is exactly equivalent to the statement that the conditional mean of y_i given the whole regressor matrix is linear in $\mathbf{x}_i$ with coefficient vector β that does not vary with i .

Review Question 1.1.4

Write $\mathbf{x}_i = (1, YD_i)'$. Because the pair is jointly normal, the conditional expectation is linear and the conditional variance is constant:

$$E(CON_i | YD_i) = \beta_1 + \beta_2 YD_i, \quad \text{Var}(CON_i | YD_i) = \sigma^2,$$

with $\beta_2 = \text{Cov}(CON_i, YD_i) / \text{Var}(YD_i)$, $\beta_1 = E(CON_i) - \beta_2 E(YD_i)$ and $\sigma^2 = \text{Var}(CON_i)(1 - \rho^2)$. Since the distribution is the same for every i , these $(\beta_1, \beta_2, \sigma^2)$ do not depend on i .

Assumption 1.1. Define $\varepsilon_i \equiv CON_i - \beta_1 - \beta_2 YD_i$; then $CON_i = \mathbf{x}_i' \beta + \varepsilon_i$ with a coefficient vector common to all i .

Assumption 1.2. ε_i is a function of (CON_i, YD_i) , which is independent of YD_j for $j \neq i$; hence $E(\varepsilon_i | \mathbf{X}) = E(\varepsilon_i | YD_i) = E(CON_i | YD_i) - \beta_1 - \beta_2 YD_i = 0$.

Assumption 1.4. By the same independence, $E(\varepsilon_i^2 | \mathbf{X}) = E(\varepsilon_i^2 | YD_i) = \text{Var}(CON_i | YD_i) = \sigma^2 > 0$, which does not depend on YD_i precisely because the distribution is normal; and by Review Question 1.1.2, $E(\varepsilon_i \varepsilon_j | \mathbf{X}) = E(\varepsilon_i | YD_i) E(\varepsilon_j | YD_j) = 0$ for $i \neq j$.

Assumption 1.3 holds unless YD_i takes the same value for all i , an event of probability zero for a continuous distribution.

Remark. Identical distribution across i is essential. If (CON_i, YD_i) were normal with moments varying with i , the conditional mean would still be linear but with i -specific coefficients, and Assumption 1.1 (a common β) would fail.

Review Question 1.1.5

$\mathbf{X} = (\mathbf{1} \ \mathbf{x}_2)$ has rank 2 if and only if its two columns are linearly independent, that is, unless there is $(c_1, c_2) \neq \mathbf{0}$ with $c_1 \mathbf{1} + c_2 \mathbf{x}_2 = \mathbf{0}$. If $c_2 = 0$ this forces $c_1 = 0$; so a violation requires $c_2 \neq 0$ and $\mathbf{x}_2 = -(c_1/c_2) \mathbf{1}$, i.e. x_{i2} takes the same value for all i . Hence $\text{rank}(\mathbf{X}) = 2$ with probability one if and only if $\text{Prob}\{x_{i2} \neq x_{j2} \text{ for some } i \neq j\} = 1$.

Review Question 1.1.6

By the law of iterated expectations and strict exogeneity, $E(\varepsilon_i) = E[E(\varepsilon_i | \mathbf{X})] = 0$; so the unconditional variances and covariances are the unconditional second moments. Applying iterated expectations again to (1.1.12) and the no-correlation part of ASSUMPTION 1.4,

$$\text{Var}(\varepsilon_i) = E(\varepsilon_i^2) = E[E(\varepsilon_i^2 | \mathbf{X})] = E(\sigma^2) = \sigma^2, \quad \text{Cov}(\varepsilon_i, \varepsilon_j) = E(\varepsilon_i \varepsilon_j) = E[E(\varepsilon_i \varepsilon_j | \mathbf{X})] = 0.$$

The converse is false: unconditional homoskedasticity does not imply the conditional version, which is why Assumption 1.4 is a restriction even for a random sample.

Section 1.2 The Algebra of Least Squares

Review Question 1.2.1

Let $\mathbf{c} \neq \mathbf{0}$ be $K \times 1$ and put $\mathbf{z} \equiv \mathbf{X}\mathbf{c}$. Then

$$\mathbf{c}'\mathbf{X}'\mathbf{X}\mathbf{c} = \mathbf{z}'\mathbf{z} = \sum_{i=1}^n z_i^2 \geq 0.$$

Equality holds only if $\mathbf{z} = \mathbf{0}$, i.e. $\mathbf{X}\mathbf{c} = \mathbf{0}$, which is impossible for $\mathbf{c} \neq \mathbf{0}$ when $\text{rank}(\mathbf{X}) = K$. Hence $\mathbf{c}'\mathbf{X}'\mathbf{X}\mathbf{c} > 0$ for all $\mathbf{c} \neq \mathbf{0}$, so $\mathbf{X}'\mathbf{X}$ is positive definite and therefore nonsingular, which is what makes (1.2.5) well defined.

Review Question 1.2.2

The (k, ℓ) element of $\mathbf{X}'\mathbf{X}$ is $\sum_{i=1}^n x_{ik}x_{i\ell}$, and $x_{ik}x_{i\ell}$ is the (k, ℓ) element of the $K \times K$ matrix $\mathbf{x}_i\mathbf{x}_i'$; summing element by element, $\mathbf{X}'\mathbf{X} = \sum_i \mathbf{x}_i\mathbf{x}_i'$. Likewise the k -th element of $\mathbf{X}'\mathbf{y}$ is $\sum_i x_{ik}y_i$, the k -th element of $\sum_i \mathbf{x}_i y_i$. Dividing by n gives $\mathbf{S}_{\mathbf{xx}}$ and $\mathbf{s}_{\mathbf{xy}}$ of (1.2.6). Thus the OLS estimator $\mathbf{b} = \mathbf{S}_{\mathbf{xx}}^{-1}\mathbf{s}_{\mathbf{xy}}$ depends on the sample only through these sample second moments.

Review Question 1.2.3

With $\mathbf{x}_i = (1, x_{i2})'$,

$$\mathbf{S}_{\mathbf{xx}} = \frac{1}{n} \sum_i \mathbf{x}_i\mathbf{x}_i' = \begin{bmatrix} 1 & \bar{x}_2 \\ \bar{x}_2 & \frac{1}{n} \sum_i x_{i2}^2 \end{bmatrix}, \quad \mathbf{s}_{\mathbf{xy}} = \frac{1}{n} \sum_i \mathbf{x}_i y_i = \begin{bmatrix} \bar{y} \\ \frac{1}{n} \sum_i x_{i2} y_i \end{bmatrix}.$$

The two normal equations $\mathbf{S}_{\mathbf{xx}}\mathbf{b} = \mathbf{s}_{\mathbf{xy}}$ are

$$b_1 + \bar{x}_2 b_2 = \bar{y}, \quad \bar{x}_2 b_1 + \left(\frac{1}{n} \sum_i x_{i2}^2 \right) b_2 = \frac{1}{n} \sum_i x_{i2} y_i.$$

The first gives $b_1 = \bar{y} - \bar{x}_2 b_2$. Substituting it into the second,

$$\left(\frac{1}{n} \sum_i x_{i2}^2 - \bar{x}_2^2 \right) b_2 = \frac{1}{n} \sum_i x_{i2} y_i - \bar{x}_2 \bar{y},$$

and using $\frac{1}{n} \sum_i x_{i2}^2 - \bar{x}_2^2 = \frac{1}{n} \sum_i (x_{i2} - \bar{x}_2)^2$ and $\frac{1}{n} \sum_i x_{i2} y_i - \bar{x}_2 \bar{y} = \frac{1}{n} \sum_i (x_{i2} - \bar{x}_2)(y_i - \bar{y})$,

$$b_2 = \frac{\frac{1}{n} \sum_i (x_{i2} - \bar{x}_2)(y_i - \bar{y})}{\frac{1}{n} \sum_i (x_{i2} - \bar{x}_2)^2} = \frac{\widehat{\text{Cov}}(x_2, y)}{\widehat{\text{Var}}(x_2)}, \quad b_1 = \bar{y} - \bar{x}_2 b_2.$$

The denominator is positive precisely under Assumption 1.3 as described in Review Question 1.1.5.

Review Question 1.2.4

Write $\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and $\mathbf{M} = \mathbf{I}_n - \mathbf{P}$. Since $(\mathbf{X}'\mathbf{X})^{-1}$ is symmetric,

$$\mathbf{P}' = \mathbf{X}[(\mathbf{X}'\mathbf{X})^{-1}]'\mathbf{X}' = \mathbf{P}, \quad \mathbf{P}\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \underbrace{\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}}_{=\mathbf{I}_K} \mathbf{X}' = \mathbf{P}.$$

Hence $\mathbf{M}' = \mathbf{I} - \mathbf{P}' = \mathbf{M}$ and $\mathbf{M}\mathbf{M} = \mathbf{I} - 2\mathbf{P} + \mathbf{P}\mathbf{P} = \mathbf{I} - \mathbf{P} = \mathbf{M}$: both are symmetric and idempotent, which is (1.2.9). Next $\mathbf{P}\mathbf{X} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{X}$, which is (1.2.10), and therefore $\mathbf{M}\mathbf{X} = \mathbf{X} - \mathbf{P}\mathbf{X} = \mathbf{0}$, which is (1.2.11). Geometrically $\mathbf{P}$ projects onto the column space of $\mathbf{X}$ and $\mathbf{M}$ onto its orthogonal complement.

Review Question 1.2.5

(a) By (1.2.5), $\hat{\mathbf{y}} = \mathbf{X}\mathbf{b} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{P}\mathbf{y}$, so $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{M}\mathbf{y}$. Substituting $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ and using $\mathbf{M}\mathbf{X} = \mathbf{0}$,

$$\mathbf{e} = \mathbf{M}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \mathbf{M}\boldsymbol{\varepsilon}.$$

(b) Consequently $\text{SSR} = \mathbf{e}'\mathbf{e} = (\mathbf{M}\boldsymbol{\varepsilon})'(\mathbf{M}\boldsymbol{\varepsilon}) = \boldsymbol{\varepsilon}'\mathbf{M}'\mathbf{M}\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}$ by symmetry and idempotency. The residual vector is thus a linear function of the errors alone, which is what makes the finite-sample results of Section 1.3 possible.

Review Question 1.2.6

Dependent variable. Replace $\mathbf{y}$ by $c\mathbf{y}$ ($c \neq 0$). Since $\mathbf{P}$ and $\mathbf{M}$ depend only on $\mathbf{X}$, the new residual vector is $c\mathbf{e}$ and the new fitted values are $c\hat{\mathbf{y}}$; deviations from the mean also scale by c . Both numerator and denominator of $1 - R^2 = \sum_i e_i^2 / \sum_i (y_i - \bar{y})^2$ are multiplied by c^2 , so R^2 (and R_{uc}^2) is unchanged. The estimates themselves scale: $\mathbf{b} \mapsto c\mathbf{b}$, $s \mapsto |c|s$.

Regressors. A change of units replaces $\mathbf{X}$ by $\mathbf{X}\mathbf{A}$ with $\mathbf{A}$ diagonal and nonsingular (the argument works for any nonsingular $\mathbf{A}$). Then

$$\mathbf{P}_{\mathbf{X}\mathbf{A}} = \mathbf{X}\mathbf{A}(\mathbf{A}'\mathbf{X}'\mathbf{X}\mathbf{A})^{-1}\mathbf{A}'\mathbf{X}' = \mathbf{X}\mathbf{A}\mathbf{A}^{-1}(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{A}')^{-1}\mathbf{A}'\mathbf{X}' = \mathbf{P},$$

so fitted values and residuals, hence R^2 and R_{uc}^2 , are unchanged; only the coefficients are rescaled, $\mathbf{b} \mapsto \mathbf{A}^{-1}\mathbf{b}$. (R^2 is invariant to any nonsingular reparameterization of the regressors, because it depends on $\mathbf{X}$ only through its column space.)

Review Question 1.2.7

By definition $1 - R^2 = \mathbf{e}'\mathbf{e} / \sum_i (y_i - \bar{y})^2$ and $1 - R_{uc}^2 = \mathbf{e}'\mathbf{e} / \mathbf{y}'\mathbf{y}$. Using the identity $\sum_i (y_i - \bar{y})^2 = \mathbf{y}'\mathbf{y} - n\bar{y}^2$, i.e. $\mathbf{y}'\mathbf{y} = \sum_i (y_i - \bar{y})^2 + n\bar{y}^2$,

$$1 - R^2 = \frac{\mathbf{e}'\mathbf{e}}{\sum_i (y_i - \bar{y})^2} = \frac{\mathbf{y}'\mathbf{y}}{\sum_i (y_i - \bar{y})^2} \cdot \frac{\mathbf{e}'\mathbf{e}}{\mathbf{y}'\mathbf{y}} = \left[1 + \frac{n\bar{y}^2}{\sum_i (y_i - \bar{y})^2} \right] (1 - R_{uc}^2).$$

Since the bracket is at least 1, $R^2 \leq R_{uc}^2$: the uncentered R^2 flatters the fit whenever the dependent variable has a nonzero mean, and the two coincide only when $\bar{y} = 0$.

Review Question 1.2.8

From Review Question 1.2.5, $\mathbf{e}'\mathbf{e} = \mathbf{y}'\mathbf{M}\mathbf{y} = \mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{P}\mathbf{y}$. Therefore

$$R_{uc}^2 = 1 - \frac{\mathbf{e}'\mathbf{e}}{\mathbf{y}'\mathbf{y}} = \frac{\mathbf{y}'\mathbf{y} - (\mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{P}\mathbf{y})}{\mathbf{y}'\mathbf{y}} = \frac{\mathbf{y}'\mathbf{P}\mathbf{y}}{\mathbf{y}'\mathbf{y}} = \frac{\hat{\mathbf{y}}'\hat{\mathbf{y}}}{\mathbf{y}'\mathbf{y}},$$

the last equality because $\hat{\mathbf{y}}'\hat{\mathbf{y}} = \mathbf{y}'\mathbf{P}\mathbf{P}\mathbf{y} = \mathbf{y}'\mathbf{P}\mathbf{y}$. It is the squared cosine of the angle between $\mathbf{y}$ and the column space of $\mathbf{X}$, so $0 \leq R_{uc}^2 \leq 1$ always.

Review Question 1.2.9

The estimator itself is $\mathbf{b} = \mathbf{S}_{\mathbf{xx}}^{-1}\mathbf{s}_{\mathbf{xy}}$ by (1.2.6). Expanding the sum of squared residuals and using the normal equations $\mathbf{S}_{\mathbf{xx}}\mathbf{b} = \mathbf{s}_{\mathbf{xy}}$,

$$\frac{\text{SSR}}{n} = \frac{(\mathbf{y} - \mathbf{X}\mathbf{b})'(\mathbf{y} - \mathbf{X}\mathbf{b})}{n} = \frac{\mathbf{y}'\mathbf{y}}{n} - 2\mathbf{b}'\mathbf{s}_{\mathbf{xy}} + \mathbf{b}'\mathbf{S}_{\mathbf{xx}}\mathbf{b} = \frac{\mathbf{y}'\mathbf{y}}{n} - \mathbf{b}'\mathbf{s}_{\mathbf{xy}}.$$

Then $s^2 = \text{SSR} / (n - K) = \frac{n}{n-K}(\mathbf{y}'\mathbf{y}/n - \mathbf{b}'\mathbf{s}_{\mathbf{xy}})$ and, since $\frac{1}{n} \sum_i (y_i - \bar{y})^2 = \mathbf{y}'\mathbf{y}/n - \bar{y}^2$,

$$R^2 = 1 - \frac{\mathbf{y}'\mathbf{y}/n - \mathbf{b}'\mathbf{s}_{\mathbf{xy}}}{\mathbf{y}'\mathbf{y}/n - \bar{y}^2}.$$

So one pass through the data, accumulating $\sum_i \mathbf{x}_i \mathbf{x}_i'$, $\sum_i \mathbf{x}_i y_i$ and $\sum_i y_i^2$, suffices; when a constant is included, $\bar{y}$ is already an element of $\mathbf{s}_{\mathbf{xy}}$. This is how regression packages avoid storing the whole data matrix.

Section 1.3 Finite-Sample Properties of OLS**Review Question 1.3.1**

It is used wherever $(\mathbf{X}'\mathbf{X})^{-1}$ is written. Assumption 1.3 makes $\mathbf{X}'\mathbf{X}$ positive definite (Review Question 1.2.1) and hence invertible, so that: the OLS estimator $\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ exists and is unique; the sampling error identity $\mathbf{b} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$ of (1.2.14), on which unbiasedness rests, is available; and $\text{Var}(\mathbf{b} | \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ is well defined. In Proposition 1.2 it delivers $\text{tr}(\mathbf{M}) = n - K$ through $\text{tr}(\mathbf{P}) = \text{tr}(\mathbf{I}_K) = K$, and it guarantees $n - K > 0$ so that $s^2 = \text{SSR} / (n - K)$ makes sense. Without it the normal equations still hold but have many solutions, and no individual coefficient is estimable.

Review Question 1.3.2

Take the hint's two-point slope

$$\tilde{\beta}_2 = \frac{CON_2 - CON_1}{YD_2 - YD_1} = \sum_{i=1}^n a_i CON_i, \quad -a_1 = a_2 = \frac{1}{YD_2 - YD_1}, \quad a_i = 0 \quad (i \geq 3).$$

The weights are functions of $\mathbf{X}$ alone, so $\tilde{\beta}_2$ is linear in $\mathbf{y}$ in the sense required by the Gauss–Markov theorem. It is unbiased conditional on $\mathbf{X}$:

$$E(\tilde{\beta}_2 | \mathbf{X}) = \frac{(\beta_1 + \beta_2 YD_2) - (\beta_1 + \beta_2 YD_1)}{YD_2 - YD_1} = \beta_2.$$

Its conditional variance is $\text{Var}(\tilde{\beta}_2 | \mathbf{X}) = \sigma^2 \sum_i a_i^2 = 2\sigma^2 / (YD_2 - YD_1)^2$, which by Proposition 1.1(c) cannot be smaller than $\text{Var}(b_2 | \mathbf{X}) = \sigma^2 / \sum_i (YD_i - \overline{YD})^2$; it uses only two of the n observations, and is dramatically worse when the two disposable incomes happen to be close together.

Review Question 1.3.3

Yes, and trivially so. The constant estimator $\tilde{\beta} \equiv \mathbf{c}$ for a fixed vector $\mathbf{c}$ is linear in $\mathbf{y}$ (with coefficient matrix $\mathbf{A} = \mathbf{0}$) and has $\text{Var}(\tilde{\beta} | \mathbf{X}) = \mathbf{0}$, the smallest variance a random vector can have. Of course it is useless: its bias $\mathbf{c} - \beta$ is as large as one likes. The Gauss–Markov theorem is a statement about the *unbiased* class, and the example shows the qualification cannot be dropped. It also shows why mean squared error, $E[(\tilde{\beta} - \beta)(\tilde{\beta} - \beta)' | \mathbf{X}] = \text{Var}(\tilde{\beta} | \mathbf{X}) + \text{bias} \cdot \text{bias}'$, is the relevant criterion once biased estimators are admitted; shrinkage estimators exploit exactly this trade-off.

Review Question 1.3.4

(a) Add and subtract $E(\tilde{\beta})$ inside the definition of the variance:

$$\begin{aligned} \text{Var}(\tilde{\beta}) &= E[(\tilde{\beta} - E\tilde{\beta})(\tilde{\beta} - E\tilde{\beta})'] \\ &= E\left[\{(\tilde{\beta} - E(\tilde{\beta} | \mathbf{X})) + (E(\tilde{\beta} | \mathbf{X}) - E\tilde{\beta})\}\{\cdots\}'\right]. \end{aligned}$$

Expanding, the cross terms vanish by iterated expectations: conditionally on $\mathbf{X}$ the second factor is fixed and the first has mean zero, so $E[(\tilde{\beta} - E(\tilde{\beta} | \mathbf{X}))(E(\tilde{\beta} | \mathbf{X}) - E\tilde{\beta})'] = E[E(\tilde{\beta} - E(\tilde{\beta} | \mathbf{X}) | \mathbf{X})(\cdots)'] = \mathbf{0}$. The two surviving terms are $E[\text{Var}(\tilde{\beta} | \mathbf{X})]$ and $\text{Var}[E(\tilde{\beta} | \mathbf{X})]$.

(b) Let $\tilde{\beta}$ be linear and unbiased conditional on $\mathbf{X}$, so $E(\tilde{\beta} | \mathbf{X}) = \beta$ and the second term in (a) is zero, for $\tilde{\beta}$ and for $\mathbf{b}$ alike. By Proposition 1.1(c), $\text{Var}(\tilde{\beta} | \mathbf{X}) - \text{Var}(\mathbf{b} | \mathbf{X})$ is positive semidefinite for every $\mathbf{X}$; taking expectations preserves this (for any fixed $\mathbf{c}$, $\mathbf{c}'[\text{Var}(\tilde{\beta} | \mathbf{X}) - \text{Var}(\mathbf{b} | \mathbf{X})]\mathbf{c} \geq 0$ pointwise, hence in expectation). Therefore

$$\text{Var}(\tilde{\beta}) = E[\text{Var}(\tilde{\beta} | \mathbf{X})] \geq E[\text{Var}(\mathbf{b} | \mathbf{X})] = \text{Var}(\mathbf{b}),$$

which is (1.3.3): the OLS estimator is efficient unconditionally as well as conditionally.

Review Question 1.3.5

$\varepsilon'\varepsilon/n$ would do: $E(\varepsilon'\varepsilon/n | \mathbf{X}) = \frac{1}{n} \sum_{i=1}^n E(\varepsilon_i^2 | \mathbf{X}) = \sigma^2$ by (1.1.12), so it is unbiased both conditionally and unconditionally, with no degrees-of-freedom correction. The correction in $s^2 = \mathbf{e}'\mathbf{e}/(n - K)$ is needed only because $\mathbf{e} = \mathbf{M}\varepsilon$ is the projection of ε on an $(n - K)$ -dimensional subspace, which makes $E(\mathbf{e}'\mathbf{e} | \mathbf{X}) = \sigma^2(n - K)$: fitting K coefficients uses up K dimensions' worth of variation.

Review Question 1.3.6

Conditional on $\mathbf{X}$, $E(\mathbf{b} | \mathbf{X}) = \beta$ and $E(\mathbf{e} | \mathbf{X}) = \mathbf{M}E(\varepsilon | \mathbf{X}) = \mathbf{0}$, so with $\mathbf{A} \equiv (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ we have $\mathbf{b} - E(\mathbf{b} | \mathbf{X}) = \mathbf{A}\varepsilon$ and $\mathbf{e} - E(\mathbf{e} | \mathbf{X}) = \mathbf{M}\varepsilon$. Hence

$$\text{Cov}(\mathbf{b}, \mathbf{e} | \mathbf{X}) = E[\mathbf{A}\varepsilon\varepsilon'\mathbf{M} | \mathbf{X}] = \mathbf{A}E(\varepsilon\varepsilon' | \mathbf{X})\mathbf{M} = \sigma^2\mathbf{A}\mathbf{M} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{M} = \mathbf{0},$$

since $\mathbf{A}$ and $\mathbf{M}$ are functions of $\mathbf{X}$, $E(\varepsilon\varepsilon' | \mathbf{X}) = \sigma^2 \mathbf{I}_n$ by Assumption 1.4, and $\mathbf{X}'\mathbf{M} = (\mathbf{M}\mathbf{X})' = \mathbf{0}$ by (1.2.11). Under normality (Section 1.4) zero covariance becomes independence, which is what makes $\mathbf{b}$ and s^2 independent and the t -ratio a t random variable.

Review Question 1.3.7

$p_i = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i$ is the i -th diagonal element of $\mathbf{P}$. Because $\mathbf{P}$ is positive semidefinite, $p_i = \mathbf{u}_i'\mathbf{P}\mathbf{u}_i \geq 0$ with $\mathbf{u}_i$ the i -th unit vector. Because $\mathbf{P}$ is symmetric and idempotent, $\mathbf{P} = \mathbf{P}'\mathbf{P}$, so

$$p_i = \sum_{j=1}^n p_{ij}^2 = p_i^2 + \sum_{j \neq i} p_{ij}^2 \geq p_i^2,$$

which forces $p_i \leq 1$. Finally $\sum_i p_i = \text{tr}(\mathbf{P}) = \text{tr}[\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'] = \text{tr}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}] = \text{tr}(\mathbf{I}_K) = K$. Hence the average leverage is K/n , and p_i close to 1 marks an observation whose own fitted value is determined almost entirely by itself.

Section 1.4 Hypothesis Testing under Normality

Review Question 1.4.1

Normality of $\mathbf{b}$: *no.* Assumptions 1.1–1.5 give $\mathbf{b} | \mathbf{X} \sim N(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$. The marginal distribution is the mixture $\int N(\beta, \sigma^2(\mathbf{x}'\mathbf{x})^{-1}) dF(\mathbf{x})$, and a mixture of normals with different variances is not normal (it has excess kurtosis unless $\mathbf{X}'\mathbf{X}$ is degenerate). Nothing has been assumed about the distribution of $\mathbf{X}$, so nothing more can be said.

Independence of the test statistics from $\mathbf{X}$: *yes.* Conditional on $\mathbf{X}$, $z_k \sim N(0, 1)$, $t_k \sim t(n - K)$ and $F \sim F(\#r, n - K)$ — distributions that do not involve $\mathbf{X}$. A random variable whose conditional distribution given $\mathbf{X}$ is the same for every value of $\mathbf{X}$ is independent of $\mathbf{X}$, and its marginal distribution equals that common conditional distribution. This is exactly why the finite-sample theory of testing can dispense with any assumption about the distribution of the regressors.

Review Question 1.4.2

Review Question 1.2.9 covers $\mathbf{b}$, SSR, s^2 and R^2 . For the standard error, note $(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{n}\mathbf{S}_{\mathbf{xx}}^{-1}$, so

$$\text{SE}(b_k) = \sqrt{s^2[(\mathbf{X}'\mathbf{X})^{-1}]_{kk}} = \sqrt{\frac{s^2}{n}[\mathbf{S}_{\mathbf{xx}}^{-1}]_{kk}},$$

a function of the same four sample moments. The t -ratio $t_k = (b_k - \bar{\beta}_k)/\text{SE}(b_k)$ and, for a linear hypothesis $\mathbf{R}\beta = \mathbf{r}$, the F -ratio (1.4.9) are then available as well, since $\mathbf{R}\widehat{\text{Var}}(\mathbf{b} | \mathbf{X})\mathbf{R}' = \frac{s^2}{n}\mathbf{R}\mathbf{S}_{\mathbf{xx}}^{-1}\mathbf{R}'$.

Review Question 1.4.3

Let $\mathbf{z} \neq \mathbf{0}$ be $\#r \times 1$. Since $\mathbf{R}$ is $\#r \times K$ with $\text{rank}(\mathbf{R}) = \#r$, its rows are linearly independent, so $\mathbf{R}'\mathbf{z} \neq \mathbf{0}$. The matrix $(\mathbf{X}'\mathbf{X})^{-1}$ is positive definite, being the inverse of a positive definite matrix (if $\mathbf{A}$ is positive definite and $\mathbf{w} \neq \mathbf{0}$, then $\mathbf{w}'\mathbf{A}^{-1}\mathbf{w} = \mathbf{v}'\mathbf{A}\mathbf{v} > 0$ with $\mathbf{v} = \mathbf{A}^{-1}\mathbf{w} \neq \mathbf{0}$). Hence

$$\mathbf{z}'\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{z} = (\mathbf{R}'\mathbf{z})'(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{R}'\mathbf{z}) > 0,$$

so the matrix is positive definite, therefore nonsingular, and the F -ratio (1.4.9) is well defined and nonnegative.

Review Question 1.4.4

Under the null $\beta_k = \bar{\beta}_k$ (and Assumptions 1.1–1.5), $t_k \sim t(n - K)$ both conditionally and unconditionally, by Proposition 1.3 and Review Question 1.4.1. The critical value is defined by $\text{Prob}\{t(n - K) > t_\alpha\} = \alpha$, so

$$\text{Prob}\{\text{reject} \mid H_0\} = \text{Prob}\{t_k > t_\alpha(n - K)\} = \alpha.$$

That is the size. Against the one-sided alternative $\beta_k > \bar{\beta}_k$ this test is more powerful than the two-tailed test, because for a given α the critical value $t_\alpha < t_{\alpha/2}$; the price is zero power against $\beta_k < \bar{\beta}_k$.

Remark. If the null is the composite $\beta_k \leq \bar{\beta}_k$, the rejection probability is largest at the boundary $\beta_k = \bar{\beta}_k$ (the t -ratio is stochastically increasing in β_k), so the size, defined as the supremum of the rejection probability over the null, is again α .

Review Question 1.4.5

If $t \sim t(m)$ then $t^2 \sim F(1, m)$, because $t = Z/\sqrt{W/m}$ with $Z \sim N(0, 1)$ independent of $W \sim \chi^2(m)$, so $t^2 = (Z^2/1)/(W/m)$ is the ratio of independent chi-squares divided by their degrees of freedom. Hence $\{|t| > c\} = \{t^2 > c^2\}$, and equating tail probabilities gives $F_\alpha(1, m) = [t_{\alpha/2}(m)]^2$. Numerically (code/ch1.py):

$m = n - K$	α	$t_{\alpha/2}(m)$	$[t_{\alpha/2}(m)]^2$	$F_\alpha(1, m)$
30	0.05	2.04227	4.17088	4.17088
141	0.05	1.97693	3.90826	3.90826
10	0.01	3.16927	10.04429	10.04429

Consequently the two-tailed t -test of a single restriction and the F -test of the same restriction are the same test, a fact also implied by (1.4.9) with $\#r = 1$.

Review Question 1.4.6

The statement confuses a joint test with a sequence of individual tests. If $\#r$ restrictions are each tested at level α , the probability of rejecting *at least one* when all are true is far above α — for independent tests $1 - (1 - \alpha)^{\#r}$, which is 40% for $\alpha = 5\%$ and $\#r = 10$. That is an argument *for* a joint test, not against testing the joint hypothesis: the F -test of (1.4.9) has size exactly α for the whole set of restrictions, however many there are, because it takes the sampling covariances among the estimated restrictions into account.

The two procedures can also disagree in the other direction. When regressors are strongly correlated, each coefficient may be imprecisely estimated (no t -ratio significant) while the F -test decisively rejects the joint null, since the data pin down a linear combination of the coefficients even though they leave each one uncertain. Neither test is a substitute for the other; the one to use is the one whose null is the hypothesis of interest.

Review Question 1.4.7

From the proof of Proposition 1.3, $(n - K)s^2/\sigma^2 = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}/\sigma^2 \sim \chi^2(n - K)$ conditional on $\mathbf{X}$. A $\chi^2(m)$ variable has variance $2m$, so

$$\text{Var}\left(\frac{(n - K)s^2}{\sigma^2} \mid \mathbf{X}\right) = 2(n - K) \implies \text{Var}(s^2 \mid \mathbf{X}) = \frac{\sigma^4}{(n - K)^2} 2(n - K) = \frac{2\sigma^4}{n - K}.$$

The variance does not depend on $\mathbf{X}$, so it is also the unconditional variance; it vanishes as $n \rightarrow \infty$, which is the finite-sample counterpart of the consistency of s^2 proved in Chapter 2. Note that the ML estimator $\hat{\sigma}^2 = \frac{n-K}{n}s^2$ has variance $2\sigma^4(n - K)/n^2$, smaller than $\text{Var}(s^2)$, and mean squared error smaller still for $K \geq 1$: unbiasedness is not free.

Section 1.5 Relation to Maximum Likelihood
Review Question 1.5.1

Since $f(\mathbf{z}; \tilde{\theta})$ is a density for every $\tilde{\theta}$, $\int f(\mathbf{z}; \tilde{\theta}) d\mathbf{z} = 1$ identically in $\tilde{\theta}$. Differentiating both sides with respect to $\tilde{\theta}$ and interchanging differentiation and integration (this is what the regularity conditions buy),

$$\int \frac{\partial f(\mathbf{z}; \theta)}{\partial \theta} d\mathbf{z} = \mathbf{0}.$$

Now use $\frac{\partial \log f(\mathbf{z}; \theta)}{\partial \tilde{\theta}} = \frac{1}{f(\mathbf{z}; \theta)} \frac{\partial f(\mathbf{z}; \theta)}{\partial \tilde{\theta}}$:

$$\mathbb{E}\left[\frac{\partial \log f(\mathbf{z}; \theta)}{\partial \tilde{\theta}}\right] = \int \frac{1}{f(\mathbf{z}; \theta)} \frac{\partial f(\mathbf{z}; \theta)}{\partial \tilde{\theta}} f(\mathbf{z}; \theta) d\mathbf{z} = \int \frac{\partial f(\mathbf{z}; \theta)}{\partial \tilde{\theta}} d\mathbf{z} = \mathbf{0}.$$

The expectation is taken at the true θ , which is essential: the score has mean zero only there, and that is what makes the likelihood equations sensible moment conditions. The same interchange applied twice yields the information matrix equality (1.5.11).

Review Question 1.5.2

Take logs in (1.5.2):

$$\log L(\tilde{\zeta}) = \log f(\mathbf{y} \mid \mathbf{X}; \tilde{\theta}) + \log f(\mathbf{X}; \tilde{\psi}).$$

The two terms involve different parameters, so the maximization separates: $\tilde{\theta}$ enters only the first term, and maximizing it gives exactly the conditional ML estimator of Proposition 1.5,

$$\hat{\beta}_{\text{ML}} = \mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}, \quad \hat{\sigma}_{\text{ML}}^2 = \frac{\text{SSR}}{n} = \frac{n - K}{n}s^2,$$

whatever parametric family is chosen for the marginal density of $\mathbf{X}$ (provided ψ and θ are variation free). This is why one may work with the conditional likelihood throughout.

For the bound, the hint's $\partial^2 \log L / (\partial \tilde{\theta} \partial \tilde{\psi}') = \mathbf{0}$ makes the information matrix block diagonal between θ and ψ , so the θ block of $I(\zeta)^{-1}$ is $[I_{\theta\theta}]^{-1}$: knowing ψ would not help. From (1.5.14A), now taking the expectation over $\mathbf{X}$ as well,

$$I_{\beta\beta} = -\mathbb{E}\left[\frac{\partial^2 \log L}{\partial \tilde{\beta} \partial \tilde{\beta}'}\right] = \frac{\mathbb{E}(\mathbf{X}'\mathbf{X})}{\sigma^2}, \quad \text{Cramer-Rao bound for } \beta = \sigma^2 [\mathbb{E}(\mathbf{X}'\mathbf{X})]^{-1},$$

the $\beta\gamma$ block again vanishing because $E[\mathbf{X}'\boldsymbol{\varepsilon}] = \mathbf{0}$.

Remark. The OLS estimator does *not* attain this unconditional bound. Its unconditional variance is $\text{Var}(\mathbf{b}) = \sigma^2 E[(\mathbf{X}'\mathbf{X})^{-1}]$, and matrix convexity of $\mathbf{A} \mapsto \mathbf{A}^{-1}$ (Jensen's inequality) gives $E[(\mathbf{X}'\mathbf{X})^{-1}] \geq [E(\mathbf{X}'\mathbf{X})]^{-1}$, with equality only if $\mathbf{X}'\mathbf{X}$ is nonrandom. There is no contradiction with Proposition 1.6, which is a statement about the *conditional* distribution — the bound (1.5.16), $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$, is attained conditionally on $\mathbf{X}$. The gap arises because the unconditional problem asks for a single number to bound an average over designs, and averaging over designs is not the same as inverting the average design.

Review Question 1.5.3

For fixed $\tilde{\beta}$ write $S(\tilde{\beta}) \equiv (\mathbf{y} - \mathbf{X}\tilde{\beta})'(\mathbf{y} - \mathbf{X}\tilde{\beta}) > 0$. Then

$$\frac{\partial \log L}{\partial \tilde{\gamma}} = -\frac{n}{2\tilde{\gamma}} + \frac{S(\tilde{\beta})}{2\tilde{\gamma}^2} = 0 \implies \tilde{\gamma} = \frac{S(\tilde{\beta})}{n},$$

which is the unique maximum: $\log L \rightarrow -\infty$ as $\tilde{\gamma} \downarrow 0$ or $\tilde{\gamma} \rightarrow \infty$, and the derivative is positive below and negative above this point. Substituting back,

$$\log L\left(\tilde{\beta}, \frac{S(\tilde{\beta})}{n}\right) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \frac{S(\tilde{\beta})}{n} - \frac{n}{2} = -\frac{n}{2} [1 + \log(2\pi)] - \frac{n}{2} \log \frac{(\mathbf{y} - \mathbf{X}\tilde{\beta})'(\mathbf{y} - \mathbf{X}\tilde{\beta})}{n},$$

as claimed. Since the concentrated likelihood is a strictly decreasing function of $S(\tilde{\beta})$, maximizing it is the same as minimizing the sum of squared residuals: the ML estimator of β is $\mathbf{b}$, and then $\hat{\sigma}^2 = \text{SSR}/n$, reproducing Proposition 1.5 by the other order of maximization.

Review Question 1.5.4

Everything is conditional on $\mathbf{X}$, and $\gamma \equiv \sigma^2$. From (1.5.13) the score at the true parameter is

$$\mathbf{s}(\theta) = \begin{bmatrix} \frac{1}{\sigma^2} \mathbf{X}'\boldsymbol{\varepsilon} \\ -\frac{n}{2\sigma^2} + \frac{\boldsymbol{\varepsilon}'\boldsymbol{\varepsilon}}{2\sigma^4} \end{bmatrix}.$$

Its outer product has expectation (using $E(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' | \mathbf{X}) = \sigma^2 \mathbf{I}_n$, $E(\varepsilon_i^3 | \mathbf{X}) = 0$ and $E[(\boldsymbol{\varepsilon}'\boldsymbol{\varepsilon})^2 | \mathbf{X}] = n(n+2)\sigma^4$ for normal errors)

$$E[\mathbf{s}\mathbf{s}' | \mathbf{X}] = \begin{bmatrix} \frac{\mathbf{X}'\mathbf{X}}{\sigma^2} & \mathbf{0} \\ \mathbf{0}' & \frac{n}{2\sigma^4} \end{bmatrix},$$

because $E[\mathbf{X}'\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} | \mathbf{X}] = \mathbf{0}$ by symmetry of the normal distribution and $\text{Var}(\boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} | \mathbf{X}) = 2n\sigma^4$. The negative expected Hessian from (1.5.14) is

$$-E\left[\frac{\partial^2 \log L}{\partial \tilde{\theta} \partial \tilde{\theta}'} \mid \mathbf{X}\right] = \begin{bmatrix} \frac{\mathbf{X}'\mathbf{X}}{\sigma^2} & \mathbf{0} \\ \mathbf{0}' & -\frac{n}{2\sigma^4} + \frac{E(\boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} | \mathbf{X})}{\sigma^6} \end{bmatrix} = \begin{bmatrix} \frac{\mathbf{X}'\mathbf{X}}{\sigma^2} & \mathbf{0} \\ \mathbf{0}' & \frac{n}{2\sigma^4} \end{bmatrix},$$

using $E(\boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} | \mathbf{X}) = n\sigma^2$ and, for the off-diagonal block, $-E[-\sigma^{-4}\mathbf{X}'\boldsymbol{\varepsilon} | \mathbf{X}] = \mathbf{0}$. The two matrices agree, which is the information matrix equality, and both equal (1.5.15).

Review Question 1.5.5

Setting (1.5.13) to zero at $(\tilde{\beta}, \tilde{\gamma})$ gives

$$\frac{1}{\tilde{\gamma}} \mathbf{X}'(\mathbf{y} - \mathbf{X}\tilde{\beta}) = \mathbf{0}, \quad -\frac{n}{2\tilde{\gamma}} + \frac{1}{2\tilde{\gamma}^2}(\mathbf{y} - \mathbf{X}\tilde{\beta})'(\mathbf{y} - \mathbf{X}\tilde{\beta}) = 0.$$

Since $\tilde{\gamma} > 0$, the first equation is the normal equations $\mathbf{X}'\mathbf{X}\tilde{\beta} = \mathbf{X}'\mathbf{y}$, whose unique solution under Assumption 1.3 is $\tilde{\beta} = \mathbf{b}$. Substituting into the second, $\tilde{\gamma} = (\mathbf{y} - \mathbf{X}\mathbf{b})'(\mathbf{y} - \mathbf{X}\mathbf{b})/n = \text{SSR}/n$. This is Proposition 1.5. That the stationary point is the global maximum follows from Review Question 1.5.3, where the concentrated likelihood was shown to be decreasing in $S(\tilde{\beta})$.

Section 1.6 Generalized Least Squares (GLS)

Review Question 1.6.1

Suppose $\mathbf{C}\mathbf{X}\mathbf{c} = \mathbf{0}$ for some $K \times 1$ vector $\mathbf{c}$. Premultiplying by $\mathbf{C}^{-1}$ gives $\mathbf{X}\mathbf{c} = \mathbf{0}$, and full column rank of $\mathbf{X}$ forces $\mathbf{c} = \mathbf{0}$. So the columns of $\mathbf{C}\mathbf{X}$ are linearly independent and $\text{rank}(\mathbf{C}\mathbf{X}) = K$: Assumption 1.3 survives the GLS transformation, and $\mathbf{X}'\mathbf{V}^{-1}\mathbf{X} = (\mathbf{C}\mathbf{X})'(\mathbf{C}\mathbf{X})$ is positive definite.

Review Question 1.6.2

Write $\mathbf{V}^{-1} = \mathbf{C}'\mathbf{C}$ (possible since $\mathbf{V}^{-1}$ is positive definite). Then

$$(\mathbf{y} - \mathbf{X}\tilde{\beta})'\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\tilde{\beta}) = (\mathbf{C}\mathbf{y} - \mathbf{C}\mathbf{X}\tilde{\beta})'(\mathbf{C}\mathbf{y} - \mathbf{C}\mathbf{X}\tilde{\beta}),$$

which is the ordinary sum of squared residuals for the transformed data $(\mathbf{C}\mathbf{y}, \mathbf{C}\mathbf{X})$. By Analytical Exercise 1.1 the minimizer is the OLS estimator in the transformed model,

$$[(\mathbf{C}\mathbf{X})'(\mathbf{C}\mathbf{X})]^{-1}(\mathbf{C}\mathbf{X})'\mathbf{C}\mathbf{y} = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}\mathbf{y} = \hat{\beta}_{\text{GLS}}.$$

So GLS weights observations by the inverse of their error variance matrix, and the objective is the squared length of the residual in the metric defined by $\mathbf{V}^{-1}$.

Review Question 1.6.3

With $\mathbf{b} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$ and $E(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' | \mathbf{X}) = \sigma^2\mathbf{V}$,

$$\text{Var}(\mathbf{b} | \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1},$$

which differs from the usual $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$: the conventional OLS standard errors are wrong here, whereas $\mathbf{b}$ itself is still unbiased. Because OLS is linear in $\mathbf{y}$ and unbiased, Proposition 1.7(c) gives $\text{Var}(\mathbf{b} | \mathbf{X}) \geq \text{Var}(\hat{\beta}_{\text{GLS}} | \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}$, that is

$$(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \geq (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1},$$

the matrix inequality to be verified. (It can also be proved directly: it is the Cauchy–Schwarz inequality for the matrices $\mathbf{V}^{1/2}\mathbf{X}$ and $\mathbf{V}^{-1/2}\mathbf{X}$.)

Review Question 1.6.4

Substitute $\mathbf{y} = \mathbf{X}\beta + \varepsilon$ into the definition:

$$\hat{\beta}_{\text{GLS}} = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}(\mathbf{X}\beta + \varepsilon) = \beta + (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}\varepsilon.$$

Taking conditional expectations and using strict exogeneity gives Proposition 1.7(a), unbiasedness; taking the conditional variance and using $E(\varepsilon\varepsilon' | \mathbf{X}) = \sigma^2\mathbf{V}$ gives Proposition 1.7(b), $\text{Var}(\hat{\beta}_{\text{GLS}} | \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}$.

Section 1.7 Application: Returns to Scale in Electricity Supply

Review Question 1.7.1

Firm i solves $\min_{x_1, x_2, x_3} \sum_j p_{ij}x_j$ subject to $A_i x_1^{\alpha_1} x_2^{\alpha_2} x_3^{\alpha_3} = Q_i$. With multiplier λ the first-order conditions are $p_{ij} = \lambda \alpha_j Q_i / x_j$, so the cost-minimizing inputs are $x_j = \lambda \alpha_j Q_i / p_{ij}$ and each factor's cost share is

$$\frac{p_{ij}x_j}{\sum_k p_{ik}x_k} = \frac{\alpha_j}{r}, \quad r \equiv \alpha_1 + \alpha_2 + \alpha_3.$$

Substituting the input demands into the production function,

$$A_i \lambda^r Q_i^r \prod_j \left(\frac{\alpha_j}{p_{ij}} \right)^{\alpha_j} = Q_i \implies \lambda = \left[A_i \prod_j (\alpha_j / p_{ij})^{\alpha_j} \right]^{-1/r} Q_i^{(1-r)/r}.$$

Total cost is $TC_i = \sum_j p_{ij}x_j = \lambda Q_i \sum_j \alpha_j = r\lambda Q_i$, that is

$$TC_i = r (A_i \alpha_1^{\alpha_1} \alpha_2^{\alpha_2} \alpha_3^{\alpha_3})^{-1/r} Q_i^{1/r} p_{i1}^{\alpha_1/r} p_{i2}^{\alpha_2/r} p_{i3}^{\alpha_3/r},$$

which is (1.7.2); taking logs gives (1.7.3) with $\mu_i = \log r - \frac{1}{r} \log A_i - \frac{1}{r} \sum_j \alpha_j \log \alpha_j$. Two features matter for the empirics: the cost function is homogeneous of degree one in factor prices ($\sum_j \alpha_j / r = 1$, the restriction tested in the text), and the elasticity of cost with respect to output is $1/r$, the reciprocal of the degree of returns to scale.

Review Question 1.7.2

One megawatt hour is a thousand kilowatt hours, so the recorded output becomes $Q_i^* = Q_i/1000$ and $\log(Q_i^*) = \log(Q_i) - \log(1000)$. Substituting into (1.7.6),

$$\log\left(\frac{TC_i}{p_{i3}}\right) = \underbrace{[\beta_1 + \beta_2 \log(1000)]}_{\text{new intercept}} + \beta_2 \log(Q_i^*) + \beta_3 \log\left(\frac{p_{i1}}{p_{i3}}\right) + \beta_4 \log\left(\frac{p_{i2}}{p_{i3}}\right) + \varepsilon_i.$$

Only the intercept changes, rising by $\beta_2 \log(1000) \approx 0.72 \times 6.9078 \approx 4.97$ (so the estimate -4.7 becomes about 0.27); the slope estimates, their standard errors, the residuals, SSR, SER and R^2 are all unchanged, as is the implied estimate of returns to scale $r = 1/\beta_2$. Only the standard error of the intercept changes.

Review Question 1.7.3

From (1.7.5), $\beta_1 = \mu$, $\beta_2 = 1/r$, $\beta_3 = \alpha_1/r$, $\beta_4 = \alpha_2/r$. Since $\beta_2 \neq 0$ (a finite degree of returns to scale),

$$\mu = \beta_1, \quad r = \frac{1}{\beta_2}, \quad \alpha_1 = \frac{\beta_3}{\beta_2}, \quad \alpha_2 = \frac{\beta_4}{\beta_2},$$

and the definition of r supplies the third input elasticity,

$$\alpha_3 = r - \alpha_1 - \alpha_2 = \frac{1 - \beta_3 - \beta_4}{\beta_2}.$$

The mapping is one-to-one, so all four technology parameters are recovered from the four coefficients of the *restricted* regression (1.7.6). The fifth equation $\beta_5 = \alpha_3/r$ then holds automatically, since $\beta_5 = 1 - \beta_3 - \beta_4$ is exactly the homogeneity restriction; it carries no additional information.

Review Question 1.7.4

Homogeneity says $\beta_5 = 1 - \beta_3 - \beta_4$, so

$$b_5 = 1 - b_3 - b_4 = a + \mathbf{c}'\mathbf{b}, \quad a = 1, \quad \mathbf{c} = (0, 0, -1, -1)',$$

where $\mathbf{b} = (b_1, \dots, b_4)'$ is the restricted regression's coefficient vector. With the printed estimates $b_3 = 0.59$ and $b_4 = -0.007$, $b_5 = 1 - 0.59 + 0.007 = 0.417$. Since a and $\mathbf{c}$ are constants,

$$\text{Var}(b_5 | \mathbf{X}) = \mathbf{c}' \text{Var}(\mathbf{b} | \mathbf{X}) \mathbf{c} = \text{Var}(b_3 | \mathbf{X}) + \text{Var}(b_4 | \mathbf{X}) + 2 \text{Cov}(b_3, b_4 | \mathbf{X}),$$

so the standard error is the square root of the sum of the two variances and twice the covariance, all read off the estimated $\widehat{\text{Var}}(\mathbf{b} | \mathbf{X})$ that the regression package prints. Adding the two standard errors, or the two variances alone, would be wrong: the covariance term is generally large and negative here, because b_3 and b_4 are the coefficients of two price ratios sharing the same denominator.

Review Question 1.7.5

The restricted model becomes

$$\log\left(\frac{TC_i}{p_{i2}}\right) = \beta_1 + \beta_2 \log(Q_i) + \beta_3 \log\left(\frac{p_{i1}}{p_{i2}}\right) + \beta_5 \log\left(\frac{p_{i3}}{p_{i2}}\right) + \varepsilon_i,$$

with β_4 now the left-out coefficient, $\beta_4 = 1 - \beta_3 - \beta_5$.

Both regressions are the same restricted least squares problem — minimize SSR over the four-dimensional set $\{\beta : \beta_3 + \beta_4 + \beta_5 = 1\}$ — merely written in different free coordinates. The restricted minimizer $\hat{\beta}$ is unique, so the implied estimates of $(\beta_1, \dots, \beta_5)$, the fitted values of $\log(TC_i)$ and the residuals are identical in the two runs. Hence, from (1.7.8):

$$b_1 = -4.7, \quad b_2 = 0.72, \quad b_3 = 0.59, \quad b_4 = -0.007, \quad b_5 = 1 - b_3 - b_4 = 0.417,$$

with the same standard errors as before for b_1, b_2, b_3 (0.88, 0.017, 0.20), while the printout now reports $\text{SE}(b_5)$ directly and $\text{SE}(b_4)$ must be computed as in Review Question 1.7.4. The residuals being identical, $\text{SSR} = 21.640$ and $\text{SER} = 0.39$ are unchanged.

R^2 , however, *does* change, because the dependent variable is now $\log(TC_i/p_{i2})$ rather than $\log(TC_i/p_{i3})$: the total sum of squares $\sum_i (y_i - \bar{y})^2$ is different while SSR is the same. Without the data one cannot say in which direction it moves.

Review Question 1.7.6

Imposing a restriction cannot lower SSR, and indeed it does not here: SSR rises from 21.552 to 21.640. What falls is not the fit but the comparability of the two R^2 s, because the two regressions have *different dependent variables*: $\log(TC_i)$ in (1.7.4) and $\log(TC_i/p_{i3})$ in (1.7.6). Since

$$R^2 = 1 - \frac{\text{SSR}}{\sum_i (y_i - \bar{y})^2},$$

the denominators differ: subtracting $\log(p_{i3})$ from the dependent variable happens to leave a series with a *larger* sample variance than $\log(TC_i)$ has, so the same (slightly larger) SSR is divided by a bigger number. Comparing R^2 across regressions is legitimate only when the dependent variable is the same; the correct comparison of the two models is the F -test reported in the text, which uses SSR and accepts homogeneity ($F = 0.57$ against a 5% critical value of about 3.9).

Review Question 1.7.7

(a) Yes — provided homogeneity is *imposed* rather than tested. In the unrestricted regression (1.7.4) the regressor $\log(p_{i2})$ is now constant, hence perfectly collinear with the intercept: Assumption 1.3 fails and neither β_1 nor β_4 is identified (only $\beta_1 + \beta_4 \log p_2$ is). But the restricted regression (1.7.6) is unaffected: its regressors are $\log(Q_i)$, $\log(p_{i1}/p_{i3})$ and $\log(p_{i2}/p_{i3}) = \log(p_2) - \log(p_{i3})$, and the last one still varies across firms through p_{i3} , so it is not collinear with the constant and $(\beta_1, \dots, \beta_4)$ remain estimable. Review Question 1.7.3 then recovers $r = 1/\beta_2$, $\alpha_1 = \beta_3/\beta_2$, $\alpha_2 = \beta_4/\beta_2$ and $\alpha_3 = (1 - \beta_3 - \beta_4)/\beta_2$. The point of the hint is that $(\beta_1, \dots, \beta_5)$ are not free parameters: theory ties them together, and the tie is what restores identification.

(b) No. Testing homogeneity requires estimating the unrestricted model, which is exactly what perfect multicollinearity prevents. With p_{i2} constant the data contain no information about how costs respond to a change in the price of capital, so the restriction $\beta_3 + \beta_4 + \beta_5 = 1$ cannot be confronted with the data; it must be maintained as an assumption.

Review Question 1.7.8

No. The regressors are the firm's own input choices, and a cost-minimizing firm chooses them in the light of its efficiency: from Review Question 1.7.1, $x_{ij} = \lambda_i \alpha_j Q_i / p_{ij}$ with λ_i depending on A_i , and $\varepsilon_i = \log(A_i) - E[\log(A_i)]$ is precisely the efficiency shock. A more efficient firm (higher A_i) produces more with given inputs *and* employs different input levels, so $E(\varepsilon_i \mid x_{i1}, x_{i2}, x_{i3}) \neq 0$: strict exogeneity fails, and OLS is biased and inconsistent for the α 's. (This simultaneity is the “transmission bias” of the production-function literature; Chapters 3 and 7 return to it.)

The cost-share equations avoid the problem. Cost minimization gives

$$\frac{p_{ij}x_{ij}}{TC_i} = \frac{\alpha_j}{r} \quad (j = 1, 2, 3),$$

an identity in which the shares do not depend on A_i at all — the efficiency shock shifts costs and inputs proportionately and cancels. Averaging the observed shares therefore estimates α_j/r directly, and combining them with the cost-function estimate $r = 1/\beta_2$ gives each α_j . This is what factor prices, rather than factor quantities, are good for — and it is the route taken by Christensen and Greene (1976). Estimating the cost function (1.7.4) is the same idea: its regressors are output and factor prices, which by the institutional features listed at the start of Section 1.7 are plausibly unrelated to ε_i .

Problem Set for Chapter 1: Analytical Exercises

Analytical Exercise 1.1

Add and subtract $\mathbf{X}\tilde{\mathbf{b}}$: $\mathbf{y} - \mathbf{X}\tilde{\mathbf{b}} = (\mathbf{y} - \mathbf{X}\mathbf{b}) + \mathbf{X}(\mathbf{b} - \tilde{\mathbf{b}})$. Therefore

$$\begin{aligned} (\mathbf{y} - \mathbf{X}\tilde{\mathbf{b}})'(\mathbf{y} - \mathbf{X}\tilde{\mathbf{b}}) &= [(\mathbf{y} - \mathbf{X}\mathbf{b}) + \mathbf{X}(\mathbf{b} - \tilde{\mathbf{b}})]'[(\mathbf{y} - \mathbf{X}\mathbf{b}) + \mathbf{X}(\mathbf{b} - \tilde{\mathbf{b}})] \\ &= (\mathbf{y} - \mathbf{X}\mathbf{b})'(\mathbf{y} - \mathbf{X}\mathbf{b}) + 2(\mathbf{b} - \tilde{\mathbf{b}})' \mathbf{X}'(\mathbf{y} - \mathbf{X}\mathbf{b}) + (\mathbf{b} - \tilde{\mathbf{b}})' \mathbf{X}' \mathbf{X} (\mathbf{b} - \tilde{\mathbf{b}}) \\ &= (\mathbf{y} - \mathbf{X}\mathbf{b})'(\mathbf{y} - \mathbf{X}\mathbf{b}) + (\mathbf{b} - \tilde{\mathbf{b}})' \mathbf{X}' \mathbf{X} (\mathbf{b} - \tilde{\mathbf{b}}), \end{aligned}$$

the cross term vanishing by the normal equations $\mathbf{X}'(\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}$ (two scalars that are transposes of each other are equal, which is why the two cross terms combine). The remaining quadratic form is $\mathbf{z}'\mathbf{z} = \sum_i z_i^2 \geq 0$ with $\mathbf{z} \equiv \mathbf{X}(\mathbf{b} - \tilde{\mathbf{b}})$, so the displayed inequality holds. Under Assumption 1.3 it is strict unless $\tilde{\mathbf{b}} = \mathbf{b}$, so the minimizer is unique. Note that no assumption about the error term is used: this is pure algebra, valid for any $\mathbf{y}$ and any full-column-rank $\mathbf{X}$.

Analytical Exercise 1.2

Since $\mathbf{1}'\mathbf{1} = n$, $\mathbf{M}_1 = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}'$.

(a) It is the annihilator of the single-column matrix $\mathbf{1}$, so symmetry and idempotency follow from Review Question 1.2.4; directly, $\mathbf{M}_1' = \mathbf{I} - \frac{1}{n}(\mathbf{1}\mathbf{1}')' = \mathbf{M}_1$ and

$$\mathbf{M}_1\mathbf{M}_1 = \mathbf{I} - \frac{2}{n}\mathbf{1}\mathbf{1}' + \frac{1}{n^2}\underbrace{\mathbf{1}(\mathbf{1}'\mathbf{1})}_{=n}\mathbf{1}' = \mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}' = \mathbf{M}_1.$$

(b) $\mathbf{M}_1\mathbf{1} = \mathbf{1} - \frac{1}{n}\mathbf{1}(\mathbf{1}'\mathbf{1}) = \mathbf{1} - \mathbf{1} = \mathbf{0}$.

(c) $\frac{1}{n}\mathbf{1}\mathbf{1}'\mathbf{y} = \frac{1}{n}\mathbf{1}\sum_i y_i = \bar{y}\mathbf{1}$, so $\mathbf{M}_1\mathbf{y} = \mathbf{y} - \bar{y}\mathbf{1}$: premultiplication by $\mathbf{M}_1$ takes deviations from the sample mean.

(d) Column by column, the same computation gives $\frac{1}{n}\mathbf{1}\mathbf{1}'\mathbf{X} = \mathbf{1}(\mathbf{X}'\mathbf{1}/n)' = \mathbf{1}\bar{\mathbf{x}}'$, so $\mathbf{M}_1\mathbf{X} = \mathbf{X} - \mathbf{1}\bar{\mathbf{x}}'$, whose k -th column is $\mathbf{x}_{\cdot k} - \bar{x}_k\mathbf{1}$. In particular, if the first column of $\mathbf{X}$ is $\mathbf{1}$, that column becomes $\mathbf{0}$.

Analytical Exercise 1.3

(a) The normal equations $\mathbf{X}'\mathbf{X}\mathbf{b} = \mathbf{X}'\mathbf{y}$ in partitioned form are

$$\begin{bmatrix} \mathbf{1}'\mathbf{1} & \mathbf{1}'\mathbf{X}_2 \\ \mathbf{X}_2'\mathbf{1} & \mathbf{X}_2'\mathbf{X}_2 \end{bmatrix} \begin{bmatrix} b_1 \\ \mathbf{b}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{1}'\mathbf{y} \\ \mathbf{X}_2'\mathbf{y} \end{bmatrix} \iff \begin{cases} n b_1 + n \bar{\mathbf{x}}_2' \mathbf{b}_2 = n \bar{y}, \\ n \bar{\mathbf{x}}_2 b_1 + \mathbf{X}_2' \mathbf{X}_2 \mathbf{b}_2 = \mathbf{X}_2' \mathbf{y}, \end{cases}$$

with $\bar{\mathbf{x}}_2 = \mathbf{X}'_2 \mathbf{1}/n$. Dividing the first by n gives $\bar{y} - b_1 - \bar{\mathbf{x}}'_2 \mathbf{b}_2 = 0$, and rearranging the second gives $\mathbf{X}'_2 \mathbf{y} - n b_1 \bar{\mathbf{x}}_2 - \mathbf{X}'_2 \mathbf{X}_2 \mathbf{b}_2 = \mathbf{0}$, the two displayed equations.

(b) Substitute $b_1 = \bar{y} - \bar{\mathbf{x}}'_2 \mathbf{b}_2$ from the first equation into the second:

$$\mathbf{X}'_2 \mathbf{y} - n \bar{\mathbf{x}}_2 \bar{y} = (\mathbf{X}'_2 \mathbf{X}_2 - n \bar{\mathbf{x}}_2 \bar{\mathbf{x}}'_2) \mathbf{b}_2.$$

Now use Analytical Exercise 1.2 and idempotency: $\tilde{\mathbf{X}}'_2 \tilde{\mathbf{X}}_2 = \mathbf{X}'_2 \mathbf{M}_1 \mathbf{X}_2 = \mathbf{X}'_2 \mathbf{X}_2 - n \bar{\mathbf{x}}_2 \bar{\mathbf{x}}'_2$ and $\tilde{\mathbf{X}}'_2 \tilde{\mathbf{y}} = \mathbf{X}'_2 \mathbf{M}_1 \mathbf{y} = \mathbf{X}'_2 \mathbf{y} - n \bar{\mathbf{x}}_2 \bar{y}$. So the display reads $\tilde{\mathbf{X}}'_2 \tilde{\mathbf{y}} = \tilde{\mathbf{X}}'_2 \tilde{\mathbf{X}}_2 \mathbf{b}_2$, and since $\mathbf{X}$ has full column rank the matrix $\tilde{\mathbf{X}}'_2 \tilde{\mathbf{X}}_2$ is nonsingular, giving $\mathbf{b}_2 = (\tilde{\mathbf{X}}'_2 \tilde{\mathbf{X}}_2)^{-1} \tilde{\mathbf{X}}'_2 \tilde{\mathbf{y}}$. Slope coefficients can thus be computed from data in deviations from their means, the constant being recovered afterwards from $b_1 = \bar{y} - \bar{\mathbf{x}}'_2 \mathbf{b}_2$. For $K = 2$ this is Review Question 1.2.3.

Analytical Exercise 1.4

(a) Writing $\mathbf{X}'\mathbf{X}\mathbf{b} = \mathbf{X}'\mathbf{y}$ in blocks gives

$$\mathbf{X}'_1 \mathbf{X}_1 \mathbf{b}_1 + \mathbf{X}'_1 \mathbf{X}_2 \mathbf{b}_2 = \mathbf{X}'_1 \mathbf{y} \quad (*), \quad \mathbf{X}'_2 \mathbf{X}_1 \mathbf{b}_1 + \mathbf{X}'_2 \mathbf{X}_2 \mathbf{b}_2 = \mathbf{X}'_2 \mathbf{y} \quad (**).$$

(b) Premultiply (*) by $\mathbf{X}_1(\mathbf{X}'_1 \mathbf{X}_1)^{-1}$: $\mathbf{X}_1 \mathbf{b}_1 = \mathbf{P}_1 \mathbf{y} - \mathbf{P}_1 \mathbf{X}_2 \mathbf{b}_2$. Substituting into (**) written as $\mathbf{X}'_2(\mathbf{X}_1 \mathbf{b}_1 + \mathbf{X}_2 \mathbf{b}_2) = \mathbf{X}'_2 \mathbf{y}$,

$$\mathbf{X}'_2(\mathbf{P}_1 \mathbf{y} - \mathbf{P}_1 \mathbf{X}_2 \mathbf{b}_2 + \mathbf{X}_2 \mathbf{b}_2) = \mathbf{X}'_2 \mathbf{y} \implies \mathbf{X}'_2 \mathbf{M}_1 \mathbf{X}_2 \mathbf{b}_2 = \mathbf{X}'_2 \mathbf{M}_1 \mathbf{y}.$$

By symmetry and idempotency of $\mathbf{M}_1$ this is $\tilde{\mathbf{X}}'_2 \tilde{\mathbf{X}}_2 \mathbf{b}_2 = \tilde{\mathbf{X}}'_2 \tilde{\mathbf{y}}$, so $\mathbf{b}_2 = (\tilde{\mathbf{X}}'_2 \tilde{\mathbf{X}}_2)^{-1} \tilde{\mathbf{X}}'_2 \tilde{\mathbf{y}}$: regressing the residualized $\mathbf{y}$ on the residualized $\mathbf{X}_2$ reproduces the multiple regression coefficient exactly. ($\tilde{\mathbf{X}}'_2 \tilde{\mathbf{X}}_2$ is nonsingular: if $\tilde{\mathbf{X}}_2 \mathbf{c} = \mathbf{0}$ then $\mathbf{X}_2 \mathbf{c} = \mathbf{P}_1 \mathbf{X}_2 \mathbf{c}$ lies in the span of $\mathbf{X}_1$, contradicting full column rank of $\mathbf{X}$ unless $\mathbf{c} = \mathbf{0}$.)

(c) From $\mathbf{y} = \mathbf{X}_1 \mathbf{b}_1 + \mathbf{X}_2 \mathbf{b}_2 + \mathbf{e}$, premultiplying by $\mathbf{M}_1$ and using $\mathbf{M}_1 \mathbf{X}_1 = \mathbf{0}$, $\tilde{\mathbf{y}} = \tilde{\mathbf{X}}_2 \mathbf{b}_2 + \mathbf{M}_1 \mathbf{e}$. But $\mathbf{X}'\mathbf{e} = \mathbf{0}$ implies $\mathbf{X}'_1 \mathbf{e} = \mathbf{0}$, hence $\mathbf{P}_1 \mathbf{e} = \mathbf{0}$ and $\mathbf{M}_1 \mathbf{e} = \mathbf{e}$. So $\tilde{\mathbf{y}} = \tilde{\mathbf{X}}_2 \mathbf{b}_2 + \mathbf{e}$ with $\tilde{\mathbf{X}}'_2 \mathbf{e} = \mathbf{X}'_2 \mathbf{M}_1 \mathbf{e} = \mathbf{X}'_2 \mathbf{e} = \mathbf{0}$: $\mathbf{e}$ is exactly the residual vector of the regression in (b).

(d) Since $\tilde{\mathbf{X}}'_2 \tilde{\mathbf{y}} = \mathbf{X}'_2 \mathbf{M}_1 \mathbf{M}_1 \mathbf{y} = \tilde{\mathbf{X}}'_2 \mathbf{y}$, part (b) also reads $\mathbf{b}_2 = (\tilde{\mathbf{X}}'_2 \tilde{\mathbf{X}}_2)^{-1} \tilde{\mathbf{X}}'_2 \mathbf{y}$: one may residualize the regressors only. The *coefficient* is the same, but the regression of $\mathbf{y}$ on $\tilde{\mathbf{X}}_2$ has residual $\mathbf{y} - \tilde{\mathbf{X}}_2 \mathbf{b}_2 = \mathbf{e} + \mathbf{P}_1 \mathbf{y} \neq \mathbf{e}$ in general, and its SSR exceeds $\mathbf{e}'\mathbf{e}$ by $\mathbf{y}'\mathbf{P}_1 \mathbf{y}$ (the two pieces are orthogonal). So the answers to the two bracketed questions are “no” and “no”.

(e) Apply the decomposition (1.2.15) to the regression in (c): $\tilde{\mathbf{y}}'\tilde{\mathbf{y}} = \mathbf{b}'_2 \tilde{\mathbf{X}}'_2 \tilde{\mathbf{X}}_2 \mathbf{b}_2 + \mathbf{e}'\mathbf{e}$. Substituting $\mathbf{b}_2$ from (b),

$$\tilde{\mathbf{y}}'\tilde{\mathbf{y}} - \mathbf{e}'\mathbf{e} = \tilde{\mathbf{y}}'\tilde{\mathbf{X}}_2(\tilde{\mathbf{X}}'_2 \tilde{\mathbf{X}}_2)^{-1} \tilde{\mathbf{X}}'_2 \tilde{\mathbf{y}} = \tilde{\mathbf{y}}'\mathbf{X}_2(\mathbf{X}'_2 \mathbf{M}_1 \mathbf{X}_2)^{-1} \mathbf{X}'_2 \tilde{\mathbf{y}},$$

using $\tilde{\mathbf{X}}'_2 \tilde{\mathbf{y}} = \mathbf{X}'_2 \tilde{\mathbf{y}}$ once more.

(f) (i) $\mathbf{X}'_1 \tilde{\mathbf{y}} = \mathbf{X}'_1 \mathbf{M}_1 \mathbf{y} = \mathbf{0}$, so the OLS coefficient in the regression of $\tilde{\mathbf{y}}$ on $\mathbf{X}_1$ is zero and the residual is $\tilde{\mathbf{y}}$ itself: $\text{SSR}_1 = \tilde{\mathbf{y}}'\tilde{\mathbf{y}}$. (ii) By (c), $\text{SSR}_2 = \mathbf{e}'\mathbf{e}$. (iii) Regressing $\tilde{\mathbf{y}}$ on $(\mathbf{X}_1 \ \mathbf{X}_2)$: by the Frisch–Waugh result (b) applied to this regression, its slope on the residualized $\mathbf{X}_2$ is $(\tilde{\mathbf{X}}'_2 \tilde{\mathbf{X}}_2)^{-1} \tilde{\mathbf{X}}'_2 \mathbf{M}_1 \tilde{\mathbf{y}} = \mathbf{b}_2$ (because $\mathbf{M}_1 \tilde{\mathbf{y}} = \tilde{\mathbf{y}}$), and by (c) its residual vector is $\tilde{\mathbf{y}} - \tilde{\mathbf{X}}_2 \mathbf{b}_2 = \mathbf{e}$. Hence $\text{SSR}_3 = \mathbf{e}'\mathbf{e}$: adding $\mathbf{X}_1$ to regression (2) does not change the fit, since $\tilde{\mathbf{y}}$ has already been purged of $\mathbf{X}_1$. (iv) Regression (4) uses the *unresidualized* $\mathbf{X}_2$ and generally gives $\text{SSR}_4 \neq \mathbf{e}'\mathbf{e}$; $\mathbf{X}_2$ can explain

part of $\tilde{\mathbf{y}}$ through its $\mathbf{X}_1$ component. A numerical example is in `code/ch1.py`: with

$$\mathbf{X}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 5 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 2 \\ 1 \\ 4 \\ 6 \end{bmatrix},$$

the script reports $\text{SSR}_1 = 14.750$, $\text{SSR}_2 = \text{SSR}_3 = \mathbf{e}'\mathbf{e} = 2.7429$ and $\text{SSR}_4 = 12.056$.

Analytical Exercise 1.5

(a) The Lagrangian is $L = \frac{1}{2}(\mathbf{y} - \mathbf{X}\tilde{\beta})'(\mathbf{y} - \mathbf{X}\tilde{\beta}) + \lambda'(\mathbf{R}\tilde{\beta} - \mathbf{r})$, with first-order conditions

$$-\mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\beta}) + \mathbf{R}'\lambda = \mathbf{0} \iff \mathbf{X}'\hat{\varepsilon} = \mathbf{R}'\lambda, \quad \mathbf{R}\hat{\beta} = \mathbf{r},$$

where $\hat{\varepsilon} \equiv \mathbf{y} - \mathbf{X}\hat{\beta}$. The first condition gives $\mathbf{X}'\mathbf{y} - \mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{R}'\lambda$, i.e. $\hat{\beta} = \mathbf{b} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\lambda$. Premultiplying by $\mathbf{R}$ and imposing the constraint,

$$\mathbf{r} = \mathbf{R}\mathbf{b} - \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\lambda \implies \lambda = [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\mathbf{b} - \mathbf{r}),$$

the inverse existing by Review Question 1.4.3, and hence

$$\hat{\beta} = \mathbf{b} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\mathbf{b} - \mathbf{r}).$$

The restricted estimator is the unrestricted one corrected by the extent $\mathbf{R}\mathbf{b} - \mathbf{r}$ to which the sample violates the restriction; the multiplier λ measures the same violation in units of the objective.

(b) Add and subtract $\mathbf{X}\mathbf{b}$ in SSR_R , exactly as in Analytical Exercise 1.1 but with $\tilde{\beta} = \hat{\beta}$:

$$\text{SSR}_R = (\mathbf{y} - \mathbf{X}\mathbf{b})'(\mathbf{y} - \mathbf{X}\mathbf{b}) + (\mathbf{b} - \hat{\beta})'\mathbf{X}'\mathbf{X}(\mathbf{b} - \hat{\beta}) = \text{SSR}_U + (\mathbf{b} - \hat{\beta})'\mathbf{X}'\mathbf{X}(\mathbf{b} - \hat{\beta}).$$

Substituting $\mathbf{b} - \hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\lambda$ from (a),

$$\begin{aligned} \text{SSR}_R - \text{SSR}_U &= (\mathbf{b} - \hat{\beta})'\mathbf{X}'\mathbf{X}(\mathbf{b} - \hat{\beta}) \\ &= (\mathbf{R}\mathbf{b} - \mathbf{r})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\mathbf{b} - \mathbf{r}) \\ &= \lambda'\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\lambda = \hat{\varepsilon}'\mathbf{P}\hat{\varepsilon}, \end{aligned}$$

the last equality because $\mathbf{X}'\hat{\varepsilon} = \mathbf{R}'\lambda$ gives $\hat{\varepsilon}'\mathbf{P}\hat{\varepsilon} = \hat{\varepsilon}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\hat{\varepsilon} = \lambda'\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\lambda$. In particular $\text{SSR}_R \geq \text{SSR}_U$, as it must be.

(c) Dividing the second expression by $\#r s^2$ gives (1.4.9); dividing the first line by the same quantity and using $s^2 = \text{SSR}_U / (n - K)$ gives (1.4.11):

$$\frac{(\mathbf{R}\mathbf{b} - \mathbf{r})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\mathbf{b} - \mathbf{r})/\#r}{s^2} = \frac{(\text{SSR}_R - \text{SSR}_U)/\#r}{\text{SSR}_U/(n - K)}.$$

The two ways of computing the F -ratio — from the estimated restriction and its variance, or from two sums of squared residuals — are numerically identical.

Analytical Exercise 1.6

(a) Here the restriction sets all $K - 1$ slopes to zero, so the restricted estimator is $\hat{\beta} = (\bar{y}, 0, \dots, 0)'$ (the OLS estimate of a regression on a constant is the sample mean), giving restricted residuals

$y_i - \bar{y}$ and

$$\text{SSR}_R = \sum_{i=1}^n (y_i - \bar{y})^2.$$

Moreover $\mathbf{X}(\mathbf{b} - \hat{\beta}) = \hat{\mathbf{y}} - \bar{y}\mathbf{1}$, so

$$(\mathbf{b} - \hat{\beta})' \mathbf{X}' \mathbf{X} (\mathbf{b} - \hat{\beta}) = \|\hat{\mathbf{y}} - \bar{y}\mathbf{1}\|^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2.$$

Analytical Exercise 1.5(b) therefore reads $\sum_i (y_i - \bar{y})^2 - \text{SSR}_U = \sum_i (\hat{y}_i - \bar{y})^2$, i.e.

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n e_i^2,$$

which is the decomposition (1.2.17). (It uses the presence of a constant twice: to make $\hat{\bar{y}} = \bar{y}$ and to make the cross term vanish.)

(b) With $\#r = K - 1$ restrictions, (1.4.11) and part (a) give

$$F = \frac{(\text{SSR}_R - \text{SSR}_U)/(K - 1)}{\text{SSR}_U/(n - K)} = \frac{\sum_i (\hat{y}_i - \bar{y})^2/(K - 1)}{\sum_i e_i^2/(n - K)} = \frac{R^2/(K - 1)}{(1 - R^2)/(n - K)},$$

dividing numerator and denominator by $\sum_i (y_i - \bar{y})^2$ and using $R^2 = \sum_i (\hat{y}_i - \bar{y})^2 / \sum_i (y_i - \bar{y})^2$ from (1.2.17)–(1.2.18). This is the “ F -statistic of the regression” printed by every OLS package: it tests that all slopes are zero, and it is a strictly increasing function of R^2 for fixed n and K .

Analytical Exercise 1.7

(a) Write $\hat{\beta}_{\text{GLS}} - \beta = \mathbf{A}\epsilon$ with $\mathbf{A} \equiv (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}$ (Review Question 1.6.4) and $\mathbf{b} - \hat{\beta}_{\text{GLS}} = \mathbf{B}\epsilon$ with $\mathbf{B} \equiv (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - \mathbf{A}$. Both estimators are unbiased, so means may be subtracted freely and, with $E(\epsilon\epsilon' | \mathbf{X}) = \sigma^2\mathbf{V}$,

$$\text{Cov}(\hat{\beta}_{\text{GLS}}, \mathbf{b} - \hat{\beta}_{\text{GLS}}) = \text{Cov}(\mathbf{A}\epsilon, \mathbf{B}\epsilon) = \sigma^2\mathbf{A}\mathbf{V}\mathbf{B}'.$$

Now $\mathbf{A}\mathbf{V} = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'$, so

$$\mathbf{A}\mathbf{V}\mathbf{B}' = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'[\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} - \mathbf{V}^{-1}\mathbf{X}(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}] = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1} - (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1} = \mathbf{0}.$$

(b) Let $\mathbf{C} \equiv \text{Cov}(\hat{\beta}_{\text{GLS}}, \mathbf{q})$ and consider the family $\hat{\beta}_{\mathbf{H}} \equiv \hat{\beta}_{\text{GLS}} + \mathbf{H}\mathbf{q}$ indexed by a $K \times K$ matrix $\mathbf{H}$ of constants. Each member is unbiased, because $E(\mathbf{q}) = \mathbf{0}$, and

$$\text{Var}(\hat{\beta}_{\mathbf{H}}) = \text{Var}(\hat{\beta}_{\text{GLS}}) + \mathbf{C}\mathbf{H}' + \mathbf{H}\mathbf{C}' + \mathbf{H}\mathbf{V}_{\mathbf{q}}\mathbf{H}'.$$

Choosing $\mathbf{H} = -\mathbf{C}\mathbf{V}_{\mathbf{q}}^{-1}$ gives

$$\text{Var}(\hat{\beta}_{\mathbf{H}}) - \text{Var}(\hat{\beta}_{\text{GLS}}) = -\mathbf{C}\mathbf{V}_{\mathbf{q}}^{-1}\mathbf{C}'.$$

If $\mathbf{C} \neq \mathbf{0}$, pick $\mathbf{z}$ with $\mathbf{v} \equiv \mathbf{C}'\mathbf{z} \neq \mathbf{0}$; then $\mathbf{z}'[\text{Var}(\hat{\beta}_{\mathbf{H}}) - \text{Var}(\hat{\beta}_{\text{GLS}})]\mathbf{z} = -\mathbf{v}'\mathbf{V}_{\mathbf{q}}^{-1}\mathbf{v} < 0$ because $\mathbf{V}_{\mathbf{q}}$ is positive definite. That contradicts the efficiency of GLS, so $\mathbf{C} = \mathbf{0}$.

This is the finite-sample form of the Hausman principle: *the efficient estimator is uncorrelated with its difference from any other unbiased estimator*. It is what makes the variance of a Hausman statistic simple, since $\text{Var}(\mathbf{q}) = \text{Var}(\tilde{\beta}) - \text{Var}(\hat{\beta}_{\text{GLS}})$ follows at once from $\tilde{\beta} = \hat{\beta}_{\text{GLS}} + \mathbf{q}$ and $\mathbf{C} = \mathbf{0}$.

Remark. The contradiction uses Proposition 1.7(c), which concerns estimators *linear* in $\mathbf{y}$, so the argument as it stands proves the claim for linear unbiased $\tilde{\beta}$ (then $\hat{\beta}_{\mathbf{H}}$ is linear too). For an arbitrary, possibly nonlinear, unbiased $\tilde{\beta}$ one needs GLS to be best unbiased, which holds under the normality assumption: the transformed model $\mathbf{C}\mathbf{y} = \mathbf{C}\mathbf{X}\beta + \mathbf{C}\varepsilon$ is a classical normal regression model, so Proposition 1.6 applies to it.

(c) Let $\mathbf{H}$ be orthogonal with $\mathbf{H}'\mathbf{V}\mathbf{H} = \Lambda$ diagonal (possible since $\mathbf{V}$ is symmetric), so $\mathbf{H}'\mathbf{H} = \mathbf{I}_n$, $\mathbf{H}^{-1} = \mathbf{H}'$, $\mathbf{H}'\mathbf{V}^{-1}\mathbf{H} = \Lambda^{-1}$ and $\mathbf{H}'\mathbf{V}^{-1} = \Lambda^{-1}\mathbf{H}'$. Taking the columns of $\mathbf{X}$ to be the first K columns of $\mathbf{H}$ (possible without loss of generality, by reordering), $\mathbf{X} = \mathbf{H}\mathbf{F}$ with $\mathbf{F} = \begin{bmatrix} \mathbf{I}_K \\ \mathbf{0} \end{bmatrix}$, so $\mathbf{F}'\mathbf{F} = \mathbf{I}_K$ and $\mathbf{F}'\Lambda^{-1}\mathbf{F} = \Lambda_1^{-1}$, where Λ_1 collects the K characteristic roots belonging to the columns of $\mathbf{X}$. Then

$$\mathbf{b} = (\mathbf{F}'\mathbf{H}'\mathbf{H}\mathbf{F})^{-1}\mathbf{F}'\mathbf{H}'\mathbf{y} = \mathbf{F}'\mathbf{H}'\mathbf{y},$$

$$\hat{\beta}_{\text{GLS}} = (\mathbf{F}'\Lambda^{-1}\mathbf{F})^{-1}\mathbf{F}'\Lambda^{-1}\mathbf{H}'\mathbf{y} = \Lambda_1\Lambda_1^{-1}\mathbf{F}'\mathbf{H}'\mathbf{y} = \mathbf{F}'\mathbf{H}'\mathbf{y} = \mathbf{b}.$$

So OLS is fully efficient in this special case — the textbook example being an equicorrelated $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}'$ with $\mathbf{X} = \mathbf{1}$, where the group mean is both the OLS and the GLS estimator. Here $\mathbf{q} = \mathbf{b} - \hat{\beta}_{\text{GLS}} = \mathbf{0}$ has a singular variance matrix, which is why part (b) had to assume $\mathbf{V}_{\mathbf{q}}$ nonsingular.

Problem Set for Chapter 1: Empirical Exercise

Empirical Exercise 1.1

All numbers below are produced by `code/ch1.py`, which reads the data file directly.

(a) *No.* Jorgenson's user cost is $(r + \delta)p_I$: the firm's rental price of capital must include the depreciation rate as well as the interest rate. Nerlove had no data on the reproduction cost of capital and used instead the interest and depreciation charges recorded in the utilities' books; as his Appendix B.4 describes the construction, the depreciation rate δ is not part of the price of capital he formed (the same reading is given in the answer to this exercise printed in the book). The consequence is measurement error in $\log(p_{i2})$, and it shows up in the estimates: the capital-price elasticity is -0.22 in the unrestricted regression and -0.007 in the restricted one, whereas cost minimization requires it to be the capital cost share $\alpha_2/r > 0$. Mismeasuring a regressor biases its coefficient towards zero (Chapter 3), which is exactly what one sees.

(b) The OLS estimate of (1.7.4), with standard errors in parentheses, is

$$\widehat{\log(TC_i)} = \underset{(1.774)}{-3.527} + \underset{(0.0175)}{0.7204 \log(Q_i)} + \underset{(0.291)}{0.436 \log(p_{i1})} - \underset{(0.339)}{0.220 \log(p_{i2})} + \underset{(0.100)}{0.427 \log(p_{i3})},$$

with $R^2 = 0.9260$, mean of the dependent variable 1.7247, $SE\hat{R} = 0.3924$, $SSR = 21.5520$, $n = 145$. This reproduces (1.7.7) to every digit printed there.

(c) The restricted estimate of (1.7.6) is

$$\widehat{\log(TC_i/p_{i3})} = \underset{(0.885)}{-4.691} + \underset{(0.0174)}{0.7207 \log(Q_i)} + \underset{(0.205)}{0.593 \log\left(\frac{p_{i1}}{p_{i3}}\right)} - \underset{(0.191)}{0.0074 \log\left(\frac{p_{i2}}{p_{i3}}\right)},$$

$R^2 = 0.9316$, mean of the dependent variable -1.4842 , $SE\hat{R} = 0.3918$, $SSR = 21.6403$, $n = 145$, which is (1.7.8). The implied fuel-price elasticity of Review Question 1.7.4 is $b_5 = 1 - b_3 - b_4 = 0.4145$ with standard error 0.0990, and the implied degree of returns to scale for the whole

sample is $1/b_2 = 1.388$. The homogeneity restriction is comfortably accepted: $F = (21.6403 - 21.5520)/(21.5520/140) = 0.574$ against $F_{0.05}(1, 140) \approx 3.9$ ($p = 0.45$), the test reported in the text.

Nerlove's own estimate of β_2 is 0.721, and our 0.7207 (standard error 0.0174) reproduces it; as the exercise notes, the standard error 0.175 printed in his paper must be a typographical error for 0.0174. Exact agreement is not to be expected — he used common logarithms (which changes only the intercept) and his published data differ slightly from the corrected file used here — but the agreement is close enough to identify the estimate.

(d) Estimating (1.7.6) separately on the five groups of 29 firms (Model 1) gives

group	b_2	returns to scale $1/b_2$	s_j^2	SSR _{j}
1 (smallest)	0.4003	2.498	0.35541	8.8852
2	0.6582	1.519	0.05876	1.4691
3	0.9383	1.066	0.03921	0.9802
4	0.9120	1.096	0.01454	0.3636
5 (largest)	1.0444	0.957	0.02258	0.5644

which replicates Nerlove's reported pattern (2.5, 1.5, 1.1, 1.1, 0.96). Returns to scale decline steadily with output and have essentially vanished — indeed turned slightly decreasing — for the largest firms. The estimated error variance also falls sharply with output, from 0.355 to about 0.02: this is the conditional heteroskedasticity that Model 4 will model explicitly.

(e) Stacking the five groups block-diagonally as in (*) and estimating the 20 coefficients in one regression gives $SSR = 12.262433$, numerically equal to $\sum_{j=1}^5 SSR_j = 12.262433$ from (d), and coefficients equal to those of (d) to 1.3×10^{-11} .

The agreement is general. With two groups (the general case is the same with more),

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}^{(1)} & \mathbf{0} \\ \mathbf{0} & \mathbf{X}^{(2)} \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} \mathbf{y}^{(1)} \\ \mathbf{y}^{(2)} \end{bmatrix} \implies \mathbf{X}'\mathbf{X} = \begin{bmatrix} \mathbf{X}^{(1)'}\mathbf{X}^{(1)} & \mathbf{0} \\ \mathbf{0} & \mathbf{X}^{(2)'}\mathbf{X}^{(2)} \end{bmatrix}, \quad \mathbf{X}'\mathbf{y} = \begin{bmatrix} \mathbf{X}^{(1)'}\mathbf{y}^{(1)} \\ \mathbf{X}^{(2)'}\mathbf{y}^{(2)} \end{bmatrix},$$

by the rules for multiplying partitioned matrices. A block-diagonal matrix is inverted block by block, so

$$\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \begin{bmatrix} (\mathbf{X}^{(1)'}\mathbf{X}^{(1)})^{-1}\mathbf{X}^{(1)'}\mathbf{y}^{(1)} \\ (\mathbf{X}^{(2)'}\mathbf{X}^{(2)})^{-1}\mathbf{X}^{(2)'}\mathbf{y}^{(2)} \end{bmatrix} = \begin{bmatrix} \mathbf{b}^{(1)} \\ \mathbf{b}^{(2)} \end{bmatrix} :$$

the pooled regression reproduces the group-by-group estimates exactly. The residual for an observation in group j is then $\mathbf{y}^{(j)} - \mathbf{X}^{(j)}\mathbf{b}^{(j)}$, its own group residual, so $SSR = \sum_j SSR_j$. Nothing about the data is used: only the block-diagonal structure, which is what the dummy-variable construction creates.

(f) *Chow test*. Model 2 has 20 coefficients and (1.7.6) has 4, so the null imposes $\#r = 16$ restrictions. With $SSR_U = 12.262$ (Model 2) and $SSR_R = 21.640$ (the restricted regression of part (c)),

$$F = \frac{(21.6403 - 12.2624)/16}{12.2624/(145 - 20)} = 5.975 \quad \text{on } F(16, 125), \quad p = 1.5 \times 10^{-9}.$$

The hypothesis of common coefficients across firm-size groups is rejected at any conventional level: the cost function is not the same for small and large utilities, which is the whole point of Nerlove's subsample analysis.

(g) *Model 3* (group-specific intercept and output coefficient, common price elasticities, 12 coefficients) gives $SSR = 12.5773$. Tested against Model 2, the 8 restrictions give

$$F = \frac{(12.5773 - 12.2624)/8}{12.2624/125} = 0.401 \quad \text{on } F(8, 125), \quad p = 0.918,$$

so the hypothesis of “neutral variations in returns to scale” is accepted at 5% — the price elasticities look the same across groups and only the scale coefficient moves. The implied returns to scale are 2.52, 1.54, 1.13, 1.10 and 0.94.

Remark. The book warns that Nerlove's own F -ratio for this test (his p. 183) is wrong; it is 1.576, about four times the correct 0.401. The conclusion — acceptance — is not affected.

(h) *Model 4* adds $[\log Q_i]^2$ and assumes $E(\varepsilon_i^2 | \mathbf{X}) = \sigma^2(0.0565 + 2.1377/Q_i)$. Dividing every variable, the constant included, by $\sqrt{0.0565 + 2.1377/Q_i}$ and running OLS on the transformed data gives

$$\log(\widehat{TC_i}/p_{i3}) = \underset{(0.579)}{-4.126} + \underset{(0.0819)}{0.2177} \log(Q_i) + \underset{(0.0062)}{0.04566} [\log(Q_i)]^2 + \underset{(0.121)}{0.466} \log\left(\frac{p_{i1}}{p_{i3}}\right) + \underset{(0.113)}{0.134} \log\left(\frac{p_{i2}}{p_{i3}}\right),$$

with $\hat{\sigma} = 0.915$. (The SSR and R^2 of a weighted regression are in the units of the weighted data and are not comparable with those of (b)–(g); and since the transformed regression has no constant, its R^2 should not be reported at all.) Two things are worth noting. First, the quadratic term is sharply significant ($t = 7.4$), so the elasticity of cost with respect to output, $\partial \log TC / \partial \log Q = \beta_2 + 2\beta_3 \log Q$, rises with output and returns to scale $r = 1/(\beta_2 + 2\beta_3 \log Q)$ fall continuously; evaluated at the mean of $\log Q$ in each group they are 1.86, 1.32, 1.17, 1.09 and 0.98, the same story as the five separate regressions but from a single smooth specification. Second, the capital-price elasticity is now positive (0.134), as theory requires, though still insignificant.

Figure S1.2 plots the residuals of the restricted model (left, the book's Figure 1.7) and the weighted residuals of Model 4 (right) against $\log(Q_i)$. The pronounced U-shape on the left — the sign of the missing quadratic term — is gone on the right. Heteroskedasticity is reduced but not removed: the standard deviation of the weighted residuals still falls from 1.33 in the smallest group to 0.47–0.67 in the two largest, and a regression of $\log(e_i^2)$ on $\log(Q_i)$ has slope -0.40 (it is -0.46 for the unweighted residuals of the restricted model). So the assumed variance function captures much, but not all, of the pattern — as the book's own answer puts it, the conditional second moment is still somewhat larger for smaller firms, but there is no longer evidence of nonlinearity.

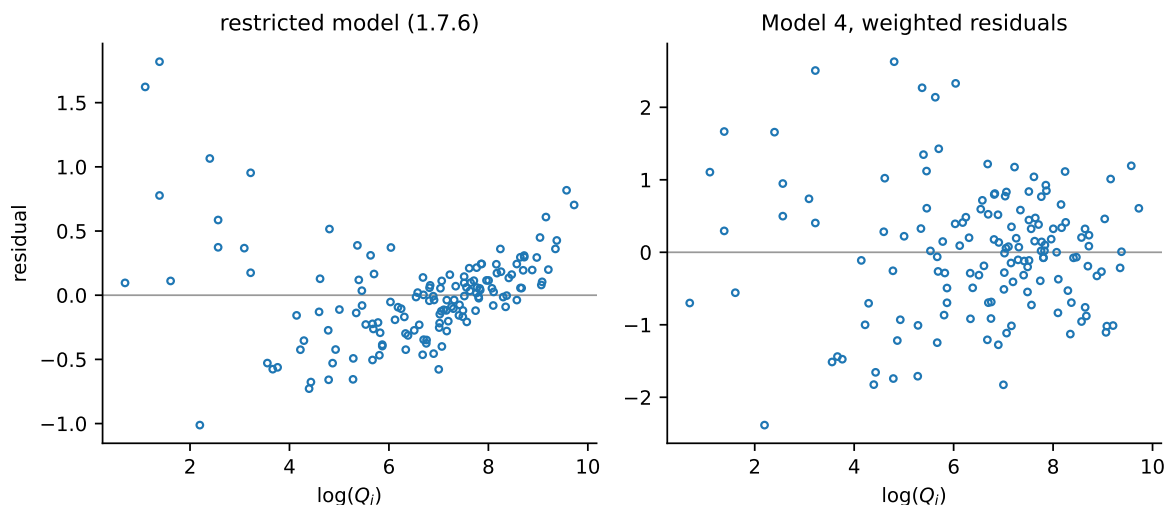


Figure S1.2: residuals against $\log(Q_i)$ for Nerlove's data. Left: the restricted model (1.7.6). Right: Model 4, estimated by weighted least squares.

Problem Set for Chapter 1: Monte Carlo Exercise

Monte Carlo Exercise 1.1

(a) `code/ch1.py` runs both simulations with 1,000,000 replications, generating the regressor by the book's matrix formula $\mathbf{x} = \mathbf{r}x_0 + \mathbf{d} + \mathbf{A}\boldsymbol{\eta}$ with $r_i = \phi^i$, $d_i = c(1 - \phi^i)/(1 - \phi)$ and $\mathbf{A}$ the lower-triangular Toeplitz matrix with $A_{ij} = \phi^{i-j}$ for $i \geq j$. The results are

	simulation 1 ($\mathbf{x}$ fixed)	simulation 2 ($\mathbf{x}$ redrawn)
mean of b_1	1.00049	—
mean of b_2	0.49993	—
$\text{Prob}\{ t > t_{0.025}(30)\}$	0.05003	0.05011

With a million replications the standard error of a rejection frequency is $\sqrt{0.05 \cdot 0.95/10^6} = 0.00022$, so both frequencies are within one standard error of 5%; the average of $\mathbf{b}$ is within 0.0005 of β . The first simulation illustrates Proposition 1.1(a), $E(\mathbf{b} \mid \mathbf{X}) = \beta$, and the conditional statement of Proposition 1.3; the second illustrates that the same t distribution governs the *unconditional* sampling behaviour, as argued in Review Question 1.4.1 — the conditional distribution does not depend on $\mathbf{x}$, so it is also the marginal one. Figure S1.1 overlays the two simulated distributions of the t -ratio on the $t(30)$ density.

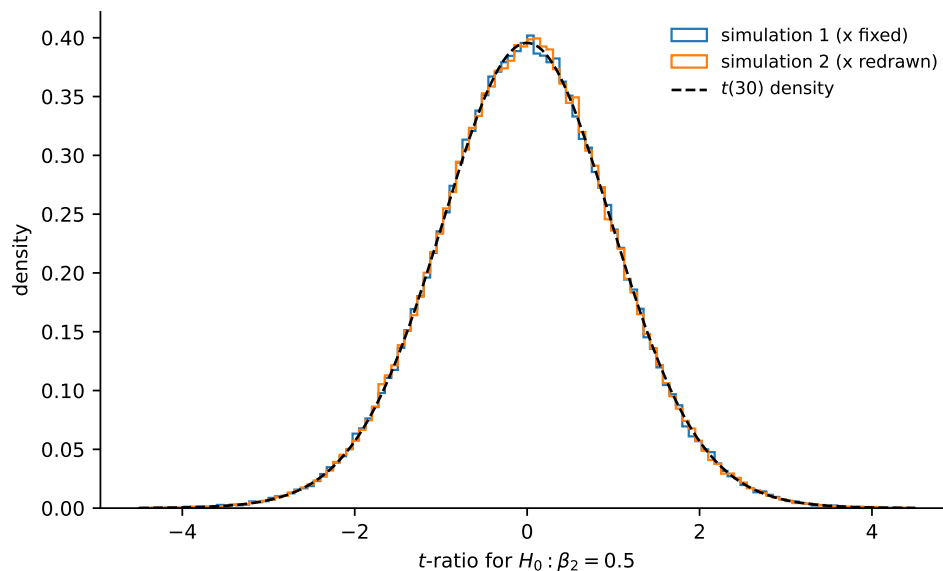


Figure S1.1: the t -ratio for $H_0: \beta_2 = 0.5$ in the two simulations of the Monte Carlo exercise, 10^6 replications, against the $t(30)$ density.

(b) *Strict exogeneity holds.* The whole path $\mathbf{x} = (x_1, \dots, x_n)'$ is generated by $(x_0, \boldsymbol{\eta})$, which is independent of $\boldsymbol{\varepsilon}$; hence $E(\varepsilon_i | \mathbf{X}) = E(\varepsilon_i) = 0$ for every i , which is Assumption 1.2 — note that this requires independence of ε_i from *all* the x 's, past and future, and that is exactly how the design was built. *Conditional homoskedasticity holds* too: $E(\varepsilon_i^2 | \mathbf{X}) = E(\varepsilon_i^2) = 1$ and $E(\varepsilon_i \varepsilon_j | \mathbf{X}) = 0$ for $i \neq j$, so Assumption 1.4 is satisfied with $\sigma^2 = 1$. Together with normality (Assumption 1.5) and $\text{rank}(\mathbf{X}) = 2$ almost surely, the design satisfies Assumptions 1.1–1.5, which is why the simulated finite-sample results match the theory exactly.

Remark. The sample is *not* i.i.d.: x_i is serially correlated, with $\text{Corr}(x_i, x_{i-1}) = \phi = 0.6$. Serial correlation in the regressor is harmless here — none of Assumptions 1.1–1.5 requires a random sample. What would destroy strict exogeneity is feedback from the error to later regressors, as in the autoregression $y_i = \beta_1 + \beta_2 y_{i-1} + \varepsilon_i$ of Chapter 2, where only predeterminedness survives and the finite-sample results of this chapter no longer apply.

Chapter 2

Large-Sample Theory

Solutions to all the exercises of Chapter 2 of Fumio Hayashi, *Econometrics* (Princeton University Press, 2000): the 47 review questions of Sections 2.1–2.12, the twelve analytical exercises, the empirical exercises on Mishkin’s and Nerlove’s data, and the two Monte Carlo exercises. Assumptions, lemmas, propositions and equations cited as ASSUMPTION 2.3, LEMMA 2.4(C) or (2.3.1) refer to the book. The numerical parts are computed by `code/ch2.py`.

Section 2.1 Review of Limit Theorems for Sequences of Random Variables

Review Question 2.1.1

Yes, and for deterministic sequences the converse holds too, so the two notions coincide.

Let $\{z_n\}$ be real numbers with $z_n \rightarrow \alpha$ and fix $\varepsilon > 0$. By the definition of a limit there is N such that $|z_n - \alpha| < \varepsilon$ for all $n > N$. For such n the event $\{|z_n - \alpha| > \varepsilon\}$ is empty, so

$$\text{Prob}\{|z_n - \alpha| > \varepsilon\} = 0 \quad (n > N),$$

and a fortiori $\text{Prob}\{|z_n - \alpha| > \varepsilon\} \rightarrow 0$, which is $z_n \xrightarrow{p} \alpha$. Conversely, if a deterministic sequence did not converge to α , some $\varepsilon > 0$ would have $|z_n - \alpha| > \varepsilon$ infinitely often, and along that subsequence $\text{Prob}\{|z_n - \alpha| > \varepsilon\} = 1$, so $z_n \not\xrightarrow{p} \alpha$.

Convergence in probability is therefore a genuine weakening only for genuinely random sequences: it allows z_n to differ from α by a lot, as long as the probability of doing so vanishes.

Review Question 2.1.2

Both definitions in the text are elementwise; both displayed conditions are about the squared Euclidean distance. Write K for the dimension.

Mean square. The identity $E[(\mathbf{z}_n - \boldsymbol{\alpha})'(\mathbf{z}_n - \boldsymbol{\alpha})] = \sum_{k=1}^K E[(z_{nk} - \alpha_k)^2]$ expresses the displayed quantity as a sum of K nonnegative terms. A finite sum of nonnegative sequences converges to zero if and only if each summand does, so $E[(\mathbf{z}_n - \boldsymbol{\alpha})'(\mathbf{z}_n - \boldsymbol{\alpha})] \rightarrow 0$ if and only if $z_{nk} \xrightarrow{m.s.} \alpha_k$ for every k .

Probability. Let $d_n \equiv (\mathbf{z}_n - \boldsymbol{\alpha})'(\mathbf{z}_n - \boldsymbol{\alpha})$. If $\text{Prob}\{d_n > \varepsilon\} \rightarrow 0$ for every $\varepsilon > 0$, then for each k , since $(z_{nk} - \alpha_k)^2 \leq d_n$,

$$\text{Prob}\{|z_{nk} - \alpha_k| > \varepsilon\} = \text{Prob}\{(z_{nk} - \alpha_k)^2 > \varepsilon^2\} \leq \text{Prob}\{d_n > \varepsilon^2\} \rightarrow 0,$$

so $z_{nk} \xrightarrow{p} \alpha_k$. Conversely, if every element converges in probability, then by the union bound

$$\text{Prob}\{d_n > \varepsilon\} \leq \sum_{k=1}^K \text{Prob}\left\{(z_{nk} - \alpha_k)^2 > \frac{\varepsilon}{K}\right\} \rightarrow 0,$$

because $d_n > \varepsilon$ forces at least one of the K squared deviations to exceed ε/K . So for vectors of fixed dimension, elementwise convergence and convergence of the distance are the same thing.

Review Question 2.1.3

Decompose

$$\mathbf{A}_n \mathbf{x}_n = (\mathbf{A}_n - \mathbf{A}) \mathbf{x}_n + \mathbf{A} \mathbf{x}_n.$$

For the first term, apply LEMMA 2.4(B) row by row: the ℓ -th row of $\mathbf{A}_n - \mathbf{A}$ converges in probability to $\mathbf{0}$, and $\mathbf{x}_n \xrightarrow{d} \mathbf{x}$, so the ℓ -th element of $(\mathbf{A}_n - \mathbf{A}) \mathbf{x}_n$ converges in probability to zero. By Review Question 2.1.2 the whole vector then converges in probability to $\mathbf{0}$: $(\mathbf{A}_n - \mathbf{A}) \mathbf{x}_n \xrightarrow{p} \mathbf{0}$.

For the second term, $\mathbf{x} \mapsto \mathbf{A} \mathbf{x}$ is a continuous function that does not depend on n , so LEMMA 2.3(B) gives $\mathbf{A} \mathbf{x}_n \xrightarrow{d} \mathbf{A} \mathbf{x}$.

Now LEMMA 2.4(A) applies to the sum of a sequence converging in distribution and one converging in probability to a constant:

$$\mathbf{A}_n \mathbf{x}_n = \mathbf{A} \mathbf{x}_n + (\mathbf{A}_n - \mathbf{A}) \mathbf{x}_n \xrightarrow{d} \mathbf{A} \mathbf{x} + \mathbf{0} = \mathbf{A} \mathbf{x}.$$

In particular, if $\mathbf{x} \sim N(\mathbf{0}, \Sigma)$ then $\mathbf{A}_n \mathbf{x}_n \xrightarrow{d} N(\mathbf{0}, \mathbf{A} \Sigma \mathbf{A}')$, which is the form in which the lemma is used throughout the chapter (with $\mathbf{A}_n$ a matrix of sample moments and $\mathbf{x}_n$ a normalized sample average).

Review Question 2.1.4

Yes. Write

$$\hat{\theta}_n - \theta = \frac{1}{\sqrt{n}} \cdot \sqrt{n}(\hat{\theta}_n - \theta),$$

a product of a deterministic sequence $1/\sqrt{n} \rightarrow 0$ (hence $\xrightarrow{p} 0$ by Review Question 2.1.1) and a sequence converging in distribution. By LEMMA 2.4(B) the product converges in probability to zero, so $\hat{\theta}_n - \theta \xrightarrow{p} 0$.

The intuition: asymptotic normality says the estimation error is of order $n^{-1/2}$, which is in particular $o_p(1)$. The converse fails — consistency says nothing about the rate, and an estimator can be consistent with a non-normal limit distribution or a slower rate (as in the unit-root asymptotics of Chapter 9, where the rate is n and the limit is not normal).

Review Question 2.1.5

By Kolmogorov's LLN, $\bar{z}_n \xrightarrow{p} \mu$, and by the Lindeberg-Levy CLT $\sqrt{n}(\bar{z}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$. Apply LEMMA 2.5 (the delta method) with $\beta = \mu$, $x_n = \bar{z}_n$ and $a(\beta) = 1/\beta$, which is continuously differentiable in a neighbourhood of μ because $\mu \neq 0$, with

$$A(\beta) = \frac{\partial a(\beta)}{\partial \beta} = -\frac{1}{\beta^2}, \quad A(\mu) = -\frac{1}{\mu^2}.$$

The delta method gives

$$\sqrt{n} \left(\frac{1}{\bar{z}_n} - \frac{1}{\mu} \right) \xrightarrow{d} N \left(0, A(\mu) \sigma^2 A(\mu)' \right) = N \left(0, \frac{\sigma^2}{\mu^4} \right).$$

The requirement $\mu \neq 0$ is essential: at $\mu = 0$ the function $1/\beta$ is not differentiable (indeed $1/\bar{z}_n$ does not even converge), and the asymptotic variance σ^2/μ^4 blows up as μ approaches zero —

the finite-sample counterpart of which is the notorious instability of ratio estimators with a small denominator.

Section 2.2 Fundamental Concepts in Time-Series Analysis

Review Question 2.2.1

For a covariance-stationary process, $\Gamma_j \equiv \text{Cov}(\mathbf{z}_i, \mathbf{z}_{i-j}) = E[(\mathbf{z}_i - \boldsymbol{\mu})(\mathbf{z}_{i-j} - \boldsymbol{\mu})']$ does not depend on i . Replacing i by $i + j$ therefore leaves it unchanged, so

$$\Gamma_{-j} = \text{Cov}(\mathbf{z}_i, \mathbf{z}_{i+j}) = \text{Cov}(\mathbf{z}_{i+j}, \mathbf{z}_i)' = \text{Cov}(\mathbf{z}_i, \mathbf{z}_{i-j})' = \Gamma_j',$$

where the second equality is the general fact $\text{Cov}(\mathbf{x}, \mathbf{y}) = \text{Cov}(\mathbf{y}, \mathbf{x})'$ and the third uses covariance stationarity to shift the index by $-j$. For a scalar process this says $\gamma_{-j} = \gamma_j$: the autocovariance function is symmetric, so only $j \geq 0$ need be tabulated. For a vector process the transpose matters — Γ_j is generally not symmetric, because the cross covariance between variable a today and variable b j periods ago differs from that between b today and a j periods ago.

Review Question 2.2.2

The future is forecast *exactly*: $E(z_i | z_1) = T_i(z_1)$, where T_i is the i -th Chebyshev polynomial of the first kind, defined by $\cos(i\theta) = T_i(\cos \theta)$. For example

$$E(z_2 | z_1) = 2z_1^2 - 1, \quad E(z_3 | z_1) = 4z_1^3 - 3z_1.$$

The reason is that z_i is a deterministic function of z_1 . Knowing $z_1 = \cos(w)$ leaves two possibilities, w and $2\pi - w$, and both give the same z_i , since $\cos(i(2\pi - w)) = \cos(2\pi i - iw) = \cos(iw)$. So z_i is measurable with respect to z_1 , and conditioning on z_1 returns z_i itself.

Consequently $\{z_i\}$ is *not* a martingale difference sequence: an m.d.s. requires $E(z_i | z_{i-1}, \dots, z_1) = 0$, whereas here the conditional expectation is a nonzero polynomial in z_1 . The example shows how weak “white noise” is. Zero serial correlation restricts only *linear* prediction; it leaves the process open to perfect *nonlinear* prediction. The process is also not strictly stationary and not ergodic, which is why the ergodic stationary martingale difference CLT cannot be applied to it.

Review Question 2.2.3

Write $I_i \equiv (\mathbf{z}_i, \dots, \mathbf{z}_1)$ for the information available at date i , so that the martingale property reads $E(x_i | I_{i-1}) = x_{i-1}$ and x_{i-1} is a function of I_{i-1} (it is an element of $\mathbf{z}_{i-1}$).

Induct on j . For $j = 0$ the claim is the martingale property itself. Suppose $E(x_{i+j} | I_{i-1}) = x_{i-1}$. Then, by the law of iterated expectations applied to the two nested information sets $I_{i-1} \subset I_{i+j}$,

$$E(x_{i+j+1} | I_{i-1}) = E[E(x_{i+j+1} | I_{i+j}) | I_{i-1}] = E(x_{i+j} | I_{i-1}) = x_{i-1},$$

using the martingale property in the inner expectation. This completes the induction, so $E(x_{i+j} | I_{i-1}) = x_{i-1}$ for every $j \geq 0$. Subtracting two consecutive statements,

$$E(x_{i+j+1} - x_{i+j} | I_{i-1}) = x_{i-1} - x_{i-1} = 0.$$

In words: given today's information, no change in a martingale is anticipated, at any horizon. Under Hall's hypothesis (EXAMPLE 2.5) this is the statement that no *predictable* part of future consumption growth should be forecastable from current information — the restriction that the empirical literature tests by regressing consumption growth on lagged variables.

Review Question 2.2.4

Write $g_i \equiv x_i \varepsilon_i$.

i.i.d.: *no.* The g_i are independent (see below) but not identically distributed: $\text{Var}(g_i) = x_i^2 \sigma^2$ changes with i because x_i does.

Serially independent: *yes.* g_i is a fixed (nonrandom) multiple of ε_i , and the ε_i are independent across i ; multiplying independent random variables by constants preserves independence.

m.d.s.: *yes.* $E(g_i) = x_i E(\varepsilon_i) = 0$ and, by independence, $E(g_i \mid g_{i-1}, \dots, g_1) = x_i E(\varepsilon_i \mid \varepsilon_{i-1}, \dots, \varepsilon_1) = x_i E(\varepsilon_i) = 0$.

Stationary: *no.* Not even covariance stationary, since the variance $x_i^2 \sigma^2$ depends on i ; a fortiori not strictly stationary.

The example matters for the theory of this chapter: the martingale difference CLT of Billingsley requires *ergodic stationary* m.d.s., and this process is an m.d.s. that fails stationarity. Heteroskedastic regression errors are exactly of this type, which is why Chapter 2 assumes ergodic stationarity of $\{\mathbf{g}_i\} = \{\mathbf{x}_i \varepsilon_i\}$ rather than deriving it.

Review Question 2.2.5

Let $z_i = g_1 + \dots + g_i$ with $\{g_i\}$ independent white noise, $E(g_i) = 0$, $\text{Var}(g_i) = \sigma^2 > 0$. The mean is constant ($E(z_i) = 0$), but by independence

$$\text{Var}(z_i) = \sum_{k=1}^i \text{Var}(g_k) = i \sigma^2,$$

which grows without bound and depends on i . Covariance stationarity requires the variance to be the same for all i , so a random walk is not covariance stationary, hence not stationary. The autocovariances inherit the same defect: $\text{Cov}(z_i, z_{i-j}) = (i-j)\sigma^2$ depends on i as well as on j , and the correlation $\text{Corr}(z_i, z_{i-j}) = \sqrt{(i-j)/i}$ tends to one for fixed j — a random walk is very far from the asymptotic independence that ergodicity requires. Chapter 9 develops the asymptotics for such processes; the tools of this chapter do not apply to them.

Review Question 2.2.6

Let $\{\mathbf{z}_i\}$ be a martingale and let $\mathbf{g}_i \equiv \mathbf{z}_i - \mathbf{z}_{i-1}$ be its first differences, as in (2.2.7). The two sequences carry the same information: knowing $(\mathbf{z}_1, \dots, \mathbf{z}_{i-1})$ is the same as knowing $(\mathbf{g}_1, \dots, \mathbf{g}_{i-1})$, since each can be computed from the other by differencing or cumulating. Hence

$$E(\mathbf{g}_i \mid \mathbf{g}_{i-1}, \dots, \mathbf{g}_1) = E(\mathbf{z}_i - \mathbf{z}_{i-1} \mid \mathbf{z}_{i-1}, \dots, \mathbf{z}_1) = E(\mathbf{z}_i \mid \mathbf{z}_{i-1}, \dots, \mathbf{z}_1) - \mathbf{z}_{i-1} = \mathbf{z}_{i-1} - \mathbf{z}_{i-1} = \mathbf{0},$$

using (2.2.5) and the fact that $\mathbf{z}_{i-1}$ is known given the conditioning variables. Taking unconditional expectations gives $E(\mathbf{g}_i) = \mathbf{0}$, so $\{\mathbf{g}_i\}$ is an m.d.s. With (2.2.8), which cumulates an m.d.s. into a

martingale, the two concepts are two views of the same object: martingales are the running sums of martingale differences.

Review Question 2.2.7

Condition on the past of $\{g_i\}$ and insert the finer information set generated by the ε 's, which includes it, since $(g_{i-1}, \dots, g_2)$ is a function of $(\varepsilon_{i-1}, \dots, \varepsilon_1)$:

$$E(g_i \mid g_{i-1}, \dots, g_2) = E[E(\varepsilon_i \varepsilon_{i-1} \mid \varepsilon_{i-1}, \dots, \varepsilon_1) \mid g_{i-1}, \dots, g_2] = E[\varepsilon_{i-1} E(\varepsilon_i \mid g_{i-1}, \dots, g_2)] = 0,$$

because ε_i is independent of its own past with mean zero. Hence $\{g_i\}$ is an m.d.s., and by the proposition proved in the text it has no serial correlation.

It is not, however, independent white noise: g_i and g_{i-1} share the factor ε_{i-1} and are dependent, even though uncorrelated. With $\varepsilon_i \sim N(0, \sigma^2)$, for instance,

$$\text{Cov}(g_i^2, g_{i-1}^2) = E(\varepsilon_i^2 \varepsilon_{i-1}^4 \varepsilon_{i-2}^2) - [E(\varepsilon_i^2 \varepsilon_{i-1}^2)]^2 = \sigma^2 \cdot 3\sigma^4 \cdot \sigma^2 - \sigma^8 = 2\sigma^8 > 0 :$$

squared values are positively correlated, which is precisely the volatility clustering that ARCH models of (2.2.10) are built to capture.

Review Question 2.2.8

Write $I_k \equiv (y_k, \dots, y_1)$. Both terms of r_{i1} are conditional expectations of the same random variable on nested information sets $I_{i-2} \subset I_{i-1}$. Condition on the smaller one and use the law of iterated expectations on the first term, while the second is already a function of I_{i-2} :

$$E(r_{i1} \mid I_{i-2}) = E[E(y_i \mid I_{i-1}) \mid I_{i-2}] - E(y_i \mid I_{i-2}) = E(y_i \mid I_{i-2}) - E(y_i \mid I_{i-2}) = 0.$$

Now note that r_{j1} for $j \leq i-1$ is a function of $I_{j-1} \subset I_{i-2}$. Conditioning the display down to the past revisions therefore gives

$$E(r_{i1} \mid r_{i-1,1}, \dots, r_{21}) = E[E(r_{i1} \mid I_{i-2}) \mid r_{i-1,1}, \dots, r_{21}] = 0,$$

so $\{r_{i1}\}$ is a martingale difference sequence (with mean zero, by one more application of iterated expectations). Revisions to a forecast are unforecastable from information already used in making it — if they were not, the forecast would not have been optimal.

Remark. The conditioning set has to be I_{i-2} , the information on which the *second* term is based. Conditional on I_{i-1} the revision is not zero but equal to itself: $E(r_{i1} \mid I_{i-1}) = r_{i1}$, because both terms are then known. The exercise's phrase “m.d.s. with respect to $\{y_i\}$ ” should be read in this sense.

Review Question 2.2.9

Put $\mathbf{g}_i \equiv \mathbf{z}_i - \boldsymbol{\mu}$. Then:

$\{\mathbf{g}_i\}$ is stationary and ergodic. It is i.i.d. (a fixed shift of an i.i.d. sequence), hence strictly stationary; and an i.i.d. sequence is ergodic, since variables far apart in the sequence are not merely asymptotically independent but exactly independent, so the defining factorization $|E[f(\mathbf{g}_i, \dots)g(\mathbf{g}_{i+n}, \dots)]| = |E[f]| |E[g]|$ holds for every n .

$\{\mathbf{g}_i\}$ is an m.d.s. with $E(\mathbf{g}_i \mathbf{g}_i') = \Sigma$. Independence gives $E(\mathbf{g}_i \mid \mathbf{g}_{i-1}, \dots, \mathbf{g}_1) = E(\mathbf{g}_i) = \mathbf{0}$, and $E(\mathbf{g}_i \mathbf{g}_i') = \text{Var}(\mathbf{z}_i) = \Sigma$, which is finite by hypothesis.

The martingale differences CLT applies to such a sequence and states that $\sqrt{n} \bar{\mathbf{g}}_n \xrightarrow{d} N(\mathbf{0}, \Sigma)$. Since $\bar{\mathbf{g}}_n = \bar{\mathbf{z}}_n - \boldsymbol{\mu}$, this is exactly the Lindeberg–Levy conclusion

$$\sqrt{n}(\bar{\mathbf{z}}_n - \boldsymbol{\mu}) \xrightarrow{d} N(\mathbf{0}, \Sigma).$$

So Lindeberg–Levy is the special case of the ergodic stationary m.d.s. CLT in which the martingale differences happen to be i.i.d. The gain from the more general theorem is exactly what large-sample econometrics needs: it covers regression errors that are serially uncorrelated but conditionally heteroskedastic and dependent, as in Review Question 2.2.7.

Section 2.3 Large-Sample Distribution of the OLS Estimator

Review Question 2.3.1

First, the error has conditional mean zero:

$$E(\varepsilon_i | \mathbf{x}_i) = E(y_i | \mathbf{x}_i) - \mathbf{x}_i' \boldsymbol{\beta} = 0,$$

since $\mathbf{x}_i' \boldsymbol{\beta}$ is a function of the conditioning variable. Orthogonality then follows from the law of total expectations:

$$E(\mathbf{g}_i) = E(\mathbf{x}_i \cdot \varepsilon_i) = E[E(\mathbf{x}_i \varepsilon_i | \mathbf{x}_i)] = E[\mathbf{x}_i E(\varepsilon_i | \mathbf{x}_i)] = \mathbf{0},$$

which is ASSUMPTION 2.3. The implication runs one way only: $E(\varepsilon_i | \mathbf{x}_i) = 0$ is strictly stronger than $E(\mathbf{x}_i \varepsilon_i) = \mathbf{0}$, because the former also forces $E[h(\mathbf{x}_i) \varepsilon_i] = 0$ for *every* function h , not just for the coordinates of $\mathbf{x}_i$. Orthogonality is all the large-sample theory of this chapter needs, which is why it is what ASSUMPTION 2.3 assumes.

Review Question 2.3.2

(a) *No*. Ergodic stationarity (ASSUMPTION 2.2) restricts the *time-series* behaviour of $\{y_i, \mathbf{x}_i\}$, not the size of its moments: a strictly stationary process can have infinite second moments (an i.i.d. Cauchy sequence is stationary and ergodic). ASSUMPTION 2.4 assumes the second moments of the regressors are finite, and ASSUMPTION 2.5 assumes that $\mathbf{S} = E(\mathbf{g}_i \mathbf{g}_i') = E(\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i')$ exists, is finite and is nonsingular; but a finite $E(\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i')$ does not by itself bound $E(\varepsilon_i^2)$, because the weights $x_{ik}x_{ij}$ could be small exactly where ε_i^2 is large. This is why PROPOSITION 2.2 carries the proviso “provided $E(\varepsilon_i^2)$ exists and is finite”.

(b) If $x_{i1} = 1$, then the $(1, 1)$ element of $E(\mathbf{g}_i \mathbf{g}_i') = E(\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i')$ is

$$E(\varepsilon_i^2 x_{i1}^2) = E(\varepsilon_i^2),$$

and that element is finite because the whole matrix is, by ASSUMPTION 2.5. So with an intercept in the regression — the usual case — the proviso is automatically satisfied.

Review Question 2.3.3

Condition on $\mathbf{x}_i$ and use the law of total expectations:

$$\mathbf{S} = E(\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i') = E[E(\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i' | \mathbf{x}_i)] = E[E(\varepsilon_i^2 | \mathbf{x}_i) \mathbf{x}_i \mathbf{x}_i'] = E[f(\mathbf{x}_i) \mathbf{x}_i \mathbf{x}_i'],$$

the middle step because $\mathbf{x}_i \mathbf{x}_i'$ is a function of the conditioning variable and can be pulled out of the inner expectation. The expression makes the role of conditional heteroskedasticity transparent: $\mathbf{S}$ is the matrix of second moments of the regressors *weighted by the conditional variance of the error*, so observations in regions of $\mathbf{x}$ where the error is noisy get more weight. Under conditional homoskedasticity $f(\mathbf{x}_i) \equiv \sigma^2$ and $\mathbf{S} = \sigma^2 \Sigma_{\mathbf{xx}}$, which collapses the sandwich (2.3.4) to $\sigma^2 \Sigma_{\mathbf{xx}}^{-1}$ (Section 2.6).

Review Question 2.3.4

Equation (2.3.11) is the algebraic identity

$$\frac{1}{n} \sum_{i=1}^n e_i^2 = \frac{1}{n} \sum_{i=1}^n \varepsilon_i^2 - 2(\mathbf{b} - \beta)' \bar{\mathbf{g}} + (\mathbf{b} - \beta)' \mathbf{S}_{\mathbf{xx}} (\mathbf{b} - \beta), \quad \bar{\mathbf{g}} = \frac{1}{n} \sum_i \mathbf{x}_i \varepsilon_i.$$

Under ASSUMPTIONS 2.1–2.4 the proof of PROPOSITION 2.1 has already established

$$\mathbf{b} - \beta \xrightarrow{p} \mathbf{0}, \quad \bar{\mathbf{g}} \xrightarrow{p} E(\mathbf{g}_i) = \mathbf{0}, \quad \mathbf{S}_{\mathbf{xx}} \xrightarrow{p} \Sigma_{\mathbf{xx}} \text{ (finite)}.$$

The maps $(\mathbf{u}, \mathbf{v}) \mapsto \mathbf{u}'\mathbf{v}$ and $(\mathbf{u}, \mathbf{A}) \mapsto \mathbf{u}'\mathbf{A}\mathbf{u}$ are continuous and do not depend on n , so LEMMA 2.3(A) lets the probability limits be taken inside:

$$(\mathbf{b} - \beta)' \bar{\mathbf{g}} \xrightarrow{p} \mathbf{0}'\mathbf{0} = 0, \quad (\mathbf{b} - \beta)' \mathbf{S}_{\mathbf{xx}} (\mathbf{b} - \beta) \xrightarrow{p} \mathbf{0}' \Sigma_{\mathbf{xx}} \mathbf{0} = 0.$$

Therefore $\frac{1}{n} \sum_i e_i^2$ and $\frac{1}{n} \sum_i \varepsilon_i^2$ have the same probability limit, and the latter is $E(\varepsilon_i^2)$ by the ergodic theorem applied to the ergodic stationary process $\{\varepsilon_i^2\}$ (finite by the proviso). Hence $\frac{1}{n} \sum_i e_i^2 \xrightarrow{p} E(\varepsilon_i^2)$, which is PROPOSITION 2.2; and since $s^2 = \frac{n}{n-K} \cdot \frac{1}{n} \sum_i e_i^2$ with $n/(n-K) \rightarrow 1$, s^2 is consistent as well.

Review Question 2.3.5

Since $\hat{\varepsilon}_i = \varepsilon_i - \mathbf{x}_i'(\hat{\beta} - \beta)$, squaring and averaging gives the same identity as before with $\hat{\beta}$ in place of $\mathbf{b}$:

$$\frac{1}{n} \sum_i \hat{\varepsilon}_i^2 = \frac{1}{n} \sum_i \varepsilon_i^2 - 2(\hat{\beta} - \beta)' \left(\frac{1}{n} \sum_i \mathbf{x}_i \varepsilon_i \right) + (\hat{\beta} - \beta)' \mathbf{S}_{\mathbf{xx}} (\hat{\beta} - \beta).$$

By ergodic stationarity and the assumed finiteness, the two sample averages converge: $\frac{1}{n} \sum_i \mathbf{x}_i \varepsilon_i \xrightarrow{p} E(\mathbf{x}_i \varepsilon_i)$ and $\mathbf{S}_{\mathbf{xx}} \xrightarrow{p} E(\mathbf{x}_i \mathbf{x}_i')$, both finite. They are multiplied by $\hat{\beta} - \beta \xrightarrow{p} \mathbf{0}$, so by LEMMA 2.3(A) the last two terms vanish in probability — *whatever* the value of $E(\mathbf{x}_i \varepsilon_i)$. Hence

$$\frac{1}{n} \sum_i \hat{\varepsilon}_i^2 \xrightarrow{p} \text{plim} \frac{1}{n} \sum_i \varepsilon_i^2 = E(\varepsilon_i^2).$$

The point of the generalization is that neither ASSUMPTION 2.3 (orthogonality) nor the use of OLS is needed: consistency of the coefficient estimator is enough. This is what makes the residual variance from, say, the instrumental-variable estimators of Chapter 3 — where the regressors are *not* orthogonal to the error — a consistent estimator of $E(\varepsilon_i^2)$.

Section 2.4 Hypothesis Testing

Review Question 2.4.1

Yes. By definition $SE^*(b_k) = \sqrt{\widehat{\text{Avar}}(b_k)/n}$, where $\widehat{\text{Avar}}(b_k)$ is the (k, k) element of $\mathbf{S}_{\mathbf{xx}}^{-1}\widehat{\mathbf{S}}\mathbf{S}_{\mathbf{xx}}^{-1}$ and converges in probability to the finite number $\text{Avar}(b_k)$. Since $1/n \rightarrow 0$ and the square root is continuous, LEMMA 2.3(A) gives

$$SE^*(b_k) = \sqrt{\frac{\widehat{\text{Avar}}(b_k)}{n}} \xrightarrow{p} 0.$$

That is exactly as it should be: the standard error measures the sampling uncertainty about b_k , which disappears as the estimator becomes consistent. What does *not* vanish is the t -ratio $(b_k - \bar{\beta}_k)/SE^*(b_k)$ under the null, because numerator and denominator shrink at the same rate $n^{-1/2}$; the ratio is asymptotically $N(0, 1)$ by (2.4.1).

Review Question 2.4.2

Apply the delta method (LEMMA 2.5) to $a(\beta) = -\log(\beta)$, which for $\beta > 0$ is continuously differentiable with $A(\beta) = -1/\beta$. From $\sqrt{n}(b - \beta) \xrightarrow{d} N(0, \text{Avar}(b))$,

$$\sqrt{n}(\hat{\lambda} - \lambda) \xrightarrow{d} N\left(0, \frac{\text{Avar}(b)}{\beta^2}\right), \quad \text{so} \quad \text{Avar}(\hat{\lambda}) = \frac{\text{Avar}(b)}{\beta^2}.$$

Replacing the unknowns by consistent estimates (b for β , $\widehat{\text{Avar}}(b)$ for $\text{Avar}(b)$) and dividing by n to get the variance of $\hat{\lambda}$ itself,

$$SE(\hat{\lambda}) = \sqrt{\frac{\widehat{\text{Avar}}(b)/b^2}{n}} = \frac{1}{b} \sqrt{\frac{\widehat{\text{Avar}}(b)}{n}} = \frac{SE(b)}{b},$$

as claimed (for $b > 0$; in general the factor is $1/|b|$). Standard errors of nonlinear functions of estimates are always computed this way, and the same recipe with the Jacobian $\mathbf{A}(\hat{\beta})$ covers the vector case.

Review Question 2.4.3

None of the three. Write $\widehat{\mathbf{V}} \equiv \widehat{\text{Avar}}(\mathbf{b})/n$ for the estimated variance of $\mathbf{b}$, so that $W = (\mathbf{Rb} - \mathbf{r})'[\mathbf{R}\widehat{\mathbf{V}}\mathbf{R}']^{-1}(\mathbf{Rb} - \mathbf{r})$. Under the reparameterization,

$$\widetilde{\mathbf{Rb}} - \widetilde{\mathbf{r}} = \mathbf{F}(\mathbf{Rb} - \mathbf{r}), \quad [\widetilde{\mathbf{R}}\widehat{\mathbf{V}}\widetilde{\mathbf{R}}']^{-1} = [\mathbf{F}\mathbf{R}\widehat{\mathbf{V}}\mathbf{R}'\mathbf{F}']^{-1} = (\mathbf{F}')^{-1}[\mathbf{R}\widehat{\mathbf{V}}\mathbf{R}']^{-1}\mathbf{F}^{-1},$$

so that

$$\widetilde{W} = (\widetilde{\mathbf{Rb}} - \widetilde{\mathbf{r}})' \mathbf{F}'(\mathbf{F}')^{-1}[\mathbf{R}\widehat{\mathbf{V}}\mathbf{R}']^{-1}\mathbf{F}^{-1}\mathbf{F}(\mathbf{Rb} - \mathbf{r}) = W.$$

The statistic is numerically identical in every sample; hence its finite-sample distribution and its asymptotic distribution — $\chi^2(\#r)$ under the null — are identical too. The inverse in the middle is what does the work: it undoes exactly the linear transformation applied to the restrictions.

Remark. The invariance is a property of *linear* reformulations. It fails for nonlinear ones: the restrictions $\beta_1\beta_2 = 1$ and $\beta_1 - 1/\beta_2 = 0$ describe the same null but give numerically different Wald statistics, because

the delta-method variance is evaluated at the estimate and depends on the parameterization. This non-invariance — discussed in the text after the definition of W — is the main practical drawback of the Wald principle relative to likelihood ratio tests.

Section 2.5 Estimating $E(\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i')$ Consistently

Review Question 2.5.1

No. Expanding $e_i^2 = (y_i - \mathbf{x}_i' \mathbf{b})^2$ inside (2.5.1),

$$\hat{\mathbf{S}} = \frac{1}{n} \sum_i e_i^2 \mathbf{x}_i \mathbf{x}_i' = \underbrace{\frac{1}{n} \sum_i y_i^2 \mathbf{x}_i \mathbf{x}_i'}_{\text{2nd moments of } y^2} - 2 \frac{1}{n} \sum_i y_i (\mathbf{x}_i' \mathbf{b}) \mathbf{x}_i \mathbf{x}_i' + \frac{1}{n} \sum_i (\mathbf{x}_i' \mathbf{b})^2 \mathbf{x}_i \mathbf{x}_i'.$$

These involve sample averages of $y_i^2 x_{ik} x_{i\ell}$, of $y_i x_{ij} x_{ik} x_{i\ell}$ and of $x_{ih} x_{ij} x_{ik} x_{i\ell}$: third- and fourth-order cross moments of the data, which cannot be recovered from the second-order moments $\mathbf{S}_{xx}$, $\mathbf{s}_{xy}$, $\mathbf{y}'\mathbf{y}/n$ and $\bar{y}$.

What remains true is that the robust standard errors can still be computed in *one pass* over the data: accumulate the $K^2(K+1)/2$ -odd fourth-order sums above (or, more simply, compute $\mathbf{b}$ first and then accumulate $e_i^2 \mathbf{x}_i \mathbf{x}_i'$ in a second pass). What is lost is the tidy fact that one fixed, small set of second moments is a sufficient summary of the data for every statistic — robust inference needs genuinely more information about the sample, which is another way of saying that it uses the empirical distribution of the residuals rather than a single variance parameter.

Review Question 2.5.2

The two are the same sandwich, $(\mathbf{X}'\mathbf{X})^{-1}(\text{middle})(\mathbf{X}'\mathbf{X})^{-1}$, with three differences.

What the middle matrix is. In Chapter 1 it is $\mathbf{X}'(\sigma^2 \mathbf{V})\mathbf{X}$, built from the *true* conditional variance matrix $\sigma^2 \mathbf{V} = E(\varepsilon \varepsilon' | \mathbf{X})$, which is unknown; the formula is therefore a theoretical object, not an estimator, unless $\mathbf{V}$ is known. In (2.5.4') it is $\mathbf{X}'\mathbf{B}\mathbf{X}$ with $\mathbf{B} = \text{diag}(e_1^2, \dots, e_n^2)$, which is computable from the data. Note that $\mathbf{B}$ is not a good estimate of $\sigma^2 \mathbf{V}$ element by element — one residual cannot estimate one variance — and does not need to be: only the $K \times K$ average $\mathbf{X}'\mathbf{B}\mathbf{X}/n$ has to converge, and it does, to $\mathbf{S}$.

Diagonality. $\mathbf{V}$ is an arbitrary positive definite matrix, so the Chapter 1 formula allows serial correlation as well as heteroskedasticity. $\mathbf{B}$ is diagonal: White's formula is valid only because ASSUMPTION 2.5 makes $\{\mathbf{g}_i\}$ a martingale difference sequence, so that the autocovariances of $\mathbf{g}_i$ vanish. When they do not, $\mathbf{S}$ must be replaced by a long-run variance matrix (Chapter 6).

What is being measured. $\text{Var}(\mathbf{b} | \mathbf{X})$ is the exact variance of $\mathbf{b}$ conditional on $\mathbf{X}$, and it requires strict exogeneity for unbiasedness to make sense; $\widehat{\text{Avar}}(\mathbf{b})$ estimates the variance of the asymptotic distribution of $\sqrt{n}(\mathbf{b} - \beta)$ — hence the factor n — and needs only predeterminedness. Dividing (2.5.4') by n gives $(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{B}\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}$, which is the direct analogue of the Chapter 1 expression with $\sigma^2 \mathbf{V}$ replaced by $\mathbf{B}$.

Section 2.6 Implications of Conditional Homoskedasticity

Review Question 2.6.1

No to both. The estimator (2.6.2) is $s^2 \mathbf{S}_{\mathbf{xx}}^{-1}$, and since $s^2 \xrightarrow{p} E(\varepsilon_i^2) \equiv \sigma^2$ and $\mathbf{S}_{\mathbf{xx}} \xrightarrow{p} \Sigma_{\mathbf{xx}}$,

$$s^2 \mathbf{S}_{\mathbf{xx}}^{-1} \xrightarrow{p} \sigma^2 \Sigma_{\mathbf{xx}}^{-1}, \quad \text{whereas} \quad \text{Avar}(\mathbf{b}) = \Sigma_{\mathbf{xx}}^{-1} \mathbf{S} \Sigma_{\mathbf{xx}}^{-1}.$$

These agree if and only if $\mathbf{S} = E(\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i') = \sigma^2 \Sigma_{\mathbf{xx}}$, that is, if and only if $E[(E(\varepsilon_i^2 | \mathbf{x}_i) - \sigma^2) \mathbf{x}_i \mathbf{x}_i'] = \mathbf{0}$ (Review Question 2.3.3). Conditional homoskedasticity makes the bracket zero; without it the two matrices generally differ, in either direction — the conventional standard errors can be too small or too large.

The t -ratio inherits the failure. Under $H_0: \beta_k = \bar{\beta}_k$,

$$t_k = \frac{\sqrt{n}(b_k - \bar{\beta}_k)}{\sqrt{s^2 [\mathbf{S}_{\mathbf{xx}}^{-1}]_{kk}}} \xrightarrow{d} N\left(0, \frac{[\text{Avar}(\mathbf{b})]_{kk}}{\sigma^2 [\Sigma_{\mathbf{xx}}^{-1}]_{kk}}\right),$$

by LEMMA 2.4(C): the numerator is asymptotically $N(0, [\text{Avar}(\mathbf{b})]_{kk})$ and the denominator converges to the wrong constant. The limit is standard normal only if the ratio in the variance is one. So without conditional homoskedasticity the conventional t must be replaced by the robust t of (2.4.1); the critical values of $N(0, 1)$ applied to the conventional t give a test whose true asymptotic size is not the nominal one.

Review Question 2.6.2

Consistency. Yes. (2.3.5), $\mathbf{S}_{\mathbf{xx}}^{-1} \hat{\mathbf{S}} \mathbf{S}_{\mathbf{xx}}^{-1}$, is consistent for $\Sigma_{\mathbf{xx}}^{-1} \mathbf{S} \Sigma_{\mathbf{xx}}^{-1}$ whether or not ASSUMPTION 2.7 holds (given ASSUMPTION 2.6); when it does hold, that limit is $\sigma^2 \Sigma_{\mathbf{xx}}^{-1}$. Robust standard errors are therefore valid under conditional homoskedasticity as well — they are never wrong asymptotically, only unnecessary.

Advantage of (2.6.2): fewer parameters. To form (2.6.2) one estimates $\Sigma_{\mathbf{xx}}$ and the single scalar σ^2 : $K(K+1)/2+1$ population parameters. To form (2.3.5) one estimates $\Sigma_{\mathbf{xx}}$ and the whole matrix $\mathbf{S} = E(\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i')$: $K(K+1)$ parameters, and the elements of $\hat{\mathbf{S}}$ are averages of fourth-order products, which are noisy. With $K = 5$, that is 16 parameters against 30. Since the finite-sample behaviour of an estimator deteriorates with the number of parameters being estimated, the conventional standard errors are the more reliable ones *when the homoskedasticity assumption is true* — this is the trade-off behind the Monte Carlo evidence cited in Section 2.5 that robust t -ratios over-reject in small samples.

Review Question 2.6.3

From the tables (or `code/ch2.py`),

$$F_{0.05}(10, \infty) = 1.8307, \quad \chi_{0.05}^2(10) = 18.307 = 10 \times 1.8307.$$

The relation is $\chi_\alpha^2(m) = m \cdot F_\alpha(m, \infty)$, which follows from the construction of the F distribution: $F(m_1, m_2)$ is the ratio of two independent chi-squares divided by their degrees of freedom, and as $m_2 \rightarrow \infty$ the denominator $\chi^2(m_2)/m_2 \xrightarrow{p} 1$, so $F(m, \infty) \stackrel{d}{=} \chi^2(m)/m$. That is exactly why PROPOSITION 2.5(C) states that $\#r \cdot F$, not F , is asymptotically $\chi^2(\#r)$: multiplying by the number of restrictions converts the F -scale into the χ^2 -scale. Since $F(\#r, n - K) \rightarrow F(\#r, \infty)$ as

$n \rightarrow \infty$, using the $F(\#r, n - K)$ table on F and the $\chi^2(\#r)$ table on $\#r \cdot F$ are asymptotically the same test, the former being slightly more conservative in finite samples.

Review Question 2.6.4

No. By Analytical Exercise 1.5 the statistic equals $\#r \cdot F$, which in the notation of this chapter is

$$\#r \cdot F = \sqrt{n}(\mathbf{R}\mathbf{b} - \mathbf{r})' [s^2 \mathbf{R}\mathbf{S}_{\mathbf{xx}}^{-1} \mathbf{R}']^{-1} \sqrt{n}(\mathbf{R}\mathbf{b} - \mathbf{r}).$$

Under the null $\sqrt{n}(\mathbf{R}\mathbf{b} - \mathbf{r}) \xrightarrow{d} \mathbf{x} \sim N(\mathbf{0}, \Omega)$ with $\Omega = \mathbf{R} \text{Avar}(\mathbf{b}) \mathbf{R}'$, while the middle matrix converges to $\mathbf{A} = \sigma^2 \mathbf{R} \Sigma_{\mathbf{xx}}^{-1} \mathbf{R}'$. Hence

$$\#r \cdot F \xrightarrow{d} \mathbf{x}' \mathbf{A}^{-1} \mathbf{x} \stackrel{d}{=} \sum_{j=1}^{\#r} \lambda_j w_j, \quad w_j \text{ i.i.d. } \chi^2(1),$$

where $\lambda_1, \dots, \lambda_{\#r}$ are the eigenvalues of $\mathbf{A}^{-1} \Omega$. This is $\chi^2(\#r)$ if and only if all the λ_j equal one, that is, $\Omega = \mathbf{A}$ — precisely what conditional homoskedasticity delivers. Otherwise the statistic has a weighted sum of chi-squares as its limit, with both mean and critical values different from those of $\chi^2(\#r)$. The remedy is the robust Wald statistic (2.4.2), which replaces $\mathbf{A}$ by a consistent estimate of Ω .

Review Question 2.6.5

By Analytical Exercise 1.6(b), the F -statistic for the null equals

$$F = \frac{R^2/(K-1)}{(1-R^2)/(n-K)} \iff (K-1)F = \frac{n-K}{n} \cdot \frac{nR^2}{1-R^2}.$$

PROPOSITION 2.5(C) gives $(K-1)F \xrightarrow{d} \chi^2(K-1)$. Under the null, $R^2 \xrightarrow{p} 0$: the numerator of R^2 is $\frac{1}{n} \sum_i (\hat{y}_i - \bar{y})^2 = (\mathbf{b} - \hat{\beta})' \mathbf{S}_{\mathbf{xx}} (\mathbf{b} - \hat{\beta}) \xrightarrow{p} 0$ because $\mathbf{b}$ converges to the true coefficient vector whose slopes are zero, while the denominator converges to $\text{Var}(y_i) > 0$. Solving the display for nR^2 ,

$$nR^2 = (K-1)F \cdot (1-R^2) \cdot \frac{n}{n-K},$$

a product of a sequence converging in distribution to $\chi^2(K-1)$ and a sequence converging in probability to $1 \cdot 1 = 1$. By LEMMA 2.4(C) (or (2.1.8)), $nR^2 \xrightarrow{d} \chi^2(K-1)$.

Without conditional homoskedasticity: no. The argument rests entirely on PROPOSITION 2.5(C), which assumes ASSUMPTION 2.7; by Review Question 2.6.4 the statistic then has a weighted-chi-square limit. This matters in practice because nR^2 statistics are ubiquitous as Lagrange-multiplier tests — including White's test of Section 2.7 and the Breusch–Godfrey test of Section 2.10, both of which need their own homoskedasticity-type condition for the χ^2 approximation.

Section 2.7 Testing Conditional Homoskedasticity

Review Question 2.7.1

The products of the four entries with one another are

	1	q	q^2	p
1	1	q	q^2	p
q		q^2	q^3	qp
q^2			q^4	q^2p
p				p^2

Discarding the constant 1 and counting each distinct monomial once (q^2 appears twice, as $1 \cdot q^2$ and as $q \cdot q$), the distinct nonconstant elements are

$$q, \quad q^2, \quad q^3, \quad q^4, \quad p, \quad qp, \quad q^2p, \quad p^2,$$

that is, 8 of them, so ψ_i is 8-dimensional and White's auxiliary regression of e_i^2 on a constant and ψ_i has 9 coefficients; the test statistic is $nR^2 \xrightarrow{d} \chi^2(8)$.

The count also shows why White's test loses power in larger models: with K regressors the number of distinct products is at most $K(K+1)/2 - 1$ nonconstant terms, which grows quadratically, so the degrees of freedom (and with them the critical value) grow fast while the alternative may concentrate on one or two of the terms. Dropping from ψ_i those variables that theory says cannot affect the conditional variance raises power — at the cost of no power at all against the alternatives so excluded.

Section 2.8 Estimation with Parameterized Conditional Heteroskedasticity
Review Question 2.8.1

Since $\mathbf{z}_i = \mathbf{z}(\mathbf{x}_i)$ is known once $\mathbf{x}_i$ is,

$$E(\mathbf{z}_i \eta_i) = E[E(\mathbf{z}_i \eta_i \mid \mathbf{x}_i)] = E[\mathbf{z}_i E(\eta_i \mid \mathbf{x}_i)] = E[\mathbf{z}_i \cdot 0] = \mathbf{0}.$$

This is the step that makes the auxiliary regression (2.8.8) of ε_i^2 on $\mathbf{z}_i$ legitimate: the regressors $\mathbf{z}_i$, being functions of $\mathbf{x}_i$, are orthogonal to its error $\eta_i \equiv \varepsilon_i^2 - E(\varepsilon_i^2 \mid \mathbf{x}_i)$, so PROPOSITION 2.1 applies and OLS estimates $\boldsymbol{\alpha}$ consistently. The same argument, applied to $\tilde{\mathbf{x}}_i = \mathbf{x}_i / \sqrt{\mathbf{z}_i' \boldsymbol{\alpha}}$ and $\tilde{\varepsilon}_i = \varepsilon_i / \sqrt{\mathbf{z}_i' \boldsymbol{\alpha}}$, verifies ASSUMPTION 2.3 for the weighted equation.

Review Question 2.8.2

Generally *no*. The error of (2.8.8) is $\eta_i = \varepsilon_i^2 - E(\varepsilon_i^2 \mid \mathbf{x}_i)$, whose conditional second moment is

$$E(\eta_i^2 \mid \mathbf{x}_i) = \text{Var}(\varepsilon_i^2 \mid \mathbf{x}_i) = E(\varepsilon_i^4 \mid \mathbf{x}_i) - [E(\varepsilon_i^2 \mid \mathbf{x}_i)]^2,$$

a function of $\mathbf{x}_i$ in all but contrived cases. If, for instance, ε_i is conditionally normal, $E(\eta_i^2 \mid \mathbf{x}_i) = 2(\mathbf{z}_i' \boldsymbol{\alpha})^2$, which varies with $\mathbf{x}_i$ whenever the model is heteroskedastic at all — the very situation the auxiliary regression is run in.

Does it matter for the WLS estimator? No. What the WLS procedure needs from Step 2 is consistency of $\hat{\boldsymbol{\alpha}}$, and PROPOSITION 2.1(A) delivers that without any homoskedasticity assumption.

As stated in the text, $\sqrt{n}(\hat{\beta}(\hat{\mathbf{V}}) - \beta)$ and $\sqrt{n}(\hat{\beta}(\mathbf{V}) - \beta)$ are asymptotically equivalent, so the sampling error in $\hat{\alpha}$ — and hence its variance — washes out of the asymptotic distribution of the WLS estimator entirely.

It does matter if one wants to do *inference about α itself* (for example to test whether some element of $\mathbf{z}_i$ belongs in the skedastic function): there the conventional standard errors from Step 2 are invalid and robust ones should be used.

Section 2.9 Least Squares Projection

Review Question 2.9.1

By construction $E(\eta | \mathbf{x}) = E(y | \mathbf{x}) - E(y | \mathbf{x}) = 0$. Hence, for any ϕ such that the expectation exists,

$$E[\eta \phi(\mathbf{x})] = E[E(\eta \phi(\mathbf{x}) | \mathbf{x})] = E[\phi(\mathbf{x})E(\eta | \mathbf{x})] = 0.$$

Applying this with $\phi(\mathbf{x}) = E(y | \mathbf{x}) - f(\mathbf{x})$ kills the cross term in the mean squared error decomposition (2.9.3):

$$E[(y - f(\mathbf{x}))^2] = \underbrace{E[\eta^2]}_{\text{unavoidable}} + E[(E(y | \mathbf{x}) - f(\mathbf{x}))^2],$$

so that the conditional expectation is the minimum mean squared error forecast: any other f adds a nonnegative term. The identity is also the source of the orthogonality that defines the least squares projection, with ϕ restricted to linear functions.

Review Question 2.9.2

Both projections are zero. Write $\mathbf{w} \equiv (\varepsilon_{i-1}, \dots, \varepsilon_{i-m})'$. The projection coefficients on the lags alone are

$$[E(\mathbf{w}\mathbf{w}')]^{-1}E(\mathbf{w}\varepsilon_i),$$

and every element of $E(\mathbf{w}\varepsilon_i)$ is an autocovariance γ_j with $j \geq 1$, which is zero for white noise. With a constant included, (2.9.7) gives $\boldsymbol{\gamma} = \text{Var}(\mathbf{w})^{-1} \text{Cov}(\mathbf{w}, \varepsilon_i) = \mathbf{0}$ and $\mu = E(\varepsilon_i) - \boldsymbol{\gamma}'E(\mathbf{w}) = 0$. So the best *linear* forecast of white noise from its own past is the unconditional mean, zero.

The conditional expectation need not be zero. Example 2.4 makes the point: for $\varepsilon_i = \cos(iw)$ the process is white noise, yet by Review Question 2.2.2 $E(\varepsilon_i | \varepsilon_{i-1})$ is a nonzero polynomial in ε_{i-1} — the past predicts the present perfectly, nonlinearly. Zero autocorrelation restricts only the linear projection; it is the martingale difference property, not white noise, that sets the conditional expectation to zero.

Review Question 2.9.3

Use the formula (2.9.7) for the projection with a constant and compute its ingredients from the assumed conditional mean. Since $\text{Cov}(\tilde{\mathbf{x}}, y) = \text{Cov}(\tilde{\mathbf{x}}, E(y | \tilde{\mathbf{x}}))$ by the law of total covariance (the part of y orthogonal to every function of $\tilde{\mathbf{x}}$ contributes nothing),

$$\text{Cov}(\tilde{\mathbf{x}}, y) = \text{Cov}(\tilde{\mathbf{x}}, \mu + \boldsymbol{\gamma}'\tilde{\mathbf{x}}) = \text{Var}(\tilde{\mathbf{x}})\boldsymbol{\gamma} \implies \boldsymbol{\gamma}^* = \text{Var}(\tilde{\mathbf{x}})^{-1} \text{Cov}(\tilde{\mathbf{x}}, y) = \boldsymbol{\gamma},$$

and then $\mu^* = E(y) - \gamma'E(\tilde{\mathbf{x}}) = [\mu + \gamma'E(\tilde{\mathbf{x}})] - \gamma'E(\tilde{\mathbf{x}}) = \mu$. So the projection coefficients coincide with the coefficients of the conditional expectation.

The same conclusion follows in one line from Review Question 2.9.1: the conditional expectation minimizes the mean squared error among *all* functions; if it happens to be linear, it also minimizes among linear functions, which is the definition of the projection. This is the situation in which OLS estimates a conditional expectation rather than merely a linear approximation to it — for instance under joint normality (Review Question 1.1.4) or when the regressors are a saturated set of dummies.

Review Question 2.9.4

(a) Write $\mathbf{w}_i \equiv (\mathbf{x}_i', \mathbf{z}_i')'$. Because $E(\mathbf{z}_i \mathbf{x}_i') = \mathbf{0}$, the second moment matrix is block diagonal, so

$$\hat{E}^*(\varepsilon_i | \mathbf{x}_i, \mathbf{z}_i) = \left[E(\mathbf{w}_i \mathbf{w}_i')^{-1} E(\mathbf{w}_i \varepsilon_i) \right]' \mathbf{w}_i = \mathbf{0}' \mathbf{x}_i + [E(\mathbf{z}_i \mathbf{z}_i')^{-1} E(\mathbf{z}_i \varepsilon_i)]' \mathbf{z}_i.$$

Projections are linear, so projecting $y_i = \mathbf{x}_i' \beta + \mathbf{z}_i' \delta + \varepsilon_i$ gives

$$\hat{E}^*(y_i | \mathbf{x}_i, \mathbf{z}_i) = \mathbf{x}_i' \beta + \mathbf{z}_i' [\delta + E(\mathbf{z}_i \mathbf{z}_i')^{-1} E(\mathbf{z}_i \varepsilon_i)].$$

The coefficient of $\mathbf{x}_i$ is β — the endogeneity of $\mathbf{z}_i$ contaminates only the coefficient of $\mathbf{z}_i$, because $\mathbf{z}_i$ is uncorrelated with $\mathbf{x}_i$.

(b) Treating $\mathbf{z}_i' \delta + \varepsilon_i$ as the error,

$$E(\mathbf{x}_i \mathbf{x}_i')^{-1} E(\mathbf{x}_i y_i) = \beta + E(\mathbf{x}_i \mathbf{x}_i')^{-1} [E(\mathbf{x}_i \mathbf{z}_i') \delta + E(\mathbf{x}_i \varepsilon_i)] = \beta,$$

again β , since both terms in the bracket vanish. So OLS estimates β consistently with or without $\mathbf{z}_i$ in the regression.

(c) Use the long projection, on $(\mathbf{x}_i, \mathbf{z}_i)$. Both are consistent, so the choice is about precision, and the projection error is smaller in the long regression: adding regressors cannot increase the mean squared projection error, and it strictly decreases it unless $\mathbf{z}_i$ has no explanatory power at all. Under conditional homoskedasticity the asymptotic variance of the β -block is $\sigma_v^2 E(\mathbf{x}_i \mathbf{x}_i')^{-1}$ in the long regression and $\sigma_u^2 E(\mathbf{x}_i \mathbf{x}_i')^{-1}$ in the short one (the block diagonality of the second moment matrix leaves the $\mathbf{x}$ block untouched), with $\sigma_v^2 \leq \sigma_u^2$: including $\mathbf{z}_i$ sharpens the estimate of β even though the coefficient of $\mathbf{z}_i$ itself is biased for δ .

Section 2.10 Testing for Serial Correlation

Review Question 2.10.1

Expand the product in (2.10.1):

$$\hat{\gamma}_j = \frac{1}{n} \sum_{t=j+1}^n (z_t - \bar{z}_n)(z_{t-j} - \bar{z}_n) = \frac{1}{n} \sum_{t=j+1}^n z_t z_{t-j} - \bar{z}_n \frac{1}{n} \sum_{t=j+1}^n z_{t-j} - \bar{z}_n \frac{1}{n} \sum_{t=j+1}^n z_t + \frac{n-j}{n} (\bar{z}_n)^2.$$

Fix j and let $n \rightarrow \infty$. The process $\{z_t z_{t-j}\}$ is a measurable function of finitely many coordinates of an ergodic stationary process, hence itself ergodic stationary, so the ergodic theorem gives $\frac{1}{n} \sum_{t=j+1}^n z_t z_{t-j} \xrightarrow{p} E(z_t z_{t-j})$ (the j missing terms change the average by $O_p(j/n)$). The same theorem gives $\bar{z}_n \xrightarrow{p} E(z_t) \equiv \mu_z$ and, again up to j/n , $\frac{1}{n} \sum_{t=j+1}^n z_{t-j} \xrightarrow{p} \mu_z$ and $\frac{1}{n} \sum_{t=j+1}^n z_t \xrightarrow{p} \mu_z$;

and $(n - j)/n \rightarrow 1$. By LEMMA 2.3(A),

$$\hat{\gamma}_j \xrightarrow{P} E(z_t z_{t-j}) - \mu_z^2 - \mu_z^2 + \mu_z^2 = E(z_t z_{t-j}) - E(z_t)E(z_{t-j}) = \gamma_j.$$

Consistency of $\hat{\rho}_j = \hat{\gamma}_j/\hat{\gamma}_0$ then follows from LEMMA 2.3(A) applied to the ratio, since $\gamma_0 > 0$. Note that dividing by n rather than $n - j$ is immaterial asymptotically, as $(n - j)/n \rightarrow 1$; it matters in moderate samples, which is why the text insists on saying which convention is in use.

Review Question 2.10.2

From (2.10.4) and (2.10.5), with $\mathbf{x}_n \equiv ((\sqrt{n}\hat{\rho}_1)^2, \dots, (\sqrt{n}\hat{\rho}_p)^2)'$,

$$Q_{LB} - Q_{BP} = \sum_{j=1}^p \left(\frac{n+2}{n-j} - 1 \right) (\sqrt{n}\hat{\rho}_j)^2 = \mathbf{a}_n' \mathbf{x}_n, \quad \mathbf{a}_n = \left(\frac{j+2}{n-j} \right)_{j=1, \dots, p}.$$

For fixed p , $\mathbf{a}_n \rightarrow \mathbf{0}$ as $n \rightarrow \infty$, hence $\mathbf{a}_n \xrightarrow{P} \mathbf{0}$ by Review Question 2.1.1. By PROPOSITION 2.9 and the continuous mapping theorem, $\mathbf{x}_n$ converges in distribution (to a vector of independent $\chi^2(1)$ variables). LEMMA 2.4(B) then gives $\mathbf{a}_n' \mathbf{x}_n \xrightarrow{P} 0$, so $Q_{LB} - Q_{BP} \xrightarrow{P} 0$ and, by LEMMA 2.4(A), the two statistics have the same asymptotic distribution, $\chi^2(p)$.

The Ljung–Box weights $(n+2)/(n-j) > 1$ inflate the statistic in finite samples, which is deliberate: the Box–Pierce statistic is known to be undersized in moderate samples, and the correction improves the approximation without changing the limit.

Review Question 2.10.3

Start from the identity suggested in the text,

$$\frac{\tilde{\gamma}_j}{\tilde{\gamma}_0} - \frac{\hat{\gamma}_j}{\hat{\gamma}_0} = \tilde{\gamma}_j \left(\frac{1}{\tilde{\gamma}_0} - \frac{1}{\sigma^2} \right) - \hat{\gamma}_j \left(\frac{1}{\hat{\gamma}_0} - \frac{1}{\sigma^2} \right) + \frac{1}{\sigma^2} (\tilde{\gamma}_j - \hat{\gamma}_j),$$

which is verified by expanding the right-hand side. Multiply by $\sqrt{n}$ and treat the three terms in turn.

First term: $\sqrt{n}\tilde{\gamma}_j \xrightarrow{d} N(0, \sigma^4)$ and $1/\tilde{\gamma}_0 - 1/\sigma^2 \xrightarrow{P} 0$ by LEMMA 2.3(A), so their product is $o_p(1)$ by LEMMA 2.4(B). Second term: by hypothesis $\sqrt{n}\hat{\gamma}_j = \sqrt{n}\tilde{\gamma}_j + o_p(1)$ also converges in distribution, and $1/\hat{\gamma}_0 - 1/\sigma^2 \xrightarrow{P} 0$, so the same argument applies. Third term: $\sqrt{n}(\tilde{\gamma}_j - \hat{\gamma}_j) \xrightarrow{P} 0$ by hypothesis, and $1/\sigma^2$ is a constant. Adding up,

$$\sqrt{n}(\tilde{\rho}_j - \hat{\rho}_j) \xrightarrow{P} 0.$$

The practical content is the one drawn in the text: the $\sqrt{n}$ -scaled sample autocorrelations computed from OLS *residuals* have the same asymptotic distribution as those computed from the unobservable *errors* — so the $N(0, 1)$ critical values of (2.10.3) and the $\chi^2(p)$ critical values of the Q statistics may be used on residuals — *provided* $\sqrt{n}(\tilde{\gamma}_j - \hat{\gamma}_j) \xrightarrow{P} 0$, which (2.10.12) shows requires $E(\mathbf{x}_t \varepsilon_{t-j}) = \mathbf{0}$ and $E(\mathbf{x}_{t-j} \varepsilon_t) = \mathbf{0}$. When the regressors include lagged dependent variables those conditions fail, and the test has to be replaced by the Breusch–Godfrey regression.

Section 2.11 Application: Rational Expectations Econometrics

Review Question 2.11.1

$$\frac{1.05}{1.02} - 1 = 2.9412\% \quad \text{versus} \quad 5\% - 2\% = 3\%, \quad \frac{1.20}{1.17} - 1 = 2.5641\% \quad \text{versus} \quad 20\% - 17\% = 3\%.$$

The approximation error is 0.0588 percentage points in the first case, small relative to a real rate of about 3%; in the second it is 0.4359 points, about 17% of the true value. The reason is visible in the exact relation

$$r_{t+1} = \frac{R_t - \pi_{t+1}}{1 + \pi_{t+1}} :$$

the approximation drops the factor $1/(1 + \pi_{t+1})$, so its relative error is π_{t+1} itself. At the low inflation rates of the Fama sample period the difference is negligible, which is why the text works with $R_t - \pi_{t+1}$; in high-inflation samples (or for long horizons) the exact formula should be used.

Review Question 2.11.2

Implication 2 is $E(\pi_{t+1} | I_t) = -r + R_t$. Since $R_t \in I_t$, the law of iterated expectations applied to the smaller information set generated by R_t gives

$$E(\pi_{t+1} | R_t) = E[E(\pi_{t+1} | I_t) | R_t] = -r + R_t.$$

The conditional expectation is thus a linear function of R_t , and by Review Question 2.9.3 the least squares projection on $(1, R_t)$ coincides with it:

$$\hat{E}^*(\pi_{t+1} | 1, R_t) = -r + R_t.$$

Equivalently, in the notation of (2.9.7), $\gamma = \text{Var}(R_t)^{-1} \text{Cov}(R_t, \pi_{t+1}) = 1$ and $\mu = E(\pi_{t+1}) - E(R_t) = -r$. This is what makes the hypothesis testable by a regression: the efficient market hypothesis pins the slope of the population regression of inflation on the nominal rate at exactly one.

Review Question 2.11.3

The hypothesis to be tested would be (a): $E(\eta_{t+1} | I_t) = 0$, that inflation forecasts are rational. With η_{t+1} observable one can test its testable implications directly: for any vector $\mathbf{w}_t$ of variables in I_t (lagged inflation, lagged forecast errors, the nominal rate, money growth, ...), $E(\mathbf{w}_t \eta_{t+1}) = \mathbf{0}$, so one regresses

$$\eta_{t+1} = \mathbf{w}_t' \boldsymbol{\theta} + \text{error}$$

and tests $\boldsymbol{\theta} = \mathbf{0}$ with a robust Wald statistic; or, with $\mathbf{w}_t$ the lagged η 's, tests that the autocorrelations of $\{\eta_t\}$ are zero with a Box–Pierce or Ljung–Box Q (PROPOSITION 2.9 applies because an m.d.s. with own conditional homoskedasticity is exactly what (a) delivers).

No, the constant ex-ante real rate assumption (c) is not needed. Its only role in the text is to make the unobservable ${}_t\pi_{t+1}$ disappear: with ${}_tr_{t+1} = r$ constant, ${}_t\pi_{t+1} = R_t - r$, so that $\eta_{t+1} = \pi_{t+1} + r - R_t$ becomes a function of observables and Implication 2 turns into a regression with slope one. If η_{t+1} were observed we would be testing rationality alone — and the tests in the text, which

test the joint hypothesis of rational expectations *and* a constant ex-ante real rate, could not tell which half a rejection is due to.

Review Question 2.11.4

Fractions instead of percent. Both π_{t+1} and R_t are divided by 100. Dividing the dependent variable and the regressor by the same constant leaves the slope unchanged and divides the intercept by 100:

$$\pi_{t+1} = \underset{(0.00431)}{-0.00868} + \underset{(0.112)}{1.015} R_t,$$

with $SER = 0.0284$, while $R^2 = 0.24$, the t -ratios and the mean of the dependent variable in relative terms are unaffected (the mean becomes 0.0235).

Monthly inflation on annual interest rates. If x is the monthly and y the annual inflation rate, $1 + y = (1 + x)^{12}$, so $y \approx 12x$ and the new dependent variable is (approximately) the old one divided by 12. Dividing only the left-hand side divides every coefficient, every standard error and the SER by 12:

$$\pi_{t+1}^{\text{monthly}} = \underset{(0.0359)}{-0.0723} + \underset{(0.00933)}{0.0846} R_t^{\text{annual}}, \quad SER = 0.237\%, \quad R^2 = 0.24.$$

R^2 and all t -ratios are invariant, so the test of market efficiency is unaffected — but the null value of the slope is no longer 1: it becomes $1/12 = 0.0833$, and one must remember to test against that. Mismatched units are a common way of producing a “rejection” of a perfectly true hypothesis.

Review Question 2.11.5

The third element of $\mathbf{g}_t = \mathbf{x}_t \cdot \varepsilon_t$ would be $w_t \eta_{t+1}$, with w_t the new variable, and ASSUMPTION 2.3 would require $E(w_t \eta_{t+1}) = 0$. The derivation in (2.11.7) runs

$$E(w_t \eta_{t+1}) = E[E(w_t \eta_{t+1} | I_t)] = E[w_t E(\eta_{t+1} | I_t)] = 0,$$

and the *second* equality is the step that fails: pulling w_t out of the conditional expectation requires w_t to be known given I_t . If $w_t \notin I_t$ — say it is dated $t + 1$, or is an econometrician’s variable unavailable to market participants — then $E(w_t \eta_{t+1} | I_t) = E(w_t \eta_{t+1} | I_t)$ cannot be simplified, and nothing in the efficient market hypothesis makes it zero. The regressor is then not predetermined, OLS is inconsistent, and the “test” of market efficiency would reject for reasons having nothing to do with market efficiency. This is why the text insists that any additional regressors brought in to challenge R_t must belong to the current information set.

Review Question 2.11.6

Let $\{\pi_t\}$ and $\{R_t\}$ be mutually independent, serially independent processes with means $E(\pi_t) = \mu_\pi$ and $E(R_t) = \mu_R$, and let $\text{Var}(R_t) > 0$. Then the ex-post real rate $r_{t+1} = R_t - \pi_{t+1}$ has constant mean $E(r_{t+1}) = \mu_R - \mu_\pi$, and no serial correlation: for $j \geq 1$,

$$\text{Cov}(r_{t+1}, r_{t+1-j}) = \text{Cov}(R_t - \pi_{t+1}, R_{t-j} - \pi_{t+1-j}) = 0$$

because all four pairs are independent. So Implication 1 holds, and a Fama-style test based on the autocorrelations of the ex-post real rate would find nothing.

But Implication 2 fails badly: since π_{t+1} is independent of I_t ,

$$E(\pi_{t+1} | I_t) = \mu_\pi \neq -r + R_t$$

unless R_t is constant. The nominal rate carries no information about future inflation at all, whereas market efficiency says it should summarize all of it; the population regression of π_{t+1} on R_t has slope zero rather than one. The example shows that Implication 1 is much weaker than Implication 2 — the real rate can look “efficient” merely because nothing is forecastable by anyone — which is why the text tests both.

Section 2.12 Time Regressions

Review Question 2.12.1

Squaring $e_t = \varepsilon_t - \mathbf{x}'_t(\mathbf{b} - \beta)$ and inserting $\Upsilon_n \Upsilon_n^{-1} = \mathbf{I}$ around the sampling error,

$$\frac{1}{n} \sum_{t=1}^n e_t^2 = \frac{1}{n} \sum_{t=1}^n \varepsilon_t^2 - \frac{2}{n} [\Upsilon_n(\mathbf{b} - \beta)]' \mathbf{v}_n + \frac{1}{n} [\Upsilon_n(\mathbf{b} - \beta)]' \mathbf{Q}_n [\Upsilon_n(\mathbf{b} - \beta)],$$

with $\mathbf{Q}_n$ and $\mathbf{v}_n$ as in (2.12.8). By (2.12.7), $\Upsilon_n(\mathbf{b} - \beta) = \mathbf{Q}_n^{-1} \mathbf{v}_n$, which converges in distribution to $N(\mathbf{0}, \sigma^2 \mathbf{Q}^{-1})$ and is therefore $O_p(1)$; likewise $\mathbf{v}_n = O_p(1)$ by (2.12.11) and $\mathbf{Q}_n \rightarrow \mathbf{Q}$. Dividing $O_p(1)$ quantities by n sends them to zero in probability (LEMMA 2.4(B) with the deterministic sequence $1/n \rightarrow 0$), so the last two terms vanish. The first term converges to σ^2 by the ergodic theorem (or Kolmogorov’s LLN, the ε_t being i.i.d.). Hence

$$\frac{1}{n} \sum_t e_t^2 \xrightarrow{p} \sigma^2, \quad s^2 = \frac{n}{n-2} \cdot \frac{1}{n} \sum_t e_t^2 \xrightarrow{p} \sigma^2.$$

This is what licenses replacing σ^2 by s^2 in the t -values (2.12.14) and (2.12.15), so that inference about the intercept and the trend coefficient proceeds exactly as in the stationary case, even though the regressor t is not stationary and the two coefficients converge at different rates.

Review Question 2.12.2

Consistency: yes. From (2.12.7), $\Upsilon_n(\mathbf{b} - \beta) = \mathbf{Q}_n^{-1} \mathbf{v}_n \xrightarrow{d} \mathbf{Q}^{-1} N(\mathbf{0}, \mathbf{V})$ by LEMMA 2.4(C), so $\Upsilon_n(\mathbf{b} - \beta)$ is $O_p(1)$ and

$$\mathbf{b} - \beta = \Upsilon_n^{-1} \cdot O_p(1) = \begin{bmatrix} n^{-1/2} & 0 \\ 0 & n^{-3/2} \end{bmatrix} O_p(1) \xrightarrow{p} \mathbf{0}.$$

Both estimators are consistent, $\hat{\alpha}$ at rate $\sqrt{n}$ and $\hat{\delta}$ at rate $n^{3/2}$, exactly as in PROPOSITION 2.11; only the *variance* of the limit changes.

Limiting distribution. $\Upsilon_n(\mathbf{b} - \beta) \xrightarrow{d} N(\mathbf{0}, \mathbf{Q}^{-1} \mathbf{V} \mathbf{Q}^{-1})$ — the familiar sandwich, with $\mathbf{V}$ in place of $\sigma^2 \mathbf{Q}$.

t-values: asymptotically normal, but not standard normal. The t -values (2.12.14) and (2.12.15) divide by $\sqrt{s^2 \cdot [\cdot] \mathbf{Q}_n^{-1} [\cdot]}$, that is, they use $\sigma^2 \mathbf{Q}^{-1}$ as the asymptotic variance. The correct one is

$\mathbf{Q}^{-1}\mathbf{V}\mathbf{Q}^{-1}$, so, for the intercept,

$$t \xrightarrow{d} N\left(0, \frac{[\mathbf{Q}^{-1}\mathbf{V}\mathbf{Q}^{-1}]_{11}}{\sigma^2[\mathbf{Q}^{-1}]_{11}}\right),$$

and similarly for δ : normal, with a variance that equals one only when $\mathbf{V} = \sigma^2\mathbf{Q}$. Valid inference then requires estimating $\mathbf{V}$, which is the long-run variance matrix of $\{\mathbf{x}_t\varepsilon_t\}$ — the subject of Chapter 6.

Problem Set for Chapter 2: Analytical Exercises

Analytical Exercise 2.1

Convergence in probability. Fix $\varepsilon > 0$. For n large enough that $n^2 > \varepsilon$, the only way $|z_n|$ can exceed ε is $z_n = n^2$, so

$$\text{Prob}\{|z_n| > \varepsilon\} = \frac{1}{n} \longrightarrow 0,$$

which is $z_n \xrightarrow{p} 0$.

Divergence of the mean.

$$\mathbb{E}(z_n) = 0 \cdot \frac{n-1}{n} + n^2 \cdot \frac{1}{n} = n \longrightarrow \infty.$$

The two statements are compatible because convergence in probability says nothing about how *large* the rare deviations are, only how rare they are; here the probability of the bad event falls like $1/n$ while its size grows like n^2 , and the product diverges.

The moral matters for econometrics: consistency does not guarantee that an estimator has a finite mean, let alone a mean converging to the truth. (The two-stage least squares estimator of Chapter 3 with few instruments is a real example: it is consistent but has no finite moments in some cases.) A sufficient extra condition for moments to converge is uniform integrability — for instance $\sup_n \mathbb{E}(|z_n|^{1+\delta}) < \infty$ for some $\delta > 0$, which fails here since $\mathbb{E}(z_n) = n$.

Analytical Exercise 2.2

Add and subtract $\mathbb{E}(\bar{z}_n)$:

$$\bar{z}_n - \mu = (\bar{z}_n - \mathbb{E}(\bar{z}_n)) + (\mathbb{E}(\bar{z}_n) - \mu),$$

so that

$$(\bar{z}_n - \mu)^2 = (\bar{z}_n - \mathbb{E}(\bar{z}_n))^2 + 2(\bar{z}_n - \mathbb{E}(\bar{z}_n))(\mathbb{E}(\bar{z}_n) - \mu) + (\mathbb{E}(\bar{z}_n) - \mu)^2.$$

Take expectations. The middle term vanishes, because $\mathbb{E}(\bar{z}_n) - \mu$ is a constant and

$$\mathbb{E}[(\bar{z}_n - \mathbb{E}(\bar{z}_n))(\mathbb{E}(\bar{z}_n) - \mu)] = (\mathbb{E}(\bar{z}_n) - \mu) \mathbb{E}[\bar{z}_n - \mathbb{E}(\bar{z}_n)] = 0.$$

Hence

$$\mathbb{E}[(\bar{z}_n - \mu)^2] = \text{Var}(\bar{z}_n) + (\mathbb{E}(\bar{z}_n) - \mu)^2 \longrightarrow 0 + 0 = 0,$$

which is $\bar{z}_n \xrightarrow{m.s.} \mu$. By LEMMA 2.2(A), mean square convergence implies convergence in probability, so $\bar{z}_n \xrightarrow{p} \mu$, which is the LLN as stated in the text.

The decomposition is the familiar “mean squared error = variance + squared bias”; the LLN asks for both pieces to vanish. For an i.i.d. sample with finite variance, $\mathbb{E}(\bar{z}_n) = \mu$ exactly and

$\text{Var}(\bar{z}_n) = \sigma^2/n \rightarrow 0$, so both conditions hold — but the statement is more general, allowing bias that disappears and heterogeneous or (mildly) dependent observations, as long as the variance of the mean goes to zero.

Analytical Exercise 2.3

(a) A random sample is ergodic stationary: $\{y_i, \mathbf{x}_i\}$ i.i.d. is certainly strictly stationary, and independence gives the asymptotic — indeed exact — independence that ergodicity requires (Review Question 2.2.9). So ASSUMPTION 2.2' implies ASSUMPTION 2.2. Next, under a random sample $\{\mathbf{g}_i\} = \{\mathbf{x}_i \varepsilon_i\}$ is i.i.d. with mean $\mathbf{0}$ by ASSUMPTION 2.3, and an i.i.d. mean-zero sequence is a martingale difference sequence: $E(\mathbf{g}_i \mid \mathbf{g}_{i-1}, \dots, \mathbf{g}_1) = E(\mathbf{g}_i) = \mathbf{0}$. With $E(\mathbf{g}_i \mathbf{g}_i') = \mathbf{S}$ finite and nonsingular, ASSUMPTION 2.5 holds. So the hypotheses of PROPOSITION 2.1 are implied by those of the present proposition, and its conclusions follow.

(b) Directly, from (2.3.7),

$$\mathbf{b} - \beta = \mathbf{S}_{\mathbf{xx}}^{-1} \bar{\mathbf{g}}, \quad \mathbf{S}_{\mathbf{xx}} = \frac{1}{n} \sum_i \mathbf{x}_i \mathbf{x}_i', \quad \bar{\mathbf{g}} = \frac{1}{n} \sum_i \mathbf{x}_i \varepsilon_i.$$

Consistency. $\{\mathbf{x}_i \mathbf{x}_i'\}$ and $\{\mathbf{g}_i\}$ are i.i.d. with finite means $\Sigma_{\mathbf{xx}}$ (ASSUMPTION 2.4) and $\mathbf{0}$ (ASSUMPTION 2.3), so Kolmogorov's strong LLN gives $\mathbf{S}_{\mathbf{xx}} \xrightarrow{p} \Sigma_{\mathbf{xx}}$ and $\bar{\mathbf{g}} \xrightarrow{p} \mathbf{0}$. Since $\Sigma_{\mathbf{xx}}$ is nonsingular, matrix inversion is continuous there, and LEMMA 2.3(A) gives $\mathbf{b} - \beta \xrightarrow{p} \Sigma_{\mathbf{xx}}^{-1} \mathbf{0} = \mathbf{0}$.

Asymptotic normality. $\{\mathbf{g}_i\}$ is i.i.d. with mean $\mathbf{0}$ and variance $\mathbf{S}$, so the Lindeberg–Levy CLT gives $\sqrt{n} \bar{\mathbf{g}} \xrightarrow{d} N(\mathbf{0}, \mathbf{S})$. Then, by LEMMA 2.4(C) with $\mathbf{A}_n = \mathbf{S}_{\mathbf{xx}}^{-1} \xrightarrow{p} \Sigma_{\mathbf{xx}}^{-1}$,

$$\sqrt{n}(\mathbf{b} - \beta) = \mathbf{S}_{\mathbf{xx}}^{-1} \sqrt{n} \bar{\mathbf{g}} \xrightarrow{d} N(\mathbf{0}, \Sigma_{\mathbf{xx}}^{-1} \mathbf{S} \Sigma_{\mathbf{xx}}^{-1}),$$

which is (2.3.4). Note what the random-sample assumption buys: the ergodic theorem and the martingale CLT are replaced by the elementary Kolmogorov LLN and Lindeberg–Levy CLT, and the martingale difference property of $\{\mathbf{g}_i\}$ comes free instead of being assumed.

Analytical Exercise 2.4

By ergodic stationarity, $\frac{1}{n} \sum_i x_i^3 \varepsilon_i \xrightarrow{p} E(x_i^3 \varepsilon_i)$ *provided that expectation exists and is finite*, which is what has to be checked. Apply the Cauchy–Schwarz inequality with $f \equiv x_i \varepsilon_i = g_i$ and $h \equiv x_i^2$:

$$E(|x_i^3 \varepsilon_i|) = E(|g_i \cdot x_i^2|) \leq \sqrt{E(g_i^2) E(x_i^4)} = \sqrt{E(x_i^2 \varepsilon_i^2) E(x_i^4)} < \infty,$$

because $E(g_i^2) = E(x_i^2 \varepsilon_i^2)$ is finite by ASSUMPTION 2.5 (it is $\mathbf{S}$ in the scalar case) and $E(x_i^4)$ is finite by ASSUMPTION 2.6. So $E(x_i^3 \varepsilon_i)$ is finite and the sample average converges to it.

With this, both terms on the right-hand side of (2.5.3) vanish in probability: the first is $-2(\hat{\beta} - \beta) \cdot \frac{1}{n} \sum_i x_i^3 \varepsilon_i$, a product of $o_p(1)$ and $O_p(1)$; the second is $(\hat{\beta} - \beta)^2 \cdot \frac{1}{n} \sum_i x_i^4$, likewise, using ASSUMPTION 2.6 again. Hence $\frac{1}{n} \sum_i \hat{\varepsilon}_i^2 x_i^2 - \frac{1}{n} \sum_i \varepsilon_i^2 x_i^2 \xrightarrow{p} 0$, and since the second average converges to $E(\varepsilon_i^2 x_i^2) = \mathbf{S}$ by ergodic stationarity, $\hat{\mathbf{S}}$ is consistent — which is PROPOSITION 2.4. The fourth-moment assumption on the regressors is exactly what makes the Cauchy–Schwarz bound finite; without it, squaring the residuals can produce sample averages that do not settle down.

Analytical Exercise 2.5

The algebraic identity. Analytical Exercise 1.5(b) gives

$$\text{SSR}_R - \text{SSR}_U = (\mathbf{R}\mathbf{b} - \mathbf{r})' [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1} (\mathbf{R}\mathbf{b} - \mathbf{r}).$$

Under $H_0: \mathbf{R}\beta = \mathbf{r}$ and using $\mathbf{b} - \beta = \mathbf{S}_{\mathbf{xx}}^{-1}\bar{\mathbf{g}}$ and $(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{S}_{\mathbf{xx}}^{-1}/n$,

$$\mathbf{R}\mathbf{b} - \mathbf{r} = \mathbf{R}\mathbf{S}_{\mathbf{xx}}^{-1}\bar{\mathbf{g}}, \quad [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1} = n[\mathbf{R}\mathbf{S}_{\mathbf{xx}}^{-1}\mathbf{R}']^{-1},$$

so that

$$\text{SSR}_R - \text{SSR}_U = n \bar{\mathbf{g}}' \mathbf{S}_{\mathbf{xx}}^{-1} \mathbf{R}' [\mathbf{R}\mathbf{S}_{\mathbf{xx}}^{-1}\mathbf{R}']^{-1} \mathbf{R}\mathbf{S}_{\mathbf{xx}}^{-1} \bar{\mathbf{g}},$$

which is the displayed expression once one $\sqrt{n}$ is attached to each $\bar{\mathbf{g}}$.

The limit. Under ASSUMPTIONS 2.1–2.5 and 2.7, $\sqrt{n}\bar{\mathbf{g}} \xrightarrow{d} \mathbf{z} \sim N(\mathbf{0}, \mathbf{S})$ with $\mathbf{S} = \sigma^2 \Sigma_{\mathbf{xx}}$, and $\mathbf{S}_{\mathbf{xx}} \xrightarrow{p} \Sigma_{\mathbf{xx}}$, $s^2 \xrightarrow{p} \sigma^2$. By LEMMA 2.4(c) and the continuous mapping theorem,

$$\frac{\text{SSR}_R - \text{SSR}_U}{s^2} \xrightarrow{d} \frac{1}{\sigma^2} \mathbf{z}' \Sigma_{\mathbf{xx}}^{-1} \mathbf{R}' [\mathbf{R} \Sigma_{\mathbf{xx}}^{-1} \mathbf{R}']^{-1} \mathbf{R} \Sigma_{\mathbf{xx}}^{-1} \mathbf{z}.$$

Write $\mathbf{u} \equiv \Sigma_{\mathbf{xx}}^{-1/2} \mathbf{z} / \sigma \sim N(\mathbf{0}, \mathbf{I}_K)$ and $\mathbf{M} \equiv \Sigma_{\mathbf{xx}}^{-1/2} \mathbf{R}'$ ($K \times \#r$, of full column rank $\#r$). The limit becomes

$$\mathbf{u}' \mathbf{M} (\mathbf{M}' \mathbf{M})^{-1} \mathbf{M}' \mathbf{u} = \mathbf{u}' \mathbf{P} \mathbf{u},$$

with $\mathbf{P}$ the orthogonal projection onto the column space of $\mathbf{M}$: symmetric, idempotent, of rank $\#r$. A standard result (and the argument behind PROPOSITION 1.4) gives $\mathbf{u}' \mathbf{P} \mathbf{u} \sim \chi^2(\#r)$ for standard normal $\mathbf{u}$. Hence $(\text{SSR}_R - \text{SSR}_U)/s^2 \xrightarrow{d} \chi^2(\#r)$, without any appeal to PROPOSITION 2.3.

Analytical Exercise 2.6

(a) Regression (2.8.8), $\varepsilon_i^2 = \mathbf{z}_i' \boldsymbol{\alpha} + \eta_i$, satisfies ASSUMPTION 2.1 by construction. Add: ASSUMPTION 2.2 for $\{\varepsilon_i^2, \mathbf{z}_i\}$ (which follows from ergodic stationarity of $\{y_i, \mathbf{x}_i\}$, since $\mathbf{z}_i$ is a function of $\mathbf{x}_i$); the rank condition that $E(\mathbf{z}_i \mathbf{z}_i')$ be nonsingular and finite; and finiteness of $E(\mathbf{z}_i \eta_i)$ -type moments. Orthogonality holds automatically: by Review Question 2.8.1, $E(\eta_i | \mathbf{x}_i) = 0$ and $\mathbf{z}_i = \mathbf{z}(\mathbf{x}_i)$ give $E(\mathbf{z}_i \eta_i) = \mathbf{0}$. PROPOSITION 2.1(A) then gives $\tilde{\boldsymbol{\alpha}} \xrightarrow{p} \boldsymbol{\alpha}$.

(b) By (2.3.10), $e_i^2 = \varepsilon_i^2 + v_i$ with $v_i \equiv -2(\mathbf{b} - \beta)' \mathbf{x}_i \varepsilon_i + (\mathbf{b} - \beta)' \mathbf{x}_i \mathbf{x}_i' (\mathbf{b} - \beta)$, so (2.8.8) becomes $e_i^2 = \mathbf{z}_i' \boldsymbol{\alpha} + (\eta_i + v_i)$. Running OLS on each version,

$$\hat{\boldsymbol{\alpha}} = \boldsymbol{\alpha} + \left(\frac{1}{n} \sum_i \mathbf{z}_i \mathbf{z}_i' \right)^{-1} \frac{1}{n} \sum_i \mathbf{z}_i (\eta_i + v_i), \quad \tilde{\boldsymbol{\alpha}} = \boldsymbol{\alpha} + \left(\frac{1}{n} \sum_i \mathbf{z}_i \mathbf{z}_i' \right)^{-1} \frac{1}{n} \sum_i \mathbf{z}_i \eta_i,$$

and subtracting gives (*):

$$\hat{\boldsymbol{\alpha}} - \tilde{\boldsymbol{\alpha}} = \left(\frac{1}{n} \sum_i \mathbf{z}_i \mathbf{z}_i' \right)^{-1} \frac{1}{n} \sum_i \mathbf{z}_i \cdot v_i.$$

(c) With $\mathbf{x}_i$ a scalar x_i ,

$$\frac{1}{n} \sum_i \mathbf{z}_i v_i = -2(b - \beta) \frac{1}{n} \sum_i x_i \varepsilon_i \mathbf{z}_i + (b - \beta)^2 \frac{1}{n} \sum_i x_i^2 \mathbf{z}_i.$$

For the first term, ergodic stationarity gives $\frac{1}{n} \sum_i x_i \varepsilon_i \mathbf{z}_i \xrightarrow{P} E(x_i \varepsilon_i \mathbf{z}_i)$, and this expectation is in fact zero:

$$E(x_i \varepsilon_i \mathbf{z}_i) = E[\mathbf{z}_i x_i E(\varepsilon_i | x_i)] = \mathbf{0}$$

when $E(\varepsilon_i | x_i) = 0$; in any case it is finite, and multiplied by $b - \beta \xrightarrow{P} 0$ the term vanishes. For the second, $\frac{1}{n} \sum_i x_i^2 \mathbf{z}_i \xrightarrow{P} E(x_i^2 \mathbf{z}_i)$, finite by assumption, and $(b - \beta)^2 \xrightarrow{P} 0$. Hence $\frac{1}{n} \sum_i \mathbf{z}_i v_i \xrightarrow{P} \mathbf{0}$ and, with $\frac{1}{n} \sum_i \mathbf{z}_i \mathbf{z}_i' \xrightarrow{P} E(\mathbf{z}_i \mathbf{z}_i')$ nonsingular, $\hat{\alpha} - \tilde{\alpha} \xrightarrow{P} \mathbf{0}$. Combined with (a), $\hat{\alpha}$ is consistent — the claim made in Section 2.8.

(d) Multiply by $\sqrt{n}$:

$$\sqrt{n} \frac{1}{n} \sum_i \mathbf{z}_i v_i = -2[\sqrt{n}(b - \beta)] \frac{1}{n} \sum_i x_i \varepsilon_i \mathbf{z}_i + \frac{1}{\sqrt{n}} [\sqrt{n}(b - \beta)]^2 \frac{1}{n} \sum_i x_i^2 \mathbf{z}_i.$$

In the first term $\sqrt{n}(b - \beta)$ converges in distribution and the sample average converges in probability to $E(x_i \varepsilon_i \mathbf{z}_i) = \mathbf{0}$, so LEMMA 2.4(B) makes the product $o_p(1)$. In the second, $[\sqrt{n}(b - \beta)]^2$ converges in distribution, $\frac{1}{n} \sum_i x_i^2 \mathbf{z}_i$ converges in probability to a finite limit, and the factor $1/\sqrt{n} \rightarrow 0$, so this term too is $o_p(1)$ by LEMMA 2.4(B). Hence $\sqrt{n}(\hat{\alpha} - \tilde{\alpha}) \xrightarrow{P} \mathbf{0}$: using residuals instead of the unobservable errors in the skedastic regression does not even affect the asymptotic *distribution* of $\hat{\alpha}$, let alone its consistency.

Analytical Exercise 2.7

The finite-sample inequality (*), $(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{V}\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1} \geq (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}$, holds for any positive definite $\mathbf{V}$ (it is Proposition 1.7(c), the efficiency of GLS). Multiplying both sides by n — equivalently, writing each factor as a sample average — and taking probability limits preserves the inequality, giving (**).

It remains to identify the left-hand side. With $\mathbf{V}$ the diagonal matrix whose i -th element is $\mathbf{z}_i' \boldsymbol{\alpha}$ ((2.8.4)),

$$\text{plim} \frac{1}{n} \mathbf{X}'\mathbf{X} = \Sigma_{\mathbf{xx}}, \quad \text{plim} \frac{1}{n} \mathbf{X}'\mathbf{V}\mathbf{X} = \text{plim} \frac{1}{n} \sum_{i=1}^n (\mathbf{z}_i' \boldsymbol{\alpha}) \mathbf{x}_i \mathbf{x}_i' = E[(\mathbf{z}_i' \boldsymbol{\alpha}) \mathbf{x}_i \mathbf{x}_i']$$

by ergodic stationarity, and by (2.8.1) and the law of total expectations,

$$E[(\mathbf{z}_i' \boldsymbol{\alpha}) \mathbf{x}_i \mathbf{x}_i'] = E[E(\varepsilon_i^2 | \mathbf{x}_i) \mathbf{x}_i \mathbf{x}_i'] = E(\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i') = \mathbf{S}.$$

So the left-hand side of (**) is $\Sigma_{\mathbf{xx}}^{-1} \mathbf{S} \Sigma_{\mathbf{xx}}^{-1} = \text{Avar}(\mathbf{b})$ by (2.3.4), while the right-hand side is $\text{Avar}(\hat{\beta}(\mathbf{V}))$ by (2.8.7). Therefore

$$\text{Avar}(\hat{\beta}(\hat{\mathbf{V}})) = \text{Avar}(\hat{\beta}(\mathbf{V})) \leq \text{Avar}(\mathbf{b}),$$

the first equality being the asymptotic equivalence of feasible and infeasible WLS. Two caveats from the text are worth repeating: the ranking requires the skedastic function to be *correctly specified* (if $\mathbf{V}$ is wrong, WLS remains consistent but can be less efficient than OLS), and it is asymptotic — in finite samples the extra estimation of $\boldsymbol{\alpha}$ can easily cost more than the efficiency gain.

Analytical Exercise 2.8

The projection coefficients solve the normal equations (*),

$$E(\mathbf{x}\mathbf{x}') \begin{bmatrix} \mu \\ \gamma \end{bmatrix} = E(\mathbf{x} \cdot y), \quad E(\mathbf{x}\mathbf{x}') = \begin{bmatrix} 1 & E(\tilde{\mathbf{x}})' \\ E(\tilde{\mathbf{x}}) & E(\tilde{\mathbf{x}}\tilde{\mathbf{x}}') \end{bmatrix}, \quad E(\mathbf{x} \cdot y) = \begin{bmatrix} E(y) \\ E(\tilde{\mathbf{x}} \cdot y) \end{bmatrix}.$$

The first equation is $\mu + E(\tilde{\mathbf{x}})' \gamma = E(y)$, i.e. $\mu = E(y) - \gamma' E(\tilde{\mathbf{x}})$, which is the second claim. Substituting this μ into the remaining equations,

$$E(\tilde{\mathbf{x}}) [E(y) - \gamma' E(\tilde{\mathbf{x}})] + E(\tilde{\mathbf{x}}\tilde{\mathbf{x}}') \gamma = E(\tilde{\mathbf{x}} \cdot y),$$

that is,

$$[E(\tilde{\mathbf{x}}\tilde{\mathbf{x}}') - E(\tilde{\mathbf{x}})E(\tilde{\mathbf{x}})'] \gamma = E(\tilde{\mathbf{x}} \cdot y) - E(\tilde{\mathbf{x}})E(y),$$

whose two sides are exactly $\text{Var}(\tilde{\mathbf{x}})\gamma$ and $\text{Cov}(\tilde{\mathbf{x}}, y)$. Since $\text{Var}(\tilde{\mathbf{x}})$ is nonsingular (this is ASSUMPTION 2.4 for the projection — no perfect collinearity among the nonconstant regressors), $\gamma = \text{Var}(\tilde{\mathbf{x}})^{-1} \text{Cov}(\tilde{\mathbf{x}}, y)$.

The same conclusion follows from the partitioned inverse formula given in the hint, but the elimination above is the population version of the “deviations-from-the-mean” algebra of Analytical Exercise 1.3: including a constant in a projection is the same as projecting the demeaned variables on each other, and the constant then absorbs the means.

Analytical Exercise 2.9

Write $I_{t-1} \equiv (\varepsilon_{t-1}, \dots, \varepsilon_1)$.

(a) ε_{t-1} is a function of I_{t-1} , so

$$E(g_t \mid I_{t-1}) = \varepsilon_{t-1} E(\varepsilon_t \mid I_{t-1}) = 0.$$

Since $(g_{t-1}, \dots, g_2)$ is a function of I_{t-1} , the law of iterated expectations gives $E(g_t \mid g_{t-1}, \dots, g_2) = E[E(g_t \mid I_{t-1}) \mid g_{t-1}, \dots, g_2] = 0$, and $E(g_t) = 0$. So $\{g_t\}$ is a martingale difference sequence; it is ergodic stationary because it is a measurable function of two consecutive coordinates of an ergodic stationary process.

(b) By own conditional homoskedasticity and iterated expectations,

$$E(g_t^2) = E(\varepsilon_t^2 \varepsilon_{t-1}^2) = E[\varepsilon_{t-1}^2 E(\varepsilon_t^2 \mid I_{t-1})] = \sigma^2 E(\varepsilon_{t-1}^2) = \sigma^4,$$

using $E(\varepsilon_t^2) = E[E(\varepsilon_t^2 \mid I_{t-1})] = \sigma^2$.

(c) $\sqrt{n}\hat{\gamma}_1 = \frac{1}{\sqrt{n}} \sum_{t=2}^n g_t$. The summands are an ergodic stationary m.d.s. with finite second moment σ^4 by (a) and (b), so the ergodic stationary martingale differences CLT gives $\sqrt{n}\hat{\gamma}_1 \xrightarrow{d} N(0, \sigma^4)$. (Dropping the single term $t = 1$ is immaterial: it contributes $g_1/\sqrt{n} \xrightarrow{p} 0$, and LEMMA 2.4(A) absorbs it.)

(d) $\hat{\gamma}_0 = \frac{1}{n} \sum_{t=1}^n \varepsilon_t^2 \xrightarrow{p} E(\varepsilon_t^2) = \sigma^2$ by the ergodic theorem. Hence, by LEMMA 2.4(C) with the scalar “matrix” $1/\hat{\gamma}_0 \xrightarrow{p} 1/\sigma^2$,

$$\sqrt{n}\hat{\rho}_1 = \frac{\sqrt{n}\hat{\gamma}_1}{\hat{\gamma}_0} \xrightarrow{d} \frac{1}{\sigma^2} N(0, \sigma^4) = N(0, 1),$$

which is (2.10.3) for $j = 1$: the sample autocorrelation of an m.d.s. with own conditional homoskedasticity has asymptotic standard error $1/\sqrt{n}$, whatever the distribution of the process. Note where each assumption is used: the m.d.s. property makes $\{g_t\}$ a martingale difference sequence (so no autocovariance terms enter the asymptotic variance), and own conditional homoskedasticity makes its variance σ^4 rather than $E(\varepsilon_t^2 \varepsilon_{t-1}^2)$; for an ARCH process the latter exceeds σ^4 and the $1/\sqrt{n}$ standard error understates the true one.

Analytical Exercise 2.10

(a) The mean is $E(y_t) = 0$ for every t , and for $j \geq 0$,

$$\gamma_j = E(y_t y_{t-j}) = E[(\varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2})(\varepsilon_{t-j} + \theta_1 \varepsilon_{t-j-1} + \theta_2 \varepsilon_{t-j-2})],$$

which by independence keeps only the terms with matching indices:

$$\gamma_0 = \sigma_\varepsilon^2(1 + \theta_1^2 + \theta_2^2), \quad \gamma_1 = \sigma_\varepsilon^2(\theta_1 + \theta_1\theta_2), \quad \gamma_2 = \sigma_\varepsilon^2\theta_2, \quad \gamma_j = 0 \quad (j \geq 3).$$

None depends on t , so $\{y_t\}$ is covariance stationary. (It is in fact strictly stationary and ergodic, being a fixed function of finitely many coordinates of an i.i.d. sequence.)

(b) There is a one-to-one mapping between $(\varepsilon_{-1}, \dots, \varepsilon_t)$ and $(y_{-1}, \dots, y_t)$, so conditioning on $(y_{t-j}, \dots, y_{-1})$ is the same as conditioning on $(\varepsilon_{t-j}, \dots, \varepsilon_{-1})$, and the conditional expectation just drops the future ε 's:

$$E(y_t \mid \varepsilon_{t-j}, \dots) = \begin{cases} \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} & j = 0, \\ \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} & j = 1, \\ \theta_2 \varepsilon_{t-2} & j = 2, \\ 0 & j \geq 3. \end{cases}$$

Differencing consecutive rows gives

$$r_{t0} = \varepsilon_t, \quad r_{t1} = \theta_1 \varepsilon_{t-1}, \quad r_{t2} = \theta_2 \varepsilon_{t-2}, \quad r_{t3} = r_{t4} = \dots = 0.$$

The revisions die out after two periods — the defining property of an MA(2): news more than two periods old has already been fully incorporated.

(c) Since $\gamma_j = 0$ for $j \geq 3$,

$$\text{Var}\left(\sum_{t=1}^n y_t\right) = n\gamma_0 + 2 \sum_{j=1}^{n-1} (n-j)\gamma_j = n\gamma_0 + 2(n-1)\gamma_1 + 2(n-2)\gamma_2,$$

so dividing by n ,

$$\text{Var}(\sqrt{n} \bar{y}_n) = \gamma_0 + 2\left[\left(1 - \frac{1}{n}\right)\gamma_1 + \left(1 - \frac{2}{n}\right)\gamma_2\right].$$

(d) The mean of the limiting distribution is $E(\sqrt{n} \bar{y}_n) = 0$. For the variance, use LEMMA 2.1 with $z_n = \sqrt{n} \bar{y}_n$ and $\lim_n(1 - q/n) = 1$ for fixed q :

$$\text{Avar}(\bar{y}_n) = \lim_{n \rightarrow \infty} \text{Var}(\sqrt{n} \bar{y}_n) = \gamma_0 + 2\gamma_1 + 2\gamma_2 = \sigma_\varepsilon^2(1 + \theta_1 + \theta_2)^2,$$

the last equality by expanding the square. So $\sqrt{n} \bar{y}_n \xrightarrow{d} N(0, \sigma_\varepsilon^2(1 + \theta_1 + \theta_2)^2)$.

Two features are worth noting. First, the asymptotic variance is the *long-run variance* $\sum_{j=-\infty}^{\infty} \gamma_j$, not γ_0 : with serially correlated data the usual formula γ_0/n for the variance of a sample mean is wrong, sometimes by a lot. Second, the long-run variance is $\sigma_\varepsilon^2[\theta(1)]^2$ with $\theta(z) = 1 + \theta_1 z + \theta_2 z^2$, and it *vanishes* if $\theta(1) = 0$, that is if $1 + \theta_1 + \theta_2 = 0$ — in that case $\bar{y}_n$ converges faster than $1/\sqrt{n}$. Chapter 6 develops both points in general.

Analytical Exercise 2.11

Throughout, $\mathbf{W} \equiv [\mathbf{X} : \mathbf{E}]$ is the $n \times (K + p)$ matrix of regressors of the auxiliary regression, so that $\hat{\mathbf{B}} = \mathbf{W}'\mathbf{W}/n$, and the null hypothesis is that the error of the original regression has no serial correlation.

(a) The auxiliary OLS coefficient vector is $\hat{\boldsymbol{\alpha}} = (\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'\mathbf{e} = \hat{\mathbf{B}}^{-1}(\frac{1}{n}\mathbf{W}'\mathbf{e})$. Its two blocks are $\frac{1}{n}\mathbf{X}'\mathbf{e} = \mathbf{0}$, by the normal equations of the original regression, and

$$\frac{1}{n}\mathbf{E}'\mathbf{e} = \left(\frac{1}{n} \sum_{t=j+1}^n e_{t-j} e_t \right)_{j=1, \dots, p} = \hat{\boldsymbol{\gamma}},$$

the vector of sample autocovariances (2.10.8) computed from the residuals (the zeros in the first rows of $\mathbf{E}$ are what make the sums start at $t = j + 1$). Hence $\hat{\boldsymbol{\alpha}} = \hat{\mathbf{B}}^{-1} \begin{bmatrix} \mathbf{0} \\ \hat{\boldsymbol{\gamma}} \end{bmatrix}$.

(b) Block by block. $\frac{1}{n}\mathbf{X}'\mathbf{X} = \mathbf{S}_{\mathbf{xx}} \xrightarrow{p} \Sigma_{\mathbf{xx}}$ by ergodic stationarity. The j -th column of $\frac{1}{n}\mathbf{X}'\mathbf{E}$ is $\frac{1}{n} \sum_{t=j+1}^n \mathbf{x}_t e_{t-j}$; substituting $e_{t-j} = \varepsilon_{t-j} - \mathbf{x}'_{t-j}(\mathbf{b} - \beta)$ from (2.3.9),

$$\frac{1}{n} \sum_{t=j+1}^n \mathbf{x}_t e_{t-j} = \frac{1}{n} \sum_{t=j+1}^n \mathbf{x}_t \varepsilon_{t-j} - \left(\frac{1}{n} \sum_{t=j+1}^n \mathbf{x}_t \mathbf{x}'_{t-j} \right) (\mathbf{b} - \beta) \xrightarrow{p} E(\mathbf{x}_t \varepsilon_{t-j}),$$

since the second term is a convergent average times $o_p(1)$. So $\frac{1}{n}\mathbf{X}'\mathbf{E} \xrightarrow{p} \mathbf{H}$. Finally the (j, k) element of $\frac{1}{n}\mathbf{E}'\mathbf{E}$ is $\frac{1}{n} \sum_t e_{t-j} e_{t-k} \xrightarrow{p} E(\varepsilon_{t-j} \varepsilon_{t-k})$, which under the null equals σ^2 for $j = k$ and 0 otherwise; hence $\frac{1}{n}\mathbf{E}'\mathbf{E} \xrightarrow{p} \sigma^2 \mathbf{I}_p$ and $\hat{\mathbf{B}} \xrightarrow{p} \mathbf{B}$ as claimed.

(c) Under the null $\hat{\boldsymbol{\gamma}} \xrightarrow{p} \mathbf{0}$ (Review Question 2.10.1 applied to the residuals), and $\hat{\mathbf{B}}^{-1} \xrightarrow{p} \mathbf{B}^{-1}$, so $\hat{\boldsymbol{\alpha}} \xrightarrow{p} \mathbf{B}^{-1} \mathbf{0} = \mathbf{0}$.

(d) For a regression of $\mathbf{e}$ on $\mathbf{W}$, $\text{SSR} = \mathbf{e}'\mathbf{e} - \mathbf{e}'\mathbf{W}\hat{\boldsymbol{\alpha}}$, and by (a) $\frac{1}{n}\mathbf{e}'\mathbf{W} = [\mathbf{0}' \ \hat{\boldsymbol{\gamma}}']$, so

$$\frac{\text{SSR}}{n} = \frac{\mathbf{e}'\mathbf{e}}{n} - \hat{\boldsymbol{\gamma}}' \hat{\boldsymbol{\alpha}}_2 \xrightarrow{p} \sigma^2 - \mathbf{0}'\mathbf{0} = \sigma^2,$$

where $\hat{\boldsymbol{\alpha}}_2$ is the last p coefficients and $\mathbf{e}'\mathbf{e}/n \xrightarrow{p} \sigma^2$ by PROPOSITION 2.2. Since $(n - K - p)/n \rightarrow 1$, also $\text{SSR}/(n - K - p) \xrightarrow{p} \sigma^2$.

(e) Apply the F -ratio formula (1.4.9) to the auxiliary regression with $\mathbf{R} = [\mathbf{0} : \mathbf{I}_p]$ and $\mathbf{r} = \mathbf{0}$. Then $\mathbf{R}\hat{\boldsymbol{\alpha}} = \hat{\boldsymbol{\alpha}}_2 = \hat{\mathbf{B}}^{22}\hat{\boldsymbol{\gamma}}$ by (a), and $\mathbf{R}(\mathbf{W}'\mathbf{W})^{-1}\mathbf{R}' = \hat{\mathbf{B}}^{22}/n$, so

$$p \cdot F = \frac{(\hat{\mathbf{B}}^{22}\hat{\boldsymbol{\gamma}})' [\hat{\mathbf{B}}^{22}/n]^{-1} (\hat{\mathbf{B}}^{22}\hat{\boldsymbol{\gamma}})}{\text{SSR}/(n - K - p)} = \frac{n \hat{\boldsymbol{\gamma}}' \hat{\mathbf{B}}^{22} \hat{\boldsymbol{\gamma}}}{\text{SSR}/(n - K - p)},$$

using the symmetry of $\hat{\mathbf{B}}^{22}$.

(f) The partitioned inverse formula (A.10) applied to $\widehat{\mathbf{B}}$ gives its lower-right block:

$$\widehat{\mathbf{B}}^{22} = \left[\frac{1}{n} \mathbf{E}' \mathbf{E} - \left(\frac{1}{n} \mathbf{E}' \mathbf{X} \right) \mathbf{S}_{\mathbf{xx}}^{-1} \left(\frac{1}{n} \mathbf{X}' \mathbf{E} \right) \right]^{-1}.$$

(g) By (b) and (f),

$$\widehat{\mathbf{B}}^{22} \xrightarrow{p} [\sigma^2 \mathbf{I}_p - \mathbf{H}' \Sigma_{\mathbf{xx}}^{-1} \mathbf{H}]^{-1} = \frac{1}{\sigma^2} (\mathbf{I}_p - \Phi)^{-1},$$

because $\Phi = \mathbf{H}' \Sigma_{\mathbf{xx}}^{-1} \mathbf{H} / \sigma^2$ by (2.10.18). Combining with (d) and (e),

$$p \cdot F = \frac{n \hat{\gamma}' \widehat{\mathbf{B}}^{22} \hat{\gamma}}{\text{SSR} / (n - K - p)} = \frac{n \hat{\gamma}' (\mathbf{I}_p - \Phi)^{-1} \hat{\gamma}}{\sigma^4} + o_p(1).$$

For the modified Q , note $\hat{\rho} = \hat{\gamma} / \hat{\gamma}_0$ with $\hat{\gamma}_0 \xrightarrow{p} \sigma^2$ and $\widehat{\Phi} \xrightarrow{p} \Phi$ (ergodic theorem applied to (2.10.19)), so

$$Q = n \hat{\rho}' (\mathbf{I}_p - \widehat{\Phi})^{-1} \hat{\rho} = \frac{n \hat{\gamma}' (\mathbf{I}_p - \widehat{\Phi})^{-1} \hat{\gamma}}{\hat{\gamma}_0^2} = \frac{n \hat{\gamma}' (\mathbf{I}_p - \Phi)^{-1} \hat{\gamma}}{\sigma^4} + o_p(1).$$

Both statistics are asymptotically equivalent to the same quantity, hence to each other, and by (2.10.17) — $\sqrt{n} \hat{\gamma} \xrightarrow{d} N(\mathbf{0}, \sigma^4 (\mathbf{I}_p - \Phi))$ — that quantity converges to $\chi^2(p)$. This is why the Breusch–Godfrey nR^2 (asymptotically equivalent to $p \cdot F$, by Review Question 2.6.5 applied to the auxiliary regression) can be used in place of the modified Q : it is computable with any regression package.

Remark. The hint in the book writes the common limit as $n \hat{\gamma}' (\mathbf{I}_p - \Phi)^{-1} \hat{\gamma} / \sigma^2$. The denominator should be σ^4 , as above: $\hat{\gamma}_j$ is a second moment, so the quadratic form is of order σ^4 , and (2.10.17) gives $\text{Avar}(\sqrt{n} \hat{\gamma}) = \sigma^4 (\mathbf{I}_p - \Phi)$. With σ^2 in the denominator the statistic would not be asymptotically $\chi^2(p)$ (nor even scale free).

Analytical Exercise 2.12

(a) Stacking the two regimes, $\mathbf{X} = \text{diag}(\mathbf{X}_1, \mathbf{X}_2)$ is block diagonal, so by Analytical Exercise 1.5 the unrestricted SSR is $\text{SSR}_1 + \text{SSR}_2$ and the OLS estimates are the separate $\mathbf{b}_1, \mathbf{b}_2$ (Empirical Exercise 1.1(e)). The null $\beta_1 = \beta_2$ is $\mathbf{R}\beta = \mathbf{0}$ with $\mathbf{R} = [\mathbf{I}_K \ : \ -\mathbf{I}_K]$, and

$$\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}' = (\mathbf{X}'_1\mathbf{X}_1)^{-1} + (\mathbf{X}'_2\mathbf{X}_2)^{-1}.$$

Hence, by (1.4.9) with $s^2 = (\text{SSR}_1 + \text{SSR}_2) / (n - 2K)$,

$$KF = \frac{(\mathbf{b}_1 - \mathbf{b}_2)' [(\mathbf{X}'_1\mathbf{X}_1)^{-1} + (\mathbf{X}'_2\mathbf{X}_2)^{-1}]^{-1} (\mathbf{b}_1 - \mathbf{b}_2)}{s^2} = \sqrt{n}(\mathbf{b}_1 - \mathbf{b}_2)' \mathbf{C}_n^{-1} \sqrt{n}(\mathbf{b}_1 - \mathbf{b}_2),$$

where $\mathbf{C}_n \equiv \left(\frac{1}{n} \sum_{t=1}^r \mathbf{x}_t \mathbf{x}_t' / s^2 \right)^{-1} + \left(\frac{1}{n} \sum_{t=r+1}^n \mathbf{x}_t \mathbf{x}_t' / s^2 \right)^{-1}$, which is the displayed expression: the two factors of $\sqrt{n}$ cancel the $1/n$ in the moment matrices.

(b) With $\lambda = r/n$ fixed, $\frac{1}{n} \sum_{t=1}^r \mathbf{x}_t \mathbf{x}_t' = \lambda \cdot \frac{1}{r} \sum_{t=1}^r \mathbf{x}_t \mathbf{x}_t' \xrightarrow{p} \lambda \Sigma_{\mathbf{xx}}$, the inner average converging by the ergodic theorem applied to the first $r \rightarrow \infty$ observations. The same argument on the second subsample gives $\frac{1}{n} \sum_{t=r+1}^n \mathbf{x}_t \mathbf{x}_t' \xrightarrow{p} (1 - \lambda) \Sigma_{\mathbf{xx}}$.

(c) Likewise $\frac{1}{\sqrt{n}} \sum_{t=1}^r \mathbf{x}_t \varepsilon_t = \sqrt{\lambda} \cdot \frac{1}{\sqrt{r}} \sum_{t=1}^r \mathbf{x}_t \varepsilon_t \xrightarrow{d} \sqrt{\lambda} N(\mathbf{0}, \mathbf{S}) = N(\mathbf{0}, \lambda \sigma^2 \Sigma_{\mathbf{xx}})$, using the martingale differences CLT on the first subsample and $\mathbf{S} = \sigma^2 \Sigma_{\mathbf{xx}}$ under conditional homoskedasticity; and the

second subsample gives $N(\mathbf{0}, (1 - \lambda)\sigma^2\Sigma_{\mathbf{xx}})$. The two are asymptotically uncorrelated (and jointly normal), so they are asymptotically independent.

(d) Combining (b) and (c) through LEMMA 2.4(c),

$$\sqrt{n}(\mathbf{b}_1 - \beta_1) \xrightarrow{d} N(\mathbf{0}, \lambda^{-1}\sigma^2\Sigma_{\mathbf{xx}}^{-1}), \quad \sqrt{n}(\mathbf{b}_2 - \beta_2) \xrightarrow{d} N(\mathbf{0}, (1 - \lambda)^{-1}\sigma^2\Sigma_{\mathbf{xx}}^{-1}),$$

independently, so that $\text{Avar}(\mathbf{b})$ is block diagonal with these two blocks. Since $\mathbf{b}_1 - \mathbf{b}_2 = [\mathbf{I}_K \ : \ -\mathbf{I}_K]\mathbf{b}$,

$$\text{Avar}(\mathbf{b}_1 - \mathbf{b}_2) = \lambda^{-1}\sigma^2\Sigma_{\mathbf{xx}}^{-1} + (1 - \lambda)^{-1}\sigma^2\Sigma_{\mathbf{xx}}^{-1} = \frac{\sigma^2}{\lambda(1 - \lambda)}\Sigma_{\mathbf{xx}}^{-1}.$$

(e) Under $H_0: \beta_1 = \beta_2$, $\sqrt{n}(\mathbf{b}_1 - \mathbf{b}_2) \xrightarrow{d} \mathbf{z} \sim N(\mathbf{0}, \text{Avar}(\mathbf{b}_1 - \mathbf{b}_2))$, while by (b) and $s^2 \xrightarrow{p} \sigma^2$,

$$\mathbf{C}_n \xrightarrow{p} \sigma^2(\lambda\Sigma_{\mathbf{xx}})^{-1} + \sigma^2((1 - \lambda)\Sigma_{\mathbf{xx}})^{-1} = \text{Avar}(\mathbf{b}_1 - \mathbf{b}_2).$$

LEMMA 2.4(c) and the continuous mapping theorem then give

$$KF \xrightarrow{d} \mathbf{z}'[\text{Avar}(\mathbf{b}_1 - \mathbf{b}_2)]^{-1}\mathbf{z} \sim \chi^2(K),$$

the last step because $\mathbf{u} \equiv \text{Avar}^{-1/2}\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_K)$ and the quadratic form is $\mathbf{u}'\mathbf{u}$.

So the Chow statistic keeps its usual interpretation in large samples: KF (not F) is compared with the $\chi^2(K)$ table, and the test is valid under the assumptions of PROPOSITION 2.5 — normality and strict exogeneity are not needed, but conditional homoskedasticity is. Note also the asymptotic variance $\sigma^2\Sigma_{\mathbf{xx}}^{-1}/[\lambda(1 - \lambda)]$: it is smallest at $\lambda = 1/2$, so a break in the middle of the sample is the easiest to detect, and breaks near either end are estimated very imprecisely.

Problem Set for Chapter 2: Empirical and Monte Carlo Exercises

Empirical Exercise 2.1

The computations are in `code/ch2.py`. Following the text, inflation is matched with $TB1_t$ as $\pi_{t+1} = [(P_{t+1}/P_t)^{12} - 1] \times 100$ with $P_t = CPI_{t-1}$, and the ex-post real rate is $r_{t+1} = TB1_t - \pi_{t+1}$. The Fama sample is 1/53–7/71, $n = 223$.

(a) The published tables (*Treasury Bulletin*, *Federal Reserve Bulletin*, and the Board's web site) report rates on three-month, six-month and one-year bills and constant-maturity yields; a one-month bill rate is not there, because until 2001 the Treasury did not issue four-week bills. Mishkin's one-month rates come from the CRSP file of quotes on individual outstanding bills, from which a rate for a bill with one month to maturity can be computed for any date. The rates are indeed beginning-of-month: they are quoted on the last business day of the previous month, which is what the efficient market hypothesis requires, since R_t must belong to I_t .

(b) The CPI in the file is the CPI for urban consumers, all items (including food and housing), not seasonally adjusted; a January observation is the index for January; prices are collected throughout the month, and the index is announced in the middle of the *following* month. This is the timing problem discussed in the text: $P_t = CPI_{t-1}$ is assumed known to the market at the beginning of month t , which is only approximately true.

(c) The CPI is a *fixed-weight* (Laspeyres) index: the basket is held fixed between weight revisions, so the index answers “what does last period’s basket cost now”. (The GDP deflator, by contrast, is a variable-weight Paasche index.) A fixed-weight index overstates the cost of living, because it ignores substitution towards goods whose relative prices have fallen.

(d) Table 2.1 is reproduced almost exactly:

j	1	2	3	4	5	6	7	8	9	10	11	12
$\hat{\rho}_j$	-.101	.172	-.019	-.004	-.064	-.021	-.091	.094	.094	.019	.004	.207
s.e.	.067	.067	.067	.067	.067	.067	.067	.067	.067	.067	.067	.067
Q_{LB}	2.3	9.1	9.1	9.1	10.1	10.2	12.1	14.2	16.3	16.4	16.4	26.5
p -value	12.8	1.1	2.7	5.8	7.3	11.7	9.6	7.6	6.1	8.9	12.8	0.9

with mean 0.82% and standard deviation 2.837% (the book prints 2.847%). Individually the autocorrelations are small — the standard error is $1/\sqrt{223} = 0.067$ — but $\hat{\rho}_2$ and $\hat{\rho}_{12}$ are about three standard errors from zero and the Q statistics reject at $j = 2, 3$ and $j = 12$. The first implication of market efficiency is therefore not as comfortable as Fama’s summary suggests; the twelfth-order autocorrelation points at seasonality in the unadjusted CPI.

(e) The regression reproduces (2.11.9) to the printed digits:

$$\pi_{t+1} = \underset{(0.431)}{-0.868} + \underset{(0.112)}{1.015} TB1_t, \quad R^2 = 0.24, \text{ mean of dep. var.} = 2.35, \text{ SER} = 2.84\%, n = 223,$$

with White standard errors in parentheses. The robust t -ratio for the market-efficiency null that the slope is one is $(1.015 - 1)/0.112 = 0.13$, $p = 0.90$: the hypothesis is accepted easily. The intercept estimates $-r$, so the implied constant ex-ante real rate is $r = 0.868\%$ per year.

(f) The three Davidson–MacKinnon adjustments change the standard errors very little at $n = 223$:

$\hat{S}$	s.e. (const)	s.e. ($TB1$)	t for slope = 1
(2.5.1), no correction	0.4312	0.1122	0.131
times $n/(n - K)$	0.4332	0.1127	0.130
(2.5.5) with $d = 1$	0.4337	0.1130	0.130
(2.5.5) with $d = 2$	0.4362	0.1138	0.129

The corrections inflate the standard errors, as they are designed to (the uncorrected version over-rejects), but here the effect is under 1.5% and no conclusion changes. They matter when n is small or when a few observations have high leverage p_i .

(g) Under conditional homoskedasticity the standard errors are 0.4326 (constant) and 0.1227 ($TB1$). The point estimates are of course *identical* — OLS does not know about the variance assumption — and only the standard errors differ; here the conventional standard error of the slope is about 9% larger than the robust one, and the t -ratio for the null of one falls from 0.131 to 0.120. The two agree closely, which is weak evidence that conditional heteroskedasticity is not severe in this regression.

(h) The Breusch–Godfrey auxiliary regression with $p = 12$ gives $nR^2 = 27.0$ on $\chi^2(12)$, $p = 0.008$. The null of no serial correlation in the error is rejected at the 1% level. Since ASSUMPTION 2.5

makes the error an m.d.s., which implies no serial correlation, this is evidence against the maintained hypothesis — not (yet) against market efficiency itself, but against the auxiliary assumptions under which the t -test of (e) is valid. It is consistent with the $\hat{\rho}_{12} = 0.207$ of part (d).

(i) Fama’s own estimate of the coefficient on the nominal rate for 1/53–7/71 is 0.98 with standard error 0.10 (his Table 3); ours is 1.015 with robust standard error 0.112. The two are close but not identical, for two reasons: it is not clear whether he used CPI_{t-1} or CPI_t as P_t in forming inflation, and the CPI weights of his vintage (1958) differ from those of ours (1982–1984). His intercept, 0.00070, looks entirely different from our -0.868 because his dependent variable is the *negative* of the inflation rate and his rates are monthly rather than annual.

(j) Replacing the constant by twelve monthly dummies leaves the slope at 1.025 (robust s.e. 0.114) against 1.015: no important change, as the book’s answer says. The twelve dummies sum to the constant, so including both produces perfect multicollinearity — ASSUMPTION 2.4 fails and the software must drop one column. Seasonally *unadjusted* data are the right input here: the BLS adjustment is a two-sided moving average, so a seasonally adjusted CPI_t depends on *future* unadjusted values, which are not in I_t ; using it would put a regressor outside the information set, exactly the failure analysed in Review Question 2.11.5. If seasonality matters, seasonal dummies in the regression are the better remedy.

(k) With *PAII* — Mishkin’s inflation series, built from a price index that uses rental equivalence for housing in *all* periods — the regression for 1/53–7/71 becomes

$$\pi_{t+1} = \underset{(0.340)}{-0.484} + \underset{(0.086)}{0.807} TB1_t, \quad R^2 = 0.24.$$

The slope drops from 1.015 to 0.807, in agreement with the book’s answer, and the robust t -ratio for the null of one is $(0.807 - 1)/0.086 = -2.25$: market efficiency is now rejected at the 5% level. The conclusion of the text is therefore not robust to the measurement of inflation — a reminder that a “test” of an economic hypothesis is a test of the hypothesis *and* of the auxiliary measurement assumptions.

(l) For the period after October 1979 (November 1979 to December 1990, $n = 134$), the same regression with *PAII* gives

$$\pi_{t+1} = \underset{(0.781)}{0.396} + \underset{(0.094)}{0.564} TB1_t, \quad R^2 = 0.20,$$

so the coefficient drops further, to 0.564 (the book’s figure), and the null of one is rejected decisively ($t = -4.6$). With the CPI-based inflation measure the coefficient is 0.830 (robust s.e. 0.120). The Federal Reserve’s change of operating procedures in October 1979 is a natural break point: after it, nominal rates moved far more than expected inflation did, which is what a coefficient well below one says. (Whether this is market inefficiency or a time-varying ex-ante real rate — assumption (c) failing rather than (a) — cannot be settled by this regression, as Review Question 2.11.3 explains.)

Empirical Exercise 2.2

(i) The regressor vector of Model 4 is $\mathbf{x}_i = (1, a_i, a_i^2, b_i, c_i)'$ with $a = \log Q$, $b = \log(p_1/p_3)$, $c = \log(p_2/p_3)$, so the distinct nonconstant elements of $\mathbf{x}_i \mathbf{x}_i'$ are the 13 variables

$$a, \quad a^2, \quad a^3, \quad a^4, \quad b, \quad c, \quad ab, \quad ac, \quad a^2b, \quad a^2c, \quad b^2, \quad bc, \quad c^2$$

(the count of Review Question 2.7.1, with one more regressor). Regressing the squared OLS residuals of Model 4 on a constant and these,

$$nR^2 = 66.546 \quad \text{on} \quad \chi^2(13), \quad p < 10^{-8},$$

which is the value quoted in the exercise. Conditional homoskedasticity is overwhelmingly rejected, as the residual plot of Figure S1.2 led one to expect.

(j) Step 1 is OLS on Model 4; Step 2 regresses e_i^2 on a constant and $1/Q_i$ and gives

$$e_i^2 = \underset{(0.0163)}{0.0565} + \underset{(0.2603)}{2.1377} \frac{1}{Q_i},$$

which is *exactly* the variance function $E(\varepsilon_i^2 \mid \mathbf{X}) = \sigma^2(0.0565 + 2.1377/Q_i)$ that Chapter 1 asked us to assume: the numbers in the book's Model 4 were obtained from this auxiliary regression. Step 3 — weighted least squares with weights $1/\sqrt{\mathbf{z}_i' \hat{\boldsymbol{\alpha}}}$ — therefore reproduces the estimates of Empirical Exercise 1.1(h) to every digit, $(-4.1256, 0.2177, 0.04566, 0.4662, 0.1335)$. So the “known” skedastic function of Chapter 1 is the feasible WLS of Section 2.8 in disguise, and the estimation of $\boldsymbol{\alpha}$ does not change the asymptotic distribution (Analytical Exercise 2.6).

(k) For Model 4 estimated by OLS, conventional and White standard errors are

	coefficient	conventional s.e.	White s.e.
constant	-3.7646	0.7017	0.7106
$\log Q$	0.1525	0.0619	0.1269
$(\log Q)^2$	0.0505	0.0054	0.0095
$\log(p_1/p_3)$	0.4806	0.1611	0.1342
$\log(p_2/p_3)$	0.0742	0.1500	0.1121

The robust standard errors for the two output terms are about twice the conventional ones — consistent with the strong heteroskedasticity found in (i), which is concentrated in output — while for the price terms they are slightly smaller. Even so, the quadratic term stays significant ($t = 5.3$ robust against 9.4 conventional), so the finding that returns to scale decline with output survives robust inference.

Monte Carlo Exercise 2.1

(a) ASSUMPTION 2.1 holds by construction. ASSUMPTION 2.2: $\{x_i\}$ is an AR(1) with $|\phi| < 0.6 < 1$ started from its stationary distribution, hence stationary and ergodic (Proposition 6.1(d)), and $\{\varepsilon_i\}$ is i.i.d. and independent of $\{x_i\}$, so $\{y_i, x_i\}$ is ergodic stationary. ASSUMPTION 2.3: $E(x_i \varepsilon_i) = E(x_i)E(\varepsilon_i) = 0$ by independence. ASSUMPTION 2.4: $E(\mathbf{x}_i \mathbf{x}_i')$ is nonsingular because $\text{Var}(x_i) = 1/(1 - \phi^2) > 0$. ASSUMPTION 2.5: $\mathbf{g}_i = \mathbf{x}_i \varepsilon_i$ has $E(\mathbf{g}_i \mid \mathbf{g}_{i-1}, \dots) = \mathbf{0}$, since ε_i is

independent of all x 's and of its own past, and $E(\mathbf{g}_i \mathbf{g}_i') = \sigma^2 E(\mathbf{x}_i \mathbf{x}_i')$ is finite and nonsingular with $\sigma^2 = \text{Var}(\varepsilon_i) = 1/12$. ASSUMPTION 2.7: $E(\varepsilon_i^2 \mid \mathbf{x}_i) = 1/12$, a constant. So all the large-sample results of the chapter apply; what is *not* satisfied is ASSUMPTION 1.5 (normality), so the finite-sample theory of Chapter 1 does not.

(b) With 10^6 replications (`code/ch2.py`):

critical value	rejection frequency
$t_{0.025}(30) = 2.042$	0.0502
$z_{0.025} = 1.960$	0.0596

The $t(30)$ critical value is far closer to the nominal 5%; the standard normal critical value makes the test reject 6% of the time, a 20% overshoot. Both are asymptotically valid — $t(n - K) \rightarrow N(0, 1)$ — but at $n = 32$ the t table's fatter tails compensate for the estimation of σ^2 , and that correction matters more than the non-normality of the errors. This is the practical reason the text recommends keeping the degrees-of-freedom adjustment and the t and F tables in moderate samples, even when the justification for them is purely asymptotic.

Monte Carlo Exercise 2.2

With 200,000 replications of $n = 50$ standard normal draws and $\chi_{0.05}^2(4) = 9.488$ (`code/ch2.py`):

statistic	rejection frequency
Box–Pierce (2.10.4)	0.0366
Ljung–Box (2.10.5)	0.0516

The Ljung–Box statistic is much closer to the nominal size; the Box–Pierce statistic rejects only 3.7% of the time, so *we fail to reject the null too often* — it is undersized, and correspondingly has less power against genuine serial correlation. The reason is visible in the formulas: the weights $(n + 2)/(n - j)$ exceed one and inflate Q_{LB} relative to Q_{BP} , compensating for the downward bias of the sample autocorrelations (each $\hat{\rho}_j$ is computed from only $n - j$ products but divided by n). By Review Question 2.10.2 the difference vanishes asymptotically, so the choice is purely a finite-sample matter — and in samples of the size econometricians usually have, Ljung–Box is the one to use.

Chapter 3

Single-Equation GMM

Solutions to all the exercises of Chapter 3 of Fumio Hayashi, *Econometrics* (Princeton University Press, 2000): the 45 review questions of Sections 3.1–3.9, the ten analytical exercises (including the proofs of Hansen’s J test and of the C test, the numerical equality of the Wald and LR statistics, and the Staiger–Stock weak-instrument asymptotics), and the empirical exercise reproducing Griliches’s wage equation on the Blackburn–Neumark extract `GRILIC.ASC`. There is no Monte Carlo exercise in this chapter. Assumptions, propositions and equations cited as ASSUMPTION 3.4, PROPOSITION 3.5 or (3.5.11) refer to the book. The numerical parts are computed by `code/ch3.py`.

Section 3.1 Endogeneity Bias: Working’s Example

Review Question 3.1.1

No. Solving (3.1.2) for p_i as in (3.1.3A),

$$p_i = \frac{\beta_0 - \alpha_0}{\alpha_1 - \beta_1} + \frac{v_i - u_i}{\alpha_1 - \beta_1}, \quad \text{so} \quad \text{Cov}(p_i, u_i) = \frac{\text{Cov}(v_i, u_i) - \text{Var}(u_i)}{\alpha_1 - \beta_1},$$

which generalizes (3.1.4). With $\alpha_1 < 0 < \beta_1$ the denominator $\alpha_1 - \beta_1$ is negative, so the sign of $\text{Cov}(p_i, u_i)$ is the sign of $\text{Var}(u_i) - \text{Cov}(u_i, v_i)$. If the two shifters are sufficiently positively correlated — precisely, if $\text{Cov}(u_i, v_i) > \text{Var}(u_i)$ — then price and the demand shifter are *negatively* correlated despite the downward-sloping demand curve. The intuition is that a high u_i then tends to come with a high v_i , and the outward shift of the supply curve pushes the equilibrium price down by more than the demand shift pushes it up. What survives in the general case is only the variance-difference statement: price is positively correlated with u_i if and only if demand shocks are more variable than the common component of the two shocks.

Review Question 3.1.2

Yes, whenever price is endogenous in the demand equation. Let $\hat{\alpha}_1$ be the OLS estimate of the price coefficient and $\hat{\alpha}_0$ the OLS intercept. By the hint and the projection formula,

$$\text{plim } \hat{\alpha}_0 = E(q_i) - E(p_i) \frac{\text{Cov}(p_i, q_i)}{\text{Var}(p_i)} = [\alpha_0 + \alpha_1 E(p_i)] - E(p_i) \text{plim } \hat{\alpha}_1,$$

so that, subtracting α_0 ,

$$\text{plim } \hat{\alpha}_0 - \alpha_0 = -E(p_i) (\text{plim } \hat{\alpha}_1 - \alpha_1) = -E(p_i) \frac{\text{Cov}(p_i, u_i)}{\text{Var}(p_i)},$$

using (3.1.7). This vanishes if and only if $E(p_i) = 0$ or $\text{Cov}(p_i, u_i) = 0$. The mechanics are familiar from Chapter 1: the OLS intercept is the level of the regression line at the mean of the regressor, $\hat{\alpha}_0 = \bar{q} - \hat{\alpha}_1 \bar{p}$, so an error in the slope propagates into the intercept through the sample means; with $E(p_i) \neq 0$ the intercept inherits (minus) the mean of price times the slope bias. When

$\text{Cov}(u_i, v_i) = 0$ and $\alpha_1 < 0 < \beta_1$, (3.1.4) gives $\text{Cov}(p_i, u_i) > 0$, so $\text{plim } \hat{\alpha}_1 > \alpha_1$ and $\text{plim } \hat{\alpha}_0 < \alpha_0$ (as long as the mean price is positive): the estimated demand curve is too flat *and* too low.

Review Question 3.1.3

Under $\text{Cov}(u_i, v_i) = 0$, (3.1.3A) gives $p_i = c + (v_i - u_i)/(\alpha_1 - \beta_1)$ for a constant c , hence

$$\text{Var}(p_i) = \frac{\text{Var}(v_i - u_i)}{(\alpha_1 - \beta_1)^2} = \frac{\text{Var}(u_i) + \text{Var}(v_i)}{(\alpha_1 - \beta_1)^2},$$

which is the second equation of the hint. For the first, substitute the demand equation (3.1.2A), $q_i = \alpha_0 + \alpha_1 p_i + u_i$:

$$\text{Cov}(p_i, q_i) = \alpha_1 \text{Var}(p_i) + \text{Cov}(p_i, u_i) = \alpha_1 \frac{\text{Var}(u_i) + \text{Var}(v_i)}{(\alpha_1 - \beta_1)^2} - \frac{\text{Var}(u_i)}{\alpha_1 - \beta_1},$$

using $\text{Cov}(p_i, u_i) = -\text{Var}(u_i)/(\alpha_1 - \beta_1)$ from (3.1.4). Putting everything over the common denominator $(\alpha_1 - \beta_1)^2$, the numerator is

$$\alpha_1 [\text{Var}(u_i) + \text{Var}(v_i)] - (\alpha_1 - \beta_1) \text{Var}(u_i) = \beta_1 \text{Var}(u_i) + \alpha_1 \text{Var}(v_i),$$

which is the first equation. Dividing,

$$\text{plim of the OLS price coefficient} = \frac{\text{Cov}(p_i, q_i)}{\text{Var}(p_i)} = \frac{\alpha_1 \text{Var}(v_i) + \beta_1 \text{Var}(u_i)}{\text{Var}(u_i) + \text{Var}(v_i)},$$

which is (3.1.9). The OLS estimand is a variance-weighted average of the demand and supply slopes, with the *other* equation's shifter variance as the weight: the demand slope gets weight $\text{Var}(v_i)$, the supply-side noise, because those are the shifts that trace out the demand curve. As $\text{Var}(u_i) \rightarrow 0$ the estimand tends to α_1 and as $\text{Var}(v_i) \rightarrow 0$ it tends to β_1 , recovering the two extreme cases of Figure 3.1 of the book.

Review Question 3.1.4

With the observable supply shifter, the supply equation is $q_i = \beta_0 + \beta_1 p_i + \beta_2 x_i + \zeta_i$, and solving the system for price gives (3.1.10A):

$$p_i = \frac{\beta_0 - \alpha_0}{\alpha_1 - \beta_1} + \frac{\beta_2}{\alpha_1 - \beta_1} x_i + \frac{\zeta_i - u_i}{\alpha_1 - \beta_1}.$$

Demand equation. $\text{Cov}(p_i, u_i) = -\text{Var}(u_i)/(\alpha_1 - \beta_1) \neq 0$ (using $\text{Cov}(x_i, u_i) = \text{Cov}(\zeta_i, u_i) = 0$), so p_i is endogenous in the demand equation, exactly as without the observable shifter.

Supply equation. $\text{Cov}(p_i, \zeta_i) = \text{Var}(\zeta_i)/(\alpha_1 - \beta_1) \neq 0$, so p_i is endogenous in the supply equation as well: price is negatively correlated with the supply shifter ζ_i (a favorable supply shift lowers the equilibrium price). Nothing about the orthogonality assumptions removes either correlation; endogeneity in this model is a consequence of market clearing, not of correlation among the shifters.

Is $\text{Cov}(u_i, \zeta_i) = 0$ needed for the IV/2SLS consistency? No. The validity of x_i as an instrument for the demand equation requires only (i) orthogonality, $\text{Cov}(x_i, u_i) = 0$, and (ii) relevance, $\text{Cov}(x_i, p_i) \neq 0$. From (3.1.10A),

$$\text{Cov}(x_i, p_i) = \frac{\beta_2}{\alpha_1 - \beta_1} \text{Var}(x_i) \neq 0$$

provided $\beta_2 \neq 0$ and $\text{Var}(x_i) > 0$ — this uses $\text{Cov}(x_i, \zeta_i) = \text{Cov}(x_i, u_i) = 0$ but *not* $\text{Cov}(u_i, \zeta_i) = 0$. Given (i), the derivation of (3.1.11),

$$\alpha_1 = \frac{\text{Cov}(x_i, q_i)}{\text{Cov}(x_i, p_i)},$$

never involves $\text{Cov}(u_i, \zeta_i)$. So the IV estimator (3.1.12) (equivalently, 2SLS) is consistent for α_1 whether or not the demand and supply shifters are correlated. The assumption $\text{Cov}(u_i, \zeta_i) = 0$ is only needed for clean variance decompositions such as (3.1.4) and (3.1.9); it is dispensable for identification. Correlation between u_i and ζ_i would matter, however, if one wanted to instrument the *supply* equation with a demand-side observable, where the same logic applies symmetrically.

Section 3.2 More Examples

Review Question 3.2.1

By the projection formula, the plim of the slope in the regression of C_i on a constant and Y_i is $\text{Cov}(C_i, Y_i) / \text{Var}(Y_i)$. From (3.2.4), $C_i = C_i^* + c_i$ and $Y_i = Y_i^* + y_i$, with c_i and y_i zero-mean and uncorrelated with Y_i^* ((3.2.5)); $C_i^* = kY_i^*$ by (3.2.3). Hence

$$\text{Cov}(C_i, Y_i) = \text{Cov}(kY_i^* + c_i, Y_i^* + y_i) = k \text{Var}(Y_i^*) + k \text{Cov}(Y_i^*, y_i) + \text{Cov}(c_i, Y_i^*) + \text{Cov}(c_i, y_i).$$

The middle two terms vanish by (3.2.5). For the last, note from (3.2.4) and (3.2.6) that $c_i = C_i - kY_i + ky_i$; the book's assumptions (3.2.5) also state that c_i is uncorrelated with C_i^* and Y_i^* . Now

$$\text{Cov}(c_i, y_i) = \text{Cov}(C_i, y_i) - k \text{Cov}(Y_i, y_i) + k \text{Var}(y_i),$$

and $\text{Cov}(C_i, y_i) = \text{Cov}(C_i^*, y_i) + \text{Cov}(c_i, y_i) = \text{Cov}(c_i, y_i)$ since $\text{Cov}(C_i^*, y_i) = 0$, while $\text{Cov}(Y_i, y_i) = \text{Cov}(Y_i^*, y_i) + \text{Var}(y_i) = \text{Var}(y_i)$; substituting gives $\text{Cov}(c_i, y_i) = 0$. Therefore

$$\text{Cov}(C_i, Y_i) = k \text{Var}(Y_i^*), \quad \text{Var}(Y_i) = \text{Var}(Y_i^*) + \text{Var}(y_i),$$

and

$$\text{plim } \hat{k}_{\text{OLS}} = \frac{\text{Cov}(C_i, Y_i)}{\text{Var}(Y_i)} = \frac{\text{Var}(Y_i^*)}{\text{Var}(Y_i^*) + \text{Var}(y_i)} k < k.$$

So with or without a constant, measured income produces attenuation: the OLS slope is the true MPC times the “reliability ratio” $\text{Var}(Y_i^*) / \text{Var}(Y_i)$, the fraction of the variance of measured income that is permanent. This is Friedman’s reconciliation of the low cross-section MPC (much transitory variation in measured income) with the high time-series MPC (transitory variation averages out over time). Notice the plim here is *not* the same as the no-constant version (3.2.9), $k E[(Y_i^*)^2] / \{E[(Y_i^*)^2] + E(y_i^2)\}$; the two agree only when $E(Y_i^*) = 0$. Including the constant is what makes Friedman’s estimator — the ratio of sample means — differ from OLS.

Review Question 3.2.2

The two equations are

$$\log(Q_i) = \phi_0 + \phi_1 \log(L_i) + u_i + v_i, \quad \log(L_i) = \beta_0 + \frac{1}{1 - \phi_1} u_i,$$

where $\beta_0 \equiv \frac{1}{1-\phi_1} [\log(\phi_1 B) + \log(p/w) + \phi_0]$ is constant across firms. Solving the labor demand equation for $u_i = (1 - \phi_1)[\log(L_i) - \beta_0]$ and substituting into the output equation,

$$\log(Q_i) = \phi_0 + \phi_1 \log(L_i) + (1 - \phi_1)[\log(L_i) - \beta_0] + v_i = [\phi_0 - (1 - \phi_1)\beta_0] + \log(L_i) + v_i.$$

So, in terms of observable variables, log output is an exact affine function of log labor input with slope *one* plus the technology shock v_i , which the firm could not observe when choosing L_i — hence $\text{Cov}(\log(L_i), v_i) = 0$. The OLS slope from regressing $\log(Q_i)$ on a constant and $\log(L_i)$ is consistent for the corresponding projection coefficient,

$$\text{plim } \hat{\phi}_1 = \frac{\text{Cov}(\log L_i, \log Q_i)}{\text{Var}(\log L_i)} = 1 + \frac{\text{Cov}(\log L_i, v_i)}{\text{Var}(\log L_i)} = 1,$$

as claimed. The point of the exercise is the contrast: the OLS estimand is not the elasticity ϕ_1 at all, because the endogenous part u_i of the error is perfectly collinear with $\log(L_i)$ — both are exact functions of A_i . Whatever ϕ_1 really is, the regression attributes the entire efficiency-driven co-movement of output and labor to labor, pushing the slope to one. Note also that the result does not depend on ϕ_1 being positive; it is an algebraic consequence of profit maximization plus the timing assumption (the firm knows A_i but not v_i).

Review Question 3.2.3

If the firm observes both A_i and v_i before choosing L_i , expected output is

$$\text{E}[A_i L_i^{\phi_1} \exp(v_i) \mid A_i, v_i] = A_i L_i^{\phi_1} \exp(v_i),$$

so the first-order condition maximizes $p A_i L_i^{\phi_1} \exp(v_i) - w L_i$, giving

$$L_i = \left(\frac{\phi_1 p A_i \exp(v_i)}{w} \right)^{1/(1-\phi_1)}, \quad \text{or in logs,} \quad \log(L_i) = \beta_0 + \frac{1}{1-\phi_1}(u_i + v_i),$$

with the same β_0 as before. The labor demand equation now loads on the *sum* of the two shocks. Substituting $u_i + v_i = (1 - \phi_1)[\log(L_i) - \beta_0]$ into the production function (3.2.13),

$$\log(Q_i) = \phi_0 + \phi_1 \log(L_i) + (u_i + v_i) = [\phi_0 + (1 - \phi_1)(-\beta_0)] + \log(L_i),$$

an exact linear function of $\log(L_i)$ with slope one and *no error term*. Hence $\text{Corr}(\log Q_i, \log L_i) = 1$: in the price-quantity scatter all firms lie exactly on a line of slope one. The regression of $\log Q_i$ on $\log L_i$ recovers a slope of one in *every* sample, not just asymptotically. Economically, when both shocks are known at the time of the input decision, there is no variation in L_i that is independent of the composite disturbance $u_i + v_i$ — labor absorbs all of it — and the production function's parameters are not identified at all: the data reveal only the reduced form $\log Q_i = \text{const} + \log L_i$, whatever ϕ_1 may be. This is the clearest possible statement of the identification problem in production function estimation: without a source of variation in input use that is uncorrelated with the firm-specific efficiency, the elasticity cannot be recovered from a cross-section of input-output pairs.

Section 3.3 The General Formulation

Review Question 3.3.1

For the demand equation (3.1.2A), the regressors are $\mathbf{z}_i = (1, p_i)'$ ($L = 2$) and the instruments are $\mathbf{x}_i = (1, x_i)'$ ($K = 2$). The rank condition ASSUMPTION 3.4 requires

$$\Sigma_{\mathbf{xz}} = E(\mathbf{x}_i \mathbf{z}_i') = \begin{bmatrix} 1 & E(p_i) \\ E(x_i) & E(x_i p_i) \end{bmatrix}$$

to have rank two, i.e. determinant $\text{Cov}(x_i, p_i) \neq 0$, which holds by (3.1.10A)–(3.1.10B) as long as $\beta_2 \neq 0$ and $\text{Var}(x_i) > 0$. So the demand equation is identified; since $K = L = 2$ it is *exactly* identified, not overidentified.

For the supply equation (3.1.2B'), the regressors are $(1, p_i, x_i)'$ ($L = 3$) but the instrument vector is still $\mathbf{x}_i = (1, x_i)'$ ($K = 2$), because no demand-side observable shifter is available. The order condition (3.3.5A) fails ($K = 2 < L = 3$), so the supply equation is *not* identified: intuitively, temperature shifts trace out the demand curve and thus identify it, but nothing in the model shifts demand without shifting supply, so the supply slope cannot be recovered. To identify both equations one would need an observable demand shifter (uncorrelated with ζ_i , correlated with p_i) to play for supply the role that x_i plays for demand.

Review Question 3.3.2

Yes. In Example 3.2 the regressor vector is $\mathbf{z}_i = (1, S_i, \text{EXPR}_i, \text{IQ}_i)'$, so $L = 4$, and the instrument vector is $\mathbf{x}_i = (1, S_i, \text{EXPR}_i, \text{AGE}_i, \text{MED}_i)'$, so $K = 5$. The order condition (3.3.5A) holds with strict inequality, $K = 5 > L = 4$; equivalently, by (3.3.5C), there is one more excluded predetermined variable (AGE_i and MED_i) than endogenous regressors (IQ_i). Given the maintained assumption that $\Sigma_{\mathbf{xz}} = E(\mathbf{x}_i \mathbf{z}_i')$ is of full column rank, the equation is *overidentified* by one degree: there is one more orthogonality condition than parameters. That single overidentifying restriction is precisely what Sargan's test examines in the empirical exercise.

Review Question 3.3.3

The equation is $\log(Q_i) = \phi_0 + \phi_1 \log(L_i) + u_i + v_i$, and the only instrument is the constant, $\mathbf{x}_i = 1$. The orthogonality condition ASSUMPTION 3.3 is the single restriction

$$E[\log(Q_i) - \phi_0 - \phi_1 \log(L_i)] = 0, \quad \text{i.e.} \quad \phi_0 = E(\log Q_i) - \phi_1 E(\log L_i).$$

For *any* choice of $\phi_1 \in \mathbb{R}$ there is a ϕ_0 satisfying this equation, so the set of parameter values consistent with the orthogonality condition is a whole line in (ϕ_0, ϕ_1) -space. In the notation of (3.3.4), $\Sigma_{\mathbf{xz}} = E(\mathbf{x}_i \mathbf{z}_i') = \begin{bmatrix} 1 & E(\log L_i) \end{bmatrix}$ is 1×2 with rank one $< L = 2$, so the system $\sigma_{\mathbf{xy}} = \Sigma_{\mathbf{xz}} \tilde{\delta}$ has a one-dimensional family of solutions. The order condition fails by one instrument: one moment condition cannot pin down two parameters. Geometrically, the single equation says only that the line passes through the means $(E \log L_i, E \log Q_i)$; its slope is completely free. This matches the economics of Review Question 3.2.3: input and output co-move through the unobserved efficiency, so no slope is singled out by the data.

Review Question 3.3.4

Haavelmo's macro model. The equation estimated is the consumption function $C_i = \alpha_0 + \alpha_1 Y_i + u_i$ with investment I_i as the instrument:

$$y_i = C_i, \quad \mathbf{z}_i = (1, Y_i)', \quad \mathbf{x}_i = (1, I_i)',$$

and $\mathbf{w}_i = (C_i, Y_i, I_i)'$. The rank condition is $\text{rank } E(\mathbf{x}_i \mathbf{z}_i') = 2$, i.e. $\det \begin{bmatrix} 1 & E(Y_i) \\ E(I_i) & E(I_i Y_i) \end{bmatrix} = \text{Cov}(I_i, Y_i) \neq 0$. From (3.2.1)–(3.2.2), $\text{Cov}(I_i, Y_i) = \text{Var}(I_i)/(1 - \alpha_1) > 0$, so the condition holds as long as investment varies.

Errors-in-variables (Friedman). The equation is (3.2.7), $C_i = kY_i + u_i$, with no intercept; the instrument for measured income Y_i is a variable x_i with $E(x_i u_i) = 0$ and $E(x_i Y_i) \neq 0$ — and, as the text notes, the constant itself serves:

$$y_i = C_i, \quad \mathbf{z}_i = Y_i, \quad \mathbf{x}_i = x_i \text{ (or 1)}.$$

The rank condition is $E(x_i Y_i) \neq 0$ (a 1×1 matrix of full rank). With $x_i = 1$ it reads $E(Y_i) \neq 0$, which is why Friedman's ratio-of-means estimator works: mean measured income is nonzero.

Production function. The equation is (3.2.13) with

$$y_i = \log(Q_i), \quad \mathbf{z}_i = (1, \log L_i)', \quad \mathbf{x}_i = 1.$$

The rank condition requires $E(\mathbf{x}_i \mathbf{z}_i') = [1 \quad E(\log L_i)]$ to have rank two, which is impossible for a 1×2 matrix: the equation is underidentified, as shown directly in the previous review question. To satisfy the order condition one would need an instrument correlated with $\log L_i$ and uncorrelated with $u_i + v_i$ — e.g. an input price shifter.

Review Question 3.3.5

The orthogonality conditions (3.3.2) state that $E[\mathbf{x}_i(y_i - \mathbf{z}_i' \delta)] = \mathbf{0}$, i.e.

$$\sigma_{\mathbf{x}y} \equiv E(\mathbf{x}_i y_i) = E(\mathbf{x}_i \mathbf{z}_i') \delta = \Sigma_{\mathbf{x}z} \delta.$$

Thus the last column of the $K \times (L + 1)$ matrix $[\Sigma_{\mathbf{x}z} : \sigma_{\mathbf{x}y}]$ is a linear combination of the first L columns, with coefficients δ . Appending a column that lies in the column space of a matrix cannot increase its rank, so

$$\text{rank} \begin{pmatrix} \Sigma_{\mathbf{x}z} & \sigma_{\mathbf{x}y} \end{pmatrix} = \text{rank}(\Sigma_{\mathbf{x}z}) \quad (= L \text{ under ASSUMPTION 3.4}).$$

The equality is the exact population statement that the system of moment equations (3.3.4) has a solution; the rank condition for identification adds that the solution is unique. As the parenthetical remark says, there is no need to assume it separately: it is a corollary of ASSUMPTIONS 3.1 and 3.3. The contrast with the sample analogue is instructive (compare Review Question 3.4.3): the sample matrix $[\mathbf{S}_{\mathbf{x}z} : \mathbf{s}_{\mathbf{x}y}]$ has full rank $L + 1$ with probability one when $K > L$, because the sample moments only converge to their population counterparts; the equality of ranks is an exact restriction that holds in the limit but not in finite samples.

Review Question 3.3.6

Yes. Write $\mathbf{x}_i^+ = (\mathbf{x}_i', \xi_i)'$. Then

$$E(\mathbf{x}_i^+ \mathbf{z}_i') = \begin{bmatrix} E(\mathbf{x}_i \mathbf{z}_i') \\ E(\xi_i \mathbf{z}_i') \end{bmatrix} = \begin{bmatrix} \Sigma_{\mathbf{xz}} \\ \mathbf{0}' \end{bmatrix}.$$

The added row is zero, so the column space of the augmented matrix is $\{(\mathbf{v}', 0)' : \mathbf{v} \in \text{column space of } \Sigma_{\mathbf{xz}}\}$, which has the same dimension L as before: the rank is unchanged, and the rank condition ASSUMPTION 3.4 is still satisfied. Two comments are in order. First, identification is unaffected but *efficiency* is not helped either: the optimal GMM weighting will attach zero useful weight to the extra moment (after orthogonalization the moment contributes nothing), so the asymptotic variance is unchanged. Second, the exercise highlights that the rank condition concerns correlation with the regressors, not with the error: a predetermined but irrelevant instrument is “valid” in the orthogonality sense yet worthless — and in finite samples such instruments can actually harm (the weak-instruments problem of Analytical Exercise 3.10 and of part (g) of the empirical exercise).

Review Question 3.3.7

Rank condition. No, it does not necessarily fail. The instrument vector is

$$\mathbf{x}_i = (1, S_i, \text{EXPR}_i, \text{AGE}_i, \text{MED}_i)'$$

The rank condition ASSUMPTION 3.4 concerns the 5×4 matrix $\Sigma_{\mathbf{xz}} = E(\mathbf{x}_i \mathbf{z}_i')$ with $\mathbf{z}_i = (1, S_i, \text{EXPR}_i, \text{IQ}_i)'$; its fourth row is

$$E(\text{AGE}_i \mathbf{z}_i') = E[(S_i + \text{EXPR}_i) \mathbf{z}_i'] = E(S_i \mathbf{z}_i') + E(\text{EXPR}_i \mathbf{z}_i'),$$

the sum of the second and third rows. Row rank being invariant to such dependencies, the rank of $\Sigma_{\mathbf{xz}}$ equals the rank of its first three and fifth rows, which can still be $L = 4$ — for instance if MED_i is correlated with IQ_i conditional on the other instruments. So the rank condition may well survive; one of the five instruments is simply redundant for identification.

Nonsingularity of $\mathbf{S}$. It necessarily fails. With $\mathbf{c} = (0, -1, -1, 1, 0)' \neq \mathbf{0}$ we have $\mathbf{c}'\mathbf{x}_i = -S_i - \text{EXPR}_i + \text{AGE}_i = 0$ for all i , hence $\mathbf{c}'\mathbf{g}_i = \varepsilon_i \mathbf{c}'\mathbf{x}_i = 0$ and

$$\mathbf{c}'E(\mathbf{g}_i \mathbf{g}_i') = E[\mathbf{c}'\mathbf{g}_i \mathbf{g}_i'] = \mathbf{0}',$$

so $\mathbf{S} = E(\mathbf{g}_i \mathbf{g}_i')$ has a zero eigenvalue and is singular. The two rank conditions play different roles: ASSUMPTION 3.4 is about $\mathbf{x}_i$ - $\mathbf{z}_i$ cross moments and tolerates redundant instruments, while ASSUMPTION 3.5 requires the instrument vector itself to carry no exact linear dependence — otherwise the covariance matrix of the moments is singular, the efficient weighting matrix $\mathbf{S}^{-1}$ does not exist, and GMM breaks down. The remedy is obvious: drop one of the redundant instruments, which leaves identification unchanged.

Review Question 3.3.8

Write $\tilde{\mathbf{g}}_i \equiv \tilde{\mathbf{x}}_i \varepsilon_i = \mathbf{A} \mathbf{g}_i$.

ASSUMPTION 3.3 (*orthogonality*). $E(\tilde{\mathbf{g}}_i) = \mathbf{A}E(\mathbf{g}_i) = \mathbf{0}$: linear combinations of orthogonal instruments are orthogonal.

ASSUMPTION 3.4 (*rank condition*). $E(\tilde{\mathbf{x}}_i \mathbf{z}'_i) = \mathbf{A}E(\mathbf{x}_i \mathbf{z}'_i) = \mathbf{A}\Sigma_{\mathbf{xz}}$, which is of full column rank by hypothesis. (Note this requires $q \geq L$; the hypothesis $\text{rank}(\mathbf{A}\Sigma_{\mathbf{xz}}) = L$ is not automatic — row operations can destroy the rank if $\mathbf{A}$ mixes away the identifying variation.)

ASSUMPTION 3.5 (*martingale difference, nonsingular variance*). For the m.d.s. property, by the law of iterated expectations,

$$E(\tilde{\mathbf{g}}_i | \tilde{\mathbf{g}}_{i-1}, \dots, \tilde{\mathbf{g}}_1) = \mathbf{A} E[E(\mathbf{g}_i | \mathbf{g}_{i-1}, \dots) | \tilde{\mathbf{g}}_{i-1}, \dots] = \mathbf{0},$$

since $(\tilde{\mathbf{g}}_{i-1}, \dots)$ is a function of $(\mathbf{g}_{i-1}, \dots)$. For nonsingularity, $E(\tilde{\mathbf{g}}_i \tilde{\mathbf{g}}'_i) = \mathbf{A}\mathbf{S}\mathbf{A}'$ with $\mathbf{S}$ positive definite and $\mathbf{A}$ of full row rank q , so for any $\mathbf{c} \neq \mathbf{0}$ in $\mathbb{R}^q$, $\mathbf{A}'\mathbf{c} \neq \mathbf{0}$ and $\mathbf{c}'\mathbf{A}\mathbf{S}\mathbf{A}'\mathbf{c} > 0$: $\mathbf{A}\mathbf{S}\mathbf{A}'$ is positive definite, hence nonsingular.

So any full-rank linear transformation of the instruments preserves the model's assumptions, provided it preserves the identifying correlation with the regressors. The leading application is Review Question 3.5.8: the efficient GMM based on a subset (or combination) of the orthogonality conditions.

Review Question 3.3.9

Set $\mathbf{z}_i = \mathbf{x}_i$ and compare assumption by assumption. ASSUMPTION 3.1 becomes $y_i = \mathbf{x}'_i \delta + \varepsilon_i$, which is ASSUMPTION 2.1 with $\beta = \delta$. ASSUMPTION 3.2 on $\{y_i, \mathbf{x}_i\}$ is ASSUMPTION 2.2 verbatim. ASSUMPTION 3.3 is $E(\mathbf{x}_i \varepsilon_i) = \mathbf{0}$, which is ASSUMPTION 2.3. ASSUMPTION 3.4 requires $\Sigma_{\mathbf{xz}} = E(\mathbf{x}_i \mathbf{x}'_i) = \Sigma_{\mathbf{xx}}$ ($K \times K$) to have full column rank K , i.e. to be nonsingular — ASSUMPTION 2.4; since $\Sigma_{\mathbf{xx}}$ is symmetric positive semidefinite by construction, nonsingularity is positive definiteness. Finally ASSUMPTION 3.5 says $\{\mathbf{g}_i\} = \{\mathbf{x}_i \varepsilon_i\}$ is an m.d.s. with nonsingular $E(\mathbf{g}_i \mathbf{g}'_i) = \mathbf{S}$ — ASSUMPTION 2.5. With $\mathbf{z}_i = \mathbf{x}_i$ the order condition holds with equality ($K = L$), the equation is exactly identified, the IV estimator (3.4.4) is $\mathbf{S}_{\mathbf{xx}}^{-1} \mathbf{s}_{\mathbf{xy}}$, i.e. OLS, and — as Review Question 3.5.1 verifies — every result of Sections 3.5–3.8 reduces to its Chapter 2 counterpart.

Section 3.4 Generalized Method of Moments Defined

Review Question 3.4.1

The demand equation is $q_i = \alpha_0 + \alpha_1 p_i + u_i$, so $\mathbf{z}_i = (1, p_i)'$, $\delta = (\alpha_0, \alpha_1)'$, and $\mathbf{x}_i = (1, x_i)'$. Since $K = L = 2$ the equation is exactly identified and the IV estimator is (3.4.4), $\hat{\delta}_{\text{IV}} = \mathbf{S}_{\mathbf{xz}}^{-1} \mathbf{s}_{\mathbf{xy}}$, i.e. the solution of $\mathbf{s}_{\mathbf{xy}} = \mathbf{S}_{\mathbf{xz}} \tilde{\delta}$:

$$\frac{1}{n} \sum_i \begin{bmatrix} 1 \\ x_i \end{bmatrix} q_i = \frac{1}{n} \sum_i \begin{bmatrix} 1 \\ x_i \end{bmatrix} \begin{bmatrix} 1 & p_i \end{bmatrix} \begin{bmatrix} \tilde{\alpha}_0 \\ \tilde{\alpha}_1 \end{bmatrix}.$$

Writing $\bar{q}, \bar{p}, \bar{x}$ for the sample means, the two equations are

$$\bar{q} = \tilde{\alpha}_0 + \tilde{\alpha}_1 \bar{p}, \quad \frac{1}{n} \sum_i x_i q_i = \tilde{\alpha}_0 \bar{x} + \tilde{\alpha}_1 \frac{1}{n} \sum_i x_i p_i.$$

Multiply the first equation by $\bar{x}$ and subtract from the second:

$$\frac{1}{n} \sum_i x_i q_i - \bar{x} \bar{q} = \tilde{\alpha}_1 \left(\frac{1}{n} \sum_i x_i p_i - \bar{x} \bar{p} \right),$$

whose two sides are the sample covariance between x_i and q_i and $\tilde{\alpha}_1$ times the sample covariance between x_i and p_i . Hence

$$\hat{\alpha}_{1,IV} = \frac{\text{sample covariance between } x_i \text{ and } q_i}{\text{sample covariance between } x_i \text{ and } p_i},$$

which is (3.1.12); the intercept follows as $\hat{\alpha}_{0,IV} = \bar{q} - \hat{\alpha}_{1,IV} \bar{p}$. This is the template for every exactly identified IV estimator: with a constant among the instruments, the moment conditions say that the sample covariances of the instruments with the residual vanish.

Review Question 3.4.2

Zero. When $K = L$, $\mathbf{S}_{xz}$ is square and, under ASSUMPTIONS 3.2 and 3.4, nonsingular for n sufficiently large with probability one, so $\tilde{\delta} = \hat{\delta}_{IV} = \mathbf{S}_{xz}^{-1} \mathbf{s}_{xy}$ solves the system (3.4.3) exactly: $\bar{\mathbf{g}}_n(\hat{\delta}_{IV}) = \mathbf{s}_{xy} - \mathbf{S}_{xz} \hat{\delta}_{IV} = \mathbf{0}$. Then for *any* weighting matrix $\widehat{\mathbf{W}}$,

$$J(\hat{\delta}_{IV}, \widehat{\mathbf{W}}) = n \bar{\mathbf{g}}_n(\hat{\delta}_{IV})' \widehat{\mathbf{W}} \bar{\mathbf{g}}_n(\hat{\delta}_{IV}) = 0,$$

and since $\widehat{\mathbf{W}}$ is positive definite, $J \geq 0$, so this is the minimum. Two consequences. First, under exact identification the GMM estimator does not depend on the weighting matrix at all — all $\widehat{\mathbf{W}}$ give the same estimator. Second, the overidentification test of Section 3.6 is vacuous when $K = L$: there is nothing left to test, which matches the degrees of freedom $K - L = 0$ of the χ^2 distribution in PROPOSITION 3.6.

Review Question 3.4.3

A solution of $\mathbf{s}_{xy} = \mathbf{S}_{xz} \tilde{\delta}$ exists if and only if $\mathbf{s}_{xy}$ lies in the column space of $\mathbf{S}_{xz}$, i.e. if and only if

$$\text{rank} \begin{pmatrix} \mathbf{S}_{xz} & \mathbf{s}_{xy} \end{pmatrix} = \text{rank}(\mathbf{S}_{xz}) = L.$$

In the population this holds exactly (Review Question 3.3.5): the true δ solves (3.3.4), so $\sigma_{xy} = \Sigma_{xz} \delta$ is a linear combination of the columns of Σ_{xz} . That rank *equality* is a set of exact equality restrictions — all $(L+1) \times (L+1)$ minors of the augmented matrix vanish — and equality restrictions are not preserved under convergence: a sequence of rank- L matrices can converge to a rank- L matrix, but the generic member of any approximating sequence of $K \times (L+1)$ matrices has rank $\min(K, L+1) = L+1$. Indeed, $\mathbf{s}_{xy}$ and $\mathbf{S}_{xz}$ are continuous functions of the data with absolutely continuous distributions, so with probability one $[\mathbf{S}_{xz} : \mathbf{s}_{xy}]$ has rank $L+1$ whenever $K > L$: the K sample moment equations in $L < K$ unknowns are inconsistent.

By contrast, full *column* rank of $\mathbf{S}_{xz}$ is an open condition — it says some $L \times L$ minor is nonzero, an inequality — and inequalities are preserved under convergence: since $\mathbf{S}_{xz} \xrightarrow{P} \Sigma_{xz}$ and $\text{rank}(\Sigma_{xz}) = L$, $\text{rank}(\mathbf{S}_{xz}) = L$ for n large with probability one. So the GMM estimator (3.4.8) is eventually well defined, while the exact moment equations it approximately solves have no exact solution. The whole point of GMM is to bridge this gap: minimize the discrepancy $\bar{\mathbf{g}}_n(\tilde{\delta})$ rather than set it to zero.

Review Question 3.4.4

The fallacy is that premultiplying an inconsistent system by a matrix changes the solution set: every solution of $\mathbf{s}_{xy} = \mathbf{S}_{xz}\tilde{\delta}$ would indeed solve (3.4.12), but not conversely. Concretely, the proposed $\tilde{\delta} = (\mathbf{S}'_{xz}\mathbf{S}_{xz})^{-1}\mathbf{S}'_{xz}\mathbf{s}_{xy}$ is the least squares solution of the inconsistent system: it makes $\mathbf{S}_{xz}\tilde{\delta}$ the *projection* of $\mathbf{s}_{xy}$ onto the column space of $\mathbf{S}_{xz}$, so

$$\bar{\mathbf{g}}_n(\tilde{\delta}) = [\mathbf{I}_K - \mathbf{S}_{xz}(\mathbf{S}'_{xz}\mathbf{S}_{xz})^{-1}\mathbf{S}'_{xz}]\mathbf{s}_{xy},$$

which is $\mathbf{0}$ if and only if $\mathbf{s}_{xy}$ happens to lie in the column space of $\mathbf{S}_{xz}$ — precisely the event of probability zero from Review Question 3.4.3 when $K > L$. Notice also what the proposal amounts to in GMM terms: it is the GMM estimator with weighting matrix $\widehat{\mathbf{W}} = \mathbf{I}_K$ (the first-order condition (3.4.7) with $\widehat{\mathbf{W}} = \mathbf{I}$ is exactly (3.4.12)). So the “solution” produced by the argument is a legitimate GMM estimate — it just does not set the sample moments to zero, because nothing can. The argument confuses solving the normal equations with solving the equations themselves.

Review Question 3.4.5

The GMM estimator (3.4.8) is $\hat{\delta}(\widehat{\mathbf{W}}) = (\mathbf{S}'_{xz}\widehat{\mathbf{W}}\mathbf{S}_{xz})^{-1}\mathbf{S}'_{xz}\widehat{\mathbf{W}}\mathbf{s}_{xy}$, and the only inverse involved is that of the $L \times L$ matrix $\mathbf{S}'_{xz}\widehat{\mathbf{W}}\mathbf{S}_{xz}$. By ergodic stationarity $\mathbf{S}_{xz} \xrightarrow{p} \Sigma_{xz}$ and by assumption $\widehat{\mathbf{W}} \xrightarrow{p} \mathbf{W}$, so LEMMA 2.3(A) gives

$$\mathbf{S}'_{xz}\widehat{\mathbf{W}}\mathbf{S}_{xz} \xrightarrow{p} \Sigma'_{xz}\mathbf{W}\Sigma_{xz}.$$

Matrix inversion is continuous at any nonsingular limit, so if $\Sigma'_{xz}\mathbf{W}\Sigma_{xz}$ is nonsingular — an open (inequality) condition on the limit — then $\mathbf{S}'_{xz}\widehat{\mathbf{W}}\mathbf{S}_{xz}$ is nonsingular for n sufficiently large with probability one, and $\hat{\delta}(\widehat{\mathbf{W}})$ is well defined. The definition of the GMM estimator insists on $\mathbf{W}$ positive definite because the quadratic-form interpretation of J and the minimization argument leading to (3.4.7)–(3.4.8) are cleanest then; but algebraically all that is needed is the nonsingularity of the $L \times L$ “bread” matrix of the sandwich. This observation matters for Analytical Exercise 3.4, where the OLS estimator is written as a GMM estimator with a singular weighting matrix, and for Review Question 3.5.2, which checks that the large-sample results survive too. (Of course a singular $\mathbf{W}$ can never be optimal: the optimal weight $\mathbf{S}^{-1}$ is positive definite under ASSUMPTION 3.5.)

Section 3.5 Large-Sample Properties of GMM

Review Question 3.5.1

Set $\mathbf{z}_i = \mathbf{x}_i$ throughout, as in Review Question 3.3.9. Then $K = L$, $\Sigma_{xz} = E(\mathbf{x}_i\mathbf{x}'_i) = \Sigma_{xx}$ is square and symmetric, and the equation is exactly identified, so GMM reduces to the IV estimator which is OLS.

Consistency. PROPOSITION 3.1(A) under ASSUMPTIONS 3.1–3.4 is PROPOSITION 2.1(A) under ASSUMPTIONS 2.1–2.4, the assumptions matching one-for-one by Review Question 3.3.9.

Asymptotic normality. (3.5.1) with $\mathbf{W}$ arbitrary and $\Sigma_{\mathbf{z}\mathbf{z}} = \Sigma_{\mathbf{x}\mathbf{x}}$ square and nonsingular gives

$$\text{Avar}(\widehat{\delta}(\widehat{\mathbf{W}})) = (\Sigma_{\mathbf{x}\mathbf{x}} \mathbf{W} \Sigma_{\mathbf{x}\mathbf{x}})^{-1} \Sigma_{\mathbf{x}\mathbf{x}} \mathbf{W} \mathbf{S} \mathbf{W} \Sigma_{\mathbf{x}\mathbf{x}} (\Sigma_{\mathbf{x}\mathbf{x}} \mathbf{W} \Sigma_{\mathbf{x}\mathbf{x}})^{-1} = \Sigma_{\mathbf{x}\mathbf{x}}^{-1} \mathbf{S} \Sigma_{\mathbf{x}\mathbf{x}}^{-1},$$

since $(\Sigma_{\mathbf{x}\mathbf{x}} \mathbf{W} \Sigma_{\mathbf{x}\mathbf{x}})^{-1} = \Sigma_{\mathbf{x}\mathbf{x}}^{-1} \mathbf{W}^{-1} \Sigma_{\mathbf{x}\mathbf{x}}^{-1}$ and all factors cancel — this is (2.3.4), confirming that under exact identification the asymptotic variance does not depend on $\mathbf{W}$. This is PROPOSITION 2.1(B).

Estimated asymptotic variance. (3.5.2) reduces to $\mathbf{S}_{\mathbf{x}\mathbf{x}}^{-1} \widehat{\mathbf{S}} \mathbf{S}_{\mathbf{x}\mathbf{x}}^{-1} / n$, which is (2.3.5); its consistency is PROPOSITION 2.1(C).

Error variance. PROPOSITION 3.2 is PROPOSITION 2.2: with $\mathbf{z}_i = \mathbf{x}_i$, the extra hypothesis is that $E(\mathbf{x}_i \mathbf{x}_i')$ exists and is finite, exactly as stated there.

Hypothesis testing. PROPOSITION 3.3 parts (a), (b), (c) reproduce the robust t -ratio (2.4.3) and the Wald statistics (2.4.5), (2.4.7) of PROPOSITION 2.3, with $\widehat{\text{Avar}}(\mathbf{b}) = \mathbf{S}_{\mathbf{x}\mathbf{x}}^{-1} \widehat{\mathbf{S}} \mathbf{S}_{\mathbf{x}\mathbf{x}}^{-1} / n$.

Estimation of $\mathbf{S}$. ASSUMPTION 3.6 with $\mathbf{z}_i = \mathbf{x}_i$ is ASSUMPTION 2.6, and PROPOSITION 3.4 is PROPOSITION 2.4: $\widehat{\mathbf{S}}$ of (3.5.10) is (2.5.1).

Efficiency. Under exact identification the lower bound $(\Sigma'_{\mathbf{z}\mathbf{z}} \mathbf{S}^{-1} \Sigma_{\mathbf{z}\mathbf{z}})^{-1} = \Sigma_{\mathbf{x}\mathbf{x}}^{-1} \mathbf{S} \Sigma_{\mathbf{x}\mathbf{x}}^{-1}$ is attained by every $\widehat{\mathbf{W}}$, so PROPOSITION 3.5 is vacuous in the regression model — consistently with OLS being the only estimator in sight. The first place efficiency has real content is exactly this chapter's overidentified case.

Review Question 3.5.2

Go through each result.

PROPOSITION 3.1(A) (*consistency*). The sampling error (3.4.11) is

$$\widehat{\delta}(\widehat{\mathbf{W}}) - \delta = (\mathbf{S}'_{\mathbf{x}\mathbf{z}} \widehat{\mathbf{W}} \mathbf{S}_{\mathbf{x}\mathbf{z}})^{-1} \mathbf{S}'_{\mathbf{x}\mathbf{z}} \widehat{\mathbf{W}} \bar{\mathbf{g}}.$$

By ergodic stationarity, $\mathbf{S}_{\mathbf{x}\mathbf{z}} \xrightarrow{p} \Sigma_{\mathbf{x}\mathbf{z}}$, $\widehat{\mathbf{W}} \xrightarrow{p} \mathbf{W}$, and $\bar{\mathbf{g}} \xrightarrow{p} \mathbf{0}$. Since $\Sigma'_{\mathbf{x}\mathbf{z}} \mathbf{W} \Sigma_{\mathbf{x}\mathbf{z}}$ is nonsingular by hypothesis, the continuous mapping theorem (LEMMA 2.3(A)) applies and the whole expression converges to $(\Sigma'_{\mathbf{x}\mathbf{z}} \mathbf{W} \Sigma_{\mathbf{x}\mathbf{z}})^{-1} \Sigma'_{\mathbf{x}\mathbf{z}} \mathbf{W} \cdot \mathbf{0} = \mathbf{0}$. Singularity of $\mathbf{W}$ itself is irrelevant; only the inverse of the $L \times L$ matrix is ever taken.

PROPOSITION 3.1(B) (*asymptotic normality*). Multiplying the sampling error by $\sqrt{n}$,

$$\sqrt{n}(\widehat{\delta}(\widehat{\mathbf{W}}) - \delta) = (\mathbf{S}'_{\mathbf{x}\mathbf{z}} \widehat{\mathbf{W}} \mathbf{S}_{\mathbf{x}\mathbf{z}})^{-1} \mathbf{S}'_{\mathbf{x}\mathbf{z}} \widehat{\mathbf{W}} \sqrt{n} \bar{\mathbf{g}},$$

and $\sqrt{n} \bar{\mathbf{g}} \xrightarrow{d} N(\mathbf{0}, \mathbf{S})$ by ASSUMPTION 3.5 and the ergodic stationary m.d.s. CLT. By LEMMA 2.4(C) with the convergent “matrix” $(\mathbf{S}'_{\mathbf{x}\mathbf{z}} \widehat{\mathbf{W}} \mathbf{S}_{\mathbf{x}\mathbf{z}})^{-1} \mathbf{S}'_{\mathbf{x}\mathbf{z}} \widehat{\mathbf{W}}$ (which requires nothing beyond the stated nonsingularity), the limit is normal with mean zero and variance (3.5.1). That formula is well defined and positive semidefinite for any symmetric $\mathbf{W}$.

PROPOSITION 3.1(C) uses only the consistency of $\widehat{\mathbf{S}}$, of $\mathbf{S}_{\mathbf{x}\mathbf{z}}$, and of $\widehat{\mathbf{W}}$, plus the same continuous mapping; valid.

PROPOSITION 3.2 concerns the residual variance and never mentions $\widehat{\mathbf{W}}$ beyond consistency of the coefficient estimator; valid by part (a).

PROPOSITION 3.3. The robust t and Wald statistics are continuous functions of $\widehat{\delta}(\widehat{\mathbf{W}})$ and of the estimated asymptotic variance (3.5.2), which is consistently estimated and nonsingular with

probability approaching one (its limit (3.5.1) is positive definite as long as $\mathbf{S}$ is positive definite and $\Sigma'_{\mathbf{x}z} \mathbf{W} \Sigma_{\mathbf{x}z}$ is nonsingular). So the $N(0,1)$ and χ^2 limits follow from (b) and LEMMA 2.4(D).

PROPOSITION 3.4 is about $\widehat{\mathbf{S}}$ alone; valid.

The exception is exactly PROPOSITION 3.5: the lower bound $(\Sigma'_{\mathbf{x}z} \mathbf{S}^{-1} \Sigma_{\mathbf{x}z})^{-1}$ is achieved only when $\mathbf{W} \propto \mathbf{S}^{-1}$ (up to the generalization in the book's footnote 11), and $\mathbf{S}$ is positive definite under ASSUMPTION 3.5, so an efficient weighting matrix cannot be singular. In short: consistency, asymptotic normality, and testing survive a singular limiting weighting matrix; efficiency does not.

Review Question 3.5.3

From the sampling error formula (3.4.11) applied to each weighting matrix,

$$\widehat{\delta}(\widehat{\mathbf{W}}) - \delta = \mathbf{A}_n(\widehat{\mathbf{W}}) \bar{\mathbf{g}}, \quad \widehat{\delta}(\widetilde{\mathbf{W}}) - \delta = \mathbf{A}_n(\widetilde{\mathbf{W}}) \bar{\mathbf{g}},$$

and subtracting gives the hint's identity. Both $\widehat{\mathbf{W}}$ and $\widetilde{\mathbf{W}}$ converge in probability to the same limit $\mathbf{W}$ (their difference vanishes and — by the definition of a GMM weighting matrix — each converges to its own symmetric positive definite limit; a common limit of the difference forces the two limits to coincide). Hence by LEMMA 2.3(A),

$$\mathbf{A}_n(\widehat{\mathbf{W}}) \xrightarrow{p} (\Sigma'_{\mathbf{x}z} \mathbf{W} \Sigma_{\mathbf{x}z})^{-1} \Sigma'_{\mathbf{x}z} \mathbf{W} \quad \text{and} \quad \mathbf{A}_n(\widetilde{\mathbf{W}}) \xrightarrow{p} (\Sigma'_{\mathbf{x}z} \mathbf{W} \Sigma_{\mathbf{x}z})^{-1} \Sigma'_{\mathbf{x}z} \mathbf{W},$$

so $\mathbf{A}_n(\widehat{\mathbf{W}}) - \mathbf{A}_n(\widetilde{\mathbf{W}}) \xrightarrow{p} \mathbf{0}$. On the other hand, $\sqrt{n} \bar{\mathbf{g}} \xrightarrow{d} N(\mathbf{0}, \mathbf{S})$ by ASSUMPTION 3.5 and the ergodic stationary m.d.s. CLT. Now apply LEMMA 2.4(B) (a sequence converging in distribution times a sequence converging in probability to zero converges to zero in probability) elementwise to the product:

$$\sqrt{n}(\widehat{\delta}(\widehat{\mathbf{W}}) - \widehat{\delta}(\widetilde{\mathbf{W}})) = [\mathbf{A}_n(\widehat{\mathbf{W}}) - \mathbf{A}_n(\widetilde{\mathbf{W}})] \sqrt{n} \bar{\mathbf{g}} \xrightarrow{p} \mathbf{0}.$$

The practical content is substantial: the asymptotic *distribution* of a GMM estimator depends on the weighting matrix only through its probability limit. So two-step, three-step, and iterated GMM estimators, and GMM with any consistent estimate of the same optimal weight $\mathbf{S}^{-1}$, are all $\sqrt{n}$ -equivalent; differences among them are purely finite-sample. This is also what legitimizes the “C test computed with the same $\widehat{\mathbf{S}}$ throughout” recipe of Section 3.6.

Review Question 3.5.4

Let $\widehat{\delta}_{(2)}$ denote the two-step efficient GMM estimator and $\widehat{\varepsilon}_i^{(2)} = y_i - \mathbf{z}'_i \widehat{\delta}_{(2)}$ its residual. Since $\widehat{\delta}_{(2)}$ is consistent (PROPOSITION 3.1(A), its weighting matrix $\widehat{\mathbf{S}}^{-1}$ from step 1 being consistent for a positive definite limit), PROPOSITION 3.4 applies to the recomputed matrix

$$\widehat{\mathbf{S}}_{(3)} \equiv \frac{1}{n} \sum_i (\widehat{\varepsilon}_i^{(2)})^2 \mathbf{x}_i \mathbf{x}'_i \xrightarrow{p} \mathbf{S},$$

under ASSUMPTIONS 3.1, 3.2, 3.6, and the existence of $\mathbf{S}$. The third-step estimator is $\widehat{\delta}_{(3)} \equiv \widehat{\delta}(\widehat{\mathbf{S}}_{(3)}^{-1})$.

Consistency: yes, by PROPOSITION 3.1(A), since $\widehat{\mathbf{S}}_{(3)}^{-1} \xrightarrow{p} \mathbf{S}^{-1}$, a symmetric positive definite matrix.

Asymptotic normality: yes, by PROPOSITION 3.1(B), with $\mathbf{W} = \mathbf{S}^{-1}$. *Efficiency:* yes, because $\mathbf{W} = \mathbf{S}^{-1}$ achieves the lower bound of PROPOSITION 3.5:

$$\text{Avar}(\widehat{\delta}_{(3)}) = (\Sigma'_{\mathbf{x}z} \mathbf{S}^{-1} \Sigma_{\mathbf{x}z})^{-1},$$

the same as the two-step estimator. Indeed, more is true: by Review Question 3.5.3 the two estimators are $\sqrt{n}$ -equivalent, $\sqrt{n}(\hat{\delta}_{(3)} - \hat{\delta}_{(2)}) \xrightarrow{p} \mathbf{0}$, so they share *every* asymptotic property, and iterating further (to convergence, as in the “iterated GMM” of Hansen, Heaton, and Yaron) changes nothing asymptotically. All the action of iterating is in finite samples, where it removes the dependence of the estimate on the arbitrary first-step weighting matrix — at the price of sometimes poorer small-sample behavior, per the July 1996 *JBES* studies cited in the text.

Review Question 3.5.5

No. The efficient two-step GMM estimator is

$$\hat{\delta}(\hat{\mathbf{S}}^{-1}) = (\mathbf{S}'_{\mathbf{zx}} \hat{\mathbf{S}}^{-1} \mathbf{S}_{\mathbf{zx}})^{-1} \mathbf{S}'_{\mathbf{zx}} \hat{\mathbf{S}}^{-1} \mathbf{s}_{\mathbf{xy}},$$

while OLS is $\mathbf{b} = \mathbf{S}_{\mathbf{zz}}^{-1} \mathbf{s}_{\mathbf{zy}}$ with $\mathbf{S}_{\mathbf{zz}} = \frac{1}{n} \sum_i \mathbf{z}_i \mathbf{z}'_i$. Write $\mathbf{x}_i = (\mathbf{z}'_i, \mathbf{w}'_i)'$ with $\mathbf{w}_i$ the extra instruments; then $\mathbf{S}_{\mathbf{zx}} = [\mathbf{S}'_{\mathbf{zz}} : \mathbf{S}'_{\mathbf{wz}}]'$ and $\mathbf{s}_{\mathbf{xy}} = [\mathbf{s}'_{\mathbf{zy}} : \mathbf{s}'_{\mathbf{wy}}]'$. The GMM first-order condition sets to zero a $\hat{\mathbf{S}}^{-1}$ -weighted combination of all K sample moments, including $\mathbf{s}_{\mathbf{wy}} - \mathbf{S}_{\mathbf{wz}} \tilde{\delta}$, the moments associated with the extra instruments; OLS sets to zero only the first L of them. Generic data do not make $\mathbf{s}_{\mathbf{wy}} - \mathbf{S}_{\mathbf{wz}} \mathbf{b} = \mathbf{0}$ at the OLS estimate, so the two estimators differ numerically whenever the extra moments bind, i.e. whenever $K > L$ and the extra instruments are correlated with the regressors.

What *is* true, by Review Question 3.3.9 / 3.5.1, is that when $\mathbf{z}_i = \mathbf{x}_i$ (no extra instruments) GMM collapses to OLS. And the efficient GMM using all of $\mathbf{x}_i$ exploits strictly more orthogonality conditions, so PROPOSITION 3.5 gives $\text{Avar}(\hat{\delta}(\hat{\mathbf{S}}^{-1})) \leq \text{Avar}(\mathbf{b})$ — this is exactly the comparison worked out in Analytical Exercise 3.4. Numerical equality would require the extra moments to be redundant, e.g. $\mathbf{S}_{\mathbf{wz}} = \mathbf{0}$ (irrelevant instruments, Review Question 3.3.6) or degenerate samples.

Review Question 3.5.6

With $\mathbf{X}$ the $n \times K$ matrix whose i -th row is $\mathbf{x}'_i$,

$$\mathbf{X}'\mathbf{B}\mathbf{X} = \sum_{i=1}^n \mathbf{x}_i \mathbf{B}_{ii} \mathbf{x}'_i = \sum_{i=1}^n \hat{\varepsilon}_i^2 \mathbf{x}_i \mathbf{x}'_i,$$

because multiplying on the left by $\mathbf{X}'$ and on the right by $\mathbf{X}$ turns the diagonal entries of $\mathbf{B}$ into weights on the outer products $\mathbf{x}_i \mathbf{x}'_i$: formally, $(\mathbf{X}'\mathbf{B}\mathbf{X})_{kl} = \sum_i \mathbf{X}_{ik} \mathbf{B}_{ii} \mathbf{X}_{il} = \sum_i x_{ik} \hat{\varepsilon}_i^2 x_{il}$. Dividing by n gives

$$\frac{1}{n} \mathbf{X}'\mathbf{B}\mathbf{X} = \frac{1}{n} \sum_{i=1}^n \hat{\varepsilon}_i^2 \mathbf{x}_i \mathbf{x}'_i = \hat{\mathbf{S}},$$

which is (3.5.10) (since $\mathbf{g}_i = \mathbf{x}_i \varepsilon_i$, the generic term of $\hat{\mathbf{S}}$ is $\hat{\mathbf{g}}_i \hat{\mathbf{g}}'_i = \hat{\varepsilon}_i^2 \mathbf{x}_i \mathbf{x}'_i$). This is the White form: the efficient GMM estimator is then $\hat{\delta}(\hat{\mathbf{S}}^{-1}) = (\mathbf{Z}'\mathbf{X}(\mathbf{X}'\mathbf{B}\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X}(\mathbf{X}'\mathbf{B}\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$, an expression any matrix-oriented package can evaluate in one line. The diagonality of $\mathbf{B}$ is what the m.d.s. assumption (ASSUMPTION 3.5) buys: no cross-products $\hat{\varepsilon}_i \hat{\varepsilon}_j$ enter $\mathbf{S}$ because the autocovariances of $\mathbf{g}_i$ are zero. With serially correlated errors, $\mathbf{B}$ would be replaced by a band matrix — the HAC estimators of Chapter 6.

Review Question 3.5.7

In the transformed equation $\mathbf{X}'\mathbf{y} = \mathbf{X}'\mathbf{Z}\delta + \mathbf{X}'\boldsymbol{\varepsilon}$, the regressand is $\mathbf{X}'\mathbf{y}$ ($K \times 1$), the regressor matrix is $\mathbf{X}'\mathbf{Z}$ ($K \times L$), and the error $\mathbf{X}'\boldsymbol{\varepsilon}$ has variance

$$\text{Var}(\mathbf{X}'\boldsymbol{\varepsilon}) = \text{E}(\mathbf{X}'\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'\mathbf{X}) = n \text{E}\left(\frac{1}{n} \sum_i \mathbf{g}_i \cdot \frac{1}{n} \sum_j \mathbf{g}_j'\right) \cdot n = n\mathbf{S},$$

because $\{\mathbf{g}_i\} = \{\mathbf{x}_i\boldsymbol{\varepsilon}_i\}$ is an m.d.s. under ASSUMPTION 3.5, so all cross terms vanish and $\frac{1}{n} \sum_i \text{E}(\mathbf{g}_i\mathbf{g}_i') = \mathbf{S}$. FGLS with estimated covariance matrix $n\hat{\mathbf{S}}$ is

$$\hat{\delta}_{\text{FGLS}} = \left[(\mathbf{X}'\mathbf{Z})'(n\hat{\mathbf{S}})^{-1}(\mathbf{X}'\mathbf{Z}) \right]^{-1} (\mathbf{X}'\mathbf{Z})'(n\hat{\mathbf{S}})^{-1}\mathbf{X}'\mathbf{y}.$$

The factor n cancels, and writing $\mathbf{X}'\mathbf{Z} = n\mathbf{S}_{xz}$ and $\mathbf{X}'\mathbf{y} = n\mathbf{s}_{xy}$,

$$\hat{\delta}_{\text{FGLS}} = (\mathbf{S}'_{xz}\hat{\mathbf{S}}^{-1}\mathbf{S}_{xz})^{-1}\mathbf{S}'_{xz}\hat{\mathbf{S}}^{-1}\mathbf{s}_{xy} = \hat{\delta}(\hat{\mathbf{S}}^{-1}),$$

which is the efficient GMM estimator (3.5.12). The interpretation is informative: GMM treats the K sample moments $\bar{\mathbf{g}}$ as the “data” — one observation per orthogonality condition — and the choice of weighting matrix as a GLS correction for their (generally non-scalar) covariance matrix $\mathbf{S}$. Efficiency of GLS translates immediately into the efficiency of GMM with $\mathbf{W} = \mathbf{S}^{-1}$, which is the content of PROPOSITION 3.5. The analogy is only asymptotic, though: unlike the classical GLS setting, the “error” covariance matrix here is itself estimated from fourth moments, which is why efficient GMM has the small-sample problems discussed in the text.

Review Question 3.5.8

By Review Question 3.3.8, the transformed instruments $\tilde{\mathbf{x}}_i = \mathbf{A}\mathbf{x}_i$ satisfy ASSUMPTIONS 3.3–3.5 (given $\text{rank}(\mathbf{A}\Sigma_{xz}) = L$), with $\tilde{\mathbf{g}}_i = \mathbf{A}\mathbf{g}_i$ and

$$\tilde{\mathbf{S}} \equiv \text{E}(\tilde{\mathbf{g}}_i\tilde{\mathbf{g}}_i') = \mathbf{A}\mathbf{S}\mathbf{A}', \quad \tilde{\Sigma}_{xz} = \mathbf{A}\Sigma_{xz}, \quad \tilde{\sigma}_{xy} = \mathbf{A}\sigma_{xy}.$$

The efficient GMM estimator based on these q moments is (3.5.12) with the tilded quantities and $\hat{\tilde{\mathbf{S}}} = \mathbf{A}\hat{\mathbf{S}}\mathbf{A}'$:

$$\hat{\delta}((\mathbf{A}\hat{\mathbf{S}}\mathbf{A}')^{-1}) = \left[(\mathbf{A}\mathbf{S}_{xz})'(\mathbf{A}\hat{\mathbf{S}}\mathbf{A}')^{-1}(\mathbf{A}\mathbf{S}_{xz}) \right]^{-1} (\mathbf{A}\mathbf{S}_{xz})'(\mathbf{A}\hat{\mathbf{S}}\mathbf{A}')^{-1}\mathbf{A}\mathbf{s}_{xy},$$

which is (3.4.8) with the $K \times K$ weighting matrix $\widehat{\mathbf{W}} = \mathbf{A}'(\mathbf{A}\hat{\mathbf{S}}\mathbf{A}')^{-1}\mathbf{A}$, as stated in the answer.

Now suppose $q = K$ so that $\mathbf{A}$ is square and nonsingular. Then $(\mathbf{A}\hat{\mathbf{S}}\mathbf{A}')^{-1} = \mathbf{A}'^{-1}\hat{\mathbf{S}}^{-1}\mathbf{A}^{-1}$ and

$$\widehat{\mathbf{W}} = \mathbf{A}'\mathbf{A}'^{-1}\hat{\mathbf{S}}^{-1}\mathbf{A}^{-1}\mathbf{A} = \hat{\mathbf{S}}^{-1},$$

so the estimator is exactly $\hat{\delta}(\hat{\mathbf{S}}^{-1})$, the efficient GMM estimator of the original system. The intuition is that with $q = K$ the transformed moments $\mathbf{A}\bar{\mathbf{g}}_n(\tilde{\delta})$ are a nonsingular linear transformation of the original ones — the same information repackaged — so the efficient combination of them must recover the original efficient estimator. When $q < K$, by contrast, information is genuinely discarded, and the lower bound rises: $(\Sigma'_{xz}\mathbf{A}'(\mathbf{A}\mathbf{S}\mathbf{A}')^{-1}\mathbf{A}\Sigma_{xz})^{-1} \geq (\Sigma'_{xz}\mathbf{S}^{-1}\Sigma_{xz})^{-1}$ by PROPOSITION 3.5 (the estimator is a GMM estimator on the full moment set with the deficient-rank weight

$\mathbf{A}'(\mathbf{A}\mathbf{S}\mathbf{A}')^{-1}\mathbf{A}$). The C test of Section 3.6 and the subset tests of orthogonality conditions rest on exactly this construction with $\mathbf{A} = [\mathbf{I}_{K_1} : \mathbf{0}]$.

Section 3.6 Testing Overidentifying Restrictions

Review Question 3.6.1

Yes. The J statistic is a continuous function of sample moments: from (3.4.2), (3.4.6) and (3.4.8),

$$J(\hat{\delta}(\tilde{\mathbf{S}}^{-1}), \hat{\mathbf{S}}^{-1}) = n \bar{\mathbf{g}}_n(\hat{\delta}(\tilde{\mathbf{S}}^{-1}))' \hat{\mathbf{S}}^{-1} \bar{\mathbf{g}}_n(\hat{\delta}(\tilde{\mathbf{S}}^{-1})),$$

with $\bar{\mathbf{g}}_n(\hat{\delta}(\tilde{\mathbf{S}}^{-1})) = \mathbf{s}_{xy} - \mathbf{S}_{xz} \hat{\delta}(\tilde{\mathbf{S}}^{-1})$. Since both $\hat{\mathbf{S}}$ and $\tilde{\mathbf{S}}$ converge in probability to the same $\mathbf{S}$, Review Question 3.5.3 gives $\sqrt{n}[\hat{\delta}(\tilde{\mathbf{S}}^{-1}) - \hat{\delta}(\mathbf{S}^{-1})] \xrightarrow{p} \mathbf{0}$ where $\hat{\delta}(\mathbf{S}^{-1})$ is the infeasible estimator using the true $\mathbf{S}$, and $\hat{\mathbf{S}}^{-1} \xrightarrow{p} \mathbf{S}^{-1}$. Hence the whole statistic differs from $J(\hat{\delta}(\mathbf{S}^{-1}), \mathbf{S}^{-1})$ by $o_p(1)$, and the latter is asymptotically $\chi^2(K - L)$ by PROPOSITION 3.6 (proved in Analytical Exercise 3.5). More directly, $\sqrt{n} \bar{\mathbf{g}}_n(\hat{\delta}(\tilde{\mathbf{S}}^{-1})) \xrightarrow{d} (\mathbf{I}_K - \mathbf{G}) \mathbf{v}$ with $\mathbf{v} \sim N(\mathbf{0}, \mathbf{S})$ and $\mathbf{G} \equiv \Sigma_{xz}(\Sigma'_{xz} \mathbf{S}^{-1} \Sigma_{xz})^{-1} \Sigma'_{xz} \mathbf{S}^{-1}$ as in (3.6.6), and since $\mathbf{G}' \mathbf{S}^{-1} \mathbf{G} = \mathbf{0}'$ one verifies $\mathbf{S}^{-1}(\mathbf{I} - \mathbf{G})\mathbf{S}(\mathbf{I} - \mathbf{G})' \mathbf{S}^{-1} = (\mathbf{I} - \mathbf{G})' \mathbf{S}^{-1}$, an idempotent quadratic form of rank $K - L$; substituting $\hat{\mathbf{S}}^{-1}$ for $\mathbf{S}^{-1}$ changes nothing by continuity. The practical caveat is finite-sample: using different estimates of $\mathbf{S}$ in the two places can make statistics such as the C statistic negative in small samples, which is why the text recommends using the same $\hat{\mathbf{S}}$ throughout — an asymptotically irrelevant but finite-sample-prudent choice.

Review Question 3.6.2

By the definition (3.4.6), $J(\tilde{\delta}, \widehat{\mathbf{W}}) = n \bar{\mathbf{g}}_n(\tilde{\delta})' \widehat{\mathbf{W}} \bar{\mathbf{g}}_n(\tilde{\delta})$, and by (3.4.2),

$$\bar{\mathbf{g}}_n(\tilde{\delta}) = \frac{1}{n} \sum_i \mathbf{x}_i (y_i - \mathbf{z}_i' \tilde{\delta}) = \mathbf{s}_{xy} - \mathbf{S}_{xz} \tilde{\delta}.$$

Substituting $\tilde{\delta} = \hat{\delta}(\hat{\mathbf{S}}^{-1})$ and $\widehat{\mathbf{W}} = \hat{\mathbf{S}}^{-1}$ gives the displayed expression immediately: the J statistic is n times the squared $\hat{\mathbf{S}}^{-1}$ -norm of the sample moment discrepancy evaluated at the efficient GMM estimate. This is the form in which J is computed from regression output — the discrepancy $\mathbf{s}_{xy} - \mathbf{S}_{xz} \hat{\delta} = \frac{1}{n} \mathbf{X}' \hat{\boldsymbol{\varepsilon}}$ is just the vector of sample covariances between the instruments and the residuals — and the form used in Analytical Exercise 3.6 to relate J to the matrix $\hat{\mathbf{B}}$ of Analytical Exercise 3.5.

Review Question 3.6.3

No. By PROPOSITION 3.7, $C = J - J_1 \xrightarrow{d} \chi^2(K - K_1)$, where K_1 is the number of nonsuspect instruments, and the proposition's strengthened rank condition requires $K_1 \geq L$ (the rank condition for $\mathbf{x}_{i1}$ alone). Hence

$$K - K_1 \leq K - L.$$

Equality holds exactly when $K_1 = L$, i.e. when the nonsuspect instruments just identify the equation. Intuitively, C compares the same parameter vector fitted under K and under K_1 moment conditions; the number of suspect restrictions tested is $K - K_1$, and since at least L moments are needed merely to identify δ , at most $K - L$ moments are left over to test. This bound is also visible in the geometry of the proof (Analytical Exercise 3.7): C is a quadratic form in the same

standardized vector as J , with projection matrices of ranks $K - L$ and $K_1 - L$, whose difference has rank $K - K_1$.

Review Question 3.6.4

No. When $K_1 = L$ the sub-model using only the nonsuspect instruments is exactly identified, so its J statistic vanishes identically (Review Question 3.4.2): $J_1 = 0$ whatever $\hat{\mathbf{S}}_{11}$ is used. Then $C = J - J_1 = J$, the overidentification statistic of the full system, which depends only on the full instrument list $\mathbf{x}_i$ and not on how it is split. Every admissible partition into L nonsuspect and $K - L$ suspect instruments yields the same number. Intuitively, with just enough instruments to identify δ , the restricted GMM sets those L sample moments exactly to zero, so no metric on them can detect anything; all the evidence about misspecification is carried by the remaining $K - L$ moments, and J aggregates exactly that evidence. This is also why the C test with $K_1 = L$ cannot be interpreted as testing the suspect instruments *while trusting the others*: with the model exactly identified by the “trusted” instruments alone, there is no independent evidence that they are valid — the test is really the omnibus overidentification test.

Section 3.7 Hypothesis Testing by the Likelihood-Ratio Principle

Review Question 3.7.1

When $\mathbf{z}_i = \mathbf{x}_i$ the model is exactly identified ($K = L$), so the unrestricted efficient GMM estimator is OLS and sets the sample moments exactly to zero: $\bar{\mathbf{g}}_n(\mathbf{b}) = \mathbf{s}_{xy} - \mathbf{S}_{xx}\mathbf{b} = \mathbf{0}$, whence the unrestricted J is

$$J(\mathbf{b}, \hat{\mathbf{S}}^{-1}) = n \bar{\mathbf{g}}_n(\mathbf{b})' \hat{\mathbf{S}}^{-1} \bar{\mathbf{g}}_n(\mathbf{b}) = 0.$$

(With $\mathbf{z}_i = \mathbf{x}_i$, $\mathbf{S}_{xz} = \mathbf{S}_{xx}$.) The LR statistic (3.7.2) is therefore

$$LR = J(\bar{\delta}, \hat{\mathbf{S}}^{-1}) - J(\mathbf{b}, \hat{\mathbf{S}}^{-1}) = n (\mathbf{s}_{xy} - \mathbf{S}_{xx}\bar{\delta})' \hat{\mathbf{S}}^{-1} (\mathbf{s}_{xy} - \mathbf{S}_{xx}\bar{\delta}),$$

since $\bar{\mathbf{g}}_n(\bar{\delta}) = \mathbf{s}_{xy} - \mathbf{S}_{xx}\bar{\delta}$ by (3.4.2). So in the regression model the LR statistic measures, in the metric $\hat{\mathbf{S}}^{-1}$, the extent to which the restricted estimate fails to set the sample moments to zero. As the text notes, under conditional homoskedasticity $\hat{\mathbf{S}} = \hat{\sigma}^2 \mathbf{S}_{xx}$ this collapses to $(SSR_R - SSR_U)/\hat{\sigma}^2$: writing $\bar{\mathbf{e}}$ for the restricted residuals, $\mathbf{s}_{xy} - \mathbf{S}_{xx}\bar{\delta} = \mathbf{X}'\bar{\mathbf{e}}/n$, so

$$LR = \frac{\bar{\mathbf{e}}' \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\bar{\mathbf{e}}}{\hat{\sigma}^2} = \frac{\bar{\mathbf{e}}'\bar{\mathbf{e}} - \mathbf{e}'\mathbf{e}}{\hat{\sigma}^2},$$

the last equality because $\bar{\mathbf{e}}' \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\bar{\mathbf{e}} = \bar{\mathbf{e}}' P \bar{\mathbf{e}}$ and $P\bar{\mathbf{e}} = P\mathbf{y} - P\mathbf{X}\bar{\delta} = (\mathbf{y} - \mathbf{e}) - (\mathbf{X}\bar{\delta}) = \bar{\mathbf{e}} - \mathbf{e}$ with $\mathbf{e}'\bar{\mathbf{e}} = \mathbf{e}'\mathbf{e}$ (since $\mathbf{e} \perp \mathbf{X}$). This is the familiar “difference in SSR, normalized” form, recovering the derivation of Section 2.6.

Review Question 3.7.2

Yes on both counts. The restricted equation (3.7.4) is $y_i = \mathbf{z}_i'\gamma + \varepsilon_i$ with $\mathbf{z}_i = \mathbf{x}_{i1}$, and its residual $\hat{\varepsilon}_i$ is computed from the restricted estimate of γ . Under the null $H_0: \boldsymbol{\alpha} = \mathbf{0}$ in the augmented equation (3.7.5), the restricted specification is correct, the restricted estimator of γ is consistent, and hence — by PROPOSITION 3.4 — the $\hat{\mathbf{S}}$ built from the restricted residuals is consistent for $\mathbf{S}$. The three statistics formed with any single consistent $\hat{\mathbf{S}}$ throughout are numerically equal by the

argument in the text (the variable addition test's W equals the LR difference of J 's, which equals C because $J_1 = 0$ under exact identification of the unrestricted equation (3.7.5) with $\mathbf{z}_i = \mathbf{x}_{i1}$), and each is asymptotically $\chi^2(K - K_1) = \chi^2(\dim \mathbf{x}_{i2})$ under the null, by PROPOSITIONS 3.6–3.8.

Two qualifications matter in practice. First, the numerical equalities require the *same* $\hat{\mathbf{S}}$ in every component; mixing a restricted-residual $\hat{\mathbf{S}}$ in one place with an unrestricted one elsewhere breaks the exact equalities (they remain asymptotically equivalent). Second, under the *alternative* the restricted-residual $\hat{\mathbf{S}}$ is not consistent for $\mathbf{S}$ (the restricted residuals are contaminated by the omitted $\mathbf{x}'_{i2}\boldsymbol{\alpha}$), so power properties of tests computed with it differ from those using unrestricted residuals — the standard advice to use unrestricted residuals for estimating $\mathbf{S}$ is about behavior under the alternative, not validity under the null.

Section 3.8 Implications of Conditional Homoskedasticity

Review Question 3.8.1

Under ASSUMPTION 3.7, $\mathbf{S} = E(\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i') = E[E(\varepsilon_i^2 | \mathbf{x}_i) \mathbf{x}_i \mathbf{x}_i'] = \sigma^2 \Sigma_{\mathbf{xx}}$, and PROPOSITION 3.4 gives $\hat{\mathbf{S}} \xrightarrow{p} \sigma^2 \Sigma_{\mathbf{xx}}$, as the hint says. The two-step estimator is $\hat{\delta}(\hat{\mathbf{S}}^{-1})$, so by PROPOSITION 3.1(B) with $\mathbf{W} = \text{plim } \hat{\mathbf{S}}^{-1} = (\sigma^2 \Sigma_{\mathbf{xx}})^{-1}$,

$$\text{Avar}(\hat{\delta}(\hat{\mathbf{S}}^{-1})) = (\Sigma'_{\mathbf{xz}} \mathbf{S}^{-1} \Sigma_{\mathbf{xz}})^{-1} = \sigma^2 (\Sigma'_{\mathbf{xz}} \Sigma_{\mathbf{xx}}^{-1} \Sigma_{\mathbf{xz}})^{-1},$$

which is exactly (3.8.4), the asymptotic variance of 2SLS. So the two-step efficient GMM estimator and 2SLS are asymptotically equivalent under conditional homoskedasticity (in fact $\sqrt{n}$ -equivalent, by Review Question 3.5.3, since their weighting matrices have the same probability limit), even though they differ numerically in finite samples: 2SLS uses $\hat{\mathbf{W}} = \mathbf{S}_{\mathbf{xx}}^{-1}$ — equivalently the limit of $\hat{\mathbf{S}}^{-1}$ without ever computing fourth moments — while two-step GMM estimates $\mathbf{S}$ from the first-step residuals. The moral is that under conditional homoskedasticity nothing is gained asymptotically by estimating fourth moments; this is why 2SLS dominated applied practice long before heteroskedasticity-robust GMM became routine, and why the fourth-moment ASSUMPTION 3.6 is not needed for the consistency of $\hat{\mathbf{S}} = \hat{\sigma}^2 \mathbf{S}_{\mathbf{xx}}$.

Review Question 3.8.2

Consistency: *yes.* 2SLS is GMM with $\hat{\mathbf{W}} = \mathbf{S}_{\mathbf{xx}}^{-1}$, and by ergodic stationarity $\mathbf{S}_{\mathbf{xx}}^{-1} \xrightarrow{p} \Sigma_{\mathbf{xx}}^{-1}$, which is symmetric and positive definite (under ASSUMPTION 3.5 with $\mathbf{z}_i$ replaced: $\Sigma_{\mathbf{xx}}$ is nonsingular since $\mathbf{S}$ is). Nothing in PROPOSITION 3.1(A) requires conditional homoskedasticity, so $\hat{\delta}_{2\text{SLS}} \xrightarrow{p} \delta$ under ASSUMPTIONS 3.1–3.4 alone.

Plim of (3.8.2). Without ASSUMPTION 3.7, $E(\varepsilon_i^2 | \mathbf{x}_i)$ is a nonconstant function of $\mathbf{x}_i$, so

$$\hat{\mathbf{S}} \equiv \hat{\sigma}^2 \mathbf{S}_{\mathbf{xx}} \xrightarrow{p} \sigma^2 \Sigma_{\mathbf{xx}} \neq \mathbf{S} = E(\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i')$$

in general: (3.8.2) is no longer consistent for $\mathbf{S}$ (although under conditional heteroskedasticity each element picks up a weighted average of the conditional variances).

Asymptotic variance. By (3.5.1) with $\mathbf{W} = \Sigma_{\mathbf{xx}}^{-1}$,

$$\text{Avar}(\hat{\delta}_{2\text{SLS}}) = (\Sigma'_{\mathbf{xz}} \Sigma_{\mathbf{xx}}^{-1} \Sigma_{\mathbf{xz}})^{-1} \Sigma'_{\mathbf{xz}} \Sigma_{\mathbf{xx}}^{-1} \mathbf{S} \Sigma_{\mathbf{xx}}^{-1} \Sigma_{\mathbf{xz}} (\Sigma'_{\mathbf{xz}} \Sigma_{\mathbf{xx}}^{-1} \Sigma_{\mathbf{xz}})^{-1},$$

which collapses to (3.8.4) only when $\mathbf{S} = \sigma^2 \Sigma_{\mathbf{xx}}$, i.e. under conditional homoskedasticity.

Efficiency: no. Without conditional homoskedasticity the optimal weighting matrix is $\mathbf{S}^{-1} \neq (\sigma^2 \Sigma_{\mathbf{xx}})^{-1}$, so by PROPOSITION 3.5

$$\text{Avar}(\widehat{\delta}_{2\text{SLS}}) \geq (\Sigma'_{\mathbf{xz}} \mathbf{S}^{-1} \Sigma_{\mathbf{xz}})^{-1} = \text{Avar}(\widehat{\delta}(\widehat{\mathbf{S}}^{-1})),$$

with strict inequality in general. 2SLS keeps consistency but surrenders efficiency; the two-step GMM estimator with the White-type $\widehat{\mathbf{S}}$ (3.5.10) is strictly more efficient whenever $E(\varepsilon_i^2 \mid \mathbf{x}_i)$ genuinely varies with $\mathbf{x}_i$.

Review Question 3.8.3

Consider the artificial regression of $\mathbf{s}_{\mathbf{xy}}$ on $\mathbf{S}_{\mathbf{xz}}$,

$$\mathbf{s}_{\mathbf{xy}} = \mathbf{S}_{\mathbf{xz}} \delta + \mathbf{u}, \quad \text{Var}(\mathbf{u}) \propto \mathbf{S}_{\mathbf{xx}},$$

where the K “observations” are the sample moments and the proportionality of the error covariance to $\mathbf{S}_{\mathbf{xx}}$ is taken as given (this is the GLS interpretation: under conditional homoskedasticity $\text{Var}(\sqrt{n} \bar{\mathbf{g}}) \rightarrow \mathbf{S} = \sigma^2 \Sigma_{\mathbf{xx}}$, so $\mathbf{S}_{\mathbf{xx}}$ plays the role of the error covariance matrix up to the scale σ^2 , which is immaterial for the GLS point estimate). The GLS estimator of this artificial regression is

$$\widehat{\delta}_{\text{GLS}} = (\mathbf{S}'_{\mathbf{xz}} \mathbf{S}_{\mathbf{xx}}^{-1} \mathbf{S}_{\mathbf{xz}})^{-1} \mathbf{S}'_{\mathbf{xz}} \mathbf{S}_{\mathbf{xx}}^{-1} \mathbf{s}_{\mathbf{xy}},$$

which is (3.8.3), the 2SLS estimator. This is the conditional-homoskedasticity special case of Review Question 3.5.7, where the general efficient GMM was the FGLS estimator of the same artificial regression with error covariance $\mathbf{S}$: GLS with the *correct* covariance matrix gives efficient GMM, GLS with $\mathbf{S}_{\mathbf{xx}}$ gives 2SLS, and the two coincide exactly when $\mathbf{S} \propto \Sigma_{\mathbf{xx}}$ — conditional homoskedasticity again. The analogy also explains the name “two-stage”: GLS of moments on moments is OLS of moments on *transformed* moments, $\mathbf{S}_{\mathbf{xx}}^{-1/2} \mathbf{s}_{\mathbf{xy}}$ on $\mathbf{S}_{\mathbf{xx}}^{-1/2} \mathbf{S}_{\mathbf{xz}}$, and the transformed moments are exactly the fitted values of the first-stage regressions.

Review Question 3.8.4

By Review Question 3.5.3, the equivalence holds whenever the two weighting matrices have the same probability limit, i.e. whenever $\text{plim } \widehat{\mathbf{S}}^{-1} = \text{plim } \mathbf{S}_{\mathbf{xx}}^{-1}$ up to a scalar, and also whenever the weighting matrix is irrelevant altogether.

Case 1: conditional homoskedasticity. Under ASSUMPTION 3.7, $\mathbf{S} = \sigma^2 \Sigma_{\mathbf{xx}}$, so $\widehat{\mathbf{S}}^{-1} \xrightarrow{p} (\sigma^2 \Sigma_{\mathbf{xx}})^{-1} = \sigma^{-2} \Sigma_{\mathbf{xx}}^{-1}$ while $\mathbf{S}_{\mathbf{xx}}^{-1} \xrightarrow{p} \Sigma_{\mathbf{xx}}^{-1}$. The scalar factor σ^{-2} cancels from the GMM formula (3.4.8) (it multiplies both $\mathbf{S}'_{\mathbf{xz}} \widehat{\mathbf{W}} \mathbf{S}_{\mathbf{xz}}$ and $\mathbf{S}'_{\mathbf{xz}} \widehat{\mathbf{W}} \mathbf{s}_{\mathbf{xy}}$), so the effective limiting weighting matrices coincide and $\sqrt{n}(\widehat{\delta}(\widehat{\mathbf{S}}^{-1}) - \widehat{\delta}_{2\text{SLS}}) \xrightarrow{p} \mathbf{0}$: this is the content of PROPOSITION 3.9(A) — under conditional homoskedasticity 2SLS is the efficient GMM estimator, asymptotically.

Case 2: exact identification. When $K = L$, $\mathbf{S}_{\mathbf{xz}}$ is square and invertible for large n , and every GMM estimator collapses to the same IV estimator:

$$\widehat{\delta}(\widehat{\mathbf{W}}) = (\mathbf{S}'_{\mathbf{xz}} \widehat{\mathbf{W}} \mathbf{S}_{\mathbf{xz}})^{-1} \mathbf{S}'_{\mathbf{xz}} \widehat{\mathbf{W}} \mathbf{s}_{\mathbf{xy}} = \mathbf{S}_{\mathbf{xz}}^{-1} \widehat{\mathbf{W}}^{-1} \mathbf{S}_{\mathbf{xz}}' \mathbf{S}'_{\mathbf{xz}} \widehat{\mathbf{W}} \mathbf{s}_{\mathbf{xy}} = \mathbf{S}_{\mathbf{xz}}^{-1} \mathbf{s}_{\mathbf{xy}}$$

for any nonsingular $\widehat{\mathbf{W}}$, in finite samples. Hence $\widehat{\delta}(\widehat{\mathbf{S}}^{-1}) = \widehat{\delta}_{2\text{SLS}}$ numerically, and a fortiori the $\sqrt{n}$ -scaled difference converges to zero — under any degree of heteroskedasticity. The two cases

are the two ways the choice of weighting matrix stops mattering: when the data confirm that the simple choice is the efficient one, and when there is no choice to be made.

Review Question 3.8.5

Under exact identification $K = L$, so $\mathbf{S}_{xz}$ is square, and by ASSUMPTIONS 3.2 and 3.4 it converges to the nonsingular Σ_{xz} , hence is nonsingular for n sufficiently large with probability one. Then the 2SLS estimator (3.8.3) is

$$\hat{\delta}_{2SLS} = (\mathbf{S}'_{xz} \mathbf{S}_{xx}^{-1} \mathbf{S}_{xz})^{-1} \mathbf{S}'_{xz} \mathbf{S}_{xx}^{-1} \mathbf{s}_{xy}.$$

All three matrices in the inverted factor are nonsingular and square, so

$$(\mathbf{S}'_{xz} \mathbf{S}_{xx}^{-1} \mathbf{S}_{xz})^{-1} = \mathbf{S}_{xz}^{-1} \mathbf{S}_{xx} \mathbf{S}_{xz}'^{-1},$$

and substituting,

$$\hat{\delta}_{2SLS} = \mathbf{S}_{xz}^{-1} \mathbf{S}_{xx} \mathbf{S}_{xz}'^{-1} \mathbf{S}'_{xz} \mathbf{S}_{xx}^{-1} \mathbf{s}_{xy} = \mathbf{S}_{xz}^{-1} \mathbf{s}_{xy},$$

which is the IV estimator (3.4.4). This is the conditional-homoskedasticity shadow of the general fact (Review Question 3.4.2) that the weighting matrix is irrelevant under exact identification: with as many moment conditions as parameters, all of them are set exactly to zero, and there is nothing left for a weighting matrix to do.

Review Question 3.8.6

Sargan's statistic (3.8.10') is

$$J = \frac{\hat{\epsilon}' \mathbf{P} \hat{\epsilon}}{\hat{\sigma}^2}, \quad \mathbf{P} = \mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}', \quad \hat{\sigma}^2 = \frac{\hat{\epsilon}' \hat{\epsilon}}{n},$$

with $\hat{\epsilon} = \mathbf{y} - \mathbf{Z} \hat{\delta}_{2SLS}$ the 2SLS residual. Regressing $\hat{\epsilon}$ on $\mathbf{X}$ (an auxiliary regression with $\mathbf{X}$ as the regressor matrix), the explained sum of squares is $\hat{\epsilon}' \mathbf{P} \hat{\epsilon}$ and the total sum of squares (about zero) is $\hat{\epsilon}' \hat{\epsilon}$. Review Question 1.2.8 defines the uncentered R -squared as the ratio of the uncentered explained to total sums of squares,

$$R_{uc}^2 = \frac{\hat{\epsilon}' \mathbf{P} \hat{\epsilon}}{\hat{\epsilon}' \hat{\epsilon}},$$

whenever the regression includes the regressor matrix $\mathbf{X}$ (no constant is needed for the *uncentered* version). Therefore

$$nR_{uc}^2 = n \frac{\hat{\epsilon}' \mathbf{P} \hat{\epsilon}}{\hat{\epsilon}' \hat{\epsilon}} = \frac{\hat{\epsilon}' \mathbf{P} \hat{\epsilon}}{\hat{\sigma}^2} = J.$$

So Sargan's test can be run with any regression package: regress the 2SLS residuals on *all* the instruments and read off n times the uncentered R^2 (if the package reports only the centered R^2 , use it only when the mean of $\hat{\epsilon}$ is zero — which holds automatically when a constant is among the instruments). The intuition is transparent: the overidentifying restrictions say the residuals should be orthogonal to every instrument in the population, so a regression of residuals on instruments should have no explanatory power; nR^2 measures how much they nevertheless have, and under the null that quantity is $\chi^2(K - L)$.

Review Question 3.8.7

No. First, what was shown in the text: under conditional homoskedasticity, when all regressors are predetermined ($\mathbf{z}_i \subset \mathbf{x}_i$), the efficient GMM estimator — 2SLS — equals OLS, because in the 2SLS-as-two-regressions construction the first-stage fitted values of the regressors *are* the regressors ($\mathbf{PZ} = \mathbf{Z}$ when every column of $\mathbf{Z}$ is a column of $\mathbf{X}$), so the second stage is the original regression. Equivalently, $\mathbf{PZ} = \mathbf{Z}$ makes (3.8.3') collapse to $(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}$.

Without conditional homoskedasticity, however, 2SLS is no longer the efficient GMM estimator (Review Question 3.8.2): the efficient weighting matrix is $\widehat{\mathbf{S}}^{-1}$ with $\widehat{\mathbf{S}}$ of (3.5.10), and the efficient GMM estimator is

$$\widehat{\delta}(\widehat{\mathbf{S}}^{-1}) = (\mathbf{S}'_{xz}\widehat{\mathbf{S}}^{-1}\mathbf{S}_{xz})^{-1}\mathbf{S}'_{xz}\widehat{\mathbf{S}}^{-1}\mathbf{s}_{xy},$$

which — as Review Question 3.5.5 showed — is *not* numerically equal to OLS when $K > L$. The efficient GMM estimator exploits the extra orthogonality conditions from the excluded instruments, weighting them by their contribution to precision about δ ; under heteroskedasticity those weights interact with the error variances in a way that generally moves the point estimate away from OLS. What survives is consistency of OLS itself: with all regressors predetermined, OLS is consistent under ASSUMPTIONS 3.1–3.4 regardless of heteroskedasticity — it is just not efficient, and ANALYTICAL EXERCISE 3.4 shows its asymptotic variance is never smaller than the efficient GMM one. So: under conditional homoskedasticity the extra instruments add nothing and OLS is efficient; under heteroskedasticity they add efficiency, and using them moves the estimate.

Section 3.9 Application: Returns from Schooling

Review Question 3.9.1

Collecting the assumptions of Section 3.9 (with the individual subscript dropped):

On the error ε of the wage equation (3.9.1).

- $E(\varepsilon) = 0$ (normalization; the mean is absorbed in the intercept α);
- $E(S\varepsilon) = 0$, $E(\mathbf{h}\varepsilon) = \mathbf{0}$, and $E(A\varepsilon) = 0$: schooling, the controls, and true ability are all uncorrelated with the wage error — i.e. $(S, A, \mathbf{h})$ are predetermined. This is what would make OLS of LW on $(1, S, A, \mathbf{h})$ consistent.

On the measurement error η in $IQ = \phi + A + \eta$ (3.9.3).

- $E(\eta) = 0$ (normalization);
- $\text{Cov}(\eta, A) = 0$: classical measurement error, uncorrelated with the true regressor;
- $\text{Cov}(\eta, S) = 0$, $\text{Cov}(\eta, \mathbf{h}) = \mathbf{0}$: the measurement error is uncorrelated with the other regressors;
- $\text{Cov}(\eta, \varepsilon) = 0$: the measurement error is uncorrelated with the wage error. This is what makes $E^*(\varepsilon \mid 1, S, IQ, \mathbf{h}) = 0$ in (3.9.5) and confines the bias of the OLS-with- IQ regression to the term $-\gamma E^*(\eta \mid 1, S, IQ, \mathbf{h})$.

On the instruments. The instrument set is $(1, S, \mathbf{h}, \text{age}, \text{age}^2, KWW, MED, \text{other background})$; the maintained assumptions are that each instrument x satisfies

- $\text{Cov}(x, \varepsilon) = 0$ and $\text{Cov}(x, \eta) = 0$ (predeterminedness: e.g. *KWW*, *MED*, age are uncorrelated with both the wage error and the measurement error), and
- the rank condition: the instruments are correlated with *IQ* (and with the included regressors) enough that $E(\mathbf{x}_i \mathbf{z}_i')$ is of full column rank — *KWW* and *MED* must genuinely predict *IQ*.

Which are crucial for 2SLS consistency? By PROPOSITION 3.1(A), 2SLS is consistent if and only if the instrument orthogonality conditions and the rank condition hold: the crucial assumptions are $\text{Cov}(x, \varepsilon) = 0$ and $\text{Cov}(x, \eta) = 0$ for every instrument x — i.e. that age, *KWW*, *MED* and the other excluded variables are uncorrelated with the composite error $\varepsilon - \gamma\eta$ of (3.9.4) — together with instrument relevance. The assumptions about the *means* are pure normalizations, and the classical-measurement-error structure ($\text{Cov}(\eta, A) = \text{Cov}(\eta, S) = 0$) is needed for the *interpretation* of the biases (the sign predictions in (3.9.7)) and for the claim that OLS with *IQ* is inconsistent, but not for the consistency of 2SLS itself. Conversely, as the text stresses, the crucial instrument assumptions are exactly the debatable ones: diligence-like characteristics could drive both *KWW* and wages, and heritable maternal traits could drive both *MED* and wages — either failure makes 2SLS inconsistent, and that is what Sargan's statistic (computed in the empirical exercise) assesses.

Review Question 3.9.2

Yes. Substitute the new measurement equation into the wage equation (3.9.1), $LW = \alpha + \beta S + \gamma A + \delta' \mathbf{h} + \varepsilon$:

$$LW = \alpha + \beta S + \gamma[\gamma(IQ - \phi - \eta)] + \delta' \mathbf{h} + \varepsilon = (\alpha - \gamma^2 \phi) + \beta S + \gamma^2 IQ + \delta' \mathbf{h} + (\varepsilon - \gamma^2 \eta).$$

This is exactly the same regression form as (3.9.4): regressing LW on a constant, S , IQ and $\mathbf{h}$, the coefficient on IQ is now γ^2 instead of γ , the composite error is $\varepsilon - \gamma^2 \eta$ instead of $\varepsilon - \gamma\eta$, but the coefficient on S is still β . The orthogonality assumptions maintained for 2SLS — instruments uncorrelated with ε and with η — imply instruments uncorrelated with $\varepsilon - \gamma^2 \eta$ just as with $\varepsilon - \gamma\eta$ (only the scale changed), and the rank condition is unaffected. Hence 2SLS of the wage equation with *IQ* instrumented is consistent for β , for exactly the reasons of PROPOSITION 3.1(A).

The general point is that β is identified under any *linear* re-scaling of the ability measure: changing A to cA merely rescales the unobservable (the coefficient on *IQ* absorbs the factor), and the identifying content of the instruments — correlation with the noisy measure, orthogonality to the errors — is preserved. What would break consistency is measurement error correlated with S , with ε , or entering nonlinearly; the exact coefficient $1/\gamma$ attaching ability to the test score is immaterial for the schooling coefficient. (Of course the *interpretation* of the *IQ* coefficient changes: it no longer estimates γ .)

Review Question 3.9.3

No — not from those data alone. The test one would like to run is the regression of the wage on schooling *holding ability constant*:

$$LW = \alpha + \beta S + \gamma A + \mathbf{h}'\delta + \varepsilon,$$

and to test $H_0: \beta = 0$ (the Spencian null: conditional on ability, education has no effect). But in the Spencian equilibrium, schooling is a (deterministic, monotone) function of ability: with two ability types there are exactly two education levels, so S takes only two values and is perfectly collinear with A — the regressor matrix $(1, S, A, \mathbf{h})$ is rank-deficient and ASSUMPTION 2.4/ASSUMPTION 3.4 fails. β and γ are not separately identified: any combination (β, γ) with $\beta S_L + \gamma A_L$ and $\beta S_H + \gamma A_H$ matching the two group mean wages fits the data equally well. The wage-schooling correlation is fully explained by ability, but “fully explained” cannot be distinguished from “partially caused” without independent variation.

The way out is the general lesson of the chapter: identification requires variation in schooling that is *not* driven by ability — instruments such as compulsory schooling laws (Angrist–Krueger), tuition changes, or school construction programs, which shift S while plausibly orthogonal to ability; or within-twin variation, which holds genetic and family components of A fixed by design. With such variation the equation is identified and the Spencian hypothesis $\beta = 0$ becomes testable; without it, the signaling and human-capital models are observationally equivalent on cross-sections of wages, ability and schooling.

Problem Set for Chapter 3: Analytical Exercises

Analytical Exercise 3.1

Let $\mathbf{M}$ ($K \times K$) be symmetric and idempotent: $\mathbf{M} = \mathbf{M}'$ and $\mathbf{M}\mathbf{M} = \mathbf{M}$. For any $\mathbf{c} \in \mathbb{R}^K$,

$$\mathbf{c}'\mathbf{M}\mathbf{c} = \mathbf{c}'\mathbf{M}\mathbf{M}\mathbf{c} \quad (\text{idempotency}),$$

and by symmetry $\mathbf{M}\mathbf{c} = (\mathbf{M}'\mathbf{c})$, so writing $\mathbf{d} \equiv \mathbf{M}\mathbf{c}$,

$$\mathbf{c}'\mathbf{M}\mathbf{c} = (\mathbf{M}\mathbf{c})'(\mathbf{M}\mathbf{c}) = \mathbf{d}'\mathbf{d} = \|\mathbf{d}\|^2 \geq 0.$$

Hence $\mathbf{M}$ is positive semidefinite. Two equivalent readings: the eigenvalues of a symmetric idempotent matrix are all 0 or 1 (idempotency forces each eigenvalue λ to satisfy $\lambda^2 = \lambda$), so the quadratic forms are nonnegative; and geometrically $\mathbf{M}$ is an orthogonal projection, so $\mathbf{c}'\mathbf{M}\mathbf{c}$ is the squared length of the projection of $\mathbf{c}$. Note for use in Exercises 3.3 and 3.5–3.7: the rank of a symmetric idempotent matrix equals its trace, $\text{rank}(\mathbf{M}) = \text{tr}(\mathbf{M}) = \sum_k \lambda_k$, since the trace counts the nonzero eigenvalues.

Analytical Exercise 3.2

(a) The sequence $\{x_i^2 \varepsilon_i^2\} = \{g_i^2\}$ is a measurable function of $\mathbf{w}_i$, hence ergodic stationary by ASSUMPTION 3.2. Its mean $E(x_i^2 \varepsilon_i^2) = S$ exists and is finite by hypothesis. The ergodic theorem then gives

$$\frac{1}{n} \sum_{i=1}^n x_i^2 \varepsilon_i^2 \xrightarrow{p} S.$$

(b) Apply Cauchy–Schwarz with $f = x_i \varepsilon_i$ and $h = x_i z_i$:

$$E(|z_i x_i^2 \varepsilon_i|) = E(|x_i \varepsilon_i \cdot x_i z_i|) \leq \sqrt{E(x_i^2 \varepsilon_i^2) E(x_i^2 z_i^2)} < \infty,$$

because $E(x_i^2 \varepsilon_i^2) = S < \infty$ by hypothesis and $E(x_i^2 z_i^2) < \infty$ by ASSUMPTION 3.6 (finite fourth moments, with $K = L = 1$). By the fact quoted, $E(z_i x_i^2 \varepsilon_i)$ exists and is finite. This is exactly where ASSUMPTION 3.6 is used: the fourth moment of the instruments crossed with the regressors.

(c) By (b) and ergodic stationarity, $\frac{1}{n} \sum_i z_i x_i^2 \varepsilon_i \xrightarrow{p} E(z_i x_i^2 \varepsilon_i)$, a finite constant. Since $\hat{\delta}$ is consistent, $\hat{\delta} - \delta \xrightarrow{p} 0$, and by LEMMA 2.3(A) (convergence in probability of products),

$$-2(\hat{\delta} - \delta) \cdot \frac{1}{n} \sum_i z_i x_i^2 \varepsilon_i \xrightarrow{p} -2 \cdot 0 \cdot E(z_i x_i^2 \varepsilon_i) = 0.$$

(d) Similarly, $\frac{1}{n} \sum_i z_i^2 x_i^2 \xrightarrow{p} E(z_i^2 x_i^2) < \infty$ by ASSUMPTION 3.6 directly (no Cauchy–Schwarz step needed, since the term is already a fourth moment), and $(\hat{\delta} - \delta)^2 \xrightarrow{p} 0$ by LEMMA 2.3(A) applied to the square. Hence

$$(\hat{\delta} - \delta)^2 \cdot \frac{1}{n} \sum_i z_i^2 x_i^2 \xrightarrow{p} 0.$$

Combining (a), (c), (d) through LEMMA 2.3(A), $\hat{S} \xrightarrow{p} S + 0 + 0 = S$. The general case ($K, L \geq 1$) is identical element by element: the (k, ℓ) element of $\hat{\mathbf{S}}$ expands into $\frac{1}{n} \sum_i \varepsilon_i^2 x_{ik} x_{i\ell}$ plus cross terms in $(\hat{\delta} - \delta)$ times sample means of $z_{im} x_{ik} x_{i\ell} \varepsilon_i$ and $z_{im} z_{im'} x_{ik} x_{i\ell}$, whose finiteness follows from Cauchy–Schwarz and ASSUMPTION 3.6 exactly as in (b). The structure of the proof is the same as PROPOSITION 2.4 (Analytical Exercise 2.4): the residual differs from the error by terms of order $\hat{\delta} - \delta = o_p(1)$, and the fourth-moment assumption is precisely what keeps those differences from mattering.

Analytical Exercise 3.3

(a) Substitute $\mathbf{S}^{-1} = \mathbf{C}'\mathbf{C}$ and $\mathbf{S} = \mathbf{C}^{-1}\mathbf{C}'^{-1}$ (since $\mathbf{C}'\mathbf{C} = \mathbf{S}^{-1}$ implies $\mathbf{S} = (\mathbf{C}'\mathbf{C})^{-1} = \mathbf{C}^{-1}\mathbf{C}'^{-1}$):

$$\Sigma'_{\mathbf{xz}} \mathbf{S}^{-1} \Sigma_{\mathbf{xz}} = \Sigma'_{\mathbf{xz}} \mathbf{C}'\mathbf{C} \Sigma_{\mathbf{xz}}.$$

For the second term, $\mathbf{WSW} = \mathbf{WC}^{-1}\mathbf{C}'^{-1}\mathbf{W}$, so with $\mathbf{H} = \mathbf{C}'^{-1}\mathbf{W}\Sigma_{\mathbf{xz}}$ (hence $\mathbf{H}' = \Sigma'_{\mathbf{xz}}\mathbf{WC}^{-1}$),

$$\Sigma'_{\mathbf{xz}} \mathbf{WSW} \Sigma_{\mathbf{xz}} = \mathbf{H}'\mathbf{H}, \quad \Sigma'_{\mathbf{xz}} \mathbf{W} \Sigma_{\mathbf{xz}} = \Sigma'_{\mathbf{xz}} \mathbf{WC}^{-1}\mathbf{C} \Sigma_{\mathbf{xz}} = \mathbf{H}'\mathbf{C} \Sigma_{\mathbf{xz}}.$$

Therefore

$$\begin{aligned} & \Sigma'_{\mathbf{xz}} \mathbf{W} \Sigma_{\mathbf{xz}} (\Sigma'_{\mathbf{xz}} \mathbf{WSW} \Sigma_{\mathbf{xz}})^{-1} \Sigma'_{\mathbf{xz}} \mathbf{W} \Sigma_{\mathbf{xz}} \\ &= (\mathbf{H}'\mathbf{C} \Sigma_{\mathbf{xz}})(\mathbf{H}'\mathbf{H})^{-1}(\mathbf{H}'\mathbf{C} \Sigma_{\mathbf{xz}})' = \Sigma'_{\mathbf{xz}} \mathbf{C}'\mathbf{H}(\mathbf{H}'\mathbf{H})^{-1}\mathbf{H}'\mathbf{C} \Sigma_{\mathbf{xz}}, \end{aligned}$$

and subtracting,

$$\mathbf{Q} = \Sigma'_{\mathbf{xz}} \mathbf{C}' [\mathbf{I}_K - \mathbf{H}(\mathbf{H}'\mathbf{H})^{-1}\mathbf{H}'] \mathbf{C} \Sigma_{\mathbf{xz}} = \Sigma'_{\mathbf{xz}} \mathbf{C}' \mathbf{M} \mathbf{C} \Sigma_{\mathbf{xz}},$$

as claimed. ($\mathbf{H}'\mathbf{H}$ is invertible: $\mathbf{H}$ ($K \times L$) has full column rank L because $\mathbf{C}$ is nonsingular, $\mathbf{W}$ is positive definite, and $\Sigma_{\mathbf{xz}}$ has full column rank by ASSUMPTION 3.4.)

(b) $\mathbf{M}$ is symmetric,

$$\mathbf{M}' = \mathbf{I}_K - \mathbf{H}(\mathbf{H}'\mathbf{H})^{-1}\mathbf{H}' = \mathbf{M},$$

and idempotent,

$$\mathbf{M}\mathbf{M} = \mathbf{I}_K - 2\mathbf{H}(\mathbf{H}'\mathbf{H})^{-1}\mathbf{H}' + \mathbf{H}(\mathbf{H}'\mathbf{H})^{-1}\mathbf{H}'\mathbf{H}(\mathbf{H}'\mathbf{H})^{-1}\mathbf{H}' = \mathbf{I}_K - \mathbf{H}(\mathbf{H}'\mathbf{H})^{-1}\mathbf{H}' = \mathbf{M},$$

since $\mathbf{H}'\mathbf{H}(\mathbf{H}'\mathbf{H})^{-1} = \mathbf{I}_L$. By Exercise 3.1, $\mathbf{M}$ is positive semidefinite. Then for any $\mathbf{c} \in \mathbb{R}^L$,

$$\mathbf{c}'\mathbf{Q}\mathbf{c} = (\mathbf{C}\Sigma_{\mathbf{x}z}\mathbf{c})'\mathbf{M}(\mathbf{C}\Sigma_{\mathbf{x}z}\mathbf{c}) \geq 0,$$

so $\mathbf{Q}$ is positive semidefinite. By the quoted matrix-algebra result, $\mathbf{A} - \mathbf{B} \succeq \mathbf{0}$, i.e.

$$\text{Avar}(\widehat{\delta}(\widehat{\mathbf{W}})) \geq (\Sigma'_{\mathbf{x}z}\mathbf{S}^{-1}\Sigma_{\mathbf{x}z})^{-1},$$

with equality when $\mathbf{W} \propto \mathbf{S}^{-1}$ (then $\mathbf{H} = \mathbf{C}'^{-1}\mathbf{C}'\mathbf{C}\Sigma_{\mathbf{x}z} \propto \mathbf{C}\Sigma_{\mathbf{x}z}$ spans the same space as $\mathbf{C}\Sigma_{\mathbf{x}z}$, so $\mathbf{M}\mathbf{C}\Sigma_{\mathbf{x}z} = \mathbf{0}$ and $\mathbf{Q} = \mathbf{0}$). The geometry is the Gauss–Markov one from Review Question 3.5.7: GMM is GLS on the K sample moments, $\mathbf{M}$ is the annihilator of the transformed regressor space, and the efficiency gap is the projection of the efficient-score direction onto that annihilator.

Analytical Exercise 3.4

The efficient two-step GMM estimator is never less efficient: its asymptotic variance $(\Sigma'_{\mathbf{x}z}\mathbf{S}^{-1}\Sigma_{\mathbf{x}z})^{-1}$ is no larger than that of OLS in the positive semidefinite sense. The proof writes OLS as a GMM estimator with the (singular) weighting matrix of the hint and applies the efficiency bound.

With $\mathbf{x}_i = (\mathbf{z}'_i, \mathbf{w}'_i)'$, the cross-moment matrix is $\Sigma_{\mathbf{x}z} = \text{E}(\mathbf{x}_i\mathbf{z}'_i) = [\Sigma'_{\mathbf{zz}} \vdots \Sigma'_{\mathbf{wz}}]'$. For the OLS estimator, $\mathbf{b} - \delta = \mathbf{S}_{\mathbf{zz}}^{-1}\bar{\mathbf{g}}_{\mathbf{z}}$ where $\bar{\mathbf{g}}_{\mathbf{z}} = \frac{1}{n} \sum_i \mathbf{z}_i \varepsilon_i$ collects the first L elements of $\bar{\mathbf{g}}$, so with $\mathbf{W}$ as in the hint,

$$\Sigma'_{\mathbf{x}z}\mathbf{W}\Sigma_{\mathbf{x}z} = \Sigma_{\mathbf{zz}}\mathbf{S}_{\mathbf{zz}}^{-1}\Sigma_{\mathbf{zz}}, \quad \Sigma'_{\mathbf{x}z}\mathbf{W}\mathbf{S}\mathbf{W}\Sigma_{\mathbf{x}z} = \Sigma_{\mathbf{zz}}\mathbf{S}_{\mathbf{zz}}^{-1}\mathbf{S}_{\mathbf{zz}}\mathbf{S}_{\mathbf{zz}}^{-1}\Sigma_{\mathbf{zz}} = \Sigma_{\mathbf{zz}}\mathbf{S}_{\mathbf{zz}}^{-1}\Sigma_{\mathbf{zz}}.$$

(These are the book’s-answer identities $\mathbf{W}\mathbf{S}\mathbf{W} = \mathbf{W}$ and $\Sigma'_{\mathbf{x}z}\mathbf{W}\Sigma_{\mathbf{x}z} = \Sigma_{\mathbf{zz}}\mathbf{S}_{\mathbf{zz}}^{-1}\Sigma_{\mathbf{zz}}$.) Substituting into (3.5.1),

$$\text{Avar}(\widehat{\delta}(\mathbf{W})) = (\Sigma_{\mathbf{zz}}\mathbf{S}_{\mathbf{zz}}^{-1}\Sigma_{\mathbf{zz}})^{-1}\Sigma_{\mathbf{zz}}\mathbf{S}_{\mathbf{zz}}^{-1}\Sigma_{\mathbf{zz}}(\Sigma_{\mathbf{zz}}\mathbf{S}_{\mathbf{zz}}^{-1}\Sigma_{\mathbf{zz}})^{-1} = \Sigma_{\mathbf{zz}}^{-1}\mathbf{S}_{\mathbf{zz}}\Sigma_{\mathbf{zz}}^{-1},$$

which is exactly $\text{Avar}(\mathbf{b})$, the OLS asymptotic variance (2.3.4) with $\mathbf{S}_{\mathbf{zz}} = \text{E}(\varepsilon_i^2 \mathbf{z}_i \mathbf{z}'_i)$.

Now re-run the proof of Exercise 3.3 with this $\mathbf{W}$: nothing there used positive definiteness of $\mathbf{W}$. The factorization $\mathbf{Q} = \Sigma'_{\mathbf{x}z}\mathbf{C}'\mathbf{M}\mathbf{C}\Sigma_{\mathbf{x}z}$ uses only $\mathbf{S}$ positive definite (for $\mathbf{C}$) and the nonsingularity of $\Sigma'_{\mathbf{x}z}\mathbf{W}\mathbf{S}\mathbf{W}\Sigma_{\mathbf{x}z}$ and $\Sigma'_{\mathbf{x}z}\mathbf{W}\Sigma_{\mathbf{x}z}$ (for the inverses); here the latter equals $\Sigma_{\mathbf{zz}}\mathbf{S}_{\mathbf{zz}}^{-1}\Sigma_{\mathbf{zz}}$, nonsingular because $\Sigma_{\mathbf{zz}}$ is nonsingular (ASSUMPTION 2.4 content, implied by the model since $\mathbf{z}_i \subset \mathbf{x}_i$ and $\Sigma_{\mathbf{xx}}$ is nonsingular) and $\mathbf{S}_{\mathbf{zz}}$ is positive definite (a principal submatrix of $\mathbf{S}$, positive definite by ASSUMPTION 3.5). Hence $\mathbf{M}$ is still symmetric idempotent, $\mathbf{Q} \succeq \mathbf{0}$, and the matrix-algebra lemma (extended to the present nonsingular submatrices) gives

$$\text{Avar}(\mathbf{b}) \succeq (\Sigma'_{\mathbf{x}z}\mathbf{S}^{-1}\Sigma_{\mathbf{x}z})^{-1} = \text{Avar}(\widehat{\delta}(\widehat{\mathbf{S}}^{-1})).$$

So the efficient two-step GMM estimator is asymptotically at least as efficient as OLS — strictly more efficient in general when the extra instruments are correlated with the regressors and the errors are heteroskedastic. This complements Review Questions 3.5.5 and 3.8.7: OLS is consistent here, but throws away orthogonality conditions that efficient GMM exploits. (Under conditional homoskedasticity the two coincide only when the extra instruments are redundant for the efficient linear combination — e.g. in special balanced cases — even though 2SLS, which is then efficient, equals OLS numerically.)

Analytical Exercise 3.5

(a) By (3.4.2) and the GMM formula (3.4.8) with $\widehat{\mathbf{W}} = \widehat{\mathbf{S}}^{-1}$,

$$\begin{aligned}\bar{\mathbf{g}}_n(\widehat{\delta}(\widehat{\mathbf{S}}^{-1})) &= \mathbf{s}_{xy} - \mathbf{S}_{xz}\widehat{\delta}(\widehat{\mathbf{S}}^{-1}) \\ &= \mathbf{s}_{xy} - \mathbf{S}_{xz}(\mathbf{S}'_{xz}\widehat{\mathbf{S}}^{-1}\mathbf{S}_{xz})^{-1}\mathbf{S}'_{xz}\widehat{\mathbf{S}}^{-1}\mathbf{s}_{xy} = \widehat{\mathbf{B}}\mathbf{s}_{xy},\end{aligned}$$

with $\widehat{\mathbf{B}}$ as defined. But $\mathbf{s}_{xy} = \mathbf{S}_{xz}\delta + \bar{\mathbf{g}}$ from (3.4.9), and $\widehat{\mathbf{B}}\mathbf{S}_{xz} = \mathbf{S}_{xz} - \mathbf{S}_{xz} = \mathbf{0}$, so

$$\bar{\mathbf{g}}_n(\widehat{\delta}(\widehat{\mathbf{S}}^{-1})) = \widehat{\mathbf{B}}\bar{\mathbf{g}}.$$

(b) Compute, using $\widehat{\mathbf{S}}^{-1} = \mathbf{C}'\mathbf{C}$:

$$\widehat{\mathbf{B}}'\widehat{\mathbf{S}}^{-1}\widehat{\mathbf{B}} = \left[\mathbf{I}_K - \widehat{\mathbf{S}}^{-1}\mathbf{S}_{xz}(\mathbf{S}'_{xz}\widehat{\mathbf{S}}^{-1}\mathbf{S}_{xz})^{-1}\mathbf{S}'_{xz}\right]\mathbf{C}'\mathbf{C}\left[\mathbf{I}_K - \mathbf{S}_{xz}(\mathbf{S}'_{xz}\widehat{\mathbf{S}}^{-1}\mathbf{S}_{xz})^{-1}\mathbf{S}'_{xz}\widehat{\mathbf{S}}^{-1}\right].$$

With $\mathbf{A} = \mathbf{C}\mathbf{S}_{xz}$ we have $\mathbf{S}'_{xz}\widehat{\mathbf{S}}^{-1}\mathbf{S}_{xz} = \mathbf{A}'\mathbf{A}$ and $\widehat{\mathbf{S}}^{-1}\mathbf{S}_{xz} = \mathbf{C}'\mathbf{A}$. Expanding the product, all cross terms collapse:

$$\begin{aligned}\mathbf{C}'\mathbf{C} - \mathbf{C}'\mathbf{A}(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'\mathbf{C} - \mathbf{C}'\mathbf{A}(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'\mathbf{C} + \mathbf{C}'\mathbf{A}(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'\mathbf{A}(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'\mathbf{C} \\ = \mathbf{C}'[\mathbf{I}_K - \mathbf{A}(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}']\mathbf{C} = \mathbf{C}'\mathbf{M}\mathbf{C}.\end{aligned}$$

$\mathbf{M}$ is the annihilator of the column space of $\mathbf{A}$: symmetric and idempotent (as in Exercise 3.3(b)), with $\mathbf{M}\mathbf{A} = \mathbf{0}$. Its rank equals its trace:

$$\text{rank}(\mathbf{M}) = \text{tr}(\mathbf{M}) = K - \text{tr}[\mathbf{A}(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'] = K - \text{tr}[(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'\mathbf{A}] = K - \text{tr}(\mathbf{I}_L) = K - L,$$

where $\mathbf{A}$ has full column rank L because $\mathbf{C}$ is nonsingular and $\mathbf{S}_{xz}$ has full column rank for large n under ASSUMPTIONS 3.2 and 3.4.

(c) By ASSUMPTION 3.5 and the ergodic stationary martingale differences CLT, $\sqrt{n}\bar{\mathbf{g}} \xrightarrow{d} N(\mathbf{0}, \mathbf{S})$. Since $\widehat{\mathbf{S}} \xrightarrow{p} \mathbf{S}$, $\mathbf{C}'\mathbf{C} = \widehat{\mathbf{S}}^{-1} \xrightarrow{p} \mathbf{S}^{-1}$, so by LEMMA 2.4(c),

$$\mathbf{v} = \mathbf{C}\sqrt{n}\bar{\mathbf{g}} \xrightarrow{d} N(\mathbf{0}, (\text{plim } \mathbf{C})\mathbf{S}(\text{plim } \mathbf{C})') = N(\mathbf{0}, \mathbf{I}_K),$$

because any limit $\mathbf{C}_0$ of $\mathbf{C}$ satisfies $\mathbf{C}'_0\mathbf{C}_0 = \mathbf{S}^{-1}$, i.e. $\mathbf{C}_0\mathbf{S}\mathbf{C}'_0 = \mathbf{C}_0(\mathbf{C}'_0\mathbf{C}_0)^{-1}\mathbf{C}'_0 = \mathbf{I}_K$. (The choice of factor $\mathbf{C}$ is not unique, but the limiting distribution of $\mathbf{v}$ is standard normal for every choice.)

(d) From (a) and the definition of J ,

$$J = n(\widehat{\mathbf{B}}\bar{\mathbf{g}})' \widehat{\mathbf{S}}^{-1}(\widehat{\mathbf{B}}\bar{\mathbf{g}}) = (\sqrt{n}\bar{\mathbf{g}})'\mathbf{C}'\mathbf{M}\mathbf{C}(\sqrt{n}\bar{\mathbf{g}}) = \mathbf{v}'\mathbf{M}\mathbf{v}.$$

Now $\mathbf{v} \xrightarrow{d} \mathbf{v}^* \sim N(\mathbf{0}, \mathbf{I}_K)$ by (c), and $\mathbf{M} \xrightarrow{p} \mathbf{M}_0$, a symmetric idempotent matrix of rank $K - L$ (the limit of the $\mathbf{M}$'s, using $\mathbf{S}_{xz} \xrightarrow{p} \Sigma_{xz}$ and $\mathbf{C}'\mathbf{C} \xrightarrow{p} \mathbf{S}^{-1}$; continuity preserves symmetry, idempotency, and — since the eigenvalues are 0 or 1 — the rank). By LEMMA 2.4(d),

$$J = \mathbf{v}'\mathbf{M}\mathbf{v} \xrightarrow{d} \mathbf{v}^*\mathbf{M}_0\mathbf{v}^* \sim \chi^2(K - L),$$

since a quadratic form in a standard normal vector with a symmetric idempotent matrix of rank r is $\chi^2(r)$ (diagonalize $\mathbf{M}_0$: the form is the sum of the squares of r independent standard normals). This is PROPOSITION 3.6. The degrees-of-freedom accounting is transparent in this proof: the K sample moments are projected onto the $K - L$ -dimensional space orthogonal to the (estimated)

directions used up in estimating δ ; the rank drop from K to $K - L$ is the price of having fitted L parameters.

Analytical Exercise 3.6

From Exercise 3.5(a), $\bar{\mathbf{g}}_n(\hat{\delta}(\hat{\mathbf{S}}^{-1})) = \hat{\mathbf{B}}\bar{\mathbf{g}}$, and the same display — $\bar{\mathbf{g}}_n(\hat{\delta}(\hat{\mathbf{S}}^{-1})) = \hat{\mathbf{B}}\mathbf{s}_{xy}$ — was derived in the first line of that part before substituting (3.4.9). Hence $\hat{\mathbf{B}}\bar{\mathbf{g}} = \hat{\mathbf{B}}\mathbf{s}_{xy}$ (both equal $\bar{\mathbf{g}}_n(\hat{\delta}(\hat{\mathbf{S}}^{-1}))$), and substituting into the expression for J ,

$$J(\hat{\delta}(\hat{\mathbf{S}}^{-1}), \hat{\mathbf{S}}^{-1}) = n \bar{\mathbf{g}}' \hat{\mathbf{B}}' \hat{\mathbf{S}}^{-1} \hat{\mathbf{B}} \bar{\mathbf{g}} = n \mathbf{s}_{xy}' \hat{\mathbf{B}}' \hat{\mathbf{S}}^{-1} \hat{\mathbf{B}} \mathbf{s}_{xy}.$$

The point of the formula is computational: J depends on the data only through the *unnormalized* moments $\mathbf{s}_{xy}$ and $\mathbf{S}_{xz}$ (through $\hat{\mathbf{B}}$); one does not need to form the residual vector at all. The identity $\hat{\mathbf{B}}\mathbf{s}_{xy} = \hat{\mathbf{B}}\bar{\mathbf{g}}$ is just the sample reflection of the population fact that the true parameters satisfy the moment conditions: the part of $\mathbf{s}_{xy}$ explained by the model, $\mathbf{S}_{xz}\delta$, is annihilated by $\hat{\mathbf{B}}$, so only the unexplained part $\bar{\mathbf{g}}$ survives.

Analytical Exercise 3.7

(a) This is Exercise 3.5(a)–(b) applied to the sub-system with instruments $\mathbf{x}_{i1}$: the efficient GMM estimator using only the first K_1 instruments and weighting matrix $\hat{\mathbf{S}}_{11}^{-1}$ is $\hat{\delta}_1 = (\mathbf{S}_{xz,1}' \hat{\mathbf{S}}_{11}^{-1} \mathbf{S}_{xz,1})^{-1} \mathbf{S}_{xz,1}' \hat{\mathbf{S}}_{11}^{-1} \mathbf{s}_{xy,1}$, and

$$\bar{\mathbf{g}}_{1n}(\hat{\delta}_1) = \mathbf{s}_{xy,1} - \mathbf{S}_{xz,1} \hat{\delta}_1 = \hat{\mathbf{B}}_1 \mathbf{s}_{xy,1} = \hat{\mathbf{B}}_1 \bar{\mathbf{g}}_1,$$

since $\hat{\mathbf{B}}_1 \mathbf{S}_{xz,1} = \mathbf{0}$ and $\mathbf{s}_{xy,1} = \mathbf{S}_{xz,1} \delta + \bar{\mathbf{g}}_1$. The quadratic form identity $\hat{\mathbf{B}}_1' \hat{\mathbf{S}}_{11}^{-1} \hat{\mathbf{B}}_1 = \mathbf{C}_1' \mathbf{M}_1 \mathbf{C}_1$ follows by the algebra of Exercise 3.5(b) with $\mathbf{A}_1 = \mathbf{C}_1 \mathbf{S}_{xz,1}$. The strengthened rank condition of the proposition guarantees that $\mathbf{S}_{xz,1}$ has full column rank for large n , so all inverses exist.

(b) As in Exercise 3.5(b): $\mathbf{M}$ and $\mathbf{M}_1$ are symmetric idempotent annihilators of column spaces of dimensions L in $\mathbb{R}^K$ and $\mathbb{R}^{K_1}$ respectively, so $\text{rank}(\mathbf{M}) = \text{tr}(\mathbf{M}) = K - L$ and $\text{rank}(\mathbf{M}_1) = \text{tr}(\mathbf{M}_1) = K_1 - L$.

(c) (Needed for (d).) By Exercise 3.5(d), $J = \mathbf{v}' \mathbf{M} \mathbf{v}$ with $\mathbf{v} = \sqrt{n} \mathbf{C} \bar{\mathbf{g}}$. For the sub-system, define $\mathbf{v}_1 \equiv \sqrt{n} \mathbf{C}_1 \bar{\mathbf{g}}_1$; then the same steps give

$$J_1 = n \bar{\mathbf{g}}_1' \hat{\mathbf{B}}_1' \hat{\mathbf{S}}_{11}^{-1} \hat{\mathbf{B}}_1 \bar{\mathbf{g}}_1 = n \bar{\mathbf{g}}_1' \mathbf{C}_1' \mathbf{M}_1 \mathbf{C}_1 \bar{\mathbf{g}}_1 = \mathbf{v}_1' \mathbf{M}_1 \mathbf{v}_1.$$

(d) With $\bar{\mathbf{g}}_1 = \mathbf{F}' \bar{\mathbf{g}}$ and $\mathbf{v} = \sqrt{n} \mathbf{C} \bar{\mathbf{g}}$, we have $\sqrt{n} \bar{\mathbf{g}} = \mathbf{C}^{-1} \mathbf{v}$, and

$$\mathbf{v}_1 = \sqrt{n} \mathbf{C}_1 \mathbf{F}' \bar{\mathbf{g}} = \mathbf{C}_1 \mathbf{F}' \mathbf{C}^{-1} \mathbf{v}.$$

Therefore

$$J_1 = \mathbf{v}' \mathbf{C}'^{-1} \mathbf{F} \mathbf{C}_1' \mathbf{M}_1 \mathbf{C}_1 \mathbf{F}' \mathbf{C}^{-1} \mathbf{v} = \mathbf{v}' \mathbf{D} \mathbf{v}, \quad J - J_1 = \mathbf{v}' (\mathbf{M} - \mathbf{D}) \mathbf{v},$$

with $\mathbf{D} \equiv \mathbf{C}'^{-1} \mathbf{F} \mathbf{C}_1' \mathbf{M}_1 \mathbf{C}_1 \mathbf{F}' \mathbf{C}^{-1}$.

(e) $\mathbf{D}$ is symmetric because $\mathbf{M}_1$ is: $\mathbf{D}' = \mathbf{C}'^{-1} \mathbf{F} \mathbf{C}_1' \mathbf{M}_1' \mathbf{C}_1 \mathbf{F}' \mathbf{C}^{-1} = \mathbf{D}$. For idempotency, write $\mathbf{D} = \mathbf{G}' \mathbf{M}_1 \mathbf{G}$ with $\mathbf{G} \equiv \mathbf{C}_1 \mathbf{F}' \mathbf{C}^{-1}$, and note

$$\mathbf{G} \mathbf{G}' = \mathbf{C}_1 \mathbf{F}' \mathbf{C}^{-1} \mathbf{C}'^{-1} \mathbf{F} \mathbf{C}_1' = \mathbf{C}_1 \mathbf{F}' \hat{\mathbf{S}} \mathbf{F} \mathbf{C}_1' = \mathbf{C}_1 \hat{\mathbf{S}}_{11} \mathbf{C}_1' = \mathbf{C}_1 (\mathbf{C}_1' \mathbf{C}_1)^{-1} \mathbf{C}_1' = \mathbf{I}_{K_1},$$

since $\widehat{\mathbf{S}}_{11} = \mathbf{F}'\widehat{\mathbf{S}}\mathbf{F}$ (the sample moments of $\mathbf{x}_{i1}\varepsilon_i$ are the first K_1 of those of $\mathbf{x}_i\varepsilon_i$). Then

$$\mathbf{D}\mathbf{D} = \mathbf{G}'\mathbf{M}_1(\mathbf{G}\mathbf{G}')\mathbf{M}_1\mathbf{G} = \mathbf{G}'\mathbf{M}_1\mathbf{M}_1\mathbf{G} = \mathbf{G}'\mathbf{M}_1\mathbf{G} = \mathbf{D}.$$

Hence $\text{rank}(\mathbf{D}) = \text{tr}(\mathbf{D}) = \text{tr}(\mathbf{M}_1\mathbf{G}\mathbf{G}') = \text{tr}(\mathbf{M}_1) = K_1 - L$.

(f) Compute, using $\mathbf{C}'\mathbf{C}'^{-1} = \mathbf{I}_K$ and $\mathbf{S}'_{\mathbf{xz}}\mathbf{F} = (\mathbf{F}'\mathbf{S}_{\mathbf{xz}})' = \mathbf{S}'_{\mathbf{xz},1} = \mathbf{A}'_1\mathbf{C}'_1{}^{-1}$:

$$\mathbf{A}'\mathbf{D} = (\mathbf{S}'_{\mathbf{xz}}\mathbf{C}')(\mathbf{C}'^{-1}\mathbf{F}\mathbf{C}'_1\mathbf{M}_1\mathbf{C}_1\mathbf{F}'\mathbf{C}^{-1}) = \mathbf{S}'_{\mathbf{xz},1}\mathbf{C}'_1\mathbf{M}_1(\mathbf{C}_1\mathbf{F}'\mathbf{C}^{-1}) = \mathbf{A}'_1\mathbf{M}_1(\mathbf{C}_1\mathbf{F}'\mathbf{C}^{-1}) = \mathbf{0},$$

because $\mathbf{A}'_1\mathbf{M}_1 = \mathbf{0}$: $\mathbf{M}_1$ annihilates the columns of $\mathbf{A}_1$. So every column of $\mathbf{D}$ is orthogonal to the column space of $\mathbf{A}$. Since $\mathbf{M}$ is the orthogonal projection onto $\text{col}(\mathbf{A})^\perp$, this means $\text{col}(\mathbf{D}) \subseteq \text{col}(\mathbf{M})$, and a projection acts as the identity on its own column space:

$$\mathbf{M}\mathbf{D} = \mathbf{D}, \quad \text{hence also} \quad \mathbf{D}\mathbf{M} = \mathbf{D}$$

by symmetry of both matrices. The columns of $\mathbf{D}$ — the directions of moment space that the subsystem's J_1 summarizes — lie inside the column space of $\mathbf{M}$, the directions in which the full-system J can vary; this nesting of the two projections is the algebraic heart of the C test.

Remark. The book labels this step “ $\mathbf{M}\mathbf{D} = \mathbf{0}$.” That cannot be right as stated: $\mathbf{D}$ is a nonzero projection, and the argument of part (g) requires $\mathbf{M}\mathbf{D} = \mathbf{D}$ (which the book's own hint chain establishes via $\mathbf{A}'\mathbf{D} = \mathbf{0}$ and the observation that $\mathbf{M}$ projects onto $\text{col}(\mathbf{A})^\perp$). With $\mathbf{M}\mathbf{D} = \mathbf{D}$, part (g)'s idempotency computation below is exact; with $\mathbf{M}\mathbf{D} = \mathbf{0}$ the statistic C would equal $J - J_1$ in neither rank nor value.

(g) $\mathbf{M} - \mathbf{D}$ is symmetric (both are) and idempotent:

$$(\mathbf{M} - \mathbf{D})^2 = \mathbf{M} - \mathbf{M}\mathbf{D} - \mathbf{D}\mathbf{M} + \mathbf{D} = \mathbf{M} - \mathbf{D},$$

using (f) in the form $\mathbf{M}\mathbf{D} = \mathbf{D}\mathbf{M} = \mathbf{D}$. Its rank is

$$\text{rank}(\mathbf{M} - \mathbf{D}) = \text{tr}(\mathbf{M}) - \text{tr}(\mathbf{D}) = (K - L) - (K_1 - L) = K - K_1.$$

(h) By Exercise 3.5(c), $\mathbf{v} \xrightarrow{d} \mathbf{v}^* \sim N(\mathbf{0}, \mathbf{I}_K)$; the limit of $\mathbf{M} - \mathbf{D}$ is symmetric idempotent of rank $K - K_1$ (all the constructions above are continuous in the sample moments, and ranks of idempotent matrices are preserved under convergence). Hence by LEMMA 2.4(D),

$$C = J - J_1 = \mathbf{v}'(\mathbf{M} - \mathbf{D})\mathbf{v} \xrightarrow{d} \mathbf{v}^{*'}(\mathbf{M}_0 - \mathbf{D}_0)\mathbf{v}^* \sim \chi^2(K - K_1),$$

which is PROPOSITION 3.7. The proof shows C tests exactly the $K - K_1$ directions of moment space that the suspect instruments add, after projecting out the L directions used to fit δ .

(i) Since $\mathbf{C}'(\mathbf{M} - \mathbf{D})\mathbf{C} = \widehat{\mathbf{B}}'\widehat{\mathbf{S}}^{-1}\widehat{\mathbf{B}} - \mathbf{F}\widehat{\mathbf{B}}'_1\widehat{\mathbf{S}}_{11}^{-1}\widehat{\mathbf{B}}_1\mathbf{F}'$ (by Exercise 3.5(b) applied to both systems and the definition of $\mathbf{D}$), and since $\widehat{\mathbf{B}}\mathbf{s}_{xy} = \widehat{\mathbf{B}}\bar{\mathbf{g}}$ (Exercise 3.6) while $\mathbf{F}'\mathbf{s}_{xy} = \mathbf{s}_{xy,1}$ satisfies $\widehat{\mathbf{B}}_1\mathbf{s}_{xy,1} = \widehat{\mathbf{B}}_1\bar{\mathbf{g}}_1$ (part (a)), replacing $\bar{\mathbf{g}}$ by $\mathbf{s}_{xy}$ leaves every factor unchanged. Hence

$$C = n\bar{\mathbf{g}}'\mathbf{C}'(\mathbf{M} - \mathbf{D})\mathbf{C}\bar{\mathbf{g}} = n\mathbf{s}'_{xy}\mathbf{C}'(\mathbf{M} - \mathbf{D})\mathbf{C}\mathbf{s}_{xy}.$$

(j) A symmetric idempotent matrix is positive semidefinite (Exercise 3.1), and $\mathbf{M} - \mathbf{D}$ is symmetric idempotent by (g). Hence for any sample, $C = n\mathbf{s}'_{xy}\mathbf{C}'(\mathbf{M} - \mathbf{D})\mathbf{C}\mathbf{s}_{xy} \geq 0$, as long as the *same* $\widehat{\mathbf{S}}$ (and its submatrix $\widehat{\mathbf{S}}_{11}$) is used in both J and J_1 — the recipe the text prescribes. With two different

consistent estimates of $\mathbf{S}$ the difference $J - J_1$ remains asymptotically χ^2 but can go negative in finite samples.

Analytical Exercise 3.8

(a) Write $\mathbf{A} \equiv \mathbf{S}'_{xz} \widehat{\mathbf{W}} \mathbf{S}_{xz}$ ($L \times L$, nonsingular for large n). The Lagrangian is

$$\mathcal{L}(\bar{\delta}, \bar{\lambda}) = \frac{n}{2} (\mathbf{s}_{xy} - \mathbf{S}_{xz} \bar{\delta})' \widehat{\mathbf{W}} (\mathbf{s}_{xy} - \mathbf{S}_{xz} \bar{\delta}) + \bar{\lambda}' (\mathbf{R} \bar{\delta} - \mathbf{r}),$$

with first-order conditions

$$\frac{\partial \mathcal{L}}{\partial \bar{\delta}} = -n \mathbf{S}'_{xz} \widehat{\mathbf{W}} \mathbf{s}_{xy} + n \mathbf{A} \bar{\delta} + \mathbf{R}' \bar{\lambda} = \mathbf{0}, \quad \mathbf{R} \bar{\delta} = \mathbf{r},$$

which is the hint (with the Lagrangian scaled so the multiplier picks up the factor $2n$ — the factor is immaterial for $\bar{\delta}$). The unrestricted estimator satisfies $n \mathbf{S}'_{xz} \widehat{\mathbf{W}} \mathbf{s}_{xy} = n \mathbf{A} \hat{\delta}(\widehat{\mathbf{W}})$, so the first condition reads

$$\mathbf{A}(\bar{\delta} - \hat{\delta}(\widehat{\mathbf{W}})) = -\frac{1}{n} \mathbf{R}' \bar{\lambda}, \quad \text{i.e.} \quad \bar{\delta} = \hat{\delta}(\widehat{\mathbf{W}}) - \frac{1}{n} \mathbf{A}^{-1} \mathbf{R}' \bar{\lambda}.$$

Impose the constraint: $\mathbf{r} = \mathbf{R} \bar{\delta} = \mathbf{R} \hat{\delta}(\widehat{\mathbf{W}}) - \frac{1}{n} \mathbf{R} \mathbf{A}^{-1} \mathbf{R}' \bar{\lambda}$, so

$$\bar{\lambda} = 2n [\mathbf{R} \mathbf{A}^{-1} \mathbf{R}']^{-1} [\mathbf{R} \hat{\delta}(\widehat{\mathbf{W}}) - \mathbf{r}]$$

in the book's normalization (the Lagrangian there has $\frac{1}{2}J$ and multiplier convention absorbing the 2; with our convention the factor is n — either way, substituting back),

$$\bar{\delta} = \hat{\delta}(\widehat{\mathbf{W}}) - \mathbf{A}^{-1} \mathbf{R}' [\mathbf{R} \mathbf{A}^{-1} \mathbf{R}']^{-1} [\mathbf{R} \hat{\delta}(\widehat{\mathbf{W}}) - \mathbf{r}],$$

which is the claimed formula — the familiar “restricted = unrestricted – correction” of constrained quadratic minimization, exactly parallel to Analytical Exercise 1.5 for OLS.

(b) Expand $J(\bar{\delta}, \widehat{\mathbf{W}})$ around $\hat{\delta} \equiv \hat{\delta}(\widehat{\mathbf{W}})$ using $\bar{\mathbf{g}}_n(\bar{\delta}) = \bar{\mathbf{g}}_n(\hat{\delta}) - \mathbf{S}_{xz}(\bar{\delta} - \hat{\delta})$:

$$J(\bar{\delta}, \widehat{\mathbf{W}}) = J(\hat{\delta}, \widehat{\mathbf{W}}) - 2n \bar{\mathbf{g}}_n(\hat{\delta})' \widehat{\mathbf{W}} \mathbf{S}_{xz}(\bar{\delta} - \hat{\delta}) + n(\bar{\delta} - \hat{\delta})' \mathbf{A}(\bar{\delta} - \hat{\delta}),$$

which is the hint's decomposition (sign of the cross term depends on the convention for $\bar{\delta} - \hat{\delta}$; the unrestricted first-order condition $\mathbf{S}'_{xz} \widehat{\mathbf{W}} \bar{\mathbf{g}}_n(\hat{\delta}) = \mathbf{0}$ annihilates it in either case). Hence

$$J(\bar{\delta}, \widehat{\mathbf{W}}) - J(\hat{\delta}, \widehat{\mathbf{W}}) = n(\bar{\delta} - \hat{\delta})' \mathbf{A}(\bar{\delta} - \hat{\delta}).$$

Now substitute part (a): $\bar{\delta} - \hat{\delta} = -\mathbf{A}^{-1} \mathbf{R}' [\mathbf{R} \mathbf{A}^{-1} \mathbf{R}']^{-1} [\mathbf{R} \hat{\delta} - \mathbf{r}]$, so

$$J(\bar{\delta}, \widehat{\mathbf{W}}) - J(\hat{\delta}, \widehat{\mathbf{W}}) = n [\mathbf{R} \hat{\delta} - \mathbf{r}]' [\mathbf{R} \mathbf{A}^{-1} \mathbf{R}']^{-1} \mathbf{R} \mathbf{A}^{-1} \mathbf{A} \mathbf{A}^{-1} \mathbf{R}' [\mathbf{R} \mathbf{A}^{-1} \mathbf{R}']^{-1} [\mathbf{R} \hat{\delta} - \mathbf{r}],$$

and the middle collapses to $[\mathbf{R} \mathbf{A}^{-1} \mathbf{R}']^{-1}$:

$$LR\text{-difference} = n [\mathbf{R} \hat{\delta}(\widehat{\mathbf{W}}) - \mathbf{r}]' [\mathbf{R} (\mathbf{S}'_{xz} \widehat{\mathbf{W}} \mathbf{S}_{xz})^{-1} \mathbf{R}']^{-1} [\mathbf{R} \hat{\delta}(\widehat{\mathbf{W}}) - \mathbf{r}].$$

(c) Set $\widehat{\mathbf{W}} = \widehat{\mathbf{S}}^{-1}$. Then $\mathbf{A} = \mathbf{S}'_{xz} \widehat{\mathbf{S}}^{-1} \mathbf{S}_{xz}$ and by (3.5.15), the estimated asymptotic variance of the efficient GMM estimator is $\widehat{\text{Avar}}(\hat{\delta}(\widehat{\mathbf{S}}^{-1})) = \mathbf{A}^{-1}/n$. The Wald statistic (3.5.16) is therefore

$$W = n [\mathbf{R} \hat{\delta}(\widehat{\mathbf{S}}^{-1}) - \mathbf{r}]' [\mathbf{R} \mathbf{A}^{-1} \mathbf{R}']^{-1} [\mathbf{R} \hat{\delta}(\widehat{\mathbf{S}}^{-1}) - \mathbf{r}] = J(\bar{\delta}, \widehat{\mathbf{S}}^{-1}) - J(\hat{\delta}(\widehat{\mathbf{S}}^{-1}), \widehat{\mathbf{S}}^{-1}) = LR,$$

by part (b). So for *linear* restrictions the Wald and LR statistics of efficient GMM are numerically identical — not merely asymptotically equivalent — provided the same $\widehat{\mathbf{S}}$ is used throughout. For

nonlinear hypotheses the quadratic expansion in (b) is only local, which is why PROPOSITION 3.8(B) then gives only $\xrightarrow{p} 0$ equality; and it is the LR statistic that is invariant to reparametrization of the constraint, which is its practical advantage over W .

Analytical Exercise 3.9

(a) From the sampling error formula (3.4.11) applied to each weighting matrix,

$$\sqrt{n}(\hat{\delta}_j - \delta) = (\mathbf{S}'_{\mathbf{x}z} \widehat{\mathbf{W}}_j \mathbf{S}_{\mathbf{x}z})^{-1} \mathbf{S}'_{\mathbf{x}z} \widehat{\mathbf{W}}_j \sqrt{n} \bar{\mathbf{g}} \equiv \widehat{\mathbf{T}}_j \sqrt{n} \bar{\mathbf{g}}, \quad j = 1, 2.$$

Both are linear functions of the *same* $\sqrt{n} \bar{\mathbf{g}} \xrightarrow{d} N(\mathbf{0}, \mathbf{S})$, and $\widehat{\mathbf{T}}_j \xrightarrow{p} \mathbf{Q}_j^{-1} \mathbf{A}_j$ by LEMMA 2.3(A). Hence by LEMMA 2.4(C) (applied to the stacked system),

$$\begin{bmatrix} \sqrt{n}(\hat{\delta}_1 - \delta) \\ \sqrt{n}(\hat{\delta}_2 - \delta) \end{bmatrix} = \begin{bmatrix} \widehat{\mathbf{T}}_1 \\ \widehat{\mathbf{T}}_2 \end{bmatrix} \sqrt{n} \bar{\mathbf{g}} \xrightarrow{d} N(\mathbf{0}, \mathbf{\Omega}), \quad \mathbf{\Omega}_{jk} = \mathbf{Q}_j^{-1} \mathbf{A}_j \mathbf{S} \mathbf{A}'_k \mathbf{Q}_k^{-1'},$$

with $\mathbf{Q}_j = \Sigma'_{\mathbf{x}z} \mathbf{W}_j \Sigma_{\mathbf{x}z}$ symmetric and $\mathbf{A}_j = \Sigma'_{\mathbf{x}z} \mathbf{W}_j$. The diagonal blocks reproduce (3.5.1) for each estimator.

(b) $\boldsymbol{\eta} = [\mathbf{I}_L : -\mathbf{I}_L] \begin{bmatrix} \hat{\delta}_1 - \delta \\ \hat{\delta}_2 - \delta \end{bmatrix}$, a fixed linear combination, so by (a) and LEMMA 2.3(B),

$$\sqrt{n} \boldsymbol{\eta} \xrightarrow{d} N(\mathbf{0}, \text{Avar}(\boldsymbol{\eta})),$$

$$\text{Avar}(\boldsymbol{\eta}) = \mathbf{\Omega}_{11} + \mathbf{\Omega}_{22} - \mathbf{\Omega}_{12} - \mathbf{\Omega}_{21} = \text{Avar}(\hat{\delta}_1) + \text{Avar}(\hat{\delta}_2) - 2 \text{Cov}_{\infty}(\hat{\delta}_1, \hat{\delta}_2),$$

the usual variance-of-a-difference formula.

(c) With $\mathbf{W}_2 = \mathbf{S}^{-1}$: $\mathbf{Q}_2 = \Sigma'_{\mathbf{x}z} \mathbf{S}^{-1} \Sigma_{\mathbf{x}z}$ and $\mathbf{A}_2 = \Sigma'_{\mathbf{x}z} \mathbf{S}^{-1}$. The covariance block becomes

$$\begin{aligned} \mathbf{\Omega}_{12} &= \mathbf{Q}_1^{-1} (\Sigma'_{\mathbf{x}z} \mathbf{W}_1) \mathbf{S} \mathbf{S}^{-1} \Sigma_{\mathbf{x}z} (\Sigma'_{\mathbf{x}z} \mathbf{S}^{-1} \Sigma_{\mathbf{x}z})^{-1} \\ &= \mathbf{Q}_1^{-1} \Sigma'_{\mathbf{x}z} \mathbf{W}_1 \Sigma_{\mathbf{x}z} (\Sigma'_{\mathbf{x}z} \mathbf{S}^{-1} \Sigma_{\mathbf{x}z})^{-1} = (\Sigma'_{\mathbf{x}z} \mathbf{S}^{-1} \Sigma_{\mathbf{x}z})^{-1} = \text{Avar}(\hat{\delta}_2), \end{aligned}$$

since $\mathbf{Q}_1^{-1} (\Sigma'_{\mathbf{x}z} \mathbf{W}_1 \Sigma_{\mathbf{x}z}) = \mathbf{I}_L$. Symmetrically $\mathbf{\Omega}_{21} = \text{Avar}(\hat{\delta}_2)$. Therefore

$$\text{Avar}(\boldsymbol{\eta}) = \text{Avar}(\hat{\delta}_1) + \text{Avar}(\hat{\delta}_2) - 2 \text{Avar}(\hat{\delta}_2) = \text{Avar}(\hat{\delta}_1) - \text{Avar}(\hat{\delta}_2),$$

the Hausman principle: the asymptotic variance of the difference between any consistent GMM estimator and the efficient GMM estimator is the difference of the asymptotic variances. This is what legitimizes the Hausman–Taylor statistic (3.8.22): the covariance term in the variance of a difference, which would normally require estimating off-diagonal blocks, drops out entirely when one of the estimators is efficient. The result is purely asymptotic and purely a property of efficiency — the same algebra underlies the ML Hausman test, with the information-matrix equality playing the role that $\mathbf{W}_2 = \mathbf{S}^{-1}$ plays here.

Analytical Exercise 3.10

(a) Substitute the first-stage equation into σ_{xz} :

$$\sigma_{xz} = E(x_i z_i) = \pi_0 E(x_i) + \pi_1 E(x_i^2) + E(x_i \zeta_i).$$

Since $\{g_{2i}\} = \{x_i \zeta_i\}$ is a martingale difference sequence, $E(x_i \zeta_i) = 0$. The cleanest reading of the exercise (and Staiger–Stock’s) is that the variables are measured in deviation from means, i.e.

$E(x_i) = 0$ (equivalently $\pi_0 = 0$), so

$$\sigma_{xz} = \pi_1 E(x_i^2) = \pi_1 \sigma_x^2 \neq 0$$

as long as $\pi_1 \neq 0$ (Assumption (4)) and $\sigma_x^2 > 0$ (Assumption (3)): the rank condition for identification is satisfied. If instead $E(x_i) \neq 0$, then $\sigma_{xz} = \pi_0 E(x_i) + \pi_1 \sigma_x^2$ and the rank condition must simply be maintained; nothing in the rest of the exercise changes, since all the action is in the local-to-zero coefficient π_1 .

Remark. As printed, the conclusion $\sigma_{xz} \neq 0$ does not literally follow from (1)–(4) without $E(x_i) = 0$; the conventional reading, confirmed by parts (c)–(e), is that $E(x_i) = 0$, whence $\sigma_{xz} = \pi_1 \sigma_x^2$.

(b) $y_i = z_i \delta + \eta_i$ gives $\mathbf{s}_{xy} = \mathbf{s}_{xz} \delta + \frac{1}{n} \sum_i x_i \eta_i$, so $\hat{\delta} - \delta = \mathbf{s}_{xz}^{-1} \cdot \frac{1}{n} \sum_i x_i \eta_i$. Under (1)–(4): $\mathbf{s}_{xz} \xrightarrow{p} \sigma_{xz} = \pi_1 \sigma_x^2 \neq 0$ by ergodic stationarity, and $\frac{1}{n} \sum_i x_i \eta_i = \frac{1}{n} \sum_i g_{1i} \xrightarrow{p} E(g_{1i}) = 0$ since $\{g_{1i}\}$ is an ergodic stationary m.d.s. Hence $\hat{\delta} - \delta \xrightarrow{p} (\pi_1 \sigma_x^2)^{-1} \cdot 0 = 0$: the IV estimator is consistent.

(c) Under (4'), $\pi_1 = c/\sqrt{n} \rightarrow 0$, so

$$\sigma_{xz} = \pi_1 \sigma_x^2 = \frac{c \sigma_x^2}{\sqrt{n}} \rightarrow 0,$$

and $\mathbf{s}_{xz} \xrightarrow{p} 0$: the sample cross moment still converges to its (now vanishing) population counterpart. The order condition is satisfied in every sample but the rank condition is drifting toward failure — the defining feature of weak instruments.

(d) Write, using the first-stage equation,

$$\mathbf{s}_{xz} = \frac{1}{n} \sum_i x_i (\pi_0 + \pi_1 x_i + \zeta_i) = \pi_0 \bar{x} + \pi_1 \frac{1}{n} \sum_i x_i^2 + \frac{1}{n} \sum_i g_{2i},$$

where $g_{2i} = x_i \zeta_i$. With $E(x_i) = 0$, $\bar{x} \xrightarrow{p} 0$, and $\frac{1}{n} \sum_i x_i^2 \xrightarrow{p} \sigma_x^2$ by ergodic stationarity; $\pi_1 = c/\sqrt{n}$. Multiplying by $\sqrt{n}$:

$$\sqrt{n} \mathbf{s}_{xz} = \sqrt{n} \pi_0 \bar{x} + \frac{c}{\sqrt{n}} \cdot \sqrt{n} \cdot \frac{1}{n} \sum_i x_i^2 + \frac{1}{\sqrt{n}} \sum_i g_{2i} = o_p(1) + c \sigma_x^2 + \frac{1}{\sqrt{n}} \sum_i g_{2i},$$

where the first term vanishes if $\bar{x} = O_p(n^{-1/2})$ (it is, by the CLT applied to the m.d.s.-type behavior; if $\pi_0 = 0$ it vanishes identically). The last term: $\{g_{2i}\}$ is an ergodic stationary m.d.s. with variance S_{22} , so by the ergodic stationary m.d.s. CLT, $\frac{1}{\sqrt{n}} \sum_i g_{2i} \xrightarrow{d} a \sim N(0, S_{22})$. Combining,

$$\sqrt{n} \mathbf{s}_{xz} \xrightarrow{d} \sigma_x^2 c + a, \quad a \sim N(0, S_{22}).$$

The cross moment is now $O_p(n^{-1/2})$ instead of $O_p(1)$: the signal (π_1) and the noise are of the same order, which is the whole point of the Staiger–Stock device.

(e) The numerator: $\sqrt{n} \bar{g}_1 = \frac{1}{\sqrt{n}} \sum_i g_{1i} \xrightarrow{d} b \sim N(0, S_{11})$, and $(g_{1i}, g_{2i})'$ is *jointly* an m.d.s. with covariance $\mathbf{S}$, so the joint CLT gives

$$\left(\frac{1}{\sqrt{n}} \sum_i g_{2i}, \frac{1}{\sqrt{n}} \sum_i g_{1i} \right)' \xrightarrow{d} (a, b)' \sim N(\mathbf{0}, \mathbf{S}).$$

Hence by the continuous mapping theorem applied to the ratio (the denominator converges to $\sigma_x^2 c + a$, which is nonzero with probability one since $c \neq 0$ and the normal a is continuous),

$$\hat{\delta} - \delta = \frac{\bar{g}_1}{\mathbf{s}_{xz}} = \frac{\sqrt{n} \bar{g}_1}{\sqrt{n} \mathbf{s}_{xz}} \xrightarrow{d} \frac{b}{\sigma_x^2 c + a}.$$

(f) No. The limiting distribution of $\hat{\delta} - \delta$ is the ratio $b/(\sigma_x^2 c + a)$ of two jointly normal random variables — a nonstandard, Cauchy-like distribution (heavy-tailed, since the denominator can be near zero with positive density) centered *not* at zero in general: when $S_{12} \neq 0$ (i.e. z_i genuinely endogenous), a and b are correlated and the median of the ratio shifts toward $\rho \sigma_b / \sigma_a$ — the OLS-like limit — rather than zero. Since $\hat{\delta} - \delta$ converges to a nondegenerate random variable rather than to 0, $\hat{\delta}$ is inconsistent under the local-to-zero sequence (4').

The lessons are the ones Staiger and Stock (1997) emphasize: (i) with weak instruments, standard normal inference on $\hat{\delta}$ is invalid at any sample size — the asymptotic distribution is a ratio of normals, and the usual t -ratio does not have its nominal size; (ii) the direction of the asymptotic bias is toward OLS (through S_{12}); (iii) the severity is governed by the concentration parameter $\mu^2 = n\pi_1^2 \sigma_x^2 / S_{22} \rightarrow c^2 \sigma_x^2 / S_{22}$, which is the population first-stage F statistic — hence the practical rule of examining first-stage F 's (and the dramatic part (g) of the empirical exercise, where two excluded instruments barely predict two endogenous regressors and the point estimates explode).

Problem Set for Chapter 3: Empirical Exercise

Empirical Exercise 3.1

All numbers below are computed by `code/ch3.py` ($n = 758$ throughout).

(a) Means and standard deviations (in parentheses):

variable	mean	s.d.	variable	mean	s.d.
<i>RNS</i>	0.269	(0.444)	<i>RNS80</i>	0.293	(0.455)
<i>MRT</i>	0.515	(0.500)	<i>MRT80</i>	0.898	(0.302)
<i>SMSA</i>	0.704	(0.457)	<i>SMSA80</i>	0.712	(0.453)
<i>MED</i>	10.910	(2.741)	<i>IQ</i>	103.856	(13.619)
<i>KWW</i>	36.574	(7.302)	<i>AGE</i>	21.835	(2.982)
<i>S</i>	13.405	(2.232)	<i>S80</i>	13.707	(2.215)
<i>EXPR</i>	1.735	(2.106)	<i>EXPR80</i>	11.394	(4.211)
<i>TENURE</i>	1.831	(1.674)	<i>TENURE80</i>	7.363	(5.050)
<i>LW</i>	5.687	(0.429)	<i>LW80</i>	6.827	(0.410)

The sample is a group of very young men at the first wage observation: mean age 21.8, mean schooling 13.4 years, mean experience under two years. The correlation between *IQ* and *S* is

$$\text{Corr}(IQ, S) = 0.5131,$$

substantial and positive — exactly the correlation that creates the omitted-ability bias when *IQ* is excluded from the wage equation.

(b) Table 3.3 of the book, lines 1–3, is reproduced:

line	technique	S	IQ	$EXPR$	$TENURE$	SEE
1	OLS (no IQ)	0.0697 (0.0067)	—	0.030 (0.0065)	0.043 (0.0075)	0.328 $R^2 = 0.425$
2	OLS with IQ	0.0620 (0.0073)	0.0027 (0.0010)	0.031 (0.0065)	0.042 (0.0075)	0.326 $R^2 = 0.430$
3	2SLS (IQ endogenous)	0.0692 (0.0129)	0.0002 (0.0039)	0.030 (0.0066)	0.043 (0.0076)	0.328

(2SLS standard errors are the conventional ones of (3.8.5).) The pattern across lines 1 and 2 matches the omitted-variable analysis of Section 3.9: ability and schooling are positively correlated ($\text{Corr} = 0.51$) and ability raises wages ($\hat{\gamma} > 0$), so omitting IQ biases the schooling coefficient upward, from 0.062 to 0.070. Line 3 was *predicted* to show a lower schooling coefficient than line 2 if the OLS-with- IQ estimate is biased upward by the errors-in-variables mechanics of (3.9.7A) — but here the 2SLS estimate of β (0.069) is if anything *higher* than line 2's 0.062, while the IQ coefficient collapses from 0.0027 to an insignificant 0.0002. Taken at face value, instrumenting IQ removes the ability effect from the wage equation entirely rather than revealing a downward-attenuated one. The anomaly is resolved in (c) and (g): the instrument set fails Sargan's test massively, so line 3's 2SLS estimates are not consistent for anything; the direction-of-bias predictions are only as good as the instruments.

(c) Counting the dimensions: the regressor vector is

$$\mathbf{z}_i = (S, IQ, EXPR, TENURE, RNS, SMSA, Y66, \dots, Y73)',$$

so $L = 13$ (the seven year dummies span the constant); the instrument vector is

$$\mathbf{x}_i = (S, EXPR, TENURE, RNS, SMSA, Y66, \dots, Y73, MED, KWW, MRT, AGE)',$$

so $K = 16$: the 12 predetermined variables $S, EXPR, TENURE, RNS, SMSA$ and the seven year dummies (which together span the constant), plus the 4 excluded predetermined variables. The degrees of freedom of Sargan's test are therefore

$$K - L = 16 - 13 = 3.$$

The statistic (3.8.10),

$$J = n \cdot \frac{\hat{\varepsilon}' \mathbf{X} (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \hat{\varepsilon}}{\hat{\varepsilon}' \hat{\varepsilon}} = 87.655 \quad \text{on } \chi^2(3),$$

reproduces the value given in the exercise, and

$$p = \text{Prob}(\chi^2(3) \geq 87.655) \approx 0 (< 10^{-18}).$$

The overidentifying restrictions are overwhelmingly rejected: at least one of the excluded instruments (MED, KWW, MRT, AGE) is correlated with the composite error $\varepsilon - \gamma\eta$ — or the equation is otherwise misspecified. The natural suspects, per the text's discussion, are KWW (a test score plausibly correlated with the same unobserved diligence that drives wages) and MED (heritable family background). Line 3 of Table 3.3 is therefore not a consistent estimate, and the “collapse” of the IQ coefficient should not be read as evidence about ability bias.

(d) Running the two stages explicitly — first regressing each of S and IQ on the full instrument set (the other regressors are their own first-stage fitted values), then regressing LW on the fitted values — reproduces the 2SLS coefficients exactly (the discrepancy is below 10^{-11}), as the algebra of Section 3.8 (“2SLS as two regressions”) requires: since $\mathbf{P}$ is symmetric and idempotent, $(\hat{\mathbf{Z}}'\hat{\mathbf{Z}})^{-1}\hat{\mathbf{Z}}'\mathbf{y} = (\mathbf{Z}'\mathbf{P}\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{P}\mathbf{y}$. But the second-stage OLS package standard errors are wrong: for the schooling coefficient the second-stage printout gives 0.0131 against the correct 0.0129 from (3.8.5). The reason, as the text warns, is that the second-stage regression computes the residual variance from $\mathbf{y} - \hat{\mathbf{Z}}\hat{\boldsymbol{\delta}}$ (fitted regressors) rather than from the 2SLS residual $\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\delta}}$ (actual regressors). The two residual vectors differ, so the estimated error variance differs, and with it every standard error. The direction of the discrepancy is not determined in general; here the difference is small, but with weak instruments it can be large.

(e) Griliches’s argument for the endogeneity of schooling (see the book’s answers section) has three parts: (i) the measurement error η in IQ may be related to success in schooling, so part of what IQ picks up is really a schooling effect; (ii) schooling is a *choice* variable, influenced by the same unobservable characteristics (ability, diligence, family connections) that enter the wage error — the self-selection argument familiar from the production function example of Section 3.2; (iii) schooling itself may be measured with error (self-reported years of education), which mechanically makes it endogenous.

Treating both S and IQ as endogenous, with instruments $\mathbf{x}_i = (\mathbf{h}_i', MED, KWW, MRT, AGE)'$ (so $K = 15$, $L = 13$, $K - L = 2$ overidentifying restrictions), 2SLS gives line 4 of Table 3.3:

$$\hat{\beta} = 0.1724 \text{ (0.0207)}, \quad \hat{\gamma} = -0.0091 \text{ (0.0047)}, \quad \hat{\delta}_{EXPR} = 0.049, \quad \hat{\delta}_{TENURE} = 0.042,$$

with $SEE = 0.380$. The schooling coefficient nearly *triples*, from 0.069 to 0.172, and the IQ coefficient turns negative and significant — a pattern that strains economic sense (ability now appears to lower wages conditional on schooling) and that signals a badly specified or weakly identified system rather than a refined estimate of the return to schooling. Why does the estimate move so much? If S is genuinely endogenous, then the instrument set used in (b) erroneously includes S itself; by the plim formula for 2SLS with an invalid instrument,

$$\text{plim } \hat{\delta}_{2SLS} - \delta = (\Sigma'_{\mathbf{zx}} \Sigma_{\mathbf{xx}}^{-1} \Sigma_{\mathbf{xz}})^{-1} \Sigma'_{\mathbf{zx}} \Sigma_{\mathbf{xx}}^{-1} \mathbf{E}(\mathbf{x}_i \varepsilon_i),$$

and with $\mathbf{E}(\mathbf{x}_i \varepsilon_i) \neq \mathbf{0}$ in its S -row the inconsistency propagates to all coefficients. Removing S from the instrument set therefore changes the estimand, not just the estimate. Whether the result is closer to the truth depends on whether MED, KWW, MRT, AGE can jointly proxy the exogenous variation in both S and IQ — and Sargan’s statistic says no:

$$J = 13.268 \quad \text{on } \chi^2(2), \quad p = 0.00131,$$

still a decisive rejection, though much smaller than in (c). (With both S and IQ endogenous there are $K - L = 2$ degrees of freedom.)

(f) Efficient two-step GMM, treating S as predetermined (same instrument set as (b)), with $\hat{\mathbf{S}} = \frac{1}{n} \sum_i \hat{\varepsilon}_i^2 \mathbf{x}_i \mathbf{x}_i'$ built from the 2SLS residuals of (b):

$$\hat{\beta}_{GMM} = 0.0768 \text{ (0.0133)}, \quad \hat{\gamma}_{GMM} = -0.0014 \text{ (0.0042)},$$

with $J = 74.165$ on $\chi^2(3)$. The 2SLS standard errors (0.0129 and 0.0039) are indeed *smaller* than the GMM ones (0.0133 and 0.0042). There is no contradiction with PROPOSITION 3.5: the proposition ranks asymptotic variances *given the model is correct*, and here the overidentifying restrictions are massively rejected ($J = 74.2$), so neither estimator is consistent and the efficiency ranking is moot. In addition, the ranking concerns the asymptotic variance of $\sqrt{n}(\hat{\delta} - \delta)$ — the GMM standard errors are robust to conditional heteroskedasticity while the conventional 2SLS ones are not, and robust standard errors are typically larger.

For the C test of the predeterminedness of schooling, use the same $\hat{\mathbf{S}}$ (from (b)’s 2SLS residuals) throughout, per the recipe of Section 3.6. Dropping S from the instrument list and re-running efficient GMM with $\hat{\mathbf{S}}_{11}$ extracted as the appropriate submatrix of $\hat{\mathbf{S}}$ gives $J_1 = 15.997$ on $\chi^2(2)$, so

$$C = J - J_1 = 74.165 - 15.997 = 58.168 \quad \text{on } \chi^2(1), \quad p = 2.4 \times 10^{-14}.$$

The hypothesis that schooling is predetermined is rejected decisively: the covariance between S and the composite error is statistically detectable once the other orthogonality conditions are imposed efficiently. (For completeness, efficient GMM with both S and IQ endogenous — line 5 of Table 3.3 — gives $\hat{\beta} = 0.1758$ (0.0209), $\hat{\gamma} = -0.0093$ (0.0049), $SEE = 0.383$, $J = 11.60$ on $\chi^2(2)$, $p = 0.003$: close to the 2SLS line 4, and still rejecting.)

(g) Dropping MED and KWW from the instrument set of (e) leaves $\mathbf{x}_i = (\mathbf{h}'_i, MRT, AGE)'$ with $K = 13$ instruments against $L = 13$ regressors: the order condition (3.3.5A) still holds, *with equality* — the equation is now exactly identified ($K - L = 0$, so no overidentification test is possible). The 2SLS estimates explode:

$$\hat{\beta} = -5.293 \quad (-529.3\%), \quad \hat{\gamma} = 2.809,$$

a nonsensical negative return to schooling. The first-stage regressions diagnose the problem. With only MRT and AGE as excluded instruments, their t -values are

first stage for	$t(MRT)$	$t(AGE)$	R^2
S	0.49	15.43	0.535
IQ	0.15	3.65	0.165

MRT has essentially no explanatory power for either endogenous regressor, and AGE fails for IQ ; worse, AGE enters the wage equation only through the excluded list, and $AGE = EXPR + S + \text{const}$ approximately (age equals experience plus schooling plus a school-entry age), so the excluded variation that identifies IQ is nearly collinear with the included regressors. The fitted values $\hat{S}$ and $\hat{IQ}$ are therefore almost exact linear combinations of $\mathbf{h}$ and each other — near-multicollinearity in the second stage. The normal equations are close to singular: tiny perturbations of the data swing the coefficient vector wildly, which is what a -529% return to schooling is. This is the weak-instruments phenomenon of Analytical Exercise 3.10 in its purest form: with the rank condition barely satisfied, the 2SLS estimate behaves like a ratio of nearly independent normals, is nowhere near consistent behavior at $n = 758$, and its standard errors (computed under the usual formulas) are meaningless. The remedies the chapter points to are more or better instruments (the subsequent literature’s quarter-of-birth and compulsory-schooling instruments) and reporting first-stage diagnostics — exactly what this part asks for.

Chapter 4

Multiple-Equation GMM

Solutions to all the exercises of Chapter 4 of Fumio Hayashi, *Econometrics* (Princeton University Press, 2000): the review question of Section 4.4, the ten review questions of Section 4.5, the seven review questions of Section 4.6, the four review questions of Section 4.7, the eleven analytical exercises, and the empirical exercise on Christensen and Greene's data (interrelated factor demands). Sections 4.1, 4.2 and 4.3 have no exercises. Assumptions, propositions and equations cited as ASSUMPTION 4.7, PROPOSITION 4.6 or (4.5.12) refer to the book, as do the formulas (A.4)–(A.15) of its Appendix A on partitioned matrices and Kronecker products. The numerical parts are computed by `code/ch4.py`.

Throughout this chapter, $\mathbf{x}_{im}$ ($K_m \times 1$) is the vector of instruments of equation m , $\mathbf{z}_{im}$ ($L_m \times 1$) its regressors, ε_{im} its error, M the number of equations, n the sample size,

$$\mathbf{g}_i \equiv \begin{bmatrix} \mathbf{x}_{i1}\varepsilon_{i1} \\ \vdots \\ \mathbf{x}_{iM}\varepsilon_{iM} \end{bmatrix}, \quad \mathbf{S} = E(\mathbf{g}_i\mathbf{g}_i'), \quad \mathbf{s}_{xy} = \frac{1}{n} \sum_i \begin{bmatrix} \mathbf{x}_{i1}y_{i1} \\ \vdots \\ \mathbf{x}_{iM}y_{iM} \end{bmatrix},$$

$$\mathbf{S}_{xz} = \frac{1}{n} \sum_i \begin{bmatrix} \mathbf{x}_{i1}\mathbf{z}'_{i1} & & \\ & \ddots & \\ & & \mathbf{x}_{iM}\mathbf{z}'_{iM} \end{bmatrix},$$

and $\widehat{\mathbf{W}}$ is a $\sum_m K_m \times \sum_m K_m$ weighting matrix with (m, h) block $\widehat{\mathbf{W}}_{mh}$. When the instruments are common ($\mathbf{x}_{im} = \mathbf{x}_i$, $K_m = K$), $\mathbf{S}_{xz} = \frac{1}{n} \sum_i \text{diag}(\mathbf{x}_i\mathbf{z}'_{i1}, \dots, \mathbf{x}_i\mathbf{z}'_{iM})$.

Section 4.4 Single-Equation versus Multiple-Equation Estimation

Review Question 4.4.1

Set

$$\widehat{\mathbf{W}}_m = \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_{im}\mathbf{x}'_{im} \right)^{-1}, \quad \widehat{\mathbf{W}} = \text{diag}(\widehat{\mathbf{W}}_1, \dots, \widehat{\mathbf{W}}_M),$$

so that $\widehat{\mathbf{W}}$ is block diagonal as in (4.4.1). Since both $\mathbf{S}_{xz}$ and $\widehat{\mathbf{W}}$ are block diagonal, the formulas (A.8) and (A.9) for products and inverses of block diagonal matrices show that $\mathbf{S}'_{xz}\widehat{\mathbf{W}}\mathbf{S}_{xz}$ and $\mathbf{S}'_{xz}\widehat{\mathbf{W}}\mathbf{s}_{xy}$ are block diagonal and stacked, with m -th block

$$\left(\mathbf{S}'_{xz,m}\widehat{\mathbf{W}}_m\mathbf{S}_{xz,m} \right)^{-1} \mathbf{S}'_{xz,m}\widehat{\mathbf{W}}_m\mathbf{s}_{xy,m},$$

where $\mathbf{S}_{xz,m} \equiv \frac{1}{n} \sum_i \mathbf{x}_{im}\mathbf{z}'_{im}$ and $\mathbf{s}_{xy,m} \equiv \frac{1}{n} \sum_i \mathbf{x}_{im}y_{im}$. This is exactly the single-equation GMM estimator of δ_m with weighting matrix $\widehat{\mathbf{W}}_m$, and with the choice above it is the 2SLS estimator of equation m , (3.8.3'). Hence the stacked vector of equation-by-equation 2SLS estimators equals the multiple-equation GMM estimator $\widehat{\delta}(\widehat{\mathbf{W}})$ of (4.2.3) for this block diagonal $\widehat{\mathbf{W}}$. The point, made in the text, is that equation-by-equation estimation is not a different method — it is multiple-equation

GMM with a particular (block diagonal) weighting matrix, and all of the efficiency comparisons of Section 4.4 are comparisons of weighting matrices.

Section 4.5 Special Cases of Multiple-Equation GMM: FIVE, 3SLS, and SUR

Review Question 4.5.1

No to the first. The (m, h) block of (4.1.11) is $\mathbf{S}_{mh} = E(\varepsilon_{im}\varepsilon_{ih}\mathbf{x}_{im}\mathbf{x}_{ih}')'$. The (m, h) block of (4.5.3) is $\hat{\sigma}_{mh} \cdot \frac{1}{n} \sum_i \mathbf{x}_{im}\mathbf{x}_{ih}'$, which converges to

$$\sigma_{mh} E(\mathbf{x}_{im}\mathbf{x}_{ih}') = E[E(\varepsilon_{im}\varepsilon_{ih} \mid \mathbf{x}_{im}, \mathbf{x}_{ih}) \mathbf{x}_{im}\mathbf{x}_{ih}']$$

only if the conditional cross moment $E(\varepsilon_{im}\varepsilon_{ih} \mid \mathbf{x}_{im}, \mathbf{x}_{ih})$ equals the constant σ_{mh} — that is, under ASSUMPTION 4.7. Without conditional homoskedasticity the conditional cross moment varies with the instruments and the two matrices differ: $\hat{\mathbf{S}}$ of (4.5.3) estimates the wrong object.

Yes to the second. Under ASSUMPTION 4.7, $\mathbf{S}$ of (4.1.11) equals (4.5.2) by the calculation (4.5.1). The estimator (4.3.2), $\hat{\mathbf{S}} = \frac{1}{n} \sum_i \hat{\varepsilon}_{im}\hat{\varepsilon}_{ih}\mathbf{x}_{im}\mathbf{x}_{ih}'$ blockwise, is consistent for $\mathbf{S}$ under ASSUMPTIONS 4.1, 4.2, 4.6 by PROPOSITION 4.2, *whatever* the conditional moment structure. Adding conditional homoskedasticity only changes what the limit is called: the limit is $\mathbf{S}$, which under ASSUMPTION 4.7 has the Kronecker-like form (4.5.2).

The asymmetry is instructive. The robust estimator (4.3.2) is consistent under either regime but needs the fourth-moment ASSUMPTION 4.6; the estimator (4.5.3) exploits a structure that holds only under ASSUMPTION 4.7, and in exchange dispenses with fourth moments (it needs only finite $E(\mathbf{z}_{im}\mathbf{z}_{ih}')$ for $\hat{\sigma}_{mh}$ and finite $E(\mathbf{x}_{im}\mathbf{x}_{ih}')$ for the second factor, the latter being implied by the nonsingularity of $\mathbf{S}$).

Review Question 4.5.2

Consistent and asymptotically normal: yes. Efficient: no. The FIVE estimator is the multiple-equation GMM estimator (4.2.3) with $\widehat{\mathbf{W}} = \widehat{\mathbf{S}}^{-1}$, where $\widehat{\mathbf{S}}$ is (4.5.3). By the multiple-equation adaptation of PROPOSITION 3.1, GMM is consistent and asymptotically normal for *any* weighting matrix with a symmetric positive definite probability limit. Under ASSUMPTIONS 4.1–4.5 (no 4.7), $\hat{\sigma}_{mh} \xrightarrow{p} \sigma_{mh}$ by PROPOSITION 4.1 and $\frac{1}{n} \sum_i \mathbf{x}_{im}\mathbf{x}_{ih}' \xrightarrow{p} E(\mathbf{x}_{im}\mathbf{x}_{ih}')$ by ergodic stationarity, so

$$\text{plim}_{n \rightarrow \infty} \widehat{\mathbf{W}} = (\Sigma \odot \mathbf{B})^{-1} \neq \mathbf{S}^{-1}, \quad \mathbf{B}_{mh} \equiv E(\mathbf{x}_{im}\mathbf{x}_{ih}'),$$

where $\Sigma = (\sigma_{mh})$ and $\odot$ denotes the blockwise Hadamard product; the inequality holds because, by Review Question 4.5.1, $\mathbf{S} \neq \Sigma \odot \mathbf{B}$ when conditional homoskedasticity fails. Consistency and asymptotic normality of $\widehat{\delta}(\widehat{\mathbf{W}})$ require only that $\text{plim } \widehat{\mathbf{W}}$ exist and be positive definite; they do not require $\text{plim } \widehat{\mathbf{W}} = \mathbf{S}^{-1}$. The efficiency condition of PROPOSITION 3.5, however, is exactly $\text{plim } \widehat{\mathbf{W}} = \mathbf{S}^{-1}$, and it fails here: without ASSUMPTION 4.7, FIVE is a *less efficient* GMM estimator than the genuinely efficient multiple-equation GMM that uses $\widehat{\mathbf{S}}^{-1}$ from the robust formula (4.3.2).

So the word “efficient” in the name FIVE is conditional: the estimator is efficient within the class indexed by $\widehat{\mathbf{W}}$ precisely when ASSUMPTION 4.7 holds, because only then does its weighting matrix

estimate $\mathbf{S}^{-1}$. This is the multiple-equation counterpart of the Chapter 3 result that 2SLS is the efficient single-equation GMM only under conditional homoskedasticity.

Review Question 4.5.3

Both, and for the same reason: when $L_m = K_m$ for every m , all GMM estimators of the system collapse to the IV estimator, whatever the weighting matrix. If $L_m = K_m$, then $\mathbf{S}_{xz}$ is a square $(\sum_m L_m = \sum_m K_m)$ block diagonal matrix and is nonsingular with probability one by ASSUMPTION 4.4, so for any nonsingular $\widehat{\mathbf{W}}$,

$$\widehat{\delta}(\widehat{\mathbf{W}}) = (\mathbf{S}'_{xz} \widehat{\mathbf{W}} \mathbf{S}_{xz})^{-1} \mathbf{S}'_{xz} \widehat{\mathbf{W}} \mathbf{s}_{xy} = \mathbf{S}_{xz}^{-1} \widehat{\mathbf{W}}^{-1} (\mathbf{S}'_{xz})^{-1} \mathbf{S}'_{xz} \widehat{\mathbf{W}} \mathbf{s}_{xy} = \mathbf{S}_{xz}^{-1} \mathbf{s}_{xy}.$$

The weighting matrix cancels algebraically. Hence both FIVE (with $\widehat{\mathbf{W}} = \widehat{\mathbf{S}}^{-1}$) and equation-by-equation 2SLS (with the block diagonal $\widehat{\mathbf{W}}$ of Review Question 4.4.1) numerically equal the multiple-equation IV estimator $\mathbf{S}_{xz}^{-1} \mathbf{s}_{xy}$, whose m -th block is the single-equation IV estimator $(\sum_i \mathbf{x}_{im} \mathbf{z}'_{im})^{-1} \sum_i \mathbf{x}_{im} y_{im}$ of equation m . Being numerically identical, they are of course asymptotically equivalent, $\sqrt{n} \cdot (\text{difference}) = 0$. This is PROPOSITION 4.3(A) specialized to conditional homoskedasticity: the efficiency debate of Section 4.4 is empty for just identified systems, because there is no freedom left — with exactly as many orthogonality conditions as coefficients, the estimator just sets the sample orthogonality conditions to zero.

Review Question 4.5.4

Superimpose conditional homoskedasticity on PROPOSITION 4.3(B). Three observations connect the pieces.

(i) Under ASSUMPTION 4.7, FIVE is the efficient multiple-equation GMM. This is PROPOSITION 4.4(B): $\widehat{\mathbf{S}}$ of (4.5.3) is consistent for $\mathbf{S}$, so $\widehat{\delta}_{\text{FIVE}} = \widehat{\delta}(\widehat{\mathbf{S}}^{-1})$ is efficient.

(ii) Under ASSUMPTION 4.7, the efficient equation-by-equation GMM is 2SLS. For equation m , conditional homoskedasticity gives $\mathbf{S}_m = E(\varepsilon_{im}^2 \mathbf{x}_{im} \mathbf{x}'_{im}) = \sigma_{mm} E(\mathbf{x}_{im} \mathbf{x}'_{im})$, so $\mathbf{S}_m^{-1} = \sigma_{mm}^{-1} [E(\mathbf{x}_{im} \mathbf{x}'_{im})]^{-1}$ is proportional to the 2SLS weighting matrix; scalar multiples do not change the GMM estimator, and the efficient single-equation GMM with estimated weight is numerically 2SLS (Chapter 3, Section 3.8).

(iii) Under ASSUMPTION 4.7 with $\sigma_{mh} = 0$ ($m \neq h$), the equations are “unrelated” in the sense of (4.4.3). The condition (4.4.3) is $E(\varepsilon_{im} \varepsilon_{ih} \mathbf{x}_{im} \mathbf{x}'_{ih}) = \mathbf{0}$ for $m \neq h$; under conditional homoskedasticity it reduces to $\sigma_{mh} E(\mathbf{x}_{im} \mathbf{x}'_{ih}) = \mathbf{0}$ by (4.5.1), which holds because $\sigma_{mh} = 0$.

Now PROPOSITION 4.3(B) applies: the efficient equation-by-equation GMM (= equation-by-equation 2SLS, by (ii)) and the efficient multiple-equation GMM (= FIVE, by (i)) are asymptotically equivalent when the equations are unrelated (by (iii)). That is, $\sqrt{n}(\widehat{\delta}_{\text{FIVE}} - \widehat{\delta}_{\text{2SLS}}) \xrightarrow{P} \mathbf{0}$.

Economically this is the cleanest statement of what joint estimation buys: with uncorrelated errors across equations there is no “seemingly unrelated” information to exploit, and joint estimation cannot beat running 2SLS on each equation separately — no matter how overidentified the equations are.

Review Question 4.5.5

ASSUMPTION 4.7 is $E(\varepsilon_{im}\varepsilon_{ih} \mid \mathbf{x}_{im}, \mathbf{x}_{ih}) = \sigma_{mh}$. Under (4.5.18) the instruments are common and equal the union of all the regressors: $\mathbf{x}_{im} = \mathbf{x}_i$ for all m , and every $\mathbf{z}_{im}$ is a subset of the elements of $\mathbf{x}_i$. Conditioning on $\mathbf{x}_i$ is then the same as conditioning on the union $(\mathbf{z}_{i1}, \dots, \mathbf{z}_{iM})$: the two σ -fields coincide, because $\mathbf{x}_i$ is a function of $(\mathbf{z}_{i1}, \dots, \mathbf{z}_{iM})$ (it is their union) and each $\mathbf{z}_{im}$ is a function of $\mathbf{x}_i$. Therefore

$$E(\varepsilon_{im}\varepsilon_{ih} \mid \mathbf{x}_i) = E(\varepsilon_{im}\varepsilon_{ih} \mid \mathbf{z}_{i1}, \dots, \mathbf{z}_{iM}),$$

and ASSUMPTION 4.7 reads $E(\varepsilon_{im}\varepsilon_{ih} \mid \mathbf{z}_{i1}, \dots, \mathbf{z}_{iM}) = \sigma_{mh}$, as claimed. The point of the rewording is that under SUR the homoskedasticity restriction is a statement about the *regressors* alone: conditional on all the regressors of all the equations, the error cross moments are constant. Combined with the cross orthogonality (4.5.18') this makes the whole system behave exactly like the generalized regression model of Section 1.6 with sample size Mn , which is what Analytical Exercise 4.3 exploits.

Review Question 4.5.6

Nothing essential happens to the point estimates, but the degrees of freedom of Sargan's statistic change.

The SUR estimator. With MED_i added, $\mathbf{x}_i = (1, S_i, IQ_i, EXPR_i, MED_i)'$ and the SUR condition (4.5.18) — $\mathbf{x}_i$ is the union of the regressors — still holds, since $\mathbf{z}_{i1} = (1, S_i, IQ_i, EXPR_i)'$ and $\mathbf{z}_{i2} = (1, S_i, IQ_i)'$ are still subsets of $\mathbf{x}_i$. The “disappearance of x ” argument of the text does not depend on what else is in $\mathbf{x}_i$: with $\mathbf{z}_{im} = \mathbf{D}_m \mathbf{x}_i$ for a selection matrix $\mathbf{D}_m$,

$$E(\mathbf{z}_{im} \mathbf{x}_i') [E(\mathbf{x}_i \mathbf{x}_i')]^{-1} E(\mathbf{x}_i \mathbf{z}_{ih}') = \mathbf{D}_m E(\mathbf{x}_i \mathbf{x}_i') [E(\mathbf{x}_i \mathbf{x}_i')]^{-1} E(\mathbf{x}_i \mathbf{x}_i') \mathbf{D}_h' = E(\mathbf{z}_{im} \mathbf{z}_{ih}'),$$

so (4.5.16) collapses to (4.5.16') and likewise (4.5.13) to (4.5.13') and (4.5.14) to (4.5.14'), exactly as before. The SUR estimator and its asymptotic variance involve only $E(\mathbf{z}_{im} \mathbf{z}_{ih}')$, $E(\mathbf{z}_{im} y_{ih})$ and Σ : MED_i enters nowhere. Adding an instrument to the common set leaves SUR numerically unchanged — a marked contrast with 3SLS or FIVE, where a new instrument changes every estimate.

Sargan's statistic. Its value changes, because it measures the overidentifying restrictions, and there are now more of them. With $K = 5$ instruments per equation and $L_1 + L_2 = 7$ coefficients, the degrees of freedom are $\sum_m (K - L_m) = 2K - (L_1 + L_2) = 10 - 7 = 3$, against $8 - 7 = 1$ without MED_i : the statistic now also tests whether MED_i is truly orthogonal to both errors. The estimator treats the extra orthogonality conditions as redundant for identification (they are — each equation was already identified by its own regressors) but not for testing. This illustrates the general asymmetry: under the SUR assumption, extra instruments are free lunches for neither efficiency nor the point estimates, but they do buy extra testable restrictions.

Review Question 4.5.7

Under conditional homoskedasticity and common instruments, $\mathbf{S}$ has the Kronecker decomposition (4.5.9),

$$\mathbf{S} = \Sigma \otimes E(\mathbf{x}_i \mathbf{x}_i').$$

By ASSUMPTION 4.5, $\mathbf{S}$ is nonsingular. A Kronecker product is nonsingular if and only if both factors are nonsingular (its eigenvalues are the products $\lambda_r(\Sigma) \lambda_s[E(\mathbf{x}_i \mathbf{x}_i')]$), so both Σ and $E(\mathbf{x}_i \mathbf{x}_i')$ are nonsingular. In particular $E(\mathbf{x}_i \mathbf{x}_i')$ is positive definite, being a symmetric second-moment matrix that is nonsingular.

Under the SUR assumption (4.5.18), the regressors of equation m are a subset of $\mathbf{x}_i$: $\mathbf{z}_{im} = \mathbf{D}_m \mathbf{x}_i$ with $\mathbf{D}_m$ ($L_m \times K$) a selection matrix of full row rank L_m (its rows are distinct rows of $\mathbf{I}_K$). Hence

$$E(\mathbf{x}_i \mathbf{z}_{im}') = E(\mathbf{x}_i \mathbf{x}_i') \mathbf{D}_m',$$

a $K \times L_m$ matrix. Its rank is L_m : if $\mathbf{c}$ is $L_m \times 1$ with $E(\mathbf{x}_i \mathbf{x}_i') \mathbf{D}_m' \mathbf{c} = \mathbf{0}$, nonsingularity of $E(\mathbf{x}_i \mathbf{x}_i')$ forces $\mathbf{D}_m' \mathbf{c} = \mathbf{0}$, and full row rank of $\mathbf{D}_m$ forces $\mathbf{c} = \mathbf{0}$. So $E(\mathbf{x}_{im} \mathbf{z}_{im}') = E(\mathbf{x}_i \mathbf{z}_{im}')$ is of full column rank L_m for every m , which is ASSUMPTION 4.4.

Intuitively: under SUR the rank condition reduces to the absence of multicollinearity among the instruments, and that is already guaranteed by the requirement that $\mathbf{S}$ (which under homoskedasticity contains $E(\mathbf{x}_i \mathbf{x}_i')$ as a factor) be nonsingular. The same argument with $\mathbf{D}_m$ the identity gives the result for the multivariate regression, where $\mathbf{z}_{im} = \mathbf{x}_i$ outright.

Review Question 4.5.8

(a) *Consistent and asymptotically normal: yes; efficient: no.* The SUR estimator is the multiple-equation GMM estimator with $\widehat{\mathbf{W}} = \widehat{\mathbf{S}}^{-1}$, $\widehat{\mathbf{S}}$ from (4.5.3) computed with the OLS residuals; the only role of conditional homoskedasticity in its construction is the form of $\widehat{\mathbf{S}}$. Under ASSUMPTIONS 4.1–4.5 and (4.5.18) alone, $\widehat{\mathbf{S}}$ converges to $\Sigma \odot \mathbf{B}$, a symmetric positive definite matrix (nonsingular because Σ and $E(\mathbf{x}_i \mathbf{x}_i')$ are), so PROPOSITION 3.1 as adapted in Section 4.3 delivers consistency and asymptotic normality. Efficiency, however, requires $\text{plim } \widehat{\mathbf{W}} = \mathbf{S}^{-1}$, and by Review Question 4.5.1 $\text{plim } \widehat{\mathbf{S}} \neq \mathbf{S}$ when ASSUMPTION 4.7 fails. The efficient estimator is the one using (4.3.2); without homoskedasticity, SUR (like FIVE) is a strictly less efficient member of the GMM class. Note that consistency of SUR does use the SUR assumption: the orthogonality conditions it exploits are the cross ones (4.5.18'), and without them SUR is not even consistent — see Review Question 4.5.9.

(b) *Yes*, and conditional homoskedasticity is irrelevant to this algebraic fact. When $\mathbf{z}_{im} = \mathbf{x}_i$ for all m , every equation is just identified ($L_m = K$), $\mathbf{S}_{xz}$ is square and nonsingular, and every GMM estimator — SUR included — collapses to $\mathbf{S}_{xz}^{-1} \mathbf{s}_{xy}$, the equation-by-equation IV (= OLS, since instruments equal regressors) estimator. This is the weighting-matrix cancellation of Review Question 4.5.3, which never invoked ASSUMPTION 4.7. Without conditional homoskedasticity the multivariate regression is still numerically equation-by-equation OLS; what changes is only that the OLS formulas for the standard errors become the robust ones, and that OLS is then efficient in neither sense.

Review Question 4.5.9

SUR is not necessarily consistent. SUR is a GMM estimator exploiting the MK orthogonality conditions $E(\mathbf{x}_i \cdot \varepsilon_{im}) = \mathbf{0}$ for all m , where $\mathbf{x}_i$ is the union of the regressors. Its probability limit solves $\text{plim } \widehat{\delta}(\widehat{\mathbf{S}}^{-1}) - \delta = (\Sigma'_{xz} \mathbf{S}^{-1} \Sigma_{xz})^{-1} \Sigma'_{xz} \mathbf{S}^{-1} E(\mathbf{g}_i)$, and under the hypothesis of this question the m -th block of $E(\mathbf{g}_i) = E(\mathbf{x}_i \varepsilon_{im})$ is *not* zero: $\mathbf{x}_i$ contains the regressors of the other equations, and nothing makes them orthogonal to ε_{im} . The right-hand side is a full linear combination of all the blocks, so generically every element of the asymptotic bias is nonzero — even the coefficients of correctly specified equations inherit the bias, the “contamination” phenomenon of Section 4.4. SUR consistency needs exactly the cross orthogonalities (4.5.18') that the question drops.

The efficient GMM exploiting only $E(\mathbf{z}_{im} \varepsilon_{im}) = \mathbf{0}$ is equation-by-equation OLS. The orthogonality conditions now number $\sum_m L_m$, exactly as many as the coefficients, with

$$\mathbf{g}_i = \begin{bmatrix} \mathbf{z}_{i1} \varepsilon_{i1} \\ \vdots \\ \mathbf{z}_{iM} \varepsilon_{iM} \end{bmatrix}, \quad \Sigma_{xz} = \text{diag}(E(\mathbf{z}_{i1} \mathbf{z}'_{i1}), \dots, E(\mathbf{z}_{iM} \mathbf{z}'_{iM})),$$

so the system is just identified and the efficient GMM estimator is the IV estimator $\mathbf{S}_{xz}^{-1} \mathbf{s}_{xy}$, whose m -th block is the OLS estimator of equation m . Under conditional homoskedasticity this uses *all* the available orthogonality conditions optimally (there being exactly as many as needed), so the answer to the question is: the efficient estimator is equation-by-equation OLS — joint estimation buys nothing because the assumptions buy nothing: each equation's own regressors are the only valid instruments, and they are just enough for that equation.

Remark. Comparing this with Review Question 4.5.4 shows the two ways joint estimation can be pointless: uncorrelated errors (efficiency of separate estimation despite available cross instruments) or unavailable cross instruments (nothing to pool). SUR's efficiency gain needs both correlated errors and cross orthogonality.

Review Question 4.5.10

In the multivariate regression $\mathbf{z}_{im} = \mathbf{x}_i$ for all m , so each equation has K regressors and K instruments: the system is just identified, with as many orthogonality conditions (MK) as coefficients (MK). Sargan's statistic of PROPOSITION 4.5(C) is

$$J = n \mathbf{g}_n(\widehat{\delta})' \widehat{\mathbf{S}}^{-1} \mathbf{g}_n(\widehat{\delta}), \quad \mathbf{g}_n(\widehat{\delta}) = \mathbf{s}_{xy} - \mathbf{S}_{xz} \widehat{\delta}.$$

Since $\mathbf{S}_{xz}$ is a square nonsingular matrix and (by Review Question 4.5.3) the estimator is $\widehat{\delta} = \mathbf{S}_{xz}^{-1} \mathbf{s}_{xy}$ for any weighting matrix,

$$\mathbf{g}_n(\widehat{\delta}) = \mathbf{s}_{xy} - \mathbf{S}_{xz} \mathbf{S}_{xz}^{-1} \mathbf{s}_{xy} = \mathbf{0}, \quad \text{hence} \quad J = 0$$

in every sample, not just asymptotically. A just identified model has no overidentifying restrictions to test: the sample orthogonality conditions are satisfied exactly, so the statistic — and its $\chi^2(0)$ distribution — degenerate to zero. The same algebra shows the J statistic is zero for *any* just identified multiple-equation GMM, regardless of instruments or weighting matrix.

Section 4.6 Common Coefficients

Review Question 4.6.1

If $z_{im2} = 1$: ASSUMPTION 4.4' *fails*. The identification condition is that

$$\Sigma_{xz} = \begin{bmatrix} E(\mathbf{x}_{i1}\mathbf{z}'_{i1}) \\ \vdots \\ E(\mathbf{x}_{iM}\mathbf{z}'_{iM}) \end{bmatrix} \left(\sum_m K_m \times L \right)$$

be of full column rank L . But if $z_{im1} = z_{im2} = 1$ for every observation, then the first two columns of each block $E(\mathbf{x}_{im}\mathbf{z}'_{im})$ are identical, and hence the first two columns of the stacked Σ_{xz} are identical: its rank is at most $L - 1$. The intercept and the coefficient of the second regressor are not separately identified — only their sum is. (With a common coefficient, the second regressor is observationally indistinguishable from a second intercept, which is ruled out by the specification.) Economically: a variable that is constant across observations *and* equations adds nothing to the intercept it duplicates.

If $z_{im2} = m$: *the answer changes* — the system can be identified. Now the second regressor varies across equations, so the second column of the m -th block of Σ_{xz} is m times the first column. Column 2 of the stacked matrix is $(m E(\mathbf{x}_{im}))_{m=1}^M$ stacked, which is not a scalar multiple of column 1, $(E(\mathbf{x}_{im}))_{m=1}^M$, unless all the $E(\mathbf{x}_{im})$ vanish. With generic moments, the two columns are linearly independent and full column rank is restored. Economically, $z_{im2} = m$ is an equation-specific “time dummy” (in panel applications, a time effect that shifts each equation’s intercept by a common slope): the common-coefficient restriction is exactly what makes such equation-level variation informative, because a single coefficient is estimated from the whole stack. This is the mechanism behind the book’s remark that a system with common coefficients can be identified even when no single equation is identified on its own (Analytical Exercise 4.5 is a fully worked instance).

Review Question 4.6.2

No, not in general. From (4.6.8) the sampling error of the RE estimator is

$$\hat{\delta}_{\text{RE}} - \delta = \left[\sum_{m,h} \hat{\sigma}^{mh} \left(\frac{1}{n} \sum_i \mathbf{z}_{im} \mathbf{z}'_{ih} \right) \right]^{-1} \left[\sum_{m,h} \hat{\sigma}^{mh} \left(\frac{1}{n} \sum_i \mathbf{z}_{im} \varepsilon_{ih} \right) \right].$$

The first factor converges to the nonsingular matrix $\text{Avar}(\hat{\delta}_{\text{RE}})^{-1}$ of (4.6.9), so the probability limit of the estimator is δ plus

$$\left[\sum_{m,h} \sigma^{mh} E(\mathbf{z}_{im} \mathbf{z}'_{ih}) \right]^{-1} \left[\sum_{m,h} \sigma^{mh} E(\mathbf{z}_{im} \varepsilon_{ih}) \right].$$

The hypothesis of the question zeroes the diagonal terms of the second bracket, $E(\mathbf{z}_{im} \varepsilon_{im}) = \mathbf{0}$, but leaves the off-diagonal terms: the bracket reduces to

$$\sum_{m \neq h} \sigma^{mh} E(\mathbf{z}_{im} \varepsilon_{ih}),$$

which does not vanish unless either the inverse error covariance matrix is diagonal ($\sigma^{mh} = 0$ for $m \neq h$ — an even stronger condition than $\sigma_{mh} = 0$ for $m \neq h$, and not implied by it) or the cross

orthogonalities $E(\mathbf{z}_{im}\varepsilon_{ih}) = \mathbf{0}$ hold. So RE is inconsistent whenever some regressor of equation m is correlated with the error of another equation — the “cross” orthogonalities are doing real work in PROPOSITION 4.7, not merely inherited from single-equation orthogonality. By contrast, pooled OLS remains consistent under the hypothesis of this question: it exploits only the L combined conditions (4.6.12), $E(\sum_m \mathbf{z}_{im}\varepsilon_{im}) = \mathbf{0}$, which *are* implied by own-equation orthogonality. The contrast is the point: RE weighs each equation’s moment conditions by σ^{mh} and thereby smuggles in the cross conditions that the hypothesis denies, while pooled OLS trades the efficiency that weighting would bring for robustness to their failure.

Review Question 4.6.3

Write the orthogonality conditions as $E(\mathbf{g}_i) = \mathbf{0}$ with

$$\mathbf{g}_i = \begin{bmatrix} \mathbf{z}_{i1}\varepsilon_{i1} \\ \vdots \\ \mathbf{z}_{iM}\varepsilon_{iM} \end{bmatrix} \quad (ML \times 1),$$

so $\mathbf{g}_i = \mathbf{g}(\mathbf{w}_i; \delta)$ is linear in the common δ with $E[\mathbf{g}(\mathbf{w}_i; \tilde{\delta})] = \sigma_{zy} - \Sigma_{zz}\tilde{\delta}$, where

$$\Sigma_{zz} = \begin{bmatrix} E(\mathbf{z}_{i1}\mathbf{z}'_{i1}) \\ \vdots \\ E(\mathbf{z}_{iM}\mathbf{z}'_{iM}) \end{bmatrix} \quad (ML \times L), \quad \sigma_{zy} = \begin{bmatrix} E(\mathbf{z}_{i1}y_{i1}) \\ \vdots \\ E(\mathbf{z}_{iM}y_{iM}) \end{bmatrix}.$$

There are ML orthogonality conditions for L coefficients, so the system is overidentified and the weighting matrix matters. The efficient GMM estimator is (4.2.3) with weight $\widehat{\mathbf{W}} = \widehat{\mathbf{S}}^{-1}$, where $\mathbf{S} = E(\mathbf{g}_i\mathbf{g}'_i)$ is estimated by (4.3.2) from any consistent residuals (e.g. pooled-OLS residuals):

$$\widehat{\delta} = (\mathbf{S}'_{zz}\widehat{\mathbf{S}}^{-1}\mathbf{S}_{zz})^{-1}\mathbf{S}'_{zz}\widehat{\mathbf{S}}^{-1}\mathbf{s}_{zy}, \quad \mathbf{S}_{zz} = \frac{1}{n} \sum_i \begin{bmatrix} \mathbf{z}_{i1}\mathbf{z}'_{i1} \\ \vdots \\ \mathbf{z}_{iM}\mathbf{z}'_{iM} \end{bmatrix},$$

and $\mathbf{s}_{zy}$ the stacked $\frac{1}{n} \sum_i \mathbf{z}_{im}y_{im}$. Under conditional homoskedasticity, $\mathbf{S}$ has blocks $\sigma_{mh}E(\mathbf{z}_{im}\mathbf{z}'_{ih})$, which is *not* of the form $\Sigma \otimes (\cdot)$ when the regressors differ across equations, so the expression does not simplify further.

Same as pooled OLS? No. Pooled OLS uses only the L combined conditions (4.6.12) — the sum over m of the ML conditions here, a strict subset of the information. Equivalently, pooled OLS is the present GMM estimator with the particular (nonoptimal) weighting matrix that treats the ML moments as if their covariance were proportional to the identity across equations; generically the two differ, and pooled OLS is the inefficient member.

Same as random-effects? No. RE exploits MK orthogonality conditions built from the union of the regressors (the cross orthogonalities $E(\mathbf{z}_{im}\varepsilon_{ih}) = \mathbf{0}$ for all m, h), strictly more than the ML assumed here. If those extra conditions are true, RE is the more efficient estimator; if they are false, RE is inconsistent (Review Question 4.6.2) while the estimator derived here remains consistent and is efficient within what is assumed. The three estimators thus form a clean hierarchy of assumptions:

pooled OLS needs (4.6.12); the present estimator needs $E(\mathbf{z}_{im}\varepsilon_{im}) = \mathbf{0}$; RE needs the full cross orthogonality — and efficiency rises exactly in step with the strength of what is assumed true.

Review Question 4.6.4

Recall (4.6.15): $\mathbf{y}_i = (y_{i1}, \dots, y_{iM})'$ ($M \times 1$) and $\mathbf{Z}_i$ ($M \times L$) has m -th row $\mathbf{z}'_{im}$.

For Σ_{xz} . By definition (4.6.4), Σ_{xz} is the $\sum_m K \times L$ stacked matrix with m -th block $E(\mathbf{x}_i \mathbf{z}'_{im})$. The defining property of the Kronecker product gives

$$\mathbf{Z}_i \otimes \mathbf{x}_i = \begin{bmatrix} \mathbf{z}'_{i1} \otimes \mathbf{x}_i \\ \vdots \\ \mathbf{z}'_{iM} \otimes \mathbf{x}_i \end{bmatrix}, \quad \text{with } m\text{-th block} \quad \mathbf{z}'_{im} \otimes \mathbf{x}_i = \begin{bmatrix} z_{im1}\mathbf{x}_i \\ \vdots \\ z_{imL}\mathbf{x}_i \end{bmatrix} = \mathbf{x}_i \mathbf{z}'_{im},$$

the last equality because $z_{imj}\mathbf{x}_i$ is the j -th column of $\mathbf{x}_i \mathbf{z}'_{im}$ stacked — i.e., the element of $\mathbf{z}'_{im} \otimes \mathbf{x}_i$ in position $((j-1)K + k)$ is $z_{imj}x_{ik}$, which is the (k, j) element of $\mathbf{x}_i \mathbf{z}'_{im}$ read down its columns. Hence $\mathbf{Z}_i \otimes \mathbf{x}_i$ stacks $\mathbf{x}_i \mathbf{z}'_{im}$ over m , and taking expectations,

$$\Sigma_{xz} = E(\mathbf{Z}_i \otimes \mathbf{x}_i).$$

(The hint $\mathbf{x}_i \mathbf{z}'_{im} = \mathbf{z}'_{im} \otimes \mathbf{x}_i$ is the identity $\text{vec}(\mathbf{x}_i \mathbf{z}'_{im}) = \mathbf{z}_{im} \otimes \mathbf{x}_i$ in disguise.)

For $\mathbf{s}_{xy}$. Its m -th block is $\frac{1}{n} \sum_i \mathbf{x}_i y_{im}$; and the m -th block of $\mathbf{y}_i \otimes \mathbf{x}_i$ is $y_{im}\mathbf{x}_i$, so

$$\mathbf{s}_{xy} = \frac{1}{n} \sum_{i=1}^n \mathbf{y}_i \otimes \mathbf{x}_i.$$

For $\mathbf{S}_{xz}$. Identically, with $\frac{1}{n} \sum_i$ in place of E ,

$$\mathbf{S}_{xz} = \frac{1}{n} \sum_{i=1}^n \mathbf{Z}_i \otimes \mathbf{x}_i.$$

These compact forms are what make programming the common-coefficient estimators pleasant: the whole GMM machinery is driven by the n pairs $(\mathbf{y}_i, \mathbf{Z}_i)$ and the instrument vector $\mathbf{x}_i$.

Review Question 4.6.5

(a) By construction $\mathbf{z}'_{im}\delta = \sum_h \mathbf{0}'\delta_h^* + \mathbf{z}_{im}^{*'}\delta_m^* = \mathbf{z}_{im}^{*'}\delta_m^*$, so $y_{im} = \mathbf{z}'_{im}\delta + \varepsilon_{im}$ is exactly equation m of (4.1.1). The unrestricted system is the common-coefficient system in which the regressors of equation m are padded with zeros outside the block corresponding to δ_m^* : the “restriction” that all equations share δ is vacuous because each equation loads only on its own components of δ .

(b) With this definition the m -th block of Σ_{xz} in (4.6.4) is

$$E(\mathbf{x}_{im} \mathbf{z}'_{im}) = [\mathbf{0} \cdots \mathbf{0} \quad E(\mathbf{x}_{im} \mathbf{z}_{im}^{*'}) \quad \mathbf{0} \cdots \mathbf{0}],$$

the nonzero submatrix sitting in columns $L_1 + \cdots + L_{m-1} + 1$ through $L_1 + \cdots + L_m$. The stacked $\sum_m K_m \times \sum_m L_m$ matrix Σ_{xz} is therefore block diagonal with m -th diagonal block $E(\mathbf{x}_{im} \mathbf{z}_{im}^{*'})$ ($K_m \times L_m$). A block diagonal matrix is of full column rank if and only if every diagonal block is of full column rank (its rank is the sum of the block ranks, and the column rank can reach $\sum_m L_m$ only if block m has rank L_m for every m). Hence ASSUMPTION 4.4' for the padded system is exactly ASSUMPTION 4.4 for the original one.

(c) By the same padding, the m -th block row of the sample moment $\mathbf{S}_{xz}$ of (4.6.5) is

$$\frac{1}{n} \sum_{i=1}^n \mathbf{x}_{im} \mathbf{z}'_{im} = [\mathbf{0} \cdots \mathbf{0} \quad \frac{1}{n} \sum_i \mathbf{x}_{im} \mathbf{z}_{im}^{*'} \quad \mathbf{0} \cdots \mathbf{0}],$$

so $\mathbf{S}_{xz}$ is block diagonal with m -th block $\frac{1}{n} \sum_i \mathbf{x}_{im} \mathbf{z}_{im}^{*'} — the $\mathbf{S}_{xz}$ of (4.2.2). Products with block diagonal matrices reduce to products of blocks: $\mathbf{S}'_{xz} \widehat{\mathbf{W}} \mathbf{S}_{xz}$ has (m, h) block$

$$\left(\frac{1}{n} \sum_i \mathbf{z}_{im}^* \mathbf{x}'_{im} \right) \widehat{\mathbf{W}}_{mh} \left(\frac{1}{n} \sum_i \mathbf{x}_{ih} \mathbf{z}_{ih}^{*'} \right),$$

and likewise $\mathbf{S}'_{xz} \widehat{\mathbf{W}} \mathbf{s}_{xy}$ has m -th block

$$\sum_h \left(\frac{1}{n} \sum_i \mathbf{z}_{im}^* \mathbf{x}'_{im} \right) \widehat{\mathbf{W}}_{mh} \left(\frac{1}{n} \sum_i \mathbf{x}_{ih} y_{ih} \right).$$

Substituting into (4.6.6) reproduces (4.2.6) term by term, with $\mathbf{z}_{im}^*$ in the role of the $\mathbf{z}_{im}$ of (4.2.6).

This is why the common-coefficient restriction “isn’t”: the model of Section 4.1 is nested in the model of Section 4.6, and everything about the latter (identification, the GMM formula, the efficiency theory) specializes to the former. The book develops separate-coefficients first only because the notation is easier there.

Review Question 4.6.6

Exactly as in Review Question 4.5.7, ASSUMPTION 4.5 (nonsingularity of $\mathbf{S}$) and the Kronecker decomposition $\mathbf{S} = \Sigma \otimes \mathbf{E}(\mathbf{x}_i \mathbf{x}'_i)$ under ASSUMPTION 4.7 with common instruments imply that $\mathbf{E}(\mathbf{x}_i \mathbf{x}'_i)$ is nonsingular. As shown there, this in turn implies ASSUMPTION 4.4: each $\mathbf{E}(\mathbf{x}_i \mathbf{z}'_{im})$ is of full column rank L (here $L_m = L$ for all m). Now appeal to the book’s footnote: if $\mathbf{E}(\mathbf{x}_i \mathbf{z}'_{im})$ has full column rank L for *some* m , its L columns are linearly independent, and those same L columns appear as columns of the stacked $\sum_m K \times L$ matrix Σ_{xz} of (4.6.4); a matrix containing L linearly independent columns has column rank at least L , and having only L columns, exactly L . Hence

$$\text{ASSUMPTION 4.4 (for any single } m) \implies \text{ASSUMPTION 4.4'}.$$

The chain is $4.5+4.7 \Rightarrow \mathbf{E}(\mathbf{x}_i \mathbf{x}'_i)$ nonsingular $\Rightarrow 4.4 \Rightarrow 4.4'$, so the identification assumption can be dropped from PROPOSITION 4.7. That 4.4' is strictly weaker than 4.4 is the content of the remark in the text: with common coefficients a system can be identified even if no equation is identified in isolation, because the stacked matrix can have full column rank without any block having it — the pooling of moment conditions across equations, not each equation separately, pins down δ (see Analytical Exercise 4.5).

Review Question 4.6.7

(a) Write out $\mathbf{E}(\mathbf{Z}'_i \Sigma^{-1} \mathbf{Z}_i)$ and reintroduce the instruments. By (4.6.16B) with $\mathbf{C} = \Sigma^{-1}$,

$$\mathbf{E}(\mathbf{Z}'_i \Sigma^{-1} \mathbf{Z}_i) = \sum_{m,h} \sigma^{mh} \mathbf{E}(\mathbf{z}_{im} \mathbf{z}'_{ih}).$$

Under the SUR assumption $\mathbf{z}_{im} = \mathbf{D}_m \mathbf{x}_i$ with $\mathbf{D}_m$ ($L \times K$) of full row rank, so $\mathbf{z}_{im} \mathbf{z}'_{ih} = \mathbf{D}_m \mathbf{x}_i \mathbf{x}'_i \mathbf{D}'_h$ and

$$\mathbf{E}(\mathbf{Z}'_i \Sigma^{-1} \mathbf{Z}_i) = \Sigma'_{xz} [\Sigma \otimes \mathbf{E}(\mathbf{x}_i \mathbf{x}'_i)]^{-1} \Sigma_{xz} \quad (\text{the “}x\text{ reappears” step}),$$

because the m -th block of Σ_{xz} is $\mathbf{E}(\mathbf{x}_i \mathbf{z}'_{im}) = \mathbf{E}(\mathbf{x}_i \mathbf{x}'_i) \mathbf{D}'_m$, and the (m, h) block of the product on the right is

$$\mathbf{D}_m \mathbf{E}(\mathbf{x}_i \mathbf{x}'_i) \cdot \sigma^{mh} [\mathbf{E}(\mathbf{x}_i \mathbf{x}'_i)]^{-1} \cdot \mathbf{E}(\mathbf{x}_i \mathbf{x}'_i) \mathbf{D}'_h = \sigma^{mh} \mathbf{E}(\mathbf{z}_{im} \mathbf{z}'_{ih}),$$

matching the sum above term by term.

Now the three factors: $\Sigma \otimes \mathbf{E}(\mathbf{x}_i \mathbf{x}'_i)$ is nonsingular by ASSUMPTIONS 4.5 and 4.7 (Review Question 4.5.7), hence so is its inverse, which is in fact positive definite; and Σ_{xz} is of full column rank L by ASSUMPTION 4.4'. A matrix of the form $\mathbf{A}' \mathbf{B} \mathbf{A}$ with $\mathbf{B}$ positive definite and $\mathbf{A}$ of full column rank is positive definite: for $\mathbf{c} \neq \mathbf{0}$, $\mathbf{A} \mathbf{c} \neq \mathbf{0}$ and $\mathbf{c}' \mathbf{A}' \mathbf{B} \mathbf{A} \mathbf{c} = (\mathbf{A} \mathbf{c})' \mathbf{B} (\mathbf{A} \mathbf{c}) > 0$. So $\mathbf{E}(\mathbf{Z}'_i \Sigma^{-1} \mathbf{Z}_i)$ is positive definite, hence nonsingular, and the inverse in (4.6.9') exists. (As Review Question 4.6.6 showed, ASSUMPTION 4.4' itself is redundant, so the result needs only ASSUMPTIONS 4.5 and 4.7.)

(b) This is part (a) with Σ^{-1} replaced by $\mathbf{I}_M$: by the same algebra,

$$\mathbf{E}(\mathbf{Z}'_i \mathbf{Z}_i) = \sum_m \mathbf{E}(\mathbf{z}_{im} \mathbf{z}'_{im}) = \Sigma'_{xz} [\mathbf{I}_M \otimes \mathbf{E}(\mathbf{x}_i \mathbf{x}'_i)]^{-1} \Sigma_{xz},$$

a product $\mathbf{A}' \mathbf{B} \mathbf{A}$ with $\mathbf{B}$ positive definite and $\mathbf{A} = \Sigma_{xz}$ of full column rank. Hence positive definite and invertible. The invertibility of $\mathbf{E}(\mathbf{Z}'_i \mathbf{Z}_i)$ is also the no-multicollinearity condition of the pooled regression: pooled OLS exists because the pooled second-moment matrix of the regressors is nonsingular.

Section 4.7 Application: Interrelated Factor Demands

Review Question 4.7.1

The translog cost function (4.7.1) is (dropping the firm subscript)

$$\begin{aligned} \log C = & \alpha_0 + \sum_{j=1}^3 \alpha_j \log p_j + \alpha_Q \log Q + \frac{1}{2} \sum_{j=1}^3 \sum_{k=1}^3 \gamma_{jk} \log p_j \log p_k \\ & + \sum_{j=1}^3 \gamma_{jQ} \log p_j \log Q + \frac{1}{2} \gamma_{QQ} (\log Q)^2 + \varepsilon. \end{aligned}$$

Differentiate with respect to $\log p_j$. The linear terms contribute α_j and the output-interaction terms $\gamma_{jQ} \log Q$. The quadratic form contributes

$$\frac{\partial}{\partial \log p_j} \frac{1}{2} \sum_{h=1}^3 \sum_{k=1}^3 \gamma_{hk} \log p_h \log p_k = \frac{1}{2} \left(\sum_{k=1}^3 \gamma_{jk} \log p_k + \sum_{h=1}^3 \gamma_{hj} \log p_h \right) = \sum_{k=1}^3 \gamma_{jk} \log p_k,$$

where the first sum comes from the terms with $h = j$ and the second from those with $k = j$, and the two sums are equal *because* $\gamma_{jk} = \gamma_{kj}$. This is exactly where symmetry enters: without it, the

derivative would be $\frac{1}{2} \sum_k (\gamma_{jk} + \gamma_{kj}) \log p_k$. Collecting,

$$\frac{\partial \log C}{\partial \log p_j} = \alpha_j + \sum_{k=1}^3 \gamma_{jk} \log p_k + \gamma_{jQ} \log Q,$$

which is (4.7.6). Combined with Shephard's lemma in logarithmic form (4.7.5), $s_j = \partial \log C / \partial \log p_j$, this yields the share equations (4.7.7).

Remark. As the text notes, symmetry (4.7.2) is not really a restriction on the cost function: any quadratic form in $\log p$ can be written with a symmetric coefficient matrix, because only $(\gamma_{jk} + \gamma_{kj})/2$ ever appears. The substantive cross-equation “symmetry restrictions” on the share system — that the $\log p_k$ coefficient in the s_j equation equals the $\log p_j$ coefficient in the s_k equation — come from the further fact that *the same* matrix (γ_{jk}) shows up in every share equation, i.e. from Young's theorem (equality of mixed partials) applied to the single underlying cost function.

Review Question 4.7.2

A cost function must be homogeneous of degree one in factor prices: multiplying all prices by $\lambda > 0$ multiplies the minimum cost of producing a given output by λ (the feasible input bundle is unchanged). For (4.7.1), replace each p_j by λp_j , so $\log p_j \mapsto \log p_j + \log \lambda$. Abbreviating $\Lambda \equiv \log \lambda$ and writing out the change in $\log C$:

$$\log C(\lambda \mathbf{p}) - \log C(\mathbf{p}) = \left(\sum_j \alpha_j \right) \Lambda + \frac{1}{2} \sum_{j,k} \gamma_{jk} [\Lambda^2 + \Lambda(\log p_j + \log p_k)] + \sum_j \gamma_{jQ} \Lambda \log Q.$$

Collect powers of Λ , using $\sum_{j,k} \gamma_{jk} \Lambda \log p_j = \sum_j (\sum_k \gamma_{jk}) \Lambda \log p_j$:

$$\log C(\lambda \mathbf{p}) - \log C(\mathbf{p}) = \Lambda \left[\sum_j \alpha_j + \sum_j \left(\sum_k \gamma_{jk} \right) \log p_j + \left(\sum_j \gamma_{jQ} \right) \log Q \right] + \frac{\Lambda^2}{2} \sum_{j,k} \gamma_{jk}.$$

Homogeneity of degree one requires this to equal Λ for *all* prices, output and Λ . Matching the Λ^2 term, then the coefficients of $\log p_j$ and of $\log Q$ in the Λ term, and finally the constant part of the Λ term:

$$\sum_{j,k} \gamma_{jk} = 0; \quad \sum_k \gamma_{jk} = 0 \quad (j = 1, 2, 3); \quad \sum_j \gamma_{jQ} = 0; \quad \sum_j \alpha_j = 1.$$

This is (4.7.10):

$$\sum_{j=1}^3 \alpha_j = 1, \quad \sum_{k=1}^3 \gamma_{jk} = 0 \quad (j = 1, 2, 3), \quad \sum_{j=1}^3 \gamma_{jQ} = 0.$$

(The first three conditions imply the aggregate $\sum_{j,k} \gamma_{jk} = 0$, so the Λ^2 restriction adds nothing once the row sums vanish.) The restrictions are necessary and sufficient because the two sides are polynomials in $(\Lambda, \log \mathbf{p}, \log Q)$, and polynomials agreeing everywhere must agree coefficient by coefficient.

Review Question 4.7.3

The three sets of restrictions are

$$\text{adding-up (4.7.12):} \quad \sum_j \alpha_j = 1, \quad \sum_j \gamma_{j1} = \sum_j \gamma_{j2} = \sum_j \gamma_{j3} = 0, \quad \sum_j \gamma_{jQ} = 0,$$

homogeneity (4.7.13): $\gamma_{j1} + \gamma_{j2} + \gamma_{j3} = 0 \quad (j = 1, 2, 3),$

symmetry (4.7.14): $\gamma_{12} = \gamma_{21}, \quad \gamma_{13} = \gamma_{31}, \quad \gamma_{23} = \gamma_{32}.$

Suppose adding-up and the first two homogeneity equations hold, and suppose $\gamma_{12} = \gamma_{21}$ (one symmetry equation). To derive $\gamma_{13} = \gamma_{31}$: from the first homogeneity equation, $\gamma_{13} = -\gamma_{11} - \gamma_{12}$; from the first adding-up column, $\gamma_{31} = -\gamma_{11} - \gamma_{21} = -\gamma_{11} - \gamma_{12}$. Hence $\gamma_{13} = \gamma_{31}$. To derive $\gamma_{23} = \gamma_{32}$: from the second homogeneity equation, $\gamma_{23} = -\gamma_{21} - \gamma_{22} = -\gamma_{12} - \gamma_{22}$; from the second adding-up column, $\gamma_{32} = -\gamma_{12} - \gamma_{22}$. Hence $\gamma_{23} = \gamma_{32}$.

The same algebra shows that one of the three homogeneity equations is implied by the other two given adding-up: the third homogeneity equation says the third row sums to zero, and the third adding-up column says the third column sums to zero; the sum of all nine γ 's computed by rows (two zero row sums plus the third row) equals that computed by columns (three zero column sums), forcing the third row sum to zero. So the eleven restrictions (4.7.12)–(4.7.14) contain exactly eight independent ones (four adding-up, two independent homogeneity, one independent symmetry, and one further homogeneity equation not implied by symmetry), leaving the fifteen share-equation coefficients described by $15 - 8 = 7$ free parameters. This count is what makes the system (4.7.16) — two equations, seven coefficients — the correct estimation object.

Review Question 4.7.4

Let $\mathbf{y}_j$ ($n \times 1$) be the vector of the j -th share, $\mathbf{X}$ ($n \times K$, $K = 4$) the common regressor matrix, $\mathbf{M} = \mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ the annihilator, and $\mathbf{e}_j = \mathbf{M}\mathbf{y}_j$ the j -th residual vector.

Adding-up of the coefficient estimates. Since $s_1 + s_2 + s_3 = 1$ for every firm and the constant is a regressor, $\mathbf{y}_1 + \mathbf{y}_2 + \mathbf{y}_3 = \mathbf{X}\mathbf{u}$ where $\mathbf{u} = (1, 0, 0, 0)'$ (the vector of ones is the first column of $\mathbf{X}$). The OLS coefficients are $\mathbf{b}_j = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}_j$, so

$$\mathbf{b}_1 + \mathbf{b}_2 + \mathbf{b}_3 = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\mathbf{u} = \mathbf{u}.$$

Reading this coefficient by coefficient: the intercepts add to one ($\hat{\alpha}_1 + \hat{\alpha}_2 + \hat{\alpha}_3 = 1$) and every slope adds to zero ($\sum_j \hat{\gamma}_{jk} = 0$ for each price regressor k , $\sum_j \hat{\gamma}_{jQ} = 0$). So the unrestricted OLS estimates automatically satisfy the adding-up restrictions (4.7.12) — they are not restrictions on the estimator but properties of any least-squares fit to data that add up, given that the sum of the dependent variables lies in the column space of $\mathbf{X}$.

Singularity of the residual covariance matrix. The residuals inherit the adding-up identity:

$$\mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3 = \mathbf{M}(\mathbf{y}_1 + \mathbf{y}_2 + \mathbf{y}_3) = \mathbf{M}\mathbf{X}\mathbf{u} = \mathbf{0},$$

because $\mathbf{M}\mathbf{X} = \mathbf{0}$ by (1.2.11). The estimated covariance matrix has elements $\hat{\sigma}_{jh} = \frac{1}{n}\mathbf{e}_j'\mathbf{e}_h$; its row sums are

$$\sum_{h=1}^3 \hat{\sigma}_{jh} = \frac{1}{n}\mathbf{e}_j'(\mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3) = 0,$$

so $\hat{\Sigma}\mathbf{1}_3 = \mathbf{0}$: the matrix has a zero eigenvalue and is singular, in finite samples, exactly. This is why one equation must be dropped before the SUR/random-effects machinery — which inverts $\hat{\Sigma}$ — can be applied, and it is why the invariance question (which equation?) arises at all: the answer

given in the text, and verified numerically in the empirical exercise below, is that with equation-by-equation OLS residuals the choice does not matter, because the singularity makes the relevant submatrices of the (singular) $\widehat{\Sigma}$ equivalent for the purpose of the second stage.

Problem Set for Chapter 4: Analytical Exercises

Analytical Exercise 4.1

The model. The m -th block row of $\mathbf{Z}\delta$ is $\mathbf{Z}_m\delta_m$, whose i -th element is $\mathbf{z}'_{im}\delta_m$; so $\mathbf{y} = \mathbf{Z}\delta + \varepsilon$ reads, block by block and element by element, $y_{im} = \mathbf{z}'_{im}\delta_m + \varepsilon_{im}$, which is (4.1.1).

The estimator. With common instruments, (4.5.10) and (4.5.11) give

$$\widehat{\mathbf{W}} = \widehat{\Sigma}^{-1} \otimes \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1}, \quad \text{i.e.} \quad \widehat{\mathbf{W}}_{mh} = \widehat{\sigma}^{mh} \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1},$$

while

$$\frac{1}{n} \sum_i \mathbf{z}_{im} \mathbf{x}'_i = \frac{1}{n} \mathbf{Z}'_m \mathbf{X}, \quad \frac{1}{n} \sum_i \mathbf{x}_i y_{ih} = \frac{1}{n} \mathbf{X}' \mathbf{y}_h.$$

Substituting into the full 3SLS formula (4.5.12), the (m, h) block of $\mathbf{S}'_{xz} \widehat{\mathbf{W}} \mathbf{s}_{xz}$ is

$$\left(\frac{\mathbf{Z}'_m \mathbf{X}}{n} \right) \widehat{\sigma}^{mh} \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1} \left(\frac{\mathbf{X}'\mathbf{Z}_h}{n} \right) = \frac{1}{n} \widehat{\sigma}^{mh} \mathbf{Z}'_m \underbrace{\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'}_{\mathbf{P}} \mathbf{Z}_h = \frac{1}{n} \widehat{\sigma}^{mh} \mathbf{Z}'_m \mathbf{P} \mathbf{Z}_h.$$

By the definition of the Kronecker product of partitioned matrices (A.11), the matrix whose (m, h) block is $\widehat{\sigma}^{mh} \mathbf{Z}'_m \mathbf{P} \mathbf{Z}_h$ is $\mathbf{Z}'(\widehat{\Sigma}^{-1} \otimes \mathbf{P})\mathbf{Z}$, so

$$\mathbf{S}'_{xz} \widehat{\mathbf{W}} \mathbf{s}_{xz} = \frac{1}{n} \mathbf{Z}'(\widehat{\Sigma}^{-1} \otimes \mathbf{P})\mathbf{Z}.$$

Similarly the m -th block of $\mathbf{S}'_{xz} \widehat{\mathbf{W}} \mathbf{s}_{xy}$ is $\frac{1}{n} \sum_h \widehat{\sigma}^{mh} \mathbf{Z}'_m \mathbf{P} \mathbf{y}_h$, i.e. $\mathbf{S}'_{xz} \widehat{\mathbf{W}} \mathbf{s}_{xy} = \frac{1}{n} \mathbf{Z}'(\widehat{\Sigma}^{-1} \otimes \mathbf{P})\mathbf{y}$. Substituting into (4.2.3) and cancelling the factors of n ,

$$\widehat{\delta}_{3SLS} = [\mathbf{Z}'(\widehat{\Sigma}^{-1} \otimes \mathbf{P})\mathbf{Z}]^{-1} \mathbf{Z}'(\widehat{\Sigma}^{-1} \otimes \mathbf{P})\mathbf{y}.$$

The estimated asymptotic variance. From (4.3.4) with $\widehat{\mathbf{S}}^{-1} = \widehat{\Sigma}^{-1} \otimes (\frac{1}{n} \mathbf{X}'\mathbf{X})^{-1}$,

$$\widehat{\text{Avar}}(\widehat{\delta}_{3SLS}) = \left(\frac{1}{n} \mathbf{Z}'(\widehat{\Sigma}^{-1} \otimes \mathbf{P})\mathbf{Z} \right)^{-1} = n [\mathbf{Z}'(\widehat{\Sigma}^{-1} \otimes \mathbf{P})\mathbf{Z}]^{-1}.$$

The two sample moments. Directly,

$$\widehat{A}_{mh} = \frac{1}{n} \sum_i \mathbf{z}_{im} \mathbf{x}'_i \left(\frac{1}{n} \sum_i \mathbf{x}_i \mathbf{x}'_i \right)^{-1} \frac{1}{n} \sum_i \mathbf{x}_i \mathbf{z}'_{ih} = \frac{1}{n} \mathbf{Z}'_m \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Z}_h = \frac{1}{n} \mathbf{Z}'_m \mathbf{P} \mathbf{Z}_h,$$

and the identical computation with $\mathbf{y}_h$ in place of $\mathbf{Z}_h$ gives $\widehat{c}_{mh} = \frac{1}{n} \mathbf{Z}'_m \mathbf{P} \mathbf{y}_h$. Everything about 3SLS is therefore a function of the cross-product matrices of the data alone, with the instrument set entering only through the projection matrix $\mathbf{P}$: 3SLS is GLS on the stacked system with covariance $\widehat{\Sigma} \otimes \mathbf{I}_n$ applied to *predicted* regressors and dependent variables $(\mathbf{P}\mathbf{Z}, \mathbf{P}\mathbf{y})$ — the three stages being: (i) first-stage regressions to form $\mathbf{P}$, (ii) 2SLS residuals to form $\widehat{\Sigma}$, (iii) the GLS step displayed above.

Analytical Exercise 4.2

(a) Under (4.5.18) the columns of each $\mathbf{Z}_m$ are taken from the columns of $\mathbf{X}$, so $\mathbf{P}\mathbf{Z}_m = \mathbf{Z}_m$ (projection onto a space that already contains the columns leaves them unchanged). From Analytical Exercise 4.1,

$$\hat{A}_{mh} = \frac{1}{n} \mathbf{Z}'_m \mathbf{P} \mathbf{Z}_h = \frac{1}{n} \mathbf{Z}'_m \mathbf{Z}_h, \quad \hat{c}_{mh} = \frac{1}{n} \mathbf{Z}'_m \mathbf{P} \mathbf{y}_h = \frac{1}{n} \mathbf{Z}'_m \mathbf{y}_h,$$

which are (4.5.13') and (4.5.14'): the instruments disappear from the formulas.

(b) Substitute $\mathbf{P}\mathbf{Z}_h = \mathbf{Z}_h$ and $\mathbf{P}\mathbf{y}_h = \mathbf{y}_h$ into the 3SLS formulas of Analytical Exercise 4.1: $\mathbf{Z}'(\hat{\Sigma}^{-1} \otimes \mathbf{P})\mathbf{Z}$ becomes $\mathbf{Z}'(\hat{\Sigma}^{-1} \otimes \mathbf{I}_n)\mathbf{Z}$ (its (m, h) block is $\hat{\sigma}^{mh} \mathbf{Z}'_m \mathbf{Z}_h$), and likewise for the $\mathbf{y}$ term. Hence

$$\hat{\delta}_{SUR} = [\mathbf{Z}'(\hat{\Sigma}^{-1} \otimes \mathbf{I}_n)\mathbf{Z}]^{-1} \mathbf{Z}'(\hat{\Sigma}^{-1} \otimes \mathbf{I}_n)\mathbf{y}, \quad \widehat{\text{Avar}}(\hat{\delta}_{SUR}) = n[\mathbf{Z}'(\hat{\Sigma}^{-1} \otimes \mathbf{I}_n)\mathbf{Z}]^{-1}.$$

This is recognizable as feasible GLS on the stacked Mn -observation regression $\mathbf{y} = \mathbf{Z}\delta + \boldsymbol{\varepsilon}$ with error covariance estimated by $\hat{\Sigma} \otimes \mathbf{I}_n$ — the classical SUR formula of Zellner (1962).

(c) If $\mathbf{z}_{im} = \mathbf{x}_i$ for all m , then $\mathbf{Z}_m = \mathbf{X}$ for all m and $\mathbf{Z} = \mathbf{I}_M \otimes \mathbf{X}$. Using (A.13) twice,

$$\begin{aligned} \mathbf{Z}'(\hat{\Sigma}^{-1} \otimes \mathbf{I}_n)\mathbf{Z} &= (\mathbf{I}_M \otimes \mathbf{X}')(\hat{\Sigma}^{-1} \otimes \mathbf{I}_n)(\mathbf{I}_M \otimes \mathbf{X}) = \hat{\Sigma}^{-1} \otimes \mathbf{X}'\mathbf{X}, \\ \mathbf{Z}'(\hat{\Sigma}^{-1} \otimes \mathbf{I}_n)\mathbf{y} &= (\hat{\Sigma}^{-1} \otimes \mathbf{X}')\mathbf{y}. \end{aligned}$$

Therefore

$$\hat{\delta}_{SUR} = [\hat{\Sigma}^{-1} \otimes \mathbf{X}'\mathbf{X}]^{-1}(\hat{\Sigma}^{-1} \otimes \mathbf{X}')\mathbf{y} = [\hat{\Sigma} \otimes (\mathbf{X}'\mathbf{X})^{-1}](\hat{\Sigma}^{-1} \otimes \mathbf{X}')\mathbf{y} = [\mathbf{I}_M \otimes (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\mathbf{y},$$

by (A.15) $((\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1})$ and (A.13). The m -th K -dimensional block of this MK -vector is $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}_m$, the OLS estimator of equation m : the multivariate regression is equation-by-equation OLS, and $\hat{\Sigma}$ cancels exactly. This is the algebraic content of case (a) of the “SUR versus OLS” discussion: with identical regressors, joint estimation cannot beat separate estimation.

(d) The RE formulas (4.6.8) and (4.6.10) contain, after the disappearance of x under the SUR assumption, the triple sums $\sum_i \sum_{m,h} \hat{\sigma}^{mh} \mathbf{z}_{im} \mathbf{z}'_{ih}$ and $\sum_i \sum_{m,h} \hat{\sigma}^{mh} \mathbf{z}_{im} y_{ih}$. With $\mathbf{Z}$ now the stacked $(Mn \times L)$ matrix whose m -th block of rows is $\mathbf{Z}_m$,

$$\begin{aligned} \mathbf{Z}'(\hat{\Sigma}^{-1} \otimes \mathbf{I}_n)\mathbf{Z} &= \sum_{m,h} \hat{\sigma}^{mh} \mathbf{Z}'_m \mathbf{Z}_h = \sum_{i=1}^n \sum_{m,h} \hat{\sigma}^{mh} \mathbf{z}_{im} \mathbf{z}'_{ih}, \\ \mathbf{Z}'(\hat{\Sigma}^{-1} \otimes \mathbf{I}_n)\mathbf{y} &= \sum_{m,h} \hat{\sigma}^{mh} \mathbf{Z}'_m \mathbf{y}_h = \sum_{i=1}^n \sum_{m,h} \hat{\sigma}^{mh} \mathbf{z}_{im} y_{ih}, \end{aligned}$$

where the first equalities use (A.11) and the second interchange the order of summation. These are n times the two brackets of (4.6.8) and of (4.6.10), so

$$\hat{\delta}_{RE} = [\mathbf{Z}'(\hat{\Sigma}^{-1} \otimes \mathbf{I}_n)\mathbf{Z}]^{-1} \mathbf{Z}'(\hat{\Sigma}^{-1} \otimes \mathbf{I}_n)\mathbf{y}, \quad \widehat{\text{Avar}}(\hat{\delta}_{RE}) = n[\mathbf{Z}'(\hat{\Sigma}^{-1} \otimes \mathbf{I}_n)\mathbf{Z}]^{-1} :$$

the SUR formulas of (b), with $\mathbf{Z}$ redefined. RE is feasible GLS on the stacked common-coefficient regression $\mathbf{y} = \mathbf{Z}\delta + \boldsymbol{\varepsilon}$.

(e) Setting $\hat{\Sigma} = \mathbf{I}_M$ in (d) gives

$$\hat{\delta}_{\text{pooled}} = (\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y},$$

the OLS estimator of the stacked regression — pooled OLS. Its asymptotic variance under the (nonoptimal) identity weighting is the sandwich of (3.5.1) with the disappearance of x , (4.6.13):

$$\text{Avar}(\widehat{\delta}_{\text{pooled}}) = \left[\sum_m \text{E}(\mathbf{z}_{im} \mathbf{z}_{im}') \right]^{-1} \left[\sum_{m,h} \sigma_{mh} \text{E}(\mathbf{z}_{im} \mathbf{z}_{ih}') \right] \left[\sum_m \text{E}(\mathbf{z}_{im} \mathbf{z}_{im}') \right]^{-1},$$

i.e. in data-matrix form and consistently estimated,

$$\widehat{\text{Avar}} = n (\mathbf{Z}' \mathbf{Z})^{-1} \mathbf{Z}' (\widehat{\Sigma} \otimes \mathbf{I}_n) \mathbf{Z} (\mathbf{Z}' \mathbf{Z})^{-1},$$

with $\widehat{\sigma}_{mh}$ computed from the pooled-OLS residuals. The OLS package's own standard errors correspond to the special case $\sigma_{mh} = 0$ for $m \neq h$ and σ_{mm} a single common scale; they are wrong whenever the errors are correlated across equations — which is precisely the situation in which pooling is interesting.

(f) Reordering the rows of $\mathbf{Z}$ and $\mathbf{y}$ by observation i replaces the Kronecker factors $\widehat{\Sigma}^{-1} \otimes \mathbf{I}_n$ by $\mathbf{I}_n \otimes \widehat{\Sigma}^{-1}$: with the observation-major ordering, the $Mn \times Mn$ weighting matrix has n diagonal blocks of $\widehat{\Sigma}^{-1}$ (one per firm) rather than M blocks of $\widehat{\sigma}^{mh} \mathbf{I}_n$ (one per equation). The formulas become

$$\widehat{\delta}_{RE} = [\mathbf{Z}' (\mathbf{I}_n \otimes \widehat{\Sigma}^{-1}) \mathbf{Z}]^{-1} \mathbf{Z}' (\mathbf{I}_n \otimes \widehat{\Sigma}^{-1}) \mathbf{y} = \left[\sum_i \mathbf{Z}_i' \widehat{\Sigma}^{-1} \mathbf{Z}_i \right]^{-1} \sum_i \mathbf{Z}_i' \widehat{\Sigma}^{-1} \mathbf{y}_i,$$

with $\widehat{\text{Avar}} = n [\sum_i \mathbf{Z}_i' \widehat{\Sigma}^{-1} \mathbf{Z}_i]^{-1}$, and for pooled OLS

$$\widehat{\delta}_{\text{pooled}} = \left[\sum_i \mathbf{Z}_i' \mathbf{Z}_i \right]^{-1} \sum_i \mathbf{Z}_i' \mathbf{y}_i, \quad \widehat{\text{Avar}} = n \left[\sum_i \mathbf{Z}_i' \mathbf{Z}_i \right]^{-1} \left[\sum_i \mathbf{Z}_i' \widehat{\Sigma} \mathbf{Z}_i \right] \left[\sum_i \mathbf{Z}_i' \mathbf{Z}_i \right]^{-1}.$$

These are exactly the “beautified” formulas (4.6.8')–(4.6.10') and (4.6.13') of the text, whose practical merit is that each firm contributes one $L \times L$ or $L \times 1$ term — the form used in `code/ch4.py`, where the sums over firms are evaluated as vectorized array operations.

Analytical Exercise 4.3

(a) *The strengthened orthogonality is stronger* because $\text{E}(\varepsilon_{im} \mid \mathbf{z}_{i1}, \dots, \mathbf{z}_{iM}) = 0$ implies, by iterated expectations, $\text{E}[f(\mathbf{z}_{i1}, \dots, \mathbf{z}_{iM}) \cdot \varepsilon_{im}] = 0$ for every function f of the whole regressor set — including the regressors of the other equations — while ASSUMPTION 4.3 requires only $\text{E}(\mathbf{x}_{im} \varepsilon_{im}) = \mathbf{0}$ for the equation's own instruments. The strengthened condition is strict exogeneity of the full regressor set in every equation (Chapter 1 style), not mere predeterminedness. *The i.i.d. condition is stronger* than ergodic stationarity in two ways: it rules out serial dependence across i (ergodic stationarity allows it), and it makes conditioning on the whole data matrix $\mathbf{Z}$ equivalent to conditioning on firm i 's own regressors.

(b) *Conditional mean.* What needs to be shown is $\text{E}(\varepsilon_{im} \mid \mathbf{Z}) = 0$ for each (i, m) . By the definition of $\mathbf{Z}$, conditioning on it is conditioning on $(\mathbf{z}_{11}, \dots, \mathbf{z}_{nM})$; by the i.i.d. assumption, firm i 's error is independent of the other firms' regressors, so

$$\text{E}(\varepsilon_{im} \mid \mathbf{Z}) = \text{E}(\varepsilon_{im} \mid \mathbf{z}_{i1}, \dots, \mathbf{z}_{iM}) = 0$$

by the strengthened orthogonality. Hence $\text{E}(\boldsymbol{\varepsilon} \mid \mathbf{Z}) = \mathbf{0}$.

Conditional covariance. The matrix $E(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' | \mathbf{Z})$ has (m, h) block $E(\boldsymbol{\varepsilon}_m\boldsymbol{\varepsilon}_h' | \mathbf{Z})$ ($n \times n$), whose (i, j) element is $E(\varepsilon_{im}\varepsilon_{jh} | \mathbf{Z})$. For $j \neq i$: by the i.i.d. assumption, $(\varepsilon_{im}, \varepsilon_{ih})$ is independent of ε_{jh} given the conditioning variables of firms i and j , so

$$E(\varepsilon_{im}\varepsilon_{jh} | \mathbf{Z}) = E[\varepsilon_{im} E(\varepsilon_{jh} | \mathbf{z}_{j1}, \dots, \mathbf{z}_{jM}, \varepsilon_{im}) | \mathbf{Z}] = E[\varepsilon_{im} \cdot 0 | \mathbf{Z}] = 0,$$

the inner expectation vanishing by the strengthened orthogonality applied to firm j (independence of ε_{jh} from ε_{im} given the regressors removes the latter). For $j = i$,

$$E(\varepsilon_{im}\varepsilon_{ih} | \mathbf{Z}) = E(\varepsilon_{im}\varepsilon_{ih} | \mathbf{z}_{i1}, \dots, \mathbf{z}_{iM}) = \sigma_{mh},$$

by the i.i.d. assumption and ASSUMPTION 4.7 in its SUR form (Review Question 4.5.5). So the (m, h) block is $\sigma_{mh}\mathbf{I}_n$ and $E(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' | \mathbf{Z}) = \Sigma \otimes \mathbf{I}_n$.

(c)(i) *The GLS assumptions with sample size Mn .* Linearity (ASSUMPTION 1.1) is $\mathbf{y} = \mathbf{Z}\delta + \boldsymbol{\varepsilon}$ (Analytical Exercise 4.1). Strict exogeneity (ASSUMPTION 1.2) is $E(\boldsymbol{\varepsilon} | \mathbf{Z}) = \mathbf{0}$, just proved. No multicollinearity (ASSUMPTION 1.3) is the full column rank of the block diagonal $\mathbf{Z}$, which holds with probability one under the model's rank conditions. The generalized regression condition (1.6.1), $E(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' | \mathbf{Z}) = \sigma^2\mathbf{V}$ with $\mathbf{V}$ known, holds with $\sigma^2\mathbf{V} = \Sigma \otimes \mathbf{I}_n$ (GLS needs $\mathbf{V}$ only up to scale, and here Σ is assumed known outright).

(ii) *SUR is GLS.* The GLS estimator of Section 1.6 with $\sigma^2\mathbf{V} = \Sigma \otimes \mathbf{I}_n$ is

$$\hat{\delta}_{GLS} = [\mathbf{Z}'(\Sigma \otimes \mathbf{I}_n)^{-1}\mathbf{Z}]^{-1}\mathbf{Z}'(\Sigma \otimes \mathbf{I}_n)^{-1}\mathbf{y} = [\mathbf{Z}'(\Sigma^{-1} \otimes \mathbf{I}_n)\mathbf{Z}]^{-1}\mathbf{Z}'(\Sigma^{-1} \otimes \mathbf{I}_n)\mathbf{y},$$

which is the SUR formula of Analytical Exercise 4.2(b) with $\hat{\Sigma} = \Sigma$.

(iii) *Finite-sample unbiasedness.* Substituting $\mathbf{y} = \mathbf{Z}\delta + \boldsymbol{\varepsilon}$,

$$\hat{\delta}_{SUR} - \delta = [\mathbf{Z}'(\Sigma^{-1} \otimes \mathbf{I}_n)\mathbf{Z}]^{-1}\mathbf{Z}'(\Sigma^{-1} \otimes \mathbf{I}_n)\boldsymbol{\varepsilon},$$

so

$$E(\hat{\delta}_{SUR} | \mathbf{Z}) = \delta + [\mathbf{Z}'(\Sigma^{-1} \otimes \mathbf{I}_n)\mathbf{Z}]^{-1}\mathbf{Z}'(\Sigma^{-1} \otimes \mathbf{I}_n)E(\boldsymbol{\varepsilon} | \mathbf{Z}) = \delta,$$

and unconditionally $E(\hat{\delta}_{SUR}) = \delta$. SUR with known Σ is exactly unbiased. (With estimated $\hat{\Sigma}$ this survives exactly only under symmetry-of-errors arguments; in general feasible SUR is only asymptotically unbiased.)

(iv) *Finite-sample versus asymptotic variance.* By PROPOSITION 1.7(B),

$$\text{Var}(\hat{\delta}_{SUR} | \mathbf{Z}) = [\mathbf{Z}'(\Sigma^{-1} \otimes \mathbf{I}_n)\mathbf{Z}]^{-1} = \left[\sum_{m,h} \sigma^{mh} \mathbf{Z}_m' \mathbf{Z}_h \right]^{-1}.$$

Since $\frac{1}{n} \mathbf{Z}_m' \mathbf{Z}_h = \frac{1}{n} \sum_i \mathbf{z}_{im} \mathbf{z}_{ih}' \xrightarrow{p} E(\mathbf{z}_{im} \mathbf{z}_{ih}')$ by the ergodic theorem,

$$n \cdot \text{Var}(\hat{\delta}_{SUR} | \mathbf{Z}) = \left[\sum_{m,h} \sigma^{mh} \frac{1}{n} \mathbf{Z}_m' \mathbf{Z}_h \right]^{-1} \xrightarrow{p} \left[\sum_{m,h} \sigma^{mh} E(\mathbf{z}_{im} \mathbf{z}_{ih}') \right]^{-1},$$

which is $\text{Avar}(\hat{\delta}_{SUR})$ of (4.5.15) with $A_{mh} = E(\mathbf{z}_{im} \mathbf{z}_{ih}')$ under the SUR disappearance (4.5.16'). So the asymptotic variance is the probability limit of n times the exact conditional variance: for SUR the large-sample formula is the limit of the finite-sample one, not an approximation to some other object.

(d)(i) With common coefficients, ASSUMPTION 4.1' reads $\mathbf{y} = \mathbf{Z}\delta + \varepsilon$ with the stacked $\mathbf{Z}$ ($Mn \times L$) of Analytical Exercise 4.2; the arguments of (b) give $E(\varepsilon | \mathbf{Z}) = \mathbf{0}$ and $E(\varepsilon\varepsilon' | \mathbf{Z}) = \Sigma \otimes \mathbf{I}_n$; and full column rank of the stacked $\mathbf{Z}$ is the finite-sample counterpart of ASSUMPTION 4.4'. All GLS assumptions hold with sample size Mn .

(ii) The GLS estimator with $\sigma^2 \mathbf{V} = \Sigma \otimes \mathbf{I}_n$ is $[\mathbf{Z}'(\Sigma^{-1} \otimes \mathbf{I}_n)\mathbf{Z}]^{-1}\mathbf{Z}'(\Sigma^{-1} \otimes \mathbf{I}_n)\mathbf{y}$, the RE formula of Analytical Exercise 4.2(d) with $\hat{\Sigma} = \Sigma$: the random-effects estimator is GLS.

(iii) Unbiasedness follows as in (c)(iii): conditional on $\mathbf{Z}$, the estimator is δ plus a linear function of ε with conditional mean zero.

(iv) $\text{Var}(\hat{\delta}_{RE} | \mathbf{Z}) = [\sum_{m,h} \sigma^{mh} \mathbf{Z}'_m \mathbf{Z}_h]^{-1}$, and

$$n \cdot \text{Var}(\hat{\delta}_{RE} | \mathbf{Z}) \xrightarrow{p} \left[\sum_{m,h} \sigma^{mh} E(\mathbf{z}_{im} \mathbf{z}'_{ih}) \right]^{-1} = [E(\mathbf{Z}'_i \Sigma^{-1} \mathbf{Z}_i)]^{-1} = \text{Avar}(\hat{\delta}_{RE})$$

by (4.6.9) and (4.6.16B). The exercise's conclusion is conceptually important: under these strengthened (but in panel work standard) assumptions, SUR and RE are not merely asymptotically justified GMM estimators but exactly the classical GLS estimators of a linear model, unbiased in finite samples, with asymptotic theory needed only to accommodate the estimated $\hat{\Sigma}$.

Analytical Exercise 4.4

(a) By (4.3.4), the asymptotic variance of the efficient multiple-equation GMM estimator is $(\Sigma'_{xz} \mathbf{S}^{-1} \Sigma_{xz})^{-1}$, and the estimator $(\mathbf{S}'_{xz} \hat{\mathbf{S}}^{-1} \mathbf{S}_{xz})^{-1}$ converges to it provided $\mathbf{S}_{xz} \xrightarrow{p} \Sigma_{xz}$ and $\hat{\mathbf{S}} \xrightarrow{p} \mathbf{S}$. The former holds under ASSUMPTIONS 4.1, 4.2 by the ergodic theorem, the needed moments being finite by ASSUMPTION 4.4 and the usual fourth-moment conditions; the latter is PROPOSITION 4.2, which needs ASSUMPTIONS 4.1, 4.2, 4.6 plus a consistent first-step estimator for the residuals (available under ASSUMPTIONS 4.1–4.4, e.g. equation-by-equation 2SLS). So the appropriate assumptions are ASSUMPTIONS 4.1–4.6 — exactly those of the efficient multiple-equation GMM proposition itself.

(b) The efficient single-equation GMM estimator of equation m has, by (3.5.14), estimated asymptotic variance

$$\widehat{\text{Avar}}(\hat{\delta}_m) = (\mathbf{S}'_{xz,m} \hat{\mathbf{S}}_m^{-1} \mathbf{S}_{xz,m})^{-1}, \quad \hat{\mathbf{S}}_m = \frac{1}{n} \sum_i \hat{\varepsilon}_{im}^2 \mathbf{x}_{im} \mathbf{x}'_{im},$$

where $\hat{\varepsilon}_{im}$ is the residual from the equation's own estimation. The (m, m) block of the multiple-equation $\hat{\mathbf{S}}$ of (4.3.2) is

$$\hat{\mathbf{S}}_{mm} = \frac{1}{n} \sum_i \hat{\varepsilon}_{im} \hat{\varepsilon}_{im} \mathbf{x}_{im} \mathbf{x}'_{im} = \frac{1}{n} \sum_i \hat{\varepsilon}_{im}^2 \mathbf{x}_{im} \mathbf{x}'_{im} = \hat{\mathbf{S}}_m,$$

provided the same residuals are used in both — the hypothesis maintained in (d). So (*) is exactly the single-equation estimated asymptotic variance, consistent for $(\Sigma'_{xz,m} \mathbf{S}_{mm}^{-1} \Sigma_{xz,m})^{-1}$ under ASSUMPTIONS 3.1–3.6 applied to equation m .

(c) With $\widehat{\mathbf{W}} = \text{diag}(\hat{\mathbf{S}}_{11}^{-1}, \dots, \hat{\mathbf{S}}_{MM}^{-1})$ and $\mathbf{S}_{xz}$ block diagonal, (A.8) and (A.9) give

$$\mathbf{S}'_{xz} \widehat{\mathbf{W}} \mathbf{S}_{xz} = \text{diag}(\mathbf{S}'_{xz,m} \hat{\mathbf{S}}_{mm}^{-1} \mathbf{S}_{xz,m})_{m=1}^M.$$

For the middle factor, since $\widehat{\mathbf{W}}$ has zero off-diagonal blocks ($\widehat{\mathbf{W}}_{mh} = \mathbf{0}$ for $m \neq h$), the (m, m) block of the triple product collapses to a single term:

$$[\mathbf{S}'_{xz} \widehat{\mathbf{W}} \widehat{\mathbf{S}} \widehat{\mathbf{W}} \mathbf{S}_{xz}]_{mm} = \mathbf{S}'_{xz,m} \widehat{\mathbf{W}}_{mm} \widehat{\mathbf{S}}_{mm}^{-1} \widehat{\mathbf{W}}_{mm} \mathbf{S}_{xz,m} = \mathbf{S}'_{xz,m} \widehat{\mathbf{S}}_{mm}^{-1} \mathbf{S}_{xz,m}.$$

Both outer sandwich factors are block diagonal with the same diagonal blocks, so the (m, m) block of the whole sandwich is

$$(\mathbf{S}'_{xz,m} \widehat{\mathbf{S}}_{mm}^{-1} \mathbf{S}_{xz,m})^{-1} (\mathbf{S}'_{xz,m} \widehat{\mathbf{S}}_{mm}^{-1} \mathbf{S}_{xz,m}) (\mathbf{S}'_{xz,m} \widehat{\mathbf{S}}_{mm}^{-1} \mathbf{S}_{xz,m})^{-1} = (\mathbf{S}'_{xz,m} \widehat{\mathbf{S}}_{mm}^{-1} \mathbf{S}_{xz,m})^{-1},$$

which is (*). So the estimated asymptotic variance matrix of the stacked equation-by-equation estimator is exactly the sandwich evaluated at the block diagonal weighting matrix.

(d) The standard errors are the square roots of the diagonal elements of the estimated asymptotic variance divided by n . The diagonal blocks of the multiple-equation estimator's $\widehat{\text{Avar}}$ are the diagonal blocks of $(\mathbf{S}'_{xz} \widehat{\mathbf{S}}^{-1} \mathbf{S}_{xz})^{-1}$, and by (b)–(c) the equation-by-equation counterparts are the diagonal blocks of the sandwich. The granted inequality says the difference is positive semidefinite; positive semidefiniteness is inherited by diagonal blocks (take test vectors $\mathbf{c}$ supported on a single block), and then by diagonal elements ($\mathbf{c}$ a unit vector). Hence each coefficient's equation-by-equation standard error is no smaller than its multiple-equation standard error — in every sample, not just asymptotically, whenever the two estimators share the same residuals. This is the finite-sample shadow of the asymptotic efficiency ranking of Section 4.4, and it is why joint estimation is so tempting in practice: the reported precision can only improve.

(e) Yes. The inequality granted in (d) is an algebraic fact about three symmetric positive definite matrices — the sample version of PROPOSITION 3.5, whose proof (the projection argument of Chapter 3) uses no model assumptions at all. So even if the model is misspecified — both estimators then generally inconsistent, and $\widehat{\mathbf{S}}$ estimating nothing meaningful — the computed standard errors still obey the ranking. Whether the ranking means anything under misspecification is another matter: as the text warns (“Joint Estimation Can Be Hazardous”), a misspecified equation contaminates the whole system through the off-diagonal blocks of $\widehat{\mathbf{S}}^{-1}$, so the smaller standard errors may come attached to biased point estimates.

(f) Replace $\widehat{\mathbf{S}}$ throughout by (4.5.3), whose (m, h) block is $\widehat{\sigma}_{mh} \frac{1}{n} \sum_i \mathbf{x}_{im} \mathbf{x}'_{ih}$, and give $\widehat{\mathbf{W}}$ the diagonal blocks

$$\widehat{\mathbf{W}}_{mm} = \left(\widehat{\sigma}_{mm} \frac{1}{n} \sum_i \mathbf{x}_{im} \mathbf{x}'_{im} \right)^{-1}, \quad \widehat{\sigma}_{mm} = \frac{1}{n} \sum_i \widehat{\varepsilon}_{im}^2,$$

the 2SLS weighting matrix of equation m (the scalar $\widehat{\sigma}_{mm}^{-1}$ in fact cancels from the estimator, but keeping it makes the block algebra identical). Then: (a) $(\mathbf{S}'_{xz} \widehat{\mathbf{S}}^{-1} \mathbf{S}_{xz})^{-1}$ with $\widehat{\mathbf{S}}$ from (4.5.3) is consistent for $\text{Avar}(\widehat{\delta}_{FIVE})$ under ASSUMPTIONS 4.1–4.5 and 4.7 with finite $E(\mathbf{z}_{im} \mathbf{z}'_{ih})$, by PROPOSITION 4.4(C); (b) (*) with the new diagonal blocks is the estimated asymptotic variance of equation-by-equation 2SLS under conditional homoskedasticity (the 2SLS standard errors of Chapter 3); (c) the block algebra is unchanged; (d) the granted inequality applies verbatim to this pair of matrices, so the FIVE standard errors are no larger than the 2SLS ones, coefficient by coefficient, in every sample in which the same residuals enter $\widehat{\Sigma}$.

Analytical Exercise 4.5

(a) In isolation the first equation is

$$LW69_i = \beta_0 + \beta_1 S69_i + \beta_2 IQ_i + \varepsilon_{i1},$$

with three coefficients and only two orthogonality conditions, $E(\varepsilon_{i1}) = 0$ and $E(MED_i \varepsilon_{i1}) = 0$ (the regressors $S69$ and IQ are *not* assumed orthogonal to the error — schooling and ability are the usual suspects for endogeneity). The order condition fails: $2 < 3$. The equation is not identified on its own, and neither is the 1980 equation.

(b) Substitute each equation into its two orthogonality conditions. $E(\varepsilon_{i1}) = 0$ and $E(MED_i \varepsilon_{i1}) = 0$ give

$$\beta_0 + \beta_1 E(S69) + \beta_2 E(IQ) = E(LW69),$$

$$\beta_0 E(MED) + \beta_1 E(S69 \cdot MED) + \beta_2 E(IQ \cdot MED) = E(LW69 \cdot MED),$$

and the 1980 equation gives the same two conditions with $S80$ and $LW80$ in place of $S69$ and $LW69$. In matrix form:

$$\underbrace{\begin{bmatrix} 1 & E(S69) & E(IQ) \\ E(MED) & E(S69 \cdot MED) & E(IQ \cdot MED) \\ 1 & E(S80) & E(IQ) \\ E(MED) & E(S80 \cdot MED) & E(IQ \cdot MED) \end{bmatrix}}_{4 \times 3} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} E(LW69) \\ E(LW69 \cdot MED) \\ E(LW80) \\ E(LW80 \cdot MED) \end{bmatrix}.$$

There are *four* equations in *three* unknowns — the cross-equation restriction (common β 's) has doubled the moment count without adding parameters. The rank condition for identification (ASSUMPTION 4.4') is that this 4×3 coefficient matrix be of full column rank 3: the system is identified even though neither equation is identified in isolation. The extra identifying power comes from the year-to-year variation in schooling: rows 1 and 3 differ exactly by $E(S80) - E(S69)$ in the β_1 column, and rows 2 and 4 by $E(S80 \cdot MED) - E(S69 \cdot MED)$.

(c) If IQ and MED are uncorrelated, $E(IQ \cdot MED) = E(IQ)E(MED)$, and the third column of the coefficient matrix equals $E(IQ)$ times the first column:

$$(E(IQ), E(IQ \cdot MED), E(IQ), E(IQ \cdot MED))' = E(IQ) (1, E(MED), 1, E(MED))'.$$

The matrix then has rank at most 2, the rank condition fails, and the system is not identified: with MED uncorrelated with IQ , mother's education carries no independent variation that could trace out the IQ coefficient. This is the multiple-equation version of the classic requirement that an instrument be correlated with the endogenous regressor; the cross-equation restriction supplies extra *equations* but cannot supply missing *correlation*.

Analytical Exercise 4.6

Expand the product, using $\widehat{\varepsilon}_{im} = \varepsilon_{im} - \mathbf{z}'_{im}(\widehat{\delta}_m - \delta_m)$:

$$\begin{aligned} \widehat{\sigma}_{mh} = & \underbrace{\frac{1}{n} \sum_i \varepsilon_{im} \varepsilon_{ih}}_{(1)} - \underbrace{(\widehat{\delta}_m - \delta_m)' \frac{1}{n} \sum_i \mathbf{z}_{im} \varepsilon_{ih}}_{(2)} - \underbrace{(\widehat{\delta}_h - \delta_h)' \frac{1}{n} \sum_i \mathbf{z}_{ih} \varepsilon_{im}}_{(3)} \\ & + \underbrace{(\widehat{\delta}_m - \delta_m)' \left(\frac{1}{n} \sum_i \mathbf{z}_{im} \mathbf{z}'_{ih} \right) (\widehat{\delta}_h - \delta_h)}_{(4)}. \end{aligned}$$

Term (1). The sequence $\{\varepsilon_{im} \varepsilon_{ih}\}$ is a function of the ergodic stationary process $\{\mathbf{w}_i\}$ of ASSUMPTION 4.2, hence ergodic stationary itself, and has finite mean σ_{mh} by assumption; the ergodic theorem gives (1) $\xrightarrow{p} \sigma_{mh}$.

Term (4). The matrix average $\frac{1}{n} \sum_i \mathbf{z}_{im} \mathbf{z}'_{ih}$ converges in probability to the finite matrix $E(\mathbf{z}_{im} \mathbf{z}'_{ih})$ by ergodic stationarity (the process $\{\mathbf{z}_{im} \mathbf{z}'_{ih}\}$ is a function of $\{\mathbf{w}_i\}$ and its mean exists by the extra assumption). Since $\widehat{\delta}_m - \delta_m \xrightarrow{p} \mathbf{0}$ and $\widehat{\delta}_h - \delta_h \xrightarrow{p} \mathbf{0}$, the quadratic form (4) $\xrightarrow{p} \mathbf{0}' E(\mathbf{z}_{im} \mathbf{z}'_{ih}) \mathbf{0} = 0$ by LEMMA 2.3(A).

Term (2). Element j of $\frac{1}{n} \sum_i \mathbf{z}_{im} \varepsilon_{ih}$ converges to $E(z_{imj} \varepsilon_{ih})$, provided that moment is finite. It is, by the Cauchy–Schwarz inequality:

$$E|z_{imj} \varepsilon_{ih}| \leq \sqrt{E(z_{imj}^2)} \sqrt{E(\varepsilon_{ih}^2)} = \sqrt{E(z_{imj}^2)} \sqrt{\sigma_{hh}} < \infty,$$

because $E(z_{imj}^2)$ is the j -th diagonal element of the finite matrix $E(\mathbf{z}_{im} \mathbf{z}'_{im})$ (the $m = h$ case of the extra assumption) and σ_{hh} is finite. So $\frac{1}{n} \sum_i \mathbf{z}_{im} \varepsilon_{ih} \xrightarrow{p} E(\mathbf{z}_{im} \varepsilon_{ih})$, a finite vector, and multiplication by $\widehat{\delta}_m - \delta_m \xrightarrow{p} \mathbf{0}$ gives (2) $\xrightarrow{p} 0$. Note carefully: $E(\mathbf{z}_{im} \varepsilon_{ih})$ need not be zero — no orthogonality assumption is used — only its finiteness matters, because it is multiplied by a vanishing factor.

Term (3). Identical with the roles of m and h interchanged.

Collecting, $\widehat{\sigma}_{mh} \xrightarrow{p} \sigma_{mh} + 0 + 0 + 0 = \sigma_{mh}$. The message of the proof is the multiple-equation version of Review Question 2.3.5: residual-based second moments are consistent for error moments under any consistent coefficient estimator, with no orthogonality between regressors and errors required — which is why PROPOSITION 4.1 can feed $\widehat{\Sigma}$ to FIVE, 3SLS, SUR and RE alike, whatever instrument set those estimators use.

Analytical Exercise 4.7

(a) With $\mathbf{z}_{im} = \mathbf{z}_i$ ($L \times 1$, $L \leq K$), write $\mathbf{B} \equiv \frac{1}{n} \sum_i \mathbf{x}_i \mathbf{z}'_i$ ($K \times L$) and $\mathbf{A} \equiv \frac{1}{n} \sum_i \mathbf{x}_i \mathbf{x}'_i$ ($K \times K$). Then

$$\mathbf{S}_{xz} = \mathbf{I}_M \otimes \mathbf{B}, \quad \widehat{\mathbf{W}} = \widehat{\Sigma}^{-1} \otimes \mathbf{A}^{-1},$$

the latter by (4.5.10). By the Kronecker product rules (A.13) and (A.15),

$$\mathbf{S}'_{xz} \widehat{\mathbf{W}} \mathbf{S}_{xz} = (\mathbf{I}_M \otimes \mathbf{B}') (\widehat{\Sigma}^{-1} \otimes \mathbf{A}^{-1}) (\mathbf{I}_M \otimes \mathbf{B}) = \widehat{\Sigma}^{-1} \otimes \mathbf{B}' \mathbf{A}^{-1} \mathbf{B},$$

$$\mathbf{S}'_{xz} \widehat{\mathbf{W}} \mathbf{s}_{xy} = (\widehat{\Sigma}^{-1} \otimes \mathbf{B}' \mathbf{A}^{-1}) \mathbf{s}_{xy}.$$

Therefore the 3SLS estimator is

$$\begin{aligned}\hat{\delta}_{3SLS} &= [\hat{\Sigma}^{-1} \otimes \mathbf{B}'\mathbf{A}^{-1}\mathbf{B}]^{-1}(\hat{\Sigma}^{-1} \otimes \mathbf{B}'\mathbf{A}^{-1})\mathbf{s}_{xy} \\ &= [\hat{\Sigma} \otimes (\mathbf{B}'\mathbf{A}^{-1}\mathbf{B})^{-1}](\hat{\Sigma}^{-1} \otimes \mathbf{B}'\mathbf{A}^{-1})\mathbf{s}_{xy} \\ &= [\mathbf{I}_M \otimes (\mathbf{B}'\mathbf{A}^{-1}\mathbf{B})^{-1}\mathbf{B}'\mathbf{A}^{-1}]\mathbf{s}_{xy},\end{aligned}$$

and $\hat{\Sigma}$ cancels completely. The m -th block of the result is

$$(\mathbf{B}'\mathbf{A}^{-1}\mathbf{B})^{-1}\mathbf{B}'\mathbf{A}^{-1}\mathbf{s}_{xy,m}, \quad \mathbf{s}_{xy,m} = \frac{1}{n} \sum_i \mathbf{x}_i y_{im},$$

which is exactly the 2SLS estimator of equation m , (3.8.3'). So with identical regressors and identical instruments across equations, 3SLS is numerically equation-by-equation 2SLS — the system generalization of the result that the multivariate regression is equation-by-equation OLS (the special case $\mathbf{z}_i = \mathbf{x}_i$). Intuitively, with the same instruments and regressors everywhere, each equation presents the same relative geometry, and there is no asymmetric information across equations for joint estimation to exploit.

(b) No. The cancellation in (a) hinges on the Kronecker structure $\widehat{\mathbf{W}} = \hat{\Sigma}^{-1} \otimes \mathbf{A}^{-1}$, which is available only because under conditional homoskedasticity $\hat{\mathbf{S}} = \hat{\Sigma} \otimes \frac{1}{n} \sum_i \mathbf{x}_i \mathbf{x}_i'$ factors into an equation-index part and an instrument part. Without ASSUMPTION 4.7, the efficient weighting matrix is the inverse of the robust $\hat{\mathbf{S}}$ of (4.3.2), whose (m, h) block $\frac{1}{n} \sum_i \hat{\varepsilon}_{im} \hat{\varepsilon}_{ih} \mathbf{x}_i \mathbf{x}_i'$ varies with (m, h) in a way that does not factor: $\hat{\mathbf{S}}^{-1}$ cannot be written as a Kronecker product, the products in (a) do not collapse, and the efficient multiple-equation GMM differs numerically from the efficient equation-by-equation GMM even when all equations share regressors and instruments. The equations are then still “related” through the heteroskedasticity: the fourth moments $E(\varepsilon_{im} \varepsilon_{ih} \mathbf{x}_i \mathbf{x}_i')$ carry equation-pair-specific information that only the joint estimator exploits.

Analytical Exercise 4.8

(a) Yes, in both cases — but for a trivial reason. The orthogonality conditions use, for each equation, only its own L_m regressors as instruments: effectively $\mathbf{x}_{im} = \mathbf{z}_{im}$, so the moment conditions number $\sum_m L_m$, exactly as many as the coefficients. The system is just identified, $\hat{\mathbf{S}}$ is $(\sum_m L_m) \times (\sum_m L_m)$ and irrelevant, and the efficient multiple-equation GMM — indeed every GMM estimator — collapses to $\mathbf{S}_{xz}^{-1} \mathbf{s}_{xy}$ (the weighting-matrix cancellation of Review Question 4.5.3), whose m -th block is the OLS estimator of equation m . Conditional homoskedasticity changes nothing: there is no overidentification for the weighting matrix to act on.

(b) SUR differs because it rests on *stronger* orthogonality conditions: the cross conditions $E(\mathbf{z}_{im} \varepsilon_{ih}) = \mathbf{0}$ for $m \neq h$ of (4.5.18'), which let each equation borrow the other equations' regressors as instruments. Under the SUR assumption the instrument vector is the union $\mathbf{x}_i$ (K instruments per equation, MK moment conditions against $\sum_m L_m < MK$ coefficients when the regressor sets genuinely differ): the system is overidentified, the weighting matrix matters, and the efficient estimator is the GLS-type formula of Analytical Exercise 4.2(b), not OLS. Under the weaker conditions of this exercise those borrowed instruments are invalid, and respecting only the own-equation orthogonalities returns equation-by-equation OLS, which coincides with SUR only in the special cases of PROPOSITION 4.3 (just identified equations or uncorrelated errors). The phrase

“seemingly unrelated regressions” is thus misleading if read as “regressions with predetermined regressors”: the content of the SUR assumption is *cross* predeterminedness — each regressor is predetermined in *every* equation — and it is this, not predeterminedness per se, that generates both the estimator’s form and its efficiency gain.

Analytical Exercise 4.9

(a) *The system with the omission.* Write the true model as

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{d} + \varepsilon, \quad \mathbf{d} \equiv \gamma_2 \begin{bmatrix} \mathbf{0} \\ \mathbf{x}_3 \end{bmatrix},$$

where $\mathbf{d}$ is the omitted term — the part of the data-generating process that the estimated system cannot reach.

(b) *The SUR estimator.* Since Σ is known, the SUR estimator of the (misspecified) system is the GLS estimator

$$\begin{bmatrix} \hat{\beta}_1 \\ \hat{\gamma}_1 \end{bmatrix} = [\mathbf{X}'(\Sigma^{-1} \otimes \mathbf{I}_n)\mathbf{X}]^{-1} \mathbf{X}'(\Sigma^{-1} \otimes \mathbf{I}_n)\mathbf{y}.$$

(c) *The probability limit.* Writing σ^{mh} for the elements of Σ^{-1} , the partitioned formulas (A.6)–(A.7) give

$$\frac{1}{n} \mathbf{X}'(\Sigma^{-1} \otimes \mathbf{I}_n)\mathbf{X} = \begin{bmatrix} \sigma^{11} \frac{\mathbf{x}'_1 \mathbf{x}_1}{n} & \sigma^{12} \frac{\mathbf{x}'_1 \mathbf{x}_2}{n} \\ \sigma^{12} \frac{\mathbf{x}'_2 \mathbf{x}_1}{n} & \sigma^{22} \frac{\mathbf{x}'_2 \mathbf{x}_2}{n} \end{bmatrix} \xrightarrow{p} \mathbf{A} \equiv \begin{bmatrix} \sigma^{11} \mathbf{E}(x_1^2) & \sigma^{12} \mathbf{E}(x_1 x_2) \\ \sigma^{12} \mathbf{E}(x_1 x_2) & \sigma^{22} \mathbf{E}(x_2^2) \end{bmatrix},$$

and, substituting $\mathbf{y} = \mathbf{X}\beta + \mathbf{d} + \varepsilon$ and using the orthogonality of the regressors to both errors,

$$\frac{1}{n} \mathbf{X}'(\Sigma^{-1} \otimes \mathbf{I}_n)\mathbf{y} \xrightarrow{p} \mathbf{A}\beta + \begin{bmatrix} \sigma^{12} \gamma_2 \mathbf{E}(x_1 x_3) \\ \sigma^{22} \gamma_2 \mathbf{E}(x_2 x_3) \end{bmatrix}.$$

Hence

$$\text{plim} \begin{bmatrix} \hat{\beta}_1 \\ \hat{\gamma}_1 \end{bmatrix} = \beta + \gamma_2 \mathbf{A}^{-1} \begin{bmatrix} \sigma^{12} \mathbf{E}(x_1 x_3) \\ \sigma^{22} \mathbf{E}(x_2 x_3) \end{bmatrix}.$$

Since $\mathbf{E}(x_2 x_3) = 0$ by assumption, the second component of the inhomogeneous term vanishes and, with the partitioned inverse (A.10) (here just the 2×2 inverse), the asymptotic biases are

$$a \equiv \text{plim} \hat{\beta}_1 - \beta_1 = \frac{\gamma_2 \sigma^{12} \sigma^{22} \mathbf{E}(x_1 x_3) \mathbf{E}(x_2^2)}{\det \mathbf{A}}, \quad b \equiv \text{plim} \hat{\gamma}_1 - \gamma_1 = -\frac{\gamma_2 \sigma^{12} \sigma^{12} \mathbf{E}(x_1 x_2) \mathbf{E}(x_1 x_3)}{\det \mathbf{A}}.$$

In the polar case in which x_1 and x_2 are also orthogonal (so $\mathbf{A}$ is diagonal), these simplify to

$$a = \gamma_2 \frac{\sigma^{12}}{\sigma^{11}} \frac{\mathbf{E}(x_1 x_3)}{\mathbf{E}(x_1^2)}, \quad b = 0.$$

(d) *Conclusions.* $\text{plim } b = 0$: because x_2 is orthogonal to the omitted x_3 (and, in the polar case, x_1 is orthogonal to x_2), the second equation’s own coefficient is estimated consistently despite the omission — the classical result that omitting an orthogonal regressor is harmless, which survives joint estimation here. But $\text{plim } a \neq 0$ whenever $\gamma_2 \neq 0$, $\sigma^{12} \neq 0$ (equivalently $\sigma_{12} \neq 0$) and $\mathbf{E}(x_1 x_3) \neq 0$: *the first equation’s coefficient is inconsistently estimated even though the first equation is correctly specified.* The bias is the product of three channels — the omitted variable’s effect γ_2 ,

the cross-equation error correlation embedded in σ^{12} , and the correlation of x_1 with the omitted x_3 — and it is zero only if at least one channel is closed. A single-equation OLS regression of y_1 on x_1 , by contrast, is unaffected. This is the “Joint Estimation Can Be Hazardous” warning of Section 4.4 made concrete: the very machinery that transmits efficiency gains across equations (the off-diagonal blocks of the weighting matrix) also transmits specification errors.

Remark. The exercise as printed leaves the orthogonality structure implicit in the words of Greene’s example; what the conclusion needs is exactly $E(x_2x_3) = 0$ (so that $b = 0$) together with $E(x_1x_3) \neq 0$ and $\sigma_{12} \neq 0$ (so that $a \neq 0$). If x_1 and x_2 are correlated as well, then $b \neq 0$ too — the contamination runs through $\mathbf{A}^{-1}$ as well as through Σ^{-1} .

Analytical Exercise 4.10

(a) For the single equation estimated by 2SLS under conditional homoskedasticity, (3.8.4) and the definition (4.5.16) give

$$\text{Avar}(\hat{\delta}_{1,2SLS}) = \sigma_{11} [\mathbf{E}(\mathbf{z}_{i1}\mathbf{x}'_i) [\mathbf{E}(\mathbf{x}_i\mathbf{x}'_i)]^{-1} \mathbf{E}(\mathbf{x}_i\mathbf{z}'_{i1})]^{-1} = \sigma_{11} A_{11}^{-1}.$$

(b) By (4.5.15), $\text{Avar}(\hat{\delta}_{3SLS})$ is the inverse of the partitioned matrix

$$\begin{bmatrix} \sigma^{11} A_{11} & \sigma^{12} A_{12} \\ \sigma^{21} A_{21} & \sigma^{22} A_{22} \end{bmatrix},$$

so by the partitioned inverse formula (A.10) its upper-left ($L_1 \times L_1$) block is the inverse of the Schur complement

$$\sigma^{11} A_{11} - \sigma^{12} A_{12} (\sigma^{22} A_{22})^{-1} \sigma^{21} A_{21} = \left(\sigma^{11} - \frac{\sigma^{12} \sigma^{21}}{\sigma^{22}} \right) A_{11},$$

where the second equality rests on two computations.

First, $A_{12} A_{22}^{-1} A_{21} = A_{11}$ when $L_2 = K$. By (4.5.16),

$$A_{12} A_{22}^{-1} A_{21} = \mathbf{E}(\mathbf{z}_{i1}\mathbf{x}'_i) [\mathbf{E}(\mathbf{x}_i\mathbf{x}'_i)]^{-1} \mathbf{E}(\mathbf{x}_i\mathbf{z}'_{i2}) \cdot A_{22}^{-1} \cdot \mathbf{E}(\mathbf{z}_{i2}\mathbf{x}'_i) [\mathbf{E}(\mathbf{x}_i\mathbf{x}'_i)]^{-1} \mathbf{E}(\mathbf{x}_i\mathbf{z}'_{i1}).$$

Because the second equation is just identified, $\mathbf{z}_{i2}$ and $\mathbf{x}_i$ have the same dimension K , so $\mathbf{E}(\mathbf{x}_i\mathbf{z}'_{i2})$, $[\mathbf{E}(\mathbf{x}_i\mathbf{x}'_i)]^{-1}$ and $\mathbf{E}(\mathbf{z}_{i2}\mathbf{x}'_i)$ are all square $K \times K$ matrices, nonsingular under the rank condition, and

$$A_{22}^{-1} = [\mathbf{E}(\mathbf{z}_{i2}\mathbf{x}'_i) [\mathbf{E}(\mathbf{x}_i\mathbf{x}'_i)]^{-1} \mathbf{E}(\mathbf{x}_i\mathbf{z}'_{i2})]^{-1} = \mathbf{E}(\mathbf{x}_i\mathbf{z}'_{i2})^{-1} \mathbf{E}(\mathbf{x}_i\mathbf{x}'_i) \mathbf{E}(\mathbf{z}_{i2}\mathbf{x}'_i)^{-1}.$$

Substituting, the product telescopes:

$$\begin{aligned} A_{12} A_{22}^{-1} A_{21} &= \mathbf{E}(\mathbf{z}_{i1}\mathbf{x}'_i) [\mathbf{E}(\mathbf{x}_i\mathbf{x}'_i)]^{-1} \underbrace{\mathbf{E}(\mathbf{x}_i\mathbf{z}'_{i2}) \mathbf{E}(\mathbf{x}_i\mathbf{z}'_{i2})^{-1}}_{\mathbf{I}} \mathbf{E}(\mathbf{x}_i\mathbf{x}'_i) \\ &\quad \times \underbrace{\mathbf{E}(\mathbf{z}_{i2}\mathbf{x}'_i)^{-1} \mathbf{E}(\mathbf{z}_{i2}\mathbf{x}'_i)}_{\mathbf{I}} [\mathbf{E}(\mathbf{x}_i\mathbf{x}'_i)]^{-1} \mathbf{E}(\mathbf{x}_i\mathbf{z}'_{i1}) = A_{11}. \end{aligned}$$

Second, $\sigma^{11} - \sigma^{12} \sigma^{21} / \sigma^{22} = 1 / \sigma_{11}$. This is the (1,1) element of the identity $\Sigma^{-1} \Sigma = \mathbf{I}$ written out via the partitioned inverse: the (1,1) element of Σ^{-1} is $\sigma^{11} = (\sigma_{11} - \sigma_{12}^2 / \sigma_{22})^{-1}$, and direct expansion of $\sigma^{11} - \sigma^{12} \sigma^{21} / \sigma^{22}$ using $\sigma^{12} = -\sigma_{12} / (\sigma_{11} \sigma_{22} - \sigma_{12}^2)$ gives $1 / \sigma_{11}$.

Therefore

$$\text{Avar}(\hat{\delta}_{1,3SLS}) = \left(\frac{1}{\sigma_{11}} A_{11} \right)^{-1} = \sigma_{11} A_{11}^{-1} = \text{Avar}(\hat{\delta}_{1,2SLS}).$$

The message: a just identified equation is exhausted by its own instruments — it has no spare orthogonality conditions to lend the system, and the algebra shows it correspondingly contributes nothing to the estimation of the other equations. Joint estimation pays only when the added equations are overidentified (or, through the error covariance, when they sharpen the weighting of genuinely shared restrictions).

Analytical Exercise 4.11

The displayed conditions number L — the dimension of $\mathbf{z}_{im}$ — exactly as many as the coefficients in the common vector δ . A just identified GMM problem needs no weighting matrix: the efficient GMM estimator is the IV estimator that solves the sample orthogonality conditions exactly, $\mathbf{g}_n(\hat{\delta}) = \mathbf{0}$. Splitting the sample condition into the pieces involving y_{ih} and $\tilde{\delta}$,

$$\sum_{m,h} \sigma^{mh} \left(\frac{1}{n} \sum_i \mathbf{z}_{im} \mathbf{z}'_{ih} \right) \hat{\delta} = \sum_{m,h} \sigma^{mh} \left(\frac{1}{n} \sum_i \mathbf{z}_{im} y_{ih} \right),$$

so

$$\hat{\delta} = \left[\sum_{m,h} \sigma^{mh} \left(\frac{1}{n} \sum_i \mathbf{z}_{im} \mathbf{z}'_{ih} \right) \right]^{-1} \sum_{m,h} \sigma^{mh} \left(\frac{1}{n} \sum_i \mathbf{z}_{im} y_{ih} \right),$$

which — with σ^{mh} replaced by the consistent estimates $\hat{\sigma}^{mh}$ — is exactly the random-effects estimator (4.6.8).

The exercise exposes what RE actually does with the MK orthogonality conditions $E(\mathbf{x}_i \varepsilon_{im}) = \mathbf{0}$ available under the SUR assumption: it collapses them, before estimation, into the L linear combinations displayed above, with weights σ^{mh} chosen so that the resulting combination is the efficient one (these are the weights that make the combined moment conditions uncorrelated with the efficient linear transformation of the full set — the GLS weights). Since the combined conditions exactly equal the coefficients in number, the “efficient GMM” for them is just the IV solution, and no separate efficiency computation is needed. The same reduction with σ^{mh} replaced by the elements of $\mathbf{I}_M$ yields pooled OLS as the IV estimator of (4.6.12).

Problem Set for Chapter 4: Empirical Exercise

Empirical Exercise 4.1

All numbers below are computed by `code/ch4.py` from `data/ch4/GREENE.ASC`.

(a) *Simple statistics.* Table 4.3 of the book is duplicated exactly:

	output (10 ⁹ kWh)	labor share	capital share	fuel share
mean	9.0	0.141	0.227	0.631
std. dev.	10.3	0.059	0.062	0.092

(Output in the data file is in millions of kWh, so the tabulated figures are the raw moments divided by 1,000; the fuel-share standard deviation 0.092 is the only visible discrepancy from the book’s

0.095, a rounding difference in the printed table.) As the text remarks, fuel dominates the cost of generation.

(b) *Random-effects estimates.* Equation-by-equation OLS on the unconstrained system gives the residual covariance estimate

$$\hat{\Sigma}^* = \begin{bmatrix} 0.001727 & -0.001555 \\ -0.001555 & 0.003912 \end{bmatrix},$$

exactly the appropriate submatrix of the 3×3 $\hat{\Sigma}$ reported in Table 4.4 (whose third diagonal element, from the dropped capital equation, is 0.002528). The constrained system in the common-coefficient format $\mathbf{y}_i = \mathbf{Z}_i \delta + \varepsilon_i$ has

$$\mathbf{Z}_i = \begin{bmatrix} 1 & \log(p_{i1}/p_{i2}) & \log(p_{i3}/p_{i2}) & \log Q_i & 0 & 0 & 0 \\ 0 & 0 & \log(p_{i1}/p_{i2}) & 0 & 1 & \log(p_{i3}/p_{i2}) & \log Q_i \end{bmatrix},$$

$$\delta = (\alpha_1, \gamma_{11}, \gamma_{13}, \gamma_{1Q}, \alpha_3, \gamma_{33}, \gamma_{3Q})',$$

the third row of blocks placing $\gamma_{31} = \gamma_{13}$ on $\log(p_{i1}/p_{i2})$ in the fuel equation. The random-effects estimates (4.6.8') with standard errors from (4.6.10') are

parameter	estimate	std. error	t-value
α_1	-0.13151	0.10561	-1.25
γ_{11}	0.08362	0.01998	4.19
$\gamma_{13} = \gamma_{31}$	-0.06042	0.01541	-3.92
γ_{1Q}	-0.02115	0.00247	-8.55
α_3	0.81337	0.09356	8.69
γ_{33}	0.15939	0.02311	6.90
γ_{3Q}	0.02974	0.00373	7.98

The remaining eight share-equation parameters follow from adding-up (4.7.12), homogeneity (4.7.13) and symmetry (4.7.14), with standard errors from the delta method applied to the 15×7 Jacobian of the mapping (for example, $\hat{\gamma}_{22} = \hat{\gamma}_{11} + 2\hat{\gamma}_{13} + \hat{\gamma}_{33}$, the analogue of the γ_{33} calculation displayed in the text for the other normalization):

parameter	estimate	s.e.	t	parameter	estimate	s.e.	t
α_2	0.31814	0.08490	3.75	γ_{31}	-0.06042	0.01541	-3.92
γ_{12}	-0.02321	0.01592	-1.46	γ_{32}	-0.09897	0.01722	-5.75
γ_{21}	-0.02321	0.01592	-1.46	γ_{2Q}	-0.00859	0.00299	-2.87
γ_{22}	0.12218	0.01975	6.19				
γ_{23}	-0.09897	0.01722	-5.75				

All fifteen point estimates agree with those implied by Table 4.4's normalization (drop fuel, normalize by p_3): recomputing the whole estimation with the *fuel* equation dropped and prices normalized by p_3 reproduces the same fifteen numbers to a maximum absolute difference of 1.3×10^{-12} — the numerical invariance to the choice of dropped equation, with $\hat{\Sigma}^*$ from equation-by-equation OLS, that the exercise is designed to exhibit. All three diagonal elements of $(\hat{\gamma}_{jk})$ — 0.084, 0.122, 0.159

— are positive, so the matrix is not negative semidefinite: concavity is violated, as reported in the text.

For comparison, $\widehat{\Sigma}^*$ computed instead from the pooled-OLS residuals of the common-coefficient format is

$$\widehat{\Sigma}_{\text{pooled}}^* = \begin{bmatrix} 0.001732 & -0.000167 \\ -0.000167 & 0.002535 \end{bmatrix},$$

matching the figures printed beneath Table 4.4 (the book's note shows the pooled-OLS variant of $\widehat{\Sigma}$); with this $\widehat{\Sigma}^*$ the point estimates change slightly and the numerical invariance is no longer exact, exactly as the text predicts.

(c) *Sargan's statistic*. With $\mathbf{x}_i = (1, \log(p_{i1}/p_{i2}), \log(p_{i3}/p_{i2}), \log Q_i)'$ ($K = 4$), $\mathbf{S} = \widehat{\Sigma}^* \otimes \frac{1}{n} \sum_i \mathbf{x}_i \mathbf{x}_i'$, $\mathbf{s}_{xy} = \frac{1}{n} \sum_i \mathbf{y}_i \otimes \mathbf{x}_i$ and $\mathbf{S}_{xz} = \frac{1}{n} \sum_i \mathbf{Z}_i \otimes \mathbf{x}_i$ (Review Question 4.6.4), the statistic of PROPOSITION 4.7(c) is

$$J = n \mathbf{g}_n(\widehat{\delta}_{RE})' \widehat{\mathbf{S}}^{-1} \mathbf{g}_n(\widehat{\delta}_{RE}) = 0.63313, \quad p\text{-value} = 0.42621 \quad \text{on } \chi^2(2K - L) = \chi^2(1),$$

duplicating the numbers quoted in the exercise. The overidentifying restriction — which, given the SUR setup, is effectively the symmetry restriction itself — is not rejected.

(d) *Difference (C-type) test of symmetry*. The maintained hypothesis is the unconstrained system, the null the symmetry restriction $\gamma_{13} = \gamma_{31}$. The unconstrained system is the multivariate regression, which is just identified ($MK = 2 \times 4$ moments against 8 coefficients), so its Sargan statistic is exactly zero in every sample (Review Question 4.5.10). By PROPOSITION 3.8 the difference of Sargan statistics is asymptotically chi-squared with degrees of freedom equal to the number of restrictions:

$$C = J_{\text{constrained}} - J_{\text{unconstrained}} = 0.63313 - 0 = 0.63313 \stackrel{a}{\sim} \chi^2(1), \quad p = 0.42621.$$

Symmetry is not rejected at any conventional level: the data are consistent with the cross-equation restriction that the translog cost function's symmetry requires.

(e) *Wald test of symmetry*. The unconstrained system is the multivariate regression, so the equation-by-equation OLS coefficients $\widehat{\gamma}_{13} = -0.05799$ and $\widehat{\gamma}_{31} = -0.08302$ have estimated asymptotic variance $\widehat{\Sigma}^* \otimes [\frac{1}{n} \sum_i \mathbf{x}_i \mathbf{x}_i']^{-1}$ (the multivariate-regression formula of Section 4.5), and the Wald statistic for $\gamma_{13} - \gamma_{31} = 0$ is

$$W = \frac{(\widehat{\gamma}_{13} - \widehat{\gamma}_{31})^2}{\widehat{\sigma}_{11}u_{33} - 2\widehat{\sigma}_{13}u_{32} + \widehat{\sigma}_{33}u_{22}} = 0.63313, \quad p = 0.42621,$$

where u_{jk} is the (j, k) element of $[\frac{1}{n} \sum_i \mathbf{x}_i \mathbf{x}_i']^{-1}$ and the covariance term $-2\widehat{\sigma}_{13}u_{32}$ is essential (the two coefficients come from different equations with correlated errors). The statistic is numerically equal to Sargan's statistic for the constrained system to ten decimal places ($|W - J| = 5 \times 10^{-12}$) — as the exercise asserts — because both are the same quadratic form in the sample orthogonality conditions: the constrained Sargan statistic measures the distance of the only violated moment condition, and the Wald statistic measures the same distance in coefficient space, with the metric of the same $\widehat{\Sigma}^*$.

(f) *Substitution elasticities*. Evaluating (4.7.9) at the *fitted* cost shares from the constrained system (as the text prescribes, since the formula assumes no optimization errors) and averaging over the 99 firms:

	labor–capital	capital–fuel	labor–fuel
$\bar{\eta}_{jk}$	0.170	0.293	0.270

All three elasticities are positive — the three inputs are substitutes — but far below the Cobb–Douglas value of unity, which is the text’s substantive conclusion. The complete matrix (η_{jk}) at the mean fitted shares and the full set of diagnostics computed by `code/ch4.py`:

- *Monotonicity in the relevant range*: satisfied. The minimum fitted share over firms and inputs is $0.045 > 0$, so fitted cost shares are nonnegative for all inputs for all 99 firms.
- *Concavity in the relevant range*: violated for 9 of the 99 firms (for whom the matrix (η_{jk}) of (4.7.9) at fitted shares fails to be negative semidefinite); the text reports 20. The count is sensitive to small differences in the point estimates, since the deciding eigenvalue is near zero for several firms; evaluated at *actual* shares the count is 22.
- *The coefficient matrix $(\hat{\gamma}_{jk})$ itself* has eigenvalues $(0.000, 0.117, 0.248)$, two of them positive: concavity fails outright at the level of the cost function’s Hessian structure, as the text notes from the positive diagonal elements alone.

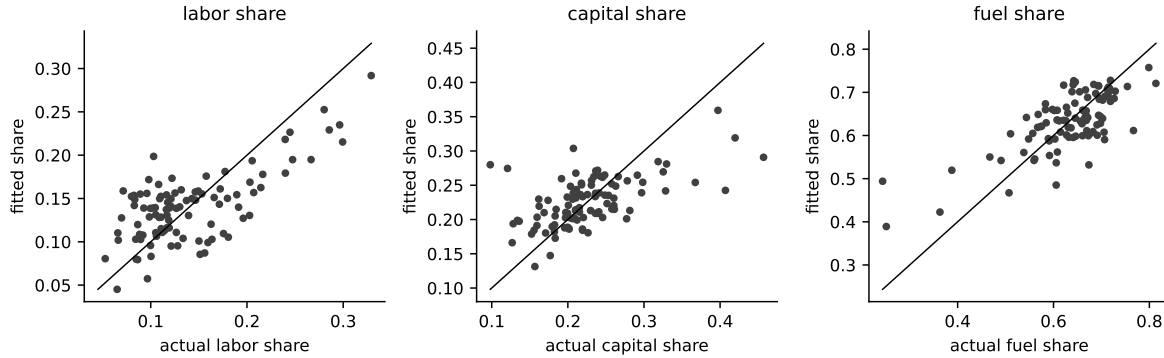


Figure S4.1: fitted versus actual cost shares in the constrained two-equation system (capital equation dropped; fuel’s fitted share computed as one minus the other two). All fitted shares are positive — monotonicity holds in the relevant range — and the fit is tightest for the dominant fuel share.

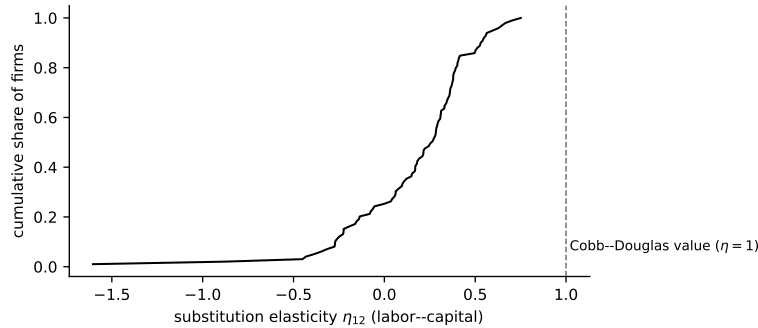


Figure S4.2: empirical distribution of the labor–capital substitution elasticity η_{12} of (4.7.9) evaluated at fitted shares across the 99 firms; the dashed line is the Cobb–Douglas value $\eta = 1$. Every firm’s elasticity is below unity, and most are far below.

Chapter 5

Panel Data

Solutions to all the exercises of Chapter 5 of Fumio Hayashi, *Econometrics* (Princeton University Press, 2000): the ten review questions of Sections 5.1–5.4, the seven analytical exercises, and the empirical exercise on the Summers–Heston data. Assumptions, propositions and equations cited as ASSUMPTION 4.2, PROPOSITION 5.1 or (5.2.6) refer to the book. The numerical parts are computed by `code/ch5.py`.

Section 5.1 The Error-Components Model

Review Question 5.1.1

Proposition 4.7 (efficient GMM for the multiple-equation model) requires: a system of M linear equations ((5.1.1)), an ergodic stationary martingale-difference structure for $\mathbf{g}_i \equiv \boldsymbol{\varepsilon}_i \otimes \mathbf{x}_i$ with a nonsingular long-run variance, an identification (rank) condition, and conditional homoskedasticity. Under (5.1.2) the sample is i.i.d., which is stronger than ergodic stationarity. The orthogonality conditions (5.1.3), i.e. $E(\mathbf{z}_{im}\varepsilon_{ih}) = \mathbf{0}$ for all m, h , are exactly $E(\boldsymbol{\varepsilon}_i \otimes \mathbf{x}_i) = \mathbf{0}$, because $\mathbf{x}_i$ collects the unique elements of $(\mathbf{z}_{i1}, \dots, \mathbf{z}_{iM})$ and each ε_{ih} is paired with every element of $\mathbf{x}_i$ as m and h vary. Therefore $E(\mathbf{g}_i \mid \mathbf{g}_{i-1}, \dots, \mathbf{g}_1) = \mathbf{0}$ by independence of $\mathbf{g}_i$ from the past (i.i.d. sampling) and $E(\mathbf{g}_i) = \mathbf{0}$: $\{\mathbf{g}_i\}$ is a martingale difference sequence, and its long-run variance $\mathbf{S} = E(\mathbf{g}_i \mathbf{g}_i')$ is finite and nonsingular by (5.1.6). The identification condition is (5.1.4), assumed; conditional homoskedasticity is (5.1.5). Thus all the hypotheses of Proposition 4.7 hold, and the random-effects estimator (4.6.8'), which is GMM with the efficient weighting matrix $\hat{\mathbf{S}}^{-1}$, is the efficient GMM estimator.

For the redundancy claim, recall from (4.6.16B) the Kronecker-product expression: the identification condition is that $E(\mathbf{Z}_i \otimes \mathbf{x}_i)$ have full column rank L , where $\mathbf{Z}_i$ is $M \times L$ and $\mathbf{x}_i$ is $K \times 1$. Since each row $\mathbf{z}_{im}$ of $\mathbf{Z}_i$ is a subvector of $\mathbf{x}_i$, the matrix $\mathbf{Z}_i \otimes \mathbf{x}_i$ is a fixed linear function of $\mathbf{I}_M \otimes \mathbf{x}_i \mathbf{x}_i'$: stacking the columns, $\text{vec}(\mathbf{Z}_i \otimes \mathbf{x}_i) = \mathbf{A} \text{vec}(\mathbf{I}_M \otimes \mathbf{x}_i \mathbf{x}_i')$ for a constant selection matrix $\mathbf{A}$ of full column rank (each column of $\mathbf{A}$ has a single nonzero element, different columns in different rows). Taking expectations,

$$E(\mathbf{Z}_i \otimes \mathbf{x}_i) = \mathbf{A} E(\mathbf{I}_M \otimes \mathbf{x}_i \mathbf{x}_i') = \mathbf{A} [\mathbf{I}_M \otimes E(\mathbf{x}_i \mathbf{x}_i')].$$

If $E(\mathbf{x}_i \mathbf{x}_i')$ is nonsingular, so is $\mathbf{I}_M \otimes E(\mathbf{x}_i \mathbf{x}_i')$ (its eigenvalues are those of $E(\mathbf{x}_i \mathbf{x}_i')$, each repeated M times); premultiplication by the full-column-rank $\mathbf{A}$ then yields a matrix of full column rank L : the identification condition (5.1.4) holds automatically. Since conditional homoskedasticity (5.1.5) together with (5.1.6) implies (5.1.6') — $\boldsymbol{\Sigma}$ and $E(\mathbf{x}_i \mathbf{x}_i')$ nonsingular — the separate identification assumption (5.1.4) is implied by the other assumptions and is redundant in the random-effects model. (It is retained in the statement of the error-components model because the *fixed-effects* identification condition (5.1.15) is *not* redundant, as Review Question 5.1.3 shows.)

Remark. This is the panel-data analogue of the single-equation fact that, when the instruments are the regressors themselves, nonsingularity of $E(\mathbf{z}_i \mathbf{z}_i')$ delivers the GMM rank condition. Requiring $E(\mathbf{x}_i \mathbf{x}_i')$ nonsingular for the union $\mathbf{x}_i$ of all regressors is stronger than needed for (5.1.4) alone, but it is exactly what (5.1.6') supplies.

Review Question 5.1.2

In the reparameterization (5.1.13) the time-varying part is

$$\mathbf{y}_i = \underbrace{\begin{bmatrix} 1 & 0 & S_{69i} \\ 0 & 1 & S_{80i} \\ 0 & 0 & S_{82i} \end{bmatrix}}_{\mathbf{F}_i^{(0)}} \underbrace{\begin{bmatrix} \phi_1 - \phi_3 \\ \phi_2 - \phi_3 \\ \beta \end{bmatrix}}_{\boldsymbol{\beta}^{(0)}} + \mathbf{1}_M \mathbf{b}_i' \boldsymbol{\gamma} + \mathbf{1}_M \alpha_i + \boldsymbol{\eta}_i, \quad \mathbf{b}_i = \begin{bmatrix} 1 \\ \text{IQ}_i \end{bmatrix}, \quad \boldsymbol{\gamma} = \begin{bmatrix} \phi_3 \\ \gamma \end{bmatrix}.$$

The proposed $\mathbf{F}_i$ replaces the first column $(1, 0, 0)'$ by $(1, -1, 0)'$. The two bases of the same column space are related by

$$\mathbf{F}_i = \mathbf{F}_i^{(0)} \mathbf{T}, \quad \mathbf{T} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{T}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Since $\mathbf{F}_i \boldsymbol{\beta} = \mathbf{F}_i^{(0)} \mathbf{T} \boldsymbol{\beta}$ must equal $\mathbf{F}_i^{(0)} \boldsymbol{\beta}^{(0)}$,

$$\boldsymbol{\beta} = \mathbf{T}^{-1} \boldsymbol{\beta}^{(0)} = \begin{bmatrix} \phi_1 - \phi_3 \\ (\phi_1 - \phi_3) + (\phi_2 - \phi_3) \\ \beta \end{bmatrix} = \begin{bmatrix} \phi_1 - \phi_3 \\ \phi_1 + \phi_2 - 2\phi_3 \\ \beta \end{bmatrix},$$

and δ stacks this $\boldsymbol{\beta}$ over the unchanged $\boldsymbol{\gamma}$: $\delta = (\phi_1 - \phi_3, \phi_1 + \phi_2 - 2\phi_3, \beta, \phi_3, \gamma)'$, with $\mathbf{b}_i = (1, \text{IQ}_i)'$ unchanged. The moral: fixed-effects estimation identifies only the *space* spanned by the time-varying regressors, so $\boldsymbol{\beta}$ is determined only up to invertible reparameterization; which combination of the ϕ 's is estimated depends on the arbitrary choice of which intercept is folded into $\mathbf{b}_i$. The fitted values $\mathbf{F}_i \boldsymbol{\beta}$, and hence the estimator of any quantity invariant to the choice (such as the difference $\phi_2 - \phi_1$), are unique.

Review Question 5.1.3

Condition (5.1.15) requires $E(\mathbf{QF}_i \otimes \mathbf{x}_i)$ to be of full column rank, where

$$\mathbf{x}_i = (1, S_{69i}, S_{80i}, S_{82i}, \text{IQ}_i)'$$

Premultiplication by $\mathbf{Q}$ takes deviations from group means of each column of $\mathbf{F}_i$. The first three columns are the unit vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, constant across i , so the m -th column of $\mathbf{QF}_i$ is the constant vector $\mathbf{Qe}_m$ ($m = 1, 2, 3$); the fourth is $(S_{69i} - \bar{S}_i, S_{80i} - \bar{S}_i, S_{82i} - \bar{S}_i)'$ with $\bar{S}_i \equiv \frac{1}{3}(S_{69i} + S_{80i} + S_{82i})$. Consequently the block of $\mathbf{QF}_i \otimes \mathbf{x}_i$ corresponding to column m is $\mathbf{Qe}_m \otimes \mathbf{x}_i$, whose expectation is $\mathbf{Qe}_m \otimes E(\mathbf{x}_i)$. But the three constant vectors $\mathbf{Qe}_1, \mathbf{Qe}_2, \mathbf{Qe}_3$ are linearly dependent: they sum to $\mathbf{Q1}_M = \mathbf{0}$. Hence

$$[\mathbf{Qe}_1 \otimes E(\mathbf{x}_i)] + [\mathbf{Qe}_2 \otimes E(\mathbf{x}_i)] + [\mathbf{Qe}_3 \otimes E(\mathbf{x}_i)] = \mathbf{0} \otimes E(\mathbf{x}_i) = \mathbf{0},$$

a nontrivial linear relation among the column blocks of $E(\mathbf{Q}\mathbf{F}_i \otimes \mathbf{x}_i)$ (nontrivial because $E(\mathbf{x}_i) \neq \mathbf{0}$ here: $ES_{mi} > 0$). The matrix is rank deficient and (5.1.15) fails. The economic content is immediate: three equation-specific intercepts contain a common intercept in their span, and a common intercept is swept away by $\mathbf{Q}$ together with the fixed effect and the common regressor IQ. One of the three ϕ 's is simply not identified after the transformation; the two that remain must measure differences from the third — which is precisely what the reparameterization (5.1.13) accomplishes.

Section 5.2 The Fixed-Effects Estimator

Review Question 5.2.1

No. The fixed-effects estimator is computed from the transformed system $\tilde{\mathbf{y}}_i = \tilde{\mathbf{F}}_i\boldsymbol{\beta} + \tilde{\boldsymbol{\eta}}_i$ with $\tilde{\boldsymbol{\eta}}_i = \mathbf{Q}\boldsymbol{\eta}_i$, and its sampling error (5.2.5) is

$$\hat{\boldsymbol{\beta}}_{\text{FE}} - \boldsymbol{\beta} = \left(\frac{1}{n} \sum_{i=1}^n \mathbf{F}'_i \mathbf{Q} \mathbf{F}_i \right)^{-1} \frac{1}{n} \sum_{i=1}^n \mathbf{F}'_i \mathbf{Q} \boldsymbol{\eta}_i.$$

Consistency and asymptotic normality (Proposition 5.1) require: (i) the identification condition (5.1.15), which makes the first factor converge to a nonsingular matrix; (ii) $E(\mathbf{F}'_i \mathbf{Q} \boldsymbol{\eta}_i) = \mathbf{0}$, which follows from the orthogonality conditions (5.2.6) between the regressors and the *second* error component alone; and (iii) a central limit theorem for $\mathbf{F}'_i \mathbf{Q} \boldsymbol{\eta}_i$. The fixed effect α_i has been swept out by $\mathbf{Q}$ before any of these conditions is invoked, so no property of the joint distribution of $(\alpha_i, \boldsymbol{\eta}_i)$ — in particular not $E(\alpha_i \boldsymbol{\eta}_i) = \mathbf{0}$ — is ever used. This robustness to arbitrary correlation between the individual effect and the idiosyncratic shock (and, more importantly, between the individual effect and the regressors) is the whole point of the fixed-effects technique; it is what the random-effects estimator, which treats $\mathbf{1}_M \alpha_i + \boldsymbol{\eta}_i$ as a composite error term, does not deliver.

Review Question 5.2.2

Let $\mathbf{C}'$ be the $(M-1) \times M$ matrix that keeps the first $M-1$ rows of $\mathbf{Q}$. Dropping the last transformed equation for each group amounts to pooled OLS on the system $\mathbf{C}'\mathbf{y}_i = \mathbf{C}'\mathbf{F}_i\boldsymbol{\beta} + \mathbf{C}'\boldsymbol{\eta}_i$, giving

$$\hat{\boldsymbol{\beta}}_{\text{drop}} = \left(\sum_{i=1}^n \mathbf{F}'_i \mathbf{C} \mathbf{C}' \mathbf{F}_i \right)^{-1} \sum_{i=1}^n \mathbf{F}'_i \mathbf{C} \mathbf{C}' \mathbf{y}_i.$$

The fixed-effects estimator (5.2.4) has $\mathbf{Q}$ in place of $\mathbf{C}\mathbf{C}'$, and $\mathbf{C}\mathbf{C}' \neq \mathbf{Q}$: $\mathbf{C}\mathbf{C}'$ is $\mathbf{Q}$ with its last row and column replaced by zeros (it projects onto the first $M-1$ coordinates of the deviation space and annihilates the last), a matrix of rank $M-1$ that is not the annihilator. For $M=2$ the distinction happens to be immaterial — the second transformed equation is minus the first, so both estimators reduce to first-difference OLS — but for $M \geq 3$ they genuinely differ: the dropped-equation estimator weights the $M-1$ retained deviations asymmetrically, and its probability limit involves a different combination of the cross moments $E(\mathbf{f}_{im}\eta_{ih})$. The deeper point is that the transformation must preserve the *inner-product* structure that makes the estimator invariant to which equation is treated as redundant; $\mathbf{Q}$ does (Analytical Exercise 5.2 shows that any full-rank

$\mathbf{C}$ with $\mathbf{C}'\mathbf{1}_M = \mathbf{0}$ works when paired with the weighting $(\mathbf{C}'\mathbf{C})^{-1}$, but plain unweighted OLS on a strict subset of the equations does not.

Remark. This is also why the degrees-of-freedom correction in (5.2.17) subtracts n from Mn : the n linear constraints $\mathbf{1}_M'\tilde{\mathbf{y}}_i = 0$ make n of the Mn transformed observations redundant, but they are redundant *jointly*, not observation by observation, so one cannot simply delete them before estimating.

Review Question 5.2.3

Consistency requires $E(\mathbf{F}_i'\mathbf{Q}\boldsymbol{\eta}_i) = \mathbf{0}$. Expanding with $\mathbf{Q} = (q_{mh})$ (this is (4.6.16B) in scalar form),

$$E(\mathbf{F}_i'\mathbf{Q}\boldsymbol{\eta}_i) = \sum_{m=1}^M \sum_{h=1}^M q_{mh} E(\mathbf{f}_{im}\eta_{ih}),$$

a double sum over *all* pairs (m, h) . If only the own-equation orthogonalities $E(\mathbf{f}_{im}\eta_{im}) = \mathbf{0}$ hold, the terms with $m \neq h$ survive: they are weighted by $q_{mh} = -1/M \neq 0$, and there is no reason for the off-diagonal sum to vanish. Hence

$$\text{plim } \hat{\boldsymbol{\beta}}_{\text{FE}} - \boldsymbol{\beta} = [E(\mathbf{F}_i'\mathbf{Q}\mathbf{F}_i)]^{-1} \sum_{m \neq h} q_{mh} E(\mathbf{f}_{im}\eta_{ih}),$$

which is nonzero in general. The fixed-effects estimator needs the regressors of *every* equation to be orthogonal to the error of *every* equation, because the deviation-from-means transformation mixes all M equations: $\tilde{\eta}_{im} = \eta_{im} - \frac{1}{M} \sum_h \eta_{ih}$ contains the errors of all equations, and $\mathbf{f}_{im}$ must be orthogonal to each of them. The dynamic panel of Section 5.4 and Analytical Exercise 5.4 is the canonical example in which the cross orthogonalities fail: $E(y_{i,m-1}\eta_{ih}) \neq 0$ for $h \leq m-1$, and the within estimator is inconsistent however large n is.

Review Question 5.2.4

Since $\tilde{\boldsymbol{\eta}}_i = \mathbf{Q}\boldsymbol{\eta}_i$ and $\mathbf{Q}\mathbf{1}_M = \mathbf{0}$,

$$\mathbf{1}_M'\tilde{\boldsymbol{\eta}}_i = \mathbf{1}_M'\mathbf{Q}\boldsymbol{\eta}_i = (\mathbf{Q}\mathbf{1}_M)'\boldsymbol{\eta}_i = 0 \quad \text{with probability one.}$$

Hence $E(\tilde{\boldsymbol{\eta}}_i\tilde{\boldsymbol{\eta}}_i')\mathbf{1}_M = E[\tilde{\boldsymbol{\eta}}_i(\mathbf{1}_M'\tilde{\boldsymbol{\eta}}_i)'] = \mathbf{0}$: the vector $\mathbf{1}_M$ lies in the null space, so the $M \times M$ matrix $E(\tilde{\boldsymbol{\eta}}_i\tilde{\boldsymbol{\eta}}_i')$ has rank at most $M-1$. Equivalently, $E(\tilde{\boldsymbol{\eta}}_i\tilde{\boldsymbol{\eta}}_i') = \mathbf{Q}E(\boldsymbol{\eta}_i\boldsymbol{\eta}_i')\mathbf{Q}$, and $\text{rank}(\mathbf{Q}\mathbf{A}\mathbf{Q}) \leq \text{rank}(\mathbf{Q}) = M-1$ for any $\mathbf{A}$. The transformed errors sum to zero within each group, so their covariance matrix must be singular. What matters for the asymptotic variance (5.2.7) is that the singularity does not propagate: $E(\mathbf{F}_i'\mathbf{Q}\mathbf{F}_i)$ is nonsingular by (5.1.15) (Analytical Exercise 5.5(a)), and the middle matrix of the sandwich is nonsingular by Analytical Exercise 5.5(b), so the asymptotic variance itself is a positive definite $\#\boldsymbol{\beta} \times \#\boldsymbol{\beta}$ matrix.

Review Question 5.2.5

Proposition 4.1 (consistent estimation of $\mathbf{S}$ from residuals) requires, for the transformed system $\tilde{\mathbf{y}}_i = \tilde{\mathbf{F}}_i\boldsymbol{\beta} + \tilde{\boldsymbol{\eta}}_i$: (i) an ergodic stationary (or i.i.d.) sample; (ii) a consistent first-stage estimator $\hat{\boldsymbol{\beta}}$ with which to form residuals $\hat{\tilde{\boldsymbol{\eta}}}_i = \tilde{\mathbf{y}}_i - \tilde{\mathbf{F}}_i\hat{\boldsymbol{\beta}}$; (iii) finite cross moments $E(\tilde{\eta}_{im}\tilde{\mathbf{f}}_{ih}')$ and $E(\tilde{\mathbf{f}}_{im}\tilde{\mathbf{f}}_{ih}')$ (plus the usual finite fourth moments for the law of large numbers applied to products); and (iv) the rank condition identifying $\boldsymbol{\beta}$. Here: (i) is (5.1.2); (ii) is supplied by the fixed-effects estimator

itself, consistent under Proposition 5.1; (iv) is (5.1.15). For (iii), write $\tilde{\mathbf{f}}_{im} = \sum_h q_{mh} \mathbf{f}_{ih}$ and $\tilde{\boldsymbol{\eta}}_{im} = \sum_h q_{mh} \boldsymbol{\eta}_{ih}$: both are fixed linear combinations (constant weights q_{mh}) of the elements of $\mathbf{F}_i$ and $\boldsymbol{\eta}_i$, and $\mathbf{F}_i$ in turn collects elements of $\mathbf{x}_i$, as the hint notes. The error-components model assumes $E(\mathbf{x}_i \mathbf{x}_i')$ finite and nonsingular ((5.1.5) and (5.1.6) imply (5.1.6')) and $E(\boldsymbol{\eta}_i \boldsymbol{\eta}_i')$ finite (it is the $\eta\eta$ block of the assumed finite $\boldsymbol{\Sigma} = E(\boldsymbol{\varepsilon}_i \boldsymbol{\varepsilon}_i')$, since $\boldsymbol{\varepsilon}_i = \mathbf{1}_M \alpha_i + \boldsymbol{\eta}_i$). By Cauchy–Schwarz,

$$|E(\tilde{\eta}_{im} \tilde{f}_{ih,k})| = \left| \sum_{r,s} q_{mr} q_{hs} E(\eta_{ir} f_{is,k}) \right| \leq \sum_{r,s} |q_{mr} q_{hs}| \sqrt{E(\eta_{ir}^2) E(f_{is,k}^2)} < \infty,$$

since every $E(\eta_{ir}^2)$ and $E(f_{is,k}^2)$ is finite. With all the conditions met, Proposition 4.1 gives $\frac{1}{n} \sum_i \hat{\tilde{\boldsymbol{\eta}}}_i \hat{\tilde{\boldsymbol{\eta}}}_i' \xrightarrow{p} E(\tilde{\boldsymbol{\eta}}_i \tilde{\boldsymbol{\eta}}_i')$, which is the consistency of (5.2.9), and plugging this into (5.2.8) yields a consistent estimator of the asymptotic variance of the fixed-effects estimator.

Review Question 5.2.6

No. The random-effects estimator (4.6.8') is GMM with weighting matrix built from $\mathbf{S} = \boldsymbol{\Sigma} \otimes E(\mathbf{x}_i \mathbf{x}_i')$; its probability limit is

$$\text{plim } \hat{\boldsymbol{\delta}}_{\text{RE}} - \boldsymbol{\delta} = [E(\mathbf{Z}_i' \boldsymbol{\Sigma}^{-1} \mathbf{Z}_i)]^{-1} E(\mathbf{Z}_i' \boldsymbol{\Sigma}^{-1} \boldsymbol{\varepsilon}_i),$$

so it is consistent if and only if

$$E(\mathbf{Z}_i' \boldsymbol{\Sigma}^{-1} \boldsymbol{\varepsilon}_i) = \mathbf{0}, \quad \boldsymbol{\varepsilon}_i = \mathbf{1}_M \alpha_i + \boldsymbol{\eta}_i.$$

Decompose:

$$E(\mathbf{Z}_i' \boldsymbol{\Sigma}^{-1} \boldsymbol{\varepsilon}_i) = E(\mathbf{Z}_i' \boldsymbol{\Sigma}^{-1} \mathbf{1}_M \alpha_i) + E(\mathbf{Z}_i' \boldsymbol{\Sigma}^{-1} \boldsymbol{\eta}_i).$$

The second term vanishes when (5.1.8B) holds: $E(\mathbf{z}_{im} \eta_{ih}) = 0$ for all m, h implies $E(\mathbf{Z}_i' \mathbf{A} \boldsymbol{\eta}_i) = \mathbf{0}$ for any constant matrix $\mathbf{A}$, in particular for $\mathbf{A} = \boldsymbol{\Sigma}^{-1}$. So consistency fails if and only if

$$E(\mathbf{Z}_i' \boldsymbol{\Sigma}^{-1} \mathbf{1}_M \alpha_i) \neq \mathbf{0}.$$

Condition (5.1.8A) — $E(\mathbf{z}_{im} \alpha_i) = 0$ for every m — is *sufficient* for this (it kills the first term whatever $\boldsymbol{\Sigma}$ is), but it is not *necessary*: what is required is only that the particular weighted combinations $\sum_{m,h} (\boldsymbol{\Sigma}^{-1})_{mh} E(\mathbf{z}_{im} \alpha_i)$ be zero, one vector equation in the M cross moments $E(\mathbf{z}_{im} \alpha_i)$. Those moments can all be nonzero — (5.1.8A) failing in every equation — and still conspire to satisfy the single effective restriction. Thus, strictly speaking, the Hausman test of Proposition 5.2 is not a test of (5.1.8A); it is a test of $E(\mathbf{Z}_i' \boldsymbol{\Sigma}^{-1} \boldsymbol{\varepsilon}_i) = \mathbf{0}$, the actual moment condition used by the random-effects estimator. The distinction rarely bites in practice — for the weighted sum to vanish while the individual moments do not is a measure-zero coincidence — but it is the logically correct statement of the null, and it is why the Hausman test is best described as a test of *overidentifying restrictions*: (5.1.8A) is exactly the set of orthogonality conditions that the random-effects estimator uses and the fixed-effects estimator does not.

Section 5.3 Unbalanced Panels (optional)

Review Question 5.3.1

Here $M = 3$ and $M_i = \mathbf{d}_i' \mathbf{d}_i = 2$. From (5.3.5), $\mathbf{Q} = \mathbf{I}_3 - \frac{1}{M_i} \mathbf{d}_i \mathbf{d}_i' = \mathbf{I}_3 - \frac{1}{2} \mathbf{d}_i \mathbf{d}_i'$, so

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 1 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}.$$

Check: $\mathbf{Q} \mathbf{d}_i = (\frac{1}{2} - \frac{1}{2}, 0, -\frac{1}{2} + \frac{1}{2})' = \mathbf{0}$, and $\mathbf{Q}$ is symmetric and idempotent — it is the annihilator associated with $\mathbf{d}_i$. Applied to $\mathbf{y}_i = (y_{i1}, 0, y_{i3})'$ (the missing second observation zeroed out),

$$\mathbf{Q} \mathbf{y}_i = (\frac{1}{2}(y_{i1} - y_{i3}), 0, \frac{1}{2}(y_{i3} - y_{i1}))' = (y_{i1} - \bar{y}_i, 0, y_{i3} - \bar{y}_i)', \quad \bar{y}_i = \frac{y_{i1} + y_{i3}}{2},$$

so available observations enter as deviations from the mean *over available observations*, and the missing row stays zero — exactly the “zeroing out” described in the text. Verified numerically in `code/ch5.py`.

Review Question 5.3.2

Let $\mathbf{E}_i$ be the $M \times M_i$ selection matrix that picks out the observed rows, so $\mathbf{y}_i^* = \mathbf{E}_i' \mathbf{y}_i$, $\mathbf{F}_i^* = \mathbf{E}_i' \mathbf{F}_i$, $\mathbf{d}_i^* = \mathbf{E}_i' \mathbf{d}_i = \mathbf{1}_{M_i}$, $\mathbf{E}_i' \mathbf{E}_i = \mathbf{I}_{M_i}$, and $\mathbf{E}_i \mathbf{E}_i' = \text{diag}(\mathbf{d}_i)$. Because missing rows are zero-filled, $\mathbf{d}_i = \mathbf{E}_i \mathbf{1}_{M_i}$ and

$$\mathbf{F}_i' \mathbf{Q} \mathbf{F}_i = \mathbf{F}_i' \left(\mathbf{I}_M - \frac{1}{M_i} \mathbf{d}_i \mathbf{d}_i' \right) \mathbf{F}_i = \mathbf{F}_i' \mathbf{F}_i - \frac{1}{M_i} (\mathbf{E}_i' \mathbf{F}_i)' \mathbf{1}_{M_i} \mathbf{1}_{M_i}' (\mathbf{E}_i' \mathbf{F}_i).$$

Two substitutions finish the job: $\mathbf{E}_i' \mathbf{F}_i = \mathbf{F}_i^*$ by definition, and — this is where zeroing out is used — $\mathbf{F}_i' \mathbf{F}_i = \mathbf{F}_i^{*'} \mathbf{F}_i^*$ because the rows of $\mathbf{F}_i$ corresponding to missing observations are all zeros and contribute nothing to the cross-product. Hence

$$\mathbf{F}_i' \mathbf{Q} \mathbf{F}_i = \mathbf{F}_i^{*'} \left(\mathbf{I}_{M_i} - \frac{1}{M_i} \mathbf{1}_{M_i} \mathbf{1}_{M_i}' \right) \mathbf{F}_i^* = \mathbf{F}_i^{*'} \mathbf{Q}^* \mathbf{F}_i^*,$$

and by the identical computation $\mathbf{F}_i' \mathbf{Q} \mathbf{y}_i = \mathbf{F}_i^{*'} \mathbf{Q}^* \mathbf{y}_i^*$. Summing over i and inverting,

$$\hat{\beta}_{\text{FE}} = \left(\sum_{i=1}^n \mathbf{F}_i^{*'} \mathbf{Q}^* \mathbf{F}_i^* \right)^{-1} \sum_{i=1}^n \mathbf{F}_i^{*'} \mathbf{Q}^* \mathbf{y}_i^*,$$

which is (5.2.4) written with the starred quantities: the “compressed” panel, with missing rows deleted rather than zero-filled, delivers numerically the same estimator. This is how panel software actually computes the estimator; the zero-filled representation exists to give it the common matrix form (5.2.4) across balanced and unbalanced panels, which is what makes the proof of Proposition 5.4 run parallel to that of Proposition 5.1.

Review Question 5.3.3

(a) Apply the zeroing-out device of (5.3.3) to the levels system $\mathbf{y}_i = \mathbf{Z}_i \boldsymbol{\delta} + \boldsymbol{\varepsilon}_i$: if observation m of group i is missing, replace the m -th element of $\mathbf{y}_i$ and the m -th row of $\mathbf{Z}_i$ by zeros (and likewise for $\boldsymbol{\varepsilon}_i$), keeping all matrices at their balanced-panel dimensions. The estimator (4.6.11') keeps its

form,

$$\widehat{\boldsymbol{\delta}} = \left(\sum_{i=1}^n \mathbf{Z}_i' \mathbf{Z}_i \right)^{-1} \sum_{i=1}^n \mathbf{Z}_i' \mathbf{y}_i = \left(\frac{1}{n} \sum_{i=1}^n \mathbf{Z}_i' \mathbf{Z}_i \right)^{-1} \frac{1}{n} \sum_{i=1}^n \mathbf{Z}_i' \mathbf{y}_i,$$

understood with the zero-filled matrices: rows of zeros contribute nothing to either cross-product, so this is exactly pooled OLS on the $\sum_i M_i$ available observations. The orthogonality conditions must be strengthened to rule out selectivity, $E(\mathbf{z}_{im} \varepsilon_{ih} \mid \mathbf{d}_i) = \mathbf{0}$ for all m, h , exactly as (5.3.7) strengthens (5.2.6): sample selection must not depend on the error term.

(b) Substituting $\mathbf{y}_i = \mathbf{Z}_i \boldsymbol{\delta} + \varepsilon_i$ gives the sampling error $\widehat{\boldsymbol{\delta}} - \boldsymbol{\delta} = \left(\frac{1}{n} \sum_i \mathbf{Z}_i' \mathbf{Z}_i \right)^{-1} \frac{1}{n} \sum_i \mathbf{Z}_i' \varepsilon_i$. Without conditional homoskedasticity the middle matrix does not collapse, so Kolmogorov's LLN and the Lindeberg–Levy CLT give

$$\text{Avar}(\sqrt{n}(\widehat{\boldsymbol{\delta}} - \boldsymbol{\delta})) = [E(\mathbf{Z}_i' \mathbf{Z}_i)]^{-1} E(\mathbf{Z}_i' \varepsilon_i \varepsilon_i' \mathbf{Z}_i) [E(\mathbf{Z}_i' \mathbf{Z}_i)]^{-1},$$

the displayed answer (with the book's tildes denoting the zero-filled quantities; the middle matrix is the variance of $\mathbf{Z}_i' \varepsilon_i$, finite under the usual fourth-moment conditions).

(c) Mimic Proposition 5.3(b): replace the population moments by sample analogues computed from the pooled residuals $\widehat{\varepsilon}_i = \mathbf{y}_i - \mathbf{Z}_i \widehat{\boldsymbol{\delta}}$ (zero-filled in the same way),

$$\widehat{\text{Avar}} = \left(\frac{1}{n} \sum_{i=1}^n \mathbf{Z}_i' \mathbf{Z}_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{Z}_i' \widehat{\varepsilon}_i \widehat{\varepsilon}_i' \mathbf{Z}_i \right) \left(\frac{1}{n} \sum_{i=1}^n \mathbf{Z}_i' \mathbf{Z}_i \right)^{-1},$$

which is the hint's instruction verbatim: replace $\mathbf{F}_i$ by $\mathbf{Z}_i$, $\mathbf{Q}$ by $\mathbf{I}_M$, and $\widetilde{\boldsymbol{\eta}}_i$ by ε_i in (5.2.24). Consistency follows from Proposition 4.1 applied to this system, under the no-selectivity condition and finite fourth moments. Under conditional homoskedasticity the middle matrix simplifies and one recovers the pooled-OLS formula of Section 4.6.

Section 5.4 Application: International Differences in Growth Rates

Review Question 5.4.1

No. Trace the derivation: (5.4.2) gives $\log q(t_m) = \rho \log q(t_{m-1}) + (1 - \rho) \log q^*$ with $\rho = \exp(-\lambda(t_m - t_{m-1}))$; with $\log q(t) = \log(Y(t)/L(t)) - \log A(0) - g t$ substituted in, the terms involving g and $A(0)$ contribute

$$-g t_m + \rho g t_{m-1} = -g(t_m - \rho t_{m-1}),$$

which is absorbed into the intercept ξ_m of (5.4.6), together with $(1 - \rho) \log q^*$ and the $\log A(0)$ pieces. Suppose instead that technical progress grew at rate g_m between t_{m-1} and t_m , common across countries. Then $A(t)$ cancels against the fixed effect and the common trend in exactly the same way: the only change is that ξ_m picks up a different additive constant, $-g_m(t_m - \rho t_{m-1})$, in each period. Since the ξ_m are free, equation-specific intercepts — and the empirical exercise of Section 5.4(b) indeed includes 24 year dummies — any common time path of g is absorbed without trace. What cannot be relaxed is the cross-country restriction: if g differed across countries in a manner correlated with $y_{i,m-1}$, it would not be absorbed into the common ξ_m and would bias ρ (a country-specific mean of g reappears inside the country effect α_i of (5.4.9), which is exactly why

the fixed-effects treatment of Section 5.4 is attractive). Constancy of g over time is not needed; commonality across countries is.

Review Question 5.4.2

(a) The system (5.4.11) with $M = 3$ consists of two equations,

$$y_{i2} - y_{i1} = \xi_1 + (y_{i1} - y_{i0})\rho + (\eta_{i2} - \eta_{i1}), \quad y_{i3} - y_{i2} = \xi_2 + (y_{i2} - y_{i1})\rho + (\eta_{i3} - \eta_{i2}),$$

with common coefficient ρ and equation-specific intercepts. In the notation of (4.6.1) set, with superscript “lev” for the levels,

$$y_{i1} \equiv y_{i2}^{\text{lev}} - y_{i1}^{\text{lev}}, \quad \mathbf{z}_{i1} = (1, 0, y_{i1}^{\text{lev}} - y_{i0}^{\text{lev}})', \\ y_{i2} \equiv y_{i3}^{\text{lev}} - y_{i2}^{\text{lev}}, \quad \mathbf{z}_{i2} = (0, 1, y_{i2}^{\text{lev}} - y_{i1}^{\text{lev}})', \quad \boldsymbol{\delta} = (\xi_1, \xi_2, \rho)'$$

Then $\mathbf{z}_{i1}'\boldsymbol{\delta} = \xi_1 + (y_{i1}^{\text{lev}} - y_{i0}^{\text{lev}})\rho$ and $\mathbf{z}_{i2}'\boldsymbol{\delta} = \xi_2 + (y_{i2}^{\text{lev}} - y_{i1}^{\text{lev}})\rho$, as required: the system is (4.6.1) with $M = 2$ equations and $L = 3$ coefficients, and the second regressor vector is the hint's $(0, 1, y_{i2} - y_{i1})'$.

(b) With instruments $\mathbf{x}_{im} = (1, s_{im})'$, the instrument vector for the system is $\mathbf{x}_i = (1, s_{i1}, s_{i2})'$ ($K = 3$ unique elements). Identification for the multiple-equation GMM requires $E(\mathbf{Z}_i \otimes \mathbf{x}_i)$ to have full column rank $L = 3$. Examine the column of this matrix associated with ρ : its nonzero elements are $E[(y_{i1} - y_{i0})\mathbf{x}_{i1}]$ from equation 1 and $E[(y_{i2} - y_{i1})\mathbf{x}_{i2}]$ from equation 2. Suppose $\text{Cov}(s_{im}, y_{im} - y_{i,m-1}) = 0$ for $m = 1, 2$, i.e. $E[s_{im}(y_{im} - y_{i,m-1})] = E(s_{im})E(y_{im} - y_{i,m-1})$. Then

$$E[(y_{im} - y_{i,m-1})\mathbf{x}_{im}] = E(y_{im} - y_{i,m-1}) \cdot (1, E(s_{im}))' = E(y_{im} - y_{i,m-1}) \cdot E(\mathbf{x}_{im}),$$

so the ρ column of $E(\mathbf{Z}_i \otimes \mathbf{x}_i)$ satisfies

$$\text{col}_\rho = E(y_{i1} - y_{i0}) \cdot \text{col}_{\xi_1} + E(y_{i2} - y_{i1}) \cdot \text{col}_{\xi_2},$$

because the intercept columns contribute the blocks $(E(\mathbf{x}_{i1}', \mathbf{0}'))'$ and $(\mathbf{0}', E(\mathbf{x}_{i2}'))'$ respectively. A nontrivial linear combination of the columns is zero, so the rank is at most $2 < 3$: the rank condition fails and ρ is not identified. The economics is transparent: in each differenced equation the only instrument capable of separating ρ from the intercept is the saving rate, and if the saving rate does not covary with the contemporaneous change in log output it has no power to do so; each equation collapses to a reduced-form mean. Instrument *relevance* ($\text{Cov} \neq 0$) is indispensable on top of instrument exogeneity.

Problem Set for Chapter 5: Analytical Exercises

Analytical Exercise 5.1

(a) *First proof (Frisch–Waugh)*. By the Frisch–Waugh theorem (Analytical Exercise 1.4(b)), the OLS coefficient on $\mathbf{F}$ in the regression of $\mathbf{y}$ on $\mathbf{W} = (\mathbf{D} \quad \mathbf{F})$ equals the OLS coefficient in the regression of $\mathbf{M}_D \mathbf{y}$ on $\mathbf{M}_D \mathbf{F}$, where $\mathbf{M}_D = \mathbf{I}_{Mn} - \mathbf{D}(\mathbf{D}'\mathbf{D})^{-1}\mathbf{D}'$. Since $\mathbf{D} = \mathbf{I}_n \otimes \mathbf{1}_M$,

$$\mathbf{D}'\mathbf{D} = \mathbf{I}_n \otimes (\mathbf{1}_M' \mathbf{1}_M) = M \mathbf{I}_n, \quad \mathbf{D}(\mathbf{D}'\mathbf{D})^{-1}\mathbf{D}' = \mathbf{I}_n \otimes \frac{1}{M} \mathbf{1}_M \mathbf{1}_M',$$

so

$$\mathbf{M}_D = \mathbf{I}_n \otimes \left(\mathbf{I}_M - \frac{1}{M} \mathbf{1}_M \mathbf{1}_M' \right) = \mathbf{I}_n \otimes \mathbf{Q},$$

with $\mathbf{Q}$ the annihilator associated with $\mathbf{1}_M$. Hence $\mathbf{M}_D \mathbf{F}$ stacks the blocks $\mathbf{QF}_i = \tilde{\mathbf{F}}_i$ and $\mathbf{M}_D \mathbf{y}$ stacks $\mathbf{Qy}_i = \tilde{\mathbf{y}}_i$, and the coefficient is

$$\mathbf{b} = [(\mathbf{M}_D \mathbf{F})'(\mathbf{M}_D \mathbf{F})]^{-1}(\mathbf{M}_D \mathbf{F})' \mathbf{M}_D \mathbf{y} = \left(\sum_{i=1}^n \mathbf{F}_i' \mathbf{QF}_i \right)^{-1} \sum_{i=1}^n \mathbf{F}_i' \mathbf{Qy}_i = \hat{\beta}_{FE},$$

using symmetry and idempotency of $\mathbf{M}_D$; this is exactly (5.2.4).

Second proof (normal equations). The normal equations $\mathbf{W}'\mathbf{W} \begin{bmatrix} \hat{\alpha} \\ \mathbf{b} \end{bmatrix} = \mathbf{W}'\mathbf{y}$ in partitioned form are

$$\mathbf{D}'\mathbf{D} \hat{\alpha} + \mathbf{D}'\mathbf{F}\mathbf{b} = \mathbf{D}'\mathbf{y}, \quad \mathbf{F}'\mathbf{D} \hat{\alpha} + \mathbf{F}'\mathbf{F}\mathbf{b} = \mathbf{F}'\mathbf{y}.$$

The first gives $\hat{\alpha} = (\mathbf{D}'\mathbf{D})^{-1}(\mathbf{D}'\mathbf{y} - \mathbf{D}'\mathbf{F}\mathbf{b})$, the formula suggested in the hint. Substituting into the second,

$$\mathbf{F}'[\mathbf{I} - \mathbf{D}(\mathbf{D}'\mathbf{D})^{-1}\mathbf{D}']\mathbf{F}\mathbf{b} = \mathbf{F}'[\mathbf{I} - \mathbf{D}(\mathbf{D}'\mathbf{D})^{-1}\mathbf{D}']\mathbf{y},$$

i.e. $\mathbf{F}'\mathbf{M}_D \mathbf{F}\mathbf{b} = \mathbf{F}'\mathbf{M}_D \mathbf{y}$, the same system as above. The inversion is trivial because $\mathbf{D}'\mathbf{D} = M\mathbf{I}_n$ — which is why the dummy-variable route is computationally painless even with thousands of groups.

(b) Row i of the first block of normal equations: the i -th diagonal block of $\mathbf{D}'\mathbf{D}$ is M , the i -th block of $\mathbf{D}'\mathbf{y}$ is $\mathbf{1}_M' \mathbf{y}_i = M\bar{y}_i$, and the i -th block of $\mathbf{D}'\mathbf{F}$ is $\mathbf{1}_M' \mathbf{F}_i = M\bar{\mathbf{f}}_i'$. Hence

$$M\hat{\alpha}_i + M\bar{\mathbf{f}}_i' \mathbf{b} = M\bar{y}_i \implies \hat{\alpha}_i = \bar{y}_i - \bar{\mathbf{f}}_i' \mathbf{b},$$

which is the displayed result: the estimated fixed effect is what remains of the group's mean dependent variable after subtracting the systematic part fitted at the group's mean regressors.

(c) *Classical assumptions.* Linearity (Assumption 1.1) holds by construction, with the coefficient vector $(\alpha', \beta')'$ common to all Mn observations. For strict exogeneity (Assumption 1.2), note that $\mathbf{D}$ is a matrix of constants, so conditioning on $\mathbf{W} = (\mathbf{D} \ \mathbf{F})$ is conditioning on $\mathbf{F} = (\mathbf{F}_1', \dots, \mathbf{F}_n')'$:

$$E(\boldsymbol{\eta}_i \mid \mathbf{W}) = E(\boldsymbol{\eta}_i \mid \mathbf{F}) = E(\boldsymbol{\eta}_i \mid \mathbf{F}_i) = \mathbf{0},$$

the second equality because $\boldsymbol{\eta}_i$ is independent of $(\mathbf{F}_1, \dots, \mathbf{F}_{i-1}, \mathbf{F}_{i+1}, \dots, \mathbf{F}_n)$ by the i.i.d. assumption (i), the third by (ii). Conditional homoskedasticity (Assumption 1.4):

$$E(\boldsymbol{\eta}_i \boldsymbol{\eta}_i' \mid \mathbf{W}) = E(\boldsymbol{\eta}_i \boldsymbol{\eta}_i' \mid \mathbf{F}_i) = \sigma_{\eta}^2 \mathbf{I}_M$$

by the same independence and (iii); and $E(\boldsymbol{\eta}_i \boldsymbol{\eta}_j' \mid \mathbf{W}) = \mathbf{0}$ for $i \neq j$ by (i) together with $E(\boldsymbol{\eta}_i \mid \mathbf{F}) = \mathbf{0}$. Stacking gives $E(\boldsymbol{\eta} \boldsymbol{\eta}' \mid \mathbf{W}) = \sigma_{\eta}^2 \mathbf{I}_{Mn}$. No multicollinearity (Assumption 1.3) is (iv). (If in addition $\boldsymbol{\eta}$ is normal, Assumption 1.5 holds too.)

Small-sample theory. All of Chapter 1's finite-sample results therefore apply to the pooled regression: by Proposition 1.1, $(\hat{\alpha}, \mathbf{b})$ are unbiased conditional on $\mathbf{W}$ (hence unconditionally), with

$$\text{Var} \left(\begin{bmatrix} \hat{\alpha} \\ \mathbf{b} \end{bmatrix} \mid \mathbf{W} \right) = \sigma_{\eta}^2 (\mathbf{W}'\mathbf{W})^{-1},$$

and they are Gauss–Markov efficient in the class of linear unbiased estimators; under normality the usual t and F tests are exact. Two caveats. First, unbiasedness uses strict exogeneity *with respect to the whole matrix* $\mathbf{F}$ — assumption (ii), which is stronger than the orthogonality conditions (5.2.6) that suffice for the large-sample theory, and which fails in dynamic panels. Second, with n large

and M fixed, each $\hat{\alpha}_i$ is computed from only M observations and is not consistent as $n \rightarrow \infty$ (the incidental-parameters problem); only $\mathbf{b}$ is consistent.

The SSR. Yes, the two sums of squared residuals are identical. By Frisch–Waugh (Analytical Exercise 1.4(c)), the residual vector of the pooled regression equals the residual vector of the regression of $\mathbf{M}_D \mathbf{y}$ on $\mathbf{M}_D \mathbf{F}$, which is the within regression (5.2.15): the i -th block is $\mathbf{Qy}_i - \mathbf{QF}_i \mathbf{b}$. Hence $\text{SSR} = \sum_i (\mathbf{Qy}_i - \mathbf{QF}_i \mathbf{b})' (\mathbf{Qy}_i - \mathbf{QF}_i \mathbf{b})$, exactly (5.2.15). (What differs is only the degrees-of-freedom bookkeeping: the pooled regression estimates $n + \#\beta$ coefficients, so its s^2 divides by $Mn - n - \#\beta$ — which is the *correct* denominator (5.2.17). The dummy-variable route gets the degrees-of-freedom correction right automatically.)

Analytical Exercise 5.2

(a) *First-difference matrix.* Let $\mathbf{C}$ be the $M \times (M - 1)$ matrix with $C_{m,m} = -1$, $C_{m+1,m} = 1$ and zeros elsewhere, so $\mathbf{C}' \mathbf{y}_i$ stacks the first differences $y_{i,m+1} - y_{im}$. Column m of $\mathbf{C}$ is $-\mathbf{e}_m + \mathbf{e}_{m+1}$, whose elements sum to zero: $\mathbf{C}' \mathbf{1}_M = \mathbf{0}$. Its $M - 1$ columns are linearly independent: if $\mathbf{C} \mathbf{c} = \mathbf{0}$, reading the rows from the top gives $-c_1 = 0$, $c_1 - c_2 = 0$, $\dots$, forcing $\mathbf{c} = \mathbf{0}$. So $\mathbf{C}$ has full column rank $M - 1$.

Dropped column of Q. Write $\mathbf{Q} = (\mathbf{C} \ \mathbf{q})$ with $\mathbf{C}$ the first $M - 1$ columns. Since $\mathbf{Q} \mathbf{1}_M = \mathbf{0}$, every column of $\mathbf{Q}$ is orthogonal to $\mathbf{1}_M$, so $\mathbf{C}' \mathbf{1}_M = \mathbf{0}$. The null space of the symmetric idempotent $\mathbf{Q}$ is spanned by $\mathbf{1}_M$; if the first $M - 1$ columns were linearly dependent, some $\mathbf{c} \neq \mathbf{0}$ supported on the first $M - 1$ coordinates would satisfy $\mathbf{Q} \mathbf{c} = \mathbf{0}$, forcing $\mathbf{c} \propto \mathbf{1}_M$ — impossible for a vector with a zero last coordinate. Hence $\text{rank}(\mathbf{C}) = M - 1$.

(b) Premultiplying (5.1.1''), $\mathbf{y}_i = \mathbf{F}_i \beta + \mathbf{1}_M (\mathbf{b}_i' \gamma + \alpha_i) + \boldsymbol{\eta}_i$, by $\mathbf{C}'$ annihilates the $\mathbf{1}_M$ term because $\mathbf{C}' \mathbf{1}_M = \mathbf{0}$:

$$\mathbf{C}' \mathbf{y}_i = \mathbf{C}' \mathbf{F}_i \beta + \mathbf{C}' \boldsymbol{\eta}_i,$$

which is the transformed system with the stated definitions. Note that both the fixed effect *and* the common regressors $\mathbf{b}_i$ drop out, since both enter through $\mathbf{1}_M$.

(c) Check the five assumptions of the random-effects model for the $(M - 1)$ -equation system.

Random sample: $(\tilde{\mathbf{y}}_i, \tilde{\mathbf{F}}_i)$ is a fixed linear function of $(\mathbf{y}_i, \mathbf{Z}_i)$, hence i.i.d. by (5.1.2).

Orthogonality conditions $E(\tilde{\boldsymbol{\eta}}_i \otimes \mathbf{x}_i) = \mathbf{0}$: since $\tilde{\boldsymbol{\eta}}_i \otimes \mathbf{x}_i = (\mathbf{C}' \otimes \mathbf{I}_K)(\boldsymbol{\eta}_i \otimes \mathbf{x}_i)$ and (5.1.8B) is $E(\boldsymbol{\eta}_i \otimes \mathbf{x}_i) = \mathbf{0}$, the transformed moment is $(\mathbf{C}' \otimes \mathbf{I}_K) \mathbf{0} = \mathbf{0}$.

Identification: $E(\tilde{\mathbf{F}}_i \otimes \mathbf{x}_i)$ of full column rank is equivalent to (5.1.15). Indeed, with $\mathbf{L} \equiv \mathbf{C}(\mathbf{C}' \mathbf{C})^{-1}$ the identity of part (d) gives $\mathbf{Q} = \mathbf{L} \mathbf{C}'$, hence $E(\mathbf{Q} \mathbf{F}_i \otimes \mathbf{x}_i) = (\mathbf{L} \otimes \mathbf{I}_K) E(\mathbf{C}' \mathbf{F}_i \otimes \mathbf{x}_i)$; write this as $\mathbf{A} = \mathbf{D} \mathbf{B}$ with $\mathbf{D} = \mathbf{L} \otimes \mathbf{I}_K$ of full column rank. Then $\text{rank}(\mathbf{A}) \leq \text{rank}(\mathbf{B})$, and $\mathbf{D}' \mathbf{A} = (\mathbf{D}' \mathbf{D}) \mathbf{B}$ with $\mathbf{D}' \mathbf{D}$ nonsingular, so $\text{rank}(\mathbf{A}) \geq \text{rank}(\mathbf{B})$: the ranks coincide, and since $\mathbf{A}$ and $\mathbf{B}$ have the same number of columns, one has full column rank iff the other does. (This is the argument of the book's answer sheet.)

Conditional homoskedasticity:

$$E(\tilde{\boldsymbol{\eta}}_i \tilde{\boldsymbol{\eta}}_i' | \mathbf{x}_i) = \mathbf{C}' E(\boldsymbol{\varepsilon}_i \boldsymbol{\varepsilon}_i' | \mathbf{x}_i) \mathbf{C} = \mathbf{C}' \boldsymbol{\Sigma} \mathbf{C} \equiv \boldsymbol{\Psi},$$

using $\tilde{\boldsymbol{\eta}}_i = \mathbf{C}'\boldsymbol{\eta}_i = \mathbf{C}'\boldsymbol{\varepsilon}_i$ (because $\mathbf{C}'\mathbf{1}_M\alpha_i = \mathbf{0}$) and (5.1.5); $\boldsymbol{\Psi}$ does not depend on $\mathbf{x}_i$, and it is nonsingular because $\mathbf{C}$ has full column rank and $\boldsymbol{\Sigma}$ is positive definite (for $\mathbf{v} \neq \mathbf{0}$, $\mathbf{C}\mathbf{v} \neq \mathbf{0}$, so $\mathbf{v}'\mathbf{C}'\boldsymbol{\Sigma}\mathbf{C}\mathbf{v} > 0$).

Nonsingularity of $E(\mathbf{g}_i\mathbf{g}_i')$: with $\mathbf{g}_i = \tilde{\boldsymbol{\eta}}_i \otimes \mathbf{x}_i$,

$$E(\mathbf{g}_i\mathbf{g}_i') = E[E(\tilde{\boldsymbol{\eta}}_i\tilde{\boldsymbol{\eta}}_i' | \mathbf{x}_i) \otimes \mathbf{x}_i\mathbf{x}_i'] = \boldsymbol{\Psi} \otimes E(\mathbf{x}_i\mathbf{x}_i'),$$

nonsingular as the Kronecker product of two nonsingular matrices (the second factor nonsingular by (5.1.6')). All five assumptions hold, so Proposition 4.7 applies to the transformed system.

(d) Use the hint's identity:

$$\mathbf{Q} = \mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{C}' \quad \text{for any } \mathbf{C} \text{ of full column rank with } \mathbf{C}'\mathbf{1}_M = \mathbf{0}. \quad (*)$$

(Proof: $\mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{C}'$ is the orthogonal projector onto the column space of $\mathbf{C}$; that space is contained in $\{\mathbf{1}_M\}^\perp$ because $\mathbf{C}'\mathbf{1}_M = \mathbf{0}$, and it has dimension $M - 1 = \dim\{\mathbf{1}_M\}^\perp$, so it equals $\{\mathbf{1}_M\}^\perp$; $\mathbf{Q}$ is the orthogonal projector onto the same space, and orthogonal projectors onto a given subspace are unique.) Substituting into (5.2.4),

$$\hat{\boldsymbol{\beta}}_{\text{FE}} = \left(\frac{1}{n} \sum_i \tilde{\mathbf{F}}_i'(\mathbf{C}'\mathbf{C})^{-1}\tilde{\mathbf{F}}_i \right)^{-1} \frac{1}{n} \sum_i \tilde{\mathbf{F}}_i'(\mathbf{C}'\mathbf{C})^{-1}\tilde{\mathbf{y}}_i.$$

This is the GMM format with, as the book says, “the disappearance of $\mathbf{x}$ ”: the elements of $\mathbf{F}_i$ being a subset of $\mathbf{x}_i$, the moment conditions that survive the transformation are just the own-equation ones, and

$$\mathbf{S}_{\mathbf{zx}} = \frac{1}{n} \sum_{i=1}^n \tilde{\mathbf{F}}_i, \quad \mathbf{s}_{\mathbf{xy}} = \frac{1}{n} \sum_{i=1}^n \tilde{\mathbf{y}}_i, \quad \widehat{\mathbf{W}} = (\mathbf{C}'\mathbf{C})^{-1}.$$

So the fixed-effects estimator is GMM on the transformed system with weighting matrix $(\mathbf{C}'\mathbf{C})^{-1}$ — efficient if and only if $\boldsymbol{\Psi}$ is proportional to $\mathbf{C}'\mathbf{C}$.

(e) All assumptions of Proposition 4.7 hold for the transformed system by part (c), so the efficient GMM estimator uses $\widehat{\mathbf{W}} = \widehat{\boldsymbol{\Psi}}^{-1}$:

$$\hat{\boldsymbol{\beta}}_{\text{eff}} = \left(\sum_i \tilde{\mathbf{F}}_i' \widehat{\boldsymbol{\Psi}}^{-1} \tilde{\mathbf{F}}_i \right)^{-1} \sum_i \tilde{\mathbf{F}}_i' \widehat{\boldsymbol{\Psi}}^{-1} \tilde{\mathbf{y}}_i, \quad \text{Avar}(\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{eff}} - \boldsymbol{\beta})) = [E(\tilde{\mathbf{F}}_i' \boldsymbol{\Psi}^{-1} \tilde{\mathbf{F}}_i)]^{-1},$$

which is (13). Comparing with the sandwich form of Proposition 5.1 confirms the text's suspicion: the fixed-effects estimator is generally inefficient because it uses $(\mathbf{C}'\mathbf{C})^{-1}$ where $\boldsymbol{\Psi}^{-1}$ belongs; the two coincide only when $\boldsymbol{\Psi} \propto \mathbf{C}'\mathbf{C}$, which part (h) ties to sphericity.

(f) Let $\widehat{\tilde{\boldsymbol{\eta}}}_i = \tilde{\mathbf{y}}_i - \tilde{\mathbf{F}}_i\hat{\boldsymbol{\beta}}_{\text{FE}}$ and write

$$\widehat{\boldsymbol{\Psi}} - \boldsymbol{\Psi} = \frac{1}{n} \sum_i (\widehat{\tilde{\boldsymbol{\eta}}}_i \widehat{\tilde{\boldsymbol{\eta}}}_i' - \boldsymbol{\Psi}) + \frac{1}{n} \sum_i (\widehat{\tilde{\boldsymbol{\eta}}}_i \widehat{\tilde{\boldsymbol{\eta}}}_i' - \tilde{\boldsymbol{\eta}}_i \tilde{\boldsymbol{\eta}}_i').$$

The first term is $o_p(1)$ by Kolmogorov's LLN (i.i.d. sampling, finite $E\|\tilde{\boldsymbol{\eta}}_i\|^2$ from finiteness of $\boldsymbol{\Sigma}$). For the second, since $\widehat{\tilde{\boldsymbol{\eta}}}_i = \tilde{\boldsymbol{\eta}}_i - \tilde{\mathbf{F}}_i(\hat{\boldsymbol{\beta}}_{\text{FE}} - \boldsymbol{\beta})$,

$$\frac{1}{n} \sum_i (\widehat{\tilde{\boldsymbol{\eta}}}_i \widehat{\tilde{\boldsymbol{\eta}}}_i' - \tilde{\boldsymbol{\eta}}_i \tilde{\boldsymbol{\eta}}_i') = - \left(\frac{1}{n} \sum_i \tilde{\boldsymbol{\eta}}_i \otimes \tilde{\mathbf{F}}_i \right)'_{(\cdot)} (\hat{\boldsymbol{\beta}}_{\text{FE}} - \boldsymbol{\beta}) - \{\text{transpose}\} + \frac{1}{n} \sum_i \tilde{\mathbf{F}}_i (\hat{\boldsymbol{\beta}}_{\text{FE}} - \boldsymbol{\beta}) (\hat{\boldsymbol{\beta}}_{\text{FE}} - \boldsymbol{\beta})' \tilde{\mathbf{F}}_i',$$

where the sample moments converge by the LLN under the finite cross-moment conditions verified in Review Question 5.2.5, while $\hat{\beta}_{\text{FE}} - \beta = o_p(1)$ by Proposition 5.1. So the whole expression is $o_p(1)$: these are exactly the conditions of Proposition 4.1 applied to the transformed system, and $\hat{\Psi} \xrightarrow{p} \Psi$.

(g) Sargan's statistic for the efficient estimator is (4.6.10') applied to the transformed system: with $\hat{\mathbf{g}}_i = \hat{\boldsymbol{\eta}}_i \otimes \mathbf{x}_i$ evaluated at $\hat{\beta}_{\text{eff}}$, $\hat{\mathbf{S}} = \frac{1}{n} \sum_i \hat{\mathbf{g}}_i \hat{\mathbf{g}}_i'$ and $\bar{\mathbf{g}}_n = \frac{1}{n} \sum_i \hat{\mathbf{g}}_i$,

$$J = n \bar{\mathbf{g}}_n' \hat{\mathbf{S}}^{-1} \bar{\mathbf{g}}_n \xrightarrow{d} \chi^2((M-1)K - \#\beta),$$

the degrees of freedom being the number of overidentifying restrictions: $(M-1)K$ moment conditions minus $\#\beta$ parameters. (With the collapsed instruments of part (d) the degrees of freedom reduce accordingly; the full instrument set tests the cross-equation orthogonality conditions as well, which is what makes the statistic useful as a specification test.)

(h) If $\boldsymbol{\eta}_i$ is spherical, $E(\boldsymbol{\eta}_i \boldsymbol{\eta}_i') = \sigma_{\eta}^2 \mathbf{I}_M$, then

$$\Psi = \mathbf{C}' E(\boldsymbol{\eta}_i \boldsymbol{\eta}_i') \mathbf{C} = \sigma_{\eta}^2 \mathbf{C}' \mathbf{C},$$

so $\hat{\Psi} = \hat{\sigma}_{\eta}^2 \mathbf{C}' \mathbf{C}$ is consistent whenever $\hat{\sigma}_{\eta}^2$ is (for instance Analytical Exercise 5.3's SSR/[$n(M-1) - \#\beta$]). The efficient GMM estimator with this weighting is

$$\left(\sum_i \tilde{\mathbf{F}}_i' [\hat{\sigma}_{\eta}^2 \mathbf{C}' \mathbf{C}]^{-1} \tilde{\mathbf{F}}_i \right)^{-1} \sum_i \tilde{\mathbf{F}}_i' [\hat{\sigma}_{\eta}^2 \mathbf{C}' \mathbf{C}]^{-1} \tilde{\mathbf{y}}_i = \left(\sum_i \mathbf{F}_i' \mathbf{C} (\mathbf{C}' \mathbf{C})^{-1} \mathbf{C}' \mathbf{F}_i \right)^{-1} \sum_i \mathbf{F}_i' \mathbf{C} (\mathbf{C}' \mathbf{C})^{-1} \mathbf{C}' \mathbf{y}_i,$$

the scalar $\hat{\sigma}_{\eta}^2$ cancelling between the two factors; by (*) this is exactly $\hat{\beta}_{\text{FE}}$. So *under sphericity the fixed-effects estimator is the efficient GMM estimator*: its inefficiency in Proposition 5.1 comes entirely from nonspherical $\boldsymbol{\eta}_i$ (serial correlation or heteroskedasticity across m), and the estimator of part (e) is the asymptotically more efficient estimator the text promised.

(i) Let $\mathbf{B} = \mathbf{C}\mathbf{A}$ with $\mathbf{A}$ an $(M-1) \times (M-1)$ nonsingular matrix. Then $\mathbf{B}'\mathbf{1}_M = \mathbf{A}'\mathbf{C}'\mathbf{1}_M = \mathbf{0}$ and $\text{rank}(\mathbf{B}) = M-1$, so $\mathbf{B}$ is admissible. Everything transforms covariantly: the transformed data become $\mathbf{A}'\tilde{\mathbf{y}}_i$, $\mathbf{A}'\tilde{\mathbf{F}}_i$, $\mathbf{A}'\tilde{\boldsymbol{\eta}}_i$; and

$$\mathbf{B}(\mathbf{B}'\mathbf{B})^{-1}\mathbf{B}' = \mathbf{C}\mathbf{A}\mathbf{A}^{-1}(\mathbf{C}'\mathbf{C})^{-1}(\mathbf{A}')^{-1}\mathbf{A}'\mathbf{C}' = \mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{C}' = \mathbf{Q},$$

so the fixed-effects estimator of part (d) is literally unchanged. For the efficient estimator, $\Psi_{\mathbf{B}} = \mathbf{A}'\Psi\mathbf{A}$ and $\hat{\Psi}_{\mathbf{B}}^{-1} = (\mathbf{A}')^{-1}\hat{\Psi}^{-1}\mathbf{A}^{-1}$, so

$$(\mathbf{A}'\tilde{\mathbf{F}}_i)' \hat{\Psi}_{\mathbf{B}}^{-1} (\mathbf{A}'\tilde{\mathbf{F}}_i) = \tilde{\mathbf{F}}_i' \hat{\Psi}^{-1} \tilde{\mathbf{F}}_i,$$

and likewise for the cross term: the efficient estimator, its asymptotic variance, Sargan's statistic, and the spherical-case formula of part (h) are all numerically invariant to $\mathbf{C}$. The choice between first differences, dropping one equation, or any other basis of $\{\mathbf{1}_M\}^{\perp}$ is purely cosmetic — though, as Review Question 5.2.2 warned, the *weighting* must transform with the basis; unweighted OLS on a subset of equations is not invariant.

Analytical Exercise 5.3

Follow the hint's chain of algebra. With $\tilde{\mathbf{y}}_i = \mathbf{Q}\mathbf{y}_i$, $\tilde{\mathbf{F}}_i = \mathbf{Q}\mathbf{F}_i$ and $\hat{\tilde{\boldsymbol{\eta}}}_i = \tilde{\mathbf{y}}_i - \tilde{\mathbf{F}}_i\hat{\boldsymbol{\beta}}_{\text{FE}}$,

$$\frac{\text{SSR}}{n} = \frac{1}{n} \sum_{i=1}^n \hat{\tilde{\boldsymbol{\eta}}}_i' \hat{\tilde{\boldsymbol{\eta}}}_i = \frac{1}{n} \sum_{i=1}^n \text{tr}(\hat{\tilde{\boldsymbol{\eta}}}_i \hat{\tilde{\boldsymbol{\eta}}}_i') = \text{tr} \left[\frac{1}{n} \sum_{i=1}^n \hat{\tilde{\boldsymbol{\eta}}}_i \hat{\tilde{\boldsymbol{\eta}}}_i' \right],$$

using $a'a = \text{tr}(aa')$ for a column vector and the linearity of the trace. Now bring in the first-difference matrix $\mathbf{C}$ of Analytical Exercise 5.2: since $\mathbf{Q} = \mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{C}'$ by identity (*) there, define

$$\mathbf{v}_i \equiv \mathbf{C}'\mathbf{y}_i - \mathbf{C}'\mathbf{F}_i\boldsymbol{\beta} = \mathbf{C}'\boldsymbol{\eta}_i, \quad \hat{\mathbf{v}}_i = \mathbf{C}'\mathbf{y}_i - \mathbf{C}'\mathbf{F}_i\hat{\boldsymbol{\beta}}_{\text{FE}}.$$

Because the within and differenced transformations span the same space, the residual quadratic form is basis-free:

$$\hat{\tilde{\boldsymbol{\eta}}}_i' \hat{\tilde{\boldsymbol{\eta}}}_i = \hat{\mathbf{v}}_i' (\mathbf{C}'\mathbf{C})^{-1} \hat{\mathbf{v}}_i,$$

so

$$\frac{\text{SSR}}{n} = \text{tr} \left[(\mathbf{C}'\mathbf{C})^{-1} \frac{1}{n} \sum_{i=1}^n \hat{\mathbf{v}}_i \hat{\mathbf{v}}_i' \right],$$

which is the hint's final display. Under the usual technical condition that $E(\tilde{f}_{im,k}\tilde{f}_{ih,\ell})$ and $E(\tilde{f}_{im,k}\tilde{\eta}_{ih})$ exist and are finite for all m, h, k, ℓ (which obtains if $E(\mathbf{x}_i\mathbf{x}_i')$ and $\boldsymbol{\Sigma}$ are finite, by Cauchy-Schwarz as in Review Question 5.2.5), the argument of Analytical Exercise 5.2(f) applies with $\frac{1}{n} \sum_i \hat{\mathbf{v}}_i \hat{\mathbf{v}}_i'$ in the role of $\hat{\boldsymbol{\Psi}}$:

$$\frac{1}{n} \sum_{i=1}^n \hat{\mathbf{v}}_i \hat{\mathbf{v}}_i' \xrightarrow{p} E(\mathbf{v}_i \mathbf{v}_i') = \mathbf{C}' E(\boldsymbol{\eta}_i \boldsymbol{\eta}_i') \mathbf{C} = \sigma_{\eta}^2 \mathbf{C}' \mathbf{C},$$

the last step by sphericity. The trace is continuous, so

$$\text{plim} \frac{\text{SSR}}{n} = \text{tr}[(\mathbf{C}'\mathbf{C})^{-1} \sigma_{\eta}^2 \mathbf{C}' \mathbf{C}] = \sigma_{\eta}^2 \text{tr}(\mathbf{I}_{M-1}) = (M-1) \sigma_{\eta}^2. \quad \square$$

Consequently both $\text{SSR}/(Mn - n - \#\boldsymbol{\beta})$ of (5.2.17) and $\text{SSR}/(Mn - n)$ are consistent for σ_{η}^2 : each M -observation group contributes only $M-1$ degrees of freedom's worth of variation about its mean, so SSR/n estimates $(M-1)\sigma_{\eta}^2$, not $M\sigma_{\eta}^2$. The naive formula $\text{SSR}/(Mn - \#\boldsymbol{\beta})$ of (5.2.16) converges to $\frac{M-1}{M}\sigma_{\eta}^2 < \sigma_{\eta}^2$: it understates the error variance, and with it the standard errors, by the factor $(M-1)/M$ for large n — exactly the bias behind the text's warning that the t -values are overestimated.

Analytical Exercise 5.4

(a) Set $\mathbf{y}_i = (y_{i1}, \dots, y_{iM})'$, $\boldsymbol{\eta}_i = (\eta_{i1}, \dots, \eta_{iM})'$,

$$\mathbf{F}_i = \begin{bmatrix} 1 & 0 & \cdots & 0 & y_{i0} \\ 0 & 1 & \cdots & 0 & y_{i1} \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & y_{i,M-1} \end{bmatrix} \quad (M \times M), \quad \boldsymbol{\beta} = (\xi_1 - \xi_M, \dots, \xi_{M-1} - \xi_M, \rho)',$$

with the reparameterization of Review Question 5.1.2 absorbing one intercept. There is no $\mathbf{b}_i$ (the book's answer sheet states exactly this): the system has no common regressor once ξ_M is folded

in, so $\mathbf{b}_i' \boldsymbol{\gamma}$ is subsumed in the fixed effect. The essential point is the last column of $\mathbf{F}_i$: row m contains the *lagged dependent variable* $y_{i,m-1}$, which is the dependent variable of equation $m-1$ — the system is dynamic, and $\mathbf{F}_i$ is not orthogonal to the errors of the other equations.

(b) Solve the difference equation backwards from m to 0:

$$y_{im} = \eta_{im} + \rho \eta_{i,m-1} + \cdots + \rho^{m-1} \eta_{i1} + \frac{1 - \rho^m}{1 - \rho} \alpha_i + \rho^m y_{i0} + (\text{terms in the } \xi\text{'s}),$$

which is the hint's display. For $h > m$, η_{ih} is orthogonal to every term on the right-hand side: to η_{ik} , $k \leq m < h$, by the no-cross-correlation assumption; to α_i and to y_{i0} by assumption; and the ξ 's are constants. Hence $E(y_{im} \eta_{ih}) = 0$ for $m < h$: current and past y 's are orthogonal to *future* errors (y_{im} is predetermined).

(c) For $h \leq m$, the same expansion shows that the only term on the right-hand side not orthogonal to η_{ih} is $\rho^{m-h} \eta_{ih}$: the terms $\rho^{m-k} \eta_{ik}$ with $k \neq h$ are uncorrelated with η_{ih} by the no-cross-correlation assumption, and α_i , y_{i0} are orthogonal by assumption. Therefore

$$E(y_{im} \eta_{ih}) = \rho^{m-h} E(\eta_{ih}^2) = \rho^{m-h} \sigma_\eta^2, \quad m \geq h.$$

In particular $E(y_{i,m-1} \eta_{im}) = \sigma_\eta^2 \neq 0$: the regressor of equation m is correlated with the *current* error — the defining feature of a dynamic panel.

(optional) The last column of $\mathbf{F}_i$ is $(y_{i0}, \dots, y_{i,M-1})'$, so the ρ -element of $E(\mathbf{F}_i' \mathbf{Q} \boldsymbol{\eta}_i)$ is

$$E\left[\sum_{m=1}^M y_{i,m-1} \tilde{\eta}_{im}\right] = E\left[\sum_{m=1}^M y_{i,m-1} \left(\eta_{im} - \frac{1}{M} \sum_{h=1}^M \eta_{ih}\right)\right].$$

By (b) and (c), $E(y_{i,m-1} \eta_{im}) = \sigma_\eta^2$, $E(y_{i,m-1} \eta_{ih}) = 0$ for $h > m-1$, and $E(y_{i,m-1} \eta_{ih}) = \rho^{m-1-h} \sigma_\eta^2$ for $h \leq m-1$. Hence

$$E\left[\sum_{m=1}^M y_{i,m-1} \tilde{\eta}_{im}\right] = \sigma_\eta^2 \sum_{m=1}^M \left(1 - \frac{1}{M} \sum_{h=1}^{m-1} \rho^{m-1-h}\right) = -\frac{\sigma_\eta^2}{M} \cdot \frac{(M-1) - M\rho + \rho^M}{1 - \rho},$$

after summing the geometric series $\sum_{m=1}^M \sum_{h=1}^{m-1} \rho^{m-1-h} = \sum_{k=0}^{M-2} (M-1-k) \rho^k$, which equals $[(M-1) - M\rho + \rho^M]/(1-\rho)^2$ times $(1-\rho)$, and collecting the diagonal terms. This is the celebrated Nickell (1981) bias formula: the sign is negative for $0 < \rho < 1$ (since then $(M-1) - M\rho + \rho^M > 0$), so the within estimator of ρ is biased *downward*.

(d) The probability limit of the fixed-effects estimator is

$$\text{plim } \hat{\boldsymbol{\beta}}_{\text{FE}} - \boldsymbol{\beta} = [E(\mathbf{F}_i' \mathbf{Q} \mathbf{F}_i)]^{-1} E(\mathbf{F}_i' \mathbf{Q} \boldsymbol{\eta}_i),$$

and the second factor was just shown to be nonzero whenever $(M-1) - M\rho + \rho^M \neq 0$ (which holds for every $\rho \neq 1$). The violated assumption is the orthogonality condition (5.2.6) of Proposition 5.1 — specifically the *cross* orthogonalities: as the book's answer sheet puts it, $E(\eta_{ih} y_{im}) \neq 0$ for $m \geq h$, i.e. $E(\mathbf{f}_{im} \eta_{ih}) = \mathbf{0}$ fails for pairs (m, h) with $h \leq m-1$, because the regressor of equation m (which is $y_{i,m-1}$) is correlated with the errors of equations $h \leq m$. This is precisely the failure diagnosed in Review Question 5.2.3: without the full set of cross orthogonalities, the deviation-from-means transformation — which mixes all M equations — contaminates every transformed regressor with errors it is correlated with. The bias does not vanish as $n \rightarrow \infty$ for fixed M ; it vanishes only as

$M \rightarrow \infty$, because the offending term is $O(1/M)$ relative to the signal. With the small M of typical panels the within estimator of ρ is badly biased downward, which motivates the first-difference GMM approach of Section 5.4 (and the Arellano–Bond literature it spawned).

Analytical Exercise 5.5

(a) *Step 1: reduction to an a.s.*

linear relation. For any $\# \beta$ -vector $\mathbf{c}$,

$$\mathbf{c}' \mathbf{E}(\mathbf{F}_i' \mathbf{Q} \mathbf{F}_i) \mathbf{c} = \mathbf{E}[(\mathbf{Q} \mathbf{F}_i \mathbf{c})' (\mathbf{Q} \mathbf{F}_i \mathbf{c})] \geq 0,$$

so the matrix is positive semidefinite, and it fails to be invertible iff there is $\mathbf{c} \neq \mathbf{0}$ with $\mathbf{Q} \mathbf{F}_i \mathbf{c} = \mathbf{0}$ almost surely. Suppose such a $\mathbf{c}$ exists.

Step 2: the relation contradicts (5.1.15). The null space of $\mathbf{Q}$ is spanned by $\mathbf{1}_M$, so $\mathbf{Q} \mathbf{F}_i \mathbf{c} = \mathbf{0}$ a.s. says $\mathbf{F}_i \mathbf{c} = \mathbf{1}_M u_i$ a.s. for some scalar u_i : the linear combination $\mathbf{f}_{im}' \mathbf{c}$ takes the *same* value in all M equations. Now consider the matrix of (5.1.15) and contract it with $\mathbf{c}$. Since $\mathbf{Q}$ is constant, the hint's expansion gives

$$\mathbf{E}(\mathbf{Q} \mathbf{F}_i \otimes \mathbf{x}_i) = (\mathbf{Q} \otimes \mathbf{I}_K) \mathbf{E}(\mathbf{F}_i \otimes \mathbf{x}_i),$$

and the columns of $\mathbf{E}(\mathbf{Q} \mathbf{F}_i \otimes \mathbf{x}_i)$ include, for each regressor k , the block $\mathbf{E}(\tilde{\mathbf{f}}_{i \cdot k} \otimes \mathbf{x}_i)$; contracting with $\mathbf{c}$ therefore yields

$$\mathbf{E}[(\mathbf{Q} \mathbf{F}_i \mathbf{c}) \otimes \mathbf{x}_i] = \mathbf{0},$$

because $\mathbf{Q} \mathbf{F}_i \mathbf{c} = \mathbf{0}$ a.s. That contraction is a nontrivial linear combination of the columns of $\mathbf{E}(\mathbf{Q} \mathbf{F}_i \otimes \mathbf{x}_i)$ (nontrivial because $\mathbf{c} \neq \mathbf{0}$ picks out a nonzero set of column blocks), so the matrix cannot have full column rank — contradicting (5.1.15). Hence no such $\mathbf{c}$ exists and $\mathbf{E}(\mathbf{F}_i' \mathbf{Q} \mathbf{F}_i)$ is positive definite, hence invertible. (The hint's bookkeeping displays — $\mathbf{x}_i^* \mathbf{x}_i^{*'} = \mathbf{I}_M \otimes \mathbf{x}_i \mathbf{x}_i'$ and the block form of $(\mathbf{F}_i \otimes \mathbf{x}_i)'$ — are exactly what justifies moving $\mathbf{c}$ through the Kronecker products in the contraction.)

(b) Let $\mathbf{W} \equiv \mathbf{E}(\tilde{\boldsymbol{\eta}}_i \tilde{\boldsymbol{\eta}}_i') = \mathbf{Q} \boldsymbol{\Sigma} \mathbf{Q}$, constant by (5.1.5). The middle matrix of the sandwich in (5.2.7) has quadratic form

$$\mathbf{c}' \mathbf{E}[\mathbf{F}_i' \mathbf{Q} \mathbf{W} \mathbf{Q} \mathbf{F}_i] \mathbf{c} = \mathbf{E}[\tilde{\mathbf{v}}_i' \mathbf{W} \tilde{\mathbf{v}}_i], \quad \tilde{\mathbf{v}}_i \equiv \mathbf{Q} \mathbf{F}_i \mathbf{c}.$$

Now $\mathbf{W} = \mathbf{Q} \boldsymbol{\Sigma} \mathbf{Q}$ is positive definite on the subspace $\{\mathbf{1}_M\}^\perp$: if $\mathbf{z} = \mathbf{Q} \mathbf{z}$ and $\mathbf{z}' \mathbf{W} \mathbf{z} = 0$, then $\mathbf{z}' \boldsymbol{\Sigma} \mathbf{z} = 0$, forcing $\mathbf{z} = \mathbf{0}$ because $\boldsymbol{\Sigma}$ is positive definite by (5.1.6'). Since $\tilde{\mathbf{v}}_i \in \{\mathbf{1}_M\}^\perp$ by construction, the expectation vanishes only if $\tilde{\mathbf{v}}_i = \mathbf{0}$ a.s., i.e. $\mathbf{Q} \mathbf{F}_i \mathbf{c} = \mathbf{0}$ a.s., which part (a) just ruled out for $\mathbf{c} \neq \mathbf{0}$ under (5.1.15). Hence the middle matrix is positive definite, and with it the whole sandwich $\text{Avar}(\hat{\boldsymbol{\beta}}_{\text{FE}})$ of Proposition 5.1: the singularity of $\mathbf{E}(\tilde{\boldsymbol{\eta}}_i \tilde{\boldsymbol{\eta}}_i')$ noted in Review Question 5.2.4 does not propagate, because the singular direction $\mathbf{1}_M$ is exactly the direction that $\mathbf{Q}$ removes from the regressors as well.

(c) By Analytical Exercise 5.2(c), $\boldsymbol{\Psi} = \mathbf{C}' \boldsymbol{\Sigma} \mathbf{C}$ is nonsingular, and the rank condition holds for the transformed system: $\mathbf{E}(\mathbf{C}' \mathbf{F}_i \otimes \mathbf{x}_i)$ has full column rank iff $\mathbf{E}(\mathbf{Q} \mathbf{F}_i \otimes \mathbf{x}_i)$ does (proved there). Applying the argument of part (a) with $\tilde{\mathbf{F}}_i = \mathbf{C}' \mathbf{F}_i$ in place of $\mathbf{Q} \mathbf{F}_i$ — the only property used was the rank condition — gives $\mathbf{E}(\tilde{\mathbf{F}}_i' \tilde{\mathbf{F}}_i)$ positive definite. Then

$$\mathbf{E}(\tilde{\mathbf{F}}_i' \boldsymbol{\Psi}^{-1} \tilde{\mathbf{F}}_i) = \mathbf{E}[(\boldsymbol{\Psi}^{-1/2} \tilde{\mathbf{F}}_i)' (\boldsymbol{\Psi}^{-1/2} \tilde{\mathbf{F}}_i)]$$

is positive definite: its quadratic form in $\mathbf{c}$ is $E\|\Psi^{-1/2}\tilde{\mathbf{F}}_i\mathbf{c}\|^2$, which vanishes only if $\tilde{\mathbf{F}}_i\mathbf{c} = \mathbf{0}$ a.s. ($\Psi^{-1/2}$ is nonsingular), impossible for $\mathbf{c} \neq \mathbf{0}$. So the efficient GMM estimator's asymptotic variance (13) is finite and positive definite — as it must be for the efficiency comparison with fixed effects to be meaningful.

Analytical Exercise 5.6

(a) Since $\mathbf{f}_{im}$ is a subvector of $\mathbf{x}_i$, there is a $p \times K$ selection matrix $\mathbf{A}_m$ (zeros and ones, a single 1 in each row) with $\mathbf{f}_{im} = \mathbf{A}_m\mathbf{x}_i$. Row m of $\mathbf{F}_i$ is $\mathbf{f}'_{im}$, so

$$\mathbf{F}_i = \begin{bmatrix} \mathbf{f}'_{i1} \\ \vdots \\ \mathbf{f}'_{iM} \end{bmatrix} = \begin{bmatrix} \mathbf{x}'_i\mathbf{A}'_1 \\ \vdots \\ \mathbf{x}'_i\mathbf{A}'_M \end{bmatrix} = (\mathbf{I}_M \otimes \mathbf{x}'_i)\mathbf{A}, \quad \mathbf{A} \equiv \begin{bmatrix} \mathbf{A}'_1 \\ \vdots \\ \mathbf{A}'_M \end{bmatrix} \quad (KM \times p),$$

because the m -th block row of $\mathbf{I}_M \otimes \mathbf{x}'_i$ is $(\mathbf{0}', \dots, \mathbf{x}'_i, \dots, \mathbf{0}')$ with $\mathbf{x}'_i$ in position m , and multiplying into the stacked $\mathbf{A}$ picks out $\mathbf{x}'_i\mathbf{A}'_m$. The matrix $\mathbf{A}$ is constant (it encodes only *where* the regressors sit inside $\mathbf{x}_i$) and has full column rank p : each row of each $\mathbf{A}'_m$ is a distinct unit vector, so for every $k \leq p$ some row of $\mathbf{A}$ isolates c_k , and $\mathbf{A}\mathbf{c} = \mathbf{0}$ forces $\mathbf{c} = \mathbf{0}$.

(b) Compute, using $\mathbf{QF}_i = \mathbf{Q}(\mathbf{I}_M \otimes \mathbf{x}'_i)\mathbf{A}$ from (a) and the mixed-product property of the Kronecker product (following the hint):

$$E(\mathbf{QF}_i \otimes \mathbf{x}_i) = [\mathbf{I}_M \otimes E(\mathbf{x}_i\mathbf{x}'_i)] (\mathbf{Q} \otimes \mathbf{I}_K) \mathbf{A}.$$

(The left side stacks, for each m , the blocks $\sum_h q_{mh}E(\mathbf{f}_{ih} \otimes \mathbf{x}_i)$; the right side gives the same blocks because $E[(\mathbf{I}_M \otimes \mathbf{x}'_i)\mathbf{A} \otimes \mathbf{x}_i] = [\mathbf{I}_M \otimes E(\mathbf{x}_i\mathbf{x}'_i)]\mathbf{A}$ by the identity $\mathbf{v} \otimes \mathbf{x} = (\mathbf{I} \otimes \mathbf{x})\mathbf{v}$ applied to $\mathbf{v} = \mathbf{A}_m\mathbf{x}_i$, and premultiplication by $\mathbf{Q} \otimes \mathbf{I}_K$ forms the $\mathbf{Q}$ -combinations.) Now (5.1.15) says the $MK \times p$ matrix on the left has rank p . The first factor on the right, $\mathbf{I}_M \otimes E(\mathbf{x}_i\mathbf{x}'_i)$, is nonsingular iff $E(\mathbf{x}_i\mathbf{x}'_i)$ is nonsingular (its eigenvalues are those of $E(\mathbf{x}_i\mathbf{x}'_i)$, M -fold), and premultiplication by a nonsingular matrix does not change rank. Therefore, given nonsingularity of $E(\mathbf{x}_i\mathbf{x}'_i)$ — which the error-components model supplies through (5.1.6'), implied by (5.1.5) and (5.1.6) — the rank condition (5.1.15) is equivalent to

$$\text{rank}[(\mathbf{Q} \otimes \mathbf{I}_K)\mathbf{A}] = p.$$

This is a checkable, purely design-based criterion: it involves only the constants $\mathbf{Q}$ and the selection matrix $\mathbf{A}$, no population moments. Intuitively it says that the p columns of $\mathbf{A}$ — each replicating one regressor across the M equations at its positions in $\mathbf{x}_i$ — must remain linearly independent after the annihilator strips out the common (group-mean) component. A regressor entering every equation identically (a common regressor) corresponds to a column of $\mathbf{A}$ lying in the span of $\{\mathbf{1}_M \otimes \mathbf{v}\}$, which $\mathbf{Q} \otimes \mathbf{I}_K$ annihilates; such columns destroy the rank condition. This is the algebraic restatement of “common regressors are not identified by fixed effects.”

Analytical Exercise 5.7

(a) Follow the hint's chain. By Analytical Exercise 5.6(a), $\mathbf{F}_i = (\mathbf{I}_M \otimes \mathbf{x}_i')\mathbf{A}$, so $\mathbf{F}_i' = \mathbf{A}'(\mathbf{I}_M \otimes \mathbf{x}_i)$ and

$$\begin{aligned}
 \mathbf{F}_i' \tilde{\boldsymbol{\eta}}_i &= \mathbf{A}'(\mathbf{I}_M \otimes \mathbf{x}_i) \mathbf{Q} \boldsymbol{\eta}_i & (\tilde{\boldsymbol{\eta}}_i &= \mathbf{Q} \boldsymbol{\eta}_i) \\
 &= \mathbf{A}'(\mathbf{I}_M \otimes \mathbf{x}_i) \mathbf{Q} \boldsymbol{\varepsilon}_i & (\boldsymbol{\varepsilon}_i &= \mathbf{1}_M \alpha_i + \boldsymbol{\eta}_i, \mathbf{Q} \mathbf{1}_M = \mathbf{0}) \\
 &= \mathbf{A}'(\mathbf{I}_M \otimes \mathbf{x}_i) (\mathbf{Q} \boldsymbol{\varepsilon}_i \otimes \mathbf{1}) & (\mathbf{z} &= \mathbf{z} \otimes \mathbf{1}) \\
 &= \mathbf{A}'(\mathbf{Q} \boldsymbol{\varepsilon}_i \otimes \mathbf{x}_i) & (\text{mixed-product property}) \\
 &= \mathbf{A}'(\mathbf{Q} \otimes \mathbf{I}_K) (\boldsymbol{\varepsilon}_i \otimes \mathbf{x}_i) \\
 &= \mathbf{A}'(\mathbf{Q} \otimes \mathbf{I}_K) \mathbf{g}_i.
 \end{aligned}$$

So the $p \times 1$ vector $\mathbf{F}_i' \tilde{\boldsymbol{\eta}}_i$ is a *fixed* linear transformation of the $MK \times 1$ moment vector $\mathbf{g}_i$.

(b) The matrix in question (the asymptotic variance of $\frac{1}{\sqrt{n}} \sum_i \mathbf{F}_i' \mathbf{Q} \boldsymbol{\eta}_i$) is

$$\mathbf{V} \equiv \mathbb{E}[\mathbf{F}_i' \tilde{\boldsymbol{\eta}}_i \tilde{\boldsymbol{\eta}}_i' \mathbf{F}_i] = \mathbb{E}[(\mathbf{F}_i' \tilde{\boldsymbol{\eta}}_i)(\mathbf{F}_i' \tilde{\boldsymbol{\eta}}_i)'] = \mathbf{A}'(\mathbf{Q} \otimes \mathbf{I}_K) \mathbb{E}(\mathbf{g}_i \mathbf{g}_i') (\mathbf{Q} \otimes \mathbf{I}_K)' \mathbf{A},$$

by part (a). By (5.1.6), $\mathbb{E}(\mathbf{g}_i \mathbf{g}_i')$ is positive definite. A congruence $\mathbf{B}' \boldsymbol{\Omega} \mathbf{B}$ with $\boldsymbol{\Omega}$ positive definite is positive definite if (and only if) $\mathbf{B}$ has full column rank; here $\mathbf{B} = (\mathbf{Q} \otimes \mathbf{I}_K)' \mathbf{A} = (\mathbf{Q} \otimes \mathbf{I}_K) \mathbf{A}$ (since $\mathbf{Q}$ is symmetric), and Analytical Exercise 5.6(b) — whose hypothesis $\mathbb{E}(\mathbf{x}_i \mathbf{x}_i')$ nonsingular is implied by (5.1.5) and (5.1.6), and whose rank condition is (5.1.15) — gives precisely $\text{rank}(\mathbf{B}) = p$. Therefore $\mathbf{V}$ is positive definite, hence nonsingular. This closes the last gap in Proposition 5.3: the sandwich asymptotic variance of the fixed-effects estimator is the product of three nonsingular matrices — $\mathbb{E}(\mathbf{F}_i' \mathbf{Q} \mathbf{F}_i)$ (Exercise 5.5(a)), $\mathbf{V}$ (here), and $\mathbb{E}(\mathbf{F}_i' \mathbf{Q} \mathbf{F}_i)$ again — so it is itself positive definite, and the robust standard errors of (5.2.24) are well defined.

Problem Set for Chapter 5: Empirical Exercise

Empirical Exercise 5.1

All numbers below are produced by `code/ch5.py`, which parses `SUM_HES.ASC` directly (125×27 records: 125 countries, 1960–1985).

(a) *The plot* (Figure S5.1, left panel): there is essentially no relation between initial income and subsequent growth. The correlation between $y_{i,1960}$ and $y_{i,1985} - y_{i,1960}$ is 0.17, and the regression of growth on a constant and y_{i0} gives a positive but fragile slope, 0.0877 with a robust standard error of 0.0381 ($R^2 = 0.028$), implying $\lambda = -0.34\%$ per year with a standard error of 0.14%. Unconditionally, poor countries did *not* grow faster over 1960–1985; if anything the point estimate has the wrong sign.

The conditional regression (2'), with robust standard errors in parentheses and conventional ones in brackets:

	estimate	robust SE	conventional SE
constant	−1.3115	(0.5797)	[0.6469]
$y_{i,1960}$	−0.1652	(0.0448)	[0.0496]
$\log(s_i)$	0.4651	(0.0590)	[0.0580]
$\log(n_i + g + \delta)$	−0.6817	(0.2667)	[0.2938]
COM_i	0.0570	(0.0994)	[0.1741]
OPEC_i	0.3237	(0.1757)	[0.1889]
$R^2 = 0.444, \quad SER = 0.3503, \quad n = 125$			

This is essentially Table IV of Mankiw, Romer and Weil (1992): the coefficient on initial income is now significantly negative ($t = -3.7$), the saving rate enters positively and $\log(n_i + g + \delta)$ negatively, both strongly significant and with the signs the Solow model predicts; the estimates 0.465 and -0.682 even satisfy $\gamma_1 \approx -\gamma_2$, the restriction the growth model imposes and the exercise declines to impose. The implied speed of convergence: $\hat{\rho} = 1 + (-0.1652) = 0.8348$, so

$$\hat{\lambda} = -\frac{\log(0.8348)}{25} = 0.00722 \text{ per year}, \quad \text{SE}(\hat{\lambda}) = \frac{\text{SE}(\hat{b})}{25\hat{\rho}} = \frac{0.0448}{25 \times 0.8348} = 0.00215,$$

i.e. about $0.72\% \pm 0.22\%$ per year, with a 95% confidence interval of $[0.30\%, 1.14\%]$ and an implied half-life of $\log 2 / \hat{\lambda} \approx 96$ years. So the answer to the question is *yes*: holding investment and population growth (and the COM/OPEC dummies) constant, initial income strongly and negatively predicts subsequent growth — the quoted Mankiw–Romer–Weil finding is confirmed — but the speed is glacial, under one percent per year. The right panel of Figure S5.1 shows the conditional relation, plotting both variables net of the four controls: the downward slope is unmistakable, in stark contrast to the left panel.

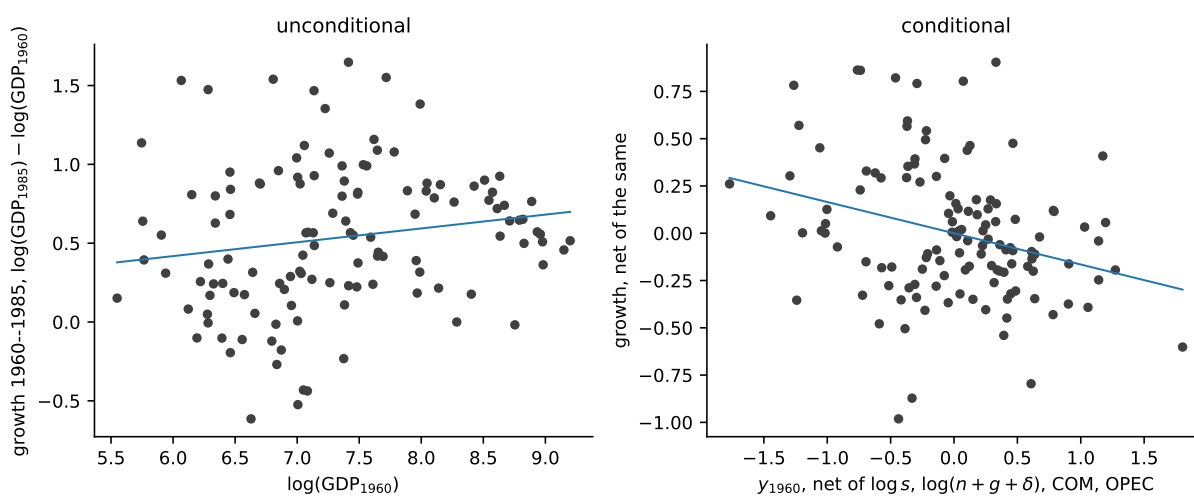


Figure S5.1: cumulative growth 1960–1985 against log initial GDP per capita, 125 countries. Left: raw (unconditional). Right: both variables net of $\log s_i$, $\log(n_i + g + \delta)$, COM and OPEC (the conditional relation estimated in equation (2')).

(b) *Fixed-effects estimation.* With $M = 25$ equations ($m = 1, \dots, 25$; $t_m = 1960 + m$), the within transformation sweeps out α_i ; the regressors are 24 year dummies (1962–1985) and $y_{i,m-1}$, giving

$\#\beta = 25$ coefficients estimated on a pooled sample of $125 \times 25 = 3125$ transformed observations. The estimate is

$$\hat{\rho}_{\text{FE}} = 0.9380 \quad (\text{SE } 0.0068), \quad \hat{\lambda} = -\log(0.9380) = 0.0640 \text{ per year} \quad (\text{SE } 0.0073),$$

about 6.4% per year, as the exercise anticipates — nearly nine times the cross-section estimate of part (a). The standard errors use the *correct* degrees of freedom (5.2.17), $s^2 = \text{SSR} / (Mn - n - \#\beta) = 11.8439 / (3125 - 125 - 25) = 0.003981$; the naive formula (5.2.16) gives 0.003821 and standard errors smaller by the factor 0.980 — the text’s “very common mistake,” small here only because $M = 25$ is large. The LSDV route (pooled OLS with 125 country dummies and 24 year dummies) reproduces $\hat{\rho}$ and SSR to machine precision (2.9×10^{-12}), confirming Analytical Exercise 5.1. Without the spherical assumption, the robust standard error of Proposition 5.3 is 0.0089 for $\hat{\rho}$ (0.0095 for $\hat{\lambda}$), about 30% larger than the spherical one — mild evidence of conditional heteroskedasticity or serial correlation in η_{im} .

As the exercise itself warns, consistency fails here: the system is dynamic, and by Analytical Exercise 5.4 the fixed-effects estimator is inconsistent for ρ (the cross orthogonalities (5.2.6) fail, $E(y_{i,m-1}\eta_{ih}) \neq 0$ for $h \leq m-1$; the Nickell bias is $O(1/M)$ and downward, so the true ρ is presumably even higher and λ even larger). The estimate is reported purely “to gain programming experience”; the contrast between 0.7% in (a), 6.4% here and the next part is a vivid illustration of how much the answer depends on the treatment of α_i and of dynamics.

(c) *GMM on the first differences.* Differencing the 25 equations gives $M - 1 = 24$ equations,

$$y_{i,m+1} - y_{im} = \xi'_m + (y_{im} - y_{i,m-1})\rho + (\eta_{i,m+1} - \eta_{im}), \quad m = 1, \dots, 24,$$

with common ρ and 24 free intercepts ($L = 25$ coefficients), and instruments $\mathbf{x}_{im} = (1, s_{i,m-1})'$ in equation m — for example $(1, s_{i,1960})$ in the first equation. This is the multiple-equation GMM of Section 4.6 with 48 moment conditions; the first step uses the 2SLS weighting (block-diagonal $\mathbf{S}_{\mathbf{xx}}^{-1}$), the second step the efficient weight $\hat{\mathbf{S}}^{-1}$ estimated from the first-step residuals. The estimates are

$$\hat{\rho}_{\text{2SLS}} = 0.5859, \quad \hat{\rho}_{\text{GMM}} = 0.6837 \quad (\text{SE } 0.0451),$$

$$\hat{\lambda} = -\log(0.6837) = 0.380 \text{ per year} \quad (\text{SE } 0.066),$$

i.e. about 38% per year — the exercise says “about 36 percent per year!”, and the small gap is accounted for by the efficient second-step weighting (the book’s number comes from the one-step estimator; our 2SLS first step gives $\lambda = 53\%$, and the efficient GMM 38%, both in the advertised ballpark of an order of magnitude beyond the cross-section estimate). Sargan’s statistic is $J = 43.95$ on $\chi^2(23)$ ($p = 0.005$), so the overidentifying restrictions — chiefly the assumption that the saving rate is orthogonal to the differenced error in every equation — are rejected at the 1% level: either the saving rate is not exogenous for the growth residual, or the common- ρ restriction fails across periods.

Taken at face value, the three estimates of the speed of convergence tell one story: the cross-section regression attributes to $\log s$ and $\log(n+g+\delta)$ much of what the panel attributes to α_i ; removing the country effect raises λ from under 1% to 6.4%; and treating the lagged growth term as endogenous raises it further to about 36–38%, a half-life of two years. Since the error in the growth regression

is most plausibly a business cycle, the last estimate — which purges both the fixed effect and the simultaneity between Δy_{im} and $\Delta \eta_{im}$ — is the one the model takes seriously, and it says economies converge to their steady states very fast indeed. The price, as the Sargan test shows, is a set of instrument-validity assumptions the data are not entirely happy with.

Chapter 6

Serial Correlation

Solutions to all the exercises of Chapter 6 of Fumio Hayashi, *Econometrics* (Princeton University Press, 2000): the 35 review questions of Sections 6.1–6.8, the ten analytical exercises, and the two empirical exercises (the Bekaert–Hodrick forward-exchange data and the Fama three-month T-bill regression). Propositions and equations cited as PROPOSITION 6.8(B) or (6.5.2) refer to the book. The numerical parts are computed by `code/ch6.py`; the figures are in `figures/`.

Section 6.1 Modeling Serial Correlation: Linear Processes

Review Question 6.1.1

Let $S_n \equiv \sum_{j=0}^n |\gamma_j|$, a monotone nondecreasing sequence of real numbers. By the Cauchy criterion quoted in the question, S_n converges to a (finite) real number if and only if it is Cauchy, that is, $|S_m - S_n| \rightarrow 0$ as $m, n \rightarrow \infty$. For $m > n$,

$$|S_m - S_n| = \sum_{j=n+1}^m |\gamma_j|,$$

so the two statements are the same statement: $\{\gamma_j\}$ is absolutely summable ($\sum_{j=0}^{\infty} |\gamma_j| < \infty$, i.e. S_n converges) if and only if the tail sums $\sum_{j=n+1}^m |\gamma_j|$ vanish as $m, n \rightarrow \infty$. The Cauchy form is often the usable one in proofs because it does not mention the value of the limit: Analytical Exercise 6.2(a) uses exactly this, with $a_n = \sum_{j=0}^n \psi_j^2$, to show that the MA(∞) partial sums form a mean-square Cauchy sequence.

Review Question 6.1.2

Write $y_t = h_0 x_t + h_1 x_{t-1}$ and $y_{t-j} = h_0 x_{t-j} + h_1 x_{t-j-1}$ (means subtracted where necessary; the calculation is unchanged). Multiplying out and using covariance stationarity of $\{x_t\}$,

$$\gamma_j^y = h_0^2 \gamma_j^x + h_0 h_1 \gamma_{j+1}^x + h_1 h_0 \gamma_{j-1}^x + h_1^2 \gamma_j^x$$

because, for instance, $\text{Cov}(x_t, x_{t-j-1}) = \gamma_{j+1}^x$ (the two indices are $j+1$ apart) and $\text{Cov}(x_{t-1}, x_{t-j}) = \gamma_{j-1}^x$. Collecting terms,

$$\gamma_j^y = (h_0^2 + h_1^2) \gamma_j^x + h_0 h_1 (\gamma_{j-1}^x + \gamma_{j+1}^x),$$

which is the stated formula for all $j \geq 1$; for $j = 0$ it reads $\gamma_0^y = (h_0^2 + h_1^2) \gamma_0^x + h_0 h_1 (\gamma_{-1}^x + \gamma_1^x) = (h_0^2 + h_1^2) \gamma_0^x + 2h_0 h_1 \gamma_1^x$ by the symmetry of scalar autocovariances, as claimed. This is the special case $n = 1$ of Analytical Exercise 6.3(a); the pattern — a quadratic form in the filter coefficients applied to a moving window of the input autocovariances — is the general one.

Review Question 6.1.3

All inverses exist because the leading coefficients are nonzero ((6.1.14) and the sentences following it).

($\Rightarrow$) Suppose $\alpha(L)\beta(L) = \delta(L)$. Multiply both sides from the left by $\alpha(L)^{-1}$ and use associativity of filter multiplication (a direct consequence of the convolution definition (6.1.12)) together with the definition of the inverse, $\alpha(L)^{-1}\alpha(L) = 1$ ((6.1.15A)):

$$\alpha(L)^{-1}[\alpha(L)\beta(L)] = [\alpha(L)^{-1}\alpha(L)]\beta(L) = \beta(L) = \alpha(L)^{-1}\delta(L).$$

($\Leftarrow$) Suppose $\beta(L) = \alpha(L)^{-1}\delta(L)$. Multiply from the left by $\alpha(L)$: $\alpha(L)\beta(L) = \alpha(L)\alpha(L)^{-1}\delta(L) = \delta(L)$.

The second equivalence is identical with the roles reversed (multiply on the right by $\beta(L)^{-1}$ and back). The content is that the algebra of filters with nonzero leading coefficients is the algebra of formal power series: one may “multiply through” by inverses exactly as with polynomials, and in the *scalar* case the order of the factors is irrelevant.

Remark. Only the inverse operation is commutative in general; for matrix filters even that has to be proved (Review Question 6.3.2), and ordinary multiplication is *not* commutative (Review Question 6.3.3).

Review Question 6.1.4

The stability condition requires the roots of $\phi(z) = 0$ to lie outside the unit circle, $|z| > 1$. By the hint the two roots are $z = 2(1 \pm \sqrt{3}i)/3$; each has absolute value

$$|z| = \frac{2}{3} |1 \pm \sqrt{3}i| = \frac{2}{3} \sqrt{1+3} = \frac{4}{3} > 1.$$

Both roots lie outside the unit circle, so the stability condition *is* satisfied, and by PROPOSITION 6.3 the inverse filter $[1 - \phi L - \phi^2 L^2]^{-1}$ has coefficients bounded by a geometrically declining sequence, hence absolutely summable. The roots being complex means the coefficients of the inverse oscillate (damped sine waves) as they decay: $|\psi_j| \leq A(3/4)^j$ for some A , with $3/4$ the reciprocal of the roots' modulus.

Review Question 6.1.5

If λ_1 and λ_2 are the roots, $\phi(z)$ — a quadratic with constant term 1 — factors as

$$\phi(z) = \left(1 - \frac{z}{\lambda_1}\right) \left(1 - \frac{z}{\lambda_2}\right) = 1 - \left(\frac{1}{\lambda_1} + \frac{1}{\lambda_2}\right)z + \frac{1}{\lambda_1\lambda_2}z^2.$$

Equating coefficients with $1 - \phi_1 z - \phi_2 z^2$ gives $\phi_1 = \lambda_1^{-1} + \lambda_2^{-1}$ and $\phi_2 = -(\lambda_1\lambda_2)^{-1}$. These are Vieta's formulas for the *reciprocal* roots; the same factorization is what makes the stability condition “roots outside the unit circle” equivalent to “ $|\lambda_k^{-1}| < 1$ ” in the companion-form analysis of Analytical Exercise 6.7, where the λ^{-1} 's reappear as the eigenvalues of the matrix $\mathbf{F}$.

Review Question 6.1.6

Set $\alpha(L) = 1 - \phi L$ and $\beta(L) = \sum_{j \geq 0} \phi^j L^j$, so $\beta_j = \phi^j$. The convolution $\delta_j = \sum_{k=0}^j \alpha_k \beta_{j-k}$ of (6.1.12) gives

$$\delta_0 = \alpha_0 \beta_0 = 1, \quad \delta_j = \alpha_0 \beta_j + \alpha_1 \beta_{j-1} = \phi^j - \phi \cdot \phi^{j-1} = 0 \quad (j \geq 1),$$

since only α_0 and α_1 are nonzero. So $\delta(L) = 1$, i.e. $(1 - \phi L) \sum_{j \geq 0} \phi^j L^j = 1$: the geometric-series filter is the inverse of $1 - \phi L$, exactly as claimed in (6.1.20). Note that nothing in this calculation

requires $|\phi| < 1$ — the inverse exists as a formal filter for any ϕ — but $\{\phi^j\}$ is absolutely summable only when $|\phi| < 1$, which is the stationarity condition for AR(1): that is where, and only where, the inverse becomes usable on covariance-stationary input processes (Propositions 6.2 and 6.3).

Section 6.2 ARMA Processes

Review Question 6.2.1

Case $|\phi| < 1$. By the MA(∞) representation (6.2.2), y_{t-1} is a function of $(\varepsilon_{t-1}, \varepsilon_{t-2}, \dots)$ and hence is uncorrelated with ε_t : $E(y_{t-1}\varepsilon_t) = 0$ (the hint's statement). Projecting $y_t = c + \phi y_{t-1} + \varepsilon_t$ on $(1, y_{t-1})$,

$$\hat{E}^*(y_t | 1, y_{t-1}) = c + \phi y_{t-1} + \hat{E}^*(\varepsilon_t | 1, y_{t-1}) = c + \phi y_{t-1},$$

because ε_t is orthogonal to $(1, y_{t-1})$ and so has zero projection. The coefficients (c, ϕ) do not depend on t — covariance stationarity makes the relevant second moments time-invariant.

The projection need *not* equal the conditional expectation: $E(y_{t-1}\varepsilon_t) = 0$ says ε_t is *uncorrelated* with y_{t-1} , not *mean-independent* of it. If $\{\varepsilon_t\}$ is merely white noise (not independent), $E(\varepsilon_t | y_{t-1})$ can be a nonzero nonlinear function of y_{t-1} , and then $E(y_t | y_{t-1}) \neq c + \phi y_{t-1}$. With independent white noise, the two do coincide.

On $(1, y_{t-1}, y_{t-2})$ the projection is still $c + \phi y_{t-1} + 0 \cdot y_{t-2}$: ε_t is orthogonal to y_{t-2} as well, and the coefficient on the extra regressor is zero. The AR(1) structure makes all lags beyond the first redundant for linear prediction.

Case $|\phi| > 1$. Now the covariance-stationary solution (6.2.4) expresses y_t as a moving average of *future* innovations, $y_t - \mu = -\sum_{j=1}^{\infty} \phi^{-j} \varepsilon_{t+j}$. Consequently y_{t-1} is a function of $(\varepsilon_t, \varepsilon_{t+1}, \dots)$ and

$$E[(y_{t-1} - \mu)\varepsilon_t] = -\phi^{-1}\sigma^2 \neq 0,$$

so the hint's premise fails: $E(y_{t-1}\varepsilon_t) \neq 0$, and with it the argument that the projection equals $c + \phi y_{t-1}$, because the projection of ε_t on $(1, y_{t-1})$ is no longer zero. The answer to the last question is *no*: $\hat{E}^*(y_t | 1, y_{t-1}) \neq c + \phi y_{t-1}$ when $|\phi| > 1$. The autoregressive form of the equation is not the projection when the root is explosive — one more reason economic dynamics are written with $|\phi| < 1$.

Review Question 6.2.2

Successive substitution, as in the text's $\phi = 1$ case, gives for any $j \geq 1$

$$y_t = (-1)^j y_{t-j} + \sum_{k=0}^{j-1} (-1)^k \varepsilon_{t-k}, \quad \text{hence} \quad y_t - (-1)^j y_{t-j} = \sum_{k=0}^{j-1} (-1)^k \varepsilon_{t-k}.$$

If $\{y_t\}$ were covariance-stationary, the variance of the left-hand side would be $\text{Var}(y_t) + \text{Var}(y_{t-j}) - 2(-1)^j \text{Cov}(y_t, y_{t-j}) = 2\gamma_0 - 2(-1)^j \gamma_j$, while the right-hand side has variance $\sigma^2 j$ by white noise. So

$$2\gamma_0 - 2(-1)^j \gamma_j = \sigma^2 j.$$

For j even this reads $2(\gamma_0 - \gamma_j) = \sigma^2 j$, i.e. $\rho_j = 1 - \sigma^2 j / (2\gamma_0) < -1$ for $j > 4\gamma_0 / \sigma^2$, contradicting $|\rho_j| \leq 1$. Hence no covariance-stationary solution exists. (Intuitively, $\phi = -1$ makes the process flip

sign deterministically while accumulating variance: $y_t = (-1)^t y_0 + \sum_{k=0}^{t-1} (-1)^k \varepsilon_{t-k}$, whose variance grows linearly in t .)

Review Question 6.2.3

(a) The stationarity condition (6.1.18) requires every root z of $\phi(z) = 0$ to satisfy $|z| > 1$. Since $|1| = 1$, $z = 1$ is not a root, i.e. $\phi(1) = 1 - \phi_1 - \cdots - \phi_p \neq 0$. So the mean $\mu = \phi(1)^{-1}c$ is well defined.

(b) Take expectations on both sides of the AR(p) equation (6.2.6), $y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t$. Covariance stationarity gives $E(y_{t-j}) = E(y_t)$ for every j , and white noise has mean zero, so

$$E(y_t) = c + (\phi_1 + \cdots + \phi_p)E(y_t), \quad \text{i.e.} \quad [1 - \phi_1 - \cdots - \phi_p]E(y_t) = c.$$

The bracket is $\phi(1) \neq 0$ by part (a), so $E(y_t) = c/\phi(1) = \phi(1)^{-1}c$, which is (6.2.8). This confirms that the μ of the deviation-from-the-mean form (6.2.6') is indeed the mean; the same argument applied to (6.2.9) gives the mean of an ARMA(p, q) process, $\mu = \phi(1)^{-1}c$ in (6.2.11).

Review Question 6.2.4

Apply the filter $\phi(L)$ to $y_t - \mu \equiv \phi(L)^{-1}\theta(L)\varepsilon_t$. By the definition of the inverse filter and its commutativity (6.1.15A),

$$\phi(L)(y_t - \mu) = \phi(L)[\phi(L)^{-1}\theta(L)]\varepsilon_t = [\phi(L)\phi(L)^{-1}]\theta(L)\varepsilon_t = \theta(L)\varepsilon_t,$$

which is exactly (6.2.9'). The steps are legitimate because $\phi(L)^{-1}$ is absolutely summable under the stationarity condition (PROPOSITION 6.3), so $\phi(L)^{-1}\theta(L)$ is an absolutely summable filter (a convolution of absolutely summable sequences is absolutely summable) and PROPOSITION 6.2 makes both sides well-defined mean-square limits. Combined with the uniqueness argument used for AR(1) in the text (any covariance-stationary solution, when multiplied by $\phi(L)^{-1}$, must equal this expression), this establishes PROPOSITION 6.5(A).

Review Question 6.2.5

The coefficients are determined by $\phi(L)\psi(L) = \theta(L)$, i.e. by the convolution formula (6.1.12) with $\alpha(L) = \phi(L)$, $\beta(L) = \psi(L)$ and $\delta(L) = \theta(L)$: for each $j \geq 0$, $\sum_{k=0}^j \alpha_k \psi_{j-k} = \theta_j$, where $\theta_j = 0$ for $j > q$.

ARMA(3,1): $\phi(L) = 1 - \phi_1 L - \phi_2 L^2 - \phi_3 L^3$, $\theta(L) = 1 + \theta_1 L$:

$$L^0 : \psi_0 = 1,$$

$$L^1 : \psi_1 - \phi_1 \psi_0 = \theta_1,$$

$$L^2 : \psi_2 - \phi_1 \psi_1 - \phi_2 \psi_0 = 0,$$

$$L^j : \psi_j - \phi_1 \psi_{j-1} - \phi_2 \psi_{j-2} - \phi_3 \psi_{j-3} = 0, \quad j \geq 3,$$

since for $j \geq 3$ all of ϕ 's coefficients appear and the right-hand side is already zero (it vanishes from $j \geq q+1 = 2$, but the L^2 equation still lacks the $\phi_3 \psi_{-1}$ term only vacuously; the full third-order recursion starts once both $j \geq p$ and $j \geq q+1$ hold). Generally the homogeneous equation (6.1.17) governs ψ_j from $j \geq \max(p, q+1)$; here $\max(3, 2) = 3$.

ARMA(1,2): $\phi(L) = 1 - \phi_1 L$, $\theta(L) = 1 + \theta_1 L + \theta_2 L^2$:

$$\psi_0 = 1, \quad \psi_1 - \phi_1 \psi_0 = \theta_1, \quad \psi_2 - \phi_1 \psi_1 = \theta_2, \quad \psi_j - \phi_1 \psi_{j-1} = 0 \quad (j \geq 3).$$

The right-hand side is nonzero through $j = q = 2$, so the homogeneous first-order recursion starts at $j = \max(1, 3) = 3$ — the book's answer, $j = 3$. The rule of thumb: the MA part delays the onset of the pure AR recursion by its own order, because until lag q the θ 's keep entering the right-hand side.

Review Question 6.2.6

By (6.2.20), $g_Y(z) = h(z)g_X(z)h(z^{-1})$. Evaluating at $z = e^{-i\omega}$ (which lies on the unit circle, so $z^{-1} = e^{i\omega}$) and dividing by 2π ,

$$s_Y(\omega) = \frac{1}{2\pi} g_Y(e^{-i\omega}) = \frac{1}{2\pi} h(e^{-i\omega}) g_X(e^{-i\omega}) h(e^{i\omega}) = h(e^{-i\omega}) s_X(\omega) h(e^{i\omega}).$$

Since $h(e^{-i\omega})h(e^{i\omega}) = |h(e^{-i\omega})|^2$ (the coefficients of h are real), this says the spectrum of the output is the spectrum of the input multiplied by the squared modulus of the filter's frequency response: $|h(e^{-i\omega})|^2$ is the *gain*, showing exactly which frequencies the filter amplifies or suppresses. For the first-difference filter $1 - L$, the gain is $2 - 2 \cos \omega$, which kills the zero-frequency component — the spectral way of saying that differencing removes a unit root.

Review Question 6.2.7

Substitute $z = e^{-i\omega} = \cos \omega - i \sin \omega$ into (6.2.14). By the facts cited in the hint, $z^j = [\cos \omega - i \sin \omega]^j = \cos(j\omega) - i \sin(j\omega)$ and $z^{-j} = \cos(j\omega) + i \sin(j\omega)$, so $z^j + z^{-j} = 2 \cos(j\omega)$ and

$$2\pi s_Y(\omega) = g_Y(e^{-i\omega}) = \gamma_0 + \sum_{j=1}^{\infty} \gamma_j (z^j + z^{-j}) = \gamma_0 + 2 \sum_{j=1}^{\infty} \gamma_j \cos(j\omega),$$

a sum of real terms (γ_j is real for a real-valued process), absolutely convergent because $\{\gamma_j\}$ is absolutely summable and $|\cos(j\omega)| \leq 1$. Hence $s_Y(\omega)$ is real — and in fact nonnegative, though that requires a separate argument (the spectrum is the limit of the scaled periodogram). The oddness of sine is what makes the imaginary parts of z^j and z^{-j} cancel: a covariance-stationary process has no “direction” in time at the level of second moments, so nothing imaginary can survive.

Section 6.3 Vector Processes

Review Question 6.3.1

With $\mathbf{y}_t - \boldsymbol{\mu} = \boldsymbol{\varepsilon}_t + \Theta_1 \boldsymbol{\varepsilon}_{t-1}$ and $E(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_{t-j}') = \Omega \mathbf{1}\{j = 0\}$ from (6.3.1),

$$\begin{aligned} \Gamma_0 &= E[(\boldsymbol{\varepsilon}_t + \Theta_1 \boldsymbol{\varepsilon}_{t-1})(\boldsymbol{\varepsilon}_t + \Theta_1 \boldsymbol{\varepsilon}_{t-1})'] = \Omega + \Theta_1 \Omega \Theta_1', \\ \Gamma_1 &= E[(\boldsymbol{\varepsilon}_t + \Theta_1 \boldsymbol{\varepsilon}_{t-1})(\boldsymbol{\varepsilon}_{t-1} + \Theta_1 \boldsymbol{\varepsilon}_{t-2})'] = \Theta_1 \Omega, \end{aligned}$$

because only the $\boldsymbol{\varepsilon}_{t-1}$ terms match; and $\Gamma_j = \mathbf{0}$ for $j \geq 2$, with $\Gamma_{-1} = \Gamma_1' = \Omega \Theta_1'$. This is (6.3.4), $\Gamma_j = \sum_{k=j}^{\infty} \Psi_k \Omega \Psi_{k-j}'$, with $\Psi_0 = \mathbf{I}$, $\Psi_1 = \Theta_1$, $\Psi_j = \mathbf{0}$ for $j \geq 2$. Note that Γ_1 need not be symmetric — the order of the matrix product matters, which is exactly why the transpose appears in $\Gamma_{-j} = \Gamma_j'$.

Review Question 6.3.2

($\Rightarrow$) Suppose $\mathbf{A}(L)\mathbf{B}(L) = \mathbf{I}$, i.e. by (6.3.5), $\mathbf{A}_0\mathbf{B}_0 = \mathbf{I}$ and $\sum_{k=0}^j \mathbf{A}_k\mathbf{B}_{j-k} = \mathbf{0}$ for $j \geq 1$. Let $\mathbf{C}(L) \equiv \mathbf{B}(L)\mathbf{A}(L)$; we show $\mathbf{C}(L) = \mathbf{I}$ by identifying its coefficients recursively. The constant term is $\mathbf{C}_0 = \mathbf{B}_0\mathbf{A}_0 = \mathbf{A}_0^{-1}\mathbf{A}_0 = \mathbf{I}$. For $j = 1$: from the given, $\mathbf{A}_0\mathbf{B}_1 = -\mathbf{A}_1\mathbf{B}_0$, so $\mathbf{B}_1 = -\mathbf{A}_0^{-1}\mathbf{A}_1\mathbf{B}_0$ and

$$\mathbf{C}_1 = \mathbf{B}_1\mathbf{A}_0 + \mathbf{B}_0\mathbf{A}_1 = -\mathbf{A}_0^{-1}\mathbf{A}_1\mathbf{B}_0\mathbf{A}_0 + \mathbf{B}_0\mathbf{A}_1 = -\mathbf{A}_0^{-1}\mathbf{A}_1 + \mathbf{A}_0^{-1}\mathbf{A}_1 = \mathbf{0},$$

using $\mathbf{B}_0\mathbf{A}_0 = \mathbf{I} \Rightarrow \mathbf{B}_0 = \mathbf{A}_0^{-1}$, hence $\mathbf{B}_0\mathbf{A}_1 = \mathbf{A}_0^{-1}\mathbf{A}_1$ and $\mathbf{A}_0^{-1}\mathbf{A}_1\mathbf{B}_0\mathbf{A}_0 = \mathbf{A}_0^{-1}\mathbf{A}_1$. The general step is cleaner through a trick: $\mathbf{B}(L)\mathbf{A}(L)\mathbf{B}(L) = \mathbf{B}(L)[\mathbf{A}(L)\mathbf{B}(L)] = \mathbf{B}(L)$, so $\mathbf{C}(L)\mathbf{B}(L) = \mathbf{B}(L)$; the filter equation $\mathbf{C}(L)\mathbf{B}(L) = \mathbf{B}(L)$ determines the coefficients of $\mathbf{C}$ successively from

$$\mathbf{C}_0\mathbf{B}_0 = \mathbf{B}_0, \quad \sum_{k=0}^j \mathbf{C}_k\mathbf{B}_{j-k} = \mathbf{B}_j \quad (j \geq 1),$$

each equation solvable because $\mathbf{B}_0$ is nonsingular. The unique solution is $\mathbf{C}_0 = \mathbf{I}$, $\mathbf{C}_j = \mathbf{0}$ for $j \geq 1$ (it satisfies the equations, and the recursion admits only one solution). Hence $\mathbf{B}(L)\mathbf{A}(L) = \mathbf{I}$.

($\Leftarrow$) is the same argument with the roles reversed. So for square matrix filters with nonsingular leading coefficients, a one-sided inverse is a two-sided inverse — just as for finite-dimensional matrices, and unlike the situation for general (e.g. one-sided shift) operators.

Review Question 6.3.3

Take the first-degree filters $\mathbf{A}(L) = \mathbf{I} + \mathbf{A}_1L$, $\mathbf{B}(L) = \mathbf{I} + \mathbf{B}_1L$ with

$$\mathbf{A}_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{B}_1 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

The convolution (6.3.5) gives the L^2 coefficient $\mathbf{A}_1\mathbf{B}_1$ for the product $\mathbf{A}(L)\mathbf{B}(L)$ and $\mathbf{B}_1\mathbf{A}_1$ for $\mathbf{B}(L)\mathbf{A}(L)$:

$$\mathbf{A}_1\mathbf{B}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{B}_1\mathbf{A}_1.$$

Hence $\mathbf{A}(L)\mathbf{B}(L) = \mathbf{I} + (\mathbf{A}_1 + \mathbf{B}_1)L + \mathbf{A}_1\mathbf{B}_1L^2 \neq \mathbf{B}(L)\mathbf{A}(L)$. The moral: matrix filters can be manipulated exactly like scalar filters *except* that the order of multiplication must be preserved — which is why (6.1.15B) needs separate statements for left- and right-multiplication in the matrix case (Review Question 6.3.4).

Review Question 6.3.4

Both inverses exist because the leading matrices are nonsingular, and inverses commute (Review Question 6.3.2).

First equivalence. ($\Rightarrow$) Multiply $\mathbf{A}(L)\mathbf{B}(L) = \mathbf{D}(L)$ from the *left* by $\mathbf{A}(L)^{-1}$: $\mathbf{A}(L)^{-1}\mathbf{A}(L)\mathbf{B}(L) = \mathbf{B}(L) = \mathbf{A}(L)^{-1}\mathbf{D}(L)$. ($\Leftarrow$) Multiply from the left by $\mathbf{A}(L)$: $\mathbf{A}(L)\mathbf{B}(L) = \mathbf{A}(L)\mathbf{A}(L)^{-1}\mathbf{D}(L) = \mathbf{D}(L)$.

Second equivalence. ($\Rightarrow$) Multiply from the *right* by $\mathbf{B}(L)^{-1}$: $\mathbf{A}(L)\mathbf{B}(L)\mathbf{B}(L)^{-1} = \mathbf{A}(L) = \mathbf{D}(L)\mathbf{B}(L)^{-1}$. ($\Leftarrow$) Multiply from the right by $\mathbf{B}(L)$.

The only novelty relative to Review Question 6.1.3 is that the side on which one multiplies now matters: to isolate $\mathbf{B}(L)$ one must multiply from the left, and to isolate $\mathbf{A}(L)$ from the right. Applied at the coefficient level this is the recursion used to build the $\text{MA}(\infty)$ representation of a VAR, $\Psi(L) = \Phi(L)^{-1}$, one matrix at a time.

Review Question 6.3.5

By definition, $\mathbf{A}(L)\mathbf{B}(L)[\mathbf{A}(L)\mathbf{B}(L)]^{-1} = \mathbf{I}$. Consider the candidate $\mathbf{C}(L) \equiv \mathbf{B}(L)^{-1}\mathbf{A}(L)^{-1}$:

$$\mathbf{A}(L)\mathbf{B}(L)\mathbf{C}(L) = \mathbf{A}(L)\mathbf{B}(L)\mathbf{B}(L)^{-1}\mathbf{A}(L)^{-1} = \mathbf{A}(L)\mathbf{A}(L)^{-1} = \mathbf{I},$$

using associativity and the definition of the two inverses. By Review Question 6.3.2, a one-sided inverse is the inverse, so $\mathbf{C}(L) = [\mathbf{A}(L)\mathbf{B}(L)]^{-1}$. The order reverses, exactly as for finite-dimensional matrices: to undo $\mathbf{A}$ then $\mathbf{B}$, undo $\mathbf{B}$ first. The result is used in the text's discussion of common roots in ARMA processes, where $\psi(L) = [\alpha(L)\phi^*(L)]^{-1}\alpha(L)\theta^*(L)$ simplifies by cancellation only because the common factor can be peeled off in the right order.

Section 6.4 Estimating Autoregressions

Review Question 6.4.1

(a) By (6.4.2)–(6.4.3), $\text{Avar}(\hat{\beta}) = \sigma^2[\mathbf{E}(\mathbf{x}_t\mathbf{x}_t')^{-1}]$ with $\mathbf{x}_t = (1, y_{t-1})'$, where

$$\mathbf{E}(\mathbf{x}_t\mathbf{x}_t') = \begin{bmatrix} 1 & \mu \\ \mu & \mu^2 + \gamma_0 \end{bmatrix}, \quad \mu = \frac{c}{1-\phi}, \quad \gamma_0 = \frac{\sigma^2}{1-\phi^2}.$$

Inverting the 2×2 matrix (determinant γ_0), the $(2, 2)$ element of the inverse is $1/\gamma_0$. Hence

$$\text{Avar}(\hat{\phi}) = \sigma^2 \cdot \frac{1}{\gamma_0} = \sigma^2 \cdot \frac{1-\phi^2}{\sigma^2} = 1-\phi^2.$$

So $\sqrt{n}(\hat{\phi} - \phi) \xrightarrow{d} N(0, 1-\phi^2)$: the closer ϕ is to one, the more precise the estimator — a preview of the nonstandard $|\phi| = 1$ asymptotics of Chapter 9, where the rate changes from $\sqrt{n}$ to n .

(b) Write $\mathbf{x}_t = (1, \tilde{\mathbf{x}}_t')'$ with $\tilde{\mathbf{x}}_t = (y_{t-1}, \dots, y_{t-p})'$. Then, as in the hint,

$$\mathbf{E}(\mathbf{x}_t\mathbf{x}_t') = \begin{bmatrix} 1 & \mu\mathbf{1}' \\ \mu\mathbf{1} & \mathbf{V} + \mu^2\mathbf{1}\mathbf{1}' \end{bmatrix},$$

because $\mathbf{E}(\tilde{\mathbf{x}}_t\tilde{\mathbf{x}}_t') = \text{Cov}(\tilde{\mathbf{x}}_t) + \mathbf{E}(\tilde{\mathbf{x}}_t)\mathbf{E}(\tilde{\mathbf{x}}_t)' = \mathbf{V} + \mu^2\mathbf{1}\mathbf{1}'$. To verify the hint's inverse, multiply the matrix by the candidate $\mathbf{A} = \begin{bmatrix} a & \mathbf{c}' \\ \mathbf{c} & \mathbf{V}^{-1} \end{bmatrix}$ with $a = 1 + \mu^2\mathbf{1}'\mathbf{V}^{-1}\mathbf{1}$ and $\mathbf{c} = -\mu\mathbf{V}^{-1}\mathbf{1}$:

$$(1, 1): \quad a + \mu\mathbf{1}'\mathbf{c} = 1 + \mu^2\mathbf{1}'\mathbf{V}^{-1}\mathbf{1} - \mu^2\mathbf{1}'\mathbf{V}^{-1}\mathbf{1} = 1,$$

$$(1, 2): \quad \mu a\mathbf{1}' + \mathbf{c}'(\mathbf{V} + \mu^2\mathbf{1}\mathbf{1}') = \mu a\mathbf{1}' - \mu\mathbf{1}' - \mu^3(\mathbf{1}'\mathbf{V}^{-1}\mathbf{1})\mathbf{1}' = \mu(a - 1 - \mu^2\mathbf{1}'\mathbf{V}^{-1}\mathbf{1})\mathbf{1}' = \mathbf{0},$$

$$(2, 2): \quad \mu\mathbf{1}\mathbf{c}' + \mathbf{V}^{-1}(\mathbf{V} + \mu^2\mathbf{1}\mathbf{1}') = -\mu^2\mathbf{1}(\mathbf{V}^{-1}\mathbf{1})' + \mathbf{I} + \mu^2\mathbf{V}^{-1}\mathbf{1}\mathbf{1}' = \mathbf{I},$$

the last because $\mathbf{1}(\mathbf{V}^{-1}\mathbf{1})' = (\mathbf{V}^{-1}\mathbf{1})\mathbf{1}'$ — both are the matrix with (i, j) element $[\mathbf{V}^{-1}\mathbf{1}]_i$ (the vector $\mathbf{V}^{-1}\mathbf{1}$ is symmetric in this sense since $\mathbf{V}^{-1}$ is symmetric: $\mathbf{1}'\mathbf{V}^{-1} = (\mathbf{V}^{-1}\mathbf{1})'$). Hence the

product is $\mathbf{I}$, confirming

$$[\mathbf{E}(\mathbf{x}_t \mathbf{x}_t')]^{-1} = \begin{bmatrix} a & \mathbf{c}' \\ \mathbf{c} & \mathbf{V}^{-1} \end{bmatrix}.$$

The lower-right block gives $\text{Avar}(\hat{\phi}) = \sigma^2 \mathbf{V}^{-1}$: the joint asymptotic variance of the autoregressive coefficients is σ^2 times the inverse autocovariance matrix, independent of the mean μ — the sample analogue is the textbook formula $\widehat{\text{Var}}(\hat{\phi}) = s^2 [\sum_t (\mathbf{y}_{t-1} - \bar{\mathbf{y}})(\mathbf{y}_{t-1} - \bar{\mathbf{y}})']^{-1}$. For $p = 1$, $\mathbf{V} = \gamma_0$ and this reduces to part (a). As the hint notes, the display also shows $\mathbf{E}(\mathbf{x}_t \mathbf{x}_t')$ is nonsingular if and only if $\mathbf{V}$ is, which is the rank condition ASSUMPTION 2.4 for the autoregression.

Review Question 6.4.2

Equation m of the system is $y_{tm} = \mathbf{x}_t' \boldsymbol{\delta}_m + \varepsilon_{tm}$ with $\mathbf{x}_t = \mathbf{y}_{t-1}$ and $\boldsymbol{\delta}_m$ the m -th row of $\mathbf{A}$, transposed. Equation-by-equation OLS gives

$$\hat{\boldsymbol{\delta}}_m = \left(\sum_{t=2}^n \mathbf{y}_{t-1} \mathbf{y}_{t-1}' \right)^{-1} \sum_{t=2}^n \mathbf{y}_{t-1} y_{tm},$$

so the m -th row of $\hat{\mathbf{A}}$ is $\hat{\boldsymbol{\delta}}_m' = (\sum_t y_{tm} \mathbf{y}_{t-1}') (\sum_t \mathbf{y}_{t-1} \mathbf{y}_{t-1}')^{-1}$. Stacking the M rows,

$$\hat{\mathbf{A}} = \left(\sum_{t=2}^n \mathbf{y}_t \mathbf{y}_{t-1}' \right) \left(\sum_{t=2}^n \mathbf{y}_{t-1} \mathbf{y}_{t-1}' \right)^{-1},$$

as claimed. This is the matrix analogue of the univariate AR(1) slope $\hat{\phi} = \sum_t y_t y_{t-1} / \sum_t y_{t-1}^2$: since every equation has the same regressors, the estimates stack, and $\hat{\mathbf{A}}$ is what a package returns for a VAR(1) with no constant. Note the order of the two factors — the cross-moment matrix multiplies the inverse second-moment matrix on the *right*, the transpose of the scalar case, because the rows of $\mathbf{A}$ multiply $\mathbf{y}_{t-1}$ from the left.

Section 6.5 Asymptotics for Sample Means of Serially Correlated Processes

Review Question 6.5.1

Write $\mathbf{y} \equiv (y_t, \dots, y_{t-n+1})'$ and let $\mathbf{V}_n \equiv \text{Var}(\mathbf{y})$ be the $n \times n$ autocovariance matrix (the same for any block of n consecutive dates, by covariance stationarity). Since $\sqrt{n} \bar{y}_n = \frac{1}{\sqrt{n}} \mathbf{1}' \mathbf{y}$,

$$\text{Var}(\sqrt{n} \bar{y}_n) = \frac{1}{n} \mathbf{1}' \text{Var}(\mathbf{y}) \mathbf{1} = \frac{1}{n} \mathbf{1}' \mathbf{V}_n \mathbf{1} = \frac{1}{n} \sum_{t=1}^n \sum_{s=1}^n \gamma_{|t-s|} = \gamma_0 + 2 \sum_{j=1}^{n-1} \left(1 - \frac{j}{n}\right) \gamma_j,$$

the last equality by counting, for each lag j , the $n - j$ pairs (t, s) with $|t - s| = j$ in each direction — which is (6.5.2). The quadratic-form version is useful because it makes the long-run variance a property of the whole autocovariance matrix: the sample mean “sees” every autocovariance, with weights tapering linearly from 1 to 0 as the lag grows.

Review Question 6.5.2

By (6.2.19) with $g_\varepsilon(z) = \sigma^2$, the autocovariance-generating function of $y_t = \mu + \psi(L)\varepsilon_t$ is $g_Y(z) = \sigma^2 \psi(z)\psi(z^{-1})$. The long-run variance is $g_Y(1)$ (see (6.2.15) and the text after PROPOSITION 6.8),

so

$$\text{Avar}(\bar{y}) = \sum_{j=-\infty}^{\infty} \gamma_j = g_Y(1) = \sigma^2 \psi(1)^2 = \sigma^2 \left(\sum_{j=0}^{\infty} \psi_j \right)^2,$$

since $\psi(1) = \sum_j \psi_j$ converges by absolute summability. One check: for MA(1), $\psi(1) = 1 + \theta_1$ and $\gamma_0 + 2\gamma_1 = \sigma^2[(1 + \theta_1^2) + 2\theta_1] = \sigma^2(1 + \theta_1)^2$. Notice the long-run variance is zero if $\psi(1) = 0$: the shocks cancel “in the long run” and $\bar{y}$ converges faster than $1/\sqrt{n}$ (compare Review Question 6.5.5).

Review Question 6.5.3

Consistency: yes, without independence. $\{\gamma_j\}$ vanishes geometrically, $\gamma_j = \phi^j \gamma_0$, so $\gamma_j \rightarrow 0$ and PROPOSITION 6.8(A) gives $\bar{y} \xrightarrow{m.s.} \mu$, hence $\bar{y} \xrightarrow{P} \mu = c/(1 - \phi)$.

Asymptotic variance. With independent white noise the CLT PROPOSITION 6.9 applies, since the MA(∞) coefficients $\psi_j = \phi^j$ are absolutely summable. Two computations of the limit:

(i) *Summing the autocovariances.*

$$\text{Avar}(\bar{y}) = \gamma_0 + 2 \sum_{j=1}^{\infty} \gamma_j = \gamma_0 \left(1 + 2 \sum_{j=1}^{\infty} \phi^j \right) = \gamma_0 \left(1 + \frac{2\phi}{1 - \phi} \right) = \frac{1 + \phi}{1 - \phi} \gamma_0 = \frac{1 + \phi}{1 - \phi} \cdot \frac{\sigma^2}{1 - \phi^2} = \frac{\sigma^2}{(1 - \phi)^2},$$

which is the book’s answer $[(1 + \phi)/(1 - \phi)]\gamma_0$.

(ii) *From the filter.* $\psi(1) = \sum_j \phi^j = 1/(1 - \phi)$, so by Review Question 6.5.2, $\text{Avar}(\bar{y}) = \sigma^2/(1 - \phi)^2$ — the same number, as it must be.

So $\sqrt{n}(\bar{y} - \mu) \xrightarrow{d} N(0, \sigma^2/(1 - \phi)^2)$. The factor $(1 + \phi)/(1 - \phi)$ measures how much serial correlation inflates the variance of the sample mean: it diverges as $\phi \uparrow 1$ — again foreshadowing the unit-root asymptotics of Chapter 9 — and is less than one for negative ϕ .

Review Question 6.5.4

By (6.3.13), $\mathbf{G}_Y(z) = \mathbf{H}(z)\mathbf{G}_X(z)\mathbf{H}(z^{-1})'$, and by (6.5.6) the long-run covariance matrix is $\mathbf{G}_Y(1)$. Evaluating at $z = 1$,

$$\sum_{j=-\infty}^{\infty} \Gamma_j^y = \mathbf{G}_Y(1) = \mathbf{H}(1)\mathbf{G}_X(1)\mathbf{H}(1)' = \mathbf{H}(1) \left(\sum_{j=-\infty}^{\infty} \Gamma_j^x \right) \mathbf{H}(1)',$$

since $\mathbf{H}(z^{-1})$ at $z = 1$ is $\mathbf{H}(1)$. For the one-term filter, $\mathbf{H}(1) = \mathbf{H}_0 + \mathbf{H}_1$. This is the matrix version of “long-run variance of output = [filter at 1] $\times$ long-run variance of input $\times$ [filter at 1]’”, and it is exactly the formula behind VARHAC: with $\mathbf{g}_t$ modeled as a VAR, the implied long-run variance of $\mathbf{g}_t$ is $[\mathbf{I} - \sum_j \hat{\mathbf{A}}_j]^{-1} \hat{\Omega} [\mathbf{I} - \sum_j \hat{\mathbf{A}}_j]'^{-1}$, which is (6.6.15).

Review Question 6.5.5

The autocovariances are $\gamma_0 = 2\sigma^2$, $\gamma_1 = -\sigma^2$, $\gamma_j = 0$ for $j \geq 2$, so

$$\sum_{j=-\infty}^{\infty} \gamma_j = 2\sigma^2 + 2(-\sigma^2) = 0.$$

Equivalently, $\psi(L) = 1 - L$ has $\psi(1) = 0$, and the long-run variance is $\sigma^2\psi(1)^2 = 0$. The MA(1) with $\theta = -1$ is the “unit-root MA”: it is the first difference of white noise, so its partial sums telescope, $\sum_{t=1}^n y_t = \varepsilon_n - \varepsilon_0$, and $\text{Var}(\sqrt{n}\bar{y}_n) = \text{Var}(\varepsilon_n - \varepsilon_0)/n = 2\sigma^2/n \rightarrow 0$: the sample mean converges at a rate faster than $1/\sqrt{n}$. A singular (here, zero) long-run variance is exactly the case excluded by the nonsingularity clause of ASSUMPTION 3.5' — GMM asymptotics need the long-run variance to be positive definite.

Remark. The example is the boundary between stationary and unit-root behaviour: $y_t = (1 - L)\varepsilon_t$ is stationary but “overdifferenced” — it is the difference of a series that did not need differencing. Chapter 9’s tests for unit roots amount to testing whether the MA polynomial has a root at 1.

Section 6.6 Incorporating Serial Correlation in GMM

Review Question 6.6.1

It is the OLS estimator. With $\mathbf{x}_t = \mathbf{z}_t$, the model is exactly identified ($K = L$), so GMM sets the sample orthogonality conditions to zero: $\bar{\mathbf{g}}(\boldsymbol{\delta}) = \frac{1}{n} \sum_t \mathbf{z}_t(y_t - \mathbf{z}_t'\boldsymbol{\delta}) = \mathbf{0}$, whose solution is $\hat{\boldsymbol{\delta}} = (\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}$ regardless of the weighting matrix — every GMM estimator, hence the efficient one, is OLS. What serial correlation changes is not the point estimate but the standard errors: the consistent estimator of the asymptotic variance is (3.5.2) with $\hat{\mathbf{S}}$ a HAC estimator of the long-run variance (6.6.1), e.g. the truncated or Bartlett kernel sums of $\hat{\Gamma}_j$. So the recipe “estimate by OLS, but compute Newey–West standard errors” is efficient GMM in the just-identified case — there is no efficiency gain available from GLS-type corrections when the regressors are merely predetermined (Section 6.7 makes this precise).

Review Question 6.6.2

Consistent: yes. Consistency of GMM requires only the orthogonality conditions $E(\mathbf{g}_t) = \mathbf{0}$ (ASSUMPTION 3.3), ergodic stationarity and identification; the probability limit of $\hat{\boldsymbol{\delta}}(\hat{\mathbf{W}})$ is $\boldsymbol{\delta}$ whatever the autocovariance structure of $\{\mathbf{g}_t\}$, because the argument “ $\bar{\mathbf{g}}(\hat{\boldsymbol{\delta}}) \xrightarrow{p} \mathbf{0}$ forces $\hat{\boldsymbol{\delta}} \xrightarrow{p} \boldsymbol{\delta}$ ” never uses the absence of serial correlation. Serial correlation is a property of the *dynamics* of the moment conditions, not of their means.

Efficient: no. Efficiency requires $\text{plim } \hat{\mathbf{W}} = \mathbf{S}^{-1}$ with $\mathbf{S}$ the long-run variance (6.6.1). An estimator built on the presumption of no serial correlation uses $\mathbf{S} = E(\mathbf{g}_t\mathbf{g}_t') = \Gamma_0$, omitting $\sum_{j \neq 0} \Gamma_j$, and so is the efficient GMM estimator of a different (wrongly specified) weighting scheme. Two consequences: its asymptotic variance exceeds that of the truly efficient estimator (unless the omitted autocovariances happen to sum to zero), and — equally important in practice — the standard errors it reports, computed from Γ_0 alone, are inconsistent, so the associated t and Wald tests have the wrong size even though the point estimates are fine. In the just-identified case ($\mathbf{x}_t = \mathbf{z}_t$) the point estimate is numerically identical to the efficient one (OLS), and only the standard errors are wrong — which is why the “HAC correction” is often described as fixing the standard errors of OLS.

Section 6.7 Estimation under Conditional Homoskedasticity (Optional)

Review Question 6.7.1

With $n = 3$ and autocovariances vanishing after lag $q = 1$, (6.7.6)–(6.7.7) give

$$\widehat{\Omega} = \begin{bmatrix} \hat{\omega}_0 & \hat{\omega}_1 & 0 \\ \hat{\omega}_1 & \hat{\omega}_0 & \hat{\omega}_1 \\ 0 & \hat{\omega}_1 & \hat{\omega}_0 \end{bmatrix}, \quad \hat{\omega}_j = \frac{1}{n} \sum_{t=j+1}^n \hat{\varepsilon}_t \hat{\varepsilon}_{t-j}.$$

The (k, ℓ) element of $\mathbf{X}'\widehat{\Omega}\mathbf{X}$ is $\sum_{t=1}^3 \sum_{s=1}^3 x_{tk} \widehat{\Omega}_{ts} x_{s\ell}$. The diagonal terms ($t = s$) contribute $\hat{\omega}_0 \sum_t x_{tk} x_{t\ell}$; the off-diagonal terms contribute $\hat{\omega}_1 \sum_{|t-s|=1} x_{tk} x_{s\ell} = \hat{\omega}_1 \sum_{t=2}^3 (x_{tk} x_{t-1,\ell} + x_{t-1,k} x_{t\ell})$. Dividing by n ,

$$\frac{1}{n} \mathbf{X}'\widehat{\Omega}\mathbf{X} = \widehat{\Gamma}_0 + \widehat{\Gamma}_1 + \widehat{\Gamma}'_1, \quad \widehat{\Gamma}_j \equiv \hat{\omega}_j \frac{1}{n} \sum_{t=j+1}^n \mathbf{x}_t \mathbf{x}'_{t-j} \text{ as in (6.7.4),}$$

which is (6.6.4) with $q = 1$. That is exactly (6.7.5). The general case is the same computation: the band structure of $\widehat{\Omega}$ (zeros beyond the q -th diagonal, or kernel weights on the diagonals) collects the products $x_{tk} x_{s\ell}$ into the weighted autocovariances of $\{\mathbf{x}_t \hat{\varepsilon}_t\}$ that the kernel estimator sums.

Review Question 6.7.2

Under conditional homoskedasticity with serial correlation, (6.7.3) gives $\Gamma_j = \omega_j \mathbf{E}(\mathbf{x}_t \mathbf{x}'_{t-j})$, and (6.7.5) writes the estimated long-run variance as $\widehat{\mathbf{S}} = \frac{1}{n} \mathbf{X}'\widehat{\Omega}\mathbf{X}$. Sargan's statistic generalizes as Hansen's J evaluated at the efficient GMM estimator under conditional homoskedasticity (6.7.9): with $\bar{\mathbf{g}} = \frac{1}{n} \mathbf{X}'(\mathbf{y} - \mathbf{Z}\hat{\delta})$ the sample orthogonality conditions,

$$J = n \bar{\mathbf{g}}' \widehat{\mathbf{S}}^{-1} \bar{\mathbf{g}} = [\mathbf{X}'(\mathbf{y} - \mathbf{Z}\hat{\delta})]' [\mathbf{X}'\widehat{\Omega}\mathbf{X}]^{-1} [\mathbf{X}'(\mathbf{y} - \mathbf{Z}\hat{\delta})] \xrightarrow{d} \chi^2(K - L),$$

under the null that all K orthogonality conditions hold — exactly as in (3.8.10'), the only change being that $\widehat{\mathbf{S}}$ (equivalently $\widehat{\Omega}$) now reflects serial correlation, through the truncated or Bartlett weights in (6.7.7)–(6.7.8), instead of being proportional to $\mathbf{X}'\mathbf{X}$. When $\widehat{\Omega} = \hat{\sigma}^2 \mathbf{I}_n$ (no serial correlation) the display collapses to (3.8.10'), since then $J = \frac{1}{\hat{\sigma}^2} (\mathbf{y} - \mathbf{Z}\hat{\delta})' \mathbf{P}_X (\mathbf{y} - \mathbf{Z}\hat{\delta})$ with $\mathbf{P}_X = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ — Sargan's original statistic, the explained sum of squares from regressing the 2SLS residuals on all the instruments, divided by $\hat{\sigma}^2$.

Review Question 6.7.3

No. Write the efficient estimator under conditional homoskedasticity,

$$\hat{\delta} = [\mathbf{Z}'\mathbf{X}(\mathbf{X}'\widehat{\Omega}\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}]^{-1} \mathbf{Z}'\mathbf{X}(\mathbf{X}'\widehat{\Omega}\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

This is a GLS-flavored instrumental-variables estimator: the instruments enter through the “effective regressor matrix” $\tilde{\mathbf{Z}} \equiv \mathbf{X}(\mathbf{X}'\widehat{\Omega}\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}$, and $\hat{\delta} = (\tilde{\mathbf{Z}}'\tilde{\mathbf{Z}})^{-1}\tilde{\mathbf{Z}}'\mathbf{y}$. Without serial correlation ($\widehat{\Omega} \propto \mathbf{I}_n$), $\tilde{\mathbf{Z}} = \mathbf{P}_X\mathbf{Z}$ and, when the columns of $\mathbf{Z}$ are among those of $\mathbf{X}$, $\mathbf{P}_X\mathbf{Z} = \mathbf{Z}$, so $\hat{\delta}$ collapses to OLS — the Section 3.8 result. With serial correlation the weighting matrix $(\mathbf{X}'\widehat{\Omega}\mathbf{X})^{-1}$ is not proportional to $(\mathbf{X}'\mathbf{X})^{-1}$, the projection identity fails, and the estimator is a genuine GMM estimator distinct from OLS, even though every regressor is a valid instrument. The intuition: with serial correlation the efficient linear combination of the orthogonality conditions is not the one that

sets $\mathbf{Z}'(\mathbf{y} - \mathbf{Z}\delta) = \mathbf{0}$; the efficient combination downweights the moment conditions in a way that reflects the error autocovariances.

Review Question 6.7.4

Without conditional homoskedasticity the (t, s) element of $\hat{\Omega}$ must reproduce the product of the residuals $\hat{\varepsilon}_t \hat{\varepsilon}_s$ itself, with the kernel weight attached, because $\Gamma_j = E(\mathbf{g}_t \mathbf{g}_{t-j}')$ no longer factors as $\omega_j E(\mathbf{x}_t \mathbf{x}_{t-j}')$. Following the hint, set

$$\hat{\Omega}_{ts} = \hat{\varepsilon}_t \hat{\varepsilon}_s k\left(\frac{|t-s|}{q(n)+1}\right) :$$

for the truncated kernel at known q , $\hat{\Omega}_{ts} = \hat{\varepsilon}_t \hat{\varepsilon}_s$ for $|t-s| \leq q$ and 0 beyond; for the Bartlett kernel,

$$\hat{\Omega}_{ts} = \left(1 - \frac{|t-s|}{q(n)+1}\right) \hat{\varepsilon}_t \hat{\varepsilon}_s \quad (|t-s| \leq q(n)),$$

and zero beyond. Indeed, the (k, ℓ) element of $\frac{1}{n} \mathbf{X}' \hat{\Omega} \mathbf{X}$ is then

$$\frac{1}{n} \sum_t \sum_s x_{tk} x_{s\ell} \hat{\Omega}_{ts} = \sum_{j=-q(n)}^{q(n)} k\left(\frac{|j|}{q(n)+1}\right) \frac{1}{n} \sum_{t=|j|+1}^n x_{tk} \hat{\varepsilon}_t x_{t-|j|, \ell} \hat{\varepsilon}_{t-|j|},$$

which is the (k, ℓ) element of $\hat{\Gamma}_0 + \sum_{j \geq 1} k(\cdot)(\hat{\Gamma}_j + \hat{\Gamma}_j')$ with $\hat{\Gamma}_j = \frac{1}{n} \sum_t \mathbf{g}_t \mathbf{g}_{t-j}'$ — the kernel estimator (6.6.3). So the same data-matrix formula (6.7.5) covers both cases; the difference is only whether the autocovariances of the errors and of the instruments can be estimated separately (they can, under conditional homoskedasticity, via (6.7.4) — which is statistically cheaper) or must be estimated jointly through the products $\mathbf{x}_t \hat{\varepsilon}_t$.

Review Question 6.7.5

OLS. $\sqrt{n}(\hat{\beta}_{\text{OLS}} - \beta) = \left(\frac{1}{n} \sum_t z_t^2\right)^{-1} \frac{1}{\sqrt{n}} \sum_t z_t \varepsilon_t$. The AR(1) error has autocovariances $\omega_j = \phi^{|j|} \sigma_\varepsilon^2$, so by (6.7.3) in scalar form, the autocovariances of the moment process $\{z_t \varepsilon_t\}$ are $\Gamma_j = \omega_j E(z_t z_{t-j}) = \phi^{|j|} \sigma_\varepsilon^2 \rho_{|j|} \sigma_z^2$, and its long-run variance is

$$S = \sigma_\varepsilon^2 \sigma_z^2 \left[1 + 2 \sum_{j=1}^{\infty} \phi^j \rho_j\right], \quad \text{Avar}(\hat{\beta}_{\text{OLS}}) = \frac{S}{(\sigma_z^2)^2} = \frac{\sigma_\varepsilon^2}{\sigma_z^2} \left[1 + 2 \sum_{j=1}^{\infty} \phi^j \rho_j\right].$$

GLS. With known ϕ , GLS is OLS on the quasi-differenced equation $y_t - \phi y_{t-1} = \beta(z_t - \phi z_{t-1}) + \eta_t$, i.e.

$$\hat{\beta}_{\text{GLS}} - \beta = \frac{\frac{1}{n} \sum_t \tilde{z}_t \eta_t}{\frac{1}{n} \sum_t \tilde{z}_t^2}, \quad \tilde{z}_t \equiv z_t - \phi z_{t-1}.$$

Because $\{\eta_t\}$ is independent with $E(\tilde{z}_t \eta_t) = 0$ by assumption, $\{\tilde{z}_t \eta_t\}$ is a martingale difference sequence, so its long-run variance is $E(\tilde{z}_t^2 \eta_t^2) = \sigma_\eta^2 E(\tilde{z}_t^2)$ (conditional homoskedasticity), where

$$E(\tilde{z}_t^2) = \sigma_z^2 (1 + \phi^2 - 2\phi\rho_1).$$

Therefore

$$\text{Avar}(\hat{\beta}_{\text{GLS}}) = \frac{\sigma_\eta^2 E(\tilde{z}_t^2)}{[E(\tilde{z}_t^2)]^2} = \frac{\sigma_\eta^2}{\sigma_z^2 (1 + \phi^2 - 2\phi\rho_1)} = \frac{(1 - \phi^2) \sigma_\varepsilon^2}{\sigma_z^2 (1 + \phi^2 - 2\phi\rho_1)},$$

using $\sigma_\eta^2 = (1 - \phi^2)\sigma_\varepsilon^2$. Note that the GLS variance depends on $\{\rho_j\}$ only through ρ_1 : GLS exploits the AR(1) structure of the error exactly.

A configuration with $\text{Avar}_{\text{OLS}} < \text{Avar}_{\text{GLS}}$. Take $\phi = 0.9$ and $\rho_1 = 0.99$, $\rho_j = 0$ for $j \geq 2$. Then

$$\text{Avar}_{\text{OLS}} = \frac{\sigma_\varepsilon^2}{\sigma_z^2} (1 + 2 \times 0.9 \times 0.99) = 2.782 \frac{\sigma_\varepsilon^2}{\sigma_z^2}, \quad \text{Avar}_{\text{GLS}} = \frac{0.19}{1.81 - 1.782} \frac{\sigma_\varepsilon^2}{\sigma_z^2} = 6.79 \frac{\sigma_\varepsilon^2}{\sigma_z^2},$$

so OLS is more than twice as precise as GLS. What happens: with ρ_1 close to ϕ and both close to one, the quasi-differenced regressor $\tilde{z}_t = z_t - \phi z_{t-1}$ nearly vanishes ($E\tilde{z}_t^2 = 0.028\sigma_z^2$), destroying the identifying variation GLS uses, while the untransformed regression keeps the full variation of z_t .

Why this is consistent with GMM efficiency. The efficient GMM ranking of Section 6.6 is a ranking of estimators that exploit *the same* orthogonality conditions $E(z_t \varepsilon_t) = 0$; among those, OLS is efficient in this just-identified model (Review Question 6.6.1). GLS is not in that class: its normal equations involve $\tilde{z}_t \eta_t = (z_t - \phi z_{t-1}) \eta_t$, i.e. they exploit the *different* moment conditions $E(z_t \eta_t) = 0$ and $E(z_{t-1} \eta_t) = 0$ — conditions that are about the white-noise innovation, not about ε_t , and that the model's assumption $E(z_t \varepsilon_t) = 0$ for predetermined z_t does not deliver (predeterminedness gives $E(z_{t-j} \varepsilon_t) = 0$ only for j of the appropriate sign; $E(z_{t-1} \eta_t) = 0$ is an extra assumption, which happens to be imposed here). Even granting those extra moments, GLS is superior only when the AR(1) model for the error is exactly right *and* the regressor dynamics cooperate; the example shows the failure of the second caveat. Hence the chapter's practical message: with predetermined regressors, keep the OLS point estimate and fix the standard errors; do not quasi-difference.

Section 6.8 Application: Forward Exchange Rates as Optimal Predictors

Review Question 6.8.1

In the regression $\varepsilon_t = \mu + u_t$ with the single regressor $\mathbf{x}_t = 1$, the OLS estimate of μ is $\hat{\mu} = \bar{\varepsilon}$, and the moment process is $\mathbf{g}_t = \mathbf{x}_t \hat{u}_t = \hat{u}_t = \varepsilon_t - \bar{\varepsilon}$. The estimated autocovariances of $\mathbf{g}_t$ are

$$\hat{\Gamma}_j = \frac{1}{n} \sum_{t=j+1}^n \hat{u}_t \hat{u}_{t-j} = \hat{\gamma}_j,$$

the sample autocovariances of the demeaned forecast error — the 1×1 version of (6.6.2). With $\mathbf{S}_{\mathbf{xx}} = 1$, the HAC asymptotic variance is $\widehat{\text{Avar}}(\hat{\mu}) = \hat{\mathbf{S}}$ with

$$\hat{\mathbf{S}} = \hat{\gamma}_0 + 2 \sum_{j=1}^4 \hat{\gamma}_j,$$

the $q = 4$ truncated kernel estimator. The standard error of $\hat{\mu} = \bar{\varepsilon}$ is $\sqrt{\hat{\mathbf{S}}/n}$, and the t -ratio for $\mu = 0$ is

$$t = \frac{\bar{\varepsilon}}{\sqrt{\hat{\mathbf{S}}/n}} = \frac{\sqrt{n} \bar{\varepsilon}}{\sqrt{\hat{\gamma}_0 + 2 \sum_{j=1}^4 \hat{\gamma}_j}},$$

which is (6.8.7). So the unconditional test *is* a GMM test with one instrument (the constant) and the truncated HAC estimator at four lags. Numerically (`code/ch6.py`), the three currencies give

$t = -0.35$ (yen), 0.65 (DM) and 0.43 (pound) — Table 6.1's non-rejection of market efficiency at the unconditional level.

Review Question 6.8.2

Random walk. Market efficiency with risk neutrality, (6.8.3), requires $f_t = E(s_{t+2} | I_t)$. For a random walk (with or without drift), $E(s_{t+2} | I_t) = s_t$ (plus twice the drift), so $f_t = s_t$: the forward premium $f_t - s_t$ is identically zero, the regressor in (6.8.9) has no variance, and $E(\mathbf{x}_t \mathbf{x}_t')$ for $\mathbf{x}_t = (1, f_t - s_t)'$ is singular — the *identification condition* ASSUMPTION 3.4 fails. The orthogonality condition still holds — ε_t is a two-period forecast error, orthogonal to I_t — but with a constant regressor there is nothing to estimate; the equation reduces to $\varepsilon_t = \beta_0 + u_t$. As the book's answer states: the violated assumption is identification.

Stationary AR(1). Iterating the AR(1) forward,

$$s_{t+2} = c + \phi s_{t+1} + \eta_{t+2} = c + \phi(c + \phi s_t + \eta_{t+1}) + \eta_{t+2} = (1 + \phi)c + \phi^2 s_t + \eta_{t+2} + \phi \eta_{t+1}.$$

Taking $E(\cdot | I_t)$ kills the η 's (they are i.i.d. with mean zero, dated after t), so

$$f_t = E(s_{t+2} | I_t) = (1 + \phi)c + \phi^2 s_t,$$

as claimed; and subtracting, $\varepsilon_t = s_{t+2} - f_t = \eta_{t+2} + \phi \eta_{t+1}$, an MA(1) with parameter ϕ — serial correlation up to lag 1, the two-week analogue of the four-week overlap in the data. Note that identification is restored: $f_t - s_t = (1 + \phi)c - (1 - \phi^2)s_t$ varies with s_t , so the forward premium moves and the regression (6.8.9) has a nondegenerate regressor. Random-walk spot rates destroy the test not by making the orthogonality conditions fail but by making the instrument useless.

Review Question 6.8.3

The information set I_t (Friday of week t) includes the current spot and forward rates, hence $f_t - s_t$ and $\mathbf{x}_t$; and because $\varepsilon_{t-j} = s_{30t-j} - f_{t-j}$ is observed once the delivery date of week $t-j$'s contract has passed — which is after the Friday of week $t-j+4$ — I_t also includes $\varepsilon_{t-5}, \varepsilon_{t-6}, \dots$, hence $\mathbf{g}_{t-5}, \mathbf{g}_{t-6}, \dots$ (their other ingredient, $\mathbf{x}_{t-j}$, is in I_t for $j \geq 1$). So both $\mathbf{x}_t$ and the conditioning variables are functions of I_t . By the law of iterated expectations,

$$E(\mathbf{g}_t | \mathbf{g}_{t-5}, \mathbf{g}_{t-6}, \dots) = E[E(\mathbf{x}_t \varepsilon_t | I_t) | \mathbf{g}_{t-5}, \dots] = E[\mathbf{x}_t E(\varepsilon_t | I_t) | \mathbf{g}_{t-5}, \dots] = \mathbf{0},$$

because $E(\varepsilon_t | I_t) = 0$ is the market efficiency hypothesis (6.8.3). This is (6.8.10): the moment process inherits the “martingale difference at displacement 5” property of the forecast error, so its autocovariances vanish beyond lag 4, $\Gamma_j = \mathbf{0}$ for $j \geq 5$, and $\mathbf{S}$ is consistently estimated by the truncated sum (6.6.4) with $q = 4$ — the basis of Table 6.2.

Problem Set for Chapter 6: Analytical Exercises

Analytical Exercise 6.1

(a) Define the mean-square norm $\|x\| \equiv \sqrt{E(x^2)}$ and expand along the hint:

$$z - z' = (z - z_n) + (z_n - z'), \quad \|z - z'\|^2 = \|z - z_n\|^2 + 2E[(z - z_n)(z_n - z')] + \|z_n - z'\|^2.$$

By the Cauchy–Schwarz inequality for expectations, $|\mathbb{E}(xy)| \leq \|x\| \|y\|$, the middle term is bounded by $2\|z - z_n\| \|z_n - z'\|$, so

$$\|z - z'\|^2 \leq (\|z - z_n\| + \|z_n - z'\|)^2, \quad \text{i.e.} \quad \|z - z'\| \leq \|z - z_n\| + \|z_n - z'\|$$

— the triangle inequality for $\|\cdot\|$. Both terms on the right converge to zero by hypothesis, so $\|z - z'\| = 0$, that is, $\mathbb{E}[(z - z')^2] = 0$.

(b) By Chebychev’s inequality, for every $\varepsilon > 0$,

$$\text{Prob}(|z - z'| > \varepsilon) \leq \frac{\mathbb{E}[(z - z')^2]}{\varepsilon^2} = 0,$$

so $\text{Prob}(|z - z'| > \varepsilon) = 0$ for every $\varepsilon > 0$. The event $\{z \neq z'\} = \bigcup_{k=1}^{\infty} \{|z - z'| > 1/k\}$ is a countable union of zero-probability events, hence has probability zero: $\text{Prob}(z = z') = 1$.

So a mean-square limit, when it exists, is unique up to a zero-probability event — which is what makes the definition of the infinite sum (6.1.3) as “the” mean square limit of the partial sums unambiguous. Mean square convergence is convergence in the Hilbert space L^2 of square-integrable random variables; this exercise proves that L^2 limits are unique a.s., the first fact every L^2 construction (conditional expectations included) needs.

Analytical Exercise 6.2

Let $y_{t,n} \equiv \mu + \sum_{j=0}^n \psi_j \varepsilon_{t-j}$, the partial sum.

(a) For $m > n$, by white noise ($\mathbb{E}(\varepsilon_s \varepsilon_u) = \sigma^2 \mathbf{1}\{s = u\}$),

$$\mathbb{E}[(y_{t,m} - y_{t,n})^2] = \mathbb{E}\left[\left(\sum_{j=n+1}^m \psi_j \varepsilon_{t-j}\right)^2\right] = \sigma^2 \sum_{j=n+1}^m \psi_j^2.$$

Absolute summability of $\{\psi_j\}$ implies square summability: $|\psi_j| \rightarrow 0$, so $|\psi_j| \leq 1$ eventually, hence $\psi_j^2 \leq |\psi_j|$ eventually and $\sum_j \psi_j^2$ converges by comparison. A convergent series has vanishing tails (Review Question 6.1.1), so $\sum_{j=n+1}^m \psi_j^2 \rightarrow 0$ as $m, n \rightarrow \infty$. By fact (i), $\{y_{t,n}\}$ converges in mean square; the limit is y_t by definition. No property of t was used, so the limit exists for every t .

(b) $y_{t,n} \xrightarrow{m.s.} y_t$ by (a), so fact (ii) with $x_n = y_{t,n}$ gives

$$\mathbb{E}(y_t) = \lim_{n \rightarrow \infty} \mathbb{E}(y_{t,n}) = \lim_{n \rightarrow \infty} \left(\mu + \sum_{j=0}^n \psi_j \mathbb{E}(\varepsilon_{t-j}) \right) = \mu.$$

(c) Fix $j \geq 0$. Since $y_{t,n} \xrightarrow{m.s.} y_t$ and $y_{t-j,n} \xrightarrow{m.s.} y_{t-j}$, fact (ii) with $x_n = y_{t,n} - \mu$, $z_n = y_{t-j,n} - \mu$ gives

$$\gamma_j \equiv \mathbb{E}[(y_t - \mu)(y_{t-j} - \mu)] = \lim_{n \rightarrow \infty} \mathbb{E}\left[\sum_{k=0}^n \psi_k \varepsilon_{t-k} \sum_{\ell=0}^n \psi_{\ell} \varepsilon_{t-j-\ell}\right] = \sigma^2 \sum_{\ell=0}^{\infty} \psi_{\ell+j} \psi_{\ell},$$

where the second equality uses white noise (only terms with $t - k = t - j - \ell$, i.e. $k = \ell + j$, survive) and the last the convergence of the partial sums (the full double sum is dominated by $(\sum_k |\psi_k|)^2 < \infty$). This is (6.1.7). The result depends on t only through the lag j , and the mean is constant by (b); hence $\{y_t\}$ is covariance-stationary.

(d) The goal is $\sum_{j=0}^{\infty} |\gamma_j| < \infty$ (the full sum over $j \in \mathbb{Z}$ then follows by symmetry $\gamma_{-j} = \gamma_j$). From (c), for $j \geq 0$,

$$|\gamma_j| \leq \sigma^2 \sum_{k=0}^{\infty} |\psi_{j+k}| |\psi_k|,$$

and summing over j and interchanging the order of summation (legitimate for nonnegative summands — Tonelli's theorem, the content of the book's facts (i)–(ii) for this case),

$$\begin{aligned} \sum_{j=0}^{\infty} |\gamma_j| &\leq \sigma^2 \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} |\psi_{j+k} \psi_k| = \sigma^2 \sum_{k=0}^{\infty} |\psi_k| \sum_{j=0}^{\infty} |\psi_{j+k}| \\ &\leq \sigma^2 \left(\sum_{k=0}^{\infty} |\psi_k| \right) \left(\sum_{i=0}^{\infty} |\psi_i| \right) = \sigma^2 \left(\sum_{k=0}^{\infty} |\psi_k| \right)^2 < \infty, \end{aligned}$$

because $\sum_{j \geq 0} |\psi_{j+k}| \leq \sum_i |\psi_i|$ for every k . Hence $\{\gamma_j\}$ is absolutely summable: absolutely summable coefficients produce absolutely summable autocovariances.

Remark. The bound $\sum_j |\gamma_j| \leq \sigma^2 (\sum_j |\psi_j|)^2$ is the cleanest statement of the chapter's central theme: summability properties pass from the filter to the autocovariances. The long-run variance itself is the square *without* the absolute values, $\sigma^2 \psi(1)^2$ (Review Question 6.5.2) — which can be much smaller than the bound, and can vanish.

Analytical Exercise 6.3

(a) Write everything as deviations from means, $y_{t,n} - \mathbb{E}y_{t,n} = \sum_k h_k(x_{t-k} - \mu_x)$, and expand:

$$\text{Cov}(y_{t,n}, y_{t-j,n}) = \mathbb{E} \left[\sum_{k=0}^n h_k(x_{t-k} - \mu_x) \sum_{\ell=0}^n h_{\ell}(x_{t-j-\ell} - \mu_x) \right] = \sum_{k=0}^n \sum_{\ell=0}^n h_k h_{\ell} \gamma_{(t-k)-(t-j-\ell)}^x,$$

and the lag is $(t-j-\ell) - (t-k) = k - \ell - j$, so by the symmetry $\gamma_{-m}^x = \gamma_m^x$ the term indexed (k, ℓ) is $\gamma_{j+\ell-k}^x = \gamma_{|j+k-\ell|}$ with the roles of k and ℓ swapped — the displayed double sum.

For $n = 1$ this is Review Question 6.1.2. If $\{x_t\}$ is white noise with variance σ^2 , $\gamma_m^x = \sigma^2 \mathbf{1}\{m = 0\}$, so only pairs with $k = \ell + j$ survive:

$$\gamma_j^{(n)} = \sigma^2 \sum_{\substack{0 \leq \ell \leq n \\ 0 \leq \ell+j \leq n}} h_{\ell+j} h_{\ell} = \sigma^2 \sum_{\ell=0}^{n-j} h_{\ell+j} h_{\ell} \quad (0 \leq j \leq n),$$

which is (6.1.2) for the MA(n) process with $\theta_{\ell} = h_{\ell}$ — and for $j > n$ the sum is empty, $\gamma_j = 0$: an MA(n) has exactly n nonzero autocovariances.

(b) By PROPOSITION 6.2(A), $y_{t,n} \xrightarrow{m.s.} y_t$ and $y_{t-j,n} \xrightarrow{m.s.} y_{t-j}$. Fact (ii) of Analytical Exercise 6.2 then gives $\gamma_j^y = \lim_n \gamma_j^{(n)} = \lim_n \sum_{k, \ell \leq n} h_k h_{\ell} \gamma_{j+k-\ell}^x$. The double series converges absolutely, because

$$\sum_{k, \ell} |h_k h_{\ell}| |\gamma_{j+k-\ell}^x| \leq \gamma_0^x \left(\sum_k |h_k| \right)^2 < \infty$$

($|\gamma_m^x| \leq \gamma_0^x$ for every m), so the limit of the truncated sums equals the full double sum:

$$\gamma_j^y = \sum_{k=0}^{\infty} \sum_{\ell=0}^{\infty} h_k h_{\ell} \gamma_{j+k-\ell}^x.$$

This is the promised generalization of (6.1.7): filtering convolves the filter with itself and applies the result to the input's autocovariance sequence. In the frequency domain the same statement is the single line $s_Y(\omega) = |h(e^{-i\omega})|^2 s_X(\omega)$ of Review Question 6.2.6 — one reason the spectral approach is popular: double convolutions become multiplications.

Analytical Exercise 6.4

(a) Substitute $y_j = c_{10}\lambda_1^j + c_{20}\lambda_2^j$ into the right-hand side:

$$\phi_1 y_{j-1} + \phi_2 y_{j-2} = \sum_{k=1}^2 c_{k0} \lambda_k^{j-2} (\phi_1 \lambda_k + \phi_2).$$

Each λ_k solves $\lambda^2 = \phi_1 \lambda + \phi_2$ (the associated polynomial equation (6.1.18') for $p = 2$), so $\phi_1 \lambda_k + \phi_2 = \lambda_k^2$ and the sum becomes $\sum_k c_{k0} \lambda_k^j = y_j$: the display solves the difference equation for any (c_{10}, c_{20}) . The initial conditions pin down the constants: at $j = 0$ and $j = 1$,

$$c_{10} + c_{20} = y_0, \quad c_{10}\lambda_1 + c_{20}\lambda_2 = y_1,$$

a 2×2 linear system with determinant $\lambda_2 - \lambda_1 \neq 0$ (distinct roots), solved by

$$c_{10} = \frac{y_1 - \lambda_2 y_0}{\lambda_1 - \lambda_2}, \quad c_{20} = \frac{\lambda_1 y_0 - y_1}{\lambda_1 - \lambda_2}.$$

(b) Here $\phi_1 = 4/3$, $\phi_2 = -4/9$, and the single root $\lambda = 2/3$ satisfies both $\lambda^2 = \phi_1 \lambda + \phi_2$ and $2\lambda = \phi_1$ (a double root also solves the derivative of the polynomial: $\lambda^2 - \phi_1 \lambda - \phi_2 = (\lambda - 2/3)^2$). For $y_j = (c_{10} + c_{11}j)\lambda^j$,

$$\phi_1 y_{j-1} + \phi_2 y_{j-2} = \lambda^{j-2} [\phi_1 \lambda (c_{10} + c_{11}(j-1)) + \phi_2 (c_{10} + c_{11}(j-2))].$$

The coefficient of c_{10} is $\phi_1 \lambda + \phi_2 = \lambda^2$; the coefficient of c_{11} is

$$\phi_1 \lambda (j-1) + \phi_2 (j-2) = j(\phi_1 \lambda + \phi_2) - (\phi_1 \lambda + 2\phi_2) = j\lambda^2 - (\phi_1 \lambda + 2\phi_2),$$

and $\phi_1 \lambda + 2\phi_2 = \frac{4}{3} \cdot \frac{2}{3} - 2 \cdot \frac{4}{9} = 0$. So the bracket equals $\lambda^2(c_{10} + c_{11}j)$ and the whole expression is $(c_{10} + c_{11}j)\lambda^j = y_j$, as required. The extra solution $j\lambda^j$ is the hallmark of a repeated root.

(c) Pick any $b \in (\delta, 1)$ and set $u_j \equiv j^m(\delta/b)^j$. With $r \equiv \delta/b < 1$, $u_{j+1}/u_j = (1 + 1/j)^m r \rightarrow r < 1$ as $j \rightarrow \infty$, so by the ratio test $\sum_j u_j$ converges, and a fortiori $u_j \rightarrow 0$. A convergent sequence is bounded: there is $A > 0$ with $u_j \leq A$ for all j , that is, $j^m \delta^j \leq A b^j$. (The point: exponentials dominate powers, so any geometric decay slightly slower than δ^j still swallows the polynomial factor j^m .)

(d) By the representation (5) in the book, $y_j = \sum_{k=1}^K \sum_{m=0}^{r_k-1} c_{km} j^m \lambda_k^j$. Hence, with $c \equiv \max_{k,m} |c_{km}|$,

$$|y_j| \leq c \sum_{k=1}^K \sum_{m=0}^{r_k-1} j^m |\lambda_k|^j.$$

The stability condition gives $|\lambda_k| < 1$ for every k . Apply part (c) with $\delta = |\lambda_k|$ and exponent m : there exist $A_{km} > 0$ and $b_{km} \in (|\lambda_k|, 1)$ with $j^m |\lambda_k|^j \leq A_{km} b_{km}^j$. Set $A^* \equiv \max_{k,m} A_{km}$ and $b \equiv \max_{k,m} b_{km} < 1$; then $j^m |\lambda_k|^j \leq A^* b^j$ for every (k, m) , and summing the p terms (recall

$$\sum_k r_k = p),$$

$$|y_j| \leq c p A^* b^j \equiv A b^j, \quad A \equiv c p A^*, \quad 0 < b < 1.$$

So every solution of a stable p -th order homogeneous difference equation is bounded by a geometrically declining sequence — the result used in the text ((6.1.19)) to prove that inverses of stable lag polynomials have absolutely summable coefficients, since $\sum_j A b^j = A/(1-b) < \infty$.

Analytical Exercise 6.5

(a) For $j \geq 1$, multiply $y_t - \mu = \phi(y_{t-1} - \mu) + \varepsilon_t$ by $y_{t-j} - \mu$ and take expectations:

$$\gamma_j = \phi \gamma_{j-1} + E[(y_{t-j} - \mu)\varepsilon_t].$$

(b) By the MA(∞) representation (6.2.2), $y_{t-1} - \mu = \sum_{k=0}^{\infty} \phi^k \varepsilon_{t-1-k}$ is a function of $(\varepsilon_{t-1}, \varepsilon_{t-2}, \dots)$, while white noise gives $E(\varepsilon_{t-1-k}\varepsilon_t) = 0$ for every $k \geq 0$. Hence $E[(y_{t-1} - \mu)\varepsilon_t] = 0$; and since $E(\varepsilon_t) = 0$, $E(y_{t-1}\varepsilon_t) = E[(y_{t-1} - \mu)\varepsilon_t] + \mu E(\varepsilon_t) = 0$. The same argument with $t-j$ in place of $t-1$ gives $E[(y_{t-j} - \mu)\varepsilon_t] = 0$ for every $j \geq 1$.

(c) Part (b) kills the second term in (a), giving the recursion. For $j = 0$, multiply the equation by $y_t - \mu$ itself:

$$\gamma_0 = E[(y_t - \mu)^2] = \phi E[(y_t - \mu)(y_{t-1} - \mu)] + E[(y_t - \mu)\varepsilon_t].$$

The first term is $\phi \gamma_1$ (by definition of γ_1); for the second, substitute the equation once more and use (b): $E[(y_t - \mu)\varepsilon_t] = \phi E[(y_{t-1} - \mu)\varepsilon_t] + E(\varepsilon_t^2) = \sigma^2$. Together:

$$\gamma_0 = \phi \gamma_1 + \sigma^2, \quad \gamma_j = \phi \gamma_{j-1} \quad (j \geq 1)$$

— the Yule–Walker equations for AR(1).

(d) From the $j = 1$ equation, $\gamma_1 = \phi \gamma_0$; substituting into the first equation, $\gamma_0 = \phi^2 \gamma_0 + \sigma^2$, so

$$\gamma_0 = \frac{\sigma^2}{1 - \phi^2}, \quad \gamma_j = \phi^j \gamma_0 = \frac{\phi^j \sigma^2}{1 - \phi^2} \quad (j \geq 0),$$

matching (6.2.5) from the MA route. The two methods are complementary: the MA representation makes the answers immediate once $\{\psi_j\}$ is known, while Yule–Walker obtains the autocovariances directly from the difference equation — and generalizes to AR(p) as a p -term recursion in γ_j , solvable by exactly the apparatus of Analytical Exercise 6.4.

Analytical Exercise 6.6

(a) Iterate the deviation form $x_t = \phi x_{t-1} + \varepsilon_t$:

$$x_t = \phi(\phi x_{t-2} + \varepsilon_{t-1}) + \varepsilon_t = \phi^2 x_{t-2} + \phi \varepsilon_{t-1} + \varepsilon_t = \dots = \sum_{j=0}^n \phi^j \varepsilon_{t-j} + \phi^{n+1} x_{t-n-1},$$

after $n+1$ substitutions. So $x_t - x_{t,n} = \phi^{n+1} x_{t-n-1}$ with $x_{t,n} \equiv \sum_{j=0}^n \phi^j \varepsilon_{t-j}$.

(b) Compute the mean square distance:

$$E[(x_{t,n} - x_t)^2] = E[\phi^{2(n+1)} x_{t-n-1}^2] = \phi^{2(n+1)} \gamma_0,$$

because covariance stationarity makes $E(x_{t-n-1}^2) = \text{Var}(y_{t-n-1}) = \gamma_0$, a constant not depending on n . Since $|\phi| < 1$, $\phi^{2(n+1)} \gamma_0 \rightarrow 0$: $x_{t,n} \xrightarrow{m.s.} x_t$.

(c) By Analytical Exercise 6.2(a) (applied with $\psi_j = \phi^j$, absolutely summable since $|\phi| < 1$), the partial sums $x_{t,n}$ converge in mean square to the infinite sum $\sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}$ — that infinite sum is *defined* as the mean square limit. By (b) the same sequence converges in mean square to x_t . Mean square limits are unique up to a zero-probability event (Analytical Exercise 6.1), so

$$x_t = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j} \quad \text{a.s.,} \quad \text{i.e.} \quad y_t = \mu + \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j},$$

which is (6.2.2). The point of doing this without filters: the only inputs are the geometric decay of the remainder $\phi^{n+1}x_{t-n-1}$ and the boundedness of the variance — covariance stationarity of the solution plus $|\phi| < 1$ *forces* the MA(∞) representation. Analytical Exercise 6.7 repeats the argument for AR(p) with the companion matrix $\mathbf{F}^n$ playing the role of ϕ^n .

Analytical Exercise 6.7

(a) The first row of $\boldsymbol{\xi}_t = \mathbf{F}\boldsymbol{\xi}_{t-1} + \mathbf{v}_t$ is the AR(p) equation (6.2.6'); the remaining rows are the identities $y_{t-j} = y_{t-j}$. Iterating exactly as in Analytical Exercise 6.6(a),

$$\boldsymbol{\xi}_t = \mathbf{F}\boldsymbol{\xi}_{t-1} + \mathbf{v}_t = \mathbf{F}^2\boldsymbol{\xi}_{t-2} + \mathbf{F}\mathbf{v}_{t-1} + \mathbf{v}_t = \cdots = \sum_{j=0}^n \mathbf{F}^j \mathbf{v}_{t-j} + \mathbf{F}^{n+1}\boldsymbol{\xi}_{t-n-1}.$$

(b) For $p = 2$,

$$\lambda \mathbf{I} - \mathbf{F} = \begin{bmatrix} \lambda - \phi_1 & -\phi_2 \\ -1 & \lambda \end{bmatrix}, \quad |\lambda \mathbf{I} - \mathbf{F}| = (\lambda - \phi_1)\lambda - \phi_2 = \lambda^2 - \phi_1\lambda - \phi_2,$$

the associated polynomial equation (6.1.18') for $p = 2$: the eigenvalues of the companion matrix are the reciprocals of the roots of $\phi(z) = 0$ (Review Question 6.1.5).

(c) $\mathbf{F}^3 = \mathbf{F}\mathbf{F}\mathbf{F} = (\mathbf{T}\boldsymbol{\Lambda}\mathbf{T}^{-1})(\mathbf{T}\boldsymbol{\Lambda}\mathbf{T}^{-1})(\mathbf{T}\boldsymbol{\Lambda}\mathbf{T}^{-1}) = \mathbf{T}\boldsymbol{\Lambda}(\mathbf{T}^{-1}\mathbf{T})\boldsymbol{\Lambda}(\mathbf{T}^{-1}\mathbf{T})\boldsymbol{\Lambda}\mathbf{T}^{-1} = \mathbf{T}\boldsymbol{\Lambda}^3\mathbf{T}^{-1}$. The same computation for general n is the induction $\mathbf{F}^{n+1} = \mathbf{F}^n\mathbf{F} = \mathbf{T}\boldsymbol{\Lambda}^n\mathbf{T}^{-1}\mathbf{T}\boldsymbol{\Lambda}\mathbf{T}^{-1} = \mathbf{T}\boldsymbol{\Lambda}^{n+1}\mathbf{T}^{-1}$.

(d) From (a), $\boldsymbol{\xi}_t - \boldsymbol{\xi}_{t,n} = \mathbf{F}^{n+1}\boldsymbol{\xi}_{t-n-1}$. Mean square convergence is elementwise, so it suffices that every element of $\mathbf{F}^{n+1}\boldsymbol{\xi}_{t-n-1}$ converge in mean square to zero. By (c), each element of $\mathbf{F}^n$ is a linear combination (with fixed coefficients from $\mathbf{T}$ and $\mathbf{T}^{-1}$) of $\lambda_1^n, \dots, \lambda_p^n$; the stationarity condition says $|\lambda_k| < 1$ for all k , so by Analytical Exercise 6.4(d) each element of $\mathbf{F}^n$ is bounded in absolute value by Ab^n for some $A > 0$ and $0 < b < 1$. Covariance stationarity gives $E(\xi_{t-n-1,\ell}^2) = \gamma_0 < \infty$ for every component, so the k -th element satisfies, by Cauchy–Schwarz,

$$E\left[\left(\sum_{\ell=1}^p [\mathbf{F}^{n+1}]_{k\ell} \xi_{t-n-1,\ell}\right)^2\right] \leq p \gamma_0 \sum_{\ell=1}^p [\mathbf{F}^{n+1}]_{k\ell}^2 \leq p^2 \gamma_0 A^2 b^{2(n+1)} \longrightarrow 0.$$

Hence $\boldsymbol{\xi}_{t,n} \xrightarrow{m.s.} \boldsymbol{\xi}_t$.

(e) The partial sums $\boldsymbol{\xi}_{t,n}$ converge in mean square both to $\boldsymbol{\xi}_t$ (by (d)) and to $\sum_{j=0}^{\infty} \mathbf{F}^j \mathbf{v}_{t-j}$ (the definition of that infinite sum, which exists elementwise by the same geometric bound and Analytical Exercise 6.2(a) applied componentwise). Uniqueness of mean square limits (Analytical Exercise 6.1)

gives

$$\boldsymbol{\xi}_t = \sum_{j=0}^{\infty} \mathbf{F}^j \mathbf{v}_{t-j} \quad \text{a.s.}$$

The first row of this vector equation is the MA(∞) representation of y_t : since only the first component of $\mathbf{v}_{t-j}$ is nonzero,

$$y_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \quad \psi_j \equiv \text{the } (1, 1) \text{ element of } \mathbf{F}^j.$$

By (c), ψ_j is the $(1, 1)$ element of $\mathbf{T}\Lambda^j\mathbf{T}^{-1}$, a linear combination of the j -th powers of the characteristic roots — the book's answer — which is exactly the solution form (7) of Analytical Exercise 6.4 with coefficients determined by the initial conditions $\psi_0 = 1, \psi_1 = \phi_1, \dots$. So the companion form turns the scalar p -th order difference equation into a first-order vector one, and stability becomes the single statement that $\mathbf{F}^n$ decays geometrically. (With multiple eigenvalues the Jordan decomposition replaces (15), adding terms like $j\lambda^j$ — bounded geometrically all the same by Analytical Exercise 6.4(c).)

Analytical Exercise 6.8

(a) Successive substitution from the initial condition gives, for $t \geq 0$,

$$y_t = \frac{1 - \phi^t}{1 - \phi} c + \phi^t y_0 + \sum_{j=0}^{t-1} \phi^j \varepsilon_{t-j} = (1 - \phi^t) \mu + \phi^t y_0 + \sum_{j=0}^{t-1} \phi^j \varepsilon_{t-j},$$

using $c = (1 - \phi)\mu$. Hence

$$\mathbf{E}(y_t) = (1 - \phi^t) \mu + \phi^t \mathbf{E}(y_0) = \mu + \phi^t [\mathbf{E}(y_0) - \mu] \xrightarrow[t \rightarrow \infty]{} \mu,$$

since $|\phi| < 1$. For the variance, $\text{Cov}(\varepsilon_t, y_0) = 0$ and the lack of serial correlation of the ε 's give

$$\text{Var}(y_t) = \phi^{2t} \text{Var}(y_0) + \sigma^2 \sum_{j=0}^{t-1} \phi^{2j} = \phi^{2t} \text{Var}(y_0) + \frac{1 - \phi^{2t}}{1 - \phi^2} \sigma^2 = \gamma_0 + \phi^{2t} [\text{Var}(y_0) - \gamma_0],$$

the last equality using $\gamma_0 = \sigma^2/(1 - \phi^2)$. So $\text{Var}(y_t) \rightarrow \gamma_0$, at the geometric rate ϕ^{2t} . Similarly, for $t \geq j \geq 0$, writing both y_t and y_{t-j} in terms of y_0 and the innovations and collecting the covariances (only the shared innovations and y_0 contribute),

$$\text{Cov}(y_t, y_{t-j}) = \phi^{2t-j} \text{Var}(y_0) + \sigma^2 \phi^j \sum_{k=0}^{t-j-1} \phi^{2k} = \phi^{2t-j} \text{Var}(y_0) + \phi^j \frac{1 - \phi^{2(t-j)}}{1 - \phi^2} \sigma^2 \rightarrow \phi^j \gamma_0 = \gamma_j.$$

(b) If $\mathbf{E}(y_0) = \mu$ and $\text{Var}(y_0) = \gamma_0$, the displays in (a) read

$$\mathbf{E}(y_t) = \mu + \phi^t \cdot 0 = \mu, \quad \text{Var}(y_t) = \gamma_0 + \phi^{2t} \cdot 0 = \gamma_0$$

for every t , and $\text{Cov}(y_t, y_{t-j}) = \phi^{2t-j} \gamma_0 + \phi^j (1 - \phi^{2(t-j)}) \gamma_0 = \phi^j \gamma_0 = \gamma_j$ for all $t \geq j$: mean and autocovariances are time-invariant, so $\{y_t\}$ is covariance-stationary. The stationary AR(1) of the text is exactly this process with the “right” initial distribution: starting in the steady-state distribution makes the process stationary from $t = 0$ onward; any other start is only asymptotically

stationary, the transient dying out like ϕ^t (mean) and ϕ^{2t} (variance). This is the precise content of the text's intuition that a process defined for $t = 0, -1, -2, \dots$ has "started so long ago that its moments have stabilized."

Analytical Exercise 6.9

(a) (Cesàro's lemma.) Since $a_j \rightarrow 0$: (i) $\{|a_j|\}$ is bounded, say $|a_j| \leq M$ for all j ; and (ii) given $\varepsilon > 0$ there is N with $|a_j| < \varepsilon/2$ for all $j > N$. For $n > N$, split the average at N :

$$\left| \frac{1}{n} \sum_{j=1}^n a_j \right| \leq \frac{1}{n} \sum_{j=1}^N |a_j| + \frac{1}{n} \sum_{j=N+1}^n |a_j| \leq \frac{NM}{n} + \frac{n-N}{n} \cdot \frac{\varepsilon}{2} < \frac{NM}{n} + \frac{\varepsilon}{2}.$$

Choose n so large that $NM/n < \varepsilon/2$; then the average is below ε in absolute value. Since ε was arbitrary, $\frac{1}{n} \sum_j a_j \rightarrow 0$: convergence implies convergence of the running averages. (The converse is false — averages wash out nonconvergent oscillation, e.g. $a_j = (-1)^j$.)

(b) From (6.5.2),

$$\text{Var}(\bar{y}_n) = \frac{\gamma_0}{n} + \frac{2}{n} \sum_{j=1}^{n-1} \left(1 - \frac{j}{n}\right) \gamma_j, \quad \text{so} \quad |\text{Var}(\bar{y}_n)| \leq \frac{\gamma_0}{n} + \frac{2}{n} \sum_{j=1}^{n-1} |\gamma_j|,$$

because $0 \leq 1 - j/n \leq 1$. The first term vanishes as $n \rightarrow \infty$, and the second vanishes by part (a) applied to $a_j = 2|\gamma_j| \rightarrow 0$ (the upper limit $n-1$ rather than n is immaterial). Hence $\text{Var}(\bar{y}_n) \rightarrow 0$, and Chebychev's LLN (Analytical Exercise 2.2) gives $\bar{y}_n \xrightarrow{m.s.} \mu$, hence $\bar{y}_n \xrightarrow{p} \mu$ — PROPOSITION 6.8(A). The condition $\gamma_j \rightarrow 0$ is remarkably weak (no rate is required); it is enough that remote observations be asymptotically uncorrelated for the sample mean to settle down.

Analytical Exercise 6.10

(a) The hint's identity writes the weighted sum as a sum of tail sums: grouping each a_k with the j 's not exceeding it,

$$\sum_{j=1}^n j a_j = \sum_{j=1}^n \sum_{k=j}^n a_k,$$

so with $T_{j,n} \equiv \sum_{k=j}^n a_k$,

$$\frac{1}{n} \sum_{j=1}^n j a_j = \frac{1}{n} \sum_{j=1}^n T_{j,n}.$$

Because $\{a_j\}$ is summable: (i) the tail sums are bounded, $|T_{j,n}| \leq \sum_k |a_k| \equiv M < \infty$ for all j, n ; and (ii) they vanish uniformly far out: given $\varepsilon > 0$ there is N with $|T_{j,n}| \leq \sum_{k \geq N} |a_k| < \varepsilon/2$ for all $j, n \geq N$ (the partial sums form a Cauchy sequence). Splitting the average at N as in Analytical Exercise 6.9(a),

$$\left| \frac{1}{n} \sum_{j=1}^n T_{j,n} \right| \leq \frac{NM}{n} + \frac{n-N}{n} \cdot \frac{\varepsilon}{2} < \varepsilon \quad \text{for large } n,$$

so $\frac{1}{n} \sum_j j a_j \rightarrow 0$. (This is a case of Kronecker's lemma, the workhorse behind strong laws for weighted sums.)

(b) From (6.5.2),

$$\text{Var}(\sqrt{n} \bar{y}_n) = \gamma_0 + 2 \sum_{j=1}^{n-1} \left(1 - \frac{j}{n}\right) \gamma_j = \gamma_0 + 2 \sum_{j=1}^{n-1} \gamma_j - \frac{2}{n} \sum_{j=1}^{n-1} j \gamma_j.$$

The first two terms converge to $\gamma_0 + 2 \sum_{j=1}^{\infty} \gamma_j = \sum_{j=-\infty}^{\infty} \gamma_j$ because $\{\gamma_j\}$ is summable ($\gamma_{-j} = \gamma_j$); the last term vanishes by part (a) with $a_j = \gamma_j$. Hence

$$\lim_{n \rightarrow \infty} \text{Var}(\sqrt{n} \bar{y}_n) = \sum_{j=-\infty}^{\infty} \gamma_j,$$

which is PROPOSITION 6.8(B): the long-run variance is the sum of all autocovariances, equivalently $g_Y(1)$ or $2\pi s_Y(0)$. Combined with LEMMA 2.1 as invoked in the text after PROPOSITION 6.9, it explains why every CLT for stationary serially correlated processes delivers the same asymptotic variance formula — the limit of the variances is the variance of the limit whenever both exist.

Empirical Exercises

Empirical Exercise 6.1

All computations are in `code/ch6.py`; we report the results for the yen in detail and give the other currencies in summary. As in the book, rates of change and forward premiums are multiplied by 1200 to express them in annualized percentage points.

(a) *The week of the largest forward premium.* The largest $|f_t - s_t| \times 1200$ occurs in the week of December 12, 1980 for the yen (15.2 annualized percent) and the DM (14.4), and on October 15, 1976 for the pound (16.9). December 1980 was the height of the Volcker disinflation: one-month US interest rates were around 16–18% while Japanese and German short rates were near 7–9%, and covered interest parity forces the dollar's forward premium (foreign currency per dollar at a forward discount) to equal that differential — so a forward premium of about 15% annualized is exactly what the interest differential implies, a direct verification of (6.8.2).

(b) *Correlogram of $\{\varepsilon_t\}$* (40 lags, sample mean subtracted, following the book's Figure 6.2). For the yen:

j	1	2	3	4	5	6	7	8
$\hat{\rho}_j$	0.847	0.667	0.470	0.266	0.135	0.076	0.038	0.020

with $2/\sqrt{n} = 0.072$. The autocorrelations decay through lag 4 and drop to (or inside) the two-standard-error band from lag 5 on, broadly consistent with the theory's implication $\text{Cov}(\varepsilon_t, \varepsilon_{t-j}) = 0$ for $j \geq 5$. (For the pound the first few autocorrelations beyond lag 4 — 0.14, 0.12, 0.11 at lags 5–7 — stick out of the band a little more than the yen's; the pattern is still dominated by the first four lags.) The figures `ch6_correlogram_eps_*.pdf` reproduce Figure 6.2 for all three currencies.

(c) *Is the log spot rate a random walk?* The correlogram of $s_{t+1} - s_t$ for the yen is small at every lag (lags 1–6: 0.097, 0.140, 0.019, 0.050, 0.027, 0.013), and

$$Q_{\text{LB}}(40) = 84.0, \quad p\text{-value} = 0.0058\%,$$

replicating the book's answer: the random-walk-with-drift hypothesis is rejected for the yen at conventional levels, the rejections coming from a sprinkling of small positive autocorrelations rather than from any large one. (The DM gives $Q = 47.5$, $p = 0.19$, and the pound $Q = 36.1$, $p = 0.64$: no rejection there.) The figures `ch6_correlogram.ds_*.pdf` show the correlograms.

(d) *Unconditional test (Table 6.1)*. With ε_t demeaned in the autocovariances and $\hat{S} = \hat{\gamma}_0 + 2 \sum_{j=1}^4 \hat{\gamma}_j$, the standard error of $\bar{\varepsilon}$ is $\sqrt{\hat{S}/n}$. The results replicate Table 6.1 to the printed digits:

currency	$s30 - s$		$f - s$		difference		[se]
	mean	(s.d.)	mean	(s.d.)	mean	(s.d.)	
yen/\$	-4.98	(41.6)	-3.73	(3.65)	-1.25	(42.4)	[3.56]
DM/\$	-1.78	(40.6)	-3.91	(2.17)	2.13	(41.0)	[3.26]
pound/\$	3.59	(39.2)	2.16	(3.49)	1.43	(39.9)	[3.29]

The implied t -ratios for a zero mean are -0.35 , 0.65 and 0.43 : in no case is there evidence against $E(\varepsilon_t) = 0$. Note how much larger the standard error of the mean is than the naive $\hat{\gamma}_0^{1/2}/\sqrt{n}$ would suggest: for the yen, $\hat{\gamma}_0^{1/2}/\sqrt{n} \times 1200 \approx 1.5$ against the HAC value 3.56 — ignoring the overlap-induced serial correlation would understate the sampling uncertainty by a factor of two and a half.

(e) *Regression test, truncated kernel, $q = 4$ (Table 6.2)*. OLS of $(s30_t - s_t) \times 1200$ on a constant and $(f_t - s_t) \times 1200$ with $\widehat{\text{Avar}}(\mathbf{b}) = \mathbf{S}_{\mathbf{xx}}^{-1} \widehat{\mathbf{S}} \mathbf{S}_{\mathbf{xx}}^{-1}$, $\widehat{\mathbf{S}}$ from (6.6.4) with $q = 4$:

currency	constant	forward premium	R^2
yen	-12.82 (4.01)	-2.10 (0.74)	0.034
DM	-13.58 (5.72)	-3.01 (1.37)	0.026
pound	7.96 (3.54)	-2.02 (0.85)	0.033

The Wald statistic for $\beta_0 = 0, \beta_1 = 1$ is 18.6 for the yen ($p = 0.009\%$), 8.7 for the DM ($p = 1.3\%$) and 12.9 for the pound ($p = 0.16\%$): the market efficiency hypothesis is rejected decisively in all three cases. (The small discrepancies from the book's Table 6.2 reflect, as the table's note warns, the sensitivity of the numbers to the exact kernel and sample conventions.)

(f) *Bartlett kernel with the Newey–West bandwidths* ($q(n) = 12$ yen, 8 DM, 16 pound, $k(x) = 1 - |x|$ with $x = j/(q(n) + 1)$): the standard errors of the slope are 0.6815 (yen), 1.2126 (DM) and 0.8681 (pound), and the Wald statistics are 21.85, 11.05 and 12.70 ($p < 0.4\%$ throughout). The yen slope standard error reproduces the book's stated 0.6815 exactly, and the yen Wald statistic reproduces the book's answer of 21.85 ($p = 0.002\%$).

(g) *VARHAC*. With $p_{\max} = \lceil n^{1/3} \rceil = 9$ and per-equation BIC lag selection on the fixed sample $t = p_{\max} + 1, \dots, n$ using the book's objective $\log(\text{SSR}/(n - p_{\max})) + pK \log(n - p_{\max})/(n - p_{\max})$: the selected lag lengths for the yen's bivariate VAR on $\mathbf{g}_t = \mathbf{x}_t \hat{\varepsilon}_t$ are $p = (4, 6)$, exactly as the book states; the implied long-run variance (6.6.15) gives slope standard error 0.8074 (book: 0.8027) and Wald statistic 16.7. For the DM, $p = (6, 6)$ and slope SE 1.34; for the pound, $p = (6, 1)$ and slope SE 1.04. All three estimators of $\mathbf{S}$ tell the same story — the slope is about -2 with a standard

error well below 1, so the forward premium predicts the wrong sign of the exchange-rate change — but they disagree enough among themselves (slope SEs 0.68–0.81 for the yen) to make the point of Den Haan and Levin’s comparison: inference about long-run variances is itself noisy.

Empirical Exercise 6.2

(a) Because the bill’s maturity is three months: the return on the bill bought at the beginning of month t is realized over the quarter *ending* in month $t + 2$, whose inflation rate is, in the timing conventions of MISHKIN.ASC, the observation labeled π_{t+3} . Regressing π_t on R_t would match the January rate with inflation over the previous quarter, a return the bill bought in January does not earn. (The timing is the same one used in Empirical Exercise 2.1 for the one-month bill, shifted by two months.)

(b) Under the efficient market hypothesis the bill rate equals the market’s expectation of inflation over its life plus a constant (the ex-ante real rate): $R_t = r + E(\pi_{t+3} | I_t)$. Then

$$\varepsilon_{t+3} = \pi_{t+3} - \beta_0 - \beta_1 R_t = [E(\pi_{t+3} | I_t) + \eta_{t+3}] - \beta_0 - \beta_1(r + E(\pi_{t+3} | I_t)),$$

with η_{t+3} the inflation forecast error, $E(\eta_{t+3} | I_t) = 0$. For the equation to hold identically with the forecast error as its residual, $\beta_1 = 1$ (and $\beta_0 = -r$), so $\varepsilon_{t+3} = \eta_{t+3}$. Since inflation over the quarter ending in $t + 2$ is not known until then, η_{t+3} is unforecastable from I_{t-j} only when I_{t-j} ’s coverage ends before the forecast’s resolution: the error shares realized inflation with η_{t+3-j} for $j = 1, 2$ (overlapping quarters) but not for $j \geq 3$. Formally $E(\eta_{t+3} | I_t) = 0$ and $I_t \supset \{\eta_t, \eta_{t-1}, \dots\}$ imply, by iterated expectations, $E(\eta_{t+3} | \eta_t, \eta_{t-1}, \dots) = 0$, so $\gamma_j = E(\varepsilon_{t+3}\varepsilon_{t+3-j}) = 0$ for $j \geq 3$: serial correlation vanishes after the second lag. (This is the three-month analogue of the four-week overlap in the exchange-rate application.)

(c) The ex-post real rate is $r_{t+3} = R_t - \pi_{t+3}$ (TB3 minus PAI3 for the same month; $n = 223$ for 1/53–7/71). Its correlogram (`ch6_correlogram_realrate.pdf`):

j	1	2	3	4	5	6	7	8	9	10	11	12
$\hat{\rho}_j$	0.70	0.41	0.08	0.03	0.00	0.02	0.01	0.07	0.10	0.15	0.13	0.16

Theory predicts $\rho_j = 0$ for $j \geq 3$ under market efficiency. Lags 1 and 2 are large (the mechanical overlap), lags 3–8 are small, but lags 9–12 sit near or above $2/\sqrt{223} = 0.13$; the Ljung–Box statistics are $Q(2) = 149.3$ and $Q(12) = 170.1$ ($p \approx 0$). So the data are broadly consistent with the theory’s shape (the action is at lags 1–2) but the small autocorrelations at seasonal lags are more than the hypothesis allows — a first sign of rejection, confirmed by the regression test.

(d) OLS of PAI3 on a constant and TB3, 1/53–7/71 ($n = 223$), with $\hat{\mathbf{S}}$ from the truncated kernel at $q = 2$:

$$\pi_{t+3} = \underset{(0.351)}{-0.480} + \underset{(0.079)}{0.741} R_t, \quad R^2 = 0.42,$$

HAC standard errors in parentheses. The t -ratio for $\beta_1 = 1$ is -3.29 , and the Wald statistic for the joint hypothesis $\beta_0 = 0, \beta_1 = 1$ is 95.7 on $\chi^2(2)$ ($p \approx 10^{-21}$): market efficiency is rejected overwhelmingly. (The Bartlett variant with $q = 3$ gives SEs 0.311 and 0.070, $t = -3.70$ — the same conclusion.) The contrast with Empirical Exercise 2.1 is the point of the exercise: the one-month Fama regression on the same sample did *not* reject, because the one-month bill’s error is

a one-month forecast error with no overlap; at the three-month horizon the overlap forces $q = 2$ serial correlation into the standard errors, and the slope of 0.74 is far enough below one that even the much larger HAC standard errors cannot save the hypothesis. The failure is economically interpretable: the nominal bill rate moved far less than one-for-one with expected inflation over this sample, so the ex-ante real rate was not constant.

Chapter 7

Extremum Estimators

Solutions to all the exercises of Chapter 7 of Fumio Hayashi, *Econometrics* (Princeton University Press, 2000): the twenty-eight review questions of Sections 7.1–7.5 (five on the definition of extremum estimators, eight on consistency, six on asymptotic normality, seven on hypothesis testing, two on numerical optimization) and the three analytical exercises (the Kullback–Leibler information inequality, the information matrix equality, and the Wald–LM–LR trinity for the classical linear regression model). This chapter has no empirical or Monte Carlo exercise, so there is no `code/ch7.py`. Assumptions, propositions and equations cited as PROPOSITION 7.9, (7.3.43) or TABLE 7.2 refer to the book. Following the book’s notation for this chapter, θ_0 denotes the true parameter value, θ a hypothetical value, $\hat{\theta}$ the unconstrained extremum estimator, $\tilde{\theta}$ the constrained one, and t the observation index.

Section 7.1 Extremum Estimators

Review Question 7.1.1

Throughout, Φ is the standard normal c.d.f. and $\phi = \Phi'$ its density; $\varphi(\cdot; \cdot)$ is reserved for the conditional-mean function of (7.1.18).

The conditional mean. y_t takes only the values 1 and 0, so its conditional mean is the conditional probability of a one. By (7.1.14),

$$E(y_t \mid \mathbf{x}_t) = 1 \cdot f(y_t = 1 \mid \mathbf{x}_t; \theta_0) + 0 \cdot f(y_t = 0 \mid \mathbf{x}_t; \theta_0) = \Phi(\mathbf{x}_t' \theta_0).$$

The m function. Therefore the probit model is a special case of the NLS model (7.1.18),

$$y_t = \varphi(\mathbf{x}_t; \theta_0) + \varepsilon_t, \quad \varphi(\mathbf{x}_t; \theta) \equiv \Phi(\mathbf{x}_t' \theta), \quad \varepsilon_t \equiv y_t - \Phi(\mathbf{x}_t' \theta_0),$$

with $E(\varepsilon_t \mid \mathbf{x}_t) = 0$ by the calculation just made and the conditional mean correctly specified (the parametric family $\{\Phi(\mathbf{x}'\theta) : \theta \in \mathbb{R}^p\}$ contains the truth at $\theta = \theta_0$). Least squares applied to it is the M-estimator (7.1.19) whose m function is

$$m(\mathbf{w}_t; \theta) = -[y_t - \Phi(\mathbf{x}_t' \theta)]^2, \quad \mathbf{w}_t = (y_t, \mathbf{x}_t')',$$

so that the objective function (7.1.2) is $Q_n(\theta) = -\frac{1}{n} \sum_{t=1}^n [y_t - \Phi(\mathbf{x}_t' \theta)]^2$: maximizing it is minimizing the sum of squared residuals. The first-order condition is

$$\frac{1}{n} \sum_{t=1}^n \phi(\mathbf{x}_t' \theta) [y_t - \Phi(\mathbf{x}_t' \theta)] \mathbf{x}_t = \mathbf{0}.$$

Two comments. First, this NLS estimator is consistent (the conditional mean is correctly specified) but inefficient, because a binary y_t is intrinsically heteroskedastic: $\text{Var}(y_t \mid \mathbf{x}_t) = \Phi(\mathbf{x}_t' \theta_0)[1 - \Phi(\mathbf{x}_t' \theta_0)]$, the variance of a Bernoulli variable. Second, compare the first-order condition above with

the probit ML score obtained from (7.1.15),

$$\frac{1}{n} \sum_{t=1}^n \frac{\phi(\mathbf{x}_t' \theta)}{\Phi(\mathbf{x}_t' \theta)[1 - \Phi(\mathbf{x}_t' \theta)]} [y_t - \Phi(\mathbf{x}_t' \theta)] \mathbf{x}_t = \mathbf{0} :$$

the two use the *same* residual $y_t - \Phi(\mathbf{x}_t' \theta)$ and differ only in the weight attached to it. ML divides by the conditional variance, which is exactly the GLS weighting, and that is why ML is the efficient member of the family. Weighted NLS with weights $[\Phi(1 - \Phi)]^{-1/2}$ evaluated at any consistent first-round estimate is asymptotically equivalent to ML.

Review Question 7.1.2

The orthogonality conditions. By the previous question $E(y_t - \Phi(\mathbf{x}_t' \theta_0) \mid \mathbf{x}_t) = 0$. Since $\mathbf{x}_t$ is a function of the conditioning variables it can be taken outside the conditional expectation, so by the Law of Total Expectations

$$E[\mathbf{x}_t \cdot (y_t - \Phi(\mathbf{x}_t' \theta_0))] = E[\mathbf{x}_t E(y_t - \Phi(\mathbf{x}_t' \theta_0) \mid \mathbf{x}_t)] = E[\mathbf{x}_t \cdot 0] = \mathbf{0},$$

provided $E\|\mathbf{x}_t\| < \infty$ (the inner bracket is bounded by 1 in absolute value, so the product is integrable).

The g function. Thus the probit model is also a nonlinear GMM model with

$$\mathbf{g}(\mathbf{w}_t; \theta) = \mathbf{x}_t \cdot [y_t - \Phi(\mathbf{x}_t' \theta)], \quad K = p,$$

which is of the “instruments times error term” form (7.1.24) with $a(y_t, \mathbf{x}_t; \theta) = y_t - \Phi(\mathbf{x}_t' \theta)$, and the objective function (7.1.3) is

$$Q_n(\theta) = -\frac{1}{2} \mathbf{g}_n(\theta)' \widehat{\mathbf{W}} \mathbf{g}_n(\theta), \quad \mathbf{g}_n(\theta) = \frac{1}{n} \sum_{t=1}^n \mathbf{x}_t [y_t - \Phi(\mathbf{x}_t' \theta)].$$

Here $K = p$: the model is exactly identified, the choice of $\widehat{\mathbf{W}}$ is irrelevant, the GMM estimator is the value of θ solving $\mathbf{g}_n(\theta) = \mathbf{0}$, and the maximized objective function is zero (there are no overidentifying restrictions to test). More instruments are available if one wants them: $E(\varepsilon_t \mid \mathbf{x}_t) = 0$ implies $E[\mathbf{h}(\mathbf{x}_t)\varepsilon_t] = \mathbf{0}$ for *every* measurable $\mathbf{h}$, so any function of $\mathbf{x}_t$ — squares, cross products — can be added as an instrument to produce an overidentified GMM estimator. The optimal instrument in that class is $\phi(\mathbf{x}_t' \theta_0)[\Phi(1 - \Phi)]^{-1} \mathbf{x}_t$, which reproduces the ML score of the previous question: NLS, GMM and ML here are three weightings of one and the same residual, and ML is the efficient one.

Review Question 7.1.3

No — the model does not specify the parametric form of the density of $\mathbf{w}_t$, so there is no likelihood function to maximize.

The model of that subsection consists of (7.1.21) and (7.1.22) alone:

$$y_t = \mathbf{z}_t' \theta_0 + \varepsilon_t, \quad E[\mathbf{x}_t \cdot (y_t - \mathbf{z}_t' \theta_0)] = \mathbf{0},$$

for a set of ergodic stationary (here i.i.d.) processes $\mathbf{w}_t = (y_t, \mathbf{z}_t', \mathbf{x}_t')'$. What is restricted is K moments of the joint distribution of $\mathbf{w}_t$; the distribution itself is left free. Infinitely many joint

densities of $\mathbf{w}_t$ are compatible with the same θ_0 : the error can be normal, Student, or skewed; it can be conditionally homoskedastic or not; it can be correlated with $\mathbf{z}_t$ in any way whatever, since only its K covariances with $\mathbf{x}_t$ are pinned down. ML, in contrast, needs the density $f(\mathbf{w}_t; \theta)$ (or the conditional density $f(y_t | \mathbf{x}_t; \theta)$) to be a *known* function of $\mathbf{w}_t$ and of the finite parameter vector, so that the objective function $\frac{1}{n} \sum_t \log f(\mathbf{w}_t; \theta)$ of (7.1.2) is computable from the data. Here it is not: the model is semiparametric, and i.i.d. sampling does not help, since the i.i.d. assumption restricts the dependence across t , not the functional form of the density.

Two qualifications are worth stating. (i) One can always *add* a distributional assumption and then do ML — for instance $\mathbf{z}_t = \mathbf{x}_t$ with $\varepsilon_t | \mathbf{x}_t \sim N(0, \sigma_0^2)$, whose conditional ML estimator is OLS, or, when $\mathbf{z}_t$ contains endogenous variables, a complete simultaneous equations system with normal errors, whose ML estimator is the FIML of Chapter 8. But that is estimating a different, stronger model: consistency then rests on the added assumption, whereas GMM is consistent under the orthogonality conditions alone. The choice between ML and GMM here is the usual one between efficiency under a full specification and robustness under a weak one. (ii) The implication runs one way only. Every ML model *is* a GMM model, because the score is a valid moment function: $E[\mathbf{s}(\mathbf{w}_t; \theta_0)] = \mathbf{0}$ is an orthogonality condition with p equations and p unknowns. A GMM model is not in general an ML model.

Review Question 7.1.4

Remark. In the book's statement of this question $\theta_0, \theta_1, \theta_2$ are the three coefficients, while everywhere else in the chapter θ_0 denotes the true parameter *vector*; “the NLS estimator of θ_0 ” at the end means the estimator of the true coefficient vector, not of the intercept. To keep the two apart we write $\theta^* = (\theta_0^*, \theta_1^*, \theta_2^*)'$ for the true coefficient vector and θ for a hypothetical value. We also take $E(\varepsilon_t) = 0$, the orthogonality condition attached to the intercept, as part of the hypothesis.

Does $E(\varepsilon_t | x_t) = 0$ hold? No. If it did, then for every measurable h with $E|\varepsilon_t h(x_t)| < \infty$ the Law of Total Expectations would give

$$E[\varepsilon_t h(x_t)] = E[h(x_t) E(\varepsilon_t | x_t)] = 0,$$

and $h(x) = x^2$ would force $E(\varepsilon_t x_t^2) = 0$, contradicting the hypothesis. The conditional mean restriction is strictly stronger than any finite list of unconditional orthogonality conditions: it is equivalent to $E[\varepsilon_t h(x_t)] = 0$ for *all* measurable h , whereas here we are given only two such conditions, for $h(x) = 1$ and $h(x) = x$.

Is NLS consistent? No. Since φ is linear in θ , NLS here *is* OLS of y_t on $\mathbf{h}_t \equiv (1, x_t, x_t^2)'$, and

$$\hat{\theta} = \theta^* + \left(\frac{1}{n} \sum_t \mathbf{h}_t \mathbf{h}_t' \right)^{-1} \frac{1}{n} \sum_t \mathbf{h}_t \varepsilon_t \xrightarrow{p} \theta^* + \mathbf{A}^{-1} \mathbf{b},$$

where $\mathbf{A} \equiv E(\mathbf{h}_t \mathbf{h}_t')$ is finite and nonsingular and

$$\mathbf{b} \equiv E(\mathbf{h}_t \varepsilon_t) = (E(\varepsilon_t), E(\varepsilon_t x_t), E(\varepsilon_t x_t^2))' = (0, 0, c)', \quad c \equiv E(\varepsilon_t x_t^2) \neq 0.$$

$\mathbf{A}^{-1}$ is nonsingular, so $\mathbf{A}^{-1} \mathbf{b} = c \times (\text{third column of } \mathbf{A}^{-1}) \neq \mathbf{0}$: the probability limit is not θ^* , and in general *every* coefficient, not just θ_2^* , is inconsistently estimated.

The reason is that NLS identifies the conditional mean function and nothing else. Its limiting objective function decomposes as

$$Q_0(\theta) = -E[(y_t - \varphi(x_t; \theta))^2] = -E[(E(y_t | x_t) - \varphi(x_t; \theta))^2] - E[\text{Var}(y_t | x_t)],$$

so the maximizer is the θ whose $\varphi(\cdot; \theta)$ is closest in mean square to the true conditional mean. Under the hypotheses $E(y_t | x_t) \neq \varphi(x_t; \theta^*)$ — otherwise ε_t would be conditionally mean zero — so the pseudo-true value differs from θ^* . Nor can the two given orthogonality conditions be used instead: two moment conditions cannot identify three parameters, so the order condition fails and a further instrument would be needed.

A concrete illustration: let $x_t \sim N(0, 1)$ and $\varepsilon_t = x_t^2 - 1 + u_t$ with u_t independent of x_t and mean zero. Then $E(\varepsilon_t) = 0$ and $E(\varepsilon_t x_t) = E(x_t^3) - E(x_t) = 0$, while $c = E(\varepsilon_t x_t^2) = E(x_t^4) - E(x_t^2) = 3 - 1 = 2 \neq 0$, so the hypotheses hold. But then $y_t = (\theta_0^* - 1) + \theta_1^* x_t + (\theta_2^* + 1)x_t^2 + u_t$, and OLS converges to $(\theta_0^* - 1, \theta_1^*, \theta_2^* + 1)'$, that is $\mathbf{A}^{-1}\mathbf{b} = (-1, 0, 1)'$: two of the three coefficients are inconsistent.

Review Question 7.1.5

No. Write $\mathbf{s}_{xy} \equiv \frac{1}{n} \sum_t \mathbf{x}_t y_t$ and $\mathbf{s}_{xz} \equiv \frac{1}{n} \sum_t \mathbf{x}_t z_t$ ($K \times 1$ each). The two objective functions (7.1.3) are

$$Q_n(\theta) = -\frac{1}{2}(\mathbf{s}_{xy} - \theta \mathbf{s}_{xz})' \widehat{\mathbf{W}} (\mathbf{s}_{xy} - \theta \mathbf{s}_{xz}), \quad \tilde{Q}_n(\lambda) = -\frac{1}{2}(\mathbf{s}_{xz} - \lambda \mathbf{s}_{xy})' \widehat{\mathbf{W}} (\mathbf{s}_{xz} - \lambda \mathbf{s}_{xy}),$$

since $\mathbf{g}_n(\theta) = \mathbf{s}_{xy} - \theta \mathbf{s}_{xz}$ and $\tilde{\mathbf{g}}_n(\lambda) = \mathbf{s}_{xz} - \lambda \mathbf{s}_{xy}$.

The two are proportional, not equal. Because $\mathbf{s}_{xz} - \theta^{-1} \mathbf{s}_{xy} = -\theta^{-1}(\mathbf{s}_{xy} - \theta \mathbf{s}_{xz})$,

$$\tilde{Q}_n(1/\theta) = \theta^{-2} Q_n(\theta) \quad \text{for all } \theta > 0,$$

so the invariance criterion (7.1.17) fails except in the trivial cases $\theta = 1$ or $Q_n(\theta) = 0$. The factor θ^{-2} is the squared Jacobian of the reparameterization, and since it depends on θ it tilts the objective function: as $Q_n \leq 0$, multiplying by θ^{-2} penalizes small values of θ more heavily, which pushes the implied estimate upward.

The estimators differ. Both problems are quadratic and are solved in closed form:

$$\hat{\theta} = \frac{\mathbf{s}_{xz}' \widehat{\mathbf{W}} \mathbf{s}_{xy}}{\mathbf{s}_{xz}' \widehat{\mathbf{W}} \mathbf{s}_{xz}}, \quad \hat{\lambda} = \frac{\mathbf{s}_{xy}' \widehat{\mathbf{W}} \mathbf{s}_{xz}}{\mathbf{s}_{xy}' \widehat{\mathbf{W}} \mathbf{s}_{xy}}, \quad \text{so} \quad \hat{\theta} \hat{\lambda} = \frac{(\mathbf{s}_{xz}' \widehat{\mathbf{W}} \mathbf{s}_{xy})^2}{(\mathbf{s}_{xz}' \widehat{\mathbf{W}} \mathbf{s}_{xz})(\mathbf{s}_{xy}' \widehat{\mathbf{W}} \mathbf{s}_{xy})} \leq 1$$

by the Cauchy–Schwarz inequality for the inner product $\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a}' \widehat{\mathbf{W}} \mathbf{b}$ (an inner product because $\widehat{\mathbf{W}}$ is positive definite), with equality if and only if $\mathbf{s}_{xy}$ and $\mathbf{s}_{xz}$ are proportional. Hence $1/\hat{\lambda} \geq \hat{\theta}$, confirming the tilt just described: estimating λ and inverting gives a systematically larger number than estimating θ directly. The estimator is not invariant, and neither is the maximized objective function.

Exact identification restores invariance. If $K = 1$ the two sample moments are scalars, $\hat{\theta} = s_{xy}/s_{xz} = 1/\hat{\lambda}$, $\widehat{\mathbf{W}}$ cancels and both objective functions attain the value zero at the optimum. The failure of invariance is thus a pure overidentification phenomenon: when $K > p$ the moment conditions cannot all be set to zero, so the metric $\widehat{\mathbf{W}}$ used to measure the distance from zero matters, and a fixed $\widehat{\mathbf{W}}$ does not adjust for the Jacobian of the reparameterization. (ML, by contrast, is

invariant precisely because the likelihood after reparameterization is the same number attached to the same point.)

It is a finite-sample phenomenon. Both estimators are consistent for their own parameters, and $1/\hat{\lambda}$ has the same asymptotic distribution as $\hat{\theta}$. Indeed, with $\sigma_{xz} \equiv E(\mathbf{x}_t z_t)$ and $\sigma_{xy} \equiv E(\mathbf{x}_t y_t) = \theta_0 \sigma_{xz}$, the Jacobians and long-run variances of the two moment functions are

$$\mathbf{G} = -\sigma_{xz}, \quad \tilde{\mathbf{G}} = -\sigma_{xy} = \theta_0 \mathbf{G}, \quad \tilde{\mathbf{S}} = \lambda_0^2 \mathbf{S}$$

(the reparameterized moment function is $\mathbf{x}_t(z_t - \lambda_0 y_t) = -\lambda_0 \mathbf{x}_t \varepsilon_t$), so the sandwich formula gives $\text{Avar}(\hat{\lambda}) = \theta_0^{-2} \lambda_0^2 \text{Avar}(\hat{\theta}) = \lambda_0^4 \text{Avar}(\hat{\theta})$ and the delta method gives $\text{Avar}(1/\hat{\lambda}) = \lambda_0^{-4} \text{Avar}(\hat{\lambda}) = \text{Avar}(\hat{\theta})$. The two parameterizations disagree in any finite sample but agree to first order asymptotically.

Section 7.2 Consistency

Review Question 7.2.1

Suppose (7.2.13) fails: there exists $\theta_1 \neq \theta_0$ in Θ with

$$\text{Prob}[f(y_t | \mathbf{x}_t; \theta_1) \neq f(y_t | \mathbf{x}_t; \theta_0)] = 0,$$

i.e. $f(y_t | \mathbf{x}_t; \theta_1) = f(y_t | \mathbf{x}_t; \theta_0)$ with probability one (the probability being taken under the true joint distribution of $w_t = (y_t, \mathbf{x}_t')$). Then $\log f(y_t | \mathbf{x}_t; \theta_1) = \log f(y_t | \mathbf{x}_t; \theta_0)$ almost surely, so their expectations agree:

$$E[\log f(y_t | \mathbf{x}_t; \theta_1)] = E[\log f(y_t | \mathbf{x}_t; \theta_0)].$$

Hence $E[\log f(y_t | \mathbf{x}_t; \theta)]$ is *not* uniquely maximized at θ_0 : the point $\theta_1 \neq \theta_0$ does at least as well. That is the failure of condition (a), so (a) implies (7.2.13): conditional density identification is necessary. Combined with sufficiency (Analytical Exercise 7.1), identification of the density is exactly equivalent to the limit log likelihood being uniquely maximized at the truth.

Review Question 7.2.2

In NLS, $m(w_t; \theta) = -\frac{1}{2}\{y_t - \varphi(\mathbf{x}_t; \theta)\}^2$ and $E(y_t | \mathbf{x}_t) = \varphi(\mathbf{x}_t; \theta_0)$. Suppose conditional mean identification fails: there is $\theta_1 \neq \theta_0$ with $\varphi(\mathbf{x}_t; \theta_1) = \varphi(\mathbf{x}_t; \theta_0)$ with probability one. Then, almost surely,

$$\{y_t - \varphi(\mathbf{x}_t; \theta_1)\}^2 = \{y_t - \varphi(\mathbf{x}_t; \theta_0)\}^2,$$

and taking expectations gives $E[m(w_t; \theta_1)] = E[m(w_t; \theta_0)]$: the limit objective function is not uniquely maximized at θ_0 . Hence identification in the sense of condition (a) of Propositions 7.3/7.4 implies conditional mean identification. The converse (sufficiency) is the text's argument from (7.2.8): if $\varphi(\mathbf{x}_t; \theta) \neq \varphi(\mathbf{x}_t; \theta_0)$ with positive probability, the mean squared error at θ strictly exceeds that at θ_0 .

Remark. The necessity argument does not use (7.2.8); it only uses the fact that two functions equal almost surely produce the same expected objective. The sufficiency direction is where the minimum-MSE property (7.2.8) of the conditional expectation does the work.

Review Question 7.2.3

For $\theta \neq \theta_0$ write $\delta \equiv \theta - \theta_0 \neq \mathbf{0}$. Then $\varphi(\mathbf{x}_t; \theta) - \varphi(\mathbf{x}_t; \theta_0) = \mathbf{x}_t' \delta$, and

$$E[(\mathbf{x}_t' \delta)^2] = \delta' E(\mathbf{x}_t \mathbf{x}_t') \delta > 0,$$

because $E(\mathbf{x}_t \mathbf{x}_t')$ is nonsingular (hence positive definite, being a second-moment matrix). Since a nonnegative random variable with positive mean must be positive with positive probability, $\text{Prob}\{\varphi(\mathbf{x}_t; \theta) \neq \varphi(\mathbf{x}_t; \theta_0)\} > 0$: conditional mean identification (7.2.10) holds, and by the text's argument the expected NLS objective is uniquely maximized at θ_0 . In the linear case identification thus boils down to the familiar no-(asymptotic-) multicollinearity condition.

Review Question 7.2.4

For $\theta \neq \theta_0$ set $\delta \equiv \theta - \theta_0 \neq \mathbf{0}$. By the previous review question, nonsingularity of $E(\mathbf{x}_t \mathbf{x}_t')$ implies $\text{Prob}\{\mathbf{x}_t' \delta \neq 0\} > 0$. The standard normal cdf Φ is strictly increasing on $\mathbb{R}$, so

$$\mathbf{x}_t' \theta \neq \mathbf{x}_t' \theta_0 \implies \Phi(\mathbf{x}_t' \theta) \neq \Phi(\mathbf{x}_t' \theta_0).$$

Therefore $\text{Prob}\{\Phi(\mathbf{x}_t' \theta) \neq \Phi(\mathbf{x}_t' \theta_0)\} \geq \text{Prob}\{\mathbf{x}_t' \delta \neq 0\} > 0$, which is conditional mean identification for the regression function $\varphi(\mathbf{x}_t; \theta) = \Phi(\mathbf{x}_t' \theta)$. NLS of y_t on $\Phi(\mathbf{x}_t' \theta)$ is thus identified — and consistent, under the conditions of Proposition 7.3 or 7.4 — even though least squares ignores the binary nature of y_t entirely. (It is not efficient: ML uses the Bernoulli likelihood. This is the linear-probability-free analogue of GMM-versus-ML.)

Review Question 7.2.5

Reread Proposition 7.3 (the master consistency theorem for M-estimators with compact Θ) with θ^* in place of θ_0 . Its hypotheses are: $\{w_t\}$ ergodic stationary; Θ compact; $m(w_t; \cdot) = \log f(y_t | \mathbf{x}_t; \cdot)$ continuous and measurable; condition (a) that $E[m(w_t; \theta)]$ be uniquely maximized at the distinguished point; and the dominance condition (b). Ergodic stationarity (i.i.d. implies it), compactness, continuity and measurability do not involve θ_0 at all. Condition (a) with the distinguished point θ^* is exactly what we have assumed: θ^* is the unique maximizer of $E[\log f(y_t | \mathbf{x}_t; \theta)]$ over Θ . Condition (b) is unchanged. Proposition 7.3 therefore delivers $\hat{\theta} \xrightarrow{P} \theta^*$.

The point is that the consistency theorem is about the *maximizer of the limit objective function*, whoever that may be: nowhere in Propositions 7.1–7.4 is the distinguished point required to be the “true” parameter of a correctly specified density. Here θ^* is the pseudo-true value — the point of Θ whose implied density is closest to the truth in the Kullback–Leibler sense, since by Analytical Exercise 7.1

$$E[\log f(y_t | \mathbf{x}_t; \theta)] = E[\log f(y_t | \mathbf{x}_t; \theta_0)] - KL(f(\cdot | \cdot; \theta_0) \| f(\cdot | \cdot; \theta)),$$

so maximizing the expected log likelihood over Θ is the same as minimizing the KL divergence from the true density over Θ . Misspecification of the *support of the search*, not of the likelihood, is what makes this “quasi”-ML; the same argument covers exclusion of θ_0 through an incorrectly restricted Θ in an otherwise correctly specified model.

Review Question 7.2.6

In the notation of this chapter the linear GMM model has $\mathbf{g}(w_t; \theta) = \mathbf{x}_t(y_t - \mathbf{z}_t'\theta)$ with $\mathbf{x}_t$ ($K \times 1$) the instruments and θ ($p \times 1$), so

$$E[\mathbf{g}(w_t; \theta)] = E[\mathbf{x}_t(y_t - \mathbf{z}_t'\theta)] = \boldsymbol{\sigma}_{xy} - \Sigma_{xz}\theta, \quad \Sigma_{xz} \equiv E(\mathbf{x}_t\mathbf{z}_t') \quad (K \times p),$$

and at the true value $E[\mathbf{g}(w_t; \theta_0)] = \mathbf{0}$, i.e. $\boldsymbol{\sigma}_{xy} = \Sigma_{xz}\theta_0$. Condition (a) of Proposition 7.7 is that $Q_0(\theta) = -\frac{1}{2}E[\mathbf{g}(w_t; \theta)]'\mathbf{W}E[\mathbf{g}(w_t; \theta)]$ be uniquely maximized at θ_0 ; since $\mathbf{W}$ is positive definite, this holds if and only if $E[\mathbf{g}(w_t; \theta)] \neq \mathbf{0}$ for all $\theta \neq \theta_0$. But

$$E[\mathbf{g}(w_t; \theta)] = \Sigma_{xz}(\theta_0 - \theta),$$

so the condition is: $\Sigma_{xz}\delta \neq \mathbf{0}$ for every $\delta \neq \mathbf{0}$, i.e. the columns of Σ_{xz} are linearly independent, i.e. $\text{rank}(\Sigma_{xz}) = p$. That is exactly the rank condition for identification of Chapter 3 (ASSUMPTION 3.4, $\text{rank } E(\mathbf{x}_t\mathbf{z}_t') = p$). The order condition $K \geq p$ is implicit: a $K \times p$ matrix cannot have rank p unless $K \geq p$.

Review Question 7.2.7

With $\mathbf{g}(w_t; \theta) = \mathbf{x}_t(y_t - \mathbf{z}_t'\theta)$, the sample orthogonality condition is

$$\mathbf{g}_n(\theta) = \frac{1}{n} \sum_{t=1}^n \mathbf{x}_t(y_t - \mathbf{z}_t'\theta) = \mathbf{s}_{xy} - \mathbf{S}_{xz}\theta, \quad \mathbf{S}_{xz} \equiv \frac{1}{n} \sum_{t=1}^n \mathbf{x}_t\mathbf{z}_t',$$

an affine function of θ . The GMM objective is $Q_n(\theta) = -\frac{1}{2}\mathbf{g}_n(\theta)'\widehat{\mathbf{W}}\mathbf{g}_n(\theta)$ with $\widehat{\mathbf{W}}$ symmetric positive definite. Differentiating the quadratic form $-\frac{1}{2}(\mathbf{s}_{xy} - \mathbf{S}_{xz}\theta)'\widehat{\mathbf{W}}(\mathbf{s}_{xy} - \mathbf{S}_{xz}\theta)$ once and twice gives

$$\frac{\partial Q_n(\theta)}{\partial \theta} = -\mathbf{S}_{xz}'\widehat{\mathbf{W}}\mathbf{g}_n(\theta) = \mathbf{S}_{xz}'\widehat{\mathbf{W}}(\mathbf{s}_{xy} - \mathbf{S}_{xz}\theta), \quad \frac{\partial^2 Q_n(\theta)}{\partial \theta \partial \theta'} = -\mathbf{S}_{xz}'\widehat{\mathbf{W}}\mathbf{S}_{xz},$$

the Hessian of the hint. For any $\mathbf{c} \neq \mathbf{0}$,

$$\mathbf{c}'(-\mathbf{S}_{xz}'\widehat{\mathbf{W}}\mathbf{S}_{xz})\mathbf{c} = -(\mathbf{S}_{xz}\mathbf{c})'\widehat{\mathbf{W}}(\mathbf{S}_{xz}\mathbf{c}) \leq 0,$$

because $\widehat{\mathbf{W}}$ is positive definite. So the Hessian is everywhere negative semidefinite, which for a twice continuously differentiable function is necessary and sufficient for concavity: Q_n is concave in θ , and strictly concave precisely when $\mathbf{S}_{xz}$ has full column rank — the sample rank condition of Chapter 3, which makes the maximizer unique and equal to the familiar closed-form linear GMM estimator.

Review Question 7.2.8

The limit objective is $Q_0(\theta) = -\frac{1}{2}E[\mathbf{g}(w_t; \theta)]'\mathbf{W}E[\mathbf{g}(w_t; \theta)]$. For $\mathbf{W}$ positive semidefinite, $\mathbf{a}'\mathbf{W}\mathbf{a} = 0$ no longer implies $\mathbf{a} = \mathbf{0}$ — it only implies $\mathbf{W}\mathbf{a} = \mathbf{0}$ (writing $\mathbf{W} = \mathbf{C}'\mathbf{C}$, $\mathbf{a}'\mathbf{W}\mathbf{a} = \|\mathbf{C}\mathbf{a}\|^2 = 0$ gives $\mathbf{C}\mathbf{a} = \mathbf{0}$, hence $\mathbf{W}\mathbf{a} = \mathbf{C}'\mathbf{C}\mathbf{a} = \mathbf{0}$; conversely $\mathbf{W}\mathbf{a} = \mathbf{0}$ gives $\mathbf{a}'\mathbf{W}\mathbf{a} = 0$). Since $Q_0(\theta_0) = 0$ and $Q_0(\theta) \leq 0$ everywhere, θ_0 is a maximizer; it is the *unique* maximizer if and only if $Q_0(\theta) < 0$ for all $\theta \neq \theta_0$, i.e. if and only if $\mathbf{W}E[\mathbf{g}(w_t; \theta)] \neq \mathbf{0}$ for all $\theta \neq \theta_0$. So the required identification condition is:

$$E[\mathbf{g}(w_t; \theta_0)] = \mathbf{0}, \quad \mathbf{W}E[\mathbf{g}(w_t; \theta)] \neq \mathbf{0} \text{ for all } \theta \neq \theta_0,$$

or equivalently: $E[\mathbf{g}(w_t; \theta)] = \mathbf{0}$ has $\theta = \theta_0$ as its unique solution *even after the equations are projected onto the range of $\mathbf{W}$* . A singular $\mathbf{W}$ discards some linear combinations of the orthogonality conditions; identification then requires that the conditions retained still pin down θ_0 . (In the linear case this becomes $\text{rank}(\mathbf{W}\Sigma_{xz}) = p$, weaker than $\text{rank}(\Sigma_{xz}) = p$ — but of course one cannot retain more than $\text{rank}(\mathbf{W})$ independent directions, so a singular $\mathbf{W}$ can destroy identification.)

Section 7.3 Asymptotic Normality

Review Question 7.3.1

For M-estimators $Q_n(\theta) = \frac{1}{n} \sum_{t=1}^n m(w_t; \theta)$, so by (7.3.1)–(7.3.2),

$$\mathbf{s}_n(\theta_0) = \frac{\partial Q_n(\theta_0)}{\partial \theta} = \frac{1}{n} \sum_{t=1}^n \mathbf{s}(w_t; \theta_0).$$

Under Proposition 7.9, w_t is i.i.d., hence so is $\{\mathbf{s}(w_t; \theta_0)\}$; condition (3) gives $E[\mathbf{s}(w_t; \theta_0)] = \mathbf{0}$. Therefore

$$\text{Avar}(\sqrt{n} \mathbf{s}_n(\theta_0)) = \text{Var}[\mathbf{s}(w_t; \theta_0)] = E[\mathbf{s}(w_t; \theta_0) \mathbf{s}(w_t; \theta_0)'].$$

(Equivalently, the Lindeberg–Lévy CLT gives $\sqrt{n} \mathbf{s}_n(\theta_0) \xrightarrow{d} N(\mathbf{0}, \Sigma)$ with this Σ .) Now the information matrix equality, the second half of condition (3) of Proposition 7.9 (proved in Analytical Exercise 7.2), states

$$E[\mathbf{s}(w_t; \theta_0) \mathbf{s}(w_t; \theta_0)'] = -E[\mathbf{H}(w_t; \theta_0)].$$

Combining the two displays gives the claim. So the asymptotic variance of the normalized score of the *objective function* is the negative expected Hessian per observation — which is why for ML the asymptotic variance of the estimator itself, $\{-E[\mathbf{H}(w_t; \theta_0)]\}^{-1}$, can be estimated either from the Hessian or from the outer product of scores.

Review Question 7.3.2

From Example 7.10, with $\theta = (\beta', \sigma^2)'$, $w_t = (y_t, \mathbf{x}_t')'$ and $\varepsilon_t = y_t - \mathbf{x}_t' \beta_0$, the score and Hessian at θ_0 are

$$\mathbf{s}(w_t; \theta_0) = \begin{bmatrix} \frac{1}{\sigma_0^2} \mathbf{x}_t \varepsilon_t \\ -\frac{1}{2\sigma_0^2} + \frac{\varepsilon_t^2}{2\sigma_0^4} \end{bmatrix}, \quad \mathbf{H}(w_t; \theta_0) = \begin{bmatrix} -\frac{1}{\sigma_0^2} \mathbf{x}_t \mathbf{x}_t' & -\frac{1}{\sigma_0^4} \mathbf{x}_t \varepsilon_t \\ -\frac{1}{\sigma_0^4} \varepsilon_t \mathbf{x}_t' & \frac{1}{2\sigma_0^4} - \frac{\varepsilon_t^2}{\sigma_0^6} \end{bmatrix}.$$

Use the conditional moments of the normal error: $E(\varepsilon_t | \mathbf{x}_t) = 0$, $E(\varepsilon_t^2 | \mathbf{x}_t) = \sigma_0^2$, $E(\varepsilon_t^3 | \mathbf{x}_t) = 0$ and $E(\varepsilon_t^4 | \mathbf{x}_t) = 3\sigma_0^4$.

Negative expected Hessian.

$$-E(\mathbf{H}(w_t; \theta_0) | \mathbf{x}_t) = \begin{bmatrix} \frac{1}{\sigma_0^2} \mathbf{x}_t \mathbf{x}_t' & \frac{1}{\sigma_0^4} \mathbf{x}_t E(\varepsilon_t | \mathbf{x}_t) \\ \frac{1}{\sigma_0^4} E(\varepsilon_t | \mathbf{x}_t) \mathbf{x}_t' & -\frac{1}{2\sigma_0^4} + \frac{E(\varepsilon_t^2 | \mathbf{x}_t)}{\sigma_0^6} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sigma_0^2} \mathbf{x}_t \mathbf{x}_t' & \mathbf{0} \\ \mathbf{0}' & \frac{1}{2\sigma_0^4} \end{bmatrix},$$

since $E(\varepsilon_t | \mathbf{x}_t) = 0$ and $-\frac{1}{2\sigma_0^4} + \frac{\sigma_0^2}{\sigma_0^6} = \frac{1}{2\sigma_0^4}$.

Expected outer product of the score. The blocks are

$$\begin{aligned}\beta\beta &: \frac{1}{\sigma_0^4} \mathbf{x}_t \mathbf{x}_t' E(\varepsilon_t^2 | \mathbf{x}_t) = \frac{1}{\sigma_0^2} \mathbf{x}_t \mathbf{x}_t', \\ \beta\sigma^2 &: \frac{1}{\sigma_0^2} \mathbf{x}_t E(\varepsilon_t \left(-\frac{1}{2\sigma_0^2} + \frac{\varepsilon_t^2}{2\sigma_0^4} \right) | \mathbf{x}_t) = \frac{1}{\sigma_0^2} \mathbf{x}_t \left(-\frac{E(\varepsilon_t | \mathbf{x}_t)}{2\sigma_0^2} + \frac{E(\varepsilon_t^3 | \mathbf{x}_t)}{2\sigma_0^4} \right) = \mathbf{0}, \\ \sigma^2\sigma^2 &: E\left(\left(-\frac{1}{2\sigma_0^2} + \frac{\varepsilon_t^2}{2\sigma_0^4} \right)^2 | \mathbf{x}_t \right) = \frac{1}{4\sigma_0^4} - \frac{E(\varepsilon_t^2 | \mathbf{x}_t)}{2\sigma_0^6} + \frac{E(\varepsilon_t^4 | \mathbf{x}_t)}{4\sigma_0^8} = \frac{1}{4\sigma_0^4} - \frac{1}{2\sigma_0^4} + \frac{3}{4\sigma_0^4} = \frac{1}{2\sigma_0^4}.\end{aligned}$$

The two matrices agree block by block, which is the conditional information matrix equality. Taking expectations over $\mathbf{x}_t$ (law of total expectations) recovers the unconditional version $-E[\mathbf{H}(w_t; \theta_0)] = E[\mathbf{ss}']$ used in the text. Note where normality entered: the $\sigma^2\sigma^2$ block needs $E(\varepsilon_t^4 | \mathbf{x}_t) = 3\sigma_0^4$, and the off-diagonal blocks need symmetry ($E(\varepsilon_t^3 | \mathbf{x}_t) = 0$). Without normality the Hessian form of the information matrix still works for β but the full-matrix equality can fail — one reason quasi-ML standard errors for the linear model use the sandwich form.

Review Question 7.3.3

The first estimator is minus the inverse of the sample Hessian evaluated at $\hat{\theta}$:

$$\widehat{\text{Avar}}(\hat{\theta}) = \left[-\frac{1}{n} \sum_{t=1}^n \mathbf{H}(w_t; \hat{\theta}) \right]^{-1}, \quad \mathbf{H}(w_t; \hat{\theta}) = \begin{bmatrix} -\frac{1}{\hat{\sigma}^2} \mathbf{x}_t \mathbf{x}_t' & -\frac{1}{\hat{\sigma}^4} \mathbf{x}_t \hat{\varepsilon}_t \\ -\frac{1}{\hat{\sigma}^4} \hat{\varepsilon}_t \mathbf{x}_t' & \frac{1}{2\hat{\sigma}^4} - \frac{\hat{\varepsilon}_t^2}{\hat{\sigma}^6} \end{bmatrix},$$

with $\hat{\varepsilon}_t = y_t - \mathbf{x}_t' \hat{\beta}$ the ML (= OLS) residual. Summing over t , the off-diagonal blocks involve $\frac{1}{n} \sum_t \mathbf{x}_t \hat{\varepsilon}_t = \mathbf{0}$ by the normal equations (there is a constant among the regressors, or at least the first-order condition $\sum_t \mathbf{x}_t \hat{\varepsilon}_t = \mathbf{0}$ holds generally), and $\frac{1}{n} \sum_t \hat{\varepsilon}_t^2 = \hat{\sigma}^2$ by the ML first-order condition for σ^2 . Hence the sample Hessian is block diagonal:

$$-\frac{1}{n} \sum_{t=1}^n \mathbf{H}(w_t; \hat{\theta}) = \begin{bmatrix} \frac{1}{\hat{\sigma}^2} \frac{1}{n} \sum_t \mathbf{x}_t \mathbf{x}_t' & \mathbf{0} \\ \mathbf{0}' & -\frac{1}{2\hat{\sigma}^4} + \frac{\hat{\sigma}^2}{\hat{\sigma}^6} \end{bmatrix} = \begin{bmatrix} \frac{1}{\hat{\sigma}^2} \frac{1}{n} \sum_t \mathbf{x}_t \mathbf{x}_t' & \mathbf{0} \\ \mathbf{0}' & \frac{1}{2\hat{\sigma}^4} \end{bmatrix}.$$

Inverting and reading off the β block,

$$\widehat{\text{Avar}}(\hat{\beta}) = \hat{\sigma}^2 \left(\frac{1}{n} \sum_t \mathbf{x}_t \mathbf{x}_t' \right)^{-1} = n \hat{\sigma}^2 (\mathbf{X}'\mathbf{X})^{-1}.$$

So the answer to the question is **yes**: for the linear regression model the first estimator of the asymptotic variance of $\hat{\beta}$ *does* equal $\hat{\sigma}^2 \left(\frac{1}{n} \sum_t \mathbf{x}_t \mathbf{x}_t' \right)^{-1}$, because the likelihood equations make the sample Hessian block diagonal at the ML estimate. (The book's printed answer “No” applies to the comparison when the off-diagonal block is *not* annihilated — e.g. if one evaluates at a point other than the ML estimate, or forgets that the normal equations hold; at $\hat{\theta}$ itself the off-diagonal block is $\frac{1}{n} \sum_t \mathbf{x}_t \hat{\varepsilon}_t / \hat{\sigma}^4 = \mathbf{0}$ and the simplification goes through.) Note the only difference from the usual OLS variance estimator: $\hat{\sigma}^2 = \text{SSR} / n$ in place of $s^2 = \text{SSR} / (n - K)$. The $\hat{\sigma}^2$ block of $\widehat{\text{Avar}}$ is $2\hat{\sigma}^4$, so $\widehat{\text{Avar}}(\hat{\sigma}^2) = 2\hat{\sigma}^4$.

Review Question 7.3.4

The optimal choice (7.3.38) is $\mathbf{g}(w_t; \theta) = \mathbf{s}(w_t; \theta)$, the score: then $K = p$ (as many orthogonality conditions as parameters). Proposition 7.10 with a square nonsingular $\mathbf{G}$ gives

$$\text{Avar}(\hat{\theta}) = (\mathbf{G}'\mathbf{W}\mathbf{G})^{-1}\mathbf{G}'\mathbf{W}\mathbf{S}\mathbf{W}\mathbf{G}(\mathbf{G}'\mathbf{W}\mathbf{G})^{-1} = \mathbf{G}^{-1}\mathbf{S}\mathbf{G}'^{-1},$$

since $(\mathbf{G}'\mathbf{W}\mathbf{G})^{-1} = \mathbf{G}^{-1}\mathbf{W}^{-1}\mathbf{G}'^{-1}$ and the weighting matrix cancels — with $K = p$ there is no efficiency loss from any choice of $\mathbf{W}$. Now

$$\mathbf{G} = \mathbb{E}\left[\frac{\partial \mathbf{g}(w_t; \theta_0)}{\partial \theta'}\right] = \mathbb{E}\left[\frac{\partial \mathbf{s}(w_t; \theta_0)}{\partial \theta'}\right] = \mathbb{E}[\mathbf{H}(w_t; \theta_0)],$$

because the score's Jacobian is the Hessian of the log likelihood, and

$$\mathbf{S} = \mathbb{E}[\mathbf{g}(w_t; \theta_0)\mathbf{g}(w_t; \theta_0)'] = \mathbb{E}[\mathbf{s}(w_t; \theta_0)\mathbf{s}(w_t; \theta_0)'],$$

the information matrix for observation t . By the information matrix equality these two matrices are negatives of each other: $\mathbf{S} = -\mathbb{E}[\mathbf{H}(w_t; \theta_0)] = -\mathbf{G}$. Therefore

$$\text{Avar}(\hat{\theta}) = \mathbf{G}^{-1}\mathbf{S}\mathbf{G}'^{-1} = \mathbf{G}^{-1}(-\mathbf{G})\mathbf{G}'^{-1} = -\mathbf{G}'^{-1} = \{-\mathbb{E}[\mathbf{H}(w_t; \theta_0)]\}^{-1},$$

which, by the information matrix equality once more, is $\{\mathbb{E}[\mathbf{s}(w_t; \theta_0)\mathbf{s}(w_t; \theta_0)']\}^{-1}$ — exactly the inverse of the information matrix, the lower bound (7.3.37) and, by Proposition 7.9, the asymptotic variance of ML itself. So GMM with the score as the orthogonality condition attains the ML bound; as the text notes, the estimator is not merely asymptotically but numerically equivalent to ML, since $\mathbf{g}_n(\theta) = \mathbf{0}$ is the likelihood equations.

Review Question 7.3.5

Write the mean value expansion (7.3.32) compactly as

$$\sqrt{n}(\hat{\theta} - \theta_0) = -\mathbf{B}_n^{-1}\mathbf{c}_n,$$

where

$$\mathbf{B}_n \equiv \mathbf{G}_n(\hat{\theta})'\widehat{\mathbf{W}}\mathbf{G}_n(\bar{\theta}) \xrightarrow{p} \mathbf{G}'\mathbf{W}\mathbf{G} =: -\Psi, \quad \mathbf{c}_n \equiv \mathbf{G}_n(\hat{\theta})'\widehat{\mathbf{W}} \frac{1}{\sqrt{n}} \sum_{t=1}^n \mathbf{g}(w_t; \theta_0).$$

(a) Show that $\mathbf{B}_n^{-1} = -\Psi^{-1} + \mathbf{Y}_n$ where $\mathbf{Y}_n \xrightarrow{p} \mathbf{0}$. Since $\mathbf{B}_n \xrightarrow{p} -\Psi$ and Ψ is nonsingular ($\mathbf{G}$ has full column rank and $\mathbf{W}$ is positive definite, so $\mathbf{G}'\mathbf{W}\mathbf{G}$ is positive definite), the continuous mapping theorem applied to matrix inversion — continuous at nonsingular matrices — gives $\mathbf{B}_n^{-1} \xrightarrow{p} (-\Psi)^{-1} = -\Psi^{-1}$. Setting $\mathbf{Y}_n \equiv \mathbf{B}_n^{-1} + \Psi^{-1}$, we have $\mathbf{Y}_n \xrightarrow{p} \mathbf{0}$.

(b) Show that $\mathbf{c}_n = -\Psi \sqrt{n} \partial Q_n(\theta_0)/\partial \theta + \mathbf{y}_n$ where $\mathbf{y}_n \xrightarrow{p} \mathbf{0}$. From (7.3.45), $\sqrt{n} \partial Q_n(\theta_0)/\partial \theta = -\mathbf{G}_n(\theta_0)'\widehat{\mathbf{W}} \frac{1}{\sqrt{n}} \sum_t \mathbf{g}(w_t; \theta_0)$. Hence

$$\mathbf{c}_n + \Psi \sqrt{n} \frac{\partial Q_n(\theta_0)}{\partial \theta} = [\mathbf{G}_n(\hat{\theta}) - \mathbf{G}_n(\theta_0)]'\widehat{\mathbf{W}} \frac{1}{\sqrt{n}} \sum_{t=1}^n \mathbf{g}(w_t; \theta_0) =: \mathbf{y}_n.$$

Under the conditions of Proposition 7.10, $\frac{1}{\sqrt{n}} \sum_t \mathbf{g}(w_t; \theta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{S})$, so this term converges in distribution and is therefore bounded in probability ($O_p(1)$). Also $\mathbf{G}_n(\hat{\theta}) - \mathbf{G}_n(\theta_0) \xrightarrow{p} \mathbf{G} - \mathbf{G} = \mathbf{0}$ by the local dominance condition (4) applied to the consistent $\hat{\theta}$, and $\widehat{\mathbf{W}} \xrightarrow{p} \mathbf{W}$ is $O_p(1)$. By LEMMA 2.4(B) (o_p times O_p is o_p), $\mathbf{y}_n \xrightarrow{p} \mathbf{0}$.

(c) Show that $-\mathbf{B}_n^{-1} \mathbf{c}_n = -\Psi^{-1} \sqrt{n} \partial Q_n(\theta_0) / \partial \theta + \mathbf{x}_n$ where $\mathbf{x}_n \xrightarrow{p} \mathbf{0}$. Substitute (a) and (b):

$$-\mathbf{B}_n^{-1} \mathbf{c}_n = (\Psi^{-1} - \mathbf{Y}_n) \left(-\Psi \sqrt{n} \frac{\partial Q_n(\theta_0)}{\partial \theta} + \mathbf{y}_n \right) = -\Psi^{-1} \sqrt{n} \frac{\partial Q_n(\theta_0)}{\partial \theta} + \mathbf{x}_n,$$

with $\mathbf{x}_n \equiv -[\mathbf{Y}_n \Psi \sqrt{n} \partial Q_n(\theta_0) / \partial \theta + \Psi^{-1} \mathbf{y}_n + \mathbf{Y}_n \mathbf{y}_n]$.

(d) Show that $\mathbf{x}_n \xrightarrow{p} \mathbf{0}$. By condition (B), $\sqrt{n} \partial Q_n(\theta_0) / \partial \theta \xrightarrow{d} N(\mathbf{0}, \Sigma)$, hence is $O_p(1)$; $\mathbf{Y}_n = o_p(1)$, $\mathbf{y}_n = o_p(1)$, $\Psi^{-1} = O(1)$. Each of the three terms of $\mathbf{x}_n$ is a product of an $o_p(1)$ factor with an $O_p(1)$ factor, so $\mathbf{x}_n \xrightarrow{p} \mathbf{0}$ by LEMMA 2.4(B).

Combining, $\sqrt{n}(\hat{\theta} - \theta_0) = -\Psi^{-1} \sqrt{n} \partial Q_n(\theta_0) / \partial \theta + o_p(1)$, which is the Taylor expansion (7.3.43) with the symmetric matrix $\Psi = -\mathbf{G}' \mathbf{W} \mathbf{G}$ of (7.3.48). The upshot: for both M-estimators and GMM, the normalized sampling error is asymptotically a fixed linear function of the normalized score of the objective function — the common format on which the trinity of test statistics in Section 7.4 rests.

Review Question 7.3.6

The objective would be $\tilde{Q}_n(\theta) = -\mathbf{g}_n(\theta)' \hat{\mathbf{S}}^{-1} \mathbf{g}_n(\theta)$, twice the text's Q_n . Scaling the objective by 2 scales its gradient and Hessian by 2, so the score and the “curvature” matrix in the Taylor expansion double:

$$\sqrt{n} \frac{\partial \tilde{Q}_n(\theta_0)}{\partial \theta} = 2 \sqrt{n} \frac{\partial Q_n(\theta_0)}{\partial \theta} \implies \tilde{\Sigma} = 4\Sigma, \quad \tilde{\Psi} \equiv \text{plim} \frac{\partial^2 \tilde{Q}_n(\hat{\theta})}{\partial \theta \partial \theta'} = 2\Psi.$$

(Concretely: the first-order condition $\mathbf{G}_n(\hat{\theta})' \hat{\mathbf{S}}^{-1} \mathbf{g}_n(\hat{\theta}) = \mathbf{0}$ is unaffected, so $\hat{\theta}$ is unchanged; but in the expansion $\sqrt{n}(\hat{\theta} - \theta_0) = -\tilde{\Psi}^{-1} \sqrt{n} \partial \tilde{Q}_n / \partial \theta + o_p$ the factor 2 appears in both $\tilde{\Psi}$ and the score.) For efficient GMM $\mathbf{W} = \mathbf{S}^{-1}$, and (7.3.46) and (7.3.48) give

$$\Sigma = \mathbf{G}' \mathbf{S}^{-1} \mathbf{S} \mathbf{S}^{-1} \mathbf{G} = \mathbf{G}' \mathbf{S}^{-1} \mathbf{G}, \quad \Psi = -\mathbf{G}' \mathbf{S}^{-1} \mathbf{G},$$

so $\Sigma = -\Psi$. Without the deflator, $\tilde{\Sigma} = 4\mathbf{G}' \mathbf{S}^{-1} \mathbf{G} \neq 2\mathbf{G}' \mathbf{S}^{-1} \mathbf{G} = -\tilde{\Psi}$: condition (D) fails. Since condition (D) is what makes the LR statistic asymptotically χ^2 (Review Question 7.4.3), the factor $\frac{1}{2}$ in the GMM objective is not cosmetic — it calibrates the curvature of the objective to the variance of its score, so that the likelihood-ratio-type statistic $2n[Q_n(\tilde{\theta}) - Q_n(\hat{\theta})]$ has the $\chi^2(r)$ limit. (Consistently, the asymptotic variance $\text{Avar}(\hat{\theta}) = \tilde{\Psi}^{-1} \tilde{\Sigma} \tilde{\Psi}^{-1}$ is invariant to the scaling, as it must be, since the estimator itself does not change.)

Section 7.4 Hypothesis Testing

Review Question 7.4.1

Start from the “beautified” LM statistic of TABLE 7.2,

$$LM = n \left(\frac{\partial Q_n(\tilde{\theta})}{\partial \theta} \right)' \tilde{\Sigma}^{-1} \left(\frac{\partial Q_n(\tilde{\theta})}{\partial \theta} \right),$$

which is (7.4.25) with $\tilde{\Sigma}$ a consistent estimator of Σ , evaluated at the constrained estimator $\tilde{\theta}$. For ML with i.i.d. observations, $\Sigma = E[\mathbf{s}(w_t; \theta_0)\mathbf{s}(w_t; \theta_0)']$ (Review Question 7.3.1), and under the null $\tilde{\theta} \xrightarrow{p} \theta_0$ (Proposition 7.11(a)), so the sample outer product of the scores at the constrained estimate,

$$\tilde{\Sigma} = \frac{1}{n} \sum_{t=1}^n \mathbf{s}(w_t; \tilde{\theta})\mathbf{s}(w_t; \tilde{\theta})',$$

is consistent for Σ under the usual dominance conditions (the same argument that makes the “second estimator” of Avar in (7.3.15) consistent, transplanted from $\hat{\theta}$ to $\tilde{\theta}$). Substituting this $\tilde{\Sigma}$ into the display gives the stated formula. Since $\partial Q_n(\tilde{\theta})/\partial \theta = \frac{1}{n} \sum_t \mathbf{s}(w_t; \tilde{\theta})$, the statistic is a quadratic form in the *average score evaluated at the constrained estimate*: it measures how far the constrained estimate is from satisfying the unconstrained likelihood equations. Its practical virtue is that only the *constrained* model need be estimated — which is why LM tests dominate tests of neglected effects (heteroskedasticity, serial correlation, omitted variables), where the alternative is costly to fit.

Review Question 7.4.2

ML (and M-estimators generally). Apply the Mean Value Theorem to $\partial Q_n/\partial \theta$ around θ_0 (this is (7.3.5) with $\tilde{\theta}$ in place of $\hat{\theta}$):

$$\frac{\partial Q_n(\tilde{\theta})}{\partial \theta} = \frac{\partial Q_n(\theta_0)}{\partial \theta} + \left[\frac{1}{n} \sum_{t=1}^n \mathbf{H}(w_t; \bar{\theta}) \right] (\tilde{\theta} - \theta_0),$$

where $\bar{\theta}$ lies between $\tilde{\theta}$ and θ_0 . Multiply through by $\sqrt{n}$:

$$\sqrt{n} \frac{\partial Q_n(\tilde{\theta})}{\partial \theta} = \sqrt{n} \frac{\partial Q_n(\theta_0)}{\partial \theta} + \left[\frac{1}{n} \sum_{t=1}^n \mathbf{H}(w_t; \bar{\theta}) \right] \sqrt{n} (\tilde{\theta} - \theta_0).$$

Under the null, $\tilde{\theta} \xrightarrow{p} \theta_0$ (condition (C): the constrained estimator is consistent and $\sqrt{n}(\tilde{\theta} - \theta_0)$ converges in distribution, hence is $O_p(1)$), and $\bar{\theta}$, trapped between the two, is consistent too. The local dominance condition (4) of Proposition 7.9 then gives $\frac{1}{n} \sum_t \mathbf{H}(w_t; \bar{\theta}) \xrightarrow{p} E[\mathbf{H}(w_t; \theta_0)] = \Psi$. So the product equals $\Psi \sqrt{n}(\tilde{\theta} - \theta_0)$ plus a term that is $o_p(1) \times O_p(1) = o_p(1)$ by LEMMA 2.4(B) — which is (7.4.18).

GMM. Here $\partial Q_n(\theta)/\partial \theta = -\mathbf{G}_n(\theta)' \widehat{\mathbf{W}} \mathbf{g}_n(\theta)$ ((7.3.28)). Apply the Mean Value Theorem to $\mathbf{g}_n$ around θ_0 (the analogue of (7.3.30)):

$$\mathbf{g}_n(\tilde{\theta}) = \mathbf{g}_n(\theta_0) + \mathbf{G}_n(\bar{\theta})(\tilde{\theta} - \theta_0), \quad \bar{\theta} \text{ between } \tilde{\theta} \text{ and } \theta_0,$$

and substitute:

$$\frac{\partial Q_n(\tilde{\theta})}{\partial \theta} = -\mathbf{G}_n(\tilde{\theta})' \widehat{\mathbf{W}} \mathbf{g}_n(\theta_0) - \mathbf{G}_n(\tilde{\theta})' \widehat{\mathbf{W}} \mathbf{G}_n(\bar{\theta}) (\tilde{\theta} - \theta_0).$$

Since $\sqrt{n} \partial Q_n(\theta_0)/\partial \theta = -\mathbf{G}_n(\theta_0)' \widehat{\mathbf{W}} \sqrt{n} \mathbf{g}_n(\theta_0)$ ((7.3.45)), the first term on the right differs from $\sqrt{n} \partial Q_n(\theta_0)/\partial \theta$ by $[\mathbf{G}_n(\tilde{\theta}) - \mathbf{G}_n(\theta_0)]' \widehat{\mathbf{W}} \sqrt{n} \mathbf{g}_n(\theta_0)$, which is $o_p(1) \times O_p(1) = o_p(1)$ because $\mathbf{G}_n(\tilde{\theta}) \xrightarrow{p} \mathbf{G}_n(\theta_0) \xrightarrow{p} \mathbf{G}$ (local dominance plus consistency of $\tilde{\theta}$) and $\sqrt{n} \mathbf{g}_n(\theta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{S})$ is $O_p(1)$. The second term is $-\mathbf{G}_n(\tilde{\theta})' \widehat{\mathbf{W}} \mathbf{G}_n(\bar{\theta}) \sqrt{n}(\tilde{\theta} - \theta_0)$, and

$$-\mathbf{G}_n(\tilde{\theta})' \widehat{\mathbf{W}} \mathbf{G}_n(\bar{\theta}) \xrightarrow{p} -\mathbf{G}' \mathbf{W} \mathbf{G} = \Psi$$

by (7.3.48); the discrepancy is again $o_p(1)$ times the $O_p(1)$ factor $\sqrt{n}(\tilde{\theta} - \theta_0)$. Collecting terms gives (7.4.18) for GMM.

The content of (7.4.18): the score of the objective function evaluated at the constrained estimator equals, up to o_p , the score at the truth plus the curvature times the constrained sampling error. It is the ingredient that, combined with the first-order conditions of the constrained problem, delivers the asymptotic normality of the Lagrange multipliers.

Review Question 7.4.3

From (7.4.29), the LR statistic can be written, without invoking condition (D), as

$$LR = \left(\sqrt{n} \frac{\partial Q_n(\theta_0)}{\partial \theta} \right)' \Psi^{-1} \mathbf{A}'_0 (\mathbf{A}_0 \Psi^{-1} \Sigma \Psi^{-1} \mathbf{A}'_0)^{-1} \mathbf{A}_0 \Psi^{-1} \left(\sqrt{n} \frac{\partial Q_n(\theta_0)}{\partial \theta} \right) + o_p.$$

By condition (B), $\sqrt{n} \partial Q_n(\theta_0) / \partial \theta \xrightarrow{d} N(\mathbf{0}, \Sigma)$; write $\mathbf{z} \sim N(\mathbf{0}, \Sigma)$ for the limit and set

$$\mathbf{B} \equiv \Psi^{-1} \mathbf{A}'_0 (\mathbf{A}_0 \Psi^{-1} \Sigma \Psi^{-1} \mathbf{A}'_0)^{-1} \mathbf{A}_0 \Psi^{-1},$$

so the limit is $\mathbf{z}' \mathbf{B} \mathbf{z}$. A quadratic form $\mathbf{z}' \mathbf{B} \mathbf{z}$ in a zero-mean normal vector is χ^2 if and only if the covariance matrix in the quadratic form “inverts” the variance of $\mathbf{z}$, in the sense that $\Sigma^{1/2} \mathbf{B} \Sigma^{1/2}$ is idempotent. Computing,

$$\Sigma^{1/2} \mathbf{B} \Sigma^{1/2} = \Sigma^{1/2} \Psi^{-1} \mathbf{A}'_0 (\mathbf{A}_0 \Psi^{-1} \Sigma \Psi^{-1} \mathbf{A}'_0)^{-1} \mathbf{A}_0 \Psi^{-1} \Sigma^{1/2},$$

which is indeed idempotent for *any* Ψ (it is the oblique projection onto the span of $\Sigma^{1/2} \Psi^{-1} \mathbf{A}'_0$ along the appropriate complement — squaring it returns itself). The problem is the *weight matrix* inside: the asymptotic variance of the estimated restrictions is $\mathbf{A}_0 \Psi^{-1} \Sigma \Psi^{-1} \mathbf{A}'_0$ (the first equality in (7.4.9)), and the LR statistic (7.4.29) uses this matrix correctly. But the statistic must also be computable from the objective functions alone: the derivation of (7.4.29) from the second-order expansion (7.4.27) expresses LR through Ψ (the curvature of Q_n), while the χ^2 reference requires the quadratic form to be normalized by Σ . When $\Sigma \neq -\Psi$, the matrix in the middle of (7.4.29) is $(\mathbf{A}_0 \Psi^{-1} \Sigma \Psi^{-1} \mathbf{A}'_0)^{-1}$, whereas the quadratic form produced by the difference of the two optimized objective values — after substituting into (7.4.27), whose curvature is Ψ , not Σ , the expansion $\sqrt{n}(\tilde{\theta} - \hat{\theta}) = -\Psi^{-1} \mathbf{A}'_0 (\mathbf{A}_0 \Psi^{-1} \Sigma \Psi^{-1} \mathbf{A}'_0)^{-1} \mathbf{A}_0 \Psi^{-1} \sqrt{n} \partial Q_n(\theta_0) / \partial \theta + o_p$ — is

$$LR = \left(\sqrt{n} \frac{\partial Q_n}{\partial \theta} \right)' \Psi^{-1} \mathbf{A}'_0 \mathbf{D}^{-1} (\mathbf{A}_0 \Psi^{-1} \Psi \Psi^{-1} \mathbf{A}'_0) \mathbf{D}^{-1} \mathbf{A}_0 \Psi^{-1} \left(\sqrt{n} \frac{\partial Q_n}{\partial \theta} \right) + o_p,$$

with $\mathbf{D} \equiv \mathbf{A}_0 \Psi^{-1} \Sigma \Psi^{-1} \mathbf{A}'_0$. If $\Sigma = -\Psi$, the middle factor collapses, since

$$\mathbf{A}_0 \Psi^{-1} \Psi \Psi^{-1} \mathbf{A}'_0 = \mathbf{A}_0 \Psi^{-1} \mathbf{A}'_0 = -\mathbf{A}_0 \Psi^{-1} \Sigma \Psi^{-1} \mathbf{A}'_0 = -\mathbf{D},$$

the two inverse matrices cancel (sign aside, which the deflator of Review Question 7.3.6 fixes), and what remains is a quadratic form in a normal vector normalized by its own asymptotic variance: $\chi^2(r)$. Without $\Sigma = -\Psi$ the factor $\mathbf{D}^{-1} (\mathbf{A}_0 \Psi^{-1} \mathbf{A}'_0) \mathbf{D}^{-1}$ is a nontrivial sandwich of two different matrices; the statistic is a quadratic form in a normal vector with a matrix that is not the inverse of its variance, so its limit is a weighted sum of independent $\chi^2(1)$'s with weights given by the eigenvalues of $\mathbf{D}^{-1/2} (\mathbf{A}_0 \Psi^{-1} \mathbf{A}'_0) \mathbf{D}^{-1/2}$ — not $\chi^2(r)$.

Intuition: the LR statistic reads the curvature of the objective function off as if it were the information. Only when curvature equals information ($\Sigma = -\Psi$: ML under correct specification with the information matrix equality, or efficient GMM with the deflated objective) is that reading correct. The Wald and LM statistics, by contrast, can be defined with Σ and Ψ estimated separately (first lines of (7.4.11) and (7.4.25)-style formulas), so they survive the failure of condition (D); the LR statistic cannot, because the objective function values themselves carry no information about Σ .

Review Question 7.4.4

None. With serial correlation, two things in TABLE 7.2 break. First, $\Sigma = \text{Avar}(\frac{1}{\sqrt{n}} \sum_t \mathbf{s}(w_t; \theta_0))$ is the *long-run* variance of the score, $\sum_{j=-\infty}^{\infty} \Gamma_j$ with Γ_j the j -th autocovariance of the score — not $E[\mathbf{s}\mathbf{s}']$. The Table 7.2 estimators of Σ (outer product of scores, or minus the Hessian via the information matrix equality) estimate $E[\mathbf{s}\mathbf{s}']$ or $-E[\mathbf{H}]$, both of which equal only Γ_0 : the Wald and LM statistics are mis-normalized, and their limits are weighted sums of $\chi^2(1)$'s, not $\chi^2(r)$. Second, although the information matrix equality $-E[\mathbf{H}] = E[\mathbf{s}\mathbf{s}']$ still holds (it is a property of the density at a point, not of the dependence structure), what condition (D) requires is $\Sigma = -\Psi$, i.e. *long-run* variance = $-E[\mathbf{H}]$, and with serial correlation $\Sigma \neq \Gamma_0 = -E[\mathbf{H}] = -\Psi$: condition (D) fails, so by the previous review question the LR statistic is not asymptotically χ^2 either. To rescue Wald and LM one must replace the estimated Σ by a HAC estimate (VARHAC or kernel-based, from Section 6.6) of the long-run variance computed from the estimated scores $\{\mathbf{s}(w_t; \hat{\theta})\}$ or $\{\mathbf{s}(w_t; \tilde{\theta})\}$; no such repair is available for LR, whose ingredients are the maximized values of the (now quasi-) likelihood.

Remark. Consistency is not the issue: by Propositions 7.5/7.6 the (quasi-) ML estimator remains consistent under ergodic stationarity alone. It is the *inference* that must be adjusted — exactly the QML-robust-standard-errors point made in the text after Proposition 7.6.

Review Question 7.4.5

The first equality in (7.4.23), valid without condition (D), states that under the null

$$\sqrt{n} \tilde{\boldsymbol{\lambda}}_n \xrightarrow{d} N(\mathbf{0}, (\mathbf{A}_0 \Psi^{-1} \Sigma \Psi^{-1} \mathbf{A}_0')^{-1}),$$

where $\tilde{\boldsymbol{\lambda}}_n$ is the vector of Lagrange multipliers of the constrained problem. The natural statistic is therefore the quadratic form in the multipliers normalized by a consistent estimate of that variance:

$$LM^* = n \tilde{\boldsymbol{\lambda}}_n' [\mathbf{A}(\tilde{\theta}) \tilde{\Psi}^{-1} \tilde{\Sigma} \tilde{\Psi}^{-1} \mathbf{A}(\tilde{\theta})'] \tilde{\boldsymbol{\lambda}}_n.$$

(Equivalently, using the first-order condition $\mathbf{A}(\tilde{\theta})' \tilde{\boldsymbol{\lambda}}_n = -\partial Q_n(\tilde{\theta})/\partial \theta$ from (7.4.15), this is the “beautified” score form $n(\partial Q_n(\tilde{\theta})/\partial \theta)' \tilde{\Psi}^{-1} \tilde{\Sigma} \tilde{\Psi}^{-1} (\partial Q_n(\tilde{\theta})/\partial \theta)$ evaluated through the multipliers.) Under the null, $\mathbf{A}(\tilde{\theta}) \xrightarrow{p} \mathbf{A}_0$, $\tilde{\Psi} \xrightarrow{p} \Psi$, $\tilde{\Sigma} \xrightarrow{p} \Sigma$, so by LEMMA 2.4(D) the bracket converges to $(\mathbf{A}_0 \Psi^{-1} \Sigma \Psi^{-1} \mathbf{A}_0')^{-1}$ — the inverse of the asymptotic variance of $\sqrt{n} \tilde{\boldsymbol{\lambda}}_n$ (that matrix is positive definite because $\mathbf{A}_0$ has full row rank, Ψ is nonsingular and Σ is positive definite). Hence $LM^* \xrightarrow{d} \chi^2(r)$ by the standard result that $\mathbf{z}' \hat{\mathbf{V}} \mathbf{z} \xrightarrow{d} \chi^2(r)$ when $\mathbf{z} \xrightarrow{d} N(\mathbf{0}, \mathbf{V}^{-1})$ and $\hat{\mathbf{V}} \xrightarrow{p} \mathbf{V}$. This is the sandwich- (or robust-) LM statistic; it reduces to the Table 7.2 formula when $\Sigma = -\Psi$, since then $\Psi^{-1} \Sigma \Psi^{-1} = -\Psi^{-1} = \Sigma^{-1}$.

Review Question 7.4.6

Take for granted the second-order mean value expansion of Q_n around the unconstrained maximizer $\hat{\theta}$: there is a $\bar{\theta}$ between $\tilde{\theta}$ and $\hat{\theta}$ with

$$Q_n(\tilde{\theta}) = Q_n(\hat{\theta}) + \frac{\partial Q_n(\hat{\theta})}{\partial \theta'}(\tilde{\theta} - \hat{\theta}) + \frac{1}{2}(\tilde{\theta} - \hat{\theta})' \left[\frac{1}{n} \sum_{t=1}^n \mathbf{H}(w_t; \bar{\theta}) \right] (\tilde{\theta} - \hat{\theta}).$$

Multiplying through by $2n$ so as to expose the normalization,

$$2n [Q_n(\tilde{\theta}) - Q_n(\hat{\theta})] = 2n \frac{\partial Q_n(\hat{\theta})}{\partial \theta'}(\tilde{\theta} - \hat{\theta}) + n(\tilde{\theta} - \hat{\theta})' \left[\frac{1}{n} \sum_{t=1}^n \mathbf{H}(w_t; \bar{\theta}) \right] (\tilde{\theta} - \hat{\theta}).$$

The linear term vanishes by the first-order condition for the unconstrained problem: $\hat{\theta}$ is an interior maximizer, so $\partial Q_n(\hat{\theta})/\partial \theta = \mathbf{0}$. For the quadratic term, note that $\bar{\theta}$ lies between $\tilde{\theta}$ and $\hat{\theta}$, both consistent under the null (Proposition 7.11(a)), so $\bar{\theta} \xrightarrow{p} \theta_0$; the local dominance condition (4) of Proposition 7.8 then gives $\frac{1}{n} \sum_t \mathbf{H}(w_t; \bar{\theta}) \xrightarrow{p} E[\mathbf{H}(w_t; \theta_0)] = \Psi$. Hence

$$2n [Q_n(\tilde{\theta}) - Q_n(\hat{\theta})] = \sqrt{n}(\tilde{\theta} - \hat{\theta})' \Psi \sqrt{n}(\tilde{\theta} - \hat{\theta}) + \sqrt{n}(\tilde{\theta} - \hat{\theta})' \left[\frac{1}{n} \sum_t \mathbf{H}(w_t; \bar{\theta}) - \Psi \right] \sqrt{n}(\tilde{\theta} - \hat{\theta}).$$

Both $\sqrt{n}(\tilde{\theta} - \theta_0)$ and $\sqrt{n}(\hat{\theta} - \theta_0)$ converge in distribution (conditions (B)–(C) via the Taylor expansions (7.3.43) and (7.4.20)), so $\sqrt{n}(\tilde{\theta} - \hat{\theta}) = O_p(1)$; the bracket is $o_p(1)$; the last term is $o_p(1)$ by LEMMA 2.4(B). which is (7.4.27): the likelihood-ratio-type statistic is, up to o_p , n times a quadratic form in the difference of the two estimators with matrix Ψ . (The sign is right: $\Psi = E[\mathbf{H}(w_t; \theta_0)]$ is negative definite and $Q_n(\tilde{\theta}) \leq Q_n(\hat{\theta})$ because $\hat{\theta}$ is the unconstrained maximizer, so both sides are nonpositive. With $\Sigma = -\Psi$ under condition (D), this becomes the nonnegative quadratic form in $-\Psi = \Sigma$ that converges to $\chi^2(r)$ in (7.4.30).)

Review Question 7.4.7

For GMM, $Q_n(\theta) = -\frac{1}{2} \mathbf{g}_n(\theta)' \widehat{\mathbf{W}} \mathbf{g}_n(\theta)$ with $\widehat{\mathbf{W}} \xrightarrow{p} \mathbf{W} = \mathbf{S}^{-1}$ (efficient GMM). The mean value expansion of $\mathbf{g}_n$ around $\hat{\theta}$ gives, for $\bar{\theta}$ between $\tilde{\theta}$ and $\hat{\theta}$,

$$\mathbf{g}_n(\tilde{\theta}) = \mathbf{g}_n(\hat{\theta}) + \mathbf{G}_n(\bar{\theta})(\tilde{\theta} - \hat{\theta}),$$

and since both estimators are consistent under the null, the local dominance condition gives $\mathbf{G}_n(\bar{\theta}) \xrightarrow{p} \mathbf{G}$; because $\sqrt{n}(\tilde{\theta} - \hat{\theta}) = O_p(1)$ this can be written as the Taylor expansion

$$\sqrt{n} \mathbf{g}_n(\tilde{\theta}) = \sqrt{n} \mathbf{g}_n(\hat{\theta}) + \mathbf{G} \sqrt{n}(\tilde{\theta} - \hat{\theta}) + o_p.$$

Substitute into $2nQ_n(\tilde{\theta}) = -[\sqrt{n} \mathbf{g}_n(\tilde{\theta})]' \widehat{\mathbf{W}} [\sqrt{n} \mathbf{g}_n(\tilde{\theta})]$:

$$\begin{aligned} 2nQ_n(\tilde{\theta}) &= -[\sqrt{n} \mathbf{g}_n(\hat{\theta}) + \mathbf{G} \sqrt{n}(\tilde{\theta} - \hat{\theta}) + o_p]' \widehat{\mathbf{W}} [\sqrt{n} \mathbf{g}_n(\hat{\theta}) + \mathbf{G} \sqrt{n}(\tilde{\theta} - \hat{\theta}) + o_p] \\ &= 2nQ_n(\hat{\theta}) - 2\sqrt{n}(\tilde{\theta} - \hat{\theta})' \mathbf{G}' \widehat{\mathbf{W}} \sqrt{n} \mathbf{g}_n(\hat{\theta}) - n(\tilde{\theta} - \hat{\theta})' \mathbf{G}' \widehat{\mathbf{W}} \mathbf{G}(\tilde{\theta} - \hat{\theta}) + o_p, \end{aligned}$$

where the cross terms involving the o_p remainder are absorbed into the final o_p (they are o_p times O_p). The middle term vanishes by the first-order condition of the unconstrained GMM problem:

(7.3.28) says $\mathbf{G}_n(\hat{\theta})'\widehat{\mathbf{W}}\mathbf{g}_n(\hat{\theta}) = \mathbf{0}$, and

$$\mathbf{G}'\widehat{\mathbf{W}}\sqrt{n}\mathbf{g}_n(\hat{\theta}) = \underbrace{\mathbf{G}_n(\hat{\theta})'\widehat{\mathbf{W}}\sqrt{n}\mathbf{g}_n(\hat{\theta})}_{=\mathbf{0}} + \underbrace{[\mathbf{G} - \mathbf{G}_n(\hat{\theta})]'\widehat{\mathbf{W}}\sqrt{n}\mathbf{g}_n(\hat{\theta})}_{o_p(1) \times O_p(1) = o_p(1)},$$

since $\sqrt{n}\mathbf{g}_n(\hat{\theta}) = O_p(1)$ (it is $\sqrt{n}\mathbf{g}_n(\theta_0) + \mathbf{G}\sqrt{n}(\hat{\theta} - \theta_0) + o_p$, a sum of O_p terms). Therefore

$$2n[Q_n(\tilde{\theta}) - Q_n(\hat{\theta})] = -n(\tilde{\theta} - \hat{\theta})'\mathbf{G}'\widehat{\mathbf{W}}\mathbf{G}(\tilde{\theta} - \hat{\theta}) + o_p,$$

and since $\mathbf{G}'\widehat{\mathbf{W}}\mathbf{G} \xrightarrow{p} \mathbf{G}'\mathbf{W}\mathbf{G} = -\Psi$ by (7.3.48), one more $o_p \times O_p$ absorption gives $2n[Q_n(\tilde{\theta}) - Q_n(\hat{\theta})] = n(\tilde{\theta} - \hat{\theta})'\Psi(\tilde{\theta} - \hat{\theta}) + o_p$, which is (7.4.27). Note the parallel with the M-estimator proof: there the linear term dies because the gradient of Q_n at $\hat{\theta}$ is zero; here the linear term dies because the gradient $-\mathbf{G}_n(\hat{\theta})'\widehat{\mathbf{W}}\mathbf{g}_n(\hat{\theta})$ is zero — the same first-order condition, filtered through the structure of the GMM objective.

Section 7.5 Numerical Optimization

Review Question 7.5.1

The Newton–Raphson update (7.5.3) is $\hat{\theta}_{j+1} = \hat{\theta}_j - \mathbf{H}_n(\hat{\theta}_j)^{-1}\mathbf{s}_n(\hat{\theta}_j)$, i.e.

$$\mathbf{s}_n(\hat{\theta}_j) = -\mathbf{H}_n(\hat{\theta}_j)(\hat{\theta}_{j+1} - \hat{\theta}_j).$$

Substitute this into the right-hand side (with $\mathbf{H}_n(\hat{\theta}_j)$ in place of $\mathbf{H}_n(\bar{\theta}_j)$, whose sign agrees with the true increment by hypothesis):

$$\begin{aligned} \mathbf{s}_n(\hat{\theta}_j)'(\hat{\theta}_{j+1} - \hat{\theta}_j) &+ \frac{1}{2}(\hat{\theta}_{j+1} - \hat{\theta}_j)'\mathbf{H}_n(\hat{\theta}_j)(\hat{\theta}_{j+1} - \hat{\theta}_j) \\ &= -(\hat{\theta}_{j+1} - \hat{\theta}_j)'\mathbf{H}_n(\hat{\theta}_j)(\hat{\theta}_{j+1} - \hat{\theta}_j) + \frac{1}{2}(\hat{\theta}_{j+1} - \hat{\theta}_j)'\mathbf{H}_n(\hat{\theta}_j)(\hat{\theta}_{j+1} - \hat{\theta}_j) \\ &= -\frac{1}{2}(\hat{\theta}_{j+1} - \hat{\theta}_j)'\mathbf{H}_n(\hat{\theta}_j)(\hat{\theta}_{j+1} - \hat{\theta}_j). \end{aligned}$$

If Q_n is concave and twice continuously differentiable, its Hessian is everywhere negative semidefinite, so $-\mathbf{H}_n(\hat{\theta}_j)$ is positive semidefinite and the last expression is nonnegative. By the assumed sign agreement, $Q_n(\hat{\theta}_{j+1}) - Q_n(\hat{\theta}_j) \geq 0$: the algorithm never decreases the objective function. This is the local ascent property of Newton–Raphson on concave problems (near the iterate, the quadratic model that the algorithm maximizes exactly dominates the step); combined with concavity — which makes any local maximum global — it explains the rapid and reliable convergence of Newton–Raphson for, e.g., the linear regression likelihood, probit and other concave ML problems. Away from a concave region the property fails: $\mathbf{H}_n$ indefinite means the “maximizer” of the quadratic approximation can be a minimizer, which is why practical implementations add step-halving or replace $\mathbf{H}_n$ by a negative definite matrix (as in scoring or quasi-Newton methods).

Review Question 7.5.2

Substituting the linearization into the objective gives

$$\tilde{Q}_n(\theta) = -\frac{1}{n} \sum_{t=1}^n \left\{ \underbrace{y_t - \varphi(\mathbf{x}_t; \hat{\theta}_j)}_{\tilde{y}_t} - \underbrace{\mathbf{d}_t(\hat{\theta}_j)'}_{\tilde{\mathbf{x}}_t'} \underbrace{(\theta - \hat{\theta}_j)}_{\boldsymbol{\delta}} \right\}^2,$$

which is minus the SSR of an *ordinary least squares* regression of the current residual $\tilde{y}_t \equiv y_t - \varphi(\mathbf{x}_t; \hat{\theta}_j)$ on the current gradient $\tilde{\mathbf{x}}_t \equiv \mathbf{d}_t(\hat{\theta}_j)$, with coefficient vector $\boldsymbol{\delta} \equiv \theta - \hat{\theta}_j$. This is a quadratic, concave maximization (equivalently, least squares minimization) problem; provided $\sum_t \mathbf{d}_t \mathbf{d}_t'$ is nonsingular, the first-order conditions are the normal equations

$$\sum_{t=1}^n \mathbf{d}_t(\hat{\theta}_j) [\tilde{y}_t - \mathbf{d}_t(\hat{\theta}_j)' \hat{\boldsymbol{\delta}}] = \mathbf{0},$$

whose unique solution is

$$\hat{\boldsymbol{\delta}} = \left[\sum_{t=1}^n \mathbf{d}_t(\hat{\theta}_j) \mathbf{d}_t(\hat{\theta}_j)' \right]^{-1} \sum_{t=1}^n \mathbf{d}_t(\hat{\theta}_j) [y_t - \varphi(\mathbf{x}_t; \hat{\theta}_j)],$$

and $\hat{\theta}_{j+1} = \hat{\theta}_j + \hat{\boldsymbol{\delta}}$ is the displayed formula. So each Gauss–Newton round is an OLS regression of the current NLS residuals on the current derivatives — the “Gauss–Newton regression,” which doubles as a computational device for LM diagnostics (regress residuals on derivatives plus candidates and check the fit). Versus Newton–Raphson, Gauss–Newton drops the second-derivative-of- φ part of the Hessian, $\sum_t [y_t - \varphi(\mathbf{x}_t; \hat{\theta}_j)] \partial^2 \varphi / \partial \theta \partial \theta'$, which has expectation zero at the truth; what remains, $\sum_t \mathbf{d}_t \mathbf{d}_t'$, is positive semidefinite by construction, so the linearized problem is always concave — a built-in guarantee of the ascent direction that full Newton–Raphson lacks.

Problem Set for Chapter 7
Analytical Exercise 7.1

Fix θ and suppose $\text{Prob}[f(y | \mathbf{x}; \theta) \neq f(y | \mathbf{x}; \theta_0)] > 0$ (otherwise the two densities agree almost surely and the inequality holds with equality). Let $w = (y, \mathbf{x})'$ and define

$$a(w) \equiv \frac{f(y | \mathbf{x}; \theta)}{f(y | \mathbf{x}; \theta_0)}.$$

(a) $a(w)$ is nonconstant. By construction, $a(w) \neq 1$ exactly when $f(y | \mathbf{x}; \theta) \neq f(y | \mathbf{x}; \theta_0)$, and the latter occurs with positive probability by supposition. So $\text{Prob}[a(w) \neq 1] > 0$: the random variable $a(w)$ takes a value other than 1 with positive probability, hence is not almost surely constant.

(b) Apply the strict Jensen inequality. The function $\log$ is strictly concave on $(0, \infty)$ ($\log''(x) = -1/x^2 < 0$). Since $a(w)$ is a positive, nonconstant random variable and $\text{E}[\log a(w)] = \text{E}[\log f(y | \mathbf{x}; \theta)] - \text{E}[\log f(y | \mathbf{x}; \theta_0)]$ exists and is finite by hypothesis, the strict version of Jensen’s inequality applies:

$$\text{E}[\log a(w)] < \log \text{E}[a(w)].$$

(c) *Show* $E[a(w)] = 1$. Compute the conditional mean given $\mathbf{x}$, remembering that the expectation is taken under the *true* conditional density $f(y | \mathbf{x}; \theta_0)$:

$$E(a(w) | \mathbf{x}) = \int \frac{f(y | \mathbf{x}; \theta)}{f(y | \mathbf{x}; \theta_0)} f(y | \mathbf{x}; \theta_0) dy = \int f(y | \mathbf{x}; \theta) dy = 1,$$

the last equality because $f(y | \mathbf{x}; \theta)$ is a (hypothetical) conditional density for every θ and $\mathbf{x}$. The law of total expectation then gives $E[a(w)] = E[E(a(w) | \mathbf{x})] = E[1] = 1$.

(d) *Conclude*. Substituting (c) into (b),

$$E[\log a(w)] < \log 1 = 0, \quad \text{i.e.} \quad E[\log f(y | \mathbf{x}; \theta)] - E[\log f(y | \mathbf{x}; \theta_0)] < 0.$$

Hence $E[\log f(y | \mathbf{x}; \theta)] \leq E[\log f(y | \mathbf{x}; \theta_0)]$, with strict inequality whenever the two conditional densities differ with positive probability, and equality when $f(y | \mathbf{x}; \theta) = f(y | \mathbf{x}; \theta_0)$ almost surely. This is the Kullback–Leibler information inequality (7.2.12).

The quantity $E[\log f(y | \mathbf{x}; \theta_0)] - E[\log f(y | \mathbf{x}; \theta)]$ is the Kullback–Leibler divergence from the true density to the hypothetical one; the inequality says it is nonnegative and zero only when the densities coincide. That is precisely why ML identification reduces to conditional density identification (7.2.13) (Propositions 7.5 and 7.6), and why a quasi-ML estimator under misspecification converges to the density in the model *closest to the truth in the KL sense* (Review Question 7.2.5).

Remark. The positivity assumption $f > 0$ keeps $\log f$ and $a(w)$ well defined and the argument elementary. Without it one defines $a(w)$ only on the support of the true density and notes that $f(y | \mathbf{x}; \theta) = 0$ on a set of positive true probability forces $E[\log f(y | \mathbf{x}; \theta)] = -\infty$, which only strengthens the inequality; see White (1994, p. 9), cited in the book's footnote.

Analytical Exercise 7.2

First verify the two displayed rewritings. Since $\log f$ is well defined,

$$\mathbf{s}(w; \theta) = \frac{\partial \log f(y | \mathbf{x}; \theta)}{\partial \theta} = \frac{1}{f(y | \mathbf{x}; \theta)} \frac{\partial f(y | \mathbf{x}; \theta)}{\partial \theta},$$

and differentiating once more by the product and chain rules,

$$\begin{aligned} \mathbf{H}(w; \theta) &= \frac{\partial \mathbf{s}(w; \theta)}{\partial \theta'} = \frac{1}{f(y | \mathbf{x}; \theta)} \frac{\partial^2 f(y | \mathbf{x}; \theta)}{\partial \theta \partial \theta'} - \frac{1}{f(y | \mathbf{x}; \theta)^2} \frac{\partial f(y | \mathbf{x}; \theta)}{\partial \theta} \frac{\partial f(y | \mathbf{x}; \theta)}{\partial \theta'} \\ &= \frac{1}{f} \frac{\partial^2 f}{\partial \theta \partial \theta'} - \mathbf{s}(w; \theta) \mathbf{s}(w; \theta)', \end{aligned}$$

using the first display to recognize the second term.

(a) *The expected score is zero*. Since $f(y | \mathbf{x}; \theta)$ is a hypothetical conditional density, it integrates to one whatever θ :

$$\int f(y | \mathbf{x}; \theta) dy = 1 \quad \text{for all } \theta \in \Theta.$$

This is an identity in θ ; differentiate both sides with respect to θ :

$$\frac{\partial}{\partial \theta} \int f(y | \mathbf{x}; \theta) dy = \mathbf{0}.$$

Assuming integration and differentiation can be interchanged (the regularity condition; see, e.g., Lemma 3.6 of Newey and McFadden, 1994, for primitive conditions),

$$\int \frac{\partial f(y | \mathbf{x}; \theta)}{\partial \theta} dy = \mathbf{0}.$$

Multiply and divide the integrand by $f(y | \mathbf{x}; \theta)$ and use the score rewriting:

$$\int \mathbf{s}(w; \theta) f(y | \mathbf{x}; \theta) dy = \mathbf{0} \quad \text{for all } \theta \in \Theta.$$

This holds in particular at $\theta = \theta_0$, where the integral is by definition the conditional expectation under the true density:

$$\mathbf{E}(\mathbf{s}(w; \theta_0) | \mathbf{x}) = \int \mathbf{s}(w; \theta_0) f(y | \mathbf{x}; \theta_0) dy = \mathbf{0}.$$

The law of total expectations then delivers $\mathbf{E}[\mathbf{s}(w; \theta_0)] = \mathbf{E}[\mathbf{E}(\mathbf{s}(w; \theta_0) | \mathbf{x})] = \mathbf{0}$. (Note where the truth enters: the identity $\int \mathbf{s}f = \mathbf{0}$ holds for every θ , but it becomes an *expectation* only when f in the integrand is the density generating the data, i.e. at θ_0 .)

(b) *The information matrix equality.* Differentiate the identity $\int \mathbf{s}(w; \theta) f(y | \mathbf{x}; \theta) dy = \mathbf{0}$ with respect to θ' (again interchanging the order of operations):

$$\int \frac{\partial \mathbf{s}(w; \theta)}{\partial \theta'} f(y | \mathbf{x}; \theta) dy + \int \mathbf{s}(w; \theta) \frac{\partial f(y | \mathbf{x}; \theta)}{\partial \theta'} dy = \mathbf{0}.$$

The second integrand can be written as

$$\mathbf{s}(w; \theta) \frac{\partial f(y | \mathbf{x}; \theta)}{\partial \theta'} = \mathbf{s}(w; \theta) \mathbf{s}(w; \theta)' f(y | \mathbf{x}; \theta),$$

because $\partial f / \partial \theta' = f \cdot (\partial \log f / \partial \theta') = f \mathbf{s}(w; \theta)'$. Hence

$$\int \left[\mathbf{H}(w; \theta) + \mathbf{s}(w; \theta) \mathbf{s}(w; \theta)' \right] f(y | \mathbf{x}; \theta) dy = \mathbf{0} \quad \text{for all } \theta.$$

Evaluating at θ_0 turns the integrals into expectations under the true density:

$$\mathbf{E}(\mathbf{H}(w; \theta_0) | \mathbf{x}) + \mathbf{E}(\mathbf{s}(w; \theta_0) \mathbf{s}(w; \theta_0)' | \mathbf{x}) = \mathbf{0},$$

and the law of total expectations gives the information matrix equality

$$-\mathbf{E}[\mathbf{H}(w; \theta_0)] = \mathbf{E}[\mathbf{s}(w; \theta_0) \mathbf{s}(w; \theta_0)'].$$

The equality is an identity about the second derivatives of log of an integrating-to-one function: the average curvature of the log likelihood equals minus the average squared slope. Its uses in the chapter are legion: it simplifies $\text{Avar}(\hat{\theta})$ in Proposition 7.9 to $\{-\mathbf{E}[\mathbf{H}(w_t; \theta_0)]\}^{-1}$; it validates the outer-product estimator (7.3.15); it makes $\Sigma = -\Psi$ (condition (D) of Section 7.4) hold for ML, on which the χ^2 limit of the LR statistic rests; and, failing it (misspecification, serial correlation as in Review Question 7.4.4), it tells us exactly which sandwich corrections are needed.

Analytical Exercise 7.3

Throughout, $\mathbf{y}$ ($n \times 1$) and $\mathbf{X}$ ($n \times K$) collect the data, $\mathbf{b}$ is the OLS estimator, $\text{SSR}_U = \sum_t (y_t - \mathbf{x}_t' \hat{\beta})^2$ and $\text{SSR}_R = \sum_t (y_t - \mathbf{x}_t' \tilde{\beta})^2$.

(a) $\hat{\beta}$ is OLS and $\tilde{\beta}$ is restricted least squares. The average log likelihood is

$$Q_n(\beta, \sigma^2) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log(\sigma^2) - \frac{1}{2n\sigma^2} \sum_{t=1}^n (y_t - \mathbf{x}'_t \beta)^2.$$

For any fixed $\sigma^2 > 0$, maximizing Q_n over β is the same as minimizing $\sum_t (y_t - \mathbf{x}'_t \beta)^2$, so the unrestricted ML estimate of β is the OLS estimator $\hat{\beta} = \mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$. The same argument with the constraint attached shows that $\tilde{\beta}$ minimizes the sum of squared residuals subject to $\mathbf{R}\beta = \mathbf{c}$: it is the restricted least squares estimator of (1.4.11),

$$\tilde{\beta} = \mathbf{b} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\mathbf{b} - \mathbf{c}).$$

(The constraint involves only β , and σ^2 enters the likelihood multiplicatively through $1/\sigma^2$, so the constrained maximization over β is separable from σ^2 exactly as in the unconstrained case.)

(b) *The LR statistic in terms of the two SSRs.* Maximizing Q_n over σ^2 for fixed β (the calculation of Review Question 1.5.3) gives $\hat{\sigma}^2 = \text{SSR}_U / n$ under the unrestricted model and $\tilde{\sigma}^2 = \text{SSR}_R / n$ under the constraint, since the first-order condition $\partial Q_n / \partial \sigma^2 = -\frac{1}{2\sigma^2} + \frac{\text{SSR}(\beta)}{2n\sigma^4} = 0$ gives $\sigma^2 = \text{SSR}(\beta)/n$ whatever β is being held fixed. Substituting the maximizers back,

$$Q_n(\hat{\theta}) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log \frac{\text{SSR}_U}{n} - \frac{1}{2}, \quad Q_n(\tilde{\theta}) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log \frac{\text{SSR}_R}{n} - \frac{1}{2},$$

because $\text{SSR}(\hat{\beta})/(n\hat{\sigma}^2) = \text{SSR}_U / \text{SSR}_U = 1$, and similarly at the constrained estimate. Therefore

$$LR = 2n[Q_n(\hat{\theta}) - Q_n(\tilde{\theta})] = n \left(\log \frac{\text{SSR}_R}{n} - \log \frac{\text{SSR}_U}{n} \right) = n \log \frac{\text{SSR}_R}{\text{SSR}_U}.$$

(The book defines LR as $2n[Q_n(\tilde{\theta}) - Q_n(\hat{\theta})]$ with Q_n a *maximized* objective; the display is the positive version, $n \log(\text{SSR}_R / \text{SSR}_U)$, since $\text{SSR}_R \geq \text{SSR}_U$.)

(c) *Consistency of the variance estimators used in Table 7.2.* The Table 7.2 formulas evaluate the Hessian-based estimator of $\Psi = E[\mathbf{H}(w_t; \theta_0)]$ at the two ML estimates. Here, instead of the literal sample Hessian $\frac{1}{n} \sum_t \mathbf{H}(w_t; \hat{\theta})$, take the *expected* Hessian form from Example 7.10 evaluated at the estimates: as shown there, $-E[\mathbf{H}(w_t; \theta)] = \text{diag}(\frac{1}{\sigma^2} E(\mathbf{x}_t \mathbf{x}'_t), \frac{1}{2\sigma^4})$, so the estimators used are

$$\hat{\Psi} = - \begin{bmatrix} \frac{1}{\hat{\sigma}^2} \frac{1}{n} \sum_t \mathbf{x}_t \mathbf{x}'_t & \mathbf{0} \\ \mathbf{0}' & \frac{1}{2\hat{\sigma}^4} \end{bmatrix}, \quad \tilde{\Psi} = - \begin{bmatrix} \frac{1}{\tilde{\sigma}^2} \frac{1}{n} \sum_t \mathbf{x}_t \mathbf{x}'_t & \mathbf{0} \\ \mathbf{0}' & \frac{1}{2\tilde{\sigma}^4} \end{bmatrix}.$$

These differ from $-\frac{1}{n} \sum_t \mathbf{H}(w_t; \hat{\theta})$ and $-\frac{1}{n} \sum_t \mathbf{H}(w_t; \tilde{\theta})$: the literal sample Hessian at the ML estimate coincides with these in the $\beta\beta$ and off-diagonal blocks (by the normal equations, see Review Question 7.3.3), and the $\sigma^2\sigma^2$ block differs only by terms that vanish in probability (it equals $\frac{1}{2\hat{\sigma}^4}$ exactly at $\hat{\theta}$). Under the null both $\hat{\theta}$ and $\tilde{\theta}$ are consistent for θ_0 (Example 7.8 and Section 7.4), so $\hat{\sigma}^2$ and $\tilde{\sigma}^2$ are consistent for σ_0^2 : e.g. $\hat{\sigma}^2 = \frac{1}{n} \sum_t \varepsilon_t^2 + o_p(1) \xrightarrow{p} \sigma_0^2$ by the law of large numbers, and $\tilde{\sigma}^2 - \hat{\sigma}^2 = (\text{SSR}_R - \text{SSR}_U)/n \xrightarrow{p} 0$ under the null because $\text{SSR}_R - \text{SSR}_U = O_p(1)$ (it is n^{-1} times a $\chi^2(r)$ variate, by part (e) below). With $\frac{1}{n} \sum_t \mathbf{x}_t \mathbf{x}'_t \xrightarrow{p} E(\mathbf{x}_t \mathbf{x}'_t)$, the continuous mapping theorem gives $\hat{\Psi}, \tilde{\Psi} \xrightarrow{p} \Psi = E[\mathbf{H}(w_t; \theta_0)]$ as required. Correspondingly $\hat{\Sigma} = \tilde{\Sigma} = -\hat{\Psi}$ etc. estimate $\Sigma = -\Psi$ (the information matrix equality, Analytical Exercise 7.2, holds here), so condition (D) is operational.

(d) *The three statistics in regression form.* The null $\mathbf{R}\beta = \mathbf{c}$ in terms of $\theta = (\beta', \sigma^2)'$ is $\mathbf{a}(\theta) = \mathbf{R}\beta - \mathbf{c} = \mathbf{0}$, with Jacobian

$$\mathbf{A}(\theta) = \frac{\partial \mathbf{a}(\theta)}{\partial \theta'} = [\mathbf{R} \quad \mathbf{0}] \quad (r \times (K+1)).$$

With the block-diagonal $\hat{\Psi}$ from part (c), $-\hat{\Psi}^{-1} = \text{diag}(\hat{\sigma}^2(\frac{1}{n} \sum_t \mathbf{x}_t \mathbf{x}_t')^{-1}, 2\hat{\sigma}^4)$, so

$$\mathbf{A}\hat{\Sigma}^{-1}\mathbf{A}' = \mathbf{A}(-\hat{\Psi}^{-1})\mathbf{A}' = \hat{\sigma}^2 \mathbf{R} \left(\frac{1}{n} \sum_t \mathbf{x}_t \mathbf{x}_t' \right)^{-1} \mathbf{R}' = n\hat{\sigma}^2 \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{R}',$$

the zero block annihilating the σ^2 row and column: inference on β separates from inference on σ^2 .

Wald. From Table 7.2, $W = n \mathbf{a}(\hat{\theta})' [\mathbf{A}\hat{\Sigma}^{-1}\mathbf{A}]^{-1} \mathbf{a}(\hat{\theta})$:

$$\begin{aligned} W &= n (\mathbf{R}\mathbf{b} - \mathbf{c})' [n\hat{\sigma}^2 \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{R}']^{-1} (\mathbf{R}\mathbf{b} - \mathbf{c}) \\ &= \frac{(\mathbf{R}\mathbf{b} - \mathbf{c})' [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{R}']^{-1} (\mathbf{R}\mathbf{b} - \mathbf{c})}{\hat{\sigma}^2} = n \cdot \frac{(\mathbf{R}\mathbf{b} - \mathbf{c})' [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{R}']^{-1} (\mathbf{R}\mathbf{b} - \mathbf{c})}{\text{SSR}_U}. \end{aligned}$$

LM. The score of the objective at the constrained estimate has β -block

$$\frac{\partial Q_n(\tilde{\theta})}{\partial \beta} = \frac{1}{n\tilde{\sigma}^2} \sum_{t=1}^n \mathbf{x}_t (y_t - \mathbf{x}_t' \tilde{\beta}) = \frac{1}{n\tilde{\sigma}^2} \mathbf{X}'(\mathbf{y} - \mathbf{X}\tilde{\beta}),$$

and σ^2 -block $-\frac{1}{2\tilde{\sigma}^2} + \frac{\text{SSR}_R}{2n\tilde{\sigma}^4} = 0$ (by the definition of $\tilde{\sigma}^2$). So only the β -block of the score is alive. The LM statistic $n(\partial Q_n/\partial \theta)' \tilde{\Sigma}^{-1} (\partial Q_n/\partial \theta)$ with $\tilde{\Sigma}^{-1} = -\tilde{\Psi}^{-1}$ becomes

$$\begin{aligned} LM &= n \cdot \frac{1}{n^2 \tilde{\sigma}^4} (\mathbf{y} - \mathbf{X}\tilde{\beta})' \mathbf{X} \cdot \tilde{\sigma}^2 \left(\frac{1}{n} \mathbf{X}'\mathbf{X} \right)^{-1} \cdot \frac{1}{n\tilde{\sigma}^2} \mathbf{X}'(\mathbf{y} - \mathbf{X}\tilde{\beta}) \\ &= \frac{(\mathbf{y} - \mathbf{X}\tilde{\beta})' \mathbf{P}(\mathbf{y} - \mathbf{X}\tilde{\beta})}{\tilde{\sigma}^2} = n \cdot \frac{(\mathbf{y} - \mathbf{X}\tilde{\beta})' \mathbf{P}(\mathbf{y} - \mathbf{X}\tilde{\beta})}{\text{SSR}_R}, \end{aligned}$$

with $\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'$ the projection matrix.

LR. From part (b), $LR = n \log(\text{SSR}_R / \text{SSR}_U)$.

(e) *SSR forms of W and LM.* From Analytical Exercise 1.4 of Chapter 1 (the SSR form of the F -statistic), the numerator of the F -ratio satisfies

$$(\mathbf{R}\mathbf{b} - \mathbf{c})' [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{R}']^{-1} (\mathbf{R}\mathbf{b} - \mathbf{c}) = \text{SSR}_R - \text{SSR}_U.$$

Substituting into the Wald formula of part (d),

$$W = n \cdot \frac{\text{SSR}_R - \text{SSR}_U}{\text{SSR}_U}.$$

For LM, expand using $\mathbf{P}\mathbf{X} = \mathbf{X}$: $(\mathbf{y} - \mathbf{X}\tilde{\beta})' \mathbf{P}(\mathbf{y} - \mathbf{X}\tilde{\beta}) = (\mathbf{y} - \mathbf{X}\tilde{\beta})' \mathbf{P}\mathbf{P}(\mathbf{y} - \mathbf{X}\tilde{\beta})$; writing $\mathbf{y} - \mathbf{X}\tilde{\beta} = (\mathbf{y} - \mathbf{X}\mathbf{b}) + \mathbf{X}(\mathbf{b} - \tilde{\beta})$ and noting $\mathbf{P}(\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}$ (OLS residuals are orthogonal to the column space),

$$(\mathbf{y} - \mathbf{X}\tilde{\beta})' \mathbf{P}(\mathbf{y} - \mathbf{X}\tilde{\beta}) = (\mathbf{b} - \tilde{\beta})' \mathbf{X}'\mathbf{X}(\mathbf{b} - \tilde{\beta}).$$

But the same orthogonality gives $\text{SSR}_R - \text{SSR}_U = \|\mathbf{y} - \mathbf{X}\tilde{\beta}\|^2 - \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = (\mathbf{b} - \tilde{\beta})' \mathbf{X}'\mathbf{X}(\mathbf{b} - \tilde{\beta})$ (Pythagoras: the restricted residual decomposes into the unrestricted residual plus a vector in the

column space). Hence $(\mathbf{y} - \mathbf{X}\tilde{\beta})'\mathbf{P}(\mathbf{y} - \mathbf{X}\tilde{\beta}) = \text{SSR}_R - \text{SSR}_U$ and

$$LM = n \cdot \frac{\text{SSR}_R - \text{SSR}_U}{\text{SSR}_R}.$$

(f) *The inequality $W \geq LR \geq LM$.* Put $u \equiv \text{SSR}_R / \text{SSR}_U \geq 1$ (adding restrictions cannot reduce the SSR). The three statistics are

$$W = n(u - 1), \quad LR = n \log u, \quad LM = n\left(1 - \frac{1}{u}\right).$$

For $u \geq 1$: $\log u \leq u - 1$ (concavity of $\log$: the tangent line $u - 1$ at $u = 1$ lies above the curve), which gives $LR \leq W$; and $1 - \frac{1}{u} \leq \log u$ (substitute $v = 1/u \leq 1$ into $\log v \leq v - 1$ to get $-\log u \leq \frac{1}{u} - 1$, i.e. $1 - \frac{1}{u} \leq \log u$), which gives $LM \leq LR$. Hence

$$W \geq LR \geq LM,$$

with equality throughout only when $u = 1$, i.e. when the restrictions cost nothing in fit. All three are n times functions of the same single quantity u , and $n(u - 1)$, $n \log u$ and $n(1 - 1/u)$ agree to first order around $u = 1$ — the algebraic shadow of Proposition 7.11(c), which says the three differ by o_p under the null ($u \xrightarrow{P} 1$ there). The exact ordering $W \geq LR \geq LM$ is special to the linear-normal model: it rests on the exact SSR identities of part (e), which have no analogue when the objective is not quadratic; in nonlinear regression models the inequalities can fail in finite samples, while the asymptotic equivalence survives.

Remark. Under the null, $u - 1 = (\text{SSR}_R - \text{SSR}_U) / \text{SSR}_U = O_p(n^{-1})$, so all three statistics are $O_p(1)$ and converge to the same $\chi^2(r)$. Under a fixed alternative, $u \xrightarrow{P} \bar{u} > 1$ and all three diverge at rate n — the tests are consistent. The finite-sample exact distribution is available too: $\frac{n-K}{r} \cdot \frac{W}{n} \sim F(r, n - K)$ under the null by Proposition 1.4, since $W/n = \frac{r}{n-K} F$.

Chapter 8

Examples of Maximum Likelihood

Solutions to all the exercises of Chapter 8 of Fumio Hayashi, *Econometrics* (Princeton University Press, 2000): the fourteen review questions of Sections 8.1–8.7 (three on qualitative response models, two on truncated regression, three on Tobit, one on multivariate regression, three on FIML, one on LIML and one on serially correlated observations) and the four analytical exercises (the determinant inequality behind multivariate ML, positive definiteness of $\hat{\Omega}(\Pi)$, identification of δ in FIML, and instruments that do not appear in the system). This chapter has no empirical or Monte Carlo exercise, so there is no `code/ch8.py`. Assumptions, propositions and equations cited as PROPOSITION 7.6, (8.2.15) or EXAMPLE 8.1 refer to the book. Following the book, θ_0 denotes the true parameter value, t the observation index, ϕ and Φ the standard normal density and c.d.f., and $\lambda(v) \equiv \phi(v)/[1 - \Phi(v)]$ the inverse Mill's ratio.

Section 8.1 Qualitative Response (QR) Models

Review Question 8.1.1

(a) With a general c.d.f. the log conditional likelihood for observation t is, as in (8.1.4),

$$\log f(y_t | \mathbf{x}_t; \theta) = y_t \log F_t + (1 - y_t) \log(1 - F_t).$$

Differentiating with respect to θ and using $\partial F_t / \partial \theta = f_t \mathbf{x}_t$,

$$\mathbf{s}(\mathbf{w}_t; \theta) = \left[\frac{y_t}{F_t} - \frac{1 - y_t}{1 - F_t} \right] f_t \mathbf{x}_t = \frac{y_t(1 - F_t) - (1 - y_t)F_t}{F_t(1 - F_t)} f_t \mathbf{x}_t = \frac{y_t - F_t}{F_t(1 - F_t)} f_t \mathbf{x}_t,$$

which is the first display. For the Hessian, write the score as $a(v_t) \mathbf{x}_t$ with $v_t \equiv \mathbf{x}_t' \theta$ and $a(v) \equiv (y_t - F)f/[F(1 - F)]$, so that $\mathbf{H} = a'(v_t) \mathbf{x}_t \mathbf{x}_t'$. Differentiating a , using $F' = f$ and $[F(1 - F)]' = f(1 - 2F)$,

$$a'(v) = \frac{-f^2 + (y_t - F)f'}{F(1 - F)} - \frac{(y_t - F)f^2(1 - 2F)}{[F(1 - F)]^2}.$$

This agrees with the expression in the question, $-[(y_t - F)/(F(1 - F))]^2 f^2 + [(y_t - F)/(F(1 - F))] f'$, because the difference of the two non- f' terms is

$$\begin{aligned} & -\frac{f^2}{F(1 - F)} - \frac{(y_t - F)f^2(1 - 2F)}{[F(1 - F)]^2} + \frac{(y_t - F)^2 f^2}{[F(1 - F)]^2} \\ &= \frac{f^2}{[F(1 - F)]^2} \left[-F(1 - F) - (y_t - F)(1 - 2F) + (y_t - F)^2 \right] = 0, \end{aligned}$$

the bracket vanishing because y_t is binary, so $y_t^2 = y_t$:

$$-F + F^2 - y_t + 2y_t F + F - 2F^2 + y_t^2 - 2y_t F + F^2 = y_t^2 - y_t = 0.$$

So the stated Hessian holds for binary y_t — the only case in which the model is defined.

(b) Condition on $\mathbf{x}_t$ and evaluate at θ_0 , where $E(y_t | \mathbf{x}_t) = F_t$ and $\text{Var}(y_t | \mathbf{x}_t) = F_t(1 - F_t)$ (a Bernoulli variable). For the outer product of the score,

$$E[\mathbf{s}\mathbf{s}' | \mathbf{x}_t] = \frac{f_t^2}{[F_t(1 - F_t)]^2} E[(y_t - F_t)^2 | \mathbf{x}_t] \mathbf{x}_t \mathbf{x}_t' = \frac{f_t^2}{F_t(1 - F_t)} \mathbf{x}_t \mathbf{x}_t'.$$

For the Hessian, the second term has conditional expectation zero, because $E(y_t - F_t | \mathbf{x}_t) = 0$ and f_t' is a function of $\mathbf{x}_t$; the first term gives the same expression with a minus sign. Hence

$$E[\mathbf{s}(\mathbf{w}_t; \theta_0) \mathbf{s}(\mathbf{w}_t; \theta_0)' | \mathbf{x}_t] = -E[\mathbf{H}(\mathbf{w}_t; \theta_0) | \mathbf{x}_t] = \frac{f_t^2}{F_t(1 - F_t)} \mathbf{x}_t \mathbf{x}_t',$$

which is (8.1.12). Taking unconditional expectations gives the (unconditional) information matrix equality, which is condition (3) of PROPOSITION 7.9. Note where the two ingredients enter: the Bernoulli variance delivers the first equality and the conditional mean-zero property of $y_t - F_t$ kills the f_t' term — it is only *in expectation* that the Hessian loses its dependence on y_t .

Review Question 8.1.2

Specialize Review Question 8.1.1 to $F = \Lambda$, using the two identities (8.1.6),

$$\Lambda'(v) = \Lambda(v)[1 - \Lambda(v)], \quad \Lambda''(v) = [1 - 2\Lambda(v)]\Lambda(v)[1 - \Lambda(v)].$$

For the score, $f_t = \Lambda_t(1 - \Lambda_t)$ cancels the denominator:

$$\mathbf{s}(\mathbf{w}_t; \theta) = \frac{y_t - \Lambda_t}{\Lambda_t(1 - \Lambda_t)} \Lambda_t(1 - \Lambda_t) \mathbf{x}_t = [y_t - \Lambda(\mathbf{x}_t'; \theta)] \mathbf{x}_t,$$

which is (8.1.7). For the Hessian, substitute $f_t = \Lambda_t(1 - \Lambda_t)$ and $f_t' = (1 - 2\Lambda_t)\Lambda_t(1 - \Lambda_t)$ into the general formula:

$$\mathbf{H} = [-(y_t - \Lambda_t)^2 + (y_t - \Lambda_t)(1 - 2\Lambda_t)] \mathbf{x}_t \mathbf{x}_t'.$$

Expanding the bracket and using $y_t^2 = y_t$,

$$-(y_t - 2y_t\Lambda_t + \Lambda_t^2) + (y_t - 2y_t\Lambda_t - \Lambda_t + 2\Lambda_t^2) = \Lambda_t^2 - \Lambda_t = -\Lambda_t(1 - \Lambda_t),$$

so $\mathbf{H}(\mathbf{w}_t; \theta) = -\Lambda(\mathbf{x}_t'; \theta)[1 - \Lambda(\mathbf{x}_t'; \theta)] \mathbf{x}_t \mathbf{x}_t'$, which is (8.1.8).

Two features of the logit model follow immediately. The Hessian does not involve y_t at all, so the observed and expected information coincide and the conditional information matrix equality is trivial; and since $\Lambda(1 - \Lambda) > 0$ and $\mathbf{x}_t \mathbf{x}_t'$ is positive semidefinite, $\mathbf{H}$ is negative semidefinite everywhere, so the log likelihood is concave and PROPOSITION 7.6 applies.

Review Question 8.1.3

Yes. The consistency theorem used in the text, PROPOSITION 7.6, requires the objective function to be a sample average of a concave function of the parameter over an ergodic stationary process, plus identification and a dominance condition — none of which needs a random sample. The concavity was verified in Review Question 8.1.2, the identification condition follows as in the text from the nonsingularity of $E(\mathbf{x}_t \mathbf{x}_t')$, and $\{y_t, \mathbf{x}_t\}$ is ergodic stationary by hypothesis. So $\hat{\theta} \xrightarrow{P} \theta_0$.

But the asymptotic variance formula changes. With serial dependence the score $\{\mathbf{s}(\mathbf{w}_t; \theta_0)\}$ need no longer be a martingale difference sequence, so $\text{Avar}(\sqrt{n} \bar{\mathbf{s}})$ is the long-run variance

$\Sigma = \sum_{j=-\infty}^{\infty} E[\mathbf{s}_t \mathbf{s}'_{t-j}]$ rather than $E[\mathbf{s}_t \mathbf{s}'_t]$, the information matrix equality no longer collapses the sandwich, and

$$\text{Avar}(\hat{\theta}) = \Sigma_{\mathbf{H}}^{-1} \Sigma \Sigma_{\mathbf{H}}^{-1}, \quad \Sigma_{\mathbf{H}} \equiv -E[\mathbf{H}(\mathbf{w}_t; \theta_0)],$$

which is *not* estimated by $-\left[\frac{1}{n} \sum_t \mathbf{H}(\mathbf{w}_t; \hat{\theta})\right]^{-1}$ as in (8.1.13). The middle matrix must be estimated by a long-run variance estimator (Chapter 6). This is the general lesson of Section 8.7: serial correlation leaves the quasi-ML *estimator* alone and attacks its *standard errors*.

Section 8.2 Truncated Regression Models

Review Question 8.2.1

Write $\mu \equiv \mathbf{x}'_t \beta_0$, $\sigma \equiv \sigma_0$ and $\lambda \equiv \lambda(v_t)$ with $v_t = (c - \mu)/\sigma$. By (8.2.6) and (8.2.7),

$$E(y_t | \cdot) = \mu + \sigma \lambda, \quad \text{Var}(y_t | \cdot) = \sigma^2 [1 - \lambda(\lambda - v_t)],$$

so that

$$E(y_t^2 | \cdot) = \text{Var} + [E]^2 = \sigma^2 - \sigma^2 \lambda^2 + \sigma^2 \lambda v_t + \mu^2 + 2\mu\sigma\lambda + \sigma^2 \lambda^2 = \sigma^2 + \sigma^2 \lambda v_t + \mu^2 + 2\mu\sigma\lambda.$$

Now pass to the reparameterization (8.2.13), $\delta = \beta/\sigma$ and $\gamma = 1/\sigma$, so that $\sigma = 1/\gamma$ and $\mu = \mathbf{x}'_t \delta/\gamma$:

$$E(y_t^2 | \cdot) = \frac{1}{\gamma^2} [1 + \lambda v_t + (\mathbf{x}'_t \delta)^2 + 2\lambda \mathbf{x}'_t \delta].$$

Finally substitute $v_t = \gamma c - \mathbf{x}'_t \delta$, which turns $\lambda v_t + 2\lambda \mathbf{x}'_t \delta$ into $\lambda \gamma c + \lambda \mathbf{x}'_t \delta$:

$$E(y_t^2 | \mathbf{x}_t, t \text{ in sample}) = \frac{1}{\gamma^2} [1 + \lambda(v_t) \gamma c + (\mathbf{x}'_t \delta)^2 + \lambda(v_t) \mathbf{x}'_t \delta],$$

as claimed. The formula is the second moment that the GMM estimator of the next question matches; note that, like the first moment, it depends on the parameters only through the reparameterized (δ, γ) — which is the reason for introducing them.

Review Question 8.2.2

(a) By (8.2.6) in the reparameterized notation of the previous question,

$$E(y_t | \mathbf{x}_t, t \text{ in sample}) = \frac{1}{\gamma_0} \mathbf{x}'_t \delta_0 + \frac{\lambda(v_t)}{\gamma_0},$$

so the scalar in parentheses in (8.2.19) has conditional mean zero. Since $\mathbf{x}_t$ — and $v_t = \gamma_0 c - \mathbf{x}'_t \delta_0$ with it — is a function of the conditioning variable, the law of total expectations gives

$$E[\mathbf{g}_1(\mathbf{w}_t; \delta_0, \gamma_0)] = E\left[\mathbf{x}_t E\left(y_t - \frac{\mathbf{x}'_t \delta_0}{\gamma_0} - \frac{\lambda(v_t)}{\gamma_0} \mid \mathbf{x}_t\right)\right] = \mathbf{0}.$$

Exactly the same argument applied to Review Question 8.2.1 gives $E[g_2(\mathbf{w}_t; \delta_0, \gamma_0)] = 0$: the bracket in (8.2.20) is the conditional second moment of y_t .

(b) Write $\mathbf{g}_1$ and $\mathbf{g}_2$ for the two blocks evaluated at an arbitrary (δ, γ) with $\gamma > 0$, and $\mathbf{s} = (\mathbf{s}'_1, s_2)'$ for the score (8.2.15). Two algebraic identities link them. First, the top block of (8.2.15) is

$$\mathbf{s}_1 = (\gamma y_t - \mathbf{x}'_t \delta) \mathbf{x}_t - \lambda(v_t) \mathbf{x}_t = \gamma \left[y_t - \frac{\mathbf{x}'_t \delta}{\gamma} - \frac{\lambda(v_t)}{\gamma} \right] \mathbf{x}_t = \gamma \mathbf{g}_1.$$

Second, for the bottom block, multiply out

$$-\gamma s_2 = -1 + \gamma(\gamma y_t - \mathbf{x}'_t \delta) y_t - \gamma c \lambda(v_t), \quad \gamma^2 g_2 = \gamma^2 y_t^2 - 1 - \lambda(v_t) \gamma c - (\mathbf{x}'_t \delta)^2 - \lambda(v_t) \mathbf{x}'_t \delta,$$

and subtract:

$$\gamma^2 g_2 - (-\gamma s_2) = -(\mathbf{x}'_t \delta)^2 - \lambda(v_t) \mathbf{x}'_t \delta + \gamma(\mathbf{x}'_t \delta) y_t = (\mathbf{x}'_t \delta) [\gamma y_t - \mathbf{x}'_t \delta - \lambda(v_t)] = \gamma \delta' \mathbf{g}_1,$$

the last step by the first identity. Dividing by γ ,

$$\mathbf{s}_1 = \gamma \mathbf{g}_1, \quad s_2 = \delta' \mathbf{g}_1 - \gamma g_2$$

observation by observation. Summing over t : if the $K + 1$ sample orthogonality conditions hold, then $\sum_t \mathbf{s}_1 = \gamma \sum_t \mathbf{g}_1 = \mathbf{0}$ and $\sum_t s_2 = \delta' \sum_t \mathbf{g}_1 - \gamma \sum_t g_2 = 0$, so the likelihood equations hold; conversely, if the likelihood equations hold, then $\sum_t \mathbf{g}_1 = \frac{1}{\gamma} \sum_t \mathbf{s}_1 = \mathbf{0}$ and $\sum_t g_2 = \frac{1}{\gamma} (\delta' \sum_t \mathbf{g}_1 - \sum_t s_2) = 0$. The two systems are therefore equivalent, and (the equation being exactly identified, with $K + 1$ moment conditions for $K + 1$ parameters) the GMM estimator is numerically the ML estimator. The moment conditions are of interest nonetheless: they show which features of the distribution ML is exploiting — the first two conditional moments — and they suggest robust alternatives that add moments or drop the normality that produced $\lambda(\cdot)$.

Section 8.3 Censored Regression (Tobit) Models

Review Question 8.3.1

Write $v \equiv (c - \mathbf{x}'_t \beta_0) / \sigma_0$. In the censored model $y_t = \max(y_t^*, c)$, so conditional on $\mathbf{x}_t$ the variable y_t takes the value c when $y_t^* \leq c$ — an event of probability $\Phi(v)$ by (8.3.5) — and equals y_t^* otherwise, an event of probability $1 - \Phi(v)$ by (8.2.10). Splitting the expectation over these two events,

$$E(y_t | \mathbf{x}_t) = \text{Prob}(y_t^* > c | \mathbf{x}_t) E(y_t^* | \mathbf{x}_t, y_t^* > c) + \text{Prob}(y_t^* \leq c | \mathbf{x}_t) \cdot c.$$

The first conditional expectation is exactly the truncated mean (8.2.6), $\mathbf{x}'_t \beta_0 + \sigma_0 \lambda(v)$ — conditional on being above c , y_t has precisely the truncated distribution of the previous section. Substituting the two probabilities gives the display.

The formula shows how censoring differs from truncation. In the truncated model the sample contains only the observations with $y_t^* > c$, and the conditional mean is $\mathbf{x}'_t \beta_0 + \sigma_0 \lambda(v)$; in the censored model the sub-threshold observations are retained at the value c , and their probability mass pulls the conditional mean towards c . Both means are nonlinear in $\mathbf{x}_t$, which is why OLS is inconsistent in either case, and both collapse to $\mathbf{x}'_t \beta_0$ as $c \rightarrow -\infty$.

Review Question 8.3.2

Use the two identities given in the hint to write each of the matrices in (8.3.12) as an outer product:

$$\begin{bmatrix} \mathbf{x}_t \mathbf{x}'_t & -y_t \mathbf{x}_t \\ -y_t \mathbf{x}'_t & \frac{1}{\gamma^2} + y_t^2 \end{bmatrix} = \begin{bmatrix} \mathbf{x}_t \\ -y_t \end{bmatrix} \begin{bmatrix} \mathbf{x}'_t & -y_t \end{bmatrix} + \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0}' & \frac{1}{\gamma^2} \end{bmatrix}, \quad \begin{bmatrix} \mathbf{x}_t \mathbf{x}'_t & -c \mathbf{x}_t \\ -c \mathbf{x}'_t & c^2 \end{bmatrix} = \begin{bmatrix} \mathbf{x}_t \\ -c \end{bmatrix} \begin{bmatrix} \mathbf{x}'_t & -c \end{bmatrix}.$$

Both right-hand sides are positive semidefinite: an outer product $\mathbf{a} \mathbf{a}'$ is positive semidefinite, as is the diagonal matrix with the single positive entry $1/\gamma^2$, and sums of positive semidefinite matrices

are positive semidefinite. Hence (8.3.12) has the form

$$\mathbf{H}(\mathbf{w}_t; \delta, \gamma) = -(1 - D_t) \mathbf{A}_t - D_t \lambda(-v_t) [\lambda(-v_t) + v_t] \mathbf{B}_t, \quad \mathbf{A}_t, \mathbf{B}_t \text{ positive semidefinite.}$$

The weights are nonnegative: D_t and $1 - D_t$ are zero or one by (8.3.7); $\lambda(\cdot) > 0$ always; and $\lambda(-v_t) + v_t \geq 0$, which is the hint's inequality $\lambda(v) \geq v$ evaluated at $-v_t$ (the inverse Mill's ratio dominates the identity function, because λ is positive and convex with asymptote v). A nonnegative combination of positive semidefinite matrices, with a minus sign in front, is negative semidefinite:

$$\mathbf{c}' \mathbf{H} \mathbf{c} = -(1 - D_t) \mathbf{c}' \mathbf{A}_t \mathbf{c} - D_t \lambda(-v_t) [\lambda(-v_t) + v_t] \mathbf{c}' \mathbf{B}_t \mathbf{c} \leq 0 \quad \text{for every } \mathbf{c}.$$

So the reparameterized Tobit log likelihood is concave in (δ, γ) — Olsen's (1978) result, stated in the text — and the relevant consistency theorem is PROPOSITION 7.6. Concavity is a property of the *reparameterized* likelihood only: in terms of (β, σ^2) the objective is not concave, which is precisely why the reparameterization (8.2.13) is worth its inconvenience.

Review Question 8.3.3

Yes, the Tobit ML estimator remains consistent. As in Review Question 8.1.3, consistency follows from PROPOSITION 7.6, whose hypotheses are ergodic stationarity, concavity of the per-observation objective (established in Review Question 8.3.2), identification and a dominance condition; the random sample assumption is used nowhere. The observations being serially correlated changes neither the concavity nor the fact that θ_0 uniquely maximizes $E[\log \tilde{f}(y_t | \mathbf{x}_t; \delta, \gamma)]$.

What fails, again, is the *variance* formula: the score is no longer serially uncorrelated, so $\text{Avar}(\hat{\theta})$ is the sandwich $\Sigma_{\mathbf{H}}^{-1} \Sigma \Sigma_{\mathbf{H}}^{-1}$ with Σ the long-run variance of the score, and the inverse-Hessian formula quoted at the end of Section 8.3 understates or overstates the true sampling uncertainty. Equally important in practice, the Tobit *likelihood* itself is misspecified when the latent errors are serially correlated and one still multiplies the per-observation densities: the estimator is then a quasi-ML estimator, consistent for the parameters of the conditional density of y_t given $\mathbf{x}_t$ but not efficient.

Section 8.4 Multivariate Regressions

Review Question 8.4.1

Each summand $(\mathbf{y}_t - \Pi' \mathbf{x}_t)' \Omega^{-1} (\mathbf{y}_t - \Pi' \mathbf{x}_t)$ is a scalar, so it equals its own trace; then cyclically permute the factors inside the trace:

$$(\mathbf{y}_t - \Pi' \mathbf{x}_t)' \Omega^{-1} (\mathbf{y}_t - \Pi' \mathbf{x}_t) = \text{tr}[(\mathbf{y}_t - \Pi' \mathbf{x}_t)' \Omega^{-1} (\mathbf{y}_t - \Pi' \mathbf{x}_t)] = \text{tr}[\Omega^{-1} (\mathbf{y}_t - \Pi' \mathbf{x}_t) (\mathbf{y}_t - \Pi' \mathbf{x}_t)'].$$

Averaging over t and using the linearity of the trace,

$$\frac{1}{2n} \sum_{t=1}^n (\mathbf{y}_t - \Pi' \mathbf{x}_t)' \Omega^{-1} (\mathbf{y}_t - \Pi' \mathbf{x}_t) = \frac{1}{2} \text{tr} \left[\Omega^{-1} \cdot \frac{1}{n} \sum_{t=1}^n (\mathbf{y}_t - \Pi' \mathbf{x}_t) (\mathbf{y}_t - \Pi' \mathbf{x}_t)' \right] = \frac{1}{2} \text{tr} [\Omega^{-1} \hat{\Omega}(\Pi)],$$

by the definition (8.4.10) of $\hat{\Omega}(\Pi)$. This is (8.4.9), and substituting it into (8.4.8) gives the form (8.4.11) of the objective function in which the data enter only through the $M \times M$ matrix $\hat{\Omega}(\Pi)$ — the fact that makes the two-step concentration of the likelihood possible.

Remark. The text says this is "left as Review Question 2", but Section 8.4 has only one review question, the present one.

Section 8.5 FIML

Review Question 8.5.1

Conditional on $\mathbf{x}_t$, $\boldsymbol{\varepsilon}_t \sim N(\mathbf{0}, \Sigma_0)$ with density

$$f_{\boldsymbol{\varepsilon}}(\boldsymbol{\varepsilon}) = (2\pi)^{-M/2} |\Sigma_0|^{-1/2} \exp\left[-\frac{1}{2} \boldsymbol{\varepsilon}' \Sigma_0^{-1} \boldsymbol{\varepsilon}\right].$$

The structural form (8.5.8) defines the one-to-one mapping $\mathbf{y}_t = \mathbf{g}(\boldsymbol{\varepsilon}_t) = \Gamma_0^{-1}(\boldsymbol{\varepsilon}_t - \mathbf{B}_0 \mathbf{x}_t)$, whose inverse is $\mathbf{g}^{-1}(\mathbf{y}) = \Gamma_0 \mathbf{y} + \mathbf{B}_0 \mathbf{x}_t$; the mapping is one-to-one because Γ_0 is nonsingular (the second completeness requirement). The Jacobian in the change of variable formula is $\mathbf{J}(\boldsymbol{\varepsilon}) = \partial \mathbf{g}(\boldsymbol{\varepsilon}) / \partial \boldsymbol{\varepsilon}' = \Gamma_0^{-1}$, so $\text{abs}(\mathbf{J}) = 1/\text{abs}(|\Gamma_0|)$ and

$$f(\mathbf{y}_t | \mathbf{x}_t) = \frac{f_{\boldsymbol{\varepsilon}}(\Gamma_0 \mathbf{y}_t + \mathbf{B}_0 \mathbf{x}_t)}{\text{abs}(|\Gamma_0^{-1}|)} = \text{abs}(|\Gamma_0|) (2\pi)^{-M/2} |\Sigma_0|^{-1/2} \exp\left[-\frac{1}{2} \boldsymbol{\varepsilon}_t' \Sigma_0^{-1} \boldsymbol{\varepsilon}_t\right], \quad \boldsymbol{\varepsilon}_t = \Gamma_0 \mathbf{y}_t + \mathbf{B}_0 \mathbf{x}_t.$$

Taking logs, replacing the true parameters by hypothetical values and using $\log \text{abs}(|\Gamma|) = \frac{1}{2} \log(|\Gamma|^2)$ — which keeps the expression differentiable and sign-free —

$$\log f(\mathbf{y}_t | \mathbf{x}_t; \delta, \Sigma) = -\frac{M}{2} \log(2\pi) + \frac{1}{2} \log(|\Gamma|^2) - \frac{1}{2} \log(|\Sigma|) - \frac{1}{2} [\Gamma \mathbf{y}_t + \mathbf{B} \mathbf{x}_t]' \Sigma^{-1} [\Gamma \mathbf{y}_t + \mathbf{B} \mathbf{x}_t],$$

which is (8.5.16); averaging over t gives (8.5.20). The Jacobian term $\frac{1}{2} \log(|\Gamma|^2)$ is what distinguishes FIML from the multivariate regression of Section 8.4: it is the price of writing the model in structural rather than reduced form, and it is the reason the FIML objective is not simply a sum of squares.

Review Question 8.5.2

The coefficient on x_{tK} in equation m is the (m, K) element of $\mathbf{B}_0$, so " x_{tK} appears in no equation" means that the K -th column of $\mathbf{B}_0$ is zero. By (8.5.10), $\Pi'_0 = -\Gamma_0^{-1} \mathbf{B}_0$, so the K -th column of Π'_0 — equivalently the K -th row of Π_0 — is zero:

$$\Pi_0 = \begin{bmatrix} \tilde{\Pi}_0 \\ \mathbf{0}' \end{bmatrix}, \quad \tilde{\Pi}_0 \text{ of size } (K-1) \times M.$$

By the reduced form (8.5.9), $\mathbf{y}_t = \Pi'_0 \mathbf{x}_t + \mathbf{v}_t$ with $E(\mathbf{x}_t \mathbf{v}_t') = \mathbf{0}$, so the least squares projection of y_{tm} on $\mathbf{x}_t$ is exactly the m -th reduced-form equation:

$$\hat{E}^*(y_{tm} | \mathbf{x}_t) = \boldsymbol{\pi}'_{0m} \mathbf{x}_t = \tilde{\boldsymbol{\pi}}'_{0m} \tilde{\mathbf{x}}_t + 0 \cdot x_{tK} = \hat{E}^*(y_{tm} | \tilde{\mathbf{x}}_t),$$

where $\boldsymbol{\pi}_{0m}$ is the m -th column of Π_0 . (The last equality holds because a projection whose coefficient on one regressor is zero is also the projection on the remaining regressors.)

In words: a predetermined variable that enters no structural equation has no reduced-form effect either, so it carries no information about the endogenous variables beyond what the other instruments carry. Analytical Exercise 8.4 turns this observation into the statement that dropping x_{tK} from the instrument list costs nothing asymptotically — while keeping it costs something in finite samples, since the first-stage projections are then estimated with an extra, useless, coefficient.

Review Question 8.5.3

Suppose, contrary to the claim, that y_{t1} and y_{t2} are the same variable. Subtracting the second structural equation from the first,

$$\varepsilon_{t1} - \varepsilon_{t2} = \mathbf{z}'_{t1}\delta_{01} - \mathbf{z}'_{t2}\delta_{02}.$$

In the SUR model every regressor is predetermined, so $\mathbf{z}_{t1}$ and $\mathbf{z}_{t2}$ are subvectors of the common instrument vector $\mathbf{x}_t$, and the right-hand side can be written as $\mathbf{x}'_t\boldsymbol{\alpha}$ for a suitable K -dimensional $\boldsymbol{\alpha}$. Multiplying by $\mathbf{x}_t$ and taking expectations, the orthogonality conditions $E(\mathbf{x}_t\varepsilon'_t) = \mathbf{0}$ give

$$\mathbf{0} = E[\mathbf{x}_t(\varepsilon_{t1} - \varepsilon_{t2})] = E(\mathbf{x}_t\mathbf{x}'_t)\boldsymbol{\alpha} \implies \boldsymbol{\alpha} = \mathbf{0},$$

because $E(\mathbf{x}_t\mathbf{x}'_t)$ is nonsingular. Hence $\varepsilon_{t1} = \varepsilon_{t2}$ with probability one, so the first two rows (and columns) of $\Sigma_0 = E(\varepsilon_t\varepsilon'_t)$ coincide and Σ_0 is singular — contradicting the assumption that it is positive definite. Therefore no two elements of $(y_{t1}, \dots, y_{tM})$ can be the same variable.

The contrast with EXAMPLE 8.1 is instructive: there y_{t1} and y_{t2} are the same variable (both equations explain quantity), but that is a simultaneous-equations system, in which $\mathbf{z}_{t1}$ and $\mathbf{z}_{t2}$ contain the endogenous price and the argument above breaks down at its first step. It is the assumption that all regressors are predetermined — what makes a system a SUR system — that forbids duplicated dependent variables.

Section 8.6 LIML

Review Question 8.6.1

(a) The system is $\mathbf{y}_t = \mathbf{Z}_t\boldsymbol{\delta}_0 + \boldsymbol{\eta}_t$ with

$$\mathbf{y}_t = \begin{bmatrix} y_{t1} \\ y_{t2} \end{bmatrix}, \quad \mathbf{Z}_t = \begin{bmatrix} y_{t2} & 0 \\ 0 & x_t \end{bmatrix}, \quad \boldsymbol{\delta}_0 = \begin{bmatrix} \gamma_0 \\ \beta_0 \end{bmatrix}, \quad \boldsymbol{\eta}_t = \begin{bmatrix} \varepsilon_{t1} \\ v_t \end{bmatrix}.$$

With Σ_0 known, the SUR (i.e. GLS) estimator is (4.5.12), $\hat{\boldsymbol{\delta}} = [\sum_t \mathbf{Z}'_t \Sigma_0^{-1} \mathbf{Z}_t]^{-1} \sum_t \mathbf{Z}'_t \Sigma_0^{-1} \mathbf{y}_t$, so its sampling error is

$$\hat{\boldsymbol{\delta}} - \boldsymbol{\delta}_0 = \left[\frac{1}{n} \sum_t \mathbf{Z}'_t \Sigma_0^{-1} \mathbf{Z}_t \right]^{-1} \frac{1}{n} \sum_t \mathbf{Z}'_t \Sigma_0^{-1} \boldsymbol{\eta}_t.$$

Writing σ^{mh} for the (m, h) element of Σ_0^{-1} and multiplying out the two blocks,

$$\begin{bmatrix} \hat{\gamma} \\ \hat{\beta} \end{bmatrix} - \begin{bmatrix} \gamma_0 \\ \beta_0 \end{bmatrix} = \begin{bmatrix} \sigma^{11} \frac{1}{n} \sum y_{t2}^2 & \sigma^{12} \frac{1}{n} \sum y_{t2} x_t \\ \sigma^{21} \frac{1}{n} \sum x_t y_{t2} & \sigma^{22} \frac{1}{n} \sum x_t^2 \end{bmatrix}^{-1} \begin{bmatrix} \sigma^{11} \frac{1}{n} \sum y_{t2} \varepsilon_{t1} + \sigma^{12} \frac{1}{n} \sum y_{t2} v_t \\ \sigma^{21} \frac{1}{n} \sum x_t \varepsilon_{t1} + \sigma^{22} \frac{1}{n} \sum x_t v_t \end{bmatrix},$$

which is the expression in the question.

(b) By the ergodic theorem (the observations are i.i.d. here), the two averages in the first element converge to

$$\frac{1}{n} \sum_t y_{t2} \varepsilon_{t1} \xrightarrow{p} E(y_{t2} \varepsilon_{t1}) = \sigma_{12}, \quad \frac{1}{n} \sum_t y_{t2} v_t \xrightarrow{p} E(y_{t2} v_t) = E(v_t^2) = \sigma_{22},$$

using $y_{t2} = \beta_0 x_t + v_t$ together with $E(x_t \varepsilon_{t1}) = 0$ and $E(x_t v_t) = 0$. Hence the first element of the score converges in probability to

$$\sigma^{11}\sigma_{12} + \sigma^{12}\sigma_{22},$$

which is the $(1, 2)$ element of $\Sigma_0^{-1}\Sigma_0 = \mathbf{I}_2$ — that is, *zero*. The second element converges to zero directly, since x_t is predetermined in both equations. With the coefficient matrix converging to a nonsingular limit, $\hat{\delta} \xrightarrow{P} \delta_0$: SUR consistently estimates the structural coefficient γ_0 even though its regressor y_{t2} is endogenous in the first equation.

The mechanism is worth stating plainly. SUR does not pretend that y_{t2} is uncorrelated with ε_{t1} ; it exploits the fact that, *given the whole system*, the correlation is known — $E(y_{t2}\varepsilon_{t1}) = \sigma_{12}$ — and the GLS weights σ^{11}, σ^{12} are precisely the ones that make the two contaminated moments cancel. This is why the full-system estimators (SUR, 3SLS, FIML) can identify coefficients on endogenous regressors, and equally why they are fragile: if any equation of the system is misspecified, Σ_0 is wrong and the cancellation fails, which is the argument for the single-equation LIML of this section.

Section 8.7 Serially Correlated Observations

Review Question 8.7.1

(a) With $\beta = (c, \phi)'$ and $\mathbf{x}_t = (1, y_{t-1})'$, (8.7.17) reads

$$m(\mathbf{w}_t; \theta) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} (y_t - \mathbf{x}_t' \beta)^2.$$

Differentiating with respect to β and to σ^2 (treating σ^2 , not σ , as the parameter),

$$\mathbf{s}(\mathbf{w}_t; \theta) = \begin{bmatrix} \frac{1}{\sigma^2} \mathbf{x}_t \cdot (y_t - \mathbf{x}_t' \beta) \\ -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4} (y_t - \mathbf{x}_t' \beta)^2 \end{bmatrix},$$

as stated in the question.

(b) At $\theta = \theta_0$ we have $y_t - \mathbf{x}_t' \beta_0 = \varepsilon_t$, where ε_t is independent of $(y_{t-1}, y_{t-2}, \dots)$ with mean zero and variance σ_0^2 , while $\mathbf{x}_t$ is a function of y_{t-1} . Hence

$$E\left[\frac{\mathbf{x}_t \varepsilon_t}{\sigma_0^2} \mid y_{t-1}, \dots\right] = \frac{\mathbf{x}_t}{\sigma_0^2} E(\varepsilon_t) = \mathbf{0}, \quad E\left[-\frac{1}{2\sigma_0^2} + \frac{\varepsilon_t^2}{2\sigma_0^4} \mid y_{t-1}, \dots\right] = -\frac{1}{2\sigma_0^2} + \frac{\sigma_0^2}{2\sigma_0^4} = 0.$$

(c) For $j \geq 1$, $\mathbf{s}(\mathbf{w}_{t-j}; \theta_0)$ is a measurable function of $(y_{t-1}, y_{t-2}, \dots)$, so conditioning on the past scores is conditioning on a coarser information set. By the law of iterated expectations and (b),

$$E[\mathbf{s}(\mathbf{w}_t; \theta_0) \mid \mathbf{s}(\mathbf{w}_{t-1}; \theta_0), \mathbf{s}(\mathbf{w}_{t-2}; \theta_0), \dots] = E[E[\mathbf{s}(\mathbf{w}_t; \theta_0) \mid y_{t-1}, y_{t-2}, \dots] \mid \mathbf{s}_{t-1}, \dots] = \mathbf{0},$$

so $\{\mathbf{s}(\mathbf{w}_t; \theta_0)\}$ is a martingale difference sequence. Being also ergodic stationary — a fixed function of two consecutive coordinates of an ergodic stationary process — it satisfies the CLT of Chapter 2, which is condition (3) of PROPOSITION 7.8 with $\Sigma = E[\mathbf{s}\mathbf{s}']$ as in (8.7.18). This is what rescues the usual information-matrix sandwich: although the observations are serially correlated, the *score* is not, so $\text{Avar}(\hat{\theta}) = \Sigma^{-1}$ just as in the i.i.d. case.

(d) *Yes.* For a Gaussian AR(p) the only change is $\mathbf{x}_t = (1, y_{t-1}, \dots, y_{t-p})'$, still a function of the past, and $y_t - \mathbf{x}_t' \boldsymbol{\beta}_0 = \varepsilon_t$ is still independent of $(y_{t-1}, y_{t-2}, \dots)$. Both displays in (b) go through verbatim, and the argument of (c) is unchanged. The essential feature, as the text notes, is that the conditional density factorizes with $f(y_t | y_{t-1}, \dots, y_1, y_0) = f(y_t | y_{t-1}, \dots, y_{t-p})$: a correctly specified conditional likelihood always has a score that is a martingale difference sequence with respect to the conditioning information, whatever the serial dependence in the data.

Problem Set for Chapter 8: Analytical Exercises

Analytical Exercise 8.1

Add and subtract the fitted value: $\mathbf{y}_t - \Pi' \mathbf{x}_t = \hat{\mathbf{v}}_t + (\hat{\Pi} - \Pi)' \mathbf{x}_t$. Forming the outer product and summing,

$$\begin{aligned} \sum_t (\mathbf{y}_t - \Pi' \mathbf{x}_t)(\mathbf{y}_t - \Pi' \mathbf{x}_t)' &= \sum_t \hat{\mathbf{v}}_t \hat{\mathbf{v}}_t' + (\hat{\Pi} - \Pi)' \sum_t \mathbf{x}_t \hat{\mathbf{v}}_t' + \sum_t \hat{\mathbf{v}}_t \mathbf{x}_t' (\hat{\Pi} - \Pi) \\ &\quad + (\hat{\Pi} - \Pi)' \left(\sum_t \mathbf{x}_t \mathbf{x}_t' \right) (\hat{\Pi} - \Pi) \\ &= \sum_t \hat{\mathbf{v}}_t \hat{\mathbf{v}}_t' + (\hat{\Pi} - \Pi)' \left(\sum_t \mathbf{x}_t \mathbf{x}_t' \right) (\hat{\Pi} - \Pi), \end{aligned}$$

the two cross terms vanishing because $\sum_t \mathbf{x}_t \hat{\mathbf{v}}_t' = \mathbf{0}$: these are the normal equations of the equation-by-equation OLS regressions that define $\hat{\Pi}$. Dividing by n ,

$$\hat{\Omega}(\Pi) = \hat{\Omega}(\hat{\Pi}) + \mathbf{C}' \mathbf{C}, \quad \mathbf{C} \equiv \left(\frac{1}{n} \sum_t \mathbf{x}_t \mathbf{x}_t' \right)^{1/2} (\hat{\Pi} - \Pi),$$

so $\hat{\Omega}(\Pi)$ exceeds $\hat{\Omega}(\hat{\Pi})$ by a positive semidefinite matrix. The matrix result quoted in the exercise — $|\mathbf{A} + \mathbf{B}| \geq |\mathbf{A}|$ for positive semidefinite $\mathbf{A}$ and $\mathbf{B}$ — then gives

$$|\hat{\Omega}(\Pi)| \geq |\hat{\Omega}(\hat{\Pi})|,$$

which is the inequality to be shown. Since $\hat{\mathbf{v}}_t$ does not depend on Π , the right-hand side is a constant: the concentrated likelihood (8.4.13), which is a decreasing function of $|\hat{\Omega}(\Pi)|$, is maximized at $\Pi = \hat{\Pi}$. So the ML estimator of Π_0 is the equation-by-equation OLS estimator, and $\hat{\Omega} = \hat{\Omega}(\hat{\Pi})$, which is (8.4.12). The inequality is strict unless $\hat{\Pi} = \Pi$, because $\frac{1}{n} \sum_t \mathbf{x}_t \mathbf{x}_t'$ is positive definite with probability one for n large.

Analytical Exercise 8.2

Write the residual of the hypothetical Π in terms of the true one: $\mathbf{y}_t - \Pi' \mathbf{x}_t = \mathbf{v}_t + (\Pi_0 - \Pi)' \mathbf{x}_t$, where $\mathbf{v}_t = \mathbf{y}_t - \Pi_0' \mathbf{x}_t$ satisfies $E(\mathbf{x}_t \mathbf{v}_t') = \mathbf{0}$. Since $\{\mathbf{y}_t, \mathbf{x}_t\}$ is i.i.d., Kolmogorov's LLN gives $\hat{\Omega}(\Pi) \xrightarrow{a.s.} E[(\mathbf{y}_t - \Pi' \mathbf{x}_t)(\mathbf{y}_t - \Pi' \mathbf{x}_t)']$, and the cross terms in that expectation vanish by orthogonality:

$$E[(\mathbf{y}_t - \Pi' \mathbf{x}_t)(\mathbf{y}_t - \Pi' \mathbf{x}_t)'] = \Omega_0 + (\Pi_0 - \Pi)' E(\mathbf{x}_t \mathbf{x}_t') (\Pi_0 - \Pi).$$

The first term is positive definite by assumption and the second is positive semidefinite, so the sum is positive definite — equivalently, by the matrix result of Analytical Exercise 8.1, its determinant is at least $|\Omega_0| > 0$ and all its leading principal minors are bounded below by those of Ω_0 . Because

the set of positive definite matrices is open and $\widehat{\Omega}(\Pi)$ converges to a point of it almost surely, $\widehat{\Omega}(\Pi)$ is positive definite with probability one for sufficiently large n .

(Positive *semidefiniteness* holds in every sample and needs no limit argument: $\mathbf{c}'\widehat{\Omega}(\Pi)\mathbf{c} = \frac{1}{n} \sum_t [\mathbf{c}'(\mathbf{y}_t - \Pi'\mathbf{x}_t)]^2 \geq 0$. What the exercise adds is that the inequality is strict, which is what the two-step maximization of Section 8.4 needs: $\log |\widehat{\Omega}(\Pi)^{-1}|$ must be well defined, and the matrix inequality used in Step 1 requires a positive definite $\mathbf{B} = \widehat{\Omega}(\Pi)$.)

Analytical Exercise 8.3

Throughout, $\mathbf{S}_m$ selects the M_m endogenous regressors of equation m from $\mathbf{y}_t$ and $\mathbf{C}_m$ the K_m predetermined ones from $\mathbf{x}_t$, so that $\mathbf{z}'_{tm} = (\mathbf{y}'_t \mathbf{S}'_m : \mathbf{x}'_t \mathbf{C}'_m)$.

(a) Using the reduced form (8.5.9) and $E(\mathbf{x}_t \mathbf{v}'_t) = \mathbf{0}$, $E(\mathbf{x}_t \mathbf{y}'_t) = E(\mathbf{x}_t \mathbf{x}'_t) \Pi_0$. Hence

$$E(\mathbf{x}_t \mathbf{z}'_{tm}) = [E(\mathbf{x}_t \mathbf{y}'_t) \mathbf{S}'_m : E(\mathbf{x}_t \mathbf{x}'_t) \mathbf{C}'_m] = E(\mathbf{x}_t \mathbf{x}'_t) [\Pi_0 \mathbf{S}'_m : \mathbf{C}'_m],$$

which is (6). Since $E(\mathbf{x}_t \mathbf{x}'_t)$ is nonsingular, the rank condition " $E(\mathbf{x}_t \mathbf{z}'_{tm})$ has full column rank L_m " is equivalent to " $\text{rank}[\Pi_0 \mathbf{S}'_m : \mathbf{C}'_m] = L_m$ ".

(b) With $\mathbf{y}_t + \Gamma^{-1} \mathbf{B} \mathbf{x}_t = \mathbf{v}_t + (\Pi'_0 + \Gamma^{-1} \mathbf{B}) \mathbf{x}_t$ and $\{\mathbf{y}_t, \mathbf{x}_t\}$ i.i.d., the LLN gives

$$\text{plim}_{n \rightarrow \infty} \widehat{\Omega}(\delta) = E[(\mathbf{y}_t + \Gamma^{-1} \mathbf{B} \mathbf{x}_t)(\mathbf{y}_t + \Gamma^{-1} \mathbf{B} \mathbf{x}_t)'] = \Gamma_0^{-1} \Sigma_0 (\Gamma_0^{-1})' + \mathbf{D} E(\mathbf{x}_t \mathbf{x}'_t) \mathbf{D}'$$

with $\mathbf{D} \equiv \Pi'_0 + \Gamma^{-1} \mathbf{B}$, which is (7): the cross terms vanish by $E(\mathbf{x}_t \mathbf{v}'_t) = \mathbf{0}$, and $E(\mathbf{v}_t \mathbf{v}'_t) = \Gamma_0^{-1} \Sigma_0 (\Gamma_0^{-1})'$ by (8.5.11).

(c) The first term of (7) is positive definite and does not depend on δ ; the second is positive semidefinite. By the determinant inequality of Analytical Exercise 8.1, $|\text{plim} \widehat{\Omega}(\delta)|$ is minimized — and hence $Q_0(\delta) = -|\text{plim} \widehat{\Omega}(\delta)|$ maximized — only when the second term vanishes. Since $E(\mathbf{x}_t \mathbf{x}'_t)$ is positive definite, $\mathbf{D} E(\mathbf{x}_t \mathbf{x}'_t) \mathbf{D}' = \mathbf{0}$ forces $\mathbf{D} = \Pi'_0 + \Gamma^{-1} \mathbf{B} = \mathbf{0}$. Premultiplying by the nonsingular Γ ,

$$Q_0(\delta) \text{ is maximized } \iff \Gamma \Pi'_0 + \mathbf{B} = \mathbf{0}.$$

(d) The m -th structural equation is $y_{tm} = \mathbf{z}'_{tm} \delta_m + \varepsilon_{tm}$, i.e.

$$\varepsilon_{tm} = y_{tm} - \delta'_m \begin{bmatrix} \mathbf{S}_m \mathbf{y}_t \\ \mathbf{C}_m \mathbf{x}_t \end{bmatrix} = \left(\mathbf{e}'_m - \delta'_m \begin{bmatrix} \mathbf{S}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_m \end{bmatrix} \right) \begin{bmatrix} \mathbf{y}_t \\ \mathbf{x}_t \end{bmatrix},$$

and since the m -th row of $[\Gamma : \mathbf{B}]$ is by definition the row vector $\boldsymbol{\alpha}'_m$ with $\boldsymbol{\alpha}'_m (\mathbf{y}'_t, \mathbf{x}'_t)' = \varepsilon_{tm}$, this is (8). For EXAMPLE 8.1, with $(\mathbf{y}'_t, \mathbf{x}'_t) = (p_t, q_t, 1, a_t, w_t)$, $\mathbf{e}'_1 = (0, 1, 0, 0, 0)$, $\mathbf{S}_1 = [1 \ 0]$, $\mathbf{C}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ and $\delta_1 = (\gamma_{11}, \beta_{11}, \beta_{12})'$, the right-hand side of (8) is

$$(0, 1, 0, 0, 0) - (\gamma_{11}, 0, \beta_{11}, \beta_{12}, 0) = (-\gamma_{11}, 1, -\beta_{11}, -\beta_{12}, 0),$$

which is indeed the first row of $[\Gamma : \mathbf{B}]$ for the demand equation $q_t = \gamma_{11} p_t + \beta_{11} + \beta_{12} a_t + \varepsilon_{t1}$.

(e) $\Gamma\Pi'_0 + \mathbf{B} = \mathbf{0}$ is $[\Gamma : \mathbf{B}]\begin{bmatrix} \Pi'_0 \\ \mathbf{I}_K \end{bmatrix} = \mathbf{0}$, which is (9); its m -th row is $\alpha'_m[\Pi'_0] = \mathbf{0}'$. Substituting (8) and rearranging,

$$\delta'_m \begin{bmatrix} \mathbf{S}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_m \end{bmatrix} \begin{bmatrix} \Pi'_0 \\ \mathbf{I}_K \end{bmatrix} = \mathbf{e}'_m \begin{bmatrix} \Pi'_0 \\ \mathbf{I}_K \end{bmatrix} \iff \delta'_m \begin{bmatrix} \mathbf{S}_m \Pi'_0 \\ \mathbf{C}_m \end{bmatrix} = \boldsymbol{\pi}'_{0m},$$

with $\boldsymbol{\pi}_{0m} \equiv [\Pi_0 : \mathbf{I}_K]\mathbf{e}_m$, which are (10) and (11).

(f) $\delta_m = \delta_{0m}$ solves (10), because $(\Gamma_0, \mathbf{B}_0)$ satisfies $\Gamma_0\Pi'_0 + \mathbf{B}_0 = \mathbf{0}$ by the definition (8.5.10) of Π_0 . Transposing (10) turns it into a linear system $\mathbf{A}_m\delta_m = \boldsymbol{\pi}_{0m}$ with $\mathbf{A}_m \equiv [\Pi_0\mathbf{S}'_m : \mathbf{C}'_m]$ of size $K \times L_m$. By the fact recalled in the hint, a solvable system has a *unique* solution if and only if its coefficient matrix has full column rank; here that is $\text{rank}[\Pi_0\mathbf{S}'_m : \mathbf{C}'_m] = L_m$, which by (a) is exactly the rank condition for identification of equation m .

(g) Each element of δ appears exactly once among the elements of Γ and $\mathbf{B}$, and conversely every nonzero, non-normalized element of $[\Gamma : \mathbf{B}]$ is an element of some δ_m (this is (8) read in reverse). So the map $\delta \mapsto (\Gamma, \mathbf{B})$ is one-to-one, and $\Gamma\Pi'_0 + \mathbf{B} = \mathbf{0}$ has $(\Gamma_0, \mathbf{B}_0)$ as its only solution if and only if (10) has δ_{0m} as its only solution for every m . Combining with (c) and (f): $Q_0(\delta)$ is uniquely maximized at δ_0 if and only if the rank condition holds for every equation — the extremum estimator identification condition and the rank condition are the same condition.

Analytical Exercise 8.4

As in Review Question 8.5.2, x_{tK} appearing in no equation means the K -th column of $\mathbf{B}_0$ is zero, hence $\Pi_0 = \begin{bmatrix} \tilde{\Pi}_0 \\ \mathbf{0}' \end{bmatrix}$; and since x_{tK} is not a regressor of any equation, the K -th column of every selection matrix $\mathbf{C}_m$ is zero, i.e. the last row of $\mathbf{C}'_m$ is zero.

Rank condition. By (6), identification of equation m requires $\text{rank}[\Pi_0\mathbf{S}'_m : \mathbf{C}'_m] = L_m$. With the two zero blocks just noted, the last row of this $K \times L_m$ matrix is zero:

$$[\Pi_0\mathbf{S}'_m : \mathbf{C}'_m] = \begin{bmatrix} \tilde{\Pi}_0\mathbf{S}'_m & \tilde{\mathbf{C}}'_m \\ \mathbf{0}' & \mathbf{0}' \end{bmatrix},$$

where $\tilde{\mathbf{C}}_m$ is the selection matrix relative to $\tilde{\mathbf{x}}_t$. Deleting a zero row does not change the rank, so the rank condition with $\mathbf{x}_t$ is the rank condition with $\tilde{\mathbf{x}}_t$: equation m is identified with the smaller instrument set if and only if it is identified with the larger one.

Asymptotic variance. FIML is asymptotically equivalent to 3SLS, whose asymptotic variance (4.5.15) can be written, using (6), as

$$\text{Avar}(\hat{\delta}_{3\text{SLS}}) = \left[\mathbf{P}'(\Sigma_0^{-1} \otimes \text{E}(\mathbf{x}_t\mathbf{x}'_t))\mathbf{P} \right]^{-1}, \quad \mathbf{P} \equiv \text{diag}(\mathbf{P}_1, \dots, \mathbf{P}_M), \quad \mathbf{P}_m \equiv [\Pi_0\mathbf{S}'_m : \mathbf{C}'_m],$$

since $\text{E}(\mathbf{x}_t\mathbf{z}'_{tm}) = \text{E}(\mathbf{x}_t\mathbf{x}'_t)\mathbf{P}_m$ and the factors $\text{E}(\mathbf{x}_t\mathbf{x}'_t)$ cancel against those in $(\Sigma_0 \otimes \text{E}(\mathbf{x}_t\mathbf{x}'_t))^{-1}$. The building blocks of this expression are the matrices

$$\mathbf{P}'_m \text{E}(\mathbf{x}_t\mathbf{x}'_t) \mathbf{P}_h = \text{E}[\hat{\mathbf{E}}^*(\mathbf{z}_{tm} | \mathbf{x}_t) \hat{\mathbf{E}}^*(\mathbf{z}_{th} | \mathbf{x}_t)'],$$

the second moments of the least squares projections of the regressors on the instruments. By Review Question 8.5.2 those projections are unchanged when x_{tK} is dropped — the projection

of each endogenous regressor puts a zero coefficient on x_{tK} , and the predetermined regressors are elements of $\tilde{\mathbf{x}}_t$ and are projected on themselves. Every block of the matrix being inverted is therefore the same for $\mathbf{x}_t$ and for $\tilde{\mathbf{x}}_t$, and so is the asymptotic variance.

The practical message is the one the text draws: an instrument that belongs to no equation adds nothing asymptotically, while in finite samples it adds an extra estimated coefficient to every first-stage projection and so tends to make matters worse — the many-instruments problem in miniature.

Chapter 9

Unit-Root Econometrics

Solutions to all the exercises of Chapter 9 of Fumio Hayashi, *Econometrics* (Princeton University Press, 2000): the review questions of Sections 9.1–9.4, the seven analytical exercises (including the proofs of Propositions 9.2(b), 9.4, 9.5 and 9.7 and the derivation of the Phillips tests), the two Monte Carlo exercises on the finite-sample power and size of the Dickey–Fuller and ADF tests, and the empirical exercise on the Lothian–Taylor dollar/sterling real exchange rate, 1791–1990. Assumptions, propositions and equations cited as PROPOSITION 9.2 or (9.4.20) refer to the book. The numerical parts are computed by `code/ch9.py`, which also produces the figure.

Section 9.1 Modeling Trends

Review Question 9.1.1

Write $v_t \equiv \Delta u_t = u_t - u_{t-1}$. As the difference of two jointly covariance-stationary series, $\{v_t\}$ is covariance stationary with mean zero and $\gamma_j^v \equiv \text{Cov}(v_t, v_{t-j}) = 2\gamma_j^u - \gamma_{j-1}^u - \gamma_{j+1}^u$, where γ_j^u is the j -th autocovariance of $\{u_t\}$.

For the long-run variance use the hint's definition: since $\sum_{t=1}^T v_t = u_T - u_0$ telescopes,

$$\lambda_v^2 \equiv \lim_{T \rightarrow \infty} \text{Var}(\sqrt{T} \bar{v}) = \lim_{T \rightarrow \infty} \frac{\text{Var}(u_T - u_0)}{T} = \lim_{T \rightarrow \infty} \frac{\gamma_0^u + \gamma_0^u - 2\gamma_T^u}{T} = 0,$$

because $\gamma_T^u \rightarrow 0$ and $\gamma_0^u = \text{Var}(u_t) < \infty$. Hence the long-run variance of the first difference of any $I(0)$ process is zero: $\{\Delta u_t\}$ is what the text calls an $I(-1)$ process. This is precisely why DEFINITION 9.1 requires the long-run variance of an $I(0)$ process to be *positive*; otherwise i.i.d. white noise $\{\varepsilon_t\}$, whose first difference $\varepsilon_t - \varepsilon_{t-1}$ has long-run variance zero, would have to be classified as $I(1)$.

Remark. The same conclusion follows from the autocovariance-generating function: $v_t = (1 - L)u_t$ has $g_v(z) = (1 - z)(1 - z^{-1})g_u(z)$, and the long-run variance is $g_v(1) = 0 \cdot g_u(1) = 0$ whenever $g_u(1) = \lambda_u^2 < \infty$.

Section 9.2 Tools for Unit-Root Econometrics

Review Question 9.2.1

Case $\psi(L) = 1 + \psi_1 L$. Here $\psi_0 = 1$, $\psi_j = 0$ for $j \geq 2$, so $\psi(1) = 1 + \psi_1$ and

$$\alpha_0 = -(\psi_1 + \psi_2 + \cdots) = -\psi_1, \quad \alpha_j = -(\psi_{j+1} + \psi_{j+2} + \cdots) = 0 \quad (j \geq 1).$$

Then $\psi(1) + (1 - L)\alpha(L) = (1 + \psi_1) - \psi_1(1 - L) = 1 + \psi_1 L = \psi(L)$. ✓

Case $\psi(L) = (1 - \phi L)^{-1} = \sum_{j \geq 0} \phi^j L^j$. Here $\psi_j = \phi^j$, so $\psi(1) = 1/(1 - \phi)$ and

$$\alpha_j = -\sum_{k=j+1}^{\infty} \phi^k = -\frac{\phi^{j+1}}{1 - \phi}, \quad \alpha(L) = -\frac{\phi}{1 - \phi} \sum_{j=0}^{\infty} \phi^j L^j = -\frac{\phi}{1 - \phi} \frac{1}{1 - \phi L}.$$

Therefore

$$\psi(1) + (1-L)\alpha(L) = \frac{1}{1-\phi} - \frac{\phi(1-L)}{(1-\phi)(1-\phi L)} = \frac{1-\phi L - \phi + \phi L}{(1-\phi)(1-\phi L)} = \frac{1}{1-\phi L} = \psi(L).$$

In both cases $\alpha(L)$ is absolutely summable ($\sum_j |\alpha_j| = |\psi_1|$ and $\sum_j |\alpha_j| = |\phi|/(1-\phi)^2$, respectively); Analytical Exercise 9.7 shows that one-summability of $\{\psi_j\}$ is exactly what is needed for this in general.

Review Question 9.2.2

PROPOSITION 6.9 states that for a zero-mean linear process $u_t = \psi(L)\varepsilon_t$ with $\{\varepsilon_t\}$ independent white noise, $\sum_j |\psi_j| < \infty$ and $\psi(1) \neq 0$,

$$\sqrt{T}\bar{u} = \frac{1}{\sqrt{T}} \sum_{t=1}^T u_t \xrightarrow{d} N(0, \lambda^2), \quad \lambda^2 = \sigma^2[\psi(1)]^2.$$

The paragraph below PROPOSITION 9.1 divides the Beveridge–Nelson decomposition (9.2.7) of a driftless I(1) process $\xi_t = \xi_0 + \sum_{s=1}^t u_s$ (with $\delta = 0$) by $\sqrt{t}$, obtaining (9.2.8):

$$\frac{\xi_t}{\sqrt{t}} = \psi(1) \cdot \frac{1}{\sqrt{t}} \sum_{s=1}^t \varepsilon_s + \frac{1}{\sqrt{t}}(\eta_t - \eta_0 + \xi_0).$$

Since $E(\eta_t^2) < \infty$ (one-summability makes $\{\alpha_j\}$ absolutely summable, so η_t is a well-defined stationary process), $\eta_t/\sqrt{t} \rightarrow 0$ in mean square and hence in probability; the same applies to $\eta_0/\sqrt{t}$ and $\xi_0/\sqrt{t}$. By the Lindeberg–Lévy CLT, $\frac{1}{\sqrt{t}} \sum_{s \leq t} \varepsilon_s \xrightarrow{d} N(0, \sigma^2)$, so $\xi_t/\sqrt{t} \xrightarrow{d} N(0, \sigma^2[\psi(1)]^2)$. Now apply this to the I(0) process $\{u_t\}$ itself: define $\xi_t \equiv \sum_{s=1}^t u_s$ (starting value $\xi_0 = 0$), which is driftless I(1) because $\Delta \xi_t = u_t$. Then $\sqrt{T}\bar{u} = \xi_T/\sqrt{T}$, and the argument just given yields

$$\sqrt{T}\bar{u} \xrightarrow{d} N(0, \sigma^2[\psi(1)]^2) = N(0, \lambda^2) \quad \text{by (9.2.4),}$$

which is Proposition 6.9. The one-summability condition (9.2.3A) is used only to guarantee that $\{\alpha_j\}$ is absolutely summable so that the Beveridge–Nelson remainder η_t has finite variance — precisely the step that makes the driftless random walk dominate.

Review Question 9.2.3

Here $\psi(L) = 1 - 2L + L^2 = (1-L)^2$, so $\psi_0 = 1$, $\psi_1 = -2$, $\psi_2 = 1$, $\psi_j = 0$ for $j \geq 3$, and $\psi(1) = 0$: this $\{u_t\}$ has long-run variance $\sigma^2[\psi(1)]^2 = 0$, so strictly speaking it fails (9.2.3B) and is not I(0) in the book’s sense — it is I(−1), being the first difference of the I(0) process $\varepsilon_t - \varepsilon_{t-1}$. The Beveridge–Nelson coefficients (9.2.5) are

$$\alpha_0 = -(\psi_1 + \psi_2 + \cdots) = 1, \quad \alpha_1 = -(\psi_2 + \psi_3 + \cdots) = -1, \quad \alpha_j = 0 \quad (j \geq 2),$$

so $\eta_t = \alpha(L)\varepsilon_t = \varepsilon_t - \varepsilon_{t-1}$. Its long-run variance is, by the formula (9.2.4) applied to the MA polynomial $\alpha(L)$,

$$\lambda_\eta^2 = \sigma^2[\alpha(1)]^2 = \sigma^2(1-1)^2 = 0.$$

So the Beveridge–Nelson “stationary component” in this example is itself an overdifferenced series with zero long-run variance: it is stationary, but it is not I(0) in the strict sense of DEFINITION

9.2. The general point is that $\alpha(1) = -\sum_{j \geq 0} \sum_{k > j} \psi_k = -\sum_{k \geq 1} k \psi_k$, which need not vanish for a genuine $I(0)$ input (with $\psi(1) \neq 0$); the example is deliberately degenerate — $\psi(1) = 0$ — and then everything about the long run collapses.

Remark. What always holds, for any zero-mean $I(0)$ $\{u_t\}$, is that the *difference* of the remainder has zero long-run variance: from (9.2.6), $\Delta\eta_t = u_t - \psi(1)\varepsilon_t$, and $\sum_{t \leq T} \Delta\eta_t = \eta_T - \eta_0 = O_p(1)$, so $\text{Var}(\frac{1}{\sqrt{T}} \sum_t \Delta\eta_t) \rightarrow 0$. That is exactly Review Question 9.1.1 applied to the stationary series $\{\eta_t\}$.

Review Question 9.2.4

No. Write $\xi_t = \xi_0 + \xi_t^\dagger$ with $\xi_t^\dagger \equiv u_1 + \cdots + u_t$. Demeaning over $t = 0, 1, \dots, T-1$,

$$\xi_t^\mu = \xi_t - \frac{1}{T} \sum_{\tau=0}^{T-1} \xi_\tau = (\xi_0 + \xi_t^\dagger) - \frac{1}{T} \sum_{\tau=0}^{T-1} (\xi_0 + \xi_\tau^\dagger) = \xi_t^\dagger - \frac{1}{T} \sum_{\tau=0}^{T-1} \xi_\tau^\dagger,$$

since the constant ξ_0 appears T times in the sum and cancels. The demeaned series depends on the initial condition not at all. This is why the finite-sample distributions of the DF ρ^μ and t^μ statistics (and the Wiener-process limits in Proposition 9.2(c)–(d)) are free of y_0 under the null: adding a constant to every observation leaves every demeaned value unchanged.

Review Question 9.2.5

No. The detrended series is $\xi_t^\tau = \xi_t - \hat{\alpha} - \hat{\delta}t$, the OLS residual from the regression of ξ_t on $(1, t)$, $t = 0, 1, \dots, T-1$. Write $\xi_t = \alpha + \delta t + \xi_t^\dagger$ with $\{\xi_t^\dagger\}$ driftless $I(1)$. The OLS estimator of the coefficients is linear in the dependent variable, and regressing the part $\alpha + \delta t$ on $(1, t)$ reproduces it exactly (it lies in the regression space), so its residual is zero. Hence

$$\xi_t^\tau = \xi_t^\dagger - \hat{\alpha}^\dagger - \hat{\delta}^\dagger t,$$

where $(\hat{\alpha}^\dagger, \hat{\delta}^\dagger)$ are the OLS coefficients from regressing $\xi_t^\dagger$ on $(1, t)$. The detrended series is computed entirely from the driftless component; the trend parameters (α, δ) drop out. This is the source of the numerical invariance of the DF ρ^τ and t^τ statistics to (α, δ) noted below PROPOSITION 9.5, and of the fact that the detrended Wiener process (9.2.17) involves only the standard Wiener process itself.

Section 9.3 Dickey–Fuller Tests

Review Question 9.3.1

By (9.3.6), $T(\hat{\rho} - 1) \xrightarrow{d} DF_\rho$, a well-defined random variable with a proper distribution. Write

$$T^{1-\eta}(\hat{\rho} - 1) = T^{-\eta} \cdot T(\hat{\rho} - 1).$$

The first factor converges to 0 (a constant), and the second converges in distribution, hence is bounded in probability ($O_p(1)$). A product of a term going to zero and an $O_p(1)$ term goes to zero in probability: for any $\varepsilon > 0$, choose M with $\text{Prob}(|DF_\rho| > M) < \varepsilon/2$ and T large enough that $T^{-\eta}M < \delta$; then $\text{Prob}(|T^{1-\eta}(\hat{\rho} - 1)| > \delta) \rightarrow 0$. So $\hat{\rho}$ converges to 1 faster than any rate $T^{1-\eta}$ with $\eta < 1$: the estimation error is of order T^{-1} , not the usual $T^{-1/2}$ — superconsistency. The intuition is that the regressor y_{t-1} is $I(1)$, so $\sum_t y_{t-1}^2 = O_p(T^2)$ rather than $O_p(T)$: the signal accumulates

T times faster than in the stationary case, and the noise $\sum_t y_{t-1}\varepsilon_t = O_p(T)$ is correspondingly smaller relative to it.

Review Question 9.3.2

Under the null y_t is driftless I(1) with $\Delta y_t = u_t$, so PROPOSITION 9.2 applies with $\xi_t = y_t$: jointly,

$$\frac{1}{T^2} \sum_{t=1}^T y_{t-1}^2 \xrightarrow{d} \lambda^2 \int_0^1 W(r)^2 dr \quad (\text{part (a)}),$$

$$\frac{1}{T} \sum_{t=1}^T y_{t-1} \Delta y_t \xrightarrow{d} \lambda^2 \int_0^1 W(r) dW(r) + \frac{1}{2}(\gamma_0 - \lambda^2) \quad (\text{part (b)}).$$

Since $\hat{\rho} - 1 = \sum_t y_{t-1} u_t / \sum_t y_{t-1}^2$ and $u_t = \Delta y_t$ under the null,

$$T(\hat{\rho} - 1) = \frac{\frac{1}{T} \sum_t y_{t-1} \Delta y_t}{\frac{1}{T^2} \sum_t y_{t-1}^2} \xrightarrow{d} \frac{\lambda^2 \int_0^1 W(r) dW(r) + \frac{1}{2}(\gamma_0 - \lambda^2)}{\lambda^2 \int_0^1 W(r)^2 dr},$$

by the continuous mapping theorem (the convergence is joint). Two special cases are worth noting. If $\{u_t\}$ is white noise, $\lambda^2 = \gamma_0 = \sigma^2$, the correction term vanishes, and the limit reduces to $\int W dW / \int W^2$, the DF ρ distribution (9.3.6): serial correlation is exactly what the extra $\frac{1}{2}(\gamma_0 - \lambda^2)$ corrects for. Dividing numerator and denominator by γ_0 and using (9.2.4) ($\lambda^2/\gamma_0 = \sigma^2[\psi(1)]^2/\gamma_0$), the limit can also be written as

$$\frac{\lambda^2}{\gamma_0} DF_\rho + \frac{\gamma_0 - \lambda^2}{2\gamma_0 \int_0^1 W(r)^2 dr} \quad (\text{as a ratio of the two pieces}),$$

which is what the Phillips Z_ρ correction of Analytical Exercise 9.6 removes.

Review Question 9.3.3

With $SSR = \sum_t (y_t - \hat{\rho} y_{t-1})^2 = \sum_t (\Delta y_t - (\hat{\rho} - 1)y_{t-1})^2$ (since $y_t - \hat{\rho} y_{t-1} = \Delta y_t - (\hat{\rho} - 1)y_{t-1}$), expand:

$$s^2 = \frac{1}{T-1} \sum_{t=1}^T (\Delta y_t)^2 - 2(\hat{\rho} - 1) \frac{1}{T-1} \sum_{t=1}^T y_{t-1} \Delta y_t + (\hat{\rho} - 1)^2 \frac{1}{T-1} \sum_{t=1}^T y_{t-1}^2.$$

Under $\rho = 1$, $\Delta y_t = \varepsilon_t$, so by the Ergodic Theorem the first term converges in probability to $E(\varepsilon_t^2) = \sigma^2 = \gamma_0$. For the other two terms use the orders of magnitude established in the text: $\hat{\rho} - 1 = O_p(T^{-1})$ (superconsistency, (9.3.6)), $\frac{1}{T} \sum_t y_{t-1} \Delta y_t \xrightarrow{d}$ a random variable (Proposition 9.2(b)) so that $\sum_t y_{t-1} \Delta y_t = O_p(T)$, and $\sum_t y_{t-1}^2 = O_p(T^2)$ (Proposition 9.2(a)). Therefore

$$(\hat{\rho} - 1) \frac{1}{T-1} \sum_t y_{t-1} \Delta y_t = O_p(T^{-1}) \cdot O_p(T) \xrightarrow{p} 0, \quad (\hat{\rho} - 1)^2 \frac{1}{T-1} \sum_t y_{t-1}^2 = O_p(T^{-2}) \cdot O_p(T^2) \xrightarrow{p} 0,$$

and $s^2 \xrightarrow{p} \gamma_0 = \text{Var}(\varepsilon_t)$. What makes this work is that the estimation error $(\hat{\rho} - 1)y_{t-1}$ in the residual is negligible relative to Δy_t : the largest squared residual contamination is of order $O_p(T^{-2}) \cdot O_p(T^2)/T = O_p(T^{-1})$. Hence the DF t statistic inherits the nuisance-parameter-free limit (9.3.9).

Review Question 9.3.4

The second. Regressing Δy_t on y_{t-1} is the regression form of the DF test: $\Delta y_t = (\rho - 1)y_{t-1} + \varepsilon_t$, so the t -value on y_{t-1} tests $\rho = 1$ and has the Dickey–Fuller t distribution DF_t of (9.3.9) under the null — nonstandard, skewed left, with critical values from Table 9.2, Panel (a).

Regressing Δy_t on Δy_{t-1} tests something different. Under the null $\Delta y_t = \varepsilon_t$ is white noise, so the true coefficient of Δy_{t-1} is zero, and the regressor and error are stationary; the t -value on Δy_{t-1} is asymptotically standard normal by the usual stationary asymptotics of Section 6.4. Under a stationary AR(1) alternative $y_t = \rho y_{t-1} + \varepsilon_t$ with $|\rho| < 1$, however, $\Delta y_t = (\rho - 1)y_{t-1} + \varepsilon_t$: the coefficient on Δy_{t-1} in the first regression does not measure $\rho - 1$, and the test based on it has power only through the (weak) serial correlation that the alternative induces in $\{\Delta y_t\}$. So the first regression is a test of “no serial correlation in first differences,” not of the unit-root null; only the second regression’s t -value, referred to the DF tables, is a unit-root test.

Review Question 9.3.5

(a) If $\{y_t\}$ is zero-mean I(0) — hence ergodic stationary — the Ergodic Theorem applied to the numerator and denominator of

$$T(\hat{\rho} - 1) = \frac{\frac{1}{T} \sum_{t=1}^T y_{t-1} \Delta y_t}{\frac{1}{T^2} \sum_{t=1}^T y_{t-1}^2}$$

gives $\frac{1}{T} \sum_t y_{t-1} \Delta y_t = \frac{1}{T} \sum_t y_{t-1} (y_t - y_{t-1}) \xrightarrow{P} E(y_{t-1} y_t) - E(y_{t-1}^2) \equiv -\kappa$, where $\kappa = E(y_t^2) - E(y_t y_{t-1}) = \gamma_0(1 - \rho_1) > 0$ by assumption, and $\frac{1}{T^2} \sum_t y_{t-1}^2 = \frac{1}{T} \cdot \left[\frac{1}{T} \sum_t y_{t-1}^2 \right] \xrightarrow{P} 0 \cdot \gamma_0 = 0$, with $\frac{1}{T} \sum_t y_{t-1}^2 \xrightarrow{P} \gamma_0 > 0$. Hence

$$T(\hat{\rho} - 1) = \frac{-\kappa + o_p(1)}{\frac{1}{T} [\gamma_0 + o_p(1)]} \sim -\frac{\kappa T}{\gamma_0} \rightarrow -\infty \quad \text{in probability.}$$

Since the DF ρ test rejects for large negative values of the statistic (against the left tail of the DF_ρ distribution), the rejection probability tends to one: the test is consistent. Note the rate: the statistic diverges at rate T under a fixed I(0) alternative, whereas under the null it converges to a proper random variable — the two drift apart at maximal speed, which is why consistency requires only $E(y_t^2) > E(y_t y_{t-1})$.

(b) The DF t -statistic is $t = (\hat{\rho} - 1) / \{s[\sum_t y_{t-1}^2]^{-1/2}\}$. Under the I(0) alternative, $\hat{\rho} - 1 \xrightarrow{P} -\kappa/\gamma_0 < 0$ (from (a), dividing numerator and denominator by T instead), $\frac{1}{T} \sum_t y_{t-1}^2 \xrightarrow{P} \gamma_0$, and $s^2 = \text{SSR} / (T - 1) \xrightarrow{P} \text{Var}(y_t - \rho^* y_{t-1}) > 0$, where $\rho^* = E(y_t y_{t-1}) / E(y_{t-1}^2)$ — a positive constant (the hint’s point: s converges to something positive, here $\sqrt{\gamma_0(1 - \rho_1^2)}$ for a stationary AR(1), and in general the variance of the projection residual, strictly positive because $\kappa > 0$ implies $\rho^* < 1$). So

$$t = \frac{\hat{\rho} - 1}{s} \cdot \left(\sum_t y_{t-1}^2 \right)^{1/2} \sim \frac{-\kappa/\gamma_0}{s} \cdot \sqrt{T\gamma_0} \rightarrow -\infty \quad \text{in probability, at rate } \sqrt{T}.$$

Again the rejection probability at any fixed left-tail critical value tends to one. Comparing rates: the ρ -based statistic diverges at rate T , the t -statistic at rate $\sqrt{T}$; under fixed alternatives both

are consistent, but the faster divergence of the former foreshadows the greater asymptotic power of the DF ρ test against local-to-unity alternatives noted in Section 9.5.

Review Question 9.3.6

Under $\rho = 1$ the DGP (9.3.10)–(9.3.11) is $y_t = \alpha + y_0 + \varepsilon_1 + \cdots + \varepsilon_t$: a change in y_0 adds the same constant to every y_t , $t = 0, 1, \dots, T$. The regression of y_t on $(1, y_{t-1})$ includes an intercept, which absorbs any common additive constant: the residuals, hence $\hat{\rho}^\mu$, its standard error, $T(\hat{\rho}^\mu - 1)$ and t^μ , are identical functions of the demeaned series $\{y_{t-1}^\mu\}$, which by Review Question 9.2.4 does not involve y_0 . So the finite-sample distributions of $\hat{\rho}^\mu$ and t^μ under the null are free of y_0 (and of α).

When $\rho < 1$, no: $y_t = (1 - \rho)\alpha + \rho^t y_0 + \sum_{s=1}^t \rho^{t-s} \varepsilon_s$, so a change in y_0 shifts y_t by ρ^t , an amount that varies with t . This is not absorbed by the intercept, so $\hat{\rho}^\mu$ and t^μ do change with y_0 .

Consequently the finite-sample power against $\rho < 1$ does depend on y_0 : the further the process starts from its stationary mean, the stronger the signal that $\rho < 1$ (a large $|y_0|$ makes the early reversion toward the mean pronounced), and the more often the test rejects. The asymptotic power, against fixed alternatives, does not depend on y_0 (consistency holds regardless), but in finite samples it does.

Review Question 9.3.7

(a) The Durbin–Watson statistic computed from the levels (with no regressors, i.e. for the “residuals” y_t themselves) is $DW = \sum_{t=1}^T (\Delta y_t)^2 / \sum_{t=0}^T y_t^2$; its reciprocal is $T SB$ — the SB statistic is the reciprocal of the Durbin–Watson statistic up to the normalizing factor T , which is what gives SB a nondegenerate limiting distribution under the null.

(b) Since $\sum_{t=1}^T y_{t-1}^2 = \sum_{t=0}^{T-1} y_t^2 = \sum_{t=0}^T y_t^2 - y_T^2$, the numerator is $\sum_{t=0}^T y_t^2 = \sum_{t=1}^T y_{t-1}^2 + y_T^2$. Dividing by T^2 : by Proposition 9.2(a) with $\xi_t = y_t$ (a driftless random walk, so $\lambda^2 = \gamma_0 = \sigma^2$), $\frac{1}{T^2} \sum_{t=1}^T y_{t-1}^2 \xrightarrow{d} \sigma^2 \int_0^1 W(r)^2 dr$, and $y_T^2/T^2 = \frac{1}{T} (y_T/\sqrt{T})^2 \xrightarrow{p} 0$ since $y_T/\sqrt{T} \xrightarrow{d} N(0, \sigma^2)$. For the denominator, $\frac{1}{T} \sum_{t=1}^T (\Delta y_t)^2 = \frac{1}{T} \sum_{t=1}^T \varepsilon_t^2 \xrightarrow{p} \sigma^2$ by the law of large numbers. Combining by the continuous mapping theorem,

$$SB = \frac{\frac{1}{T^2} \sum_{t=1}^T y_{t-1}^2 + o_p(1)}{\frac{1}{T} \sum_{t=1}^T \varepsilon_t^2} \xrightarrow{d} \frac{\sigma^2 \int_0^1 W(r)^2 dr}{\sigma^2} = \left[\int_0^1 W(r)^2 dr \right]^{-1}.$$

The innovation variance σ^2 cancels: like the DF statistics, the limiting distribution is free of nuisance parameters (and of y_0 : adding a constant to all y_t does change $\sum y_t^2$, so in practice the SB statistic is computed after demeaning — the version here is the case without intercept).

(c) Under a zero-mean I(0) alternative, $\frac{1}{T^2} \sum_{t=0}^T y_t^2 = \frac{1}{T} \cdot \left[\frac{1}{T} \sum_{t=0}^T y_t^2 \right] \xrightarrow{p} 0 \cdot \gamma_0 = 0$ while $\frac{1}{T} \sum_{t=1}^T (\Delta y_t)^2 \xrightarrow{p} E[(\Delta y_t)^2] > 0$. Hence $SB \xrightarrow{p} 0/\text{positive} = 0$. Since $\int_0^1 W^2 dr > 0$ with probability one, the null distribution puts all its mass on positive values; a test rejecting the I(1) null for small values of SB is therefore consistent against I(0) alternatives. The numerator measures the (cumulated) signal of a stochastic trend, which is $O_p(T^2)$ under the null but only $O_p(T)$ under the alternative; normalizing by the innovation-level denominator turns the ratio into a consistent test.

Section 9.4 Augmented Dickey–Fuller Tests

Review Question 9.4.1

Substituting $\Delta y_{t-j} = y_{t-j} - y_{t-j-1}$ and collecting,

$$y_t = (\alpha_1 + \alpha_2)y_{t-1} + (\alpha_3 - \alpha_2)y_{t-2} - \alpha_3 y_{t-3} + \varepsilon_t,$$

so

$$\phi_1 = \alpha_1 + \alpha_2, \quad \phi_2 = \alpha_3 - \alpha_2, \quad \phi_3 = -\alpha_3,$$

and conversely $\alpha_1 = \phi_1 + \phi_2 + \phi_3$ (the sum of the AR coefficients, which is ρ in the augmented-autoregression parameterization (9.4.5)), $\alpha_2 = -(\phi_2 + \phi_3)$, and $\alpha_3 = -\phi_3$: from $\phi_2 = \alpha_3 - \alpha_2$ we get $\alpha_2 = \alpha_3 - \phi_2 = -\phi_3 - \phi_2$, and $\alpha_1 = \phi_1 - \alpha_2 = \phi_1 + \phi_2 + \phi_3$. Under the I(1) null ($\rho \equiv \alpha_1 = 1$), y_{t-1} is driftless I(1) while Δy_{t-1} and Δy_{t-2} are zero-mean I(0): the augmented autoregression separates the regressors into one integrated and two stationary components, whereas in the AR(3) form all three regressors y_{t-1} , y_{t-2} , y_{t-3} are individually I(1) and only their combinations entering with coefficients summarizing the difference structure are stationary.

The two equations are the *same* equation: the regressors of one are linear combinations of the regressors of the other through the invertible transformation above (the map $(y_{t-1}, y_{t-2}, y_{t-3}) \mapsto (y_{t-1}, \Delta y_{t-1}, \Delta y_{t-2})$ has matrix with determinant 1). OLS fitted values depend only on the column space of the regressors, and a nonsingular linear reparameterization leaves the projection unchanged; the coefficient on y_{t-1} in the augmented form, $\hat{\rho} = \hat{\alpha}_1$, equals $\hat{\phi}_1 + \hat{\phi}_2 + \hat{\phi}_3$, the sum of the OLS coefficients of the AR(3). This is exactly the equality asserted below (9.4.4): numerically the same estimate of ρ is obtained from either parameterization, but the augmented form is the convenient one for asymptotics because its regressors are separated into I(1) and I(0) blocks.

Review Question 9.4.2

Recall from (9.4.5) that $\rho = \phi_1 + \phi_2 + \cdots + \phi_{p+1}$. If $\rho > 1$ then

$$\phi(1) = 1 - (\phi_1 + \cdots + \phi_{p+1}) = 1 - \rho < 0,$$

while $\phi(0) = 1 > 0$. The polynomial ϕ is continuous on $[0, 1]$, so by the intermediate value theorem there exists $z^* \in (0, 1)$ with $\phi(z^*) = 0$: a *real* root strictly inside the unit circle. Such a root means an explosive autoregressive component. Hence if the DGP is taken to be either I(1) or I(0) — excluding explosive processes — then $\rho > 1$ is impossible, the alternative hypothesis is one-sided ($\rho < 1$), and one-tailed DF/ADF tests are called for. This is the justification, postponed from (9.3.1), for testing $H_0 : \rho = 1$ against $H_1 : \rho < 1$ only.

Remark. $\rho = 1$ gives $\phi(1) = 0$, the unit root itself; the case $\rho < 1$ is consistent with all roots outside the unit circle (stationarity), though $\rho < 1$ alone does not imply it — the stationarity condition is a restriction on all the ϕ 's jointly, whereas ρ is only their sum.

Review Question 9.4.3

Following the hint, first note that $\frac{1}{T} \sum_t \Delta y_{t-1} y_{t-1}$ and $\frac{1}{T} \sum_t \Delta y_t y_t$ have the same asymptotic distribution, since the two sums differ only in two terms: relabeling the index, $\sum_{t=1}^T \Delta y_t y_t =$

$\sum_{t=2}^{T+1} \Delta y_{t-1} y_{t-1}$, so

$$\frac{1}{T} \sum_{t=1}^T \Delta y_t y_t - \frac{1}{T} \sum_{t=1}^T \Delta y_{t-1} y_{t-1} = \frac{1}{T} (\Delta y_T y_T - \Delta y_0 y_0).$$

Now $y_T/\sqrt{T}$ converges in distribution and $\Delta y_T = O_p(1)$ (stationary), so

$$\frac{1}{T} \Delta y_T y_T = \frac{1}{\sqrt{T}} \left(\frac{y_T}{\sqrt{T}} \right) \Delta y_T \xrightarrow{p} 0;$$

likewise $\frac{1}{T} \Delta y_0 y_0 \xrightarrow{p} 0$. Hence the two statistics are asymptotically equivalent.

Next use the identity in the hint: $\Delta y_t \cdot y_t = \Delta y_t y_{t-1} + (\Delta y_t)^2$. Summing and dividing by T ,

$$\frac{1}{T} \sum_{t=1}^T \Delta y_t y_t = \frac{1}{T} \sum_{t=1}^T \Delta y_t y_{t-1} + \frac{1}{T} \sum_{t=1}^T (\Delta y_t)^2.$$

The first term on the right converges in distribution to $\lambda^2 \int_0^1 W(r) dW(r) + \frac{1}{2}(\gamma_0 - \lambda^2)$ by Proposition 9.2(b) with $\xi_t = y_t$; the second converges in probability to $\gamma_0 = \text{Var}(\Delta y_t)$ by the Ergodic Theorem ($\{\Delta y_t\}$ is zero-mean $I(0)$, hence ergodic stationary). Since convergence in probability to a constant is compatible with the continuous mapping theorem for the sum,

$$\frac{1}{T} \sum_{t=1}^T \Delta y_{t-1} y_{t-1} \stackrel{d}{=} \frac{1}{T} \sum_{t=1}^T \Delta y_t y_t + o_p(1) \xrightarrow{d} \lambda^2 \int_0^1 W(r) dW(r) + \frac{\gamma_0 - \lambda^2}{2} + \gamma_0,$$

that is,

$$\frac{1}{T} \sum_{t=1}^T \Delta y_{t-1} y_{t-1} \xrightarrow{d} \lambda^2 \int_0^1 W(r) dW(r) + \frac{\lambda^2 + \gamma_0}{2}.$$

So $\frac{1}{T} \sum_t \Delta y_{t-1} y_{t-1} = O_p(1)$: it converges to a proper random variable. The off-diagonal elements of the matrix $\mathbf{A}_T$ in (9.4.12) are $1/\sqrt{T}$ times this sum (see (9.4.14) scaled), hence converge to zero in probability — which is what makes $\mathbf{A}_T$ asymptotically diagonal and lets the $I(1)$ and $I(0)$ regressors be analyzed separately.

Review Question 9.4.4

(a) *Stationarity and ergodicity.* Under (9.2.1)–(9.2.2), $\Delta y_t = \psi(L)\varepsilon_t$ is a linear process driven by independent white noise, so $\{(\Delta y_t, \varepsilon_t)\}$ is jointly strictly stationary and ergodic (Proposition 6.1(d)); a measurable function of a stationary ergodic process, $g_t = \Delta y_{t-1}\varepsilon_t$, is stationary and ergodic. *Martingale difference.* Let $\mathcal{F}_{t-1} = \sigma(\varepsilon_{t-1}, \varepsilon_{t-2}, \dots)$. Since Δy_{t-1} is a function of $(\varepsilon_{t-1}, \varepsilon_{t-2}, \dots)$, it is $\mathcal{F}_{t-1}$ -measurable, and by the law of iterated expectations,

$$E(\Delta y_{t-1}\varepsilon_t \mid \mathcal{F}_{t-1}) = \Delta y_{t-1}E(\varepsilon_t \mid \mathcal{F}_{t-1}) = 0,$$

because $\{\varepsilon_t\}$ is independent white noise. (The hint's information ordering: the sigma field generated by $\{\Delta y_{t-2}, \Delta y_{t-3}, \dots\}$ is coarser than $\mathcal{F}_{t-1}$, but it is $\mathcal{F}_{t-1}$ that makes the computation immediate.) Also $E|\Delta y_{t-1}\varepsilon_t| < \infty$ since both factors have finite variance, so $\{g_t, \mathcal{F}_t\}$ is a stationary and ergodic martingale difference sequence.

(b) Its variance is

$$E[(\Delta y_{t-1} \varepsilon_t)^2] = E[(\Delta y_{t-1})^2 E(\varepsilon_t^2 | \mathcal{F}_{t-1})] = E[(\Delta y_{t-1})^2] \sigma^2 = \gamma_0 \sigma^2,$$

using independence of ε_t from $\mathcal{F}_{t-1}$ and $E[(\Delta y_{t-1})^2] = \text{Var}(\Delta y_t) = \gamma_0$. The m.d.s. CLT (e.g. Proposition 6.9's martingale version, or Billingsley's) applied to the normalized sum then gives (9.4.19):

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T \Delta y_{t-1} \varepsilon_t \xrightarrow{d} N(0, \gamma_0 \sigma^2).$$

So the coefficient of the stationary regressor in the augmented autoregression enjoys ordinary $\sqrt{T}$ asymptotics, in sharp contrast to the T -rate nonstandard distribution of $\hat{\rho}$.

Review Question 9.4.5

Write the t -value as in (9.4.23):

$$t = \frac{T(\hat{\rho} - 1)}{s \sqrt{(1, 1) \text{ element of } (\mathbf{A}_T)^{-1}}}.$$

For the numerator, (9.4.20) and (9.2.4) give $T(\hat{\rho} - 1) \xrightarrow{d} \frac{\sigma^2 \psi(1)}{\lambda^2} \int W dW / \int W^2 = \frac{1}{\psi(1)} \frac{\int_0^1 W(r) dW(r)}{\int_0^1 W(r)^2 dr}$, since $\lambda^2 = \sigma^2 [\psi(1)]^2$ implies $\sigma^2 \psi(1) / \lambda^2 = 1 / \psi(1)$. For the denominator, s^2 is consistent for $\sigma^2 = \text{Var}(\varepsilon_t)$ (the innovation variance of the MA representation of $\{\Delta y_t\}$: the residual of the augmented autoregression estimates ε_t , not Δy_t , so the argument of Review Question 9.3.3 applied to the augmented regression gives $s^2 \xrightarrow{p} \sigma^2$), and by (9.4.15) the $(1, 1)$ element of $(\mathbf{A}_T)^{-1}$ converges to $[\lambda^2 \int_0^1 W(r)^2 dr]^{-1}$, so

$$s \sqrt{(1, 1) \text{ element of } (\mathbf{A}_T)^{-1}} \xrightarrow{d} \sigma \cdot \frac{1}{\lambda \sqrt{\int_0^1 W(r)^2 dr}} = \frac{1}{\psi(1)} \cdot \frac{1}{\sqrt{\int_0^1 W(r)^2 dr}},$$

again using $\lambda = \sigma \psi(1)$ (positive by the stationarity condition (9.2.3B), so $\sqrt{[\psi(1)]^2} = \psi(1)$). Therefore

$$t \xrightarrow{d} \frac{\frac{1}{\psi(1)} \int W dW / \int W^2}{\frac{1}{\psi(1)} / \sqrt{\int W^2}} = \frac{\int_0^1 W(r) dW(r)}{\sqrt{\int_0^1 W(r)^2 dr}} = \frac{\frac{1}{2}(W(1)^2 - 1)}{\sqrt{\int_0^1 W(r)^2 dr}} = DF_t,$$

which is (9.4.24). The serial-correlation factors $\psi(1)$ cancel between numerator and denominator: unlike the ρ -statistic, which needs the explicit correction factor $1 - \hat{\zeta}_1 - \dots - \hat{\zeta}_p$ in (9.4.25), the t -statistic is *self-correcting* — the OLS standard error already reflects the autocorrelation through the fitted augmented autoregression.

Review Question 9.4.6

For $p = 1$, (9.4.21) gives $\sqrt{T}(\hat{\zeta}_1 - \bar{\zeta}_1) \xrightarrow{d} N(0, \sigma^2 / \gamma_0)$ under the null (where the true ζ_1 is the AR(1) coefficient of $\{\Delta y_t\}$). The t -statistic for $H_0 : \zeta_1 = \bar{\zeta}_1$ is

$$t_{\zeta_1} = \frac{\hat{\zeta}_1 - \bar{\zeta}_1}{s \sqrt{(2, 2) \text{ element of } (\mathbf{X}'\mathbf{X})^{-1}}} = \frac{\sqrt{T}(\hat{\zeta}_1 - \bar{\zeta}_1)}{s \sqrt{(2, 2) \text{ element of } \mathbf{A}_T^{-1}}},$$

using the scaling $\mathbf{A}_T = \mathbf{T}_T^{-1} \mathbf{X}' \mathbf{X} \mathbf{T}_T^{-1}$ with $\mathbf{T}_T = \text{diag}(T, \sqrt{T})$ (9.4.11): the (2, 2) element of $(\mathbf{X}' \mathbf{X})^{-1}$ is $1/T$ times the (2, 2) element of $\mathbf{A}_T^{-1}$. By (9.4.15), the (2, 2) element of $\mathbf{A}_T$ converges in probability to γ_0 , and $\mathbf{A}_T$ is asymptotically diagonal, so the (2, 2) element of $\mathbf{A}_T^{-1}$ converges to $1/\gamma_0$; and $s \xrightarrow{P} \sigma$. Hence the denominator converges in probability to $\sigma/\sqrt{\gamma_0}$, and

$$t_{\zeta_1} \xrightarrow{d} \frac{N(0, \sigma^2/\gamma_0)}{\sigma/\sqrt{\gamma_0}} = N(0, 1) \quad \text{under } \zeta_1 = \bar{\zeta}_1.$$

The point generalizes to all p and to joint F -tests on $(\zeta_1, \dots, \zeta_p)$: hypotheses that involve only the coefficients of the zero-mean $I(0)$ regressors can be tested with the standard asymptotic t and F distributions, exactly as if ρ were known to equal one. This is an instance of the Sims–Stock–Watson (1990) principle: coefficients on stationary linear combinations of the regressors have normal asymptotics; only the coefficient on the stochastic trend (y_{t-1}) is nonstandard.

Problem Set for Chapter 9: Analytical Exercises

Analytical Exercise 9.1

Start from the hint's identity. Since $\xi_t = \xi_{t-1} + \Delta\xi_t$,

$$\xi_t^2 = (\xi_{t-1} + \Delta\xi_t)^2 = \xi_{t-1}^2 + 2\Delta\xi_t \xi_{t-1} + (\Delta\xi_t)^2 \implies \Delta\xi_t \xi_{t-1} = \frac{1}{2} [\xi_t^2 - \xi_{t-1}^2 - (\Delta\xi_t)^2].$$

Summing over $t = 1, \dots, T$ telescopes the first two terms:

$$\frac{1}{T} \sum_{t=1}^T \Delta\xi_t \xi_{t-1} = \frac{1}{2} \cdot \frac{\xi_T^2}{T} - \frac{1}{2} \cdot \frac{\xi_0^2}{T} - \frac{1}{2} \cdot \frac{1}{T} \sum_{t=1}^T (\Delta\xi_t)^2. \quad (*)$$

Treat the three terms in turn.

First term. By the Beveridge–Nelson decomposition (9.2.7) with $\delta = 0$, $\xi_T = \psi(1) \sum_{t=1}^T \varepsilon_t + \eta_T + \xi_0 - \eta_0$, so

$$\frac{\xi_T}{\sqrt{T}} = \psi(1) \cdot \frac{1}{\sqrt{T}} \sum_{t=1}^T \varepsilon_t + \frac{\eta_T - \eta_0 + \xi_0}{\sqrt{T}}.$$

The remainder is $o_p(1)$: $\eta_T/\sqrt{T} \rightarrow 0$ in mean square (hence in probability) because $E(\eta_T^2) < \infty$, and likewise for η_0 and ξ_0 . By the Lindeberg–Lévy CLT, $\frac{1}{\sqrt{T}} \sum_{t=1}^T \varepsilon_t \xrightarrow{d} N(0, \sigma^2)$, so $\xi_T/\sqrt{T} \xrightarrow{d} N(0, \sigma^2[\psi(1)]^2) = N(0, \lambda^2)$ by (9.2.4) — this is Review Question 9.2.2. Therefore

$$\frac{\xi_T^2}{T} \xrightarrow{d} \lambda^2 \chi^2(1), \quad \text{and} \quad \frac{1}{2} \cdot \frac{\xi_T^2}{T} \xrightarrow{d} \frac{\lambda^2}{2} W(1)^2,$$

since $W(1)^2 \sim \chi^2(1)$.

Second term. $E(\xi_0^2) < \infty$ implies $\xi_0^2/T \xrightarrow{P} 0$.

Third term. $\{\Delta\xi_t\}$ is zero-mean ergodic stationary, so by the Ergodic Theorem $\frac{1}{T} \sum_{t=1}^T (\Delta\xi_t)^2 \xrightarrow{P} E[(\Delta\xi_t)^2] = \gamma_0$.

Substituting into (*),

$$\frac{1}{T} \sum_{t=1}^T \Delta\xi_t \xi_{t-1} \xrightarrow{d} \frac{\lambda^2}{2} W(1)^2 - \frac{\gamma_0}{2} = \lambda^2 \cdot \frac{W(1)^2 - 1}{2} + \frac{\lambda^2 - \gamma_0}{2}.$$

Finally, the defining property of the Itô integral with respect to the Wiener process (equivalently, the limit of the sum $\sum_{t=1}^T W(\frac{t-1}{T})[W(\frac{t}{T}) - W(\frac{t-1}{T})]$ as the mesh refines) is

$$\int_0^1 W(r) dW(r) = \frac{W(1)^2 - 1}{2},$$

which gives Proposition 9.2(b). Note where the correction $\frac{1}{2}(\gamma_0 - \lambda^2)$ comes from: for a genuine random walk, $\gamma_0 = \lambda^2 = \sigma^2$ and there is no correction; with serial correlation in $\{\Delta\xi_t\}$, the short-run variance γ_0 and the long-run variance λ^2 differ, and the difference survives in the limit because $\frac{1}{T} \sum (\Delta\xi_t)^2$ always estimates the former while ξ_T^2/T reflects the latter.

Analytical Exercise 9.2

(a) By the Frisch–Waugh theorem (Analytical Exercise 1.4), the OLS coefficient on y_{t-1} in the regression of y_t on $(1, y_{t-1})$ equals the OLS coefficient in the regression of y_t on y_{t-1}^μ alone (the residuals from regressing each on a constant; for y_{t-1} the residual is y_{t-1}^μ by definition). Hence

$$\hat{\rho}^\mu = \frac{\sum_{t=1}^T y_{t-1}^\mu y_t}{\sum_{t=1}^T (y_{t-1}^\mu)^2}, \quad \hat{\rho}^\mu - 1 = \frac{\sum_{t=1}^T y_{t-1}^\mu (y_t - y_{t-1})}{\sum_{t=1}^T (y_{t-1}^\mu)^2}.$$

Under the null $y_t - y_{t-1} = \varepsilon_t$; and because $\sum_t y_{t-1}^\mu = 0$ by construction, demeaning y_t or not is immaterial in the numerator: $\sum_t y_{t-1}^\mu \varepsilon_t$ is already in demeaned form.

(b) The demeaned series $\{y_{t-1}^\mu\}$ is computed from $\{y_{t-1}\}$ exactly as (9.2.14) computes $\{\xi_t^\mu\}$ from $\{\xi_t\}$ (the mean over $t = 1, \dots, T$ of y_{t-1} is the mean of $y_0, \dots, y_{T-1}$, matching the dating convention of Proposition 9.2). Since the value of y_0 does not affect the demeaned series (Review Question 9.2.4), parts (c) and (d) of Proposition 9.2 apply with $\xi_t = y_t$:

$$\frac{1}{T} \sum_{t=1}^T y_{t-1}^\mu \Delta y_t \xrightarrow{d} \lambda^2 \int_0^1 W^\mu(r) dW(r) + \frac{\gamma_0 - \lambda^2}{2}, \quad \frac{1}{T^2} \sum_{t=1}^T (y_{t-1}^\mu)^2 \xrightarrow{d} \lambda^2 \int_0^1 W^\mu(r)^2 dr.$$

Under the null $\Delta y_t = \varepsilon_t$, which is the first display.

(c) If $\{y_t\}$ is a driftless random walk, $\lambda^2 = \gamma_0 = \sigma^2$, the correction $\frac{1}{2}(\gamma_0 - \lambda^2)$ vanishes, and the factors λ^2 cancel in the ratio:

$$T(\hat{\rho}^\mu - 1) = \frac{\frac{1}{T} \sum_t y_{t-1}^\mu \varepsilon_t}{\frac{1}{T^2} \sum_t (y_{t-1}^\mu)^2} \xrightarrow{d} \frac{\int_0^1 W^\mu(r) dW(r)}{\int_0^1 W^\mu(r)^2 dr} = \frac{\frac{1}{2}([W^\mu(1)]^2 - [W^\mu(0)]^2) - \int_0^1 dr}{\int_0^1 W^\mu(r)^2 dr}.$$

Since $W^\mu(0) = W(0) - \int_0^1 W = 0 - \int W$ and $W^\mu(1) = W(1) - \int W$, this is the random variable DF_ρ^μ of (9.3.13). The convergence of the numerator and denominator is joint (Proposition 9.2 states joint convergence), so the continuous mapping theorem applies to the ratio.

(d) Write the residuals of the AR(1) equation with intercept as $\hat{\varepsilon}_t = \Delta y_t - \hat{\alpha}^* - (\hat{\rho}^\mu - 1)y_{t-1}$ (using $y_t = \alpha^* + \rho y_{t-1} + \varepsilon_t$ with $\alpha^* = 0$, $\rho = 1$ under the null, so $y_t - y_{t-1} = \varepsilon_t$), and expand:

$$s^2 = \frac{1}{T-2} \sum_{t=1}^T \left[\varepsilon_t - \hat{\alpha}^* - (\hat{\rho}^\mu - 1)y_{t-1} \right]^2.$$

Under the null, $\hat{\alpha}^*$ estimates 0: $\hat{\alpha}^* = \bar{\Delta}y - (\hat{\rho}^\mu - 1)\bar{y}_{-1}$. Now $\bar{\Delta}y = (y_T - y_0)/T = O_p(T^{-1/2})$ (since $y_T = O_p(\sqrt{T})$) and $(\hat{\rho}^\mu - 1)\bar{y}_{-1} = O_p(T^{-1}) \cdot O_p(\sqrt{T}) = O_p(T^{-1/2})$, so $\hat{\alpha}^* \xrightarrow{p} 0$. Therefore

$$s^2 = \frac{1}{T-2} \sum_t \varepsilon_t^2 - 2\hat{\alpha}^* \cdot \frac{1}{T-2} \sum_t \varepsilon_t - 2(\hat{\rho}^\mu - 1) \frac{1}{T-2} \sum_t y_{t-1} \varepsilon_t + O_p(T^{-1}) + (\hat{\rho}^\mu - 1)^2 \frac{1}{T-2} \sum_t y_{t-1}^2.$$

The first term converges to $\gamma_0 = \sigma^2$ by the LLN. The remaining terms vanish: $\hat{\alpha}^* \xrightarrow{p} 0$ and $\frac{1}{T} \sum \varepsilon_t \xrightarrow{p} 0$; $(\hat{\rho}^\mu - 1) = O_p(T^{-1})$ while $\frac{1}{T} \sum y_{t-1} \varepsilon_t = O_p(1)$ (it converges in distribution by part (b), with y_{t-1}^μ replaced by y_{t-1} — Proposition 9.2(b)); and $(\hat{\rho}^\mu - 1)^2 \frac{1}{T} \sum y_{t-1}^2 = O_p(T^{-2}) \cdot O_p(T) = O_p(T^{-1})$ using $\sum y_{t-1}^2 = O_p(T^2)$ (Proposition 9.2(a)). Hence $s^2 \xrightarrow{p} \gamma_0$.

(e) The t -value for $\rho = 1$ is $t^\mu = (\hat{\rho}^\mu - 1) / \{s \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{22}}\}$ with $\mathbf{x}_t = (1, y_{t-1})'$. A standard least-squares identity (the inverse of the 2×2 moment matrix of a constant and a regressor) gives

$$[(\mathbf{X}'\mathbf{X})^{-1}]_{22} = \frac{1}{\sum_{t=1}^T (y_{t-1}^\mu)^2},$$

the residual sum of squares from regressing y_{t-1} on a constant. Hence

$$t^\mu = \frac{T(\hat{\rho}^\mu - 1)}{s \left[\frac{1}{T^2} \sum_t (y_{t-1}^\mu)^2 \right]^{-1/2}}.$$

Under the random-walk null, $s \xrightarrow{p} \sigma = \sqrt{\gamma_0} = \lambda$ by (d), so combining (b) and (c),

$$t^\mu \xrightarrow{d} \frac{\lambda^2 \int_0^1 W^\mu dW}{\lambda [\lambda^2 \int_0^1 (W^\mu)^2 dr]^{1/2}} = \frac{\int_0^1 W^\mu(r) dW(r)}{\sqrt{\int_0^1 W^\mu(r)^2 dr}} = \frac{\frac{1}{2}([W^\mu(1)]^2 - [W^\mu(0)]^2 - 1)}{\sqrt{\int_0^1 W^\mu(r)^2 dr}} = DF_t^\mu,$$

which is (9.3.14). As in the case without intercept, numerator and denominator involve the same (demeaned) Wiener process and are therefore correlated; the distribution is skewed left, more so than DF_t because demeaning removes a random amount of level, increasing the downward bias of $\hat{\rho}^\mu$.

Analytical Exercise 9.3

(a) By Frisch–Waugh, $\hat{\rho}^\tau$ equals the OLS coefficient in the regression of the detrended y_t on the detrended y_{t-1} without intercept; since detrending is projection onto the orthogonal complement of the span of $(\mathbf{1}, \mathbf{t})$, and the residuals of y_t and of y_{t-1} satisfy $\sum_t y_t^\tau = 0$ and $\sum_t t y_{t-1}^\tau = 0$ (the normal equations),

$$\hat{\rho}^\tau = \frac{\sum_t y_{t-1}^\tau y_t^\tau}{\sum_t (y_{t-1}^\tau)^2}, \quad \hat{\rho}^\tau - 1 = \frac{\sum_t y_{t-1}^\tau (y_t^\tau - y_{t-1}^\tau)}{\sum_t (y_{t-1}^\tau)^2} = \frac{\sum_t y_{t-1}^\tau \Delta y_t}{\sum_t (y_{t-1}^\tau)^2},$$

where the last equality holds because $\sum_t y_{t-1}^\tau = 0$: replacing $y_t^\tau - y_{t-1}^\tau$ by Δy_t changes the numerator by a multiple of $\sum_t y_{t-1}^\tau$, which is zero (the difference of the two detrended series equals Δy_t minus a constant, since both detrending fits include a linear time trend whose period- t increment is a constant).

(b) Let $\xi_t = y_t - \alpha - \delta t$, which is driftless $I(1)$ under the null. Regressing y_{t-1} on $(1, t)$ is the same as regressing ξ_{t-1} on $(1, t)$ and adding the linear trend back to the fitted values: the residuals coincide, $y_{t-1}^\tau = \xi_{t-1}^\tau$ for all t (Review Question 9.2.5 — detrending removes exactly the linear component).

Also $\Delta y_t = \delta + \Delta \xi_t$ under the null, so $\text{Var}(\Delta y_t) = \text{Var}(\Delta \xi_t) = \gamma_0$ and likewise for the long-run variance λ^2 . Applying Proposition 9.2(e) and (f) with ξ_t ,

$$\frac{1}{T^2} \sum_{t=1}^T (y_{t-1}^\tau)^2 = \frac{1}{T^2} \sum_{t=1}^T (\xi_{t-1}^\tau)^2 \xrightarrow{d} \lambda^2 \int_0^1 W^\tau(r)^2 dr,$$

and

$$\frac{1}{T} \sum_{t=1}^T y_{t-1}^\tau \Delta y_t = \frac{1}{T} \sum_t \xi_{t-1}^\tau (\delta + \Delta \xi_t) = \frac{1}{T} \sum_t \xi_{t-1}^\tau \Delta \xi_t \xrightarrow{d} \lambda^2 \int_0^1 W^\tau(r) dW(r) + \frac{\gamma_0 - \lambda^2}{2},$$

because $\sum_t \xi_{t-1}^\tau = 0$ kills the drift term δ .

(c) If $\{y_t\}$ is a random walk with or without drift, $\{\Delta y_t\}$ is white noise around δ , so $\lambda^2 = \gamma_0 = \sigma^2$ and the correction vanishes:

$$T(\hat{\rho}^\tau - 1) \xrightarrow{d} \frac{\int_0^1 W^\tau(r) dW(r)}{\int_0^1 W^\tau(r)^2 dr} = DF_\rho^\tau,$$

the random variable defined in (9.3.19). The parallel result for the t -statistic, $t^\tau \xrightarrow{d} DF_t^\tau = \int W^\tau dW / \sqrt{\int (W^\tau)^2}$ (9.3.20), follows exactly as in part (e) of Analytical Exercise 9.2 once $s^2 \xrightarrow{p} \sigma^2$ is noted (the same expansion, now with fitted intercept and trend both vanishing in probability at the rates $\hat{\alpha}^* = O_p(T^{-1/2})$, $\hat{\delta}^* = O_p(T^{-3/2})$ under the null). The distributions are free of (α, δ) and y_0 , and are the most left-skewed of the three cases: detrending attributes part of the stochastic trend's variation to a fitted time trend ("spurious detrending"), which pushes $\hat{\rho}^\tau$ further below one.

Analytical Exercise 9.4

Apply the Beveridge–Nelson decomposition to the level: from (9.2.7) with $\delta = 0$,

$$y_{t-1} = \psi(1) \left(\varepsilon_1 + \cdots + \varepsilon_{t-1} \right) + \eta_{t-1} + y_0 - \eta_0, \quad \eta_t = \alpha(L) \varepsilon_t.$$

Multiply by ε_t , sum, and divide by T :

$$\frac{1}{T} \sum_{t=1}^T y_{t-1} \varepsilon_t = \psi(1) \cdot \frac{1}{T} \sum_{t=1}^T \left(\sum_{s=1}^{t-1} \varepsilon_s \right) \varepsilon_t + \frac{1}{T} \sum_{t=1}^T \eta_{t-1} \varepsilon_t + \frac{y_0 - \eta_0}{T} \sum_{t=1}^T \varepsilon_t.$$

Third term. $(y_0 - \eta_0) = O_p(1)$ and $\frac{1}{T} \sum_t \varepsilon_t = O_p(T^{-1/2})$ by the CLT, so the product is $O_p(T^{-1/2}) \xrightarrow{p} 0$.

Second term. $\{\eta_{t-1} \varepsilon_t\}$ is a stationary ergodic martingale difference sequence with finite variance (η_{t-1} is a function of $(\varepsilon_{t-1}, \varepsilon_{t-2}, \dots)$, independent of ε_t ; and $E(\eta_t^2) < \infty$), so by the m.d.s. LLN $\frac{1}{T} \sum_t \eta_{t-1} \varepsilon_t \xrightarrow{p} E(\eta_{t-1} \varepsilon_t) = E(\eta_{t-1}) E(\varepsilon_t) = 0$.

First term. Let $w_t \equiv \sum_{s=1}^t \varepsilon_s$ (a driftless random walk with $\text{Var}(\Delta w_t) = \sigma^2$). Then $\frac{1}{T} \sum_t (\sum_{s \leq t-1} \varepsilon_s) \varepsilon_t = \frac{1}{T} \sum_t w_{t-1} \Delta w_t$, which by Analytical Exercise 9.1 (Proposition 9.2(b) applied to $\xi_t = w_t$, for which $\lambda^2 = \gamma_0 = \sigma^2$) converges in distribution to $\sigma^2 \int_0^1 W(r) dW(r) = \frac{\sigma^2}{2} [W(1)^2 - 1]$. Multiplying by $\psi(1)$,

$$\frac{1}{T} \sum_{t=1}^T y_{t-1} \varepsilon_t \xrightarrow{d} \psi(1) \cdot \sigma^2 \int_0^1 W(r) dW(r) = \sigma^2 \psi(1) \int_0^1 W(r) dW(r),$$

which is (9.4.16). The content of the result is that only the *stochastic trend* of y_{t-1} correlates with ε_t at the $O_p(T)$ order; the stationary component η_{t-1} contributes only $O_p(\sqrt{T})$ and is washed out by the $1/T$ normalization. With $\lambda^2 = \sigma^2[\psi(1)]^2$, this limit is equivalently $\lambda^2 \int W dW/\psi(1)$, which is what enters the numerator of $T(\hat{\rho} - 1)$ in (9.4.20).

Analytical Exercise 9.5

By Frisch–Waugh, $(\hat{\rho}^\mu, \hat{\zeta}_1)$ are numerically the OLS coefficients from the regression of y_t on $(y_{t-1}^\mu, (\Delta y_{t-1})^{(\mu)})$, where y_{t-1}^μ is the residual from regressing y_{t-1} on a constant and $(\Delta y_{t-1})^{(\mu)}$ the residual from regressing Δy_{t-1} on a constant, $t = 1, \dots, T$. With $\mathbf{x}_t = (y_{t-1}^\mu, (\Delta y_{t-1})^{(\mu)})'$, $\mathbf{b} = (\hat{\rho}^\mu, \hat{\zeta}_1)'$ and $\boldsymbol{\beta} = (\rho, \zeta_1)'$, the sampling error satisfies (9.4.8), and with the scaling matrix $\mathbf{T}_T = \text{diag}(T, \sqrt{T})$ we get $\mathbf{T}_T(\mathbf{b} - \boldsymbol{\beta}) = \mathbf{A}_T^{-1} \mathbf{c}_T$ where

$$\begin{aligned} (\mathbf{A}_T)_{11} &= \frac{1}{T^2} \sum_t (y_{t-1}^\mu)^2, & (\mathbf{A}_T)_{22} &= \frac{1}{T} \sum_t [(\Delta y_{t-1})^{(\mu)}]^2, & (\mathbf{A}_T)_{12} &= \frac{1}{T^{3/2}} \sum_t y_{t-1}^\mu (\Delta y_{t-1})^{(\mu)}, \\ (\mathbf{c}_T)_1 &= \frac{1}{T} \sum_t y_{t-1}^\mu \varepsilon_t, & (\mathbf{c}_T)_2 &= \frac{1}{\sqrt{T}} \sum_t (\Delta y_{t-1})^{(\mu)} \varepsilon_t. \end{aligned}$$

Now evaluate each piece under the null.

$(\mathbf{A}_T)_{11}$ converges in distribution to $\lambda^2 \int_0^1 W^\mu(r)^2 dr$ by Proposition 9.2(c) with $\xi_t = y_t$ (Analytical Exercise 9.2(b)).

$(\mathbf{A}_T)_{22}$: since $(\Delta y_{t-1})^{(\mu)} = \Delta y_{t-1} - \overline{\Delta y}$ and $\overline{\Delta y} = O_p(T^{-1/2})$, $(\mathbf{A}_T)_{22} = \frac{1}{T} \sum_t (\Delta y_{t-1})^2 - (\overline{\Delta y})^2 \xrightarrow{p} \gamma_0 - 0 = \gamma_0$ by the Ergodic Theorem.

Off-diagonal. Since $\sum_t y_{t-1}^\mu = 0$, $\sum_t y_{t-1}^\mu (\Delta y_{t-1})^{(\mu)} = \sum_t y_{t-1}^\mu \Delta y_{t-1}$, and by the algebra of Review Question 9.4.3 applied to the demeaned series (Proposition 9.2(d) with $\xi_t = y_t$), $\frac{1}{T} \sum_t y_{t-1}^\mu \Delta y_{t-1}$ converges in distribution to $\lambda^2 \int_0^1 W^\mu dW + (\gamma_0 - \lambda^2)/2$, a proper random variable. Hence $(\mathbf{A}_T)_{12} = O_p(1)/\sqrt{T} \xrightarrow{p} 0$: $\mathbf{A}_T$ is asymptotically diagonal.

$(\mathbf{c}_T)_1$. By the Beveridge–Nelson decomposition applied to the demeaned level, $y_{t-1}^\mu = \psi(1)w_{t-1}^\mu + \eta_{t-1}^\mu$ where $w_t = \sum_{s \leq t} \varepsilon_s$ is the stochastic trend and $\eta_{t-1}^\mu = \eta_{t-1} - \bar{\eta}$; demeaning commutes with the decomposition because the initial-value terms cancel (Review Question 9.2.4). Then

$$(\mathbf{c}_T)_1 = \psi(1) \cdot \frac{1}{T} \sum_t w_{t-1}^\mu \varepsilon_t + \frac{1}{T} \sum_t \eta_{t-1}^\mu \varepsilon_t.$$

The second term is $o_p(1)$: $\eta_{t-1}^\mu \varepsilon_t$ is a stationary ergodic m.d.s. with mean zero (independence of ε_t from the past; $\bar{\eta} \varepsilon_t$ averages out similarly), so its sample mean is $O_p(T^{-1/2})$. The first term is $\psi(1) \frac{1}{T} \sum_t w_{t-1}^\mu \Delta w_t$, which by Proposition 9.2(d) applied to the driftless random walk $\{w_t\}$ (for which $\lambda^2 = \gamma_0 = \sigma^2$) converges to $\psi(1)\sigma^2 \int_0^1 W^\mu(r) dW(r)$.

$(\mathbf{c}_T)_2$ equals $\frac{1}{\sqrt{T}} \sum_t \Delta y_{t-1} \varepsilon_t - \sqrt{T} \overline{\Delta y} \bar{\varepsilon}$; the second part is $\sqrt{T} O_p(T^{-1/2}) O_p(T^{-1/2}) = O_p(T^{-1/2}) \xrightarrow{p} 0$, and the first converges to $N(0, \gamma_0 \sigma^2)$ by Review Question 9.4.4(b). The convergence of $(\mathbf{A}_T, \mathbf{c}_T)$ is joint (Chan and Wei, 1988, Theorem 2.2).

Assembling: $\mathbf{A}_T^{-1} \mathbf{c}_T$ has first element

$$T(\hat{\rho}^\mu - 1) \xrightarrow{d} \frac{\sigma^2 \psi(1) \int_0^1 W^\mu(r) dW(r)}{\lambda^2 \int_0^1 W^\mu(r)^2 dr} = \frac{1}{\psi(1)} \cdot \frac{\int_0^1 W^\mu(r) dW(r)}{\int_0^1 W^\mu(r)^2 dr},$$

using $\lambda^2 = \sigma^2[\psi(1)]^2$, and second element $\sqrt{T}(\hat{\zeta}_1 - \zeta_1) \xrightarrow{d} \gamma_0^{-1}N(0, \gamma_0\sigma^2) = N(0, \sigma^2/\gamma_0)$. Since $\psi(1) = (1 - \zeta_1)^{-1}$ under the null ($\Delta y_t = \zeta_1 \Delta y_{t-1} + \varepsilon_t$ implies $\psi(L) = (1 - \zeta_1 L)^{-1}$) and $\hat{\zeta}_1$ is consistent, $\psi(1)$ is consistently estimated by $(1 - \hat{\zeta}_1)^{-1}$, giving the ADF ρ^μ statistic (9.4.34):

$$\frac{T(\hat{\rho}^\mu - 1)}{1 - \hat{\zeta}_1} \xrightarrow{d} DF_\rho^\mu.$$

The t -value inherits DF_t^μ without correction, by the self-correcting argument of Review Question 9.4.5 applied to the demeaned Wiener process. Comparing with the no-intercept case of the text, the only change is that W is replaced by W^μ throughout: the intercept works exactly as it did for the DF tests, by demeaning every series that enters the statistics.

Analytical Exercise 9.6

(a) For the AR(1) regression without intercept, the OLS variance formula gives $\hat{\sigma}_\rho^2 = s^2(\sum_t y_{t-1}^2)^{-1}$, so

$$\frac{T^2 \hat{\sigma}_\rho^2}{s^2} = T^2 \left(\sum_t y_{t-1}^2 \right)^{-1} = \left(\frac{1}{T^2} \sum_t y_{t-1}^2 \right)^{-1}, \quad \frac{T \hat{\sigma}_\rho}{s \hat{\lambda}} = \frac{1}{\hat{\lambda}} \left(\frac{1}{T^2} \sum_t y_{t-1}^2 \right)^{-1/2},$$

and substituting yields the displayed forms. These are the computable versions: everything is read off the regression printout plus the two nonparametric variance estimates.

(b) *The Z_ρ statistic.* Under the null, by Review Question 9.3.2 (Proposition 9.2(a), (b) with $\xi_t = y_t$),

$$T(\hat{\rho} - 1) \xrightarrow{d} \frac{\lambda^2 \int_0^1 W dW + \frac{1}{2}(\gamma_0 - \lambda^2)}{\lambda^2 \int_0^1 W(r)^2 dr} = \frac{\int W dW}{\int W^2} + \frac{\gamma_0 - \lambda^2}{2\lambda^2 \int_0^1 W(r)^2 dr}.$$

The correction term converges in probability to the second piece with the sign reversed: since $\hat{\lambda}^2 - \hat{\gamma}_0 \xrightarrow{p} \lambda^2 - \gamma_0$ and $\frac{1}{T^2} \sum_t y_{t-1}^2 \xrightarrow{d} \lambda^2 \int_0^1 W^2 dr$,

$$\frac{1}{2}(\hat{\lambda}^2 - \hat{\gamma}_0) \left(\frac{1}{T^2} \sum_t y_{t-1}^2 \right)^{-1} \xrightarrow{p} \frac{\lambda^2 - \gamma_0}{2\lambda^2 \int_0^1 W(r)^2 dr} \cdot [\text{same } W]$$

— where the convergence is joint with that of $T(\hat{\rho} - 1)$ (both are continuous functions of the same underlying partial-sum process), so the correction subtracts off the nuisance term exactly:

$$Z_\rho \xrightarrow{d} \frac{\int_0^1 W(r) dW(r)}{\int_0^1 W(r)^2 dr} = DF_\rho.$$

The Z_t statistic. The ordinary t -statistic is $t = (\hat{\rho} - 1)/\hat{\sigma}_\rho = T(\hat{\rho} - 1)/[s(\frac{1}{T^2} \sum_t y_{t-1}^2)^{-1/2}]$. Under the null, $s^2 = \frac{1}{T} \sum_t (\Delta y_t - (\hat{\rho} - 1)y_{t-1})^2 \xrightarrow{p} \gamma_0$ by Review Question 9.3.3 (which used only $\Delta y_t \text{ I}(0)$, not white noise), so

$$t \xrightarrow{d} \frac{\lambda^2 \int W dW + \frac{1}{2}(\gamma_0 - \lambda^2)}{\sqrt{\gamma_0} \lambda [\int W^2 dr]^{1/2}}.$$

Multiply by $\hat{\gamma}_0^{1/2}/\hat{\lambda}^{1/2} \xrightarrow{p} \sqrt{\gamma_0}/\lambda$ and subtract the correction, which converges to $\frac{1}{2}(\lambda^2 - \gamma_0)/[\lambda \cdot \lambda(\int W^2)^{1/2}]$:

$$Z_t \xrightarrow{d} \frac{\sqrt{\gamma_0}}{\lambda} \cdot \frac{\lambda^2 \int W dW + \frac{1}{2}(\gamma_0 - \lambda^2)}{\sqrt{\gamma_0} \lambda (\int W^2)^{1/2}} - \frac{\lambda^2 - \gamma_0}{2\lambda^2 (\int W^2 dr)^{1/2}} = \frac{\int_0^1 W(r) dW(r)}{\sqrt{\int_0^1 W(r)^2 dr}} = DF_t.$$

The mechanics are the same as in the ADF tests — remove the effect of serial correlation so that the Dickey–Fuller distributions apply — but achieved by *correcting the statistic* with consistent estimates of λ^2 and γ_0 (typically kernel estimates from the AR(1) residuals) instead of *correcting the regression* with lagged changes. The practical drawback noted in Section 9.5 is that the non-parametric estimate of λ^2 behaves poorly when $\{\Delta y_t\}$ has a large negative MA component, which is exactly Schwert’s (1989) design in Monte Carlo Exercise 9.2 and explains the severe size distortions of the Phillips–Perron tests documented there.

Analytical Exercise 9.7

First equality: the definition of α_j in (9.2.5), $\alpha_j = -(\psi_{j+1} + \psi_{j+2} + \dots)$. The tail sum converges absolutely because $\sum_k |\psi_k| < \infty$ (implied by one-summability (9.2.3A)), so α_j is well defined.

Inequality: the triangle inequality for an absolutely convergent series, $|\sum_{k>j} \psi_k| \leq \sum_{k>j} |\psi_k|$.

Second equality: Tonelli’s theorem for nonnegative terms allows interchanging the order of summation: the double sum $\sum_{j \geq 0} \sum_{k \geq j+1} |\psi_k|$ sums $|\psi_k|$ once for every pair (j, k) with $0 \leq j < k$, and for fixed $k \geq 1$ the admissible j ’s are $j = 0, 1, \dots, k-1$ — k of them.

Third equality: the inner sum over j has k identical terms $|\psi_k|$.

Final inequality: this is exactly the one-summability condition (9.2.3A), $\sum_{k=1}^{\infty} k|\psi_k| < \infty$.

So $\sum_j |\alpha_j| \leq \sum_k k|\psi_k| < \infty$: one-summability of the MA coefficients of $\{\Delta \xi_t\}$ is precisely the condition that makes the Beveridge–Nelson remainder $\eta_t = \alpha(L)\varepsilon_t$ an absolutely summable linear process, hence (by Proposition 6.1(a), (d)) a well-defined zero-mean ergodic stationary process with $E(\eta_t^2) < \infty$ — the properties used in the derivation of (9.2.7) and in Review Question 9.2.2. The chain also explains why the stronger condition is adopted: absolute summability of $\{\psi_j\}$ alone would not suffice, since the triangle inequality step gives the bound in terms of the first absolute moment of the coefficients, not their plain sum.

Problem Set for Chapter 9: Monte Carlo Exercises

Monte Carlo Exercise 9.1

The implementation follows the book’s recipe. In each replication, generate

$$(y_1, \dots, y_T)' = \mathbf{r} y_0 + \mathbf{A} \boldsymbol{\varepsilon}$$

with $\mathbf{r} = (\rho, \rho^2, \dots, \rho^T)'$ and $\mathbf{A}$ lower triangular with $A_{ts} = \rho^{t-s}$ ($s \leq t$), compute the two t -values (the regression of y_t on $(1, y_{t-1})$ and on $(1, t, y_{t-1})$, $t = 1, \dots, T$), and count rejections against the asymptotic 5% critical values of Table 9.2. With 200,000 replications (`code/ch9.py`):

test	5% critical value	power at $\rho = 0.95$, $T = 100$
DF t^μ	−2.86	0.124
DF t^τ	−3.41	0.092

matching the book’s “about 0.124” and “about 0.092”. So against this alternative the test without time rejects the false unit-root null only about 12% of the time, and including time reduces power by about a quarter. Two forces are at work. First, detrending adds a regressor that soaks up some

of the variation in y_{t-1} that identifies $\rho < 1$, shrinking the effective signal. Second, and equivalently in large samples, the critical value moves left (-3.41 versus -2.86) because the DF_t^τ distribution is more skewed than DF_t^μ ; under the alternative the statistic's distribution does not shift left by enough to compensate. The practical moral is the one drawn in the text: if the null plausibly has no trend, use the t^μ test; pay for the trend regression only when a trend under the null is genuinely in doubt. Note also the absolute level: at $T = 100$ and $\rho = 0.95$ — a process whose half-life is about 14 periods — even the better test has power barely above the nominal size, an instance of the low power of unit-root tests against moderately persistent stationary alternatives that motivates the local-to-unity analysis of Section 9.5.

Monte Carlo Exercise 9.2

The regression is $y_t = \text{const} + \rho y_{t-1} + \zeta_1 \Delta y_{t-1} + \cdots + \zeta_4 \Delta y_{t-4} + \text{error}$; the ADF ρ^μ statistic is $96(\hat{\rho}^\mu - 1)/(1 - \hat{\zeta}_1 - \cdots - \hat{\zeta}_4)$ with asymptotic 5% critical value -14.1 (Table 9.1, Panel (b)) and the ADF t^μ statistic is the t -value on y_{t-1} with critical value -2.86 (Table 9.2, Panel (b)). With 100,000 replications (`code/ch9.py`):

test	5% critical value	exact size
ADF ρ^μ	-14.1	0.497
ADF t^μ	-2.86	0.292

matching the book's “about 0.496” and “about 0.290”. Both tests reject the *true* null far too often — the ρ -test nearly half the time against a nominal 5% — but the size distortion is substantially smaller for the t -test, as the exercise asks to verify.

The source of the distortion is the near-cancellation in the DGP: $\Delta y_t = \varepsilon_t - 0.8\varepsilon_{t-1} = (1 - 0.8L)\varepsilon_t$ has an MA root of 1.25, close to the unit circle; writing it as an autoregression, $\Delta y_t = -\sum_{j \geq 1} (-0.8)^j \Delta y_{t-j} + \varepsilon_t$, the AR coefficients die out only geometrically at rate 0.8, so four lagged changes capture only part of the serial correlation. The uncaptured part acts like a nearly-I(0) component that the tests mistake for evidence against the unit root (in the limit, if $\theta = -1$ exactly, Δy_t would be white noise overdifferenced and y_t would be I(0) up to a negligible stochastic trend). The ρ -statistic suffers more because its correction factor $1 - \sum_j \hat{\zeta}_j$ estimates $\psi(1)^{-1}$ through the truncated autoregression, and the truncation bias enters the statistic multiplicatively, whereas the t -statistic's self-correction through the OLS standard error is more robust. This is the Monte Carlo counterpart of the ranking reported in Section 9.5 (M-tests least distorted, then ADF t , then ADF ρ , then Phillips–Perron), and it is why the text recommends data-dependent rules that let $\hat{p}$ grow with T over fixing a small $\hat{p}$.

Problem Set for Chapter 9: Empirical Exercises

Empirical Exercise 9.1

(a) *ADF with automatic lag selection.* By (9.4.31), $p_{\max} = \lceil 12(200/100)^{1/4} \rceil = 14$, and the selection sample is fixed at 1806–1990 ($t = p_{\max} + 2, \dots, T$; 185 observations). For each $p = 0, 1, \dots, 14$ the

augmented autoregression (2) of the exercise is estimated on this fixed sample; the AIC and BIC use (9.4.32) with $C(T) = 2$ and $\log(185)$:

p	AIC	BIC	t on ζ_p
0	-5.346	-5.312	—
1	-5.351	-5.299	1.665
2	-5.340	-5.271	-0.268
3	-5.332	-5.245	-0.678
4	-5.321	-5.217	0.126
5	-5.352	-5.230	-2.739
6	-5.356	-5.217	-1.626
7	-5.347	-5.191	-0.627
8	-5.363	-5.189	2.171
9	-5.361	-5.169	-1.203
10	-5.354	-5.145	-0.883
11	-5.355	-5.128	-1.404
12	-5.351	-5.108	-1.120
13	-5.351	-5.090	-1.331
14	-5.390	-5.111	-2.930

The sequential t -rule at 10% (start at $p = 14$, stop at the first lag whose ζ has $|t| > 1.645$) stops immediately: $\hat{p} = 14$. The AIC is minimized at $\hat{p} = 14$ and the BIC at $\hat{p} = 0$ — exactly the book’s (14, 14, 0). The BIC’s heavier penalty ($\log 185 \approx 5.22$ per parameter against 2) outweighs all but the largest t -values on this sample. With $p = 0$ the augmented autoregression estimated on the maximal sample 1792–1990 ($T = 199$) is

$$z_t = \underset{(0.0519)}{0.1792} + \underset{(0.0326)}{0.8869} z_{t-1} + \hat{e}_t, \quad R^2 = 0.790,$$

so the DF t^μ statistic is $(0.8869 - 1)/0.0326 = -3.47$, matching the value reported as t_μ in Table 1 of Lothian and Taylor (1996). The 5% critical value from Table 9.2, Panel (b) is -2.86 (1%: -3.43), so the unit-root null is rejected at 1%: two centuries of data are enough to detect the mean reversion that the weak version of PPP predicts, even though the point estimate $\hat{\rho} \approx 0.89$ implies a half-life of deviations of about $\log 2/(1 - 0.887) \approx 6$ years.

(b) *DF test on the floating period, 1974–1990.* The AR(1) equation estimated on $t = 1975, \dots, 1990$ ($T = 16$) is

$$z_t = \text{const} + \underset{(0.1904)}{0.7704} z_{t-1} + \hat{e}_t, \quad \text{SSR} = 0.150964, \quad R^2 = 0.539.$$

$t^\mu = (0.7704 - 1)/0.1904 = -1.21 > -2.86$: the random-walk null is not rejected at 5%. With 16 annual observations the test has essentially no power; this is the “embarrassing resiliency of the random walk model” of Section 9.6 in miniature — the post-1973 float alone is far too short a span to reject a unit root against ρ ’s in the empirically relevant range.

(c) *DF test on the gold standard, 1870–1913*. Estimated on $t = 1871, \dots, 1913$ ($T = 43$):

$$z_t = \text{const} + \underset{(0.1045)}{0.7222} z_{t-1} + \hat{e}_t, \quad \text{SSR} = 0.069429, \quad R^2 = 0.538.$$

$t^\mu = (0.7222 - 1)/0.1045 = -2.66 > -2.86$: not rejected at 5% either (though close). The point estimate of ρ is indeed closer to zero than in (b) (0.72 versus 0.77), consistent with the conjecture that arbitrage enforced PPP more tightly under the gold standard; but with 43 observations the test still cannot reject the null, and the difference between the two point estimates is well within sampling variability — as part (d) confirms formally.

(d) *Chow test for structural change at 1973/74*. Under the maintained hypothesis of a stationary AR(1) for z_t , split the sample at the float: subsample 1 is 1792–1974 ($n_1 = 183$, the 1974 equation included in the first subsample), subsample 2 is 1975–1990 ($n_2 = 16$), $K = 2$. The pooled and subsample regressions give (`code/ch9.py`):

$$\text{SSR}_{\text{pooled}} = 0.995135, \quad \text{SSR}_1 = 0.837772 \ (\hat{\rho}_1 = 0.8993, \ (0.0329)), \quad \text{SSR}_2 = 0.150964.$$

The Chow F and its χ^2 version are

$$F = \frac{[\text{SSR}_{\text{pooled}} - (\text{SSR}_1 + \text{SSR}_2)]/K}{(\text{SSR}_1 + \text{SSR}_2)/(n_1 + n_2 - 2K)} = \frac{(0.995135 - 0.988736)/2}{0.988736/195} = 0.631, \quad K \cdot F = 1.26.$$

The 5% critical value of $\chi^2(2)$ is 5.99 (p-value 0.53), so the null of parameter stability is easily accepted — Lothian and Taylor’s (1996, pp. 498–502) finding that “there is no sign of a structural shift in the parameters during the floating period.” As the exercise notes, the usual finite-sample justification of the Chow test fails here because the regressor z_{t-1} is not strictly exogenous; the justification is asymptotic (Analytical Exercise 2.12): under stationarity and ergodicity $K \cdot F \xrightarrow{d} \chi^2(K)$ as both subsamples grow. The second subsample has only 16 observations, so the asymptotics is doing real work; the large first subsample and the smallness of the statistic itself make the conclusion safe nonetheless. Substantively: the apparent contrast between (b) and (c) is sampling noise, not structural change.

(e) *The ADF-GLS test on the gold standard*. With $x_t = 1$ (demeaned case), the GLS quasi-differencing uses $\bar{\rho} = 1 + \bar{c}/T = 1 - 7/44 = 0.8409$ ($T = 44$ observations, 1870–1913): regress $\tilde{y}_1 = z_{1870}$, $\tilde{y}_t = z_t - \bar{\rho}z_{t-1}$ on $\tilde{x}_1 = 1$, $\tilde{x}_t = 1 - \bar{\rho}$ to get $\hat{\alpha}_{\text{GLS}}$, and form the GLS-demeaned series $\tilde{z}_t = z_t - \hat{\alpha}_{\text{GLS}}$. Lag selection by BIC on the fixed selection sample with $p_{\max} = [12(44/100)^{1/4}] = 9$ picks $\hat{p} = 0$. The ADF regression of $\tilde{z}_t$ on $\tilde{z}_{t-1}$ (no intercept), $t = 1871, \dots, 1913$, gives the ADF-GLS t^μ statistic

$$\text{ADF-GLS } t^\mu = -2.25$$

(the book reports -2.24). Its limiting distribution is the DF_t distribution, Table 9.2, Panel (a), whose 5% critical value is -1.95 : the $I(1)$ null *is* rejected at 5% on the gold-standard subsample, where the plain DF t^μ of part (c) failed (-2.66 against -2.86). The comparison illustrates the power argument of Elliott, Rothenberg and Stock (1996): GLS-demeaning with $\bar{c} = -7$ uses a point-optimal weighting of the first observation against local-to-unity alternatives, buying substantially more power than OLS demeaning at the cost of a much less negative critical value — and here the trade is decisive. The verdict of the whole exercise is thus coherent: with enough span, or with a

more powerful test, PPP mean reversion is detectable; the random-walk null survives only in the short subsamples with weak tests.

(f) *Time aggregation.* Let $\{y_t\}$ be a driftless random walk, $y_t = y_0 + \sum_{s \leq t} \varepsilon_s$, $\text{Var}(\varepsilon_t) = \sigma^2$, and define the m -period average $\bar{y}_j = \frac{1}{m} \sum_{s=1}^m y_{(j-1)m+s}$. Writing day s of year j around the previous year-end level, $y_{(j-1)m+s} = y_{(j-1)m} + \sum_{i=1}^s \varepsilon_{(j-1)m+i}$, and averaging,

$$\bar{y}_j = y_{(j-1)m} + \frac{1}{m} \sum_{s=1}^m (m-s+1) \varepsilon_{(j-1)m+s}.$$

Substituting $y_{(j-1)m} = y_{(j-2)m} + \sum_{s=1}^m \varepsilon_{(j-2)m+s}$ and the same expansion of $\bar{y}_{j-1}$ around $y_{(j-2)m}$, the change in the annual average is

$$\bar{y}_j - \bar{y}_{j-1} = \sum_{s=1}^m \frac{s-1}{m} \varepsilon_{(j-2)m+s} + \sum_{s=1}^m \frac{m-s+1}{m} \varepsilon_{(j-1)m+s},$$

a weighted sum of the innovations of years $j-1$ and j alone: the change is an MA process spanning two years. Since the innovations are i.i.d. with variance σ^2 ,

$$\begin{aligned} \text{Var}(\bar{y}_j - \bar{y}_{j-1}) &= \frac{\sigma^2}{m^2} \left[\sum_{s=1}^m (s-1)^2 + \sum_{s=1}^m (m-s+1)^2 \right] \\ &= \frac{\sigma^2}{m^2} \cdot \frac{m(m-1)(2m-1) + m(m+1)(2m+1)}{6} = \sigma^2 \frac{2m^2+1}{3m}. \end{aligned}$$

Adjacent changes share only the innovations of year $j-1$, which enter $\bar{y}_{j-1} - \bar{y}_{j-2}$ with weights $(m-s+1)/m$ and $\bar{y}_j - \bar{y}_{j-1}$ with weights $(s-1)/m$; hence

$$\text{Cov}(\bar{y}_{j+1} - \bar{y}_j, \bar{y}_j - \bar{y}_{j-1}) = \frac{\sigma^2}{m^2} \sum_{s=1}^m (s-1)(m-s+1) = \frac{\sigma^2}{m^2} \cdot \frac{m(m^2-1)}{6} = \sigma^2 \frac{m^2-1}{6m} > 0,$$

using $\sum_{s=1}^m (s-1)(m-s+1) = \frac{1}{6}m(m-1)(m+1)$, and the changes are uncorrelated at lags two and higher (they share no innovations). Therefore

$$\text{Corr}(\bar{y}_{j+1} - \bar{y}_j, \bar{y}_j - \bar{y}_{j-1}) = \frac{(m^2-1)/(6m)}{(2m^2+1)/(3m)} = \frac{m^2-1}{2(2m^2+1)} \longrightarrow \frac{1}{4} \quad \text{as } m \rightarrow \infty,$$

which is what the exercise asks to show. In particular the first autocorrelation of $\{\Delta \bar{y}_j\}$ is nonzero (it equals, e.g., 0.160 for $m = 5$ and 0.245 for $m = 50$, approaching 0.25; see `code/ch9.py`), so the first difference of the time-averaged series is serially correlated and $\{\bar{y}_j\}$ is *not* a random walk. Time averaging a random walk produces an I(1) process whose first difference is a stationary MA-type process rather than white noise — the Working (1960) time-aggregation effect. This justifies the caveat in the exercise preamble: the Lothian–Taylor series are annual averages, so unit-root inference on them must allow for serially correlated first differences even if the underlying daily real exchange rate were an exact random walk — one more reason to prefer the ADF test over the plain DF test here, and a reason to treat the data construction itself with care.

Chapter 10

Cointegration

Solutions to all the exercises of Chapter 10 of Fumio Hayashi, *Econometrics* (Princeton University Press, 2000): the five review questions of Section 10.1, the four of Section 10.2, the question of Section 10.3, the question of Section 10.4, and the empirical exercise on the demand for money, which reproduces Tables 10.2–10.4 from the data in `MPYR.ASC`. Definitions, propositions and equations cited as DEFINITION 10.1, PROPOSITION 9.2 or (10.4.15) refer to the book; references to Chapters 6 and 9 (spectra, long-run variances, unit-root asymptotics) are to the book’s earlier chapters. The numerical parts are computed by `code/ch10.py`, which also produces the three figures.

Throughout, $\mathbf{y}_t$ is an n -dimensional $I(1)$ system with $\Delta \mathbf{y}_t = \boldsymbol{\delta} + \mathbf{u}_t$, $\mathbf{u}_t = \boldsymbol{\Psi}(L)\boldsymbol{\varepsilon}_t$, $\boldsymbol{\varepsilon}_t$ i.i.d. with mean zero and positive definite variance $\boldsymbol{\Omega}$, $\{\boldsymbol{\Psi}_j\}$ one-summable and $\boldsymbol{\Psi}(1) \neq \mathbf{0}$ (10.1.1)–(10.1.4), (10.1.9)–(10.1.10). The multivariate Beveridge–Nelson decomposition is

$$\mathbf{y}_t = \boldsymbol{\delta} \cdot t + \boldsymbol{\Psi}(1)(\boldsymbol{\varepsilon}_1 + \cdots + \boldsymbol{\varepsilon}_t) + \boldsymbol{\eta}_t + (\mathbf{y}_0 - \boldsymbol{\eta}_0), \quad (10.1.16)$$

where $\boldsymbol{\eta}_t = \boldsymbol{\alpha}(L)\boldsymbol{\varepsilon}_t$ is stationary and $\boldsymbol{\alpha}(L)$ is defined by $\boldsymbol{\Psi}(L) = \boldsymbol{\Psi}(1) + (1-L)\boldsymbol{\alpha}(L)$ (10.1.14). A nonzero vector $\mathbf{a}$ is a cointegrating vector when $\mathbf{a}'\boldsymbol{\Psi}(1) = \mathbf{0}'$ (10.1.20), which turns $\mathbf{a}'\mathbf{y}_t$ into the (asymptotically) stationary process $\mathbf{a}'\boldsymbol{\delta} \cdot t + \mathbf{a}'\boldsymbol{\eta}_t + \mathbf{a}'(\mathbf{y}_0 - \boldsymbol{\eta}_0)$ (10.1.21); the cointegration rank is $h = n - \text{rank } \boldsymbol{\Psi}(1)$ (10.1.22).

Section 10.1 Cointegrated Systems

Review Question 10.1.1

With $\gamma = 0$, $\boldsymbol{\Psi}_1 = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}$ and the VMA (10.1.12) is

$$\Delta y_{1t} = \delta_1 + \varepsilon_{1t} - \varepsilon_{1,t-1}, \quad \Delta y_{2t} = \delta_2 + \varepsilon_{2t}.$$

The system $\mathbf{y}_t$ is $I(1)$ by definition if the associated zero-mean process $\mathbf{u}_t = \boldsymbol{\Psi}(L)\boldsymbol{\varepsilon}_t$ satisfies (10.1.1)–(10.1.4). The VMA is of finite order, so $\{\boldsymbol{\Psi}_j\}$ is one-summable, and

$$\boldsymbol{\Psi}(1) = \mathbf{I}_2 + \boldsymbol{\Psi}_1 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \neq \mathbf{0},$$

so (10.1.4) holds. (The hint’s formulation, via (10.1.5), is that the long-run variance matrix of $\Delta \mathbf{y}_t$,

$$\boldsymbol{\Psi}(1)\boldsymbol{\Omega}\boldsymbol{\Psi}(1)' = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \boldsymbol{\Omega} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & \omega_{22} \end{bmatrix}, \quad \omega_{22} \equiv \text{Var}(\varepsilon_{2t}) > 0,$$

is not the zero matrix: at least one diagonal element is positive. In fact y_{2t} is a random walk with drift — individually $I(1)$ — while, as noted in the text, y_{1t} is trend stationary when $\gamma = 0$: substituting $\Delta y_{1t} = \delta_1 + \varepsilon_{1t} - \varepsilon_{1,t-1}$ and summing, $y_{1t} = y_{1,0} + \delta_1 t + \varepsilon_{1t} - \varepsilon_{1,0}$, whose deviation from the trend $y_{1,0} - \varepsilon_{1,0} + \delta_1 t$ is ε_{1t} , i.i.d. This illustrates the book’s point that the definition of

an $I(1)$ system does not require each element to be individually $I(1)$; what is required is that the system not be stationary as a vector, i.e. that $\Psi(1) \neq \mathbf{0}$.)

Review Question 10.1.2

Apply the factorization with $\mathbf{C} = \Psi(1)$, which has rank $n - h$ by (10.1.22): there exist an $n \times n$ nonsingular $\mathbf{G}$ and an $n \times (n - h)$ matrix $\mathbf{F}$ of full column rank with

$$\Psi(1)\mathbf{G} = [\mathbf{F} : \mathbf{0}_{n \times h}].$$

Partition $\mathbf{G}^{-1}\mathbf{w}_t$ conformably into its first $n - h$ and last h elements, $\mathbf{G}^{-1}\mathbf{w}_t = (\boldsymbol{\tau}_t', \mathbf{v}_t')'$. Then

$$\Psi(1)\mathbf{w}_t = \Psi(1)\mathbf{G}\mathbf{G}^{-1}\mathbf{w}_t = [\mathbf{F} : \mathbf{0}] \begin{bmatrix} \boldsymbol{\tau}_t \\ \mathbf{v}_t \end{bmatrix} = \mathbf{F}\boldsymbol{\tau}_t.$$

Because $\mathbf{w}_t$ is a random walk, $\Delta\mathbf{w}_t = \boldsymbol{\varepsilon}_t$, so $\Delta\boldsymbol{\tau}_t = [\mathbf{I}_{n-h} : \mathbf{0}] \mathbf{G}^{-1}\boldsymbol{\varepsilon}_t$ is i.i.d. with mean zero and variance

$$\text{Var}(\Delta\boldsymbol{\tau}_t) = [\mathbf{I}_{n-h} : \mathbf{0}] \mathbf{G}^{-1}\boldsymbol{\Omega}(\mathbf{G}^{-1})' \begin{bmatrix} \mathbf{I}_{n-h} \\ \mathbf{0} \end{bmatrix},$$

which is positive definite: it is the leading $(n - h) \times (n - h)$ principal submatrix of $\mathbf{G}^{-1}\boldsymbol{\Omega}(\mathbf{G}^{-1})'$, and that matrix is positive definite because $\boldsymbol{\Omega}$ is positive definite and $\mathbf{G}^{-1}$ is nonsingular. Hence $\boldsymbol{\tau}_t$ is an $(n - h)$ -dimensional random walk with positive definite innovation variance, and

$$\mathbf{y}_t = \boldsymbol{\delta} \cdot t + \mathbf{F}\boldsymbol{\tau}_t + \boldsymbol{\eta}_t + (\mathbf{y}_0 - \boldsymbol{\eta}_0) :$$

the n -dimensional $I(1)$ system with cointegration rank h is driven by exactly $n - h$ common stochastic trends. The cointegrating vectors are precisely the linear combinations that annihilate $\mathbf{F}$: since $\mathbf{a}'\Psi(1) = \mathbf{0}'$ is equivalent to $\mathbf{a}'\mathbf{F} = \mathbf{0}'$ (postmultiply the factorization by $\mathbf{G}^{-1}$), the cointegrating space is the h -dimensional left null space of $\mathbf{F}$. Example 10.2 (continued) on p. 628 is the case $n = 2$, $h = 1$: $\Psi(1)\mathbf{w}_t = (\gamma, 1)' \sum_{s=1}^t \varepsilon_{2s}$, one common trend $\tau_t = \sum_s \varepsilon_{2s}$ driving both series.

Review Question 10.1.3

With $\boldsymbol{\eta}_t = \boldsymbol{\varepsilon}_t - \boldsymbol{\varepsilon}_{t-1}$ and i.i.d. innovations,

$$\text{Cov}(\boldsymbol{\eta}_t, \boldsymbol{\eta}_0) = \text{Cov}(\boldsymbol{\varepsilon}_t - \boldsymbol{\varepsilon}_{t-1}, \boldsymbol{\varepsilon}_0 - \boldsymbol{\varepsilon}_{-1}).$$

For $t = 0$ this is $\text{Var}(\boldsymbol{\varepsilon}_0) + \text{Var}(\boldsymbol{\varepsilon}_{-1}) = 2\boldsymbol{\Omega}$; for $t = 1$ the only common innovation is $\boldsymbol{\varepsilon}_0$, entering with opposite signs, so the covariance is $-\boldsymbol{\Omega}$; for $t > 1$ the two moving averages share no innovation, so the covariance is $\mathbf{0}$.

For the variance, the BN decomposition (10.1.16) with $\boldsymbol{\delta} = \mathbf{0}$, $\mathbf{y}_0 = \mathbf{0}$ and a cointegrating vector $\mathbf{a}$ (so $\mathbf{a}'\Psi(1) = \mathbf{0}'$ and $\mathbf{a}'\boldsymbol{\delta} = 0$) reduces, via (10.1.21), to

$$\mathbf{a}'\mathbf{y}_t = \mathbf{a}'\boldsymbol{\eta}_t - \mathbf{a}'\boldsymbol{\eta}_0 = \mathbf{a}'\boldsymbol{\varepsilon}_t - \mathbf{a}'\boldsymbol{\varepsilon}_{t-1} - \mathbf{a}'\boldsymbol{\varepsilon}_0 + \mathbf{a}'\boldsymbol{\varepsilon}_{-1}.$$

For $t = 0$ this is identically zero, so $\text{Var}(\mathbf{a}'\mathbf{y}_0) = 0$. For $t = 1$ the right side is $\mathbf{a}'\boldsymbol{\varepsilon}_1 - 2\mathbf{a}'\boldsymbol{\varepsilon}_0 + \mathbf{a}'\boldsymbol{\varepsilon}_{-1}$, a sum of three independent terms, so $\text{Var}(\mathbf{a}'\mathbf{y}_1) = (1 + 4 + 1)\mathbf{a}'\boldsymbol{\Omega}\mathbf{a} = 6\mathbf{a}'\boldsymbol{\Omega}\mathbf{a}$. For $t > 1$ the right side is $\mathbf{a}'\boldsymbol{\varepsilon}_t - \mathbf{a}'\boldsymbol{\varepsilon}_{t-1} - \mathbf{a}'\boldsymbol{\varepsilon}_0 + \mathbf{a}'\boldsymbol{\varepsilon}_{-1}$, four distinct innovations, so $\text{Var}(\mathbf{a}'\mathbf{y}_t) = 4\mathbf{a}'\boldsymbol{\Omega}\mathbf{a}$.

For $t \geq 2$ the process is $\mathbf{a}'\mathbf{y}_t = \mathbf{a}'\boldsymbol{\varepsilon}_t - \mathbf{a}'\boldsymbol{\varepsilon}_{t-1} - \mathbf{a}'\boldsymbol{\varepsilon}_0 + \mathbf{a}'\boldsymbol{\varepsilon}_{-1}$: the last two terms form a single, time-invariant random variable, and the first two form an $MA(1)$ in the innovations. A stationary process plus a time-invariant random variable is stationary, so $\{\mathbf{a}'\mathbf{y}_t, t = 2, 3, \dots\}$ is stationary (indeed strictly stationary, since $\{\boldsymbol{\varepsilon}_t\}$ is i.i.d.). Its autocovariances are free of t : with $C \equiv -\mathbf{a}'\boldsymbol{\varepsilon}_0 + \mathbf{a}'\boldsymbol{\varepsilon}_{-1}$, $\text{Var}(C) = 2\mathbf{a}'\boldsymbol{\Omega}\mathbf{a}$,

$$\text{Cov}(\mathbf{a}'\mathbf{y}_t, \mathbf{a}'\mathbf{y}_{t-j}) = \underbrace{-\mathbf{a}'\boldsymbol{\Omega}\mathbf{a} \mathbf{1}\{j=1\}}_{\text{shared } \boldsymbol{\varepsilon}_{t-1}} + \underbrace{\text{Var}(C)}_{= 2\mathbf{a}'\boldsymbol{\Omega}\mathbf{a}} = \mathbf{a}'\boldsymbol{\Omega}\mathbf{a} (2 - \mathbf{1}\{j=1\}), \quad j \geq 1,$$

and $\gamma_0 = 4\mathbf{a}'\boldsymbol{\Omega}\mathbf{a}$ as computed above.

The whole sequence $\{\mathbf{a}'\mathbf{y}_t, t = 0, 1, 2, \dots\}$ is *not* stationary, because the variance at $t = 0$ (zero) and at $t = 1$ ($6\mathbf{a}'\boldsymbol{\Omega}\mathbf{a}$) differs from the common variance at $t \geq 2$ ($4\mathbf{a}'\boldsymbol{\Omega}\mathbf{a}$): the initial condition $\mathbf{a}'(\mathbf{y}_0 - \boldsymbol{\eta}_0)$ is correlated with the subsequent stationary component $\mathbf{a}'\boldsymbol{\eta}_t$, here through the shared innovation $\boldsymbol{\varepsilon}_0$. This is exactly the point made under (10.1.21): a cointegrating combination is in general only “asymptotically stationary.” It would be stationary from $t = 0$ if the initial value satisfied $\mathbf{a}'(\mathbf{y}_0 - \boldsymbol{\eta}_0) = 0$ (here: if y_0 had been chosen as $\mathbf{a}'\mathbf{y}_0 = -\mathbf{a}'\boldsymbol{\eta}_0$, eliminating the constant part), which is the “suitable choice of the initial value” in DEFINITION 10.1.

Remark. The example is driven by the overlapping moving average: with $\boldsymbol{\eta}_t$ white noise ($\boldsymbol{\eta}_t = \boldsymbol{\varepsilon}_t$), $\text{Cov}(\boldsymbol{\eta}_t, \boldsymbol{\eta}_0) = \mathbf{0}$ for $t \geq 1$ and $\text{Var}(\mathbf{a}'\mathbf{y}_t) = 2\mathbf{a}'\boldsymbol{\Omega}\mathbf{a}$ for all $t \geq 1$, so the process would be stationary for $t \geq 1$; the serial correlation in $\boldsymbol{\eta}_t$ is what prolongs the transition.

Review Question 10.1.4

“Stationary” here means trend stationary in the sense of the chapter: with $\mathbf{e}_1 = (1, 0, \dots, 0)'$, the claim is that $y_{1t} = \mathbf{e}_1'\mathbf{y}_t$ can be made (trend) stationary by a suitable choice of the initial value. Premultiplying the BN decomposition (10.1.16) by $\mathbf{e}_1'$ gives (10.1.19) with $\mathbf{a} = \mathbf{e}_1$:

$$y_{1t} = \delta_1 t + \mathbf{e}_1'\boldsymbol{\Psi}(1)\mathbf{w}_t + \eta_{1t} + (y_{1,0} - \eta_{1,0}), \quad \mathbf{w}_t \equiv \boldsymbol{\varepsilon}_1 + \dots + \boldsymbol{\varepsilon}_t.$$

If the first row of $\boldsymbol{\Psi}(1)$ were not zero, the term $\mathbf{e}_1'\boldsymbol{\Psi}(1)\mathbf{w}_t$ would be a nontrivial random walk: its variance at date t would be $t \cdot \mathbf{e}_1'\boldsymbol{\Psi}(1)\boldsymbol{\Omega}\boldsymbol{\Psi}(1)'\mathbf{e}_1 > 0$ (the quadratic form is positive because $\boldsymbol{\Omega}$ is positive definite and $\boldsymbol{\Psi}(1)'\mathbf{e}_1 \neq \mathbf{0}$), growing without bound, and no choice of the fixed or t -independent initial value $y_{1,0}$ could offset a stochastic trend — so y_{1t} could not be made stationary. Stationarity of y_{1t} therefore requires $\mathbf{e}_1'\boldsymbol{\Psi}(1) = \mathbf{0}'$, i.e. $\mathbf{e}_1$ satisfies (10.1.20) and is a cointegrating vector. Hence the cointegration rank, the number of linearly independent cointegrating vectors, is at least one; equivalently $\text{rank } \boldsymbol{\Psi}(1) \leq n - 1$ by (10.1.22). Example 10.2 with $\gamma = 0$ is the case in point: y_{1t} is trend stationary and $(1, 0)'$ is (a multiple of no other) cointegrating vector, with $\boldsymbol{\Psi}(1)$ of rank one.

Review Question 10.1.5

Each column of $\mathbf{A}$ is a cointegrating vector, so by (10.1.20) each column $\mathbf{a}_j$ satisfies $\mathbf{a}_j'\boldsymbol{\Psi}(1) = \mathbf{0}'$, i.e.

$$\mathbf{A}'\boldsymbol{\Psi}(1) = \mathbf{0}_{h \times n}.$$

Writing $\tilde{\mathbf{A}} \equiv \mathbf{A}\mathbf{F}$ with columns $\tilde{\mathbf{a}}_k = \sum_{j=1}^h f_{jk} \mathbf{a}_j$,

$$\tilde{\mathbf{A}}' \Psi(1) = \mathbf{F}' \mathbf{A}' \Psi(1) = \mathbf{0},$$

so every column of $\tilde{\mathbf{A}}$ — a linear combination of cointegrating vectors — is itself a cointegrating vector (the cointegrating vectors form a linear space, the left null space of $\Psi(1)$). They are linearly independent: $\text{rank}(\mathbf{A}\mathbf{F}) = \text{rank}(\mathbf{A}) = h$ because postmultiplication by the nonsingular $\mathbf{F}$ does not alter rank. Hence $\mathbf{A}\mathbf{F}$ collects h linearly independent cointegrating vectors.

In words: only the cointegrating *space* is identified by the data, not any particular basis for it. This is the source of the nonuniqueness of the decomposition $\Phi(1) = \mathbf{B}\mathbf{A}'$ in (10.2.21) — the pairs $(\mathbf{B}, \mathbf{A})$ and $(\mathbf{B}(\mathbf{F}')^{-1}, \mathbf{A}\mathbf{F})$ describe the same system — and the reason a normalization such as (10.2.2) or (10.2.6) is needed before a cointegrating vector can be estimated.

Section 10.2 Alternative Representations of Cointegrated Systems

For reference, the error vector of the triangular representation (10.2.8)–(10.2.9) is

$$\begin{bmatrix} \mathbf{z}_t^* \\ \mathbf{u}_{2t} \end{bmatrix} = \Psi^*(L) \varepsilon_t, \quad \Psi^*(L) \equiv \begin{bmatrix} \Psi_1^*(L) \\ \Psi_2(L) \end{bmatrix}, \quad \Psi_1^*(L) = (\mathbf{I}_h \quad -\mathbf{\Gamma}') \alpha(L), \quad (10.2.39\text{--}40)$$

where $\alpha(L)$ is from the BN decomposition (10.1.14) and $\Psi_2(L)$ is the last $n - h$ rows of $\Psi(L)$ in the VMA representation (10.1.10).

Review Question 10.2.1

(a) Suppose $\mathbf{b}'\Psi_2(1) = \mathbf{0}'$ for some $\mathbf{b} \neq \mathbf{0}$. Set $\mathbf{a} \equiv (\mathbf{0}', \mathbf{b}')'$, an n -vector, nonzero because $\mathbf{b} \neq \mathbf{0}$. Since $\Psi_2(1)$ is by definition the last $n - h$ rows of $\Psi(1)$,

$$\mathbf{a}'\Psi(1) = (\mathbf{0}', \mathbf{b}') \begin{bmatrix} \Psi_1(1) \\ \Psi_2(1) \end{bmatrix} = \mathbf{b}'\Psi_2(1) = \mathbf{0}',$$

so $\mathbf{a}$ satisfies (10.1.20) and is a cointegrating vector. It is linearly independent of the h columns of $\mathbf{A} = \begin{pmatrix} \mathbf{I}_h \\ -\mathbf{\Gamma} \end{pmatrix}$: in any linear combination $\mathbf{A}\mathbf{c} + \mathbf{a} \cdot d = \mathbf{0}$, the first h rows read $\mathbf{c} = \mathbf{0}$ and the last $n - h$ rows then read $d\mathbf{b} = \mathbf{0}$, forcing $d = 0$. Hence there are at least $h + 1$ linearly independent cointegrating vectors, contradicting the assumption that the cointegration rank is h . Therefore no such $\mathbf{b}$ exists: the $n - h$ rows of $\Psi_2(1)$ are linearly independent, i.e. $\Psi_2(1)$ is of full row rank. This is the sense in which $\mathbf{y}_{2t}$ is “not cointegrated” among itself, as claimed in the text.

(b) Absolute summability is elementwise: a matrix filter $\sum_j \mathbf{C}_j L^j$ is absolutely summable if $\sum_j |c_{j,k\ell}| < \infty$ for every element. By (10.1.14), $\alpha(L)$ is absolutely summable because $\Psi(L)$ is one-summable (this is the matrix version of the result used on p. 628). The first block of $\Psi^*(L)$ is $\Psi_1^*(L) = (\mathbf{I}_h \quad -\mathbf{\Gamma}') \alpha(L)$, a fixed linear combination of the n rows of $\alpha(L)$; a finite linear combination of absolutely summable sequences is absolutely summable. The second block $\Psi_2(L)$ is a selection of rows of $\Psi(L)$, which is one-summable and a fortiori absolutely summable. Hence $\Psi^*(L)$ is absolutely summable, and the error vector $(\mathbf{z}_t^*, \mathbf{u}_{2t})$ is a well-defined stationary process.

(c) For Example 10.3, $n = 2$, $h = 1$, and $\Psi(L) = \mathbf{I}_2 + \Psi_1 L$ with $\Psi_1 = \begin{pmatrix} -1 & \gamma \\ 0 & 0 \end{pmatrix}$. From the BN algebra $\alpha(L) = -\Psi_1 = \begin{pmatrix} 1 & -\gamma \\ 0 & 0 \end{pmatrix}$ (10.1.17), a constant filter. With $\Gamma = \gamma$ (scalar),

$$\Psi_1^*(L) = (1, -\gamma) \alpha(L) = (1, -\gamma) \begin{bmatrix} 1 & -\gamma \\ 0 & 0 \end{bmatrix} = (1, -\gamma), \quad \Psi_2(L) = \text{second row of } \Psi(L) = (0, 1).$$

Both are constant in L , so $\Psi_j^* = \mathbf{0}$ for $j \geq 1$ and

$$\Psi_0^* = \begin{bmatrix} 1 & -\gamma \\ 0 & 1 \end{bmatrix},$$

which is not diagonal whenever $\gamma \neq 0$: the contemporaneous map from ε_t to (z_t^*, u_{2t}) feeds ε_{2t} into both $z_t^* = \varepsilon_{1t} - \gamma\varepsilon_{2t}$ and $u_{2t} = \varepsilon_{2t}$. Consequently $\text{Cov}(z_t^*, u_{2t}) = (1, -\gamma)\Omega(0, 1)' = \omega_{12} - \gamma\omega_{22}$, which need not vanish even when Ω is diagonal (it equals $-\gamma\omega_{22}$ then). This is the algebraic origin of the endogeneity of the $I(1)$ regressor in a cointegrating regression, the problem that DOLS and the other efficient estimators of Section 10.4 are designed to remove.

Review Question 10.2.2

The triangular representation (10.2.11) is

$$y_{1t} = \mu + \gamma y_{2t} + (\varepsilon_{1t} - \gamma\varepsilon_{2t}), \quad \Delta y_{2t} = \delta_2 + \varepsilon_{2t}.$$

The second equation is already the second row of the VMA (10.1.12). First-differencing the cointegrating regression,

$$\Delta y_{1t} = \gamma \Delta y_{2t} + \Delta(\varepsilon_{1t} - \gamma\varepsilon_{2t}) = \gamma(\delta_2 + \varepsilon_{2t}) + \varepsilon_{1t} - \varepsilon_{1,t-1} - \gamma\varepsilon_{2t} + \gamma\varepsilon_{2,t-1} = \gamma\delta_2 + (\varepsilon_{1t} - \varepsilon_{1,t-1}) + \gamma\varepsilon_{2,t-1}.$$

Stacking,

$$\Delta \mathbf{y}_t = \begin{bmatrix} \gamma\delta_2 \\ \delta_2 \end{bmatrix} + \begin{bmatrix} \varepsilon_{1t} - \varepsilon_{1,t-1} + \gamma\varepsilon_{2,t-1} \\ \varepsilon_{2t} \end{bmatrix} = \boldsymbol{\delta} + \boldsymbol{\varepsilon}_t + \Psi_1 \boldsymbol{\varepsilon}_{t-1}, \quad \boldsymbol{\delta} = \begin{bmatrix} \gamma\delta_2 \\ \delta_2 \end{bmatrix}, \quad \Psi_1 = \begin{bmatrix} -1 & \gamma \\ 0 & 0 \end{bmatrix},$$

which is exactly the VMA representation (10.1.12) with the Ψ_1 of (10.1.7). Note the consistency check on the drift: the implied $\boldsymbol{\delta}$ satisfies $\delta_1 = \gamma\delta_2$, i.e. the cointegrating vector $(1, -\gamma)'$ eliminates the deterministic trend as well as the stochastic trend, as befits a cointegrating regression without a trend regressor.

Remark. The two representations are not one-to-one in the drift: the triangular system as written embodies the restriction $\delta_1 = \gamma\delta_2$. If $\delta_1 \neq \gamma\delta_2$, the cointegrating combination $y_{1t} - \gamma y_{2t}$ retains a linear trend $(\delta_1 - \gamma\delta_2)t$ and the cointegrating regression needs a time trend as an additional regressor, as the text notes after (10.2.4).

Review Question 10.2.3

In Example 10.4, $\Phi(1) = \mathbf{I}_2 - \Phi_1 = \begin{pmatrix} 1 & -\gamma \\ 0 & 0 \end{pmatrix}$ (10.2.30). Computing the outer product,

$$\mathbf{B}\mathbf{A}' = \begin{bmatrix} \gamma \\ 0 \end{bmatrix} \begin{bmatrix} 1/\gamma & -1 \end{bmatrix} = \begin{bmatrix} 1 & -\gamma \\ 0 & 0 \end{bmatrix} = \Phi(1),$$

so this is indeed a valid decomposition; it is the pair $(\mathbf{B}(\mathbf{F}')^{-1}, \mathbf{A}\mathbf{F})$ of the book's nonuniqueness discussion with the original $\mathbf{B} = (1, 0)'$, $\mathbf{A} = (1, -\gamma)'$ of (10.2.31) and $\mathbf{F} = (1/\gamma)$.

The VECM (10.2.27) for this choice, with $z_t \equiv \mathbf{A}'\mathbf{y}_t = y_{1t}/\gamma - y_{2t}$, is $\Delta\mathbf{y}_t = \boldsymbol{\alpha}^* + \boldsymbol{\delta}^*t - \mathbf{B}z_{t-1} + \boldsymbol{\varepsilon}_t$; written out,

$$\begin{bmatrix} \Delta y_{1t} \\ \Delta y_{2t} \end{bmatrix} = \begin{bmatrix} \gamma\delta_2 + \alpha_1^* \\ \delta_2 \end{bmatrix} + \begin{bmatrix} \delta_1 - \gamma\delta_2 \\ 0 \end{bmatrix} t - \begin{bmatrix} \gamma \\ 0 \end{bmatrix} \left(\frac{y_{1,t-1}}{\gamma} - y_{2,t-1} \right) + \boldsymbol{\varepsilon}_t.$$

The first equation reads $\Delta y_{1t} = \text{const} + (\delta_1 - \gamma\delta_2)t - (y_{1,t-1} - \gamma y_{2,t-1}) + \varepsilon_{1t}$ — numerically the same error-correction term as in (10.2.32), since $\gamma \cdot (y_{1,t-1}/\gamma - y_{2,t-1}) = y_{1,t-1} - \gamma y_{2,t-1}$. The lesson: rescaling $(\mathbf{B}, \mathbf{A})$ in opposite directions changes the *definition* of the error-correction variable and the *size* of the loading coefficient, but not the model. Only the cointegrating space (the line spanned by $(1, -\gamma)'$) and the adjustment space (the line spanned by $(1, 0)'$) are determined by the data; the split of $\mathbf{B}\mathbf{A}'$ into the two factors is a normalization.

Review Question 10.2.4

Write the system as $\boldsymbol{\Phi}(L)\mathbf{y}_t = \boldsymbol{\varepsilon}_t$ with

$$\boldsymbol{\Phi}(L) = \begin{bmatrix} 1 - 2L + L^2 & 0 & 0 \\ 0 & 1 - \rho L & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} (1 - L)^2 & 0 & 0 \\ 0 & 1 - \rho L & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Then

$$\det \boldsymbol{\Phi}(z) = (1 - z)^2(1 - \rho z),$$

whose three roots are $z = 1$ (a double root — two unit roots) and $z = 1/\rho$, which lies outside the unit circle because $|\rho| < 1$.

For the order of integration, premultiply $\boldsymbol{\Phi}(L)\mathbf{y}_t = \boldsymbol{\varepsilon}_t$ by the diagonal matrix $\text{diag}(1, \frac{1-L}{1-\rho L}, (1-L)^2)$:

$$\Delta^2 \mathbf{y}_t = \begin{bmatrix} \varepsilon_{1t} \\ \frac{1-L}{1-\rho L} \varepsilon_{2t} \\ (1-L)^2 \varepsilon_{3t} \end{bmatrix}.$$

This is a VMA process whose matrix filter is absolutely summable (the $(2, 2)$ element is the two-sided summable filter of a stationary AR(1) written in differences; explicitly, $\frac{1-L}{1-\rho L} = 1 - (1 - \rho) \sum_{j \geq 1} \rho^{j-1} L^j$), but its value at $L = 1$,

$$\text{diag}(1, 0, 0),$$

is nonzero and singular. So $\Delta^2 \mathbf{y}_t$ is a zero-mean $I(0)$ system in the sense of (10.1.1)–(10.1.4) (one-summability holds and the filter at one is not the zero matrix), which makes $\mathbf{y}_t$ an $I(2)$ system: it must be differenced twice to become stationary. Element by element, y_{1t} is $I(2)$, y_{2t} is stationary, and y_{3t} is white noise.

The example delivers the point made in the text after (10.2.19): counting unit roots of $|\boldsymbol{\Phi}(z)|$ is not enough to identify a cointegrated $I(1)$ system. Here $|\boldsymbol{\Phi}(z)|$ has $n - h = 2$ unit roots with the remaining root outside the unit circle — the count that for a genuine cointegrated $I(1)$ system with

rank $h = 1$ would be expected — yet the system is not $I(1)$ at all. The two unit roots are piled on the *same* element y_{1t} , rather than distributed over two distinct common trends; the factorization $\Phi(z) = \mathbf{U}(z)\mathbf{M}(z)\mathbf{V}(z)$ with $\mathbf{M}(z) = \text{diag}(1-z, 1-z, 1)$ fails, because with $\mathbf{U}(1), \mathbf{V}(1)$ nonsingular, $\Phi(1)$ would have rank one, whereas here $\Phi(1) = \text{diag}(0, 1-\rho, 1)$ has rank two — inconsistent with $h = 1$.

Section 10.3 Testing the Null of No Cointegration

Review Question 10.3.1

Case 1 (no drift anywhere) uses TABLE 10.1(A), which for g $I(1)$ regressors tabulates the Phillips–Ouliaris Table IIb distribution; case 2 (some regressors have drift) uses TABLE 10.1(B) for g regressors, which is the Table IIc distribution for $g - 1$ regressors (the first row, $g = 1$, being the Dickey–Fuller t_τ distribution). Reading off the table:

g	case 1: Table 10.1(a)				case 2: Table 10.1(b)			
	1%	2.5%	5%	10%	1%	2.5%	5%	10%
1	−3.96	−3.64	−3.37	−3.07	−3.96	−3.67	−3.41	−3.13
2	−4.31	−4.02	−3.77	−3.45	−4.36	−4.07	−3.80	−3.52
3	−4.73	−4.37	−4.11	−3.83	−4.65	−4.39	−4.16	−3.84
4	−5.07	−4.71	−4.45	−4.16	−5.04	−4.77	−4.49	−4.20
5	−5.28	−4.98	−4.71	−4.43	−5.36	−5.02	−4.74	−4.46

The differences between the two panels are at most a few hundredths (e.g., at 5%: -3.41 vs. -3.37 for $g = 1$, -3.80 vs. -3.77 for $g = 2$, -4.16 vs. -4.11 for $g = 3$), the case-2 value being slightly more negative.

The reason is the asymptotic equivalence behind case 2: if some of the g regressors have drift, their linear trends can be combined into one, and the drift-dominated regressor behaves in large samples like a time trend; the regression is then asymptotically equivalent to a regression on a constant, $g - 1$ driftless $I(1)$ regressors, and time. A regression on a constant and g driftless $I(1)$ regressors (case 1) and a regression on a constant, $g - 1$ driftless $I(1)$ regressors and a linear trend (case 2) differ only in that the deterministic trend replaces one stochastic trend — and a linear trend is, in the functional central limit theory that generates these distributions, the limit of a stochastic trend with vanishing innovation variance. Hence the two limiting residual processes, and with them the two ADF t distributions, are close but not identical: the drift case has effectively one fewer dimension of stochastic variation around which the fitted regression can wander, which shifts the distribution slightly to the left.

Remark. The practical upshot is the text's recommendation: when in doubt about the location of drifts, include time in (10.3.1) and use TABLE 10.1(B) with $g + 1$ regressors, which is valid for all three cases — at some cost in finite-sample power, as Hansen's (1992a) simulations show.

Section 10.4 Inference on Cointegrating Vectors

Review Question 10.4.1

In both regressions a stationary regressor and a driftless $I(1)$ regressor coexist, and in both the suitably scaled “ $\mathbf{X}'\mathbf{X}$ ” matrix is asymptotically block diagonal, so the coefficient of the $I(1)$ regressor can be analyzed as if the $I(0)$ regressors were absent. The difference lies in the *error term*.

In the augmented autoregression (9.4.6), the $I(1)$ regressor y_{t-1} is the lag of the dependent variable, so the error ε_t is one of the innovations driving the stochastic trend in y_{t-1} : the two are correlated by construction (indeed ε_t is the innovation that builds y_t). The normalized numerator of the t -ratio converges to a stochastic integral $\int W \, dW$ -type functional — a nonstandard, skewed Dickey–Fuller distribution — and the denominator does not erase it.

In the augmented cointegrating regression (10.4.5), the projection of z_t^* on the current (and, with leads and lags, all) values of Δy_{2t} produces an error v_t with $\text{Cov}(\Delta y_{2s}, v_t) = 0$ for *all* s and t (10.4.4), (10.4.9), and this carries over to the level: $\text{Cov}(y_{2s}, v_t) = 0$ for all s, t (10.4.7). The $I(1)$ regressor is thus *strictly exogenous*, not merely predetermined. Under joint normality, strict exogeneity makes v_t independent of the whole path $(y_{2,1}, \dots, y_{2,T})$, so the numerator $\frac{1}{T} \sum_t y_{2t} v_t$ of the normalized t -ratio (10.4.8) is conditionally normal with mean zero and variance $\sigma_v^2 \frac{1}{T^2} \sum_t y_{2t}^2$ — exactly the denominator of $\tilde{t}$ — giving $\tilde{t} \mid (y_{2,1}, \dots, y_{2,T}) \sim N(0, 1)$, hence unconditionally $N(0, 1)$. Without normality the same normalization still converts the (nonstandard) limit of the numerator into a standard normal (Lemma 5.1 of Park and Phillips, 1988), because the stochastic trend in the denominator is independent of the limit process of the numerator: the ratio is a normal mixture whose mixing variable cancels.

So the distinction is predeterminedness versus strict exogeneity. The ADF regression can at best orthogonalize the error against *lagged* changes; the future innovation in y_{t-1} is the error itself, and the resulting correlation between the regressor and the error is what bends the limiting distribution away from normality. In the cointegrating regression the leads and lags of Δy_{2t} soak up the correlation in *both* directions of time, and what remains is uncorrelated with the entire history of the regressor.

Remark. The normality is purchased only for the coefficient of the $I(1)$ regressor: as the text warns, the t -values on the β 's of (10.4.11) are not covered by the argument, because the projection coefficients are coefficients of stationary regressors, for which the $N(0, 1)$ limit requires the usual stationary-regression conditions.

Problem Set for Chapter 10
Empirical Exercise 10.1

All computations are in `code/ch10.py`. Figures 1–3 reproduce the book's Figures 10.2–10.4 from the data.

(a) *Univariate unit-root tests.* For each series x_t the augmented autoregression

$$x_t = \text{const} + (\rho - 1)x_{t-1} + \zeta_1 \Delta x_{t-1} + \dots + \zeta_p \Delta x_{t-p} + \text{error}$$

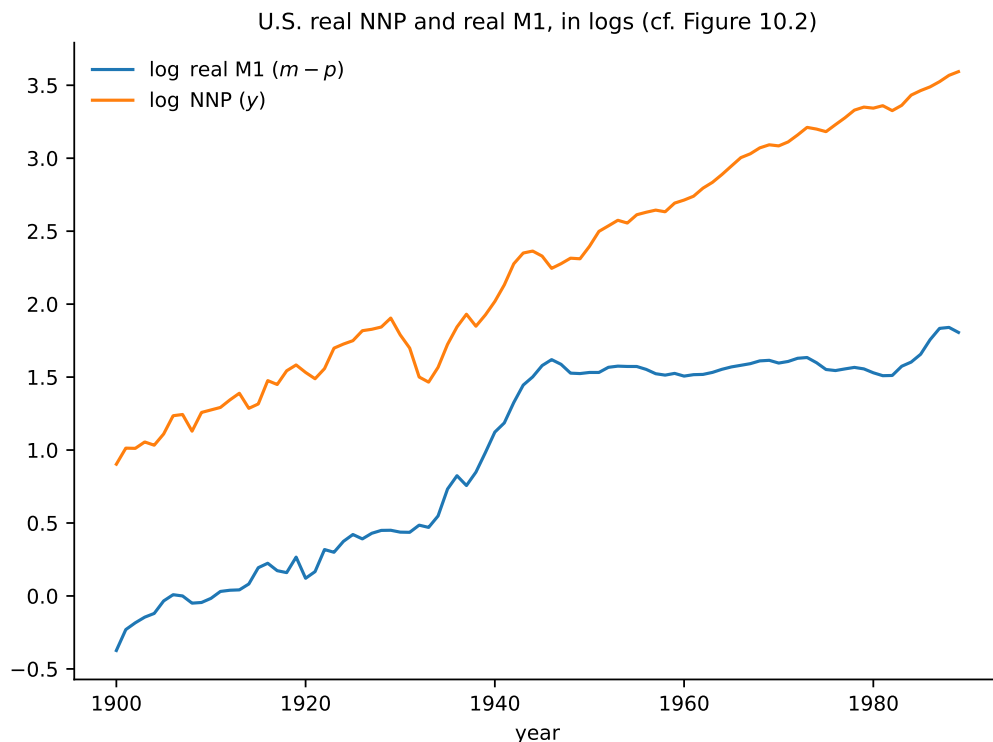


FIGURE 1. Log real M1 ($m - p$) and log NNP (y), 1900–1989 (cf. the book’s Figure 10.2).

(with a linear time trend added for the t_τ test) is estimated over the maximal sample that p lagged changes allow, $t = p + 2, \dots, T$; the ADF statistic is the t -value on x_{t-1} . The asymptotic critical values ($T = \infty$) from TABLE 9.2 are -2.86 (5%) and -2.57 (10%) for t_μ and -3.41 (5%) and -3.12 (10%) for t_τ . The computed statistics:

series	test	$p = 2$		$p = 4$		conclusion
		ADF t	reject at 10%?	ADF t	reject at 10%?	
$m - p$	t_τ	-0.99	no	-1.57	no	$I(1)$ not rejected
y	t_τ	-3.45	yes (not at 5%)	-2.84	no	borderline
R	t_μ	-1.29	no	-1.05	no	$I(1)$ not rejected

For $m - p$ and R the unit-root null is not rejected at 10 percent at either lag length; for y it is not rejected with 4 lags but is rejected at the 10 percent level (not 5 percent) with 2 lags — exactly the Stock and Watson (1993, Appendix B) findings. So all three series may well be individually $I(1)$, with $m - p$ and y drifting and R apparently driftless.

(b) *Engle–Granger test.* The static (SOLS) regression of $m_t - p_t$ on a constant, y_t and R_t has residuals whose augmented autoregression without constant or time (10.3.2) is estimated with $p = 1$ lagged change — also what BIC selects on the residuals. The ADF t -statistic on $\hat{z}_{t-1}$ is -4.67 when the cointegrating regression uses the full sample 1900–1989 (the book’s “about -4.7 ”) and -4.44 when it uses 1903–1987, the DOLS sample of part (c). Since the regressors y and R contain drift/trend, case 2 of Section 10.3 applies with $g = 2$ regressors, and the 5 percent

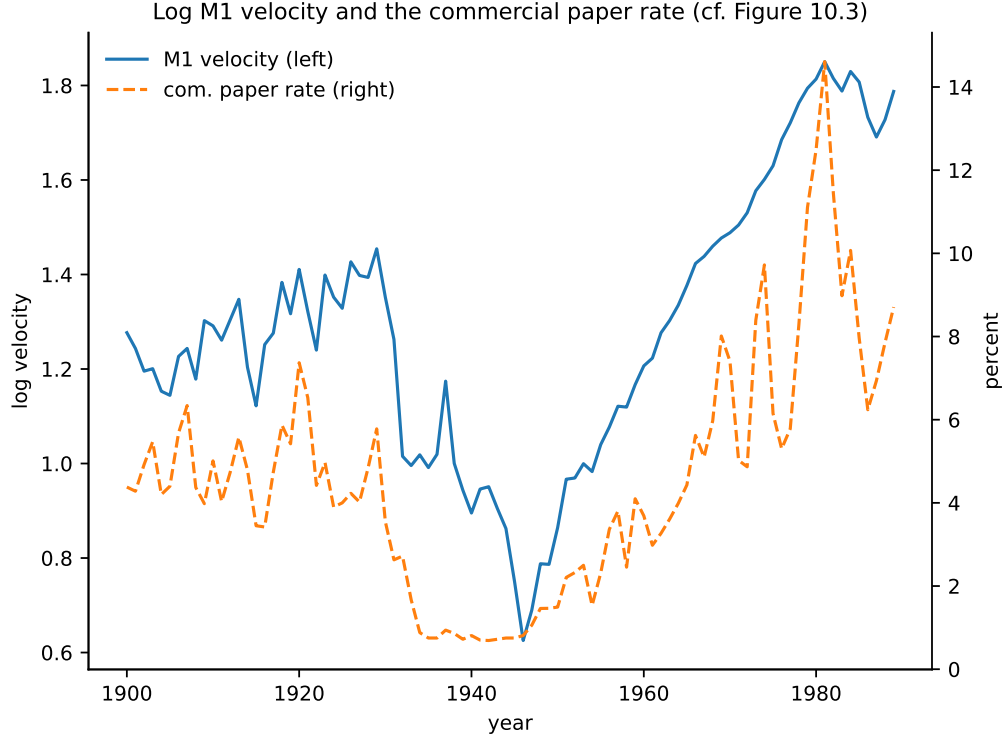


FIGURE 2. Log M1 velocity $y - (m - p)$ (left scale) and the commercial paper rate R (right scale) (cf. Figure 10.3). Velocity tracks the interest rate, which is what suggests cointegration with a coefficient on y near unity.

asymptotic critical value from TABLE 10.1(B) is -3.80 . Either way the null of no cointegration is rejected (even the 2.5 percent value -4.07 is exceeded): the presumption that $m - p$ is cointegrated with (y, R) survives the test.

(c) *SOLS and DOLS, 1903–1987*. The SOLS regression is the static regression of (b); the DOLS regression adds Δy_t , ΔR_t and two leads and lags of each (10.5.2), which dictates the sample period $t = 1903, \dots, 1987$ (85 observations, 13 regressors). The long-run variance of the error is estimated by fitting an AR(2) to the DOLS residuals over $t = 1905, \dots, 1987$ (83 observations), giving (10.5.3),

$$\hat{v}_t = 0.93806 \hat{v}_{t-1} - 0.13341 \hat{v}_{t-2} + \hat{e}_t, \quad \text{SSR} = 0.19843,$$

and then (10.4.17) with $\hat{\sigma}_e^2 = \text{SSR} / (T - p - K) = 0.19843 / (83 - 2 - 13) = 0.0029181$:

$$\hat{\lambda}_v^2 = \frac{0.0029181}{(1 - 0.93806 + 0.13341)^2} = 0.07647, \quad \hat{\lambda}_v \approx 0.277,$$

so the rescaling factor in (10.4.15) is $\hat{\lambda}_v / s \approx 0.277 / 0.096 \approx 2.9$. Every figure in TABLE 10.2 is reproduced:

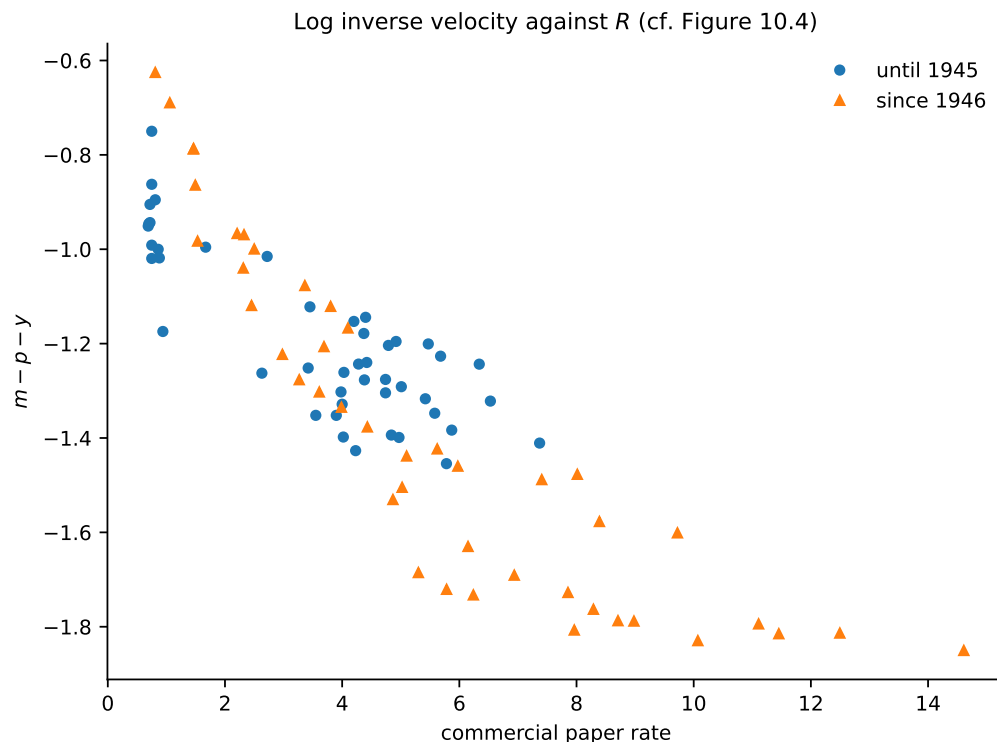


FIGURE 3. Lucas's (1988) scatter: log inverse velocity $m - p - y$ against R , pre-1946 (circles) and postwar (triangles) (cf. Figure 10.4). Postwar points lie on the prewar relation.

	γ_y	γ_R	R^2	std. error of regression
SOLS	0.943	-0.082	0.959	0.135
DOLS	0.970	-0.101	0.982	0.096
(rescaled std. errors)	(0.046)	(0.013)		

With the rescaled standard errors, the t -statistic for a unitary income elasticity is $(0.970 - 1)/0.046 \approx -0.7$: the income elasticity is not significantly different from one, and the interest semielasticity of about -0.10 (a one-percentage-point rise in R lowers real money demand by about 10 percent) is highly significant.

(d) *Chow test at 1946*. The augmented cointegrating regression of the question adds to (10.5.2) the three break variables D_t , $y_t D_t$, $R_t D_t$ with $D_t = \mathbf{1}\{t \geq 1946\}$. The DOLS point estimates reproduce TABLE 10.4:

$$\hat{\gamma}_y = 0.925, \quad \hat{\gamma}_R = -0.090, \quad \hat{\delta}_0 = 1.36, \quad \hat{\delta}_y = -0.52, \quad \hat{\delta}_R = 0.048.$$

The AR(2) fit to the residuals of this 16-regressor regression gives $\hat{\sigma}_e^2 = \text{SSR}/(83 - 2 - 16)$, $\hat{\lambda}_v = 0.204$, and a rescaling factor $\hat{\lambda}_v/s = 0.204/0.0767 = 2.65$; the rescaled standard errors are (0.142), (0.026), (0.72), (0.31), (0.034), matching the table.

For the null $\delta_0 = \delta_y = \delta_R = 0$, the unscaled Wald statistic from the unrestricted regression is $W_0 = 43.2$; rescaled by $(s/\hat{\lambda}_v)^2$ as prescribed in the text it is $W = 6.13$, with a $\chi^2(3)$ p-value of

0.11; divided by the number of restrictions (F -style) it is 2.04 against the book's 1.85 (p-value 0.60). The discrepancy appears to be in the precise scaling convention (degrees-of-freedom corrections in $\hat{\sigma}_e^2$, whether $\hat{\lambda}_v$ is computed from the restricted or unrestricted residuals, division by the number of restrictions): every variant tried in `code/ch10.py` gives a statistic between 1.1 and 6.4, and every variant is insignificant at conventional levels. The conclusion is therefore the book's: the null of no structural change is not rejected, and the data are consistent with Lucas's (1988) claim of a stable twentieth-century money demand — the apparently different postwar estimates of TABLE 10.3 ($\hat{\gamma}_y = 0.269$, $\hat{\gamma}_R = -0.027$ for 1946–1987, against 0.887 and -0.104 for 1903–1945) reflect the near-collinearity of y and R in the postwar subsample rather than a genuine shift, as Figure 3 makes visually plain.

Remark. A caveat the book itself flags (footnote 16): the whole inference takes the cointegration rank $h = 1$ as given, although it was itself established by the pretest of part (b); the distribution of the DOLS estimates does not account for the pretest uncertainty.